

THE BUSINESS OF SHIPPING

FIFTH EDITION

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Fifth Edition

by LANE C. KENDALL

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For my sister

ELISABETH KENDALL THOMPSON

Fellow, American Institute of Architects

in appreciation of

her lifelong friendship and unswerving support

in many times of crisis

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Preface

THIS VOLUME has been written to describe the business side of a commercial enterprise whose field is the entire civilized world. Historically, the theory and knowledge of shipping management, as distinguished from the practical skills of seamanship, have been transmitted from one generation to the next by word of mouth. Little has been put on paper, primarily because the finest exponents of the art of steamship management have been too busy with their day-to-day concerns to do so. The "working level" personnel often are superbly competent, but rarely qualify as literary craftsmen.

It has been my aim, in preparing this analysis of the principles of the "business" of commercial shipping, to describe that which transpires in the various divisions of a shipowning and operating organization. Insofar as possible, the procedures followed in the offices have been described and explained, as well as the underlying principles of management by which their decisions are reached.

In the process of learning the principles and practices that are set forth in these pages, I have spent a joy-filled lifetime in association with ships. It has been my good fortune to work in large and small American steamship offices, to operate a major cargo terminal, to participate in establishing and putting into effect the policies of a world-girdling American steamship organization, and to teach young men these principles learned from experience as well as from precept.

My years with ships and seamen have been filled with much toil and many strains. They also have been crowded with pleasant associations and fascinating experiences. I have discovered that with increased knowledge of the endless ramifications of the steamship business my interest in each day's work has deepened and grown more rewarding. May I hope that those who read this book

also will gain greater understanding of a worthwhile way of life, and thus heighten their appreciation of the achievements of each passing day?

This volume is the fifth edition; the first appeared in 1973. While much of the material found in the fourth edition has been retained, a significant amount of new information has been added in order to reflect new developments in the industry and to bring the book up to date. It is my hope that the depth of detail and analysis which characterize this present volume will prove to be of value to the readers of these pages.

The names of those who have assisted me throughout the decades of my association with the steamship industry are far too numerous to list, even if all of them were known to me. Some who have contributed importantly to my knowledge were nameless to me, but I have not forgotten their generous sharing of experience and hard-earned wisdom. Friends and correspondents in every type of shipping office around the world have been more than gracious in their willingness to contribute in some measure to my professional edification. To them all, I extend my heartfelt thanks.

Any errors, of course, are my own, and I assume full responsibility.

—Lane Carter Kendall

THE BUSINESS OF SHIPPING

Introduction

SHIPS and the management of ships represent one of man's oldest economic enterprises. There is no time in recorded history when man was not using watercraft of some sort to provide communication, and with communication inevitably came trade. Trade began first between tribes, rapidly extending to nations over routes that were pioneered before the dawn of history. During the great Age of Exploration, the epochal voyages of Columbus, Magellan, Vasco da Gama, and the illustrious company of "gentlemen adventurers" established the principle that man could plan where he would market his goods, and could build the ships needed to transport those goods.

Today, approximately 25,000 ships of oceangoing size ply the sea-routes. There are no seas that have not been traversed and there are no shores untouched; there is no time of day or night when scores of ships are not pushing their toilsome way across the broad expanses of the oceans. There is, likewise, no hour when someone ashore is not concerned with the welfare of the ships at sea. The need to find cargoes and passengers; the requirement to provide berthing space and representatives to assure that the ship is attended promptly, or supplied with fuel, stores, and water; the endless search for crew members to replace those unavoidable casualties of the sea; and the problems of repairing damage suffered in the course of sea-travel—these are the responsibilities of the owners, managers, and agents of commercial vessels.

The business of shipping, then, is a world-encircling enterprise whose devotees must have a vast knowledge of matters great and small, an enthusiasm for adventure even when it is vicarious and experienced through the medium of telegrams and letters, and a breadth of vision that stretches beyond the horizons of the seas. It is a world removed in some degree from the ordinary routines of

commerce, for each ship becomes a personality; each voyage a separate undertaking; each day a time of challenge; each event in far-off lands of possible significance in the movement of the ships across the waters. No other human activity is so important to so many people, and no other enterprise is so subject to conditions over which it has no control. To those whose lives are engaged in the ramified operations of shipping, it is the most rewarding of business ventures because there is the awareness that without ships, life would lack much of its variety and vitality.

For those who heed the call of the sea; who face the rigors of life in constant conflict with the great forces of nature; who find the romance of far places not just across the horizons but also in the bales and boxes, the “black gold” pumped into vast tanks, and the people who travel from land to land, there is a peculiar and deep satisfaction. Their lives, anonymous though they may be to the world, are devoted to the service of their fellows, to making the world better, to creating new ideas, and to binding together the affairs of humanity. These men—and increasingly these women—are the professionals who find true happiness only when they are “off soundings,” away from the humdrum routine of life ashore. They represent a special breed of heroic stature who ridicule any notion of romance and laugh at the sweet words of the poet who writes of far places and entrancing shores. They complain that they never have a chance to make friends ashore, but return voyage after voyage to the irresistible ships and the lure of the unknown beyond the horizon. They serve, and that is their reason for living.

It is about this fascinating world of shipping, of international commerce, of strange and exciting names and places, of unending variety, as well as rigorous competition, that this book is written. Its purpose is to lift the veil of mystery just enough to indicate why this fascination exists; to introduce those who would learn about a specialized world to the ideas and the thinking of this ancient yet new and exciting dominion of the seas.

Liner Service and Tramp Shipping

MERCHANT SHIPPING, considered from the standpoint of types of service provided, may be divided into two major categories: *Liner Service* and *Tramp Shipping*. While there are some similarities, the differences in the theory and techniques of management of these two types of marine transportation are notable. The service rendered, the geographic area covered, the operating problems, the relationship between vessel owner and vessel user, and the actual employment of the ship, vary markedly between the two categories. It is important, therefore, to be aware of those areas where the management procedures are congruent; it is equally significant that the differences be comprehended.

Liner Service

1. Sailings are regular and repeated from and to designated ports on a trade route, at intervals established in response to the quantity of cargo generated along that route. True liner service is distinguished by the repetition of voyages and the consistent advertising of such voyages. Once the service is established, the operator must conform, within narrow time limits, to the published schedule. Although the frequency of sailings is related directly to the amount of business available, it is general practice to dispatch at least one ship each month. Vessels engaged in liner service may be owned or chartered;

Tramp Shipping

1. Sailings are based on cargo commitments that vary with the vessel's employment, and are usually different for every voyage. There is no expected repetition of voyages as a normal part of tramp operation. Each trip is scheduled individually, subject to the requirements of the cargo to be carried and the particular route to be followed. In certain trades, such as oil and coal, owners often agree to make a number of repetitious voyages carrying the same commodity. These "consecutive voyages" are arranged expressly to fit the charterer's convenience, and do not establish a "liner service."



Top: A conventional break-bulk cargo liner, the *California* (built 1962) has a speed of 20 knots and a deadweight of 14,439 tons. *Bottom:* A combination break-bulk and containerized cargo carrier, the 18.2-knot *Menelaus* (built 1977) was 164.6 meters long, and had a deadweight of 21,242 tons.

Photos: Georgia Ports Authority.

it is the regularity and repetitious nature of the operation, rather than the proprietorship, which is crucial.

2. Liners are common (public) carriers, required by law to accept without discrimination between offerers any legal cargo which the ship is able to transport. Some liner operators stipulate the minimum quantity of cargo which must be presented by a single shipper; so long as the limitation is reasonable,

2. Tramps are contract (private) carriers, and normally carry full shiploads of a single commodity, usually in bulk. In most cases, there is only one shipper, but two or more shippers of the same kind of cargo occasionally may use a single ship.

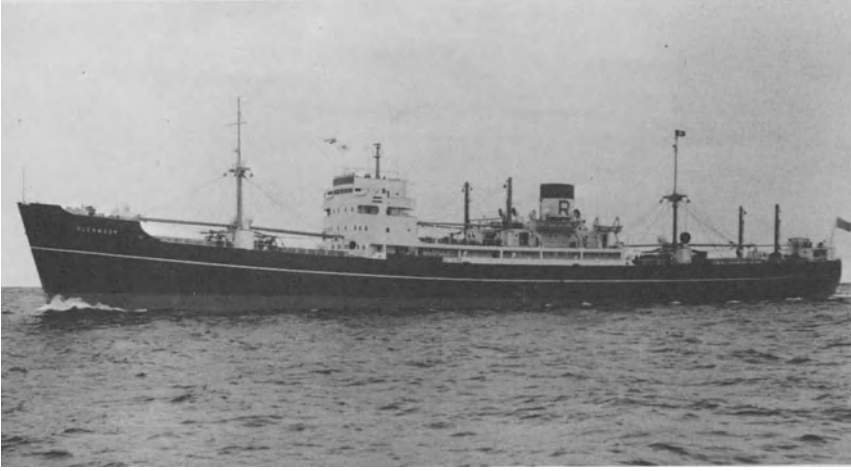
this is permissible. Cargo usually is varied, and is called either “general” or “package.” In container transportation, everything is packed into the large boxes before they are placed aboard ship. Operators accept small shipments for consolidation, i.e., packing into containers with other small lots until the boxes are filled.

3. Goods carried in liner-service ships usually are of higher value than the cargo hauled in tramps, and are charged higher freight rates. The fact that common carriers accept less than shipload lots, nearly always in packages (including containers) requiring special care in stowage, also influences the establishment of liner rates. Handling (“stevedoring”) charges always are included in the freight rate. Because of the variety of commodities loaded in every port of call, great care must be exercised by ship operators to assure delivery in good condition. Freight rates are identical for all shippers of a given item transported in the same ship.

4. A liner-service company issues a standard (or uniform) contract of carriage or *bill of lading*. Regardless of the size of the shipment, or the number of different commodities or items comprising a given lot of cargo, the provisions of the contract apply equally to all shippers who use any one vessel. These provisions are not subject to negotiation, but are unilaterally imposed by the carrier. Only in very exceptional cases will a senior

3. Cargoes carried in tramps generally are those which can be transported in bulk (“homogeneous cargoes”) and have low intrinsic value. Typical cargoes are coal, ores, grain, lumber, sugar, and phosphate rock. The cost of loading and unloading the ship in most cases is paid by the charterer, but this is subject to negotiation between shipowner and charterer. Freight rates for tramps reflect the fact that movements frequently are from a single port of loading to a single port of discharge, with minimum expense involved in the care of the cargo while in transit.

4. The owner of a tramp ship must negotiate a separate contract for each employment of his vessel, and the terms of the charter-party vary from ship to ship, depending upon the bargaining abilities of owner and charterer, and the general trend of the market. The terms of the agreement are applicable only to the ship named in the charter-party. Although the basic charter-parties are printed and follow a set form, they may be changed in any



Glenmoor, an economical, rugged, simple, general-purpose tramp ship built in 1954. Photo: Moor Line.

executive of the common carrier alter the terms of the bill of lading to accommodate an individual shipper.

5. Freight rates in the liner services are stabilized by setting identical charges for all shippers of the same item aboard a certain ship. Rates may vary, however, from one sailing to another, but increases are announced in advance. Rates are compiled into detailed listings ("freight tariffs") which are made available to shippers on demand. Frequently, two or more carriers serving a particular trade route form an association ("conference") intended to stabilize rates and regulate competition. Conference rates apply uniformly to all member lines, but are subject to review. (In the United States, the reviewing authority is the Federal Maritime

manner desired by the contracting parties. Since the changes apply to a particular ship for a particular voyage or period of time, the alterations are not publicized widely.

5. Freight rates for tramps vary according to the supply of and demand for ships. The charterers' position is strong, and rates are low, when there are comparatively few cargoes being offered and many ships are competing for the business. The shipowners' position is strong, and rates are high, when there are many cargoes and a small number of vessels. Competition among owners is keen, and often a difference of five cents a ton on a shipload lot will determine which ship is chartered. Wide extremes in rates, and abrupt changes in the level of charter rates, are evident whenever there is a major event of international significance—the



Deck petty officer's quarters in a well-equipped British tramp. Photo: Moor Line.

Commission.) The necessity to conform to government regulations imposed upon common carriers tends to prevent the freight rates charged by liner-service operators from making sudden sharp changes.

6. Services—frequency of sailings and ports of call, as well as the capabilities of the ships themselves—are adjusted to meet the demands of shippers. Many liner operators arrange their schedules to meet minimum needs during the

outbreak of a war, a major crop failure, widespread strikes in some country, for example. No freight tariffs are compiled by owners of tramp vessels, and no associations exist for the purpose of setting rates and stabilizing competition. Summaries of the freight market are published regularly and include quotations of the rates at which ships have been chartered.

6. Rates and services are determined by negotiation between shipowner and charterer, and reflect the specific requirements of the contracting parties. Regular and repeated voyages on the same route are not part of tramp opera-

year, and then augment sailings when seasonal increases are experienced. Changes in liner service often are influenced as much by political and technological considerations as they are by economic factors. Drastic changes in established liner operations are infrequent; carriers' intentions, especially relating to withdrawals from the route, usually are well publicized in advance. This is essential to the dependability of the liner trade.

7. Liner-service vessels often reflect in their design the special requirements encountered in their employment. Refrigerated fruit and meat carriers, heavy-lift ships, roll-on/roll-off vessels, container-ships, and break-bulk ships may be found on many routes, depending upon the demand for these specialized capabilities. Because more or less identical cargoes move in all the dry-cargo liners on a given trade route, many of the ships employed on that route, regardless of owner, are similar in capability. Containerships vary in their cruising speed and the number of containers they carry, but otherwise are very similar, wherever they are used. Since 1945, liner ships of all types have been growing in size and increasing in speed; inevitably the cost of their acquisition and operation has risen steeply.

8. Liner-service companies have a large and somewhat complex organization in the shore establishment, especially in the home

tion, and therefore no conferences exist. Supervision in the public interest by a regulatory authority is unnecessary; the natural working of the laws of supply and demand assure adequate control.

7. Most tramp ships are intended for worldwide service, and are of moderate size and draft. Although used primarily to transport cargoes in bulk, many tramps have a single 'tween deck and sufficient equipment and speed to permit them to be chartered for use in the break-bulk liner trades. Additionally, many modern, very large carriers designed to transport a single commodity such as ore or coal, have been placed in tramp-type operation. Compared with vessels constructed for the liner trades, tramps still are simpler in design and less costly to construct.

8. Tramp owners usually have small staffs in the home office, with little division of functions. No traffic department is needed; charters

office. Normally there are several divisions (i.e., traffic, operations, financial, and managerial) with an appropriate staff. Outport offices may duplicate this organization on a smaller scale. Liner-service operations entail contact with shippers, maintenance of an active cargo-handling terminal, and processing a great amount of detail work inherent in general cargo service on a repetitious schedule.

9. Procurement of cargo is the responsibility of the traffic department, which includes salesmen ("solicitors") to call on regular as well as prospective shippers. Advertising is extensive and continuous, and major efforts are made to disseminate information concerning the capabilities of the line. Arrival and departure times of ships are widely publicized. Shippers are assisted in the development of markets for their goods as a means of increasing cargo offerings.

10. Passengers sometimes are carried in cargo liners, but by international agreement the number is limited to twelve.

are negotiated by telephone or cable, and face-to-face contact with charterers is unusual. Because agents are employed to service the ships in ports of call, and are paid on a fee basis for each task performed, there is no need for an operating department. If the size of the owner's fleet justifies the expense, home office personnel may include specialists to assure satisfactory performance of the ships. Stevedoring very rarely is the responsibility of the owner, and therefore no terminal department is included in the home office.

9. Procurement of cargo is handled through brokers who represent the tramp shipowner in negotiations with other brokers representing cargo interests. There is no advertising, and no promotional activity. Ship movements normally appear only in the newspaper listings of vessels which have arrived or sailed.

10. Passengers are not carried aboard tramp ships, and no provision for their accommodation is made in the vessels' design.

Tramp Shipping

DURING the half century which ended in 1914, most of the transoceanic cargoes of coal, ores, and grain were carried in comparatively small, general-purpose cargo ships. Moving as they did on routings which varied from voyage to voyage, hauling now one commodity and then another, entering and leaving port without fanfare or publicity, and often in need of paint, these vessels gained the appellation of “tramp steamers.” To the extent that their wanderings and unkempt appearance paralleled those of that fraternity of homeless men known as “tramps,” the sobriquet may have been appropriate. What was not apparent was the careful and constant concern exercised over the ships by their owners, contesting for the available business with each other and with the shippers of goods needing ocean transportation, and taking pride in their ability to deliver the goods in accordance with contracts drawn up with consummate skill and negotiated with great vigor.

The purpose of tramp ships may be stated quite simply: to provide the efficient, convenient, timely, and economic transportation required by the many kinds of goods needed for a complex, industrialized society. Converting this declaration of purpose into the reality of cross-ocean movement of millions of tons of commodities typified by ores, coal, grain, and fertilizers is a task requiring enormous skill and a multitude of resources. Underlying the dedicated effort of the people afloat and ashore who comprise the tramp shipping industry is the necessity to meet constantly changing demands, and therefore to find the ways and means of assuring the flexibility of ship operation which is one of the principal reasons for the existence of the tramp.

I wish to acknowledge my debt to Mr. J. Bes, of Hilversum, Netherlands, author of *Chartering and Shipping Terms*, for his invaluable assistance in the preparation of the three chapters on tramp shipping.



A modern general-purpose tramp of 26,592 tons deadweight, the 15-knot *Mario G. L.* was built in 1974. Photo: Jeff Blinn, courtesy Moran Towing Co.

It is an acknowledged fact of both geography and economics that the raw materials so essential to the life of industrialized nations rarely are located close to the places where they are needed. Chrome ore is vital as an ingredient in high-grade steel, but it is found in large quantities only in comparatively underdeveloped areas. The availability of inexpensive and dependable transoceanic transportation is of crucial importance to both producing and consuming regions. More dramatic, perhaps, because of the greater volatility of demand, is the growth and distribution of grain. From the vast and fertile plains of the United States, Canada, Argentina, and Australia comes a never-ending stream of the golden staple. For generations, there has been a steady and predictable movement of grain from these four producing regions to England, Northern Europe, and the Mediterranean nations. Irregularly, depending upon climatic and meteorological conditions beyond the control of man, there have been urgent appeals from India, China, Russia, and those areas of Africa where drought and overgrazing have combined to make very real the terrifying threat of famine. Tramp ships fit into both delivery patterns, providing the means of transportation in the established trades and, whenever

required, shifting to new routes to alleviate critical shortages wherever they may occur. Let a crop fail in Argentina, however, and the distribution of ships reflects immediately the reduction in the cargoes available for export. Should the unlikely combination of a crop failure in Argentina coincide with a sharply curtailed harvest in China, the rearrangement of tramp shipping routes to meet the new requirements for transportation would be on a world-wide basis.

This flexibility in adapting to unanticipated and major changes in patterns of shipping has made the tramping industry disproportionately significant in the economic life of the world. No other type of shipping service can claim to be so sensitive to demand. The very nature of tramp shipping is to seek cargoes where they may be found, rather than to operate on a fixed route with a rigid schedule of departures and arrivals regardless of the quantity of goods which may be offered for carriage.

Another reason for the continued existence of tramp shipping lies in the fact that it provides the indispensable over-ocean transportation at the lowest possible cost. Metallic ores, grains, forest products, fertilizers, and related commodities are carried on surprisingly long voyages for freights which often aggregate only one-tenth to one-third of the sales prices of the goods. A measure of what alternative modes of transport might charge was revealed by a study made in 1975 which showed that the expense of sea-carriage was about one percent of the cost of moving the same item the same distance by air, five percent of the charge to haul by motortruck, and ten percent of the railroad freight bill. This same study also brought out that the principal routes and average length of voyage for the major tramp cargoes were:

Iron ore: Australia to Japan; Australia to Europe; Scandinavia to Northwest Europe; West Africa to Europe; Brazil to Japan; West Coast of South America to Japan; Indian Ocean ports to Japan; Indian Ocean ports to Europe; and Venezuela to the United States. The average voyage was 4,800 nautical miles in length.

Grain: United States and Canada to Europe; United States and Canada to the Far East; United States and Canada to the Soviet Union; Argentina to Europe; Argentina to the Far East; and Australia to the Far East. The length of the average voyage was 5,150 nautical miles.

Coal: United States to Japan; Australia to Japan; United States to Europe; Canada to Japan. There also was significant movement within the European area. The average voyage extended over 4,700 nautical miles.

Forest Products (including logs, sawn lumber, wood-pulp, and paper): North America to Japan; Northwest Coast of the United States and British Columbia to Gulf of Mexico and Atlantic ports, or to Europe; Scandinavia to Northwest Europe; Southeast Asia to Japan. Voyages on the average extended to about 2,700 nautical miles.

To generate the total of over three trillion ton-miles of ocean transportation which the movement of these commodities required, on January 1, 1983, there was a fleet of 19,495 ships. Based on an analysis of shipping statistics completed in 1975, the deadweight of the typical tramp steamer would have been not more than 20,000 tons. Using this criterion, approximately one third of the 14,280 vessels classified as "freighters" would have been operated in the tramp trades, supplementing the 5,215 ships designated as bulk carriers. These carriers' capacities would have ranged from 20,000 tons to over 250,000 tons deadweight.

In a rather technical sense, the ships exceeding 20,000 gross tons should not be considered as true tramps because they were designed, built, and operated to carry only ores, or coal, or grain. Many of these large ships—their average deadweight was 70,000 tons—were committed to haul one commodity exclusively, and might have spent a major portion of their operational lives in this one trade.¹ An unusual example of combination ships being used in different trades as a routine matter is seen in the case of Wallenius Lines, of Sweden, which took delivery in 1973 and 1974 of the *Aida* and *Otello*, each of approximately 51,500 tons deadweight. Built in Gdynia, Poland, these vessels were designed to carry simultaneously 45,000 tons of iron ore in four holds (Nos. 1, 3, 5, and 7) and about 3,500 compact-sized automobiles in the other four holds. These "garage holds" were built with thirteen decks. Normal employment of these two ships was from Japan to Europe with Japanese cars, Europe to North America with European cars, and North America to Far East and Japan carrying coal or iron ore.² The quantity of ore, coal, and grain which must be delivered to ports too shallow to accommodate the giants assures that the modern general-purpose tramp of 14,000 to 20,000 tons and the bulkers of

between 35,000 and 55,000 tons deadweight capacity will remain important units in the world's fleets.

Tramp ships, to repeat, contribute to the economic life of the world in an absolutely indispensable manner and, as has been noted, do so at low cost. There are times when the freight rate, as compared to the value of the goods transported, actually is lower than the ten to thirty percent figure previously quoted, while occasionally and inevitably the proportion is greater. The explanation for these variations lies in the fact of competition. When circumstances warrant, freight rates decline, as when there is a glut of wheat worldwide. The need for transportation is diminished, and the demand for ships is reduced correspondingly. As in any marketplace where the laws of supply and demand prevail, cutthroat competition among the suppliers commences in the attempt to find charters, and rates drop swiftly. Contrariwise, if there should be a major crop failure (as there was in the Soviet Union in 1974), enormous shipments of wheat from many producing areas suddenly will be required to replace that which could not be harvested. The demand for vessels exceeds the number available, and freight rates climb precipitately.

The example of the unexpected crop failure and the resultant effect upon the charter market focuses attention on one of the more interesting and significant aspects of the tramp shipping industry. Shipowners engaged in this trade constantly search for opportunities to capitalize upon situations similar to that described. When these enterprising individuals learn of such events, usually through their ramified intelligence system of brokers, agents, and representatives worldwide, to say nothing of their own experience in gauging the market day by day, they react promptly and decisively. Orders are dispatched to vessels not otherwise committed to make their way to the critical area, and negotiations with potential charterers are initiated. If the demand for transportation is very great, the owner whose ships have the greatest speed will stand to reap the maximum rewards.

It is, however, a paradox of the tramping business that the very action of these audacious capitalists, who will risk a multi-thousand-mile voyage on the chance of winning a single lucrative cargo, is the guarantor that the high freight rates will be reduced substantially in a relatively short period. Only so long as there is a shortage of ships

for the requirements of the moment will the rates be inflated; as the concentration of vessels in the exporting area increases, the scarcity is alleviated and freight rates plummet. All too often, instead of the gamble paying a premium to the speculator, it becomes necessary for the owner-speculator to accept substandard freight rates as a means of reducing the losses incident to working the ship to other ports where conditions seem more favorable.

Tramp shipping not only is extremely flexible in responding to the demands of commerce but also is very sensitive to changes in the patterns of sea transportation. These characteristics may be explained in part by the fact that the world fleet of several thousand tramp vessels is owned by persons of at least a score of nationalities. Domiciled in many different parts of the world, these owners keep in touch with the demands of commerce through a complex and fascinating network of representatives which spreads throughout the world. Included in the huge fleet are dozens of handsome, well maintained, expensive new ships which are large, fast, and very efficient. There are also many smaller, slower, less productive vessels significantly past their prime which remain active only so long as shortages of carriers make it possible for them to find employment. This fleet is scattered over the fabled seven seas, with typical craft in almost every port of the world.

Many ships of the tramp fleet belong to individuals. In some cases, one person may own as many as twenty, but it is more usual to find "fleets" of two or three or four units. This is especially true among Greek owners, whose loyalty to family shipping enterprises is a distinguishing characteristic. These small family groups are noted for their sturdy independence; few of them have affiliated with the larger consortia of tramp operators.

There also are a few great shipowning enterprises which include within their corporate structures tramp shipping activities. This comes about for several reasons, one of which is to participate in the profits to be earned. Another reason is to develop new trade routes, an interesting exercise in which tramp ships explore the true potential of business. If charters are to be had again and yet again between a range of ports in one region (as, for example, the United States north of Cape Hatteras but not east of New York) and another area such as the countries at the eastern end of the Mediterranean, carrying substantially the same cargoes trip after trip,

the possibility of regular liner service deserves to be probed with ships actually competing for this type of business. Often it is found that the new liner route is both desirable and economically feasible. An additional, but considerably less compelling, reason for owning tramps is to assure that vessels which become surplus to the liner service can be kept sailing, even when not returning a profit, at no greater cost than laying them up in idleness. Liner companies also may find it advantageous to have a number of ships engaged in tramping because, as temporary shortages in liner-type tonnage develop, vessels may be transferred from tramp to liner operations by a simple company directive. Only those tramps with the speed, hull design, and installed cargo-handling equipment needed in liner service can be used in this manner. There also is the possibility for these large corporations to phase out slightly obsolete ships from their liner fleets by transferring them to the tramp segment for the last years of their operational lives.

These reasons have lost much of their force with the advent of containerization in the major liner services. Whereas in the days of what is now described as "conventional break-bulk operations," the structural differences between liners and tramps were not so great as to make them incompatible, this no longer is true. Ships carrying full cargoes of containers must be built for that purpose, and cannot be switched into tramp trading for a while and then returned to liner operations. Consequently, few large steamship companies today have subsidiaries to operate tramp fleets.

It is an interesting fact of economic history that tramp shipping, as it is known today, did not come into being until the second half of the nineteenth century. By this time, steam had been harnessed to machines, and the Industrial Revolution was in full swing. The demand for raw materials to supply the machines had begun to be felt worldwide. Cotton and wool for the textile industry of England were brought from distant shores, and to those points were returned the manufactured goods of English mills and factories. During these same years of major change, steam power was adapted to ships. The insatiable demand for coal to fire their relatively primitive, low-pressure boilers necessitated the establishment of refueling points which were not too far apart geographically, but which did have some commercial activity or promise of develop-

ment. These coaling stations grew in number and, as trade routes expanded, were located farther and farther from England. A brisk export business in "best Welsh coals" evolved rather quickly and made for a satisfactory balance in ship operation, especially when the coaling stations were comparatively close to those ports where return cargoes could be obtained. The sturdy but unglamorous tramps brought back to England whatever they could find, be it cotton or wool or grain or ore. Always assured of outbound cargoes of coal should there be no requirement for their services to haul loads of manufactured goods to the distant market places, the tramps soon were the visible reminders of the vast reach of the British empire.

Admittedly, the steamships did cost more to operate than the wind-driven sailing ships, but their increased reliability and speed, and particularly the possibility of predicting within a few days the exact time of arrival of a cargo, erased that pecuniary disadvantage. Furthermore, the decreased time needed to accomplish a voyage, as compared to the sailing ship, meant that a single steamship could carry more tonnage during a year than could its more beautiful rival. If proof of the economy of the coal-fired steamer's great speed were needed, this was provided by performance. In a given period, the faster, engine-powered vessel could deliver more goods than the sailing ship, and so created greater carrying capacity without the need of building and operating more ships. Finally, the opening of the Suez Canal in 1869 sounded the death knell of the sailing ship because only powered vessels could negotiate the Red Sea on any commercially acceptable schedule.

England was the great industrial center of the Western European area, and the combination of her need for raw materials, her exports of manufactured goods, and the constant flow of "best Welsh coals" to outlying coaling stations resulted in the requirement for a large tramp fleet. Just as English capital had responded to the opportunities ashore which had been afforded by the Industrial Revolution, so English financiers in the last quarter of the nineteenth century saw the chance for rich profits in shipping. By 1900, half of the oceangoing steamship tonnage of the world was under British registry. While this numerical superiority gradually was lost as other national fleets were expanded, it remained true at

least until 1920 that more ships flew the Union Jack—familiarily known in maritime circles as the “Red Duster”—than any other flag.

In 1900, the best and most modern tramp steamer, which was not necessarily the typical ship, had a cargo capacity of about seven thousand tons. When operated with a clean bottom and in fair weather (Force 2 on the Beaufort scale of winds), she had a service speed of eleven knots. As described by the novelist William McFee, who served for years as an engineer in such a ship, the hull was 110.0 meters (370 feet) long, 14.4 meters (48 feet) in the beam, and had a full, rounded bow which facilitated stowage of cargo at the expense of speed. There were three boilers which collectively burned twenty-five tons of “best Welsh coals” each day as they supplied steam to the triple expansion reciprocating engine of 1,800 indicated horsepower. One set of cargo booms was installed over each of the four or five hatches.

A ship of this type would expect to find regular employment carrying coal outward from England, and would return in due course with a cargo of ore in bulk or cotton in bales or sugar in bags. A year’s voyages would earn a profit of twelve percent for her owners, a figure supported by the fact that her acquisition cost was £42,000. This cost must be considered in the light of contemporary values. The full cargo of coal was worth about £2,000. Crew wages were comparably low: the master was paid £25 a month; the chief engineer earned £20, while the senior mate received £10. A junior engineer worked all month for £6.³

From 1900 to 1914, the size of general-purpose dry cargo ships was increased slightly to a maximum of 7,500 tons deadweight capacity. Greater engineering efficiency reduced the consumption of coal while improving average speeds to approximately ten knots. Until the end of World War I, coal remained the predominant fuel. Between 1919 and 1939, these ships were enlarged to about ten thousand tons deadweight, and oil fuel, both for boilers and diesel engines, began to achieve popularity.

The titanic ship construction program undertaken by the United States during World War II produced thousands of ships of a single standardized design. The Liberty ship was sold worldwide beginning in 1946, and quickly became the universal general-purpose tramp. Unlike the earlier ships of this class, the Liberty had a

single 'tween deck, which enhanced her adaptability for both bulk and packaged cargoes.⁴ Until they were twenty years old (1962 and thereafter), Liberties were used on every international trade route.

It was about 1950 that dry-cargo ship sizes began to grow significantly. As the ravages of war were repaired, and European shipyards returned to full productivity, the requirements of sea-borne commerce were met with new construction. Hulls were lengthened, widened, and deepened, and cargo deadweights increased to about 14,000 or 15,000 tons. In 1968, a British shipyard, Austin & Pickersgill, introduced its replacement for the aging Liberties. The S.D. 14 (derived from the technical description of "Shelter Decker, 14,000 tons deadweight") gained widespread acceptance, and by 1981 a total of 204 ships of this design had been built, making this vessel the most successful series produced ship in peacetime history.⁵

While the general-purpose tramp was growing, the bulk carrier had been developed into a specialized vehicle for the transportation of coal, grains, and ores. Evolving from the distinctive iron ore carriers which had been in existence since early in the century, the new bulk carrier was intended to haul with equal efficiency and economy any one of those three commodities, all of which moved in enormous quantities. The first of these multipurpose bulkers could lift 40,000 tons, and demonstrated conclusively the economy of great scale. In 1981, bulkers up to 280,000 tons deadweight were in routine service on many trade routes.

Excellent reasons underlay this trend toward specialization. A vessel designed to transport a particular commodity will incorporate those features which make her ideal for that trade: overall size, draft, ratio of cargo stowage space to deadweight tonnage, dimensions of hatch openings, type of cargo-handling gear, speed, and economy of operation, to name some of the most important. For hauling the intended commodity, the specialized ship should be the most efficient and convenient that can be built. So long as the trade in that commodity provides regular employment for the vessel, she will bring benefits to her owners because of her efficiency, and will gratify her charterers because of the lower freight rates and the faster delivery of cargoes she makes possible. The entire commercial community will gain because she permits the most efficacious use of capital.

Although they are restricted in their choice of cargoes, these specialized carriers often, if not predominantly, are operated as conventional tramp ships—which is to say, they are offered for time or voyage charter on the world market, they seek out cargoes and carry them where it suits the convenience of the charterer to have his goods discharged, and they compete both among themselves and with the general-purpose tramps for what business there may be. Because of their great size, however, they are limited as to the ports they can enter, but this is only a minor problem for them.

The emergence in recent years of these huge specialized carriers has intensified the normal struggle for employment on the part of the conventional general-purpose tramp, but has not eliminated that ship from the maritime scene. It still is true that the general-purpose tramp, unlike its big competitor, is able to enter most ports of any commercial significance, and can be berthed at existing port facilities. Furthermore, the smaller size of the general-purpose vessel, as compared with the large specialized carrier, may serve the needs of a charterer very well indeed.

A manufacturer of fertilizer, for example, normally processes 15,000 tons of phosphate rock each week, and has storage facilities for a maximum of 40,000 tons of this material. By chartering several 15,000-ton tramps to establish and maintain a schedule to deliver one shipload of phosphate rock approximately every seven to ten days, the manufacturer keeps his current supplies at a satisfactory level, and has ample reserves if the ships are delayed for no more than two weeks. In effect, the ships have been integrated into the manufacturer's supply system at the cost of the slightly higher freight rate which they charge as compared to that of the larger bulkers.

It is almost axiomatic that those shippers and receivers of goods who have the physical facilities to handle and to dispose of immense quantities of a particular cargo, and the financial resources to handle transactions of great magnitude, will demand that they be furnished carriers which will give them the economies of large scale. The big bulk carriers have been the logical response to these demands and, within the limitations on their employment imposed by their length and draft, provide the required transportation at the lowest possible cost.

This type of ship enjoys a well earned and important place in



Top: A modern tramp with unusual gantry cranes, the *Bulk Eagle* is well-adapted to handle commodities like coal and iron ore. *Bottom:* Shipboard gantry cranes are paired to handle extra-long items. Photos: Munck International.

oceanborne commerce. It meets a defined need, and has demonstrated the soundness of the principle that the larger the vessel, the lower the unit cost of carriage. Only those organizations which are geared to handle, both physically and financially, enormous quantities of specific commodities are able to take advantage of the benefits which accrue from chartering these big ships.

The modern general-purpose tramp, on the other hand, is

intended to serve, and does so very successfully, a number of different trades. In one year, a fleet of four of these tramps lifted cargoes of pig iron, black iron, wheat middling pellets, scrap iron, logs, timber, tapioca, wood chips, poonac (coconut oil cake), ilmenite sand, iron ore, grain, coal, bauxite, and cement.⁶ Designed to operate into and out of ports which have relatively shallow draft and only a minimum of port facilities typified by wharves, piers, and transit sheds with their supporting upland areas, these vessels are equipped with sufficient booms, derricks, or cranes to make them "self-sustaining," i.e., capable of being loaded and unloaded by use of their own gear. Finally, the carrying capacity of the general-purpose tramp is established at least in part with reference to the quantities of whatever is offered for transportation in the different trades. Often this level of business matches exactly the capability of the ship. It is never a simple coincidence.

It is axiomatic that the controlling depth of water and the peculiarities of the channels in a port fix the limits of ship sizes which can be accommodated in a given harbor. If a ship drawing 10.5 meters (35 feet) of water, fully laden at summer load line, is offered for a trip to a harbor with a minimum depth of 9.6 meters (32 feet), there are two alternatives open to the potential charterer. One is to reject the offered ship because she is too large. The other is to accept the ship, load her only to the minimum depth, and make the best of the transaction. Circumstances prevailing in the freight market at the time when a decision must be reached will determine the alternative that is chosen.

Accommodations for the personnel staffing tramp ships have been improved steadily through the years. At the turn of the century, most of the marine superintendents ashore as well as the licensed officers aboard the tramps had served as apprentices in sailing vessels, and expected no more comfort in the steamers than they had found in the windjammers. Quarters were stark and utilitarian. Novelist William McFee, a veteran of years of service in British-flag tramps in pre-1914 days, described the conditions he encountered: small rooms allotted to licensed personnel about as comfortable as cells in a penitentiary, no electric lights, no heat, no running water, and no bathing facilities.⁷ All this has changed, and today living arrangements for all crew members are excellent. At the same time, much more attention is given to assure that the food

is well prepared and served in attractive surroundings.

Competition, however, remains unaltered in its harshness. The law of supply and demand determines which ships earn a profit. If there are many cargoes and few bottoms, there is rivalry among shippers and the freight rates climb. When the opposite condition prevails, and too many ships are chasing too few cargoes, the shipowners fight among themselves to obtain what they can, often accepting business at suicidally low rates—which explains the old saying in the tramp trade that “one good year is balanced by seven bad ones.”

Until 1966, when international transportation of cargoes loaded in containers began, tramps engaged in the foreign trade of the United States vied with the cargo liners for bulk cargoes such as grain, lumber, and some types of ore. The liners would seek “parcels” of three to six thousand tons to fill otherwise unused space, and would quote freight rates that were barely adequate to cover the cost of stevedoring. This practice hurt the tramps because they had to accept these same low rates for full loads of eight to ten thousand tons. The virtual disappearance of the break-bulk cargo liner on most of these trade routes in many instances has been accompanied by the entry of the specialized large bulkers, thus perpetuating the competition with the general-purpose tramp.

Depreciation—the annual provision made by a prudent owner to write off the value of the units of his fleet—is based usually on a theoretical twenty-year life for a ship. In the tramp trade, however, earnings are not so consistent that it is feasible each year to put into the depreciation account even five percent of the purchase price, as good financial practice might suggest. Some owners strive to cut the book value of new ships by as much as they can afford during the first two or three years of ownership, and thereafter as earnings permit. The sooner a ship's value is written off, the more quickly may the owner compete for cargoes in the bad years. Even if it should take a full twenty years to amortize the purchase of a vessel, it does not follow automatically that she will be replaced. So long as cargoes can be found, it is likely that the old vessel will be continued in service.

Competition for cargoes, slower speeds than those of the cargo liners, and nonrepetitive operations combine to restrict the tramp owner to a particular segment of commercial activity. In practice,

he transports, usually on a resupply and stockpile basis, materials typified by cement, coal, grain, fertilizers, lumber, and ores of various kinds. The unit sales price of these commodities is small; the purchase price at the loading port is low, and the difference between purchase price and sales price does not permit very high freight rates. This becomes obvious when a given item is available from two widely separated geographical regions. If that which originates at the more distant point cannot be sold in the same marketplace for no more than the price of the goods from the nearby source, the probability is that there will be no sale. Despite this pressure to keep freight rates low, it can happen in some instances that the cost of transportation equals the value of the raw material, with the predictable result that the market for that item is restricted.

To succeed in the tramping enterprise requires that the operator know something about practically all the raw materials moving in the world's commerce. At the same time, he must be acquainted with the physical characteristics of, and the cargo-handling equipment available in, those places where these materials are loaded and discharged. Always, too, he must be the optimistic speculator.

In addition to the personal attributes, the owner of tramp ships must have access to a dependable system which maintains contact with potential charterers anywhere in the world. It would be prohibitively expensive as well as administratively impractical to attempt to maintain an office in every port, or to arrange for permanent representation in those cities. To meet this demand for a low cost, worldwide agency which will provide commercial intelligence and advice when it is needed, the institution of the ship broker evolved through the centuries. The members of this branch of the shipping fraternity are the specialists who find cargoes for ships and ships for cargoes.

The center of this fascinating activity is in London, and the headquarters are located in the Baltic Exchange, not far from the Thames river and famous Tower Bridge. Here at the "Baltic," brokers who have been elected individually to membership on the basis of their personal integrity, knowledge, and resourcefulness, meet to work out charters for their principals. One man represents a shipowner and another stands in the place of a shipper. For example, suppose that Angus MacAndrew, of Glasgow, is seeking a

load for his ship, the *Highlander*, now at anchor in Hampton Roads. MacAndrew notifies John Lawson, his broker in London, who consults other brokers on the floor of the Baltic. In short order, he learns from Basil Smith that Li Wong in Shanghai needs a ship to carry grain to China from either an Atlantic or a Gulf Coast port of the United States. Following brief discussion between Lawson and Smith, an agreement as to price is reached and, after approval from the principals, the contract is made. Before the sun sets in London, the *Highlander* has been notified of her next employment and is on her way to the loading port.

The advantage of the Baltic is that here are to be found daily the representatives of practically all the owners of tramp ships in the world as well as brokers whose principals are seeking tonnage for their goods. It is much more expeditious to obtain the information and then to negotiate a charter on the floor of the Baltic than to canvass the brokers' offices by telephone, as is done in New York. Once a vessel is offered on option to a potential charterer, that ship is unavailable to other bidders, no matter how attractive their offers may be. Until the option is exercised or expires, the broker is bound by his word that the offerer will be protected. Only after the transaction has been completed is the contract drawn up. To prevent confusion, especially when international telephone conversations are involved, it is common practice to confirm offers and counteroffers by cable or telex, but these written messages do not replace the spoken pledges made by the brokers.

Shipowners as well as charterers must be aware of what charter hire rates are being offered or accepted, in precisely the same way that traders in stocks and bonds must keep abreast of swings in the securities markets. Reports of vessel charters, or "fixtures," are published in shipping journals in a technical language which conveys the essentials of the transactions in a minimum of words. The fixture of the *Highlander*, for example, would be publicized in a paragraph like this:

U. S. Northern Range to coast of China. Motorship *Highlander*, 27,000 tons, five percent, heavy grain, \$36.50; option two ports China \$37.50; 60¢ extra for U. S. Gulf loading option; FIOT; Feb. 1-12.

Interpreted for the layman, this reports that the motorship *Highlander* has been chartered to carry about 27,000 tons of heavy

grain, plus or minus five percent at the master's discretion to obtain a seaworthy load and still store and fuel his ship adequately without losing too much revenue. The grain will be loaded in a port between Cape Hatteras and Portland, Maine, and transported to the coast of China at a rate of \$36.50 for each ton loaded. If the charterer elects to discharge the cargo in two ports, the rate will be increased to \$37.50 per ton. Should the ship have to proceed to a United States port on the Gulf of Mexico to take on her cargo, the charterer will pay 60¢ additional per ton. FIOT (free in and out and trimmed) means that all expenses connected with loading, discharging, and trimming are for the account of the charterer. The *Highlander* must report at the designated loading port not earlier than February 1 and not later than February 12.

In the case described above, the *Highlander* has been obtained on a basis which suits the convenience of only one person, the shipper. It is much the same as hiring a truck to haul a load of goods, rather than sending the goods by railroad train. The truck will proceed to the pickup point set by the shipper, and will arrive at an hour arranged between truckman and shipper. It will be routed either as directed by the shipper or by the driver using the shortest route. If a railroad train is used, the goods will move over the tracks on a fixed schedule, and will be picked up and delivered according to the railroad's practice. The tramp is like the truck, whereas the seagoing counterpart to the railroad train is the cargo liner, which sails between advertised ports on a relatively inflexible schedule to which the user must conform.

How does the shipper go about obtaining the exclusive use of a tramp vessel? What arrangements are made to safeguard the interests of the shipowner? What guarantee does the shipper have that the vessel will do what is required? What sorts of agreements are made? The answers to these and related questions constitute the essence of the chartering business, and are based upon the maritime and commercial experience of thousands of men of past generations as well as of the present time.

The written agreement setting forth the terms and conditions under which the vessel owner makes a ship available to the shipper on an exclusive basis is a specialized form of contract known in maritime circles as a "charter-party," a name derived from the Latin words, *charta partita*, which mean a paper divided so that each signatory may keep an identical copy.⁸

Charter-parties are diversified into three principal categories: voyage, time, and bareboat.

The voyage charter is the most frequently employed because it places the lightest burden upon the charterer. In essence, it is a contract of affreightment covering the movement of a particular cargo from one designated area to another, at a stipulated rate of compensation (charter hire) for each ton of the goods actually transported. The shipowner is responsible for all details of vessel operation. The charterer, in addition to putting the cargo within reach of the loading gear (which may be the ship's equipment or dock-side cranes or chutes for ores and grains), must arrange for the berthing places at ports of loading and discharge, and quite often also must contract with stevedores to handle the cargo. Payment for the ship's services must be made in accordance with the provisions of the charter-party.

The time charter, while somewhat similar to the voyage charter in keeping the shipowner responsible for operation of the vessel, differs primarily in that the charterer obtains the use of the ship for a period of time to carry cargoes not barred by the contract anywhere within the geographical limitations set forth in the charter party.

Under the terms of the bareboat (or demise) charter, the owner transfers operational control of the ship to the charterer. The vessel must be seaworthy and fully equipped at the time she is delivered to the charterer, but the burden and responsibility of operation are assumed by the charterer. Except for approving the nomination of master and chief engineer, the owner does not influence the selection of officers and crew. The charterer must purchase fuel, stores, and food, pay all maintenance and drydocking expenses, and become the *de facto* manager of the vessel. The owner is kept informed of the movements of the ship, and periodically receives the agreed payment which, as in the time charter, is based upon the deadweight tonnage of the ship.

Both time and bareboat charters normally contain certain restrictions concerning the types of cargo which may be carried. At the minimum, the stipulation is that "any lawful merchandise not injurious to the vessel" may be put aboard. Some contracts are more specific; one document states that "No live stock [or] sulphur and pitch in bulk to be shipped." Should the owner wish to protect his ship from damage by certain goods, an "exclusionary clause" would

be inserted to limit the quantity of such goods to be transported. A typical clause reads, "Injurious, inflammable, or dangerous goods (such as acids, explosives, calcium carbide, ferro-silicon, naphtha, motor spirit, tar, or any of their products) to be limited" to an agreed maximum, and such dangerous goods must be "packed, loaded, stowed, and discharged in accordance with the regulations" of appropriate authorities. If these regulations entail any expenses, including the cost of erecting and dismantling magazines, those expenses shall be for the account of the charterers.

The time or bareboat charterer directs where the ship shall sail. Because marine underwriters are very definite as to the areas of the world where they will accept responsibility for damage sustained by ships they insure, the time and bareboat charters contain a stipulation generally to the effect that "the charterers are to give the necessary sailing instructions, subject to the limit of the charter, but always within the trading limits set by the hull underwriters." The newest forms of preprinted time charters contain a special blank space in which the limits of the vessel's voyages are set forth with the greatest care. A general stipulation is that the charterer shall not order the ship to a port that is hazardous to health by reason of fever or epidemic. A further restriction is imposed against sending the ship into places where there is danger of being frozen in for the duration of the winter. A typical "ice clause" follows this pattern:

The Vessel is not to be ordered to nor bound to enter any ice-bound place or any place where lights, lightships, marks and buoys are or are likely to be withdrawn by reason of ice on the Vessel's arrival or where there is risk that ordinarily the Vessel will not be able on account of ice to reach the place or to get out after having completed loading or discharging. The Vessel is not to be obliged to force ice, nor to follow icebreakers when inward bound. If on account of ice the Master considers it dangerous to remain at the loading or discharging place for fear of the Vessel being frozen in and/or damaged, he has liberty to sail to a convenient open place and await the Charterers' fresh instructions. Detention through any of the above causes to be for the Charterers' account.

If the contemplated employment of the ship entails voyages to ports where these conditions may be encountered, the terms and conditions under which that activity will be performed must be negotiated and set forth in minute detail in the charter-party.



The owner invoked the “ice clause” when his ice-strengthened bulk carrier encountered these conditions. Photo: A/S Hydraulik Brattvaag.

Any charterer who takes ships for a period of time becomes involved, as has been indicated, in considerations related to the expenditure of large sums of money (as for fuel) or the actual scheduling of voyages and the unavoidable responsibility for loss of time resulting from impaired performance of the ship. To protect the charterer in both these typical instances, there must be agreement before the contract is signed as to the amount of fuel the ship's power plant will consume in twenty-four hours at normal service speed in calm seas, and this statement must be contained in the charter-party. The speed of the ship and her carrying capacity, expressed both in cubic feet of enclosed space and in deadweight tons, are stipulated by the owner, along with numerous other details of the ship's physical characteristics, in what are called the “warranties.” These are statements of fact which are expected to be borne out by the performance of the vessel. Failure to live up to the warranties can make the owner liable for breach of contract.

Most of the warranties are provable without difficulty, but speed and fuel consumption are affected noticeably by weather conditions. Should a dispute arise over either or both of these stipulations, there will be discussions between the contracting parties until a satisfactory adjustment is made. Should no compromise be reached in these areas or in any other part of the charter-party, the dispute may be referred to arbitrators. Their decision will have the same force as a judgment handed down by a court of law.

The charterer always stipulates the port in which the ship is to be made available to him—or, in the terms of the trade, “tendered” by the owner. The charterer also sets forth the beginning and the ending of the period of days during which tender (or delivery) of the ship may be made. This period is known as “lay days.” The charterer also states the date beyond which he reserves the right to cancel the agreement and to refuse to accept the tender of the vessel. The owner protects himself from arbitrary action by including this clause in the charter-party:

If the vessel cannot be delivered by the cancelling date, the Charterers, if required [by the owners], are to declare within 48 hours (Sundays and holidays excluded) after receiving notice thereof whether they cancel or will take delivery of the Vessel.

Should the contract be for a single voyage, the exact commodity to be carried and the charter hire payable per ton of that commodity actually loaded, are designated, as are the ports of loading and discharge. These details are important to the owner not only for the obvious reasons but also because they permit him to look ahead for other employment following the completion of the present charter. Experience and knowledge will guide him in estimating the number of days the ship will need to carry out the contract; a judicious guess must be made as to the possible delays, and long-range commitments thereafter can be consummated. During those periods when there is reasonably good demand for ships, an astute owner may be able to keep his vessel profitably employed for a full year with hardly any movements in ballast.

Among provisions of the voyage charter are the number of days allowed for loading and discharge (also known as “lay days”)⁹, the penalty for holding the ship in port beyond the allowed period for working cargo (“demurrage”), and the premium for reducing the

time spent in port ("dispatch"). For instance, a 20,000-ton tanker is fixed for a load of petroleum from a Caribbean port to a range of ports on the Atlantic coast of the United States north of Cape Hatteras (generally referred to by the initials USNH); alternatively, it may be assigned specific ports such as Aruba and New York. The ship is allowed a certain number of hours for loading and discharging; demurrage is set at \$2,200 per day and dispatch at half that rate, prorated to the actual hours.

Because the voyage charter imposes the least operational responsibility upon the charterer and is convenient for the movement of bulk commodities, it has become the most common form of contract. The owner of the ship, under the "gross" form, pays for every item of expense, including loading, discharging, port fees, and all expenses in connection with the cargo, as well as the operating charges such as crew wages, subsistence, and fuel.¹⁰ Under the "net" form, the owner pays all normal ship operating costs, and the charterer is responsible for charges accrued for loading and discharging the cargo as well as for port fees (except those related directly and solely to the crew) exacted against ship and cargo. The owner retains the full burden of managing the ship.

A voyage charter, as has been pointed out, is concerned with moving a particular cargo between designated ports. It is understood that the charterer will provide sufficient cargo to fill the entire ship unless the contract stipulates otherwise. The master must assure that the loading of the vessel is correct and seaworthy; almost always he has authority to decrease or increase the quantity of cargo offered by the shipper, as may be required to give the ship proper trim and stability.

The charterer, for his part, must pay the agreed sum for each ton of the commodity he has contracted to ship; if he loads less than the ship's capacity, he is liable for the difference between what should have been loaded and what actually was put into the hull ("dead freight"). The owner intends for the master to accept as much as the safety of the ship will permit, in order to obtain the maximum revenue from the voyage. It is expected that the charterer, having set the date when he will be ready to receive the ship, will have the cargo waiting so that no time will be lost once the ship has been tendered. Any delay on the charterer's part subjects him to the possibility of paying demurrage.

Unlike the revenue earned under the voyage charter, which is based on the payment per ton of cargo loaded, the time charter hire is predicated upon the carrying capacity of the ship. That is to say, the ship has a deadweight capacity, always determined at the summer load line, and so much enclosed cargo space. The charterer has the full use of that capacity and space for the duration of the charter. He may load any cargoes not prohibited by the charter-party, and may direct the ship to proceed wherever he desires within the limits fixed by the contract. The charterer has no connection with the navigation and operation of the vessel; he provides cargoes for the ship, keeps her busy, and pays the owner at the agreed rate per deadweight ton per month. If he fails to find employment for the vessel, he still must pay the owner the stipulated hire. The owner provides the crew, stores, and all necessary equipment for the vessel; pays all port charges incurred for and in behalf of the crew; and is responsible in all respects for the navigation and maintenance of the vessel, and the stowage, care, and safety of the cargo. The charterer provides the fuel; pays port entry, clearance, and agency fees; pilotage charges, tug hire, costs of loading and discharging cargo, including wharfage and dockage; and arranges for appropriate representation in the ports during voyages.¹¹

To illustrate the theory of the time charter, here is a simplified form of agreement for the use of a ship:

This charter-party is made and concluded in New York the first day of October 1981 between Marine Carriers Company, of London, England, owners of the good British motorship *Sea Merchant*, and Ocean Trading Corporation, of Newark, New Jersey, charterers of the said motorship (described as being of the highest class at Lloyd's Register of Shipping, and of about 711,000 cubic feet grain capacity, 10,300 tons gross register, about 6,200 tons net register, and about 15,150 tons of 2,240 pounds deadweight capacity on a summer draft of 30 feet, with a service speed of 16½ knots on a consumption for all purposes of about 25 tons of best grade fuel oil), now in service and trading. These particulars are not guaranteed, but are supplied in good faith and are believed to be correct.

It is agreed that the vessel is tight, staunch, and strong, and in every way fitted for the contemplated service. She is to be delivered to the charterers in Galveston, Texas, not earlier than October 13, 1981, and not later than October 30, 1981, and redelivered to the owners in a

safe United States Atlantic port north of Cape Hatteras but not east of New York in about fifteen months from date of delivery.

Should the vessel not be delivered by October 30, 1981, the charterers have the option to cancel. If the vessel cannot be delivered by the cancelling date, the charterers are to declare, within 48 hours (Sundays and holidays excluded) of receiving notice from the owners, whether they will cancel or take delivery of the vessel.

The vessel is to be employed in lawful trades to carry only lawful merchandise, excluding explosives, livestock, sulphur, and phosphate in bulk, between good and safe ports where she can lie always afloat, within the limits of the marine underwriters' insurance policy, subject to the following conditions:

1. The owners shall provide and pay for all provisions, wages, insurance of the vessel, and consular shipping and discharging fees of the crew, as well as for all deck and engine room stores. The owners shall maintain the hull and machinery of the vessel in a thoroughly efficient state throughout the duration of this charter.
2. The charterers shall pay all dock, harbor, light, and tonnage dues at the ports of delivery and redelivery. While on hire, the charterers shall provide and pay for all fuel, port charges, pilotage (whether compulsory or not), tug assistance, canal, dock, and other dues and charges and taxes, and will arrange and pay for all loading and unloading of cargo, including trimming, stowing, weighing, tallying, and delivery of cargo.
3. The charterers shall pay for the use and hire of the vessel at the rate of twelve dollars and twenty eight cents per deadweight ton per month, including overtime for the crew. Payments shall be made in United States currency not less frequently than once every thirty days, in advance. It is agreed that a "month" shall consist of thirty days.
4. The whole reach and burden of the vessel's holds, decks, and usual places of loading shall be at the charterers' disposal.
5. The master shall prosecute his voyages with all reasonable speed, and all vessel logs shall be shown to the charterers on demand. The master shall be diligent in his care of the cargo, and shall supervise the loading, stowing, trimming, and discharge of cargo. He shall sign bills of lading as presented, in conformity with mate's or tally clerks' receipts.
6. Should the vessel be hindered or prevented from performing under this charter-party for longer than thirty six consecutive hours by reason of drydocking, or other measures necessary to maintain the ship, or as a result of breakdowns of machinery, strikes of ship's personnel, or damage to hull or machinery, hire shall be suspended

from the time the difficulty began until the vessel is returned to full service.

7. The owners shall have a lien upon all cargoes and subfreights belonging to the charterers and any freight payable under bills of lading, and the charterers shall have a lien upon the vessel for all moneys paid in advance and not earned.

8. The vessel is to work day and night if required. The owners shall pay any overtime compensation earned by officers and crew as a result of working beyond their agreed hours.

9. Any dispute arising under this charter is to be referred to arbitration in New York, N. Y., U.S.A. One arbitrator shall be nominated by owners and one by charterers, with the third to be chosen by agreement of both arbitrators.

10. A commission of 2½ percent is payable by the owners to the brokers on any hire paid under this charter.

Signed for the owners
Marine Carriers Company
by Harold T. Seaman
as broker only

Signed for the charterers
Oceanic Trading Corporation
by William T. Merchant
as broker only

The bareboat charter is similar to the time charter in that the use of the ship is acquired for a period of time, but the full responsibility of operation is transferred from owner to charterer. Obviously, a heavy burden is placed upon any person taking a ship under such an agreement. Few deals of this sort are consummated; for the most part they are the result of complex financing transactions in which the legal owner is not a ship-operating organization but a bank or similar institution. The charterer must have a complete staff in existence to assume the duties of running the ship. Nearly always steamship companies which charter ships on a bareboat basis do so to augment their fleet capabilities either for a short period of peak cargoes, or over the long term in lieu of purchasing a ship.

Perusal of the different charter-parties is rewarding, for the numerous provisions of the documents reflect the experience of shipowners and charterers through the years. Drawn by practical shipping men working alongside experts in maritime and commercial law, the charter-parties represent a working compromise between the demands of the shipowners and the insistent requirements of the charterers. Negotiations for a particular transaction usually start with one of these standardized, preprinted forms, with

appropriate insertions of agreed terms. If necessary, the clauses are modified to fit the circumstances of the proposed charter.

“Working the market” requires that both charterer and shipowner be alert to many factors. The charterer needs to know a great deal about the availability of ships and the consequent fluctuations of the freight and charter markets. He must note how rates change for the same commodity on different trade routes and at varying times of the year. The operator has to be aware of practically everything the charterer should know while keeping at his fingertips a wide assortment of facts and details which are part and parcel of his business life. He must be cognizant of conditions in ports around the world, the charges made for handling his ships, the efficiency of longshoremen, and the availability of cargo-working equipment. He has to be alert to the peculiarities of harbors—their depths, tides and winds, ice and fog anchorages, and local customs concerning the use of pilots and tugs. He is wise also to be informed of ship repair facilities in those ports. He needs contacts in anticipated ports of call which will furnish current, reliable, and prompt reports of all developments affecting shipping in those waters. Where to procure fuel is a never-ending cause of concern, especially as the quality of that which is offered today is subject to wide variation.

When it comes to cargoes, the tramp shipowner must be intimately acquainted with the characteristics of practically every commodity in the world that moves in bulk, so that when he sets a rate it will be fairly certain to earn a reasonable revenue from each ton carried. Finally, he must know how these bulk commodities are handled, for in some ports everything is mechanized and therefore assures fast loading or unloading, while in other ports the process is altogether by hand.

It is small wonder that mastering the intricacies of the tramp shipping business is a full-time career!

The Management of Tramp Shipping

TO PROVIDE timely and efficient transportation from a producing area separated from a consuming region by thousands of miles of ocean is both a highly complex business and a very severe test of the business judgment and skill of those responsible for the effective employment of tramp ships. Not every vessel is adapted to carry every cargo, and not every port has a constant stream of ships entering and leaving. Some commodities are seasonal in their movements, while others provide a steady flow month after month and year after year. Some items are easy to load and discharge, and cause no structural damage, whereas others are troublesome and dangerous. On some routes, competition between tramps is intense, which means that the margins of profit are narrow, if they exist at all. Other routes, for a variety of reasons, offer more acceptable rates of compensation, but fewer opportunities. Certain ports are renowned for prompt dispatch, while others seem to be afflicted with insoluble problems of delays and the resulting congestion. Matching ships to cargoes, selecting the most advantageous routings, ascertaining accurately the costs of performance under different types of contracts, and balancing present opportunities against future prospects, are a few of the areas of concern in the profitable and effective management of tramp ships.

As is true in any form of commercial enterprise, the motivating purpose must dominate the decision-making process. In tramp shipping, the purpose is to transport cargoes at the most profitable level of compensation. The commodities that move in greatest quantity, and therefore offer the best opportunities for employment of ships, are ores, grains, and coal. Of these, grain is the most likely to experience fluctuations in demand, and therefore to have a significant impact upon the tramp shipping industry. This fact helps to explain why grain—a term which includes a wide variety of

cereals such as wheat, corn, oats, barley, rye, sorghums, and soyas—has been called “the great leveler” of the shipping business. As a basic requirement in feeding the population of the world, grain is in constant demand. It is grown in many regions, but only the United States, Canada, Argentina, and Australia (in that descending order of magnitude) have significant quantities for export.¹ As a cargo, grain can be carried in many different types of ships, and because it moves almost exclusively in bulk, it is susceptible to mechanized handling, with resultant short loading and discharging times. Until the advent of containerships and the introduction of the very large bulk carriers, grain was an attractive “filler” cargo for ships in the liner service, and therefore became the target of intense competition between “regular traders” and tramps.

A different aspect of competition became manifest when the large, fast, specialized containerships were introduced to these routes and, at about the same time, the tankers and bulkers of at least 50,000 tons deadweight capacity began actively to seek grain cargoes. Both these types were able to offer freight rates which, because of economies of scale, were significantly below those which the conventional 12,000- to 16,000-ton tramp could accept. It is worthy of note that the movement of grain on the North Atlantic trade route is dominated by large-capacity vessels.

The tramp ship, however, has not been eliminated from the North Atlantic. Despite rigorous competition, the versatility of this vessel, with the advantage of being able to enter ports too shallow for the deep draft bulkers, has made possible satisfactory employment in a variety of cargo movements. Grain continues to be a commodity which these smaller ships can and do transport as they fight a successful battle to survive in an increasingly competitive environment. Many other dry bulk cargoes move in international commerce, and provide year-round opportunities for bold and aggressive owners of tramp ships. Metallic ores, fertilizers, coal, and forest products, to name only a few, account for a substantial tonnage which must be transported across the seas. To meet the incessant demand for cargo space entails vigorous action on the part of all concerned: producers of these materials, traffic managers, ship brokers, and shipowners. Each of these groups needs to have, in varying degrees, a world-encompassing store of knowledge on which to base decisions.



The “Fortune” class of 20,000-ton, 15.5-knot general-purpose carriers was built in Japan to replace World War II Liberty ships. Photo: Ishikawajima Harima Heavy Industries.

It is not surprising, therefore, to discover that both the users of tramp ships and the owners of those hard-working carriers become avid readers of the daily news, with special emphasis upon accounts of weather in different areas, economic conditions, political events, and crop reports. A sudden flood in midsummer, when crops are ripening, and only minor quantities of rainfall are needed, may have devastating effects on the final harvest. Not only is the farmer involved immediately, but the sellers and importers will have to begin to make adjustments in their marketing plans. Carriers—both land and oceangoing—may discover that the tonnage requiring their services is reduced to a trickle. The worldwide patterns of tramp ship operations may be altered drastically as a result of this untimely excess of water in the growing region. The ability to relate news events to transportation activities is a prime requisite for those who would participate in this fascinating and ever changing activity.



The "S. D. 14" was designed in England, and was the most successful peacetime series-built ship in history. Over 200 had been delivered by 1982. Photo: Turner, courtesy of Austin & Pickersgill, Shipbuilders.

For the owner and operator of tramp-type shipping, the goal toward which all efforts are directed is the most effective employment of the units of the fleet. The ultimate criterion of success is the profit or loss sustained from the voyages undertaken during a given accounting period. Whether the color of the ink used by the book-keeper is black, announcing a profit, or red, admitting a loss, depends upon the ability of the managers of the fleet to reduce nonrevenue-producing voyages in ballast to an acceptable minimum, to control the number of idle days between active voyages, to find the most suitable cargoes for the ships, and to obtain the best freight rates and the most favorable terms. Unquestionably, the burdens placed upon the persons who must make the final decisions are very great, but so are the rewards when the results show that the decisions were sound.

To be able to arrange for a series of charter commitments which will follow one upon another with little or no intervening time during which the ships do not earn any revenue is the ambition of every manager. Should the policy of the owner be to keep the fleet

in the market for voyage charters, where the returns are higher but the risks of finding continuous employment are greater than in the longer term time charter activity, it is necessary to be extremely selective in the commitments made. For example, a high rate may be offered to carry a cargo of grain from New Orleans to Calcutta, but unless there is some assurance that a load can be found within a comparatively short distance from Calcutta, the business may be a money loser because of the high cost of sending the ship to the area of her next employment. Ideally, as each voyage charter is ended and the ship is redelivered to the owner, a new assignment is waiting, and the time between revenue-producing voyages is reduced to the absolute minimum. In practice, this happens only occasionally; all too often, it is necessary to send the ship on a trip of several thousand miles to pick up the next load. The profits, if any, from the first voyage may be absorbed in the cost of repositioning.

To illustrate the various possibilities for employment of a tramp ship, this hypothetical example is offered: A 15,000-ton general-purpose vessel was assigned to lift grain from New Orleans to India. By the time she had completed discharge at the port of destination, these were the possibilities under consideration by the owner:

1. To proceed in ballast to Australia to pick up wheat for delivery in either the United Kingdom or the Bordeaux-Hamburg range of the European continent; or
2. To proceed in ballast from India to Mauritius to load a full cargo of sugar for a port on the Atlantic coast of either Canada or the northern range of the United States; or
3. To proceed to Murmugao, India, to take on a load of ore for Japan. From Japan, there were these potential opportunities for employment: (a) to sail in ballast to the Philippines to load copra for Europe; (b) to go to the Philippines for a cargo of sugar for a United States port on the Gulf of Mexico or the Atlantic coast; or (c) to proceed in ballast to British Columbia for either a cargo of grain for Europe, or a full load of lumber for the Atlantic coast of the United States.

In the real world of tramp shipping, there must be assurance that the terms and conditions demanded by the charterer are fair and reasonable from the viewpoint of the shipowner. What at first

may seem to be a very profitable rate may turn out to be a guarantee of a heavy loss if there is no protection against excessive delays incident to port congestion. For instance, if a cargo of grain is offered from Churchill, a Canadian port on Hudson Bay where the weather is notoriously uncertain and often shuts down the port altogether for several consecutive days, the owner should insist upon the charterer agreeing to compensate him for time lost in such an eventuality. Otherwise, despite a very attractive rate, the voyage might result in a deficit. This is especially true in the case of the modern large-capacity ships which have operating expenses amounting to many thousands of dollars each day.

As a tool with which to formulate decisions relating to the future employment of a tramp steamer, the electronic computer is of great value, gaining greater recognition year by year. In no way, however, does it supersede the commercial wisdom which tramp ship managers have acquired through years of experience in their profession. The machine is only as good as the programs by which it is governed, and the programs of greatest utility are those which reflect the realities of tramp shipping. Combining electronic wizardry with business experience in this computer age makes it possible to obtain quickly a wide variety of almost unbelievably detailed calculations. From these products of the machine the owner can determine which of the various choices available to him will result in the most efficient employment of his ship.

Despite the great amount of data which the computer can provide, the final decision as to the type of charter to accept depends upon the owner's ability to evaluate the calculations and to foresee the future demands for ships of the type he controls. If it is likely that there will be a significant increase in cargo movements in the near future, relatively short-term voyage charters are preferable in order to have the ships available when the expected higher rates are being paid. Should the indications point to a long period when rates will be held at the current level, a series of voyage charters of moderate length might be desirable—provided the ships will be kept busy. If, however, the forecast is for a severely depressed market during which freights will be reduced, it would be prudent to accept time charters paying enough to meet capital and daily operating costs with either no profit or a very small difference between revenue and expenses.

Although any forecasts regarding the trend of the world freight market contain a high degree of conjecture, there are certain developments that serve, when they occur, as guideposts on which considerable reliance may be placed. As soon as prominent grain-trading organizations begin to charter ships for some months in the future, or to accept ships under long-term time charter, it is an almost assured indication that freight rates will remain at their current levels; if they do drop, the new rates will be very close to those prevailing on the current market. Major importers of grain who are informed about future deals are very likely to attempt to cover their requirements before the news is made public. Their actions therefore are the subject of constant scrutiny in the never-ending effort to see into the future. Similar comments might be made about Chinese charterers; the perspicacity and long-range views of these businessmen are almost legendary, and have been proved to be amazingly accurate.

The necessity to make comparisons between the advantages of a voyage charter and a time charter is part of the daily routine of the tramp ship owner. In the normal course of business, there frequently are occasions when the owner must decide between two offers, one to place the ship under a voyage charter, in which the compensation is payable for the actual tonnage of cargo loaded, and the other to commit her to a time charter on which the return is predicated upon a sum paid per deadweight ton, summer draft, per month. An example of this sort of comparative computation will explain the process.

A voyage charter paying \$12.00 a ton is under consideration for a ship of 64,000 tons deadweight capacity. The estimated time to complete the charter is forty days, and operating expenses for the contemplated voyage (including fuel, canal tolls, and port fees) amount to \$16,400 a day. After deducting requirements for fuel, water, and stores, there is available for cargo a total of 61,800 tons. The calculations would set forth these details:

Cargo revenue, 61,800 tons @ \$12.00	\$741,600
Voyage expense, 40 days @ \$16,400	<u>656,000</u>
Earnings from the voyage	\$ 85,600

The alternative is to place the ship under time charter at a rate which takes cognizance of the charterer's responsibility for fuel,

canal tolls, and port fees. To determine what the time charter hire should be to produce the same earnings, this calculation would be made:

Net voyage expense, 40 days @ \$12,000	\$480,000
Profit from voyage charter	<u>85,600</u>
Revenue required from 40-day time charter	\$565,600
Charter hire per day	\$ 14,140
Rate per deadweight ton per month (30 days)	\$6.63

The owner, in deciding whether to accept a voyage or a time charter, must take into account the advantages and the risks inherent in each type of charter.

A ship under voyage charter is always under the total control of her owner. The contract calls for transporting an agreed quantity of a certain cargo between named ports, is limited to the time required to load, to travel from port to port, and to discharge, and is fulfilled when delivery of the cargo is completed. Should the voyage be lengthened by unexpected heavy weather, the added expense is borne by the owner. Unless otherwise provided by the contract, interruptions to cargo-working operations attributable to adverse weather conditions such as heavy rains, snow, or exceptionally high winds are charged to the owner. Notwithstanding the aforementioned hazards, which are not all-inclusive, the great inducement for an owner to offer his ship for a voyage charter is that the hire usually is greater than would be that of a time charter, and the opportunity remains with the owner to take advantage of increases in freight rates.

The owner of a vessel who is working the voyage charter market must be always on the alert for new opportunities. Ideally, as one charter is terminated, another is waiting for the ship, and the ports in which the ship discharges and loads are separated by very short distances, so that little time is lost between assignments. The reality is far removed from this ideal, and quite often an interval of several days elapses between completion of one charter and commencement of the next. This time of idleness is almost unavoidable because of the necessity to allow for possible (but not inevitable) contingencies. If they do not materialize, the ship lies idle.

A notable example of excellent management is to be seen in the case of the *Scandic Wasa*, a motorship of 20,280 tons deadweight.

On fuel consumption of 25 tons per day of intermediate (maximum 1,500 seconds Redwood) fuel oil, the ship had a speed of 14 to 14½ knots. In addition, she used an average of 1½ tons of diesel fuel per day.

The first charter of the *Scandic Wasa* took effect at Amsterdam, when she went under time charter for a voyage to the Arabian Gulf at a rate of \$7,350 per day. She was tendered in September. When this charter expired in November, the vessel was placed under another time charter from an Arabian Gulf port to Japan via India. The rate was \$12.50 per deadweight ton per month, equivalent to \$8,300 per day. The duration of the charter was the time needed to accomplish the single voyage.

On her arrival in Japan, the owners of the *Scandic Wasa* greeted their ship with instructions concerning another time charter. This called for a voyage from Japan to a grain loading port on the Rio de la Plata of Argentina; the routing was by way of the Cape of Good Hope. The time charter hire was \$10.70 per deadweight ton per month, equal to \$7,100 per day.

The voyage from Japan to the Argentine was accomplished in due course, but some delays were experienced on arrival in the river. The next employment for the *Scandic Wasa* was a time charter from Argentina to Italy, at a rate of \$13.75 per deadweight ton per month (\$9,100 per day). Discharged in Naples, the hard-working tramp proceeded in ballast to the nearby port of Casablanca, Morocco, to pick up a cargo of phosphate destined to Avonmouth, England. On completion of that assignment, she sailed in ballast for Detroit, Michigan, where further employment awaited her.

A frequent experience in tramp shipping is the necessity to send the ship on a fairly long voyage (as in this case, when the *Scandic Wasa* went to Detroit) at the owner's expense to reach the port where the next charter becomes effective. Sometimes the charterer can be induced to pay a "ballast bonus" to assist in meeting the cost of the voyage to the loading port. This desirable circumstance normally occurs, however, only when there is an existing or anticipated shortage of ships at the loading point, or the voyage is to commence at a place off the normal routes.

A time charter has the advantage that, once the ship is delivered to the charterer, the costs of fuel, port fees, canal tolls, and stevedoring, as well as all ship time, are for the charterer's account.



This self-unloading collier can top off a deep-draft bulk carrier at the continuous rate of 6,000 tons an hour. Photo: Canada Steamship Lines.

Delays that can make the difference between profit and loss for the owner under a voyage charter are allocated to the charterer; only in the event of a breakdown and consequent inability to perform under the charter would the owner be held responsible for ship time. Variations in fuel prices also are the concern of the charterer, a consideration of significant importance since the drastic price increases that began in 1973. The ship is assured of employment for the period stipulated in the charter-party, which protects the owner in times of reduced demand for ships, but at the same time precludes the possibility of capitalizing on a rising market.

Owners of tramp ships must be alert at all times to the direction in which the charter market is going. If freight rates are believed to be at or near the peak, it is the proper time to fix as many ships as possible for the longest periods to which charterers are agreeable. If, however, charter rates are low, but the owners are optimistic that

there will be a marked improvement in the near future, they would seek short-term contracts in order to take advantage of the higher rates when they become effective. Owners also must watch developments worldwide to position their ships for the best opportunities. If, for example, in one geographic area the principal export commands a small market, and numerous ships are vying for the available business, the competition will drive the rates down. In another part of the world, at this same time, shippers of a commodity that moves in very large quantities are unable to attract as many ships as they need, and are paying premium freight rates. The astute tramp owner will send his ship to the second region, hoping to be among the early arrivals who will get the highest payments.

It is true, however, that there are many periods when all tramp trades are depressed, freight rates are extremely low, and many ships are unable to find employment. When this situation arises, the oldest and least competitive ships are withdrawn from service and placed in what the owners hope will be temporary lay-up. This action is taken only after careful consideration because the cost of returning a ship to full commission can be very great. It is a melancholy fact of shipping that never are all the laid-up ships put back into service. When circumstances do justify bringing ships out of lay-up, at least some of the oldest, slowest, smallest, and, therefore, the most expensive, ships are sold to ship-breakers, because reactivation is found then to be too costly.

Whereas the liner service operator has a regular route and carries a predictable variety of cargoes offered, in the main, by a comparatively small number of shippers who use his vessels repeatedly, the tramp shipowner is required to find new clients for almost every voyage.² Given the world-ranging nature of tramp ship employment, this would be an impossible task for an individual owner acting by himself. The difficulty is resolved, as has been described in another chapter, by the brokers who are found in almost every port of commercial importance.

In the complex world of tramp shipping, there are brokers who have principals whose ships are engaged in many different trades, and therefore must have an extensive knowledge of the entire field of marine transportation. Inevitably, there also are individuals who handle one type of business more and more frequently, and who find themselves becoming specialists in a particular trade. In this

manner evolved those experts in the movement of liquid petroleum cargoes or in finding charters for tankers; similarly, concentrating upon fixing ships to handle full loads of grain led other experts to devote their attention exclusively to ships seeking grain cargoes, or the reciprocal relationship. There are, as might be expected, many large ship brokerage houses which serve the entire industry but have subdivided their activities and assigned specialists appropriately so that the greatest efficiency in negotiating charters can be attained.

Most chartering procedures commence with discussion of the acceptability of the standard charter-party for the particular trade to which the ship is to be committed. These standard agreements have been drawn up, primarily by shipowners' groups such as the Baltic and International Maritime Council, a multinational association of many tramp ship owners with headquarters in Copenhagen, and incorporate the experience gained from hundreds of voyages in individual trades. While they are revised and brought up to date as frequently as possible, it is true that often circumstances prevailing at a particular moment will make it desirable for one or the other party to the negotiation to suggest minor modifications in some clauses, changes of substance in other clauses, and even deletion of inapplicable or unwanted terms. Great care must be exercised in these circumstances to assure that the changes do not run counter to court rulings, established practices of the trade, or other clauses in the contract. The burden assumed by the broker to safeguard the interests of his principal is obvious.

Not only must brokers be well informed on those practices of the trade which are reflected in the terms and conditions of the charter-party, but they also must be aware of the level of freight rates prevailing day by day in the trades in which their principals desire to participate. Fortunately for all concerned, the great majority of fixtures of ships under charters is published daily in bulletins compiled by large brokerage houses or in trade journals. Although the specific terms which may have influenced the acceptability of a particular rate may not be known, the experience of the broker will permit him to make extremely accurate guesses about these matters. Constant and careful analyses of these reports are an integral part of the broker's daily routine.

In addition to these "open market" quotations, a number of ship

fixtures never receive publicity. This situation arises most often in the case of long-term charters where unusual or almost unique circumstances have made special consideration necessary, with accompanying adjustment of rates, terms, and conditions. The other possibility for desiring to keep the deal out of the news is that neither party wishes to reveal the figures—the charterer for fear it will indicate his profit, and the shipowner because he does not want to disclose how low the rate was. Complete secrecy rarely is maintained for any great length of time; the constant interchange of information about the market which occurs between brokers will serve, willy-nilly, as the medium of communication about these private deals.

Some mention already has been made of the fact that there are standard charter-parties in existence covering a great number of trades. These documents have been criticized occasionally by charterers as being the exclusive handiwork of the shipowners, and therefore they do not provide adequate protection for the interests of the parties contracting to use the ships. The explanation of this criticism lies in the fact that the shipowning and ship brokerage professions are rather closely knit and are engaged in constant interchange of ideas and information. Basically concerned with the problems of operating ships in a great variety of circumstances, the owners have in common the need to find solutions and to protect their financial positions. Shipowner organizations have been formed as a result, and these groups draw up contracts acceptable to the operators.

The users of tramp ships, on the other hand, are not organized, nor is it likely that the wide disparity of their interests can be reconciled to the extent that a particular position can be taken and held against opposition. This is not to disparage attempts which have been made to bring charterers together to form proposals to present to the shipowners; in a number of instances, they have been successful and better relations between the two interests have come about. The limited successes, however, emphasize the fact that individual charterers often have problems that can be solved only by paying attention to those acknowledged difficulties. Frequently, what is satisfactory for one charterer is not, and cannot be, acceptable in any form to another charterer.

So it comes about that the standard charter-parties drawn up by

the shipowners serve as the basis for negotiation of a contract. Every clause must be examined to be sure that it applies in its entirety to the contemplated use of the ship; if it does not, it is common practice for the principals, advised by their brokers, to make mutually acceptable adjustments. In the case of one grain charter which was used for many years for exports from Argentina, a large number of substitute clauses was drawn up to reflect changes in business practices. These typewritten changes were pasted routinely in all the preprinted forms. Eventually, these amendments were incorporated in a completely new contract, to the relief of all concerned.

This vignette of shipping history is cited to show that shipowners, by choice, operate under charter-parties which are well known and have withstood the analyses and interpretations of courts of law and decisions of arbitrators. They believe, with considerable reason, that it is better to make a few changes, as each employment requires, in the familiar contract, than to draw up a completely new document which invites interpretation by a court if a dispute should arise. When new charter-parties are proposed, they receive the most careful and searching examination on the part of the owners, their lawyers, and the brokerage profession, to be sure that, insofar as possible, the interests of the shipowners are safeguarded. It is a highly commendable fact, therefore, that new, rewritten, or substantially revised charter-parties emanate from many sources on a fairly regular basis. The tramp-shipping industry may be called conservative, but is not stagnant.

It is fitting, in rounding out this sketch of how tramp shipping is managed, to study a number of case histories of ships which were placed under time and voyage charters. The examples illustrate many of the principles discussed in the preceding pages.

Western Australia to Antwerp/Hamburg Range. Motorship *Scotspark*, 25,000 metric tons, \$21.50; 3,000 tons discharge free, barley. April 13-25.

This extremely brief report of an uncomplicated voyage charter is an example of the shorthand used in trade journals. The details, derived from other sources, make an interesting story.

The motorship *Scotspark* was a bulk carrier of 27,175 tons deadweight, with a below-decks capacity for grain of 1,400,000 cubic

feet (42,000 cubic meters). On a daily consumption of 40 tons of heavy oil (3,500 seconds Redwood) and 2 tons of diesel fuel, she had a service speed of 15 knots. Her fuel tanks had a capacity of 2,152 tons, adequate for a 51-day voyage without refueling.

In the grain trade from Australia to Europe, it is customary for charterers and owners to agree that the cargo will be loaded at the expense of the ship, at a minimum rate of 1,000 to 1,500 tons per weather working day. The cost of discharging in Europe is to be borne by the charterers, unless some specific contractual provision is made to a different effect. In the case of the *Scotspark*, the shipowners undertook to defray the expense of unloading 3,000 tons of the barley cargo.

Barley is one of the lighter grains, and has a stowage factor of 56 cubic feet (1.68 meters) to the long ton. The entire cargo space of the *Scotspark* would be filled when 25,000 tons had been taken aboard. To determine the tonnage available for fuel, the following computation was made:

Deadweight capacity at summer load line	27,175 tons
Barley cargo loaded	<u>25,000 tons</u>
Available for fuel, water, stores, etc.	2,175 tons
Estimated weight of water, stores, etc.	<u>500 tons</u>
Net tonnage available for fuel	1,675 tons

The record of this voyage shows that the ship arrived in Esperance Bay, Australia, on April 12, and tendered under the charter at the earliest permissible date, April 13. She departed Esperance Bay on April 14, arriving in Fremantle two days later. Cargo operations were finished and the *Scotspark* cleared for Antwerp via Table Bay (Capetown, South Africa) on April 20. She arrived in Antwerp on May 31, after a 41-day voyage. Her average speed was 11 knots on a daily consumption of 30 tons of fuel. Her owners apparently decided to travel at the most economical speed for the dual purpose of reducing the cost of fuel and also eliminating the need to purchase replenishment fuel at Capetown.

Indicative of how shipowners attempt to have employment arranged for their ships as charters expire is the fact that on completion of discharge the *Scotspark* was placed on time charter to Continental Lines for a voyage from Antwerp to the United States

north of Cape Hatteras, at a rate of \$2,500 a day. This charter was made possible by the fact that almost all major steamship lines in the North Atlantic service today are fully containerized, and therefore are unable to carry certain kinds of cargo. As a result, "handy size" bulkers such as the *Scotspark* occasionally find employment hauling cargoes of commodities booked to the liners but not susceptible of being carried in containers.

Vancouver to Antwerp/Hamburg Range. Motorship *Regina Oldendorff*, 37,150 tons deadweight cargo capacity; 1,620,000 cubic feet grain space. Lump sum \$330,000, basis one discharging port, with option for two discharging ports \$10,500 extra. Free in and out and trimmed. Eleven weather working days all purposes, Sundays and holidays excepted. Grain/seeds. February/March.

The *Regina Oldendorff* was a diesel-powered bulker of 37,150 tons deadweight on her summer load line. The charterer planned to use her to transport any kind of grain or a cargo of oil-seeds, or a mixture of both, from Vancouver, British Columbia (Canada), to a port on the western European coast between Antwerp and Hamburg. From the fixture report, it may be assumed that at the time the ship was chartered the precise nature of the cargo was not known, and therefore the owner demanded that the charterer pay a fixed sum for the hire of his ship. The agreed lump sum payment was \$330,000, which was owed regardless of how much grain or oil-seeds might be put aboard. To meet possible requirements to deliver the cargo in two ports, the charter-party stipulated that this could be done for an additional \$10,500. With these arrangements, the owner of the *Regina Oldendorff* was guaranteed a satisfactory revenue regardless of what kind of grain or seeds, or both, might be loaded.

It is appropriate to note that there are various kinds of grain which have quite different stowage factors, some being light and others heavy. The oil-seeds also vary in weight from light to heavy. Unless the charterer can assert positively what the tonnage and type of grain or seeds will be, the shipowner has no alternative to asking for a lump sum payment. It is reasonable to assume that the charterer would apportion the grain and oil-seeds to make the most effective use of the ship's spaces and deadweight.

The report of the *Regina Oldendorff's* performance under the charter-party shows that she arrived in Vancouver on March 1 and sailed on March 9. She made the transit of the Panama Canal on March 21, and on April 5 arrived in Rotterdam, where she discharged her entire cargo. The charterer did not exercise his right, by paying the additional fee, to have the vessel call at another port. Loading and discharging consumed a total of seventeen calendar days, including Sundays and holidays, which were not counted as part of the "all purposes" allowance of eleven days. The charter-party also provided that only "weather working days" would be debited against cargo-working time. The available data do not indicate whether adverse weather was experienced, and therefore it is impossible to establish from the record whether demurrage was incurred.

Had the *Regina Oldendorff* been used to carry a full cargo of wheat ("heavy grain") which stowed at 45 cubic feet (1.32 cubic meters) per long ton, her entire capacity of 1,620,000 cubic feet (48,600 cubic meters) would have been filled, and a total of approximately 36,000 tons of grain would have constituted the load. To explain how this figure was derived, it is necessary to point out that the Panama Canal lies in the tropical load line zone, which means that the ship could be loaded to her deepest marks. When the ship crossed latitude 20° North, a distance equal to three days steaming time from Panama, however, she was in the seasonal summer zone, which requires a higher freeboard and consequently a lighter load. On leaving Panama, the *Regina Oldendorff's* deadweight was that which was permitted at the summer load line zone plus whatever fuel she might burn on passage to 20° North. Consumption of fuel was about 40 tons a day; the excess on departure from the Canal Zone would have been 120 tons. The computation of the tonnage available for cargo was as follows:

Deadweight capacity on summer load line	37,150 tons
Extra deadweight permitted in tropical zone	<u>120 tons</u>
Total deadweight departing Panama	37,270 tons
Fuel for passage Cristobal/Rotterdam (15 days plus safety reserve equal to 5 days consumption)	800 tons
Water, stores, lubricating oil, etc.	<u>500 tons</u>

Total deductions	<u>1,300 tons</u>
Deadweight available for cargo	35,970 tons

It is an interesting operational footnote that the light weight of the cargo left ample deadweight for the ship to be fueled to capacity at Panama, a relatively cheap port in which to obtain oil.

Toledo, completing St. Lawrence, to Antwerp/Rotterdam/Amsterdam. Motorship *Moordrecht*. Heavy grain, 15,500 tons from Toledo, \$15.50, completing in St. Lawrence up to 21,500 tons, \$8.50; option to load only in Toledo, \$2.50 extra; option to discharge German North Sea port, 25 cents extra; option to discharge Avonmouth/Glasgow/Belfast, 50 cents extra. Free in and out and trimmed; 7 days all purposes, basis 4 days load if "top off" or 3 days without "topping off." 3,000 tons discharge. April.

The *Moordrecht* was a bulker with a deadweight capacity on summer load line of 22,430 tons. Her beam was less than the 75½ feet maximum width of the locks on the St. Lawrence Seaway. Because the greatest depth in the channels of the Great Lakes is about 26 feet, the cargo to be loaded in Toledo was limited to 15,500 tons. The actual capacity of the ship, after deductions for fuel, water, stores, and equipment, was 21,500 tons.

This charter-party contains several interesting options. The rate on the heavy grain (wheat) loaded in Toledo, Ohio, was \$15.50 and on a full cargo of 15,500 tons would produce revenue to the ship of \$240,250. By "topping off" in a St. Lawrence river port with 6,000 tons of wheat freighted at \$8.50, the ship's earnings would be increased by \$51,000 for a total of \$291,250. If only the 15,500 tons lifted in Toledo were carried to Europe, the freight rate would be raised by \$2.50 to \$18.00 per ton, and would bring \$279,000 to the ship. This would be \$12,250 less than if the topping off cargo were taken aboard. A measurable part of this difference would be offset by the saving in ship time attained by not stopping in the St. Lawrence river port.

At the time the ship was fixed for this business, the charterer was uncertain where delivery would be required. Flexibility, therefore, was written into the contract: the ship could be sent to a German port on the North Sea, a day's steaming distance from Amsterdam, on payment of twenty-five cents a ton above the stipulated freight rate. Should delivery be required in a port on the West Coast of the

British Isles or in Belfast, Northern Ireland, the additional payment would be fifty cents a ton.

An essential part of the contract was the statement that the charterer pay the cost of loading, trimming, and discharging the grain ("free in and out and trimmed"). Seven days were allowed for both loading and unloading; if more port time were used, the charterer would be subject to a bill for demurrage. Finally, the charterer guaranteed that the wheat would be discharged at the minimum rate of 3,000 tons a day.

The record of the voyage revealed that the *Moordrecht* sailed from Liverpool in ballast on April 7, and arrived in Toledo on April 28. She loaded her cargo and sailed for Montreal on April 30. Topping off was accomplished between May 2, when the *Moordrecht* arrived, and May 5, when she was cleared for Europe. Delivery of the cargo actually was made at Leith, on the east coast of Scotland; obviously special arrangements between the charterer and the owner were required for this development.

River Plate to U. K./Gibraltar-Hamburg Range/Mediterranean not East of the Adriatic. Motorship *North Emperor*, 46,427 tons deadweight, and 1,893,000 cu. ft. grain space. 15 knots on 43 tons fuel oil, 1,000 seconds, plus 2 tons diesel oil. \$3,100 per day plus \$127,500 ballast bonus. Trip, delivery River Plate, redelivery dropping pilot, U. K., Gibraltar-Hamburg range, or Mediterranean not east of the Adriatic. Charterers: Continental Grain Corporation. April.

To put the ship in position to carry out this time charter, it was necessary for the owner to send the *North Emperor* in ballast from Koper, near Trieste, Yugoslavia, where her previous charter had terminated, to Argentina. The full cost of the voyage was for the owner's account, but the circumstances of the charter were somewhat unusual. From the statement in the fixture report that a payment of \$127,500 would be made to the owner for the ballast voyage, it may be deduced that the market for ships at the time of the fixture was very competitive, and only a few vessels were available. To meet its own corporate commitments, Continental Grain Corporation was willing to assume part of the expense of the transatlantic voyage. The agreed charter hire of \$3,100 per day was not to be paid until the ship had passed Recalada lightship (off Montevideo, Uruguay) on the way to Rosario, 191 miles up the Rio de la Plata (River Plate) from Buenos Aires.

The ship was time chartered for that period required to make the trip from the loading port in the River Plate to a port in the United Kingdom, or in Continental Europe between Gibraltar and the eastern shore of the Adriatic Sea. As is customary in time charters, the charterer was to pay not only the daily hire of \$3,100 but also the cost of fuel consumed, which the owner warranted would not exceed 43 tons of intermediate grade fuel oil (1,000 seconds Redwood), plus two tons of diesel fuel, for each 24 hours under way. The deadweight on her summer load line was stipulated to be 46,427 tons, and 1,893,000 cubic feet (56,790 cubic meters) of space below decks were guaranteed for a grain cargo.

As a device to establish the precise moment when the ship would go "off hire," the charter-party contained the provision that when the pilot was dropped as the ship was commencing her outbound voyage from the destination port, payments under the charter would cease. Similarly, but not included in the fixture report quoted above, it was agreed between the contracting parties that the ship would go "on hire" when she passed Recalada lightship, on her way up the River Plate to Rosario, where the first part of the load would be taken aboard. Because of the shallow draft of the river at that point, the *North Emperor* was brought back to Buenos Aires to top off her load, and thus bring her down to her summer load line.

The record of this voyage shows that the *North Emperor* departed from Koper on March 24, called at St. Vincent for bunkers on April 4, and arrived in Rosario on April 16. She was unable to depart from that port until April 23, because of congestion. On arrival in Buenos Aires, she experienced the same type of difficulty, and was delayed in finding a berth until May 5. Topping off operations were finished in time for the ship to sail for a port in Italy on May 7. The delays incurred in connection with loading were for the account of the time charterer, whose responsibility according to the terms of the charter-party commenced when the ship passed Recalada lightship and continued until she was redelivered at the end of the voyage in Italy.

Tramp shipping can be highly remunerative to owners when there is a brisk demand for transportation of cargoes moving in bulk, and can be disastrous when, for reasons beyond any possibility of control by the owner, there is a sudden reduction in requirements. Inherent in this situation is the fact that ships normally working in a certain trade will be reassigned to different routes

whenever the need for carriers diminishes. In such circumstances, the competition becomes very intense, freight rates drop precipitately, and the older, less efficient ships may have to be laid up because of no employment.

In a sense, the tramp ship's owner nearly always is at a disadvantage in the market place, because only very rarely is there so small a supply of ships that owners can set their prices on a "take or leave" basis. In the words of one economist, the shipowner is a "price-taker," which means that the charterer usually dictates the maximum for ocean freights. Even granting that the bold and venturesome carrier may send his ship to new geographical regions, or reassign her to a different trade, he will find that for the most part his freedom to demand high rates is circumscribed by a variety of limitations.

Despite the problems which the owner of tramp ships must face, it remains true that it is an enterprise in which the rewards can be extremely good, and therefore it attracts both individuals and corporations to invest their capital. Whereas in the years before World War II it was possible to purchase a ship for a sum not beyond the capability of one man (or a small group of investors) to raise, the great increase in the cost of ships resulting from their larger size, higher speed, more elaborate and hence more expensive accommodations for the officers and crew, and often the provision of costly cargo-working gear such as deck cranes, mechanically operated hatch covers, and better cargo ventilating equipment, have combined to reduce the number of individual adventurers. The actual number of ships has been diminished in the postwar years, but total carrying capacity has been increased as vessel sizes have been magnified and speeds have been augmented in the ceaseless effort to achieve greater efficiency and therefore greater profit.

The emergence, in the decades following the second world war, of the large bulk carrier has had a profound impact upon the tramp shipping business. Formerly iron ore, coal, grain, and lumber moved in comparatively small ships which were interchangeable between trades. When demand for steel suddenly exploded, the limited capacities of the general-purpose tramp, even though this ship had grown by almost one hundred percent in its deadweight tonnage, proved to be unacceptable to industry, and the semispe-



The size of the 163,760-tons deadweight ore-oil carrier *Garden Green* (built 1973) is clearly evident. She was 280 meters long, 47.4 meters wide, with a draft of 17.7 meters. Photo: Overseas Shipbuilding Group.

cialized bulk carriers became more and more prominent. Today there are hundreds of bulkers which have very nearly preempted the trade in metallic ores. Many are able to lift more than 75,000 tons of cargo but require facilities in both loading and discharge ports because they have no cargo-handling gear installed. Some of these big ships are designed to carry alternate cargoes of commodities such as grain or fertilizers, and have become formidable competitors of the small dry-cargo tramps.

A further handicap which the conventional tramp must overcome is the invasion of the grain trade by tankers. It was only after the conclusion of hostilities in 1945 that it became practical to transport wheat in the cargo spaces of a tanker. This eventuated from the perfecting of a portable suction device which would remove wheat and other grain by pneumatic action rather than having to rely upon the conventional discharging equipment of the "marine leg" with its endless chain of buckets and a conveyor belt leading to the shoreside receptacles. Once the "vacuveyors" had demonstrated their effectiveness, literally scores of tankers were



With all hatches open, the 121,500-tons deadweight bulk carrier *Endeavor* (built 1975) awaits her cargo. Photo: Overseas Shipholding Group.

assigned to carrying grain. They were attractive to shippers because the cargo tanks were comparatively small, the hatches were of minimum size, and neither shifting boards nor feeders were needed. These advantages were translated into actual cash savings not only in the cost of these fittings, but also in ship time. Modern dry-cargo bulkers have compromised by installing quickly portable shifting boards which are part of the ship's outfit, and can be set up or removed in a short time, usually by the ship's crew.

Between the loss of iron ore and the keen competition for grain cargoes, the owner of the conventional tramp has found it more difficult to keep his ships employed, but has earned higher profits during those times when the demand has been strong for the type of tonnage he controls. It remains true that most ports of the world have a limiting depth of between thirty and thirty-five feet, and it is to these shallow harbors that the modern, efficient, and economical tramps make their voyages.

Tramp shipping today requires, as does the entire steamship industry, a higher degree of managerial skill than formerly, as well as the tools of good management such as the computer, the satellite-relayed telephone, and the complex automation found more and

more aboard ships as owners seek to enhance their profit-making capability by the use of sleepless and tireless monitoring devices. This sophisticated type of supervision permits more effective control of ship use, more accurate cost determination, and overall superior performance. The essential requirement remains for a keen awareness of where the business can be found and when the freight rates are favorable for time charters rather than for voyage charters, and only the skilled and careful owner is able to place his ship where it will meet the demands of the industrial world. There is now, and there always will be, a vital role for the tramp ship to play in the continuing drama of the commercial community.

Chartering and Tramp Ship Operation

TO MORE than one maritime country, tramp shipping provides a substantial portion of national income. This is especially true of Great Britain, the Scandinavian countries, Greece, Italy, and Japan. No single country today dominates the world's shipping trade as did England before 1914, but the British, Norwegians, and Greeks are still among the leaders. To these nations, sea carriage is an important source of foreign credits, and seafaring a traditional occupation for their people.

This tradition of life at sea helps to explain how men—and increasingly women, as careers at sea become available to them—can face patiently, courageously, and with spiritual satisfaction the labors involved in tramp ship operation. A typical voyage is marked by hard work and prolonged absence from the home port as the ship finds employment in the “cross trades,” i.e., on routes that cross the oceans without touching the shores of home. In fact, it is rather exceptional when the vessel is assigned a voyage back to the land of her registry.

That protracted voyages are anything but unusual gives rise to a variety of problems in management. There are, for instance, the ever present necessities of fueling, provisioning, and putting the ship into ports never before visited. The master must cope with personnel changes occurring by reason of death, desertion, or illness; if crews are repatriated periodically, there are significant alterations in the composition of the ship's company. Although what is carried nearly always is one of the commodities moving in bulk, there still are cargo-loading problems to be solved. Port entrance and clearance routines for remote or infrequently visited places must be envisioned and anticipated. While some of these details may be handled from the owner's office in the distant homeland, many of them inevitably are delegated to the master of the

ship. Throughout the voyage, he must serve as the representative of the owner, and thus becomes to a measurable extent the custodian of that often shadowy figure's fortunes.

The competitive nature of the tramp trade, previously described in some detail, is such that no permanent system of government subsidy has been developed in any country. A very interesting and quite successful venture into this field was made by the British government in 1935 when world shipping was critically depressed. A law was passed setting up a fund of two million pounds sterling to help British tramp operators, on condition that the industry devise the means to administer the scheme. A Tramp Shipping Administrative Committee was formed to work toward the goals of minimizing domestic competition, improving freight rates and chartering conditions, and promoting the fullest employment of British ships in competition with vessels which received some form of governmental assistance. British tramps were rationed to prevent uneconomic competition, and through the efforts of cooperating owners the rates of charter hire were fixed at levels which prevailed in 1934. These were sufficient only to keep the ship in operation, without allowance for depreciation or interest. The stability that resulted did improve market rates about fifteen percent in the first year, even though the volume of cargo moving remained unchanged. The scheme was continued until 1938. No similar comprehensive plan has been instituted in the post-World War II decades.¹

No undisguised subsidy is paid to tramp ships today; there are no regulations governing where the ships shall go, or how many voyages must be completed each year. Fleets vary in size from one to many ships, depending upon the financial status of the individual owners. In the Baltic region, small operators sometimes form an association and empower a single manager to handle the ships as though they belonged to one corporation.

Inasmuch as tramps spend most of their economic lives far away from their ports of registry, owners do not maintain large offices, crowded with hurrying clerks and noisy with business machines. It is more likely that the typical tramp operator will have a small headquarters where a few persons handle the details of ship movement and arrange future commitments. Although it is not precisely applicable today, the old maxim of the Golden Age does indicate

the approximate ratio of shoreside personnel to ships at sea: "One man in the office for each ship in the fleet."

The home office of the tramp relies upon brokers around the world for representation in the constant search for cargoes. Daily, and sometimes hourly, cables come from many regions with information or offers of business. Decisions are made and shipmasters, agents, brokers, and governments are informed of what the owner deems best in a wide variety of circumstances. Modern communications have changed the trade to the extent that owners now make all the major decisions, and the master no longer is expected to find employment for his ship when a charter expires. The master, however, must be a shrewd businessman as well as an excellent seaman, a good administrator, an expert in personnel management, and a conscientious representative of his owners.

Tramp owners depend upon ship brokers to provide opportunities for employment. To describe the relationship between them, it is fitting to simulate the daily drama of chartering vessels, and to follow the actions of the participants. The following situation is fictitious, but the methods depicted are those in current use.

A wheat exporting firm in Chicago, King Corporation, purchases grain in the American market and then sells it to European customers. King Corporation owns no ships, preferring to leave the burdens of vessel operation to specialists, and relies upon brokers to find suitable ships, invariably taken on voyage charters.

King is committed to deliver 200,000 tons of American wheat (heavy grain) to a French importer who has directed that delivery of the first shipload of at least 60,000 tons must be made by March 15. King must make and pay for all arrangements to transport the wheat. The vessel to be used has to hold the highest rating from the ship classification organization in order to meet the standards of the insurance underwriter.

King Corporation directs its broker, James Sharp, in New York, to find an acceptable ship at the lowest possible freight rate. Sharp canvasses the other brokerage houses, but finds nothing listed in New York which meets King's requirements. Sharp therefore cables his London counterpart, Thomas White, to ask for assistance, well aware that White can go to the Baltic Exchange and in short order be able to offer a suitable ship. The cable to White is phrased in these words:

RELIABLE PRINCIPAL OFFERS FIRM CARGO OF 60,000 TO 70,000 TONS HEAVY GRAIN NEW ORLEANS TO ONE PORT BORDEAUX-HAMBURG RANGE. LOADING NEW ORLEANS FIRST HALF MARCH. WILL USE NORGRAIN CHARTER.

White understands from this cable that a ship able to carry not less than 60,000 tons of cargo in addition to fuel, stores, and water must be in New Orleans ready to load between the first and fifteenth of March. The charter-party to be used is a well known document developed particularly to meet the demands of American grain shipment. White has a copy of the "Norgrain" charter-party in his office.

Perusal of the list of ships represented by White discloses that none of these vessels will be ready within the time limits. White therefore proceeds to the floor of the Baltic Exchange, where he meets brokers seeking cargoes for their principals' ships and other brokers looking for ships to transport cargoes. White quickly makes contact with Harrison Burnett, who represents a Norwegian owner domiciled in Oslo, Olaf Berg. Burnett tells White that Berg's big motorship *Great Viking* will be off-hire in a Texas port about the end of February. White requests a firm offer of this ship, and Burnett accordingly cables his principal, as follows:

FIRST CLASS CHARTERER MAKES FIRM OFFER OF CARGO OF 60,000 TO 70,000 TONS HEAVY GRAIN NEW ORLEANS TO ONE PORT BORDEAUX-HAMBURG RANGE. LOADING NEW ORLEANS MARCH ONE THROUGH FIFTEEN. USING NORGRAIN CHARTER. PLEASE OFFER FIRM BY FIFTEEN HUNDRED HOURS LONDON TIME TODAY.

When Olaf Berg reads this cable, he decides that a voyage of the *Great Viking* to a North European port is desirable at this time because of certain shipyard repairs he has planned for her. Berg immediately sends this cable to Burnett:

NORWEGIAN MOTORSHIP GREAT VIKING, SERVICE SPEED 15 KNOTS, 64,000 TONS DEADWEIGHT, 79,500 CUBIC METERS GRAIN CAPACITY, SEVEN HOLDS, GEARLESS, CLASSED HIGHEST AT DET NORSKE VERITAS, NOW DISCHARGING HOUSTON AND EXPECTED READY TO LOAD AFTER MARCH SECOND. SUBJECT TO REPLY REACHING US BY SEVENTEEN HUNDRED HOURS TOMORROW OSLO TIME, AUTHORIZE YOU TO NEGOTIATE FOR FULL CARGO HEAVY GRAIN 62,000 METRIC TONS FIVE PERCENT MORE OR LESS

AT VESSEL'S OPTION FROM ONE US GULF PORT TO ONE PORT BORDEAUX-HAMBURG RANGE. DISCHARGING PORT DECLARABLE ON SIGNING BILLS OF LADING. CANCELLING DATE MARCH THIRTEENTH. FREIGHT FIFTEEN DOLLARS PER METRIC TON, FREE IN AND OUT AND TRIMMED. LOADING AND UNLOADING SIX WEATHER WORKING DAYS, SUNDAYS AND HOLIDAYS INCLUDED, LAYTIME REVERSIBLE. DEMURRAGE FOURTEEN THOUSAND THREE HUNDRED DOLLARS PER DAY, DISPATCH HALF DEMURRAGE. ALL DOLLAR QUOTATIONS PREDICATED ON EXCHANGE RATE US DOLLARS TO NORWEGIAN KRONER PREVAILING CLOSE OF BUSINESS ON DAY BILLS OF LADING ARE SIGNED. NORGRAIN CHARTER ACCEPTABLE EXCEPT LAYTIME FOR LOADING TO COMMENCE AT NINETEEN HUNDRED HOURS IF NOTICE OF READINESS TENDERED BETWEEN THIRTEEN AND SEVENTEEN HUNDRED HOURS. DEMURRAGE TO BE PAID PER DAY OR PER HALF DAY AND NOT OTHERWISE PRO-RATED. BROKERAGE COMMISSIONS PAYABLE MAXIMUM THREE AND THREE QUARTERS PERCENT OF CHARTER HIRE.

Burnett immediately visits White's office, and the two brokers analyze the message from Berg. The ship can lift the minimum quantity of grain specified by Sharp, and she can be in New Orleans within the time limit. The obligation to declare the port of discharge at the time the bills of lading are signed by the master is consistent with good chartering practice: the master must know the destination of the voyage he is about to undertake. The freight rate assigns all costs of handling the grain at both loading and unloading ports to the charterer. Time allowed to take grain aboard and to discharge it is adequate, given the facilities at the port of New Orleans and the probable destination ports, and takes cognizance of the possibility that one port may be superior to another in its ability to work ships rapidly. By permitting whatever time is saved in one port to be credited toward the allowed working days in the other port, the charterer is encouraged to use the hours effectively. Demurrage—the charge for keeping the ship in port beyond the permitted number of working days—is an essential part of any voyage charter, and usually represents the owner's costs and a very small margin of profit. Berg obviously is eager for the business since he offers to pay brokerage commissions somewhat higher than the normal rate of $2\frac{1}{2}$ percent.

White considers that the terms of the offer are fair and reasonable when compared to the current charter market, and sends a cable to Sharp in New York relaying everything except the rate of commission, which for Sharp's information is quoted as 1¼ per cent. For his part, Sharp notifies King Corporation of the offer and is instructed to make a counter proposal, which is presented to White in this cablegram:

REFERENCE YOUR CABLE OF YESTERDAY OFFERING GREAT VIKING. SUBJECT TO REPLY REACHING US BY SEVENTEEN HUNDRED HOURS TODAY NEW YORK TIME, CHARTERERS ACCEPT OWNER'S OFFER EXCEPT LAY DAYS TO BE MARCH FOUR TO TWELVE INCLUSIVE. FREIGHT RATE TO BE THIRTEEN DOLLARS AND TWENTY FIVE CENTS. LAY TIME TO BE SEVEN DAYS SUNDAYS AND HOLIDAYS EXCLUDED. DEMURRAGE TO BE THIRTEEN THOUSAND SEVEN HUNDRED DOLLARS PER DAY PAYABLE DAY BY DAY AND PRO-RATED FOR ANY PART OF A DAY.

On receipt of this message, White goes to Burnett's office to present the proposed modifications. Burnett, as a broker, has no authority to accept or reject these changes, and therefore relays the cable to Berg without delay.

Berg in the meantime has noted that the charter market is weak, and that his original offer of fifteen dollars probably is out of competition. A recomputation of the cost of the anticipated voyage shows that a rate of thirteen dollars would yield an acceptable profit. He knows that once grain loading and discharging operations commence, they proceed without stopping. He is aware that he is under pressure to pay dispatch money when the charterer asks for seven days cargo-working time. There also is the demand for the ship to stand idle on Sundays and holidays, which would cut into the profit remaining when the rate is reduced to the proposed figure per ton. All these factors having been considered, Berg sends this cable to Burnett:

REFERENCE YOUR CABLE REGARDING GREAT VIKING. CHARTERER'S COUNTER OFFERS ACCEPTABLE EXCEPT LAYTIME REMAINS AT SIX WEATHER WORKING DAYS SUNDAYS AND HOLIDAYS INCLUDED.

Burnett notifies White of this response, and the information is transmitted immediately to Sharp in New York, who recommends

to King Corporation that the terms set by the shipowner be accepted without further negotiation. King is pleased that the freight rate has been reduced, and concedes the other modifications. Sharp therefore is instructed to close the deal, which is done by the following cable to White:

REFERENCE YOUR CABLE. GREAT VIKING ACCEPTED. CHARTERERS ARE KING CORPORATION OF CHICAGO. AUTHORIZE YOU TO SIGN CHARTER ON CHARTERER'S BEHALF BUT AS BROKER ONLY. AIRMAIL SIX COPIES OF COMPLETED CHARTER. THANK YOU FOR YOUR EFFORTS.

White passes the word to Burnett, who notifies Berg that his ship has been fixed for the voyage. Berg acknowledges this message in this cable:

REFERENCE YOUR CABLE CONCERNING GREAT VIKING. THIS CONFIRMS THAT TERMS AS APPROVED BY CHARTERER ARE ACCEPTABLE. SHIP WILL BE ORDERED TO NEW ORLEANS TO ARRIVE MARCH FOURTH. AUTHORIZE YOU TO SIGN CHARTER AS MY BROKER. SEND ME THREE COPIES OF CHARTER. THANK YOU FOR THIS BUSINESS.

Burnett relays this confirmation of the charter to White, who closes his file with this cablegram to Sharp:

RECONFIRM FIXTURE OF GREAT VIKING. SHIP ORDERED TO ARRIVE NEW ORLEANS MARCH FOURTH. THANK YOU FOR THIS BUSINESS.

Negotiations having been brought to a successful conclusion, the charter-party is completed by White's staff, and copies are distributed as directed by the principals. The original document, duly signed by Burnett and White, is filed in the vault at White's office and is available to authorized persons at any time.

Berg composes a brief message to the master of the *Great Viking* instructing him to sail for New Orleans in time to arrive in that city during the forenoon of March 4, so that loading can commence that same day. This cablegram is followed by an airmail letter enclosing a copy of the charter-party and specific instructions for the conduct of the voyage. In the gracious tradition of shipping, this letter is concluded with the statement, "We take this opportunity to express the hope that you will have a pleasant and satisfactory voyage."

For their part in the transaction, the three brokers will each receive $1\frac{1}{4}$ percent of the net earnings of the *Great Viking* under the charter, after demurrage or dispatch is paid.

A new development in chartering made its debut in 1985 when the Baltic International Freight Futures Exchange (BIFFEX) began operation. Through buying and selling "index points," it now is possible for a shipowner or charterer to protect himself against the likelihood that charter hire will decline or rise before the ship is available to lift a cargo, or the cargo is ready for transportation.

Thirteen significant ocean trading routes on which dry cargo moves in chartered vessels have been chosen by the exchange, and weighted according to their relative importance in the world tramp market. Each day, the members of a panel of ship brokers who belong to the Baltic Exchange, individually and without consultation, submit the spot rates which they consider applicable to the thirteen routes. The panel's recommended rates are entered into a computer which has been appropriately programmed. From the computer is derived the so-called Baltic Freight Index, which is published in maritime and commercial journals:

Baltic Freight Index

954.50 (Previous Day — 986.00)

The Baltic International Freight Futures Exchange in London and the International Futures Exchange (Bermuda) Ltd. offer owners and charterers in the bulk shipping community a forum for futures trading of a freight rate index. The contracts are based on the Baltic Freight Index, announced daily, of 13 routes listed below.

Routes	Commodity	Weight	Dollars/ Metric Ton	Index
U.S. Gulf-North Continent	Grain	20%	9.100	1002.5
U.S. Gulf-Japan	Grain	20%	14.000	980.0
U.S. North Pacific Japan	Grain	15%	8.492	920.5
U.S. Gulf-Venezuela	Grain	5%	11.417	978.0
Antwerp-Red Sea	Grain	5%	18.867	912.5
Hampton Roads-Richards Bay-Japan	Coal	5%	9.850	955.0
Hampton Roads-North Continent	Coal	5%	5.383	931.5
Queensland-Rotterdam	Coal	5%	9.175	810.5
Vancouver-San Diego-Rotterdam	Petcoke	5%	10.633	809.0
Monrovia-Rotterdam	Iron Ore	5%	4.717	1048.0
Recife-U.S. East Coast	Sugar	5%	15.375	936.5
Hamburg-West Coast India	Potash	2.5%	28.000	968.0
Aqaba-West Coast India	Phosrock	2.5%	12.125	895.5

Each Baltic Freight Index point has been assigned an arbitrary value of \$10.00. Any shipowner or charterer may purchase or sell points by arranging with a member of the BIFFEX to represent

him on the floor of the Baltic Exchange. A deposit from the trader is required, and if the market moves contrary to expectations, the broker will require payments equal to the decline in the market value of the index. As each transaction is accomplished, the information is transmitted for inclusion in the computer of the International Commodities Clearing House, Limited, which daily produces a record of all dealings. The dealings are then confirmed by the members of the BIFFEX. The clearing house also serves as the banker for the BIFFEX, and provides a guarantee that the contracts will be fulfilled. Contracts are fulfilled by either purchase or sale of the futures contracts, or by a cash payment at stipulated times during the year.

Simplified examples of the use of futures to protect against fluctuations in charter hire will demonstrate the working of the plan.

A ship of 100,000 tons capacity will be available in three months for a voyage from Hampton Roads to Japan, carrying high grade coal. On October 1, the freight rate for this trip is \$12.00 per ton, producing a revenue of \$1,200,000 for the voyage. The BIFFEX index on October 1 is 1,000, and the index forecast for January 1 also is 1,000. One contract on the BIFFEX at \$10.00 per point has a corresponding value of \$10,000. The owner sells 120 contracts at the rate of \$10.00 per point, and gains the equivalent of the ship's earning. In October, the owner was pessimistic, and predicted that the market would decline by January 1. This expectation was realized, when rates dropped by 10 percent to \$10.80 per ton. The owner bought back the 120 contracts for \$1,080,000. This gave him a net profit on the futures transaction of \$120,000, which offset the loss of an equal amount of the vessel's charter hire.

In a similar manner, a charterer might hedge his commitment for future movement of cargo. On October 1, the charter rate for heavy grain moving from New Orleans to Rotterdam was \$9.00 per ton, with reliable indications that the level would rise before January 1. The shipper of grain planned to send a parcel of 50,000 tons to Rotterdam, loading on January 1. On October 1, the BIFFEX index stood at 1,000. The charterer therefore bought 50 contracts at the price of \$9.00 per point, paying \$450,000 for the lot. As anticipated, the market did rise, and on January 1 the quoted rate was \$9.90 per ton of grain. The contracts were sold at \$9.90, and

realized \$495,000 for the charterer, who thus had adequate funds to pay the increased freight rate.²

The competition mentioned earlier in the analysis of tramp shipping manifests itself principally in the rates charged by different shipowners for transporting the same commodities. A superficial glance at the reports of ship fixtures suggests that these rates are established in a casual fashion. To the contrary, much thought and skill are required in developing the rates of charter hire. Fluctuations reflect the circumstances facing both owners and charterers at the precise moment that a decision must be reached. The first step in determining the price is to estimate the cost of performing under the charter. Success or failure—both in the sense of obtaining the business and in turning a profit from the transaction—may hinge upon the difference of a few cents a ton. The importance of the computer in establishing an acceptable rate of charter hire has been mentioned already but deserves repeated emphasis.

Of the numerous factors influencing the formation of freight rates, it is generally acknowledged that the value of the commodity has the greatest significance. In theory, the shipper of a ton of gold is able to pay \$1,000 for its transportation, while the exporter of coal, which has a value at the mine of only \$10 a ton, often cannot accept a rate of \$3 a ton. The economic reason for tramp shipping is to provide the transportation required by those articles of low value exemplified by coal. To do this requires that daily the tramp shipowner set freight rates which are low enough to encourage the movement of goods yet high enough to cover the expense of operating his ship, and ideally to yield at least a small profit.

Knowing his costs down to the smallest detail is one way in which the owner attempts to achieve this objective. "Costs," in this connection, means more than simply the obvious out-of-pocket expenses involved in ship operation; also included must be the less conspicuous items such as amortization, interest, and overhead. For example, a general-purpose tramp with operating costs of \$200 an hour is chartered to carry a load of 15,000 tons of coal. Loading at the rate of 500 tons an hour will take 30 hours. Only 200 tons are discharged per hour, for a total of 75 working hours. The charter rate must include a charge for 105 hours at \$200 per hour (\$1.40 per ton) to recover costs.

Some owners develop comprehensive tables of ship costs per deadweight ton which include crew wages and subsistence, vessel maintenance and repair, insurance, amortization, interest, and a contribution toward overhead. Whether the ship is lying in port waiting for assignment or steaming across the ocean, these items do not vary substantially. The costs that do change with the employment of the vessel are fuel, port fees, pilotage, towage, dockage and wharfage, and cargo handling. When all costs are computed, the owner should add a sum sufficient to cover unforeseen contingencies and to provide a profit. If the market is "strong," meaning that there are many cargoes and few ships, these cost figures can be used without trimming when negotiating for a charter. On the other hand, if there are many ships and few cargoes, and the market therefore is considered to be "weak," the owner must exercise the most exquisite judgment in cutting his estimates of cost to meet the figures quoted by his competitors. Sometimes it is necessary to accept business at an actual loss rather than to incur the expense of laying up the ship.

Under a gross form voyage charter, for example, the owner of a 50,000-ton vessel chartered to carry coal from Hampton Roads to Rotterdam for \$14.00 a ton pays all costs out of that sum. Expenses for loading and discharging the coal are \$5.00 per ton. Actual ship operating costs, including port time, are \$7.00 per ton, so that on the \$14.00 rate, the owner's profit is \$2.00, or \$100,000 for the voyage. Vessel costs are \$12,000 per day; any delays in the ship's passage, or unexpected increase in the cost of fuel or cargo handling reduce the margin of profit significantly.

The advent of the computer has given the modern shipowner the capability of sophisticated examination of all the elements of cost and potential profit involved in a proposed voyage. Each element of expense can be identified as a component of the venture. For instance, the speed of the ship can be related to the smoothness of her hull and to the cost of fuel when purchased in various bunkering ports. The rapidity with which the proposed cargo can be handled in the ports of loading and discharge can have a major bearing on the determination of the length of the voyage. The actual number of tons of a given commodity which can be stowed below decks is related to the amount of space needed to accommodate one weight ton. Detentions may be individually unimportant,

but cumulatively can be disastrous. The willingness of longshoremen to work overtime or on holidays and religious festivals, delays in putting the ship alongside the cargo-working berth, loss of time from shortages of pilots or tugs in the harbor, and hours needed to prepare the ship to accept the proposed load, can have a profound effect on the profitability of the voyage. Providing this kind of detailed information, set forth in terms of so much money per hour, is justification for using the computer in making voyage estimates.

In working out the finances of the voyage or voyages, the computer can develop in any desired detail such things as these: costs of the ship lying in port at anchor and at the berth working cargo; costs of the ship steaming fully loaded at designed speed; fuel consumption at different speeds and drafts; costs of fuel when purchased at different points along the route, compared with the time needed for refueling; port and harbor charges, with particular reference to harbor dues on entering the port and accrued day by day until departure; costs of loading and discharging the cargoes contemplated for the complete voyage; and the costs of overtime when cargo is being worked.

Once these data have been made available, it is possible to develop a program by which the computer will determine facts of this type concerning the proposed employment of the vessel: the itinerary of the ship, showing for each port the arrival time, waiting time, predictable lost time, and anticipated departure time; costs for each port where cargo is worked, including stevedoring and overtime expenses; and total costs of harbor dues and fees. Using these basic figures, the computer can be programmed to provide management with summaries of the cost to transport one ton or a full shipload of cargo, the total revenue to be earned on the contemplated voyage, and an analysis of the financial results of the planned employment. This last presentation would include depreciation, interest, overhead, ship operating costs, port charges, cargo-handling expense, and the net profit (or loss) at the offered rate of compensation.

Just as the computer has made management much more systematic and less dependent upon instinct and experience, so the advances in naval architecture and marine engineering have influenced profoundly the development of the tramp fleets of the



This shipboard gantry crane can be fitted with different specialized cargo-handling devices to expedite loading and unloading. Photo: Munck International.

world. The importance and versatility of the general-purpose tramp, typified by the S.D. 14 design of ship, already have been set forth. The preferred carrier in the major bulk trades in 1985 was one which could load between 25,000 and 35,000 tons of cargo, had adequate cargo gear to be self-sustaining, and had a sufficiently shallow draft to be able to enter almost any port in the world. For all the bulk commodity trades except iron ore, this size ship was very satisfactory.

The medium-sized bulker is the vessel of approximately 50,000 tons deadweight. This ship is assigned principally to haul coal and grain. Without any cargo-hoisting gear, she has proven popular in both the northern transatlantic and the northern transpacific routes.³

In the specialized iron ore service, the ship most favored in 1985 was the bulker with a capacity of between 100,000 and 125,000 tons of cargo. By assigning these ships to voyages with two ports of discharge, it was possible to send them into relatively shallow-draft ports without significant loss of revenue. This practice has rendered somewhat uncertain the long-range future of the smaller bulkers, particularly those between 60,000 and 80,000 tons dead-weight.⁴

TANKER CHARTERING

At first glance, the process by which tankers are chartered appears to be identical with that followed for dry cargo ships. That is to say, rates, terms, and conditions are negotiated among the interested parties, and when there is a meeting of minds, the deal is closed. The pressures exerted by the availability of cargoes and ships undeniably are the same, and therefore charter hire rates fluctuate from day to day. Likewise, the ship operators must be concerned with overall earnings if they are to cover out of pocket as well as capital and overhead expenses.

When the actual practice of ship brokers specializing in tanker chartering is examined, some interesting variations in the procedure emerge. The first, and by all odds the most conspicuous, is that the ship fixture reports rarely, if ever, specify the rate in monetary terms. Instead, the reports will read somewhat as follows:

Caribbean to United Kingdom/Continent, 60,000 tonner, dirty, Japanese flag, built 1979, WS 82.5.

Chiriqui Grande, Panama, to U. S. Gulf, May, 77,000 tons, dirty (Tanker *Copper Mountain*), American flag, built 1964, AR 97.5.

“WS” refers to *Worldscale*, a compendium of thousands of rates for voyages between ports, as for instance from Houston to Rotterdam or from Bahrein Island to Bayonne, New Jersey. Each rate has been calculated to return \$1,800 per voyage day for a standard vessel, defined as a tanker of 19,500 deadweight tons, traveling at 14 knots and consuming 28 tons of high viscosity fuel oil per day at sea. The \$1,800 is a purely nominal figure, and does not represent

an actual level of operating costs, nor does it produce a profit. All Worldscale (or WS) rates are based on 100; if the fixture report shows WS 40, it means a reduction from the calculated rate of 60 percent. Should the ship be fixed at higher than the scale, it would be reported as WS 140, to indicate that the charter hire was 40 percent above the published level.

Worldscale is an organization established in 1969 which has offices in New York and London. Twice each year it compiles the list of rates for 65,000 different voyages, those in the eastern hemisphere being processed by the London office, while voyages terminating in the western hemisphere are handled by the New York office. In addition to the base of 65,000 voyages, Worldscale has the computer capability to determine the rate between any two points not included in the base group. This is done by adding to the existing data base bank the distance involved and the port and canal (if any) charges which may be incurred for the nominal ship.⁵

The sample fixture report quoted above deals with a 60,000-ton ship chartered for the voyage from a Caribbean loading port to a port in the United Kingdom or, alternatively, to the Atlantic coast of the European continent. The ship is 3.178 times larger than the nominal vessel, for which the rate is \$11.53. The actual monetary payment would be determined by multiplying the rate by 3.178, and then taking 82.5 percent of the result, thus:

$$\$11.53 \times 3.178 = \$36.642$$

$$\$36.642 \times 82.5\% = \$30.23.$$

Worldscale was an outgrowth of the World War II project developed by the British and American governments to set freight rates for the tankers they had requisitioned for the military effort. The principle on which the resultant schedules of rates was erected is that, irrespective of the voyage performed, the governments would receive the same net return per day after making allowances for voyage expenses such as port charges, canal dues, and fuel costs. The convenience of the system, as contrasted with the prewar problem of negotiating each and every voyage individually, including establishing an actual monetary rate, was recognized, and chartering practice was modified accordingly when the war ended. Using the schedule, which applied to a standard T-2 tanker of 16,500 tons, with a speed of 14.5 knots and burning 30 tons of

heavy fuel oil per day, was much more efficient and a great time saver. As ships increased in deadweight tonnage and in performance characteristics, these new factors were worked into the negotiations for individual voyages of particular vessels.

The tanker trade between American ports is restricted by law to ships built, owned, and registered in the United States. The convenience of the rate scale in international movements was sufficient to justify developing a similar system for the domestic trade. When it was introduced in 1956, the compendium was known as the American Tanker Rate Schedule (ATRS). With periodic updating and appropriate modifications, including shortening the name to American Rates (AR), the schedule is still in use. (Worldscale does include many American coastwise voyages, but is rarely invoked by brokers.)

In practice, therefore, brokers for cargo interests and the shipowner will negotiate from the Worldscale or the American schedules, and, depending upon market conditions, will agree upon a rate which is either the published figure or a deviation, upward or downward, from that level. Variations from the base rate reflect the demand for ships, the availability of cargoes, the particular voyage proposed, the ships to be used, and any unusual charges the shipowner may be called upon to defray.

Because of the demonstrated economy of using a single large ship to carry a load as compared with assigning the cargo to two or more smaller ships, it is not uncommon in tanker chartering for the rate per ton of petroleum to be somewhat lower for the big ships. In mid-1985, for instance, a tanker of 240,000 tons deadweight was placed under charter to carry a cargo of black oil from West Africa to the United Kingdom-European continent at Worldscale 23. The next day, a tanker of 70,000 tons deadweight was fixed for the same voyage at WS 56. It is worthy of note, in this regard, that few of the very large crude carriers (those over 200,000 tons deadweight) are chartered at the nominal Worldscale rate. During the shipping recession of 1975-1980, these big vessels were fixed at WS 18 and WS 25, meaning that the owners were accepting reductions of 82 percent and 75 percent from the published schedule of rates. These variances are cited to indicate how competition and the economy of scale are reflected in the working of the tanker charter market.

When the negotiating parties have agreed, the rate is set in terms of the appropriate schedule, plus or minus a percentage, or at the scheduled rate. Allowances always are made for the type of cargo to be carried—dark oil (often called “dirty”), gasoline, or any other grade of petroleum—in order to compensate for the wear and tear on the ship, and the labor entailed to make the ship ready for the cargo.

It should be noted that it is not feasible to prepare similar rate schedules for dry cargoes because of the great variety of commodities which move in international commerce, the enormous number of potential voyages, and the wide disparity in cargo-han-



Specialized equipment is used to transfer coal from inland waterways' barges to a seagoing bulk carrier anchored in the Mississippi River. Photo: Cooper/T. Smith Stevedoring.

dling costs depending upon the individual ship's capabilities and port customs and charges. Ship-operating characteristics also differ so greatly in the dry cargo tramp trade that no "standard" vessel has been devised so far which fits into the picture as does the standard tanker of 19,500 tons consuming 28 tons of fuel oil per day at sea while steaming at 14 knots.

Although tramp ship operation continues today as an indispensable adjunct of international oceanborne commerce, the ships which carry the cargoes are very different indeed from the plodding coal-burners of that fabled Golden Age of Steam which ended in 1914. Today not only are the vessels larger and faster, but they are superior in every way to their progenitors. They also are very much more costly, and therefore management of these ships is more exacting and requires the support of the finest managerial tools. No matter how much the design and supervision of the ships may have changed, however, the burden of performance rests today, as it always has rested, upon the hard working personnel who take the ships across the seas and assure that the terms of the charter-parties are carried out efficiently and effectively. Although the salt-rimed and sea-weary tramp described so eloquently by Joseph Conrad, William McFee, and H. M. Tomlinson has disappeared into the mists of history, the seamen who make tramp shipping possible are still of that same heroic and honorable cast which these great writers so justly celebrated.

Organization of A Liner-Service Company

EVERY steamship company, regardless of nationality or location of corporate headquarters, is organized to meet its own particular needs and its own specialized functions, and therefore no uniform pattern of organization exists in the workaday world of shipping. Nevertheless, the functions of the different officials, by whatever title they may be known, and however they may fit into their individual companies, are sufficiently similar to permit their identification within the corporate structure.

At the top of the pyramid is the *Board of Directors*, usually made up of persons who have a financial or managerial interest in the company. The board may consist of from five to twenty or more people elected by the stockholders and responsible to them for proper management of the business. Always included as a voting member of this board is the chairman. Increasingly, in American corporate practice, this person also is the chief executive officer, meaning that he takes an active part in the affairs of the company and makes major decisions within the authority conferred upon him by the board. The president of the company, who in modern American business also serves as the chief operating officer, is exactly what that descriptive title indicates, and therefore is a member of the board. The treasurer, as the financial advisor, and the secretary, who keeps the minutes of the meetings and authenticates all official documents, very often serve as members of the board. In all matters of magnitude, especially those involving massive expenditures (as, for instance, the purchase or construction of ships, or entering into or withdrawing from a trade route) and the selection of senior executives, the board makes the final decisions.

The *President*, as the chief operating officer, is responsible to the board for the proper functioning of the entire organization. He may have come up through the ranks of the company, or he may

have been brought in at or near the top. Because he is the supervisor of all activities, he does not become involved for any protracted period of time in the working of any specific division, although his advice in those areas where he has specialized knowledge may be sought by the staff. As ships and their supporting operations have become more expensive through the years, the office of the president inexorably has been concerned primarily with the financial aspects of the company, and a primary consideration in the choice of a president is the soundness of financial judgments he has made in difficult circumstances.

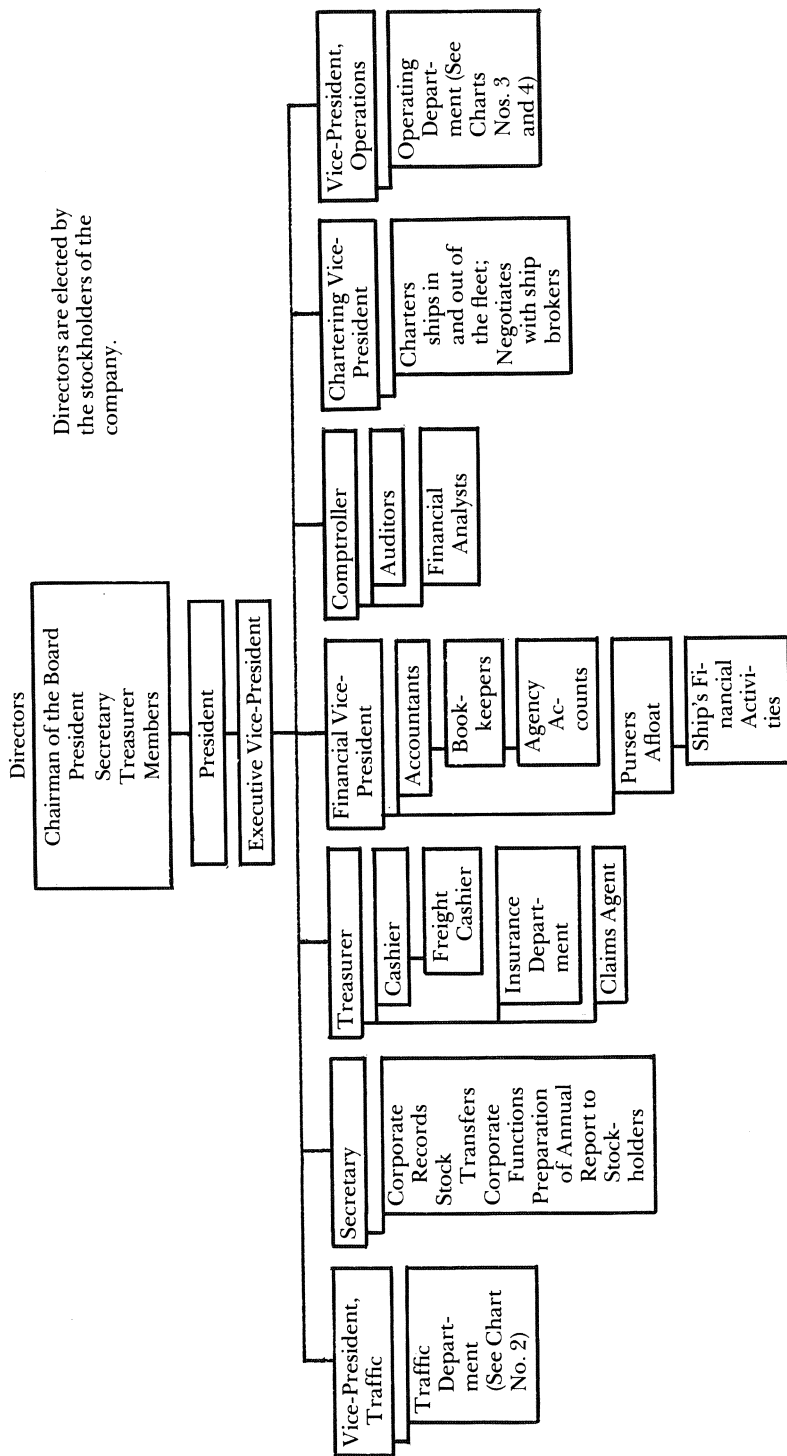
The demands upon his time imposed by these fiscal burdens have resulted in the president delegating much of the supervision of daily activity to the *Executive Vice-President*, who may be described accurately as the chief of staff. It is his responsibility to see that the policies promulgated by the board and the president are carried out. Within the scope of his authority, he makes all appropriate decisions and sends to the president's desk only those matters requiring special consideration. He is able to assume the duties of the president whenever that official is absent from his office, and in this capacity as "alternate president" must be as well informed as the president about the affairs of the company.

The *Secretary* maintains the official records of the corporation, supervises transfers and issuance of stock certificates (if the company is publicly held), accepts service of legal papers drawn against the corporation, and can represent it in any court when identification of records or other similar data is an issue. He also is responsible for preparing the annual report to the stockholders, usually in conjunction with the treasurer and the executive vice-president.

Many of the secretary's duties involve legal matters, and consequently it is not unusual that he (or she) is a lawyer. Should it be consistent with company policy, the legal section (sometimes known as the *house counsel*) may be placed under his (or her) supervision. This important segment of the organization usually consists of one or more lawyers who provide legal advice, comment on the significance of court decisions and the impact of new laws, assist trial attorneys in preparing court cases, and take part in negotiations where interpretation of contract terms may be needed.

Acting primarily as financial advisor to the chairman and the president, the *Treasurer* obtains the requisite information from the

CHART NO. 1 ORGANIZATION OF A TYPICAL AMERICAN LINER-SERVICE COMPANY



comptroller, cashier, accountants, and auditors. He is able at any time to report on the complete financial position of the company and to discuss areas of fiscal strength and weakness. In some steamship organizations, the offices of secretary and treasurer are merged into one.

To keep control over the multitude of financial records which must be maintained by a steamship company (especially if it holds a subsidy contract with the United States government), some organizations include in the executive structure a vice-president for finance. The auditors, accountants, and bookkeepers, as well as pursers aboard ship, come under this officer's supervision. As a master accountant, he cooperates with the treasurer in keeping the president, the board of directors, and stockholders informed on financial matters. Because of the obvious overlap in functions, it is not unusual for the offices of treasurer and vice-president for finance to be merged.

Analyzing the significance of accounts and supervising the operating budget are rather exact sciences, especially as computer technology is applied to financial management. Larger companies today recognize the need for qualified specialists in this area, and have established the office of *Comptroller* to handle all aspects of fiscal control. This officer is assisted by the auditors, who oversee accounting procedures and inspect financial records. In general terms, the comptroller supervises expenditures in accordance with the budget while overseeing routines of fiscal administration. In smaller companies, the treasurer performs the duties of the comptroller, assisted by the auditor and chief accountant.

From time to time, liner-service operators have occasion to charter ships into and out of their fleets. As has been described in detail elsewhere in this volume, chartering is a specialized form of endeavor, and in those organizations where it occurs with some frequency, full responsibility for this activity, with authority to make final commitments, is assigned to a vice-president. Regardless of title or location within the corporate structure, the official handling charters must keep abreast of the employment of the company fleet and the potential demand for additional tonnage or the development of a surplus. As circumstances warrant, he provides extra ships or finds charters for those which are temporarily redundant.

Aside from chartering, the functions of the officials described above are those of the senior executives of any corporation. Essential to their success are high moral character, inspiring leadership, and sound business judgment. It is an advantage if these officials have the widest possible familiarity with the maritime industry, since this will assure them of respect in the business community while providing a sound foundation for their decisions.

Maritime transportation is a business with specialized aspects, but is neither unique as an economic enterprise nor is it without a commercial counterpart. It exists to make a profit, and must sell its product (transportation) to the public in precisely the same way that the tire manufacturing industry must find markets for automobile and truck tires. Terminology differs, of course, but the realities of competition, cost, plant modernization, and promotion are as clearly defined in the shipping firm as in any manufacturing company. The techniques of management are, in their essentials, the same as in other forms of business. Circumstances and factors influencing the activities of a steamship company often are remote from, and completely beyond the control of, the maritime community, but even here there are similarities with other types of commercial enterprise.

As a case in point, the traffic department of the liner-service company has the same goals as the sales division of a tire manufacturer. Both must sell their respective products, transportation and tires. The techniques of salesmanship, adapted to the maritime industry, can be applied successfully to "merchandizing" transportation. The vice-president, traffic, is responsible for keeping a constant flow of cargo coming to the ships just as the director of sales must find buyers for the tires.

The operating department of a shipowning corporation should be viewed in a similar manner, for it bears striking resemblance to the production division of a manufacturing concern. Just as the production vice-president is required to oversee all details relating to the manufacture of the rubber tires previously mentioned, and must adjust output to meet the requirements of the sales force, so the vice-president in charge of marine operations must insure that the ships are kept in good condition, on schedule, and in all respects able to provide the service promised by the salesmen to the shippers.

In shipping as in manufacturing, the traffic (sales) and operating

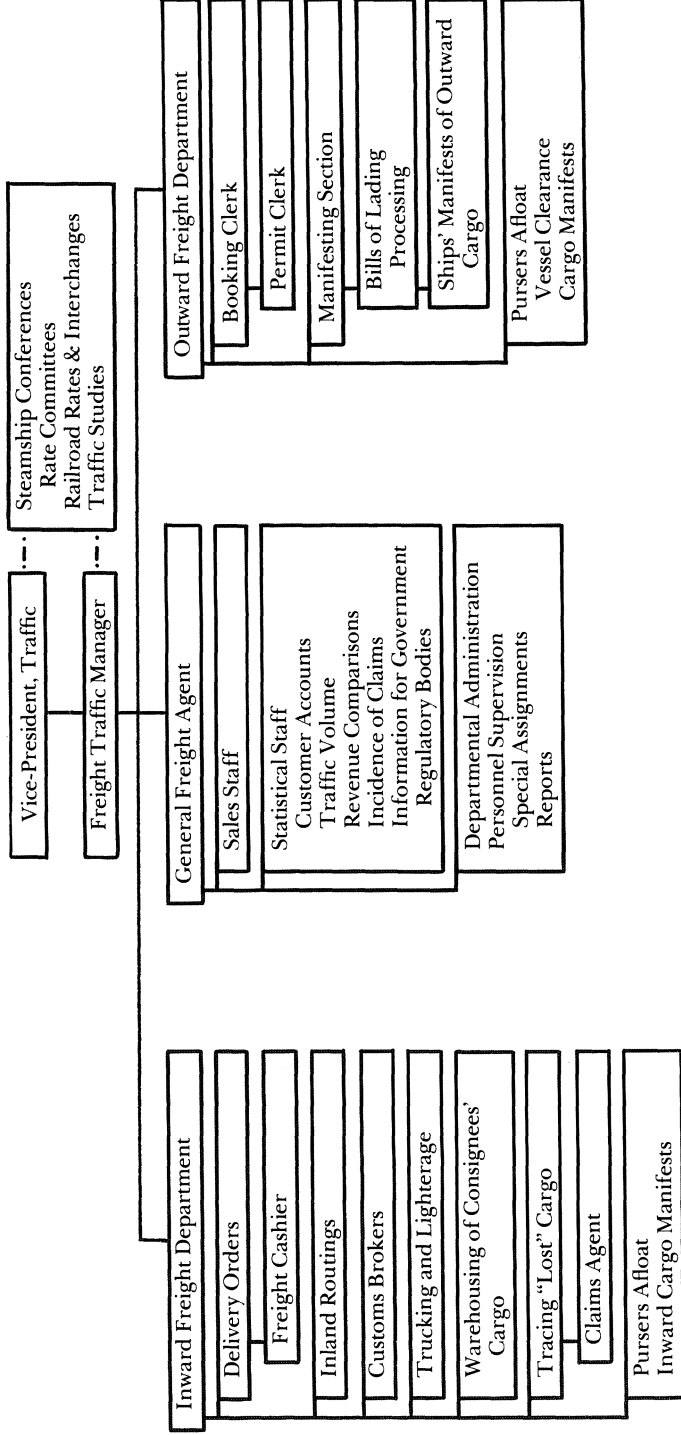
(production) departments are mutually complementary and inter-dependent. The transportation salesmen must be assured that ships will be available to lift the cargoes they procure; the operating department must have the cargoes to fill the ships they have brought into port. As one executive explained: "The best traffic department in the world cannot survive without a good operating department; the finest operating group has no reason for existence unless it is supported by a good sales force."

TRAFFIC DEPARTMENT

The *Vice-President, Traffic*, directs the department charged with generating the cargo to fill the company ships. He reports to the president through the executive vice-president, keeping both these officials informed on all matters related to both outbound and inbound cargo. Within his department, the vice-president, traffic, sets policies to achieve the goals assigned by the president and the board of directors. As an expert on traffic, he may be called upon to work with carriers to solve matters of mutual concern, or to consult with important shippers whose problems are of magnitude so great that they merit attention from an executive of his stature.

It is difficult to describe completely the responsibilities of the vice-president, traffic. He must spend much time conferring with his fellow executives, especially from the operating and chartering departments, as to the proper scheduling of ships, seasonal changes in demand for shipping space, better employment of the fleet, improvements in design for projected new or modernized tonnage, plans for the acquisition or disposition of vessels, and ways and means of minimizing stevedoring expense. He also deals with the claims agent, the insurance manager, and the financial managers to develop better ways to handle claims for compensation for damaged or lost cargo. From time to time he will confer with the terminal manager and the stevedore to consider techniques of cargo handling which might lead to fewer claims. Now and again he will be involved in formulating, with the comptroller, freight cashier, and vice-president for finance a policy dealing with the collection of freight charges. In a company handling all its cargo in

CHART NO. 2 TRAFFIC DEPARTMENT



containers, he will confer with the terminal manager to ascertain that access to the terminal is satisfactory, that control procedures are working properly, and that receipts and deliveries are meeting company promises to shippers and consignees. As the patterns of business place greater emphasis on intermodal transportation, he must devote time to the development of this aspect, especially in establishing freight rates payable to railroad and motor carriers.

Outside of the company, the vice-president, traffic, may serve as a member of an industry group studying ways to improve packaging of goods or to standardize practices of booking full container loads of cargo. Making rates which will cover the costs of inland transportation, and which therefore must be acceptable to railroads, trucking contractors, and the government regulatory bodies, consumes much of his time.

There are occasions when the vice-president, traffic, will be part of a group from the steamship industry seeking ways to prevent a disastrous freight rate war. Perhaps a new set of freight charges is under discussion, and the need to increase revenue must be considered in the light of pronouncements that higher rates will result in these commodities being priced out of the market.

If the company belongs to an association of steamship lines which regulates competition and sets uniform freight rates, the vice-president, traffic, may represent his organization at times when a firm commitment to a certain principle or policy is required. He also may be part of a special committee of the association responsible for adjustment of freight rates to meet changing conditions and shippers' requests. In addition to these "troubleshooting" assignments, the vice-president must always be alert to better ways to carry out his mission, to gain larger shares of the available business, and to augment company profits.

The *Freight Traffic Manager* is the deputy to the vice-president, traffic, and is qualified to substitute for that executive when he is absent for any reason. He is the supervisor of the freight traffic department and assures that departmental policies are carried out in the most effective manner. He considers shippers' requests for special calls at ports not on the regular schedule or to transport large quantities of unusual cargo, and makes appropriate recommendations to higher authority when necessary. Troublesome bookings, as, for example, unexpectedly large offerings for a ship

already almost filled, or proposals to delay a ship's sailing to accommodate a customer, also are referred to him.

Immediately under the freight traffic manager is the *General Freight Agent*, who directs the salesmen (sometimes called solicitors) and handles departmental administration. Salesmen are the contacts between the carrier and the shipping public; upon their ability to convince these clients that the company deserves their patronage depends its prosperity. The coordination of their efforts, and the establishment of policies with regard to sales, are of great importance, and represent a major part of the general freight agent's work. He supervises the day-to-day visitations made by the salesmen, suggests approaches to difficult customers, and makes sure that the capabilities of the company are set forth correctly. When his subordinates make suggestions which have merit, but which involve changes in departmental or company policy, he transmits these suggestions, together with his recommendations, to the appropriate executives. As he obtains information which may be useful in selling the services of the company, he calls in his sales staff and develops with them ways to use this knowledge to induce shippers to give their business to the company. Clients who benefit by this effort may become supporters of the line since they appreciate that these salesmen often bring suggestions which, if accepted, can be financially advantageous to that shipper.

In addition to these responsibilities, the general freight agent directs the keeping of records which show the business received from individual shippers, the seasonal variations in the volume of cargo offered by shippers and in the tonnage carried by the ships, and also comparisons between voyages and calendar periods. These records have many uses. They may furnish data for some governmental agency concerned with international shipping. Perhaps the president has directed that certain aspects of traffic activity be analyzed; the most readily available information would be in these statistical records. Should a traffic study (a forecast of future business trends, based on past experience and present practice) be in progress, the data from these records could be of major importance. They also may suggest to the reviewing official trends in traffic which have not been obvious to the less diligent examiner, and which have great potential for increasing business.

The statistical section's reports show the volume of business handled, revenue earned, demands for service to named ports, the desirability of certain commodities in the light of the cost of handling them, and the frequency with which claims for damage are made. These reports are circulated to the senior executives, the comptroller, the operating department, and any other officials who may require them, and are useful as the basis for judgments relating to all aspects of freight traffic.

An important activity of the general freight agent is to maintain close contact with the claims division to learn what commodities and what shippers are involved in claims for damages and the probable causes of the alleged damage as well as the validity of the claims.

As the departmental administrative officer, the general freight agent makes routine and special assignments, sets up training programs for new members of the department, transfers persons within the department for most efficient use of their capabilities, recommends promotions when earned, and dismissals when necessary. Incidental to these supervisory functions, the general freight agent maintains appropriate personnel records.

The freight traffic department is divided into two major segments, the *Outward Freight Department* and the *Inward Freight Department*, of which the outward is by far the larger. The chief of each branch usually holds the rank of assistant freight traffic manager, and reports to the freight traffic manager.

The *Booking Clerk* is a senior member of the outward department and controls the space in a break-bulk cargo ship, allocating that space to individual shippers as they make their requests. He obtains his basic information from the operating department which informs him of the amount of fuel, water, stores, and equipment to be aboard the ship on the proposed voyage, and the actual space, in cubic feet, and capacity, in tons, available for cargo. As reservations for space in a break-bulk vessel are received, the booking clerk computes the stowage factors and allots the required number of cubic feet. The salesmen work closely with the booking clerk to assure that customers are extended the treatment they have been promised. Conversely, when the booking clerk finds that he needs either high-revenue or dense cargo, he often appeals to the salesmen to assist him in making a balanced load by arranging visits to

shippers of the kind of cargo desired. On the basis of the booking clerk's daily reports, the stevedore plans the stowage of the ship, and the terminal manager tentatively lays out the transit shed.

For fully containerized ships, it is not necessary to compute stowage factors for individual shipments, nor is the stevedore dependent upon the booking reports. Instead, the booking clerk reserves undesignated "cells" for the containers offered by the shippers, and relies upon a carefully programmed computer to determine exactly where the box will be stowed. He also notifies the terminal of anticipated receipts of containers, making certain that all necessary data are furnished for proper identification of the trailer, and is in frequent contact with the gatehouse of the terminal to update and clarify earlier information.

Working directly under the booking clerk is the *Permit Clerk*, who notifies shippers of large lots of cargo (usually defined as ten thousand pounds or more from a single originator) directing them when to send their consignments to the terminal. The "permit system" is designed to favor the shipper of large lots and, by spacing the arrivals of trucks at the terminal, to eliminate the heavy expense of truck waiting time. In companies carrying nothing but containerized cargo, the receiving process at the terminal is highly sophisticated and computer oriented. The same principles noted in break-bulk shipping apply, however, and the goal is to save both customer and carrier time and money.

Cargo which moves in small lots (often described among break-bulk carriers by the railroad term of "less than carload lot" consignments) normally does not come under the permit system. Experience has shown that little, if any, time is saved for the small lot shipper as a result of attempting to schedule truck movements to the terminal, and a good deal of clerical work can be eliminated. Cargo which requires special handling (a heavy-lift item, for instance) or must be brought to the ship's side by barge or lighter, has to be scheduled to prevent congestion, confusion in the loading operations, and perhaps great expense to the shipper who would be charged for the delays at the terminal.

Patrons of steamship lines handling all cargo in containers but whose individual offerings are in comparatively small lots (referred to by the carriers as "less than container loads") are encouraged to send their consignments to a section of the terminal reserved for

this type of traffic. At this "consolidation station," small lots are processed and loaded into containers by terminal personnel. There is no significant difference between the handling of these shipments on a break-bulk terminal and at a container yard.

The manifesting section, under the *Chief Bill of Lading Clerk*, is the busiest and largest of the segments of the outward freight department. Here the shipper delivers for processing the bills of lading he had prepared. These are matched with the dock receipts from the terminal to obtain the actual measurements of the parcels in the shipment, together with their actual count and condition (or the size, number, and weight of the container(s) in the shipment), and are freighted by the *Rate Clerk*, listed on the ship's outbound cargo manifest, signed by the chief clerk for the master, and routed to the freight cashier to hold until the shipper prepays the freight charges and picks up the "freighted" bills of lading.

Pursers serve as the shipboard agents of the freight department and often have a great deal to do with the manifests and other papers prepared by the outward freight department. They usually maintain liaison with the assistant freight traffic manager in charge of billing to assure that the work aboard ship harmonizes with that performed ashore.

The *Inward Freight Department* is a service agency responsible for the consignee obtaining his cargo without delay and in an orderly manner. The carrier requires that the original bill of lading be surrendered before the cargo is released. This procedure necessitates the timely notification of the consignee of the anticipated arrival of the vessel. The consignee also is informed of those pertinent regulations with which he must comply prior to taking delivery of his goods, and is assisted appropriately by personnel of the inward freight department. When the goods are removed from the ship, the consignee calls at the inward department and exchanges the original bill of lading for a "delivery order," which authorizes the terminal to release the cargo to the consignee or his designated representative. Should the goods be destined to an inland city, the department makes arrangements for transportation pursuant to the consignee's instructions. If required, cargo will be warehoused until this transportation is available. No charge is made for this assistance; the expense of trucking, warehousing, customs broker's services, and transportation to the ultimate destination is for the

customer's account. If the consignee believes that some or all of his cargo has gone astray, the inward department initiates the search and keeps the claims agent at the loading point informed of its efforts. Finally, since pursers on the homeward voyage often make up the inbound cargo manifest, the inward department maintains constant liaison with them to make certain that their work satisfies company and governmental requirements.

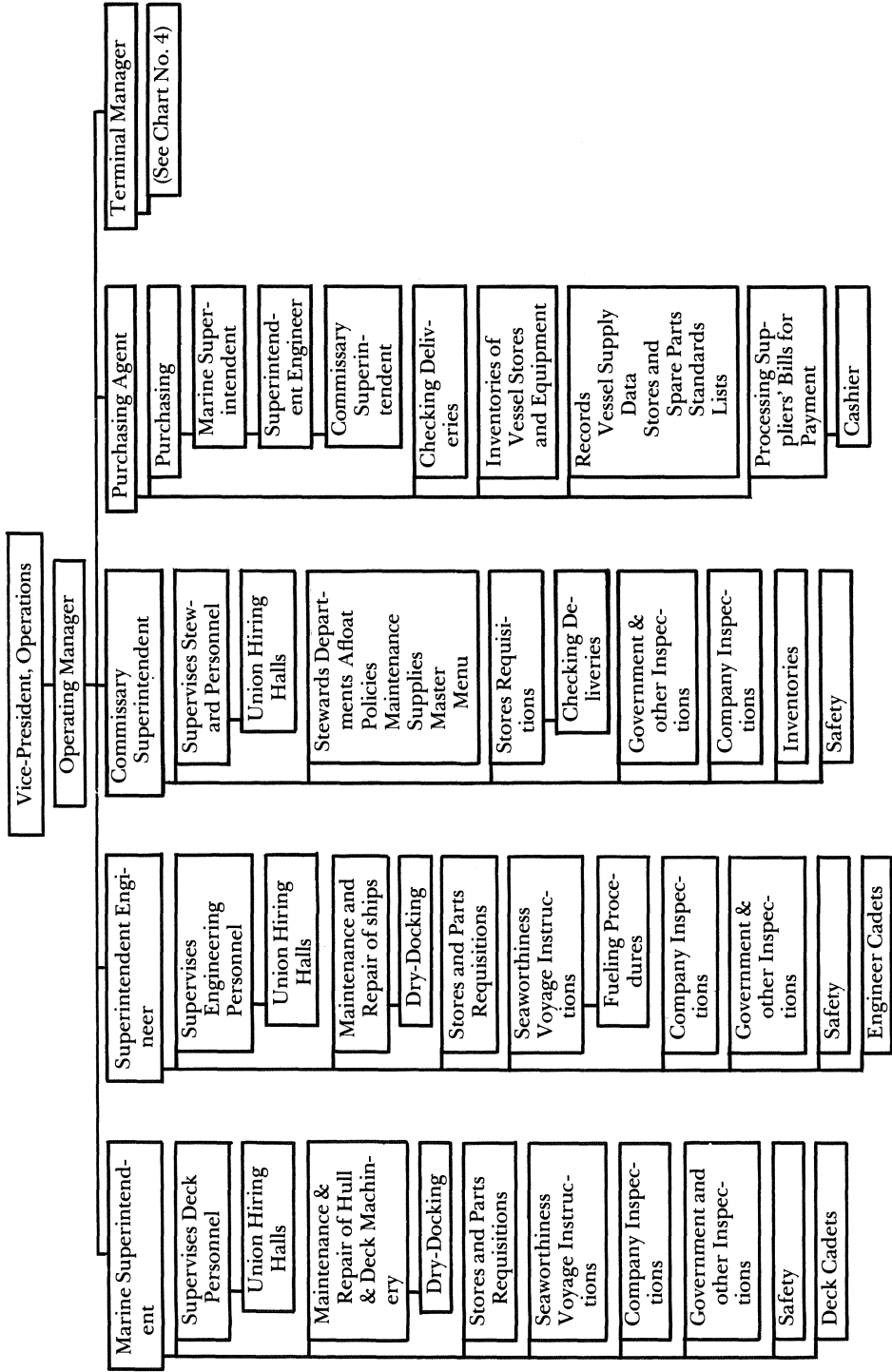
OPERATING DEPARTMENT

Supervising the operation of the fleet is the *Vice-President, Operations*. While a seafaring background is not mandatory for this official, it is highly desirable because of the intimate contact he has with the personnel who run the ships, perform the essential stevedoring functions, and make the repairs. Broad experience in management directly related to ship operation may be a satisfactory alternative to service in merchant vessels.

The vice-president is responsible to the president, through the executive vice-president, for all matters concerning ship construction, operations, stevedoring, and labor relations. If the company is small, the vice-president usually is the negotiator with shipyards for building and repair contracts, with stevedores on details of handling cargo in the various ports served by his fleet, with suppliers of ship stores and outfit, and with representatives of all types of labor unions on many aspects of the working agreements. The larger the company, the greater is the likelihood that specialists will handle each of these activities, but final decisions on these matters are the vice-president's. To assist him in his manifold duties and to assume the direction of the department whenever he is absent, the vice-president appoints an *Operating Manager*, who administers the department.

In many companies, final appointment of masters and chief engineers of the ships is made by the vice-president himself. The marine superintendent initiates the recommendation for promotion to the rank of master and forwards this through the operating manager to the vice-president. Similarly, the superintendent engineer selects the person he considers best qualified for the position of chief engineer. After study of these proposals, the operating

CHART NO. 3 OPERATING DEPARTMENT



manager lays them on the desk of the vice-president. The reason for this action is to assert, in unmistakable terms, that these two seafarers enjoy the full confidence of the highest echelons of management. It also authorizes them to talk to anyone below the vice-president on terms of equality in representing the interests of the ship.

The operating manager is the departmental executive officer and coordinator of all activities in the department. Within the very broad limits of his authority, he makes those decisions needed to carry out company and departmental policies. A most important facet of his duties is the evaluation of the periodic reports prepared by the various divisions of the department. Covering as they do the actual performance of assigned functions, these reports afford a means of determining the efficiency with which the work is accomplished, and offer opportunities for improvement in the techniques of supervision and management. The results of these evaluations always are made known to the vice-president.

Marine Division

At the head of the Marine (or Deck) Division is the *Marine Superintendent*, who is responsible to the vice-president for the seaworthiness of the ships in all particulars except for the engineering department. The marine superintendent necessarily is both a qualified shipmaster with long experience and a man with demonstrated ability as an executive. He supervises the performance of deck personnel of the fleet through study of vessel logs, conferences with ships' licensed officers, and personal observation. Before each voyage is begun, he issues instructions to the master which include the latest information relating to conditions on the route, appropriate details of special requirements, and pertinent comments on carrying out company policies as they may apply to the forthcoming voyage. He also inspects ships on their arrival in the home port to insure that governmental and company standards of maintenance are upheld. He hires all licensed deck officers except the masters, and supervises through these officers the performance of the unlicensed personnel. Qualified and meritorious deck officers and seamen are promoted on his recommendation and approval. Because of these personnel activities, the marine superin-

tendent must establish and maintain a good working relationship with officials of the maritime unions.

Safety of personnel afloat is of major importance. Some of the larger companies have *Safety Directors*; smaller organizations often designate the marine superintendent as the official in charge of the safety program for the deck department.

Requisitions from the ships for repairs to hull and deck machinery within the purview of the deck department, and plans for scheduled drydockings, inspections, surveys, and other examinations of the hull pass through the marine superintendent's office. He (or an assistant sometimes known as a *Port Captain*) always is present when the bottom is sighted, and accompanies all inspection parties on their tours. Requisitions for spare parts and consumable supplies for the deck department must be approved by the marine superintendent before they are forwarded to the purchasing agent.

Training of deck cadet-officers is under the supervision of the marine superintendent, who sets company policy concerning their duties and responsibilities while aboard ship.

Engineering Division

Occupying a position of importance equal to that of the marine superintendent, but concerned solely with the engineering activities of the fleet, is the *Superintendent Engineer*. Like his deck counterpart, this official has a background of years of service afloat and has demonstrated the qualifications desirable in an executive. Many of his responsibilities parallel those of the marine superintendent: seaworthiness of the engineering department; supervision of personnel afloat, including training of cadet-engineer officers; relations with the maritime unions; inspections and surveys; safety; and oversight of repairs and procurement of spare parts. Engine abstract logs and voyage reports are submitted to him by the chief engineers as their ships return to the home port. Examination of these documents reveals with notable accuracy the efficiency of engineering practices of the ship, and serves also to delineate those areas where improvements are in order. From the listings of repairs accomplished by the ships during their voyages, both by ship's company and contractors in ports of call, the superintendent compiles individual histories of each ship's maintenance programs.

Some large steamship companies have a maintenance and repair section which is charged with all details relating to accomplishing repairs beyond the capability of shipboard personnel. The superintendent engineer in this case serves as the liaison between the ship and the specialists of this section.

The chief engineer of a ship is the maintenance and repair officer, and has overall responsibility for the physical condition of the vessel during a voyage. All requisitions for work exceeding the capability of the ship's crew are channeled through him, and are delivered to the superintendent engineer in the home port. Once contracts for the requisite repairs have been let, actual performance thereunder is monitored by the chief engineer and the department heads concerned (e.g., deck repairs by the chief mate, commissary department by the chief steward).

When a vessel is scheduled for drydocking, the marine superintendent and the superintendent engineer cooperate fully to take maximum advantage of the time on drydock and in shipyard. Repair lists from the ship serve as the basis for writing the specifications for the work to be performed. The importance of verifying that the specifications have been carried out fully and competently cannot be exaggerated; these superintendents are the technicians responsible to the highest echelons of management for the seaworthy condition of the ship. The protection of the owners' interests, especially in regard to meeting standards of the classification societies, underwriters, and governmental agencies, must always be a paramount consideration. To this end, appropriate liaison must be maintained with the technical representatives of these bodies to assure that their inspectors will be on hand at the appropriate times.

Supervision of vessel fueling, instructions to the ship concerning keeping records of fuel consumption, and establishing the tests to be conducted on fuel taken aboard elsewhere than in the home port are particular responsibilities of the superintendent engineer. These aspects of his duties attained new significance as the uniformity and quality of marine fuel and diesel oil became questionable following the petroleum crisis of 1973. In some instances, the office of the superintendent engineer may prepare the plans for bunkering the ship during a voyage; in any event, the bunkering

records kept by this office may be consulted by staff members whenever necessary.

Commissary (Steward's) Division

Supervising all aspects of activities related to the care and feeding of shipboard personnel is the *Commissary Superintendent*. This official, who invariably has a background of seafaring experience as a ship's chief steward, establishes policy for the operation of all aspects of the steward's department (often designated as the catering department) afloat. As ships come into the home port, he inspects them to assure that they meet company standards of cleanliness, storage, and record-keeping, and that the physical condition of the accommodations and food-preparation spaces is satisfactory. Ships' chief stewards are guided and instructed in the most efficient and effective ways of preparing and serving food; frequently a master menu will be furnished by the commissary superintendent for use during the voyage. All requisitions for foodstuffs are submitted to him for approval before being sent to the purchasing department. Inventories of nonconsumable items within his jurisdiction are conducted under the direction of the commissary superintendent, and replacements for items of this type are subject to his scrutiny before being filled. All matters relating to safe operating practice are of major concern, as are instructions for the proper and safe performance of duties of steward's department personnel.

A major activity of the commissary division is the procurement of competent seafaring personnel, and therefore the superintendent maintains good relations with the maritime unions. In the absence of government certifications of competence and standards of proficiency in the performance of the duties of steward personnel, the commissary superintendent is free to establish (within the framework of the agreement with the maritime union) appropriate qualifications for selection, promotion, and dismissal.

A very important part of the routine of the commissary superintendent's office is checking supplies when they are delivered to the ship. This entails coordination of schedules with the purchasing division and the assignment of inspectors to examine goods as they are brought to the ship. It is standard practice in United States shipping

companies to have a qualified inspector from the United States Department of Agriculture examine and approve all meat, poultry, eggs, butter, and vegetables before they are accepted by the ship.

Purchasing Division

The duties of this office are set forth in detail in Chapter 9.

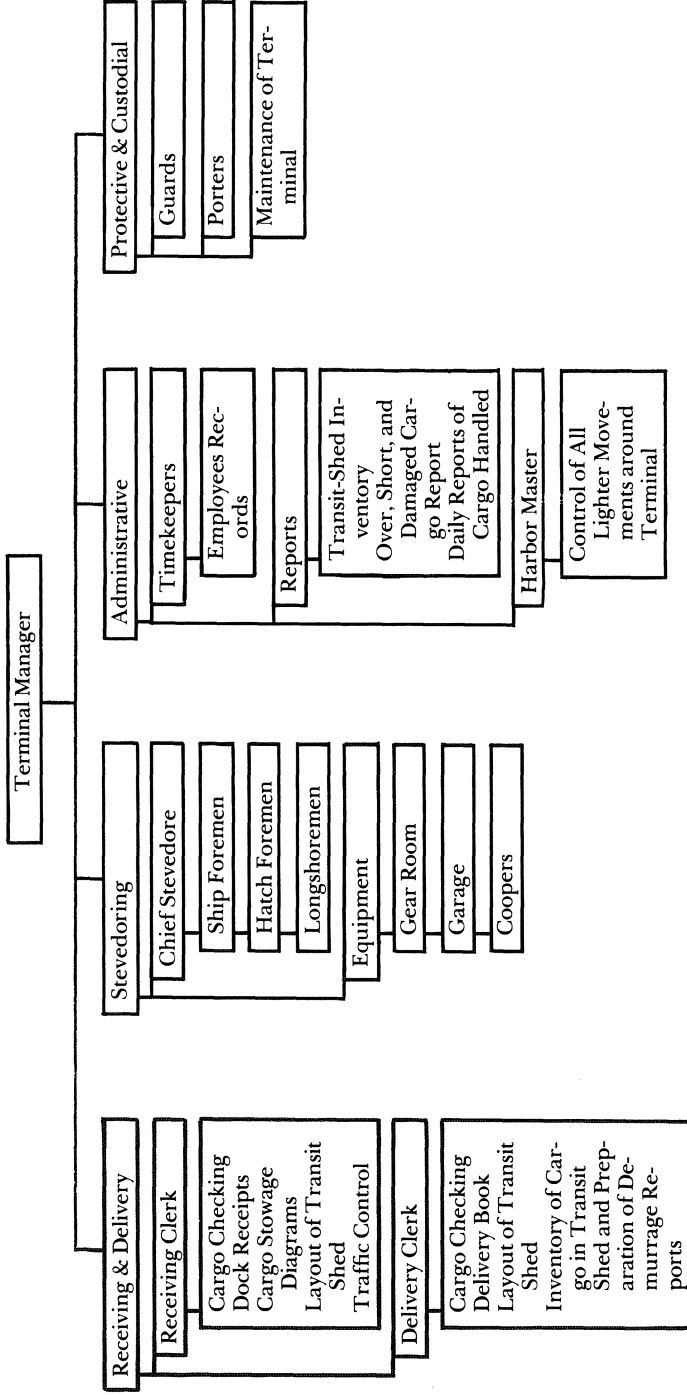
Terminal Division

The *Terminal Superintendent* is in charge of all activities directly related to handling the cargo into and out of the ships. Whether the carrier be engaged in the break-bulk or container trade, the terminal division must employ scores of skilled artisans as well as numerous clerks, and expends large amounts of money for rent of facilities, operation of equipment, and compensation of personnel. Necessarily, the terminal superintendent has had extensive experience in all phases of cargo handling, and has demonstrated as well unusual competence as an administrator and supervisor. He is the technical advisor to the president on all matters dealing with cargo handling, stevedoring, and management of the terminals operated by the company. His immediate seniors in the executive structure are the operating manager and the vice-president, operations.

Functionally, the terminal division has four sections: receiving and delivery of cargo, stevedoring, administrative, and protective and custodial. Chart No. 4 shows the assignments of responsibility and control within the division.

The *Receiving Clerk* is one of the key assistants to the terminal manager. He is in charge of the office through which pass all papers connected with outward cargo brought to the terminal for loading aboard ship. Under him are clerks who prepare the necessary reports and draw the diagrams showing how and where the cargo has been stowed in the ship. The actual examination of the cargo is performed by *checkers*, who are hired by the day as they are needed. When cargo arrives at the terminal, a checker is assigned by the receiving clerk's office to inspect, count, and measure the packages, and to insert the appropriate data on the dock receipt. The completed form, initialed by the checker, is returned to the receiving clerk for authentication attested by his signature. The original goes to the delivering carrier as proof that the cargo has been accepted in the condition noted on the dock receipt. Throughout the process of

CHART NO. 4 TERMINAL DEPARTMENT



loading the ship, a member of the receiving office staff makes frequent surveys of the cargo spaces to ascertain the exact area in which lots of cargo are stowed. From this information, the ship stowage diagrams are prepared.

In addition to handling the documentation described above, the receiving clerk assists the terminal manager to lay out the transit shed for the next ship. Copies of the booking clerk's daily reports serve to indicate the quantity of cargo for each port of call, and thereby suggest the amount of floor space to reserve for those ports. Truck traffic into the terminal area is controlled by the receiving clerk, who issues gate passes to drivers in the order of their arrival and consistent with the number of checkers available to oversee the discharge of the cargo from the vehicles.

Terminals operated by containership companies follow almost the same routine, except that the role of the checker is minimized. The exterior condition of the container is inspected carefully, the condition of the seal on the container's door latch is examined, and appropriate data to identify the container are recorded. Containers are weatherproof, and are stored in "marshalling yards," sometimes on wheeled chassis and sometimes dismounted and piled two high. Thousands of containers are assembled in a large terminal, and control of their receipt, inspection, locations in the yard, movements within the yard and to the ship's side, and stowage aboard ship depends upon the existence and constant use of a highly sophisticated system based on an electronic computer. The basic responsibility for the condition and location of containerized cargo is identical to that of the break-bulk carrier whose longshoremen handle each package, no matter how small.

The *Delivery Clerk* has functions similar to those of the receiving clerk, except that he is concerned entirely with incoming cargo. As the consignees claim their goods, a checker is assigned to inspect the packages to ascertain their condition at the moment of transfer to the consignee. The checker's report is filed in the delivery office, and the truckman or other inland carrier signs for the cargo in the "delivery book," which shows the date and hour when the ocean carrier released the goods. The delivery clerk also aids the terminal manager in laying out the terminal for incoming cargo, using information transmitted from the overseas ports to the inward freight department. The delivery clerk takes periodic inventory of

the transit shed to determine what cargo has not been claimed, and what lots have been on the transit shed's deck in excess of the allowed free storage time. From these records, charges for storage ("demurrage") are compiled.

For containership terminals, the responsibility imposed upon the delivery clerk is the same as in the break-bulk terminals, but the details of operation are incorporated into the computer system.

Loading and unloading ships employed in break-bulk service is a specialized activity normally performed by contractors rather than by the steamship company. Once awarded the contract, the stevedoring firm assigns a supervisor to the terminal. This *Chief Stevedore*, although on the contractor's payroll, functions as a de facto member of the terminal manager's staff. He has a small group of assistants who actually oversee the performance of the foremen and the longshoremen. The details of each day's work, as required to meet the schedule set forth by the terminal manager—number of gangs to be employed, supporting equipment needed, specialized workers to be assigned, and safety precautions to be followed if unusual cargo is to be handled—are the concerns of the chief stevedore, not of the terminal manager. The terminal manager assures that the terms and conditions of the contract are carried out, that longshoremen are not idle, and that the responsibility for damage to cargo which occurs in the terminal is fixed at the earliest practical moment.

Many stevedoring contracts stipulate that the contractor shall furnish all gear and equipment. Usually he also supplies the mechanics and carpenters necessary to maintain these items in proper condition. The terminal provides the space for shops. Where the contract requires the shipowner to provide and maintain the gear and equipment, the terminal manager must exercise positive supervision to make certain that all requisite materiel is in dependable, operating condition.

Container terminals represent very large investments of capital because of the space needed to store and to maneuver the containers, the multimillion-dollar cranes used to hoist the containers into and out of the ships, and the highly specialized equipment employed to shift containers from chassis to the ground and vice versa, or to stack them in the marshalling yard. Few container companies own these terminals, preferring to lease them from public bodies

(port authorities, cities, or state agencies). Relatively few laborers are needed, and those who are hired are considered as skilled operators rather than manual workers. Some containership operators manage their terminals completely with their own personnel, while others contract for everything: terminal space, labor, equipment, and supervision.

Administratively, a great deal of "paperwork" is processed by the terminal division. Among the numerous reports submitted in connection with each loading or unloading operation of a break-bulk ship, those prepared by the *Timekeeper*, the *Safety Engineer*, and the ship foremen are most useful in evaluating the competence of the labor force. The timekeeper's records are compiled daily and show how many men were employed at any time, the number of hours of straight time and overtime work, and the number of hours and the men assigned to handle premium or penalty cargo (commodities for which longshoremen receive extra compensation). Reports filed by the safety engineer (or the official charged with responsibility for safety of personnel) supply details of any accidents, injuries, or equipment failures which occurred and have been placed under investigation. From the ship foremen come data on the actual performance of longshoremen in terms of tons of cargo handled per hour, stoppages and the reasons therefor, problems encountered in the stowage or discharge operation, and related details.

Each morning, a report is drawn up to show the amount of cargo handled by the terminal during the preceding twenty-four hours. As a vessel is finished, a report of any cargo which has been over-carried, short-delivered, or damaged is prepared. Inventories of unclaimed lots of cargo are inspected and brought up to date. All these documents are channeled through the terminal superintendent's office before being disseminated to the divisions concerned. By inspection of these papers, the terminal manager is able to gain a summary of activity during the periods covered.

In containership terminals, although the computer is used to bring together all the details, the same types of reports must be prepared regularly to provide the basis for evaluating the performance of the terminal organization. The enormous cost of the ships makes rapid turnaround essential, and maintaining control of the terminal activity is a vital part of attaining that goal.

Some ports have a significant quantity of cargo moved within their limits by lighters. Any terminal which has a steady flow of lighter traffic may have among its supervisors a *Harbor Master*, who controls the arrivals, moorings, and departures of these nonself-propelled craft. All lighters come under his supervision as soon as they are moored in terminal waters and are shifted when and where the harbor master directs as he carries out the plans of the chief stevedore. When the lighter is emptied, the harbor master notifies the owner of this fact, and orders the lighter to be moved to a point where it can await its owner without interfering with other operations, and can be picked up expeditiously. It is important that prompt notification of the release of the lighter be given to that craft's owner to fix the exact time when the steamship terminal's responsibility comes to an end.

Maintenance of the physical plant is a necessary and important activity. Security guards often are assigned the duty of reporting to the maintenance office whenever they observe something that needs attention: a deck plank which should be replaced; an electrical conduit which has been damaged; a break in the protective fencing around the terminal; or piles of debris ready for removal. The porter force and the maintenance workers who react to these reports sometimes are placed under the supervision of the protective division for administrative control and routine assignments of work. Large-scale projects, such as repainting the interior of the transit shed, resurfacing the working area, or installing new lights, automatically invite the terminal manager's personal attention since they entail major financial expenditures and a possible rearrangement of all terminal activity during the job.

In his position as the overall supervisor, the terminal manager must be alert at all times to poor working practices which reduce efficiency or increase the risk of injury to employees. He must always be seeking better and more economical ways, both in terms of time and of money, to handle cargo. As appropriate, he makes suggestions for modification or construction of terminal facilities. Finally, and perhaps most importantly, he participates in the formulation of all programs and policies relating to the personnel under his supervision.

Terminal Management

A MARINE general cargo, break-bulk terminal exists for the sole purpose of effecting the efficient transfer of goods between the vessel and other carriers. Its operations are centered on this objective, and constant care is exercised to insure that cargoes are handled safely, at lowest cost, and in the least possible time. The responsibility for achieving this objective is placed on the terminal manager.

Underlying every consideration of good terminal management must be an awareness of the need for safety. Not only is safety important for the good morale of employees, but it is vital if the goodwill of shippers and consignees of goods is to be retained. Working practices which endanger the lives of the laborers are intolerable both morally and financially. Equally intolerable are habits of handling cargo which result in damage or loss of the customer's goods.

The search never ends for better and less costly methods of handling the packages constantly crossing the marine terminal. There is a pressing need for economy, because the cost of running the terminal often approaches the total of all other ship operating expenses. A saving of only one cent per ton can make a significant contribution to overall company profits.

Speed is essential in all the processes of receiving, stevedoring, sorting, and delivery of the cargo, and it is almost axiomatic that an ideal terminal manager never accepts the report that work was accomplished at "maximum speed." He must find ways to reduce the time the ship has to spend in port, for when vessels cost \$500 or more an hour to keep in commission, there are few operators who can afford the luxury of idle ships alongside a terminal. There must be a constant awareness that ships can earn money only when they are actually transporting goods from one port to another. The

shorter the port time, the more voyages a ship can make in a year, and the greater her earnings.

Terminal management depends upon an efficient organization of people, and a systematic assignment of duties to the staff so that the many functions of the terminal may be accomplished efficiently. At the head is the terminal manager, who is in general charge of all activities. His principal assistants are the chief receiving clerk, who supervises all routines relating to cargo being brought to the terminal for loading on board outbound vessels; the chief delivery clerk, who performs the same duties for cargo being turned over to consignees; and the chief stevedore, who directs vessel loading and unloading operations. Often the chief stevedore is employed by the stevedore contractor, rather than the terminal; regardless, he always serves as a member of the terminal manager's staff and advises on all matters directly connected with the labor force and actual handling of cargo into and out of ships.

The *security chief* is responsible to the terminal manager for the physical safety of the plant and its contents. He directs a force of guards who control access to the terminal area, and a number of patrolmen who routinely visit all parts of the terminal at periodic, but irregularly scheduled, intervals. In addition to the primary duties of providing security, which are concerned with prevention of theft and fire, it is the practice of some terminals to assign one or more of these ancillary functions to the security forces:

1. Inspecting refrigerated containers to assure that their machinery is operating and the proper temperatures are being maintained.
2. Connecting and disconnecting refrigerated containers when they are in the storage area.
3. Supervising the operations of tying up or letting go the mooring lines of a ship.
4. Operating an intraterminal personnel bus service to obviate pedestrian traffic.
5. Controlling the maintenance and porter units to assure that hazards to safety resulting from damage or accumulation of waste are removed quickly and efficiently.

The logic behind these additional assignments is that security

guards constantly patrol the transit sheds, storage yards, container marshalling yards, and the piers and wharves. They must be alert to any deviations from safe and accepted practices which endanger the physical safety of the plant or cargo.

Refrigerated cargo requires closely controlled temperatures, and the frequent rounds of the patrolmen provide an excellent means by which the performance of the cooling units on each container can be monitored. If a unit is observed to be functioning improperly, the patrolman notifies the appropriate office of this fact, and necessary repairs will be effected with a minimum loss of time.

Supervising the tying-up or letting-go of lines between the ship and the berth is an outgrowth of preventing theft. All too often, smugglers among the ship's company will attempt to pass their contraband to shoreside confederates during the confusion which can exist when a ship is being moored or unmoored. The security forces do not handle the lines; they observe and monitor performance by the assigned personnel. (Actual responsibility for handling the lines belongs to the ship's master and the marine superintendent.)

Because modern container terminals usually cover a very large area, considerable time of highly paid employees can be wasted in walking from one section to another. A regular bus service connecting major locations therefore is an efficient and economic provision. A ban on any pedestrian traffic within the terminal enclosure enhances safety while concurrently reducing hazards for equipment operators. It also eliminates the possibility that unauthorized persons might infiltrate those groups of workers walking from place to place. Assigning security guards as drivers serves a dual function, in that the guard is trained as an observer, and on the bus route will be alert to any unusual circumstances. The guard also has the authority to take action if any deviation from accepted practice and routine is encountered.

Finally, giving the security chief supervisory responsibility for cleanliness and maintenance of the terminal assures that when trash accumulations are reported by the patrolmen, a cleanup gang will be dispatched immediately. Burned-out light bulbs, breaks in the transit shed deck, pot holes in truck lanes, and damage to the chain link fence all pose hazards to personnel or cargo, and prompt

attention can be assured if the information from the patrolmen is converted into action by the security chief.

None of the foregoing comments, however, should suggest that the primary mission of the security chief is other than the safety and security of the terminal premises and the cargo stored therein. Routine guard patrols must be established and monitored constantly; contingency plans have to be prepared to cope with emergencies such as fire, hurricane damage, and major equipment failure which could lead to loss of cargo or injury to personnel; and methods must be developed to control intraterminal vehicular traffic. To prevent theft and pilferage, continuous vigilance and resourcefulness are needed. For instance, because truck drivers might be collaborating in cargo-stealing activities, it would be good preventive technique to study the movement of trucks within the terminal, the persons with whom the driver has a legitimate reason to do business, and the possibilities for theft which might exist. The procedures at each stop in these "danger zones" therefore should be scrutinized:

1. *Gatehouse.* The driver surrenders his documents to the gatekeeper, who verifies that this truck and cargo are expected.

2. *Inspection area.* For break-bulk shipments, the truck stops to pick up a checker who will supervise the unloading and the condition of the cargo at the moment it is accepted by the terminal. For containerized shipments, the checker inspects the box for evidence of actual or potential damage to the contents. The condition and number of the seal on the container doors are observed and recorded by the checker.

3. *Unloading area.* Break-bulk cargo is examined and tallied by the checker, who initials the dock receipt. The exact location in the transit shed is identified, so that plans can be made for the most efficient use of time and equipment in loading the cargo aboard ship. Containers brought to the terminal for outward loading in ships are trucked to the designated slots to await transfer to the ship. Once the container is released from the truck-tractor, the driver returns to the gatehouse.

4. *Gatehouse.* The driver presents all his papers to the gatekeeper, who inspects the truck (if it carried break-bulk cargo) before releasing the vehicle.¹

Other measures which the security chief should take to protect the cargo include liaison with police units, and contact with major shippers to learn if the protection system devised imposes undue delay or expense, or is in any way objectionable to these customers. It is obvious that, in the highly competitive environment of international shipping, the reaction of the patrons to procedures adopted by the carriers must be appreciated in order to obtain their co-operation and support.

The *timekeeper* is an important member of the terminal manager's staff. He keeps detailed records of the employment of every laborer hired on an hourly basis. Longshoremen work in gangs hired for four-hour shifts; their names must be inscribed individually on the time sheets at the beginning of each shift. If they handle obnoxious commodities ("penalty cargo") and are entitled to additional compensation for such work, the timekeeper is to note the actual time during which the extra money is payable. Men employed to do the many tasks around the terminal, but not members of longshoremen's gangs, are listed under the heading of "dock labor," and are paid according to the number of hours they work. It is essential that the timekeeper determine periodically that he has an accurate record of everyone who is working and that his pay accounts be consistent with the labor actually performed.

In the previous chapter the functions of the various major officials in the terminal organization have been described. These same functions must be carried out in every port where cargo is handled, whether the responsibility be delegated to a branch office of the carrier, an agent, or a contractor. None of the terminal divisions is superfluous, and all the activities of all the components of the terminal are interrelated. The terminal manager, as the representative of the highest echelons of company management, is responsible for coordinating and controlling the day-to-day activities of his subordinates.

Many steamship companies make contracts with large stevedoring organizations under which the entire terminal operation is handled by the stevedore. In these circumstances, the shipowner actually has no responsibility for the work of the terminal, but nevertheless must assume full liability for any cargo damaged by the stevedore's laborers since these men are, in the eyes of the law, agents of the steamship company. To safeguard the carrier, it is

customary for at least one representative of the steamship company to be assigned to the terminal. His function is largely that of liaison between stevedore and carrier.

In the United States, modern break-bulk terminal operation is predicated upon two facts: manpower is very expensive, and machines can do more work than men. There are many types of mechanical equipment available to use in handling cargo, but the determination of how many pieces of machinery shall be purchased and operated, and the ratio of men to machines, is the responsibility of the terminal manager. In his ceaseless efforts to reduce vessel port time and the costs of handling cargo, it is imperative that the manager be alert to the possibility of replacing obsolescent machinery with more suitable equipment. At the same time, he must be aware of the division of costs between investment in equipment, expense of equipment operation, and wages of laborers. Necessarily, every decision as to replacement of equipment must be based on the determination that more work will be done at lower cost if new machines are procured.

The proper use of cargo-handling machines is of major concern to the terminal manager, who is guided by these basic principles of materials handling:

1. *Safety of personnel is of first importance.* Every piece of machinery and every work technique must be analyzed to determine what, if any, danger there may be for the men assigned to the job. If danger is detected, the machine must be altered or the technique corrected to eliminate, insofar as is humanly possible, the threat to the worker. Experience has shown that these analyses are worthwhile because safe work practices permit faster and better performance by the employees.

2. *Operations must be balanced between men and machines.* To take maximum advantage of the capabilities of a machine, one or more men must be assigned to work with it. The judicious determination of the exact number of men required to produce maximum efficiency from a machine is a major responsibility of the terminal manager. He must guard against the possibility that too many—or too few—men are put on a job at the expense of efficiency. It often happens that, even with the will to work, too many men in a gang will interfere with each other, and the result of their collective

labors is not as good as would have been the case had fewer men been assigned.

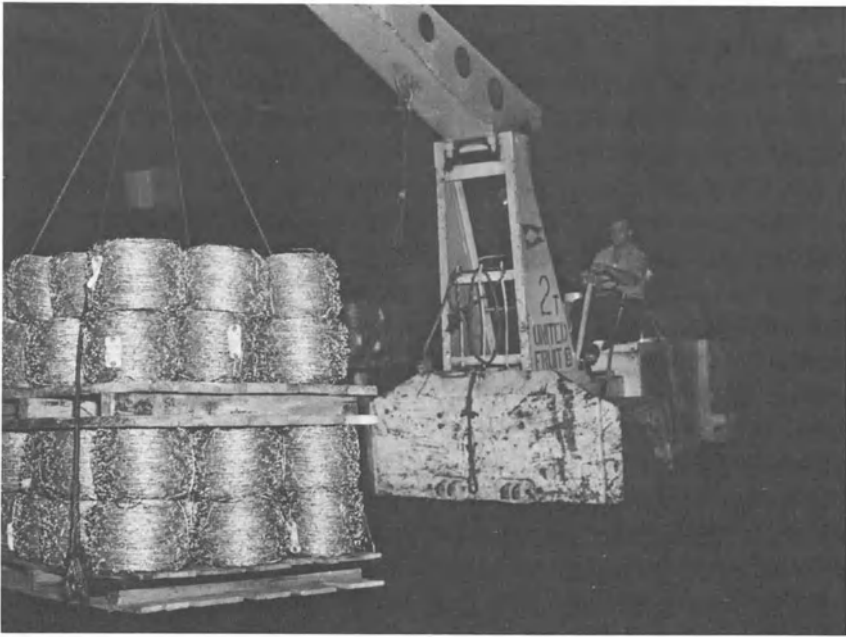
3. *Work practices must be standardized.* To calculate costs in terms of money, manpower, time, and machinery, it is necessary that standard work practices be followed throughout the terminal. There will be minor variations depending upon the personality and experience of the individual supervisor, but the pattern of activity in handling a particular commodity should be reasonably close to identical in all parts of the terminal. The manager must guard against complacency and has to be alert to suggestions for improvements in the techniques employed. Until a new procedure is approved, however, all subordinate leaders should be required to perform their tasks in the accepted manner, the purpose of this standardization being to assist management in controlling output and expense.

4. *The proper type of equipment must be assigned to the job.* It is almost axiomatic that it is inefficient to use heavy equipment for light work, and light equipment for heavy work; it is the function of management to select the machinery to be employed by the long-shoremen and to direct the types of equipment to be used for designated types of work.

5. *Movements must be direct.* Circuitous routes are longer and require more time, both for operator and machine, than direct routes. From management's viewpoint, the routes followed by wheeled materials-handling equipment should be the shortest possible to keep the time involved in transporting loads to the minimum. Lengthening of the distance inevitably increases machine time and hence requires additional equipment if the same rate of tonnage movement is to be continued.

In laying out ideal traffic patterns within the terminal, all routes should be planned to avoid cross traffic. The elimination of the hazard of collision at a crossing makes possible greater vehicle speed. With no crossings, no fields of observation need be provided at the expense of cargo storage space.

Ideally, materials-handling equipment hauls payloads in both directions, but the average marine general cargo terminal affords little opportunity for such efficiency. Consideration should be given to the possibility of back-hauls when routes are laid out. In all



Spools of barbed wire, stacked on wooden pallets, are transported by a crane truck. Lifting bridles and spreaders and the design of the pallet are clearly visible.

cases, the goal of the routing must be the maximum movement of tonnage per machine-hour.

The principal machines used in terminal operations in United States ports today are the forklift truck (sometimes referred to on the waterfront as the "hi-lo"), the mobile crane, the straddle truck, and the tractor-trailer trains. Of these, the forklift truck, traditionally powered by either a gasoline or diesel engine, but increasingly found with battery-operated electric motors to reduce environmental pollution and to enhance safety from fires, is the most versatile and most widely used. Much of this materials-handling equipment has pneumatic tires; all of it has been designed to provide maximum simplicity of operation combined with safety and comfort for the driver. Experience has shown that employee morale and efficiency decrease in direct ratio to operator fatigue.

Materials handling of break-bulk cargo on the waterfront today depends almost entirely on the cargo pallet. This invaluable piece

of equipment is a light, double-faced platform of wood, usually four feet long by forty inches wide. It is inexpensive considering the number of times it is reused. As cargo is received at the terminal, it is general practice to place the goods on pallets to build up a "handling unit" weighing one or two tons. This "palletized" unit must be moved by machine and is not broken up until the individual containers are removed for stowage in the ship.

The principle underlying the use of pallets is simple: it provides a more economical method of employing the resources of the terminal (space, manpower, and money). The greatest expense in terminal operations lies in the requirement to handle literally thousands of packages (bales, boxes, bags, crates, and loose items such as automobile tires) each day. By combining many packages into a single large handling unit at the time they are received by the terminal, and then using a machine operated by one man to perform all movements within the terminal, obvious economies are effected. Concurrently, increased efficiency is obtained by the use of terminal shed space, always critically short of requirements, and better service can be rendered the shippers by reducing the time needed for either loading or unloading cargo.

Many ocean carriers have been able to convince shippers that this "palletization" of their cargoes should be accomplished at the warehouse or factory where the shipment originates. The shipper builds up and straps the load into a single unit which remains unbroken until the consignee has it in his own place of business. There are notable benefits from this practice:

1. Individual packages are not subject to handling at the terminals, and therefore breakage and other types of damage are reduced, if not eliminated altogether.
2. The opportunities for pilferage and loss of individual packages are minimized, especially if the entire shipping unit is enclosed in a protective cover. A sheet of heavy plastic serves very well.
3. Cargo checking by all parties is simpler, more rapid, and more accurate because fewer units have to be counted.
4. Economies of time can be effected at every transfer point.

Over the years, pallet sizes have become more or less standardized, but no formal agreements as to dimensions have ever been

reached between carriers or shippers. It is true, however, that as the use of pallets has been extended beyond the confines of the terminal, the size and type of pallet used for transportation has become more and more conventional, until today the most generally used pallet is forty-eight inches long by forty inches wide, as previously mentioned. The material used for facings is almost invariably lumber one inch thick, while the so-called stringers are four inches square.

Where pallets are used extensively in the terminal, a number of important determinations must be made by the manager. Roadways and passages for the forklift trucks are basic requirements, and space for them must be allocated first in the general layout of the shed. Then the load-bearing capacity of the shed's deck must be ascertained, and rules promulgated as to the number of tiers of pallets which may be permissible. The tiers will vary with the weight of the commodity piled on the pallet, and therefore guidance must be given to the labor force if structural damage is to be avoided. The nature of the packages in each pallet also will influence the number of tiers because the bottom pallets must be protected from crushing. If cargo is palletized by the terminal, then adequate quantities (often numbering in the thousands) must be kept on hand. Provision for storing, repairing, and issuing or receiving pallets becomes part of the routine of cargo operations.

After completing these arrangements, it is necessary to select the sizes, types, and capacities of materials-handling equipment to be used in moving pallets economically and efficiently around the terminal. This machinery is expensive; only those units required, plus spares to insure good maintenance, should be procured.

An efficient, economical marine terminal is well maintained, clean, and orderly. It should be the intent of the manager to make it a place where the workers are considered as human beings deserving good treatment. The office should be neat and in good order; cargo gear should be stacked neatly; and dunnage should be conveniently located, but not in the main stream of traffic. Debris and dirt should be removed promptly, as a matter of safety rather than esthetics.

The provision of good, well lighted, and spacious toilet and washroom facilities for terminal employees reflects excellent management. In this regard, it is good practice to have, somewhere in



Forklift truck with special clamps places roll of kraft paper on a tractor-trailer train. Photo: North Carolina State Ports Authority.

the terminal, proper shelter for the laborers, so that in times of bad weather they may be protected from the elements. Not only is this humane; it is also efficient, because it keeps the labor force together, and permits immediate resumption of work when conditions are propitious. It eliminates having a large number of men wandering aimlessly through the transit shed, with the possibility of injury. A further advantage in providing a shelter is that smoking can be controlled, and the chance of a fire being started by a carelessly discarded cigar or cigarette is reduced.

Enforcing safety rules and preventing accidents among long-shoremen and other terminal workers are major responsibilities of terminal management. Three areas of concern require constant scrutiny on the part of supervisors: personnel and vehicular traffic patterns; physical safety (i. e., structural and plant conditions); and operation of equipment and vehicles.

Attention already has been drawn to the importance of controlling the circulation of vehicles within the terminal enclosure. In break-bulk transit sheds, essential pedestrian traffic should be restricted to conspicuously marked walkways. Container terminals,

partly because of their size and partly because of the density of truck traffic, should exclude pedestrians from the area in which containers are stored. For workers and others authorized to be within the terminal enclosure, transportation should be provided and its use made mandatory for everyone, without any exceptions. These measures enhance personnel safety as well as the security of cargo.

A clean and neat terminal demonstrates that a good safety program is in effect. In cold climates where snow and ice may accumulate on wharf or pier aprons or in marshalling yards, it is good management practice to have appropriate plans and equipment available to remove these hazards. Oil spills on working surfaces must be cleaned up immediately; otherwise, the oil will become a serious hazard as it is spread throughout a wide area by vehicles operating within the terminal. All motorized equipment therefore should be inspected daily for leaks, and defective units withdrawn until needed repairs have been effected.

All equipment, both aboard ship and in the terminal, should be inspected before work begins to assure that detectable deficiencies are remedied, and all unsafe machinery is barred from use. The most rigid standards must be established and adhered to in the inspection of cables, wire rope, cargo nets, slings, and any other devices employed to lift cargo. Questionable items must be replaced as soon as a potential weakness is detected. Supervisors at every level—gang bosses, foremen, supervisors, and management personnel—must be aware at all times that they are personally responsible for safe working practices within their sphere of authority, and must understand that they will be held individually accountable for accidents.²

Safety is not limited to personnel; it applies with considerable economic force to cargo entrusted to the carrier. The ship operator is obliged to handle a consignment, throughout the time it is in his custody, so that the condition at the moment of delivery to the consignee is no different from what it was when presented by the shipper. Unless he can show that the damage or loss occurred as a result of a cause for which the law grants him an exemption, the carrier is responsible and must pay the shipper or consignee the agreed value of the goods. Often the settlement of a single claim for one broken article can absorb the profit earned on the transporta-

tion of several hundred tons of other cargo. The terminal manager must strive constantly to insure that cargo moving through his terminal receives the best possible treatment.

Care of the cargo requires two things: skillful and gentle handling of packages to prevent even superficial damage, and thorough inspection of packages to verify the condition in which the cargo is presented by the shipper to the carrier. The terminal manager must assume that the longshoremen and their supervisors are competent to handle the individual packages properly, and periodic inspections are made to assure that no one deviates from approved standards of performance. The inspection of goods, it must be emphasized, is a task of utmost importance since the carrier will base his defense against a claim for damage or loss on the completeness and accuracy of the examination of each package. It is mandatory that the receiving clerk and his staff of checkers be instructed in desired procedures, that inspections be performed conscientiously and efficiently, and that adequate records of the inspections be entered on the dock receipt. The same significance attaches to the careful inspection of cargo removed from the ship. The delivery clerk is responsible for this activity. At the time the consignee accepts the shipment, the condition of each identifiable unit must be noted and appropriately recorded.

Checkers are very important in terminal operation. In most ports, they are employed by the hour. In United States ports, they receive the same hourly compensation as longshoremen but are guaranteed eight hours' pay, rather than the four hours assured longshoremen when they report for work. Usually there is a checker assigned to each gang of longshoremen, and thus employment opportunities for checkers vary with those of the longshoremen. In those terminals where a ship is on berth practically all the time, a fixed number of longshoremen and checkers is certain of steady employment. These "regulars" can be instructed in the terminal's procedures of checking, and their integrity and accuracy can be established in fact. Additional checkers are obtained, in United States ports, by notifying the longshoremen's union hiring hall. Although these casual employees are competent to inspect, measure, and weigh packages, the accuracy and completeness of their records may not be in accordance with established terminal practice. Furthermore, there is no satisfactory system for determining



Checker measures an item of cargo with his four-foot rule.

the personal integrity of these individuals before they begin their work. This aspect of operation poses a serious, recurring, and well-nigh insoluble problem in the economy of terminal management.

Efficient operation of a break-bulk cargo terminal requires care in planning each stage of activity, and nowhere is this more true than in the layout of the transit shed(s). The booking clerk keeps the receiving clerk informed as cargo commitments are made, so

that adequate space may be allotted to handle the consignments to the different ports. In the case of a company maintaining a service to a number of ports of call, as, for example, through the Mediterranean, or to the southeast coast of Africa, or the Pacific coast of South America, not only must there be an area reserved for each port, but there must be a plan for properly segregating the various types of cargo destined to each port. The following categories must be so placed within each port space that they do not contaminate any goods:

1. Wet cargo (liquids in barrels or drums) to be stowed in the bottom of the ship's holds.
2. Long steel, timber, or other pieces to be stowed on the bottom or in the wings of the cargo holds.
3. Odoriferous or obnoxious commodities requiring special stowage.
4. Hazardous articles (acids, gases, inflammables, and explosives) which must be stowed on deck or in special lockers aboard ship.
5. Packages containing commodities like lamp black, flour, and cement which sift out and may damage other cargo.
6. Refrigerated or other perishable cargo.
7. Fragile goods, such as glassware.
8. Valuable packages requiring security stowage (this sometimes is known as "mate's cargo").
9. Heavy-lift items. Port equipment or the ship's heavy-lift gear must be used to handle this type of cargo, and therefore careful scheduling of details incident to the use of this specialized equipment is essential to eliminate lost time.

Of equal importance to the terminal for planning purposes is information concerning cargo on incoming ships. The quantity, nature, and disposition of the load must be known since space for storage has to be provided somewhere within the terminal enclosure. When a considerable portion of the inbound cargo is to be transhipped into barges or lighters working on the offshore side of the vessel, there is an immediate easing of the requirement for accommodation in the transit sheds. If the cargo must be brought into the shed for sorting and delivery to consignees, a correspond-

ing amount of space will be needed. These arrangements depend upon careful coordination of all the divisions of the terminal to achieve the desired goal of efficient and expeditious turnaround of the ship.

Once the terminal has been laid out for the incoming and outgoing cargoes, the receiving clerk becomes responsible for placing the outbound cargo in the designated areas within the shed or terminal enclosure. When a consignment of cargo is delivered to the terminal, it must be accompanied by the dock receipt, which is prepared by the shipper and contains these data: name of shipper and consignee; port of destination; description of the goods, including type(s) of package and weight; and the booking number. This form, in some steamship company practice, may be prepared with five carbon copies. When he arrives at the gatehouse, the truckman surrenders the dock receipt to the receiving clerk, who stamps it with the date and time of arrival, and inserts the name and voyage number of the vessel in which the cargo is to be loaded. The original of the five copies is the official dock receipt, and on it are made appropriate notations concerning the condition of the cargo or the container, and the actual measurement and weight of the consignment. The second copy, after processing, is returned to the truckman as his proof that the cargo was delivered as ordered. The third and fifth copies are placed with the cargo; from this customary usage, the third copy is known as the "pile tag." As the consignment either is packed into a container or is stowed aboard ship, this tag is given to the chief mate for his records and for planning the outturn of cargo at the port of destination. The fourth copy remains with the receiving clerk, where it is available to the stevedore for his use in preparing the stowage plan. After the cargo has been placed aboard ship, the fifth copy, with all appropriate notations, is returned to the receiving clerk's files, where it is kept as a permanent record of the handling of the consignment.³

The receiving clerk assures that outgoing cargo is so marked that everything for one port will be easy to locate in the holds of the ship. Usually this is accomplished by means of "port marks" or symbols applied to every box, bag, crate, or other container received for shipment. For example, Genoa might be designated by a red circle, while Istanbul would be symbolized by a green square. There are no fixed rules for port marks, except that they be simple

in design, distinctive, and easy to see. They may be affixed at the convenience of the terminal, sometimes in the transit shed and sometimes after the cargo is stowed in the ship. Affixing the mark after stowage makes it certain that it will be visible during discharging operations. Regardless of where or when put on, the mark must not harm the container, nor rub nor wash off. Unpackaged articles such as lumber or steel are marked with quick-drying paint; some bagged or baled goods on which painted marks are not practical are tagged, using the same color scheme as for painted marks.

Cargo discharged by the ship is picked up by the consignee on notification by the carrier that it is available. Normal routine of delivery is in accordance with the following pattern.

The truckman presents the steamship carrier's delivery order to the gatekeeper, who verifies that the papers are in order. This process includes obtaining clearances from the customs officers on duty, and also ascertaining that all freight, terminal, and demurrage charges have been paid. If everything is in order, the truckman is dispatched to the slot where the container is parked, or to the container freight station for smaller lots. A checker supervises the hookup of the truck and trailer for full container shipments. At the container freight station, a checker will count the packages, to assure that the number matches what is designated on the delivery order. Discrepancies are noted, and reconciled if possible. When the truck is loaded, the driver returns to the gatehouse, acknowledges receipt of the cargo, and is given his gate pass.

Checking cargo in a typically busy two-way break-bulk general cargo terminal is of the utmost importance. The carrier must inspect the cargo as it is received, and record any defects noted. In terminal parlance, this is known as "taking exceptions," and refers to the phrase in the bill of lading that cargo is accepted for shipment "in apparent good order and condition, except as noted hereon." The exception is the carrier's defense against the charge that the damage was inflicted after the package was turned over to him. Ideally, the shipowner should refuse to accept any damaged cargo, and should insist that perfect packages be substituted for damaged ones, or the bill of lading would be corrected to show the number of perfect packages actually received. Practice is short of the ideal, and the carrier may have to accept articles which have been damaged somewhat before they come into his custody. He is responsible only



“Taking exceptions”: A cargo checker examines steel pipe on arrival at a marine terminal. Photo: North Carolina State Ports Authority.

for damage in excess of that noted at the time of receipt of the goods. Unfortunately for the shipowner, it often transpires that the shipper or consignee blames him for damage that was sustained prior to its receipt by the marine terminal. The more complete the carrier's record of exceptions, the better will be his legal position.

Efficient terminal management requires that notations made by checkers be accurate, definite, and complete. It is not necessary to define further the word *accurate*. The term *definite* refers to the identification of the particular item, the exact description of the damage, and the reason therefor, if readily detectable. *Complete* means that when the checker turns in the notation, there will be no need to seek further information. A few examples follow:

A bag of coffee is noted to be only partially filled (“slack”). Further examination shows that the sack has been torn and resewn. There is no ready explanation of how the bag was torn or when it was resewn. The checker's entry should be: “SACK No. 4, torn and resewn. Stamped weight, 110. Actual weight, 78 lbs.”

A sack of flour is observed to have a large brown stain on the lower portion of the bag. It is not possible to determine what damage has been caused to the flour. The entry should be: “SACK No. 22. Water-stained, lower half, front of bag.”



Pallets of bagged rice are loaded into a break-bulk ship. Photo: *Via Pensacola, Port of Pensacola Magazine.*

An unboxed automobile is discharged, and one fender is found to have been scratched and dented. The entry should be: "Ford Sedan No. 332. Right front fender, deep scratch, 9 inches long, beginning 14 inches from headlamp, running diagonally to bottom beading. Small dent, 3 inches in diameter, on top of fender 8 inches behind headlamp. Paint not broken."

A cardinal principle of terminal management is that transit sheds shall not be converted into warehouses. One device widely employed to accomplish this is to charge consignees who are dilatory in picking up their shipments a punitive storage fee, called demurrage. In the United States, the general rule is that the consignee has five working days after the ship completes discharge



A pallet load of drummed cargo is hoisted aboard a modern cargo-liner.
Photo: Delta Lines.

during which he may take delivery of his goods. After the expiration of this "free time," demurrage is charged on an accelerating basis. After two days, for instance, the rate is doubled; after four days, it is tripled; after five days it is quintupled. A necessary corollary of the demurrage plan is the efficient operation of an inventory system that will reveal daily what cargoes have been left on the terminal beyond the allowed free time, and what packages have been picked up. An incidental benefit to the terminal operator is that the systematic examination of cargo piles within the terminal enclosure often uncovers items discharged at the port by mistake. A box destined to Baltimore that is removed from the ship in New York will be "short delivered" in Baltimore, and a claim will be filed.

Locating the box in New York removes the basis for the claim. Occasionally a box, previously reported as missing or lost, will be found to have been placed on the wrong pile in the transit shed. Only as deliveries are accomplished is the missing package discovered.

The terminal manager is the director of cargo operations for the steamship company. He is directly responsible for the efficiency and economy of all terminal activities, and is expected to keep the ships on schedule insofar as port working time is concerned. The actual supervision of loading and unloading vessels is delegated to the chief stevedore and his staff. The manager, however, is called upon to make decisions on specialized aspects of cargo handling, the number and skills of laborers needed to perform an unusual job, and the procedures to be followed by clerks, checkers, long-shoremen, watchmen, and other personnel in the performance of their assigned duties. A primary concern of management is that the ship sail on time, and the entire working schedule of the terminal is adjusted toward that end.

A guiding principle in determining the financial advantage of using labor in overtime periods is derived from accurate knowledge of the true cost of ship's time. This is a complex subject; it involves not only the tangible, out-of-pocket costs of ownership, such as wages, insurance, and fuel oil, but also the intangible "earning potential" of the vessel on her assigned trade route. To fix the earning potential, it is necessary to compute the actual demand for the services of the ship, represented by the number of tons of cargo offered at each port of call, as well as to estimate the number of voyages which, in actuality, can be made in the time saved by working after normal hours. Unless it can be shown that the ship-owner derives a measurable gain, good management dictates that the vessel lie idle rather than pay laborers at the overtime rate.

For example, a break-bulk cargo ship of about 16,000 tons deadweight might have operating costs of \$350 an hour and in port would be charged wharfage and dockage fees to bring the aggregate cost of port time to \$400 an hour. The differential between straight time and overtime costs of a single longshore gang in an Atlantic coast port of the United States is \$246 an hour. Additionally, checkers and the clerical staff supporting the overtime operation would have to be paid at the prevailing rates. If the total cost of

the proposed overtime work did not exceed \$400 per hour, and quicker turnaround of the ship, with the consequential benefit of more productive voyage days, is assured, the decision to work the ship in overtime hours would be justified.

Large modern containerships may have operating costs of \$2,500 or more per hour. Because the loading and unloading process involves the employment of a very small number of long-shoremen, it is good management to operate the terminal around the clock to expedite turnaround of the vessel.

Decisions governing the practice of working cargo in overtime periods for ships in liner service require careful and systematic evaluation of the total performance of the vessels. If routine use of overtime makes possible one or more additional profit-generating voyages, it is appropriate to adopt this pattern. Should it eventuate that the habitual use of overtime made it practicable for three ships to do the work of four, the decision would be justified.

Effectively to control the flow of cargo, and to establish a system for locating a fifty-pound package in the midst of fifteen thousand tons of assorted commodities, the terminal maintains a variety of records. The first of these, in order of compilation, is the receiving clerk's *List of Cargo Received*, on which are recorded, by dock receipt numbers, the various consignments of cargo delivered by truck. A separate list is opened for each ship to be worked and stays open while that vessel is in port. When loading is completed and the ship is ready to sail, the list is closed. In appearance, the list is very simple since it shows only the serial number of the dock receipt in one column and the cubic footage and weight of the shipment in adjacent columns. In normal circumstances, it is used only by the receiving clerk and his staff to keep informed on what consignments have been brought to the terminal. It has no circulation outside this office.

A similar list for cargo delivered by barge or lighter is maintained by the receiving clerk. It differs slightly from the truck record, because it is the permanent notation of when the lighter was delivered by the harbor carrier, when work was started, and when the lighter was released to its owner. There is a column in this *Lighter Record* to show where the cargo is stored in the terminal, or stowed in the ship, as appropriate.

Each morning at eight o'clock, the receiving clerk compiles a

report of cargo work completed during the preceding twenty-four hours. A separate report is submitted for each ship and contains these details: number of tons of cargo booked into the ship, tons of cargo actually delivered by shippers during the day, tons loaded, and the quantity still to be worked. The *Daily Cargo Report* also shows hatches worked, number of men engaged, actual time they were employed, and commodities handled during the period.

The daily cargo report can be used for many purposes: it is the basis for forecasts of commodity-handling speeds, and the comprehensive record of the actual gang-hours required to unload and reload the ship. If there has been any period of "detention" (time lost from the cargo-working period), the length of the detention and the reason for it are shown. The state of the weather also is set down, as well as any unusual incidents that might explain the rate at which cargo is handled. The report provides information on both discharging and loading operations. It is circulated to the vice-president, operations, the vice-president, traffic, the freight traffic managers, and any other persons concerned with the data contained in this document.

The *Wharfage Report* is the inventory of all cargo on which demurrage is payable because it has been left on the pier beyond the allowed free time. It is compiled for each ship by the delivery clerk and shows for every shipment the commodity, consignee, marks, bill of lading number, and the reason for nondelivery. The report is made initially at the expiration of allowed free time and periodically thereafter as each demurrage period ends, until the cargo is removed from the terminal.

When a vessel's discharge has been completed, the terminal manager gathers into one report all data relating to irregularities in the handling of cargo. For instance, there will be some packages for which no documents are on hand and which investigation shows should have been delivered to another port. These are known as "over-carried" or "short-delivered" cargo. There will be consignments in which shortages have been noted. Inevitably there will be some damage, to be followed by claims from consignees for restitution of the loss. These data are assembled into the *Over, Short and Damaged Cargo Report*, commonly referred to as the O. S. and D. Report. This document is of great value in determining the actual efficiency of the terminal and the carrier's operations, for it helps to

pinpoint actual damage; it puts losses, "overs,"/"shorts," into their proper perspective; and it serves as a convenient means of notifying all other ports on the company's route that certain cargo is missing or on hand seeking an owner. The O. S. and D. Report is disseminated to the master of the ship, the terminals in every port of call, the claims department, the vice-presidents for operations and traffic, and the terminal's receiving and delivery clerks.

CONTAINERSHIP TERMINALS

Since 1965, steamship operators in the international trades have converted, to a greater or lesser extent, to transporting their cargoes in the vans used for inland distribution.⁴ Containers were offered initially by the carriers as a means of reducing cargo handling costs and port turnaround time. The boxes became popular with shippers because their goods were not handled so many times, and the condition in which they were delivered was correspondingly better. The impact of the new mode of handling cargo upon terminal management practices has been noticeable, but not revolutionary.

Major changes in the physical plant of the terminal resulted from this conversion, and new techniques were developed for mechanical handling of the containers. Worldwide, the financial cost of providing the required port facilities amounts to billions of dollars. There also was a dramatic shift in the location of the newly constructed terminals with significant consequences to the communities which either lost or gained the terminals.

A break-bulk marine terminal might consist of a berthing space equal to the length of the ship, a transit shed providing approximately 100,000 square feet of covered floor space for each ship to be worked, and an uncovered "upland" area of perhaps ten acres where nonperishable cargo could be held before loading or following discharge. Early container terminals were designed to accommodate ships 600 feet long, and offered a marshalling area of about 12.5 acres per vessel. To serve the newest generation of container-ships, which were just under 1,000 feet from bow to stern and could carry about 4,000 20-foot containers, it was necessary to have a berth at least the length of the ship and supported by between 75 and 400 acres of storage space for containers. The whole container

terminal must be surrounded by high chain link fences and equipped with towering cranes and sophisticated materials handling machines.⁵

In place of the familiar head house, truck dock, and transit shed, there now is a gatehouse with a minimum of 10 truck lanes (very large terminals may have as many as 25 lanes), electronically controlled scales, and a communications network of telephones, short-range radios, and pneumatic tubes which links the gatekeepers with the terminal control room and the carrier's booking clerk and cashier.

Most prominent because of their huge size are the container cranes which load and unload the ships. Mounted on railroad-type wheels with a lateral span between legs of at least 50 and often 100 feet, these machines weigh well over 150 tons, and in 1985 cost a minimum of \$4 million apiece. Some terminals have 6 or more of these engineering marvels.

Container terminals depend upon an elaborate system of computer controls. From the shipper's initial booking of the container to the final delivery to the consignee at the port of discharge, the computer is the critical element in the handling process. No changes in the basic principles of terminal management which have been described in connection with the break-bulk terminal have come about as a result of the container revolution, but the methods of assuring control and of keeping the necessary records have been brought into line with the almost unlimited capabilities of the electronic computer. It is noteworthy that continuous monitoring of the location of containers is a routine aspect of terminal operations. At any time, it is possible for shipper, consignee, booking clerk, cashier, terminal manager, or anyone who may have an interest in the exact position of a particular container to be informed. The value of this commercial intelligence is obvious.

In most instances, the waterproof and weathertight container is packed and sealed by the shipper at his warehouse and sent directly to the marine terminal, either by truck or a combination of railroad and highway transportation. A significant volume of cargo, however, is offered in small lots (designated as "less than container lots," or LCL), and delivered by shippers to the terminal's "container freight station" (often referred to as the CFS), where a number of these consignments will be consolidated and packed into a single

container before being placed aboard ship. A complete report of everything packed ("stuffed") into the container is compiled by a vigilant checker, who turns over the compilation to the receiving clerk for processing in the established manner. When a container filled with LCL shipments is unloaded, it is moved immediately to the container freight station, opened, and the contents taken out ("stripped") and prepared for delivery to the designated consignees. As in the stuffing process, a checker oversees the stripping, and the records are submitted to the receiving clerk.

Except as noted above, only fully loaded, sealed containers are accepted by and handled at the containership terminal. Instead of the tedious, time-consuming process of inspecting thousands of items, trained checkers examine the containers to detect any evidence of damage to the box which might reflect injury to the contents. Special attention is given to assure that the seal securing the lock on the door of the container is intact and has not been subjected to tampering. Once this inspection is completed and the container accepted for transportation, the carrier becomes responsible for the proper care and satisfactory delivery of the container to the overseas consignee.

Most liner services on the principal trade routes today handle the greater part of their cargoes in containers; some carriers accept nothing that cannot be put into one of the many types of containers now in existence. This development has modified the detailed procedures of terminal management in many ways, but has not altered the fundamental responsibilities of those charged with the care and custody of cargo. The typical containership terminal has no transit sheds (except for the container freight station, which may be spacious but still is much smaller than the break-bulk terminal's covered area) because cargo is received in waterproof containers.⁶ Transit sheds have been supplanted by hard surfaced marshalling yards where the containers are stored either before loading or after discharge. The container itself is accepted and processed as a unit, and only its unique number, weight, and exterior condition have to be recorded. No attempt is made to verify the contents unless there is evidence of damage or tampering. Cargo segregations have been eliminated; everything being sealed within the container, other consignments cannot be contaminated. The number of gangs of longshoremen has been reduced significantly, with a correspond-

ing increase in the use of extremely costly and complicated machinery. Security remains an important aspect of management, but procedures followed to protect the cargo vans are markedly different from those of the break-bulk terminal.

The traditional checker has been replaced by a clerk who notes the individual numbers of the containers, together with full identification of the delivering or receiving truckmen. The vessel on which the container is to be loaded, the gross weight and size of the container, the port of discharge, and the general classification of the contents are entered on the Equipment Interchange Report (EIR), which serves as a receipt for the container and an inspection record covering the condition of the container, chassis, or trailer at the time of receipt. The clerk also examines the seal on the container doorlock, and enters the number. If there is evidence of tampering with the seal, immediate notification is given to the supervisors for appropriate action. If no seal is in place, the terminal installs one, and notifies the booking clerk at the carrier's office of this fact.

Information obtained at the entrance to the terminal is fed into an electronic computer which matches the arrival notice with data previously supplied by the booking clerk. The computer determines the slot in the marshalling yard where the container is to be held until moved to the ship's side. The computer also is programmed to allocate the actual space ("cell") in which the container is to be stowed in the ship.

Maximum effort is demanded from the terminal organization to unload, reload, and sail the ships in the least possible number of hours. When a containership ties up to her berth, the terminal must be prepared to work around the clock, regardless of holidays or other conventional interruptions. The actual work of moving containers around the marshalling yard and stowing them aboard ship is performed entirely by machine, and only a small number of longshoremen is needed.

The role of manager of a container terminal differs from that of the conventional break-bulk terminal manager principally in the number of men and machines for whose productive employment he is responsible, and in the nature and detail of the records kept and processed.⁷ The essential principles remain unchanged: the ships must be handled with maximum speed; the cargo must be

safeguarded while in the terminal; arrangements must be made for rapid receipt and delivery of cargo and for prompt detection of "over," "short," and damaged containers; and the terminal must be laid out so that containers can be maneuvered by trucks, over-the-road truck tractors, and yard tractors ("hustlers"). Above all, safety of personnel and cargo must be paramount considerations at every stage of operation.

Assuring that the containers in its custody are protected from theft or pilferage is a major concern of terminal management. As a result of constant effort to enhance the security of the containers, these practices should be followed:

1. Restrict access to documents which reveal the contents of the boxes to the personnel whose duties require such knowledge.

2. Establish and enforce rigid rules concerning inspection of incoming containers to ascertain the condition of the boxes and to verify the unique numbers of containers, cargo seals, and chassis. If a seal is discovered to have been damaged or broken, a full record of that fact must be made. The questionable container should be opened only by authorized supervisory personnel and inspected for evidence of tampering or pilferage, and the results of this inspection entered in the record. At the conclusion of the inspection, if it appears that there is no loss or damage, a new cargo seal must be attached to the container, and its number inserted on all cargo documents. Where evidence is uncovered that pilferage or significant theft has occurred, the container must be closed and secured, placed in isolation under surveillance, and the shipper notified. The container must be kept untouched until the shipper gives instructions as to the disposition of the box. In no circumstances should the container be cleared for loading aboard ship until and unless authorized in writing by the shipper.

3. Verify that the unique numbers of the container, cargo seal, and chassis are in accordance with the documentation furnished by the shipper, and that the correct booking number appears on all documents.

4. Assure that truck drivers stop their vehicles only at points designated by the gatekeeper and that their movements are overseen by authorized terminal employees.

5. Maintain the terminal area in a clean condition, with all

containers correctly positioned in the designated slots. The marshalling yard must be brightly illuminated at night. Container slots must be set back from the fence line far enough to prevent any contact from the outside.

6. Establish uniform surveillance routines for guard patrols. For maximum efficiency, all guards should be equipped with portable two-way radios for immediate contact with supervisors whenever anything unusual is detected.

7. Inspect outgoing trucks, including the driver's cab, at the gatehouse to assure that no pilfered items are aboard, and that all the papers needed by the driver are in order.⁸

Terminal Operation

GENERAL cargo break-bulk marine terminals are operated on these principles: (1) cargo must be handled efficiently and economically between the ship and a variety of inland transportation media; (2) personnel and cargo must be protected from injury while in the terminal area; and (3) the ship must be unloaded, reloaded, and made ready for sea in the least possible time. It is almost impossible to exaggerate the magnitude of the task of the terminal manager in meeting these objectives. It is enormously complex because the nature of the cargoes to be handled varies daily and every variation presents a new problem. The whole terminal operation depends upon the availability of a large force of workers whose skills and productivity fix the level of terminal efficiency. These workers are proud of their craft, and they are fanatically loyal to their unions. Few terminals in the world today can be operated without due appreciation of the role of the labor unions and their members.

Complete mechanization of the handling process, in the manner found in a soft drink bottling plant, has not yet come to the break-bulk general cargo terminal, nor does it appear to be economically feasible to attempt to develop such systems because of the nature of general cargoes. Since packages are uniform neither in size nor in weight, they cannot be handled as are the bottles and cases in a bottling plant. In addition, vessel stowage must be planned to give the ship adequate stability and proper trim when she is at sea. The constantly changing makeup of a vessel's load means that practically never is there the same distribution of weight from one voyage to the next. Additionally, packages must be stowed compactly below decks so that they will not be damaged when the ship rolls or pitches in a heavy sea. A major problem is to keep commodities which could cause damage to each other, such as marble and redwood, from coming into contact. Finally, lots of



Skilled longshoremen complete securing a heavy locomotive against the perils of sea passage. Photo: Delta Lines.

cargo must be segregated by ports and consignees to prevent mis-delivery. So far, a mechanical installation with adequate flexibility, ruggedness, and economy has not been devised to solve all these problems of cargo stowage.

Present-day mechanization in the general cargo terminal depends upon a few machines, rather than upon a complex and expensive system of mechanical handling. The forklift truck is the best known of these machines and is the most important as well as the oldest and most flexible. The type usually encountered in American terminals is a compact, gasoline or diesel powered, cush-

ion-tired unit that can pick up a package and carry it any distance. It is able to lift its burden high in the air to stack it for storage, or it may deposit it on the ground. The versatility of the forklift truck is amazing, and a competent operator can accomplish near miracles with it. It depends for much of its operation on the pallet, a twin-faced wooden platform approximately forty-eight inches long by forty inches wide, under which the projecting steel fingers ("forks") can be fitted to provide the lift. The truck most favored by terminal operators has a capacity of two tons and can lift as high as seven feet. Alternate models also are available with lift capacity of fifteen or more tons and a stacking height in excess of sixteen feet.

The mobile truck crane has a gasoline engine and pneumatic tires. At the rear of the four-wheel vehicle is a winch, actuated by a power takeoff unit. A wire rope, shackled to a three-foot length of chain terminating in a cargo hook, runs from the winch along a steel boom projecting several feet beyond the front wheels, and over a roller sheave. When carrying a load, the hook is pulled up until it is almost in contact with the sheave; the draft of cargo hangs from the hook. From his driver's seat, the operator controls the winch, which has a capacity of three tons. The truck crane can travel any distance and is almost as maneuverable as its shorter wheelbase partner, the forklift truck. For handling commodities like bagged mail, loose rubber tires, sacked beans, and other items which can be placed in a cargo net, the truck crane is unrivalled.

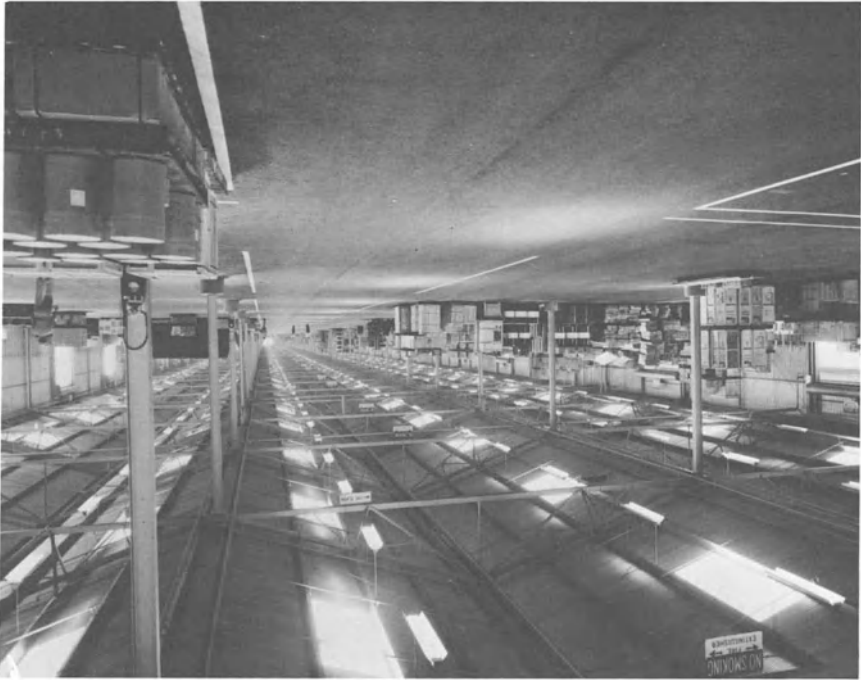
The tractor-trailer train provides an excellent means of moving cargoes over a considerable distance. It consists of a power unit, or tractor (colloquially known as a "jitney"), which tows a train of two or more trailers loaded with either palletized or loose packages or sacks. Although it can be used to advantage in any terminal where hauling distances are rather great (generally, in excess of 400 feet), it is most efficient where wide aprons extend beyond the sidewalls of the transit shed, and trailers can be brought directly under the ship's cargo hook. Most authorities agree that the optimum benefit from this type of equipment is obtained where the design of the terminal includes both wide aprons and long distances. The train also is useful in moving cargo from the truck dock to a place of rest in the shed, and from that place to the ship's side. A forklift truck usually is needed to handle the pallets. Trailers at the ship's side are loaded or emptied by means of the ship's cargo gear.



A modern, well-designed terminal built on a wharf. Wide aprons, numerous doors, and skylights enhance efficiency. Photo: Sutton, courtesy The Port of New Orleans.

Trailers are comparatively inexpensive, and large numbers are found in terminals utilizing the tractor-trailer system. The tractor, an expensive unit with a highly paid driver, should be kept moving as much of the time as possible. Ideally, a train of loaded trailers is delivered under the ship's hook and uncoupled from the tractor. The tractor then is moved to a point where it hooks on a train of empty trailers and tows them to the loading point, where another exchange is made between "empties" and "loads."

Some terminals have found that they handle enough timber or long lengths of steel, piping, or packages to justify procuring one or more straddle trucks. These specialized carriers were built originally to handle stacks of lumber, but have been proved useful for other commodities of considerable length and rather great weight. As is true in the case of the forklift truck, the straddle truck depends for efficient operation upon having its "lift" on a special type of pallet which its mechanism can grip. It is a costly machine, and investment in so specialized a piece of equipment can be war-



Interior of a well-designed and well-lighted transit shed. Photo: Port Authority of New York and New Jersey.

ranted only by the volume of appropriate cargo which must be handled routinely.

In conjunction with the equipment just described, powered and nonpowered gravity roller conveyors are found in many terminals. They are excellent for transferring packages of uniform size over relatively short distances, such as from a delivery truck to a cargo stack. They also are useful in expediting the movement of small packages from the wings of a hold to the square of the hatch, or in the reverse direction. Conveyors in service on marine terminals are easily portable and are shifted as required. The more extensive employment of these two types of conveyors in general cargo operations is not deemed practical for the reasons advanced to explain the lack of complete mechanization in a break-bulk terminal.

Terminal operation is affected by the fact that longshore labor is almost universally unionized, and union customs and working agreements must be taken into account in planning how the cargoes

are to be handled. A few examples from the port of New York will make this clear.

By long-standing agreement between employers and union, longshoremen working a break-bulk ship are hired in units of 21 men. These units ("gangs") are worked intact and may not be divided. Even when insufficient work exists for all 21 men in No. 3 hold, it is not permissible to leave 10 men there and send the other 11 men to No. 2 hold, where enlargement of the gang would expedite cargo outturn. Any gang, however, may be augmented by individually hiring additional men.

The period of work for longshoremen also is governed by union agreement. These workers are hired for a minimum of four hours, except in carefully defined circumstances. If a ship is not expected to berth until nine o'clock in the morning, and work is to commence immediately, the gangs must be hired and paid from eight o'clock. During the waiting time, the gangs may be used for any work the terminal may assign, such as piling cargo, shifting equipment, or sweeping the decks of the transit shed.

No work of any sort, except that required to berth a ship and to handle the mail, or in emergencies, will be performed on Labor Day, and no work is done after five o'clock on Saturday afternoons, except to "complete" a ship. To complete, in this sense, means finishing a job if it can be accomplished in a maximum of four hours time.

The other laborers on the terminal belong to a group collectively known as "dock labor," hired from the ranks of union members as individuals, rather than as a gang. Varied skills such as those of the carpenter, cooper, wire splicer, and cargo checker are found in the dock labor force. When it is determined that four carpenters, two wire splicers, three equipment operators, and six checkers are needed for the day's work, these 15 men are hired individually. Their wages, hours, and conditions of working are set forth in detail in the union agreements.

Many terminals, especially in the larger ports, have sufficient activity to keep a nucleus of longshoremen and dock laborers regularly employed. This group comes to the terminal each day, and although unionized and subject to the working agreement, is not compelled to apply each morning and afternoon for work. The very nature of terminal operation, however, makes it necessary to

hire additional men of those grades and skills which the particular needs of the different ships make apparent. These supernumeraries are hired for periods of four hours and may be released at the end of any shift. In most American ports, it is customary for supernumeraries to be obtained through some sort of employment center. The normal routine calls for notifying the center, which may be operated by the union (as in San Francisco) or by a governmental agency (as in the port of New York) of the needs for the next day. The center, in turn, has its own arrangements for making contact with gang foremen and individual craftsmen. At the appointed hour, the desired contingent of longshoremen and craftsmen reports to the terminal for assignment.

The terminal manager who seeks to improve the overall efficiency of his operations can capitalize upon the longshoremen's loyalty to the gang and the importance attached to the position of gang foreman. The punctilious observance of protocol by management is necessary if the authority and dignity of the foreman are to be maintained. The manager, for instance, may be dissatisfied with the performance of some members of a gang. Rather than correcting the laggards personally, the manager lodges his complaint with the chief stevedore. The chief stevedore discusses the matter with his ship supervisor ("ship foreman") and directs him to handle the problem. The ship foreman explains the matter to the gang foreman, who takes the necessary action.

It is essential to recall that the gang members look to the foreman for practically everything: the place and time of the next job, the rotation of work within the gang, promotion to less arduous and more responsible work, the acceptability of excuses for absences, and recommendations for retention in or dismissal from the gang. The foreman has earned this peculiarly privileged position by superior ability to perform the work of each man in the gang and to enforce discipline. He has no assigned duties when his gang is working; he is the supervisor. Usually he is on deck where he can observe all that transpires. He ordinarily oversees the work of the gangwayman, who directs the exact spot where a draft of cargo is to be placed; he also makes certain that his men understand where and how they are to stow the individual packages of cargo. If a difficult parcel is being handled, the foreman may be in the hold to coordinate the men's efforts. As the need arises for dunnage lum-

ber, separation paper, and other supplies, he notifies the ship foreman.

The gangwayman is the signalman who directs the winch operators and serves as the means of communication between them and the hold or the apron, using a code of hand signals to transmit orders to the winchmen. He must have excellent depth perception since he always works on the uppermost deck of the ship and has to judge accurately when a draft of cargo is approaching the last inches above the landing point perhaps fifty feet below. He should be of a placid temperament, not easily excited, and able to concentrate his undivided attention on the movement of a single cargo draft. On his hand signals and decisions depend the safety of his fellow workers and that of the cargo.

Winch operators are chosen from members of the gang on the basis of their ability to work with the gangwayman. They control the motors turning the drums on which the cables connected to the cargo hook are wound. In United States Atlantic coast ports, there is an operator for each of the two winches of a conventional rig; in Pacific coast ports, one operator controls both motors if they are close to each other. Perfect timing and teamwork are mandatory when two operators are employed. In any event, undeviating attention to the travel of the hook and the signals of the gangwayman is absolutely mandatory.

Piermen are those members of the gang designated by the foreman to work alongside the ship. In the port of New York, there are eight piermen; their counterparts are the eight holdmen. The piermen make up the drafts of cargo and hook them to the crane for loading aboard ship. If the ship is discharging, they guide drafts to the platform, and after unhooking the gear, distribute the packages to appropriate piles. Holdmen perform the same duties, except that they work in the ship's cargo spaces. Comparatively, the hold work is more arduous and requires more skill, but since men are rotated from pier to hold at the discretion of the foreman, there is no difference in wages.

In the East Coast gangs, there is a "relief man" in addition to the foreman, gangwayman, two winch operators, eight piermen, and eight holdmen. To permit continuous work yet allow men a rest period during the four-hour shift, the relief man takes over the winch operator's duties. When this functionary returns from his

rest, he becomes the acting gangwayman, while that worker moves up temporarily to foreman. Holdmen and piermen arrange their own reliefs within their respective groups. Since the relieving operation goes on continuously, the relief man is a regular member of the gang and is employed and paid on that basis.

Terminal efficiency is gauged generally by the number of tons of cargo handled in one hour by a gang of longshoremen (almost invariably supported by one or more pieces of motorized cargo-handling equipment). This rating is not based upon individual performance, nor does it show how fast or how slowly particular commodities are loaded or discharged. It does, however, set up a reasonable standard by which to compare terminals in a port and is useful in evaluating relative efficiency in the different cities where a steamship company may operate general cargo terminals. Comparisons, to be valid, must take into account the use of machines by the gang. All too often, expensive tools become simply "effort savers." In the port of New York, for instance, at least one forklift truck is required to support a gang, but the number of tons of cargo handled per hour is not much greater than it was in the days when manpower alone was used.

Whether he thinks in terms of men or machines, the terminal manager has to remember that working cargo is the sole reason for his organization. Ideally, he takes part in forming the plans for receiving cargo which is to be unloaded from the ship and also to be brought to the terminal for movement overseas. The assignment of precise locations for designated lots of outgoing cargo usually is a function of the chief stevedore, who recommends to the terminal manager certain patterns for laying out the transit shed. This layout is planned to assure that there will be sufficient space in the shed for those goods to be discharged from the ship which are most likely to be held for several days before delivery. When the terminal handles cargo in only one direction, either outgoing or incoming, it may be possible permanently to allocate certain spaces in the shed to cargo for each port of call. This arrangement is satisfactory where the quantity of cargo moving does not vary greatly from voyage to voyage, where there is not much pressure resulting from frequent arrivals or departures of ships, and where floor space so assigned, and not used, can be spared.



Paired cranes of the *Raseltin* (built 1977) serve the four hatches, handling break-bulk and containerized cargo. Photo: Jeff Blinn, courtesy Moran Towing Co.

A more flexible plan usually is followed in sheds where two-way traffic must be processed. Outbound requirements are ascertained from study of the cargo booking sheets to determine how much space is needed to accommodate consignments to each of the ports of call. Examination of the incoming ship's cargo stowage plan and manifest reveals approximately how many square feet must be allocated to each of the major blocks of her cargo. With this information, the available space in the terminal enclosure can be assigned for most effective use. It must be noted, in this connection, that the layout must be adapted to any congestion which may exist in the transit shed because consignees of cargo delivered by earlier arrivals have not been able to pick up their goods.

The actual layout of the transit shed necessarily depends upon its design. If the shed is long, narrow, and cluttered with many stanchions, there will be a significant loss of working area because of these stanchions. Modern terminal design provides for wide sheds



Break-bulk cargo-handling operations on a wide-apron wharf.

and stanchions as far apart as possible in order that the available floor space may be put to the most efficient use.

A wharf lies parallel to the shoreline and is easier to work than a finger pier. It has two advantages over the pier: there is accommodation for ships on one side only, and traffic has access to the transit shed at each end as well as at intermediate points of the wharf structure. The finger pier, as the name suggests, projects into the waterway at an angle to the shore. Ships usually are berthed on both sides, and some long piers provide moorings for two or even three vessels in tandem. Cargo operations in such circumstances are impeded because all wheeled traffic must enter and leave through a single gateway at the land end. A major cause of lost floor space in this type of installation is the need to provide clear areas where trucks can turn around.

The rule of thumb for the amount of transit shed floor space needed to work a ship's cargo is nine square feet per ton. Ideally the shed should have only one level in order to use modern terminal and materials-handling techniques to their highest levels of effi-

ciency. Wheeled vehicles of any description are able to move anywhere, and traffic control is comparatively simple. Routine inspection and inventory procedures are made easy. An important, but occasionally overlooked, gain is that desirable illumination of the working area may be obtained through installing skylights in the roof.

Nevertheless, many of the older transit sheds are two-story structures, built to obtain the largest possible number of ship berths on limited water frontage. These sheds usually are narrow and not well adapted to mechanized cargo-handling operations. The disadvantage of the upper deck lies partly in the restricted weight allowance per square foot on the upper deck, and partly in the difficulty of access for large trucks and the consequent necessity of rehandling packages. Using the second deck for incoming cargo, however, does eliminate confusion with the outbound shipments.

As a device to offset the higher cost of handling cargo on double-deck piers, a few of these sheds have been built with truck ramps connecting the street level with the upper level. The theory is that trucks can be driven from the street to the upper level to discharge or to load. This obviates the need to move cargo from the upper deck before it can be put into trucks. The floor space devoted to the ramp, however, does interfere with efficient operation of the entire terminal.

No double-deck transit sheds are equipped with elevators to lift trucks to the upper level. Installing them would be prohibitively expensive, and while they would take up less space than a ramp, they still would absorb room which can be used more profitably for other purposes.

A major problem in terminal operation is truck traffic. When large motor vehicles are permitted to enter the transit shed, roadways and turnaround spaces must be provided. Some terminal managers exclude trucks altogether, and receive and deliver cargo at the headhouse. By excluding trucks, maximum deck area is reserved for cargo. The forklift trucks and other motorized equipment of the terminal require only narrow alleys, and can move more rapidly when they do not have to be careful of moving trucks. Only authorized workmen are allowed within the transit shed, thereby increasing the security of cargo. If trucks are unloaded at the headhouse, maximum benefit is obtained from the labor of the

men assigned to this job because they spend their working time on the truck line, handling cargo as fast as trucks can be put into position. Cargo is discharged from trucks to pallet boards or cargo nets and is taken directly to the ship for loading or is transported to the place where it is to rest until loading time. Goods are delivered to the trucks of the consignees after being sorted in the transit shed and placed on pallets or in nets. Appropriate equipment is used to transfer cargo to and from the truck loading platform.

This same pattern of traffic control can be followed on the wharf. Instead of bringing trucks to the headhouse, they are driven to the loading platform which parallels, on the land side, the ship's berth. The shed is reserved exclusively for materials-handling equipment and cargo. An advantage the wharf enjoys over the typical finger pier is that trucks may be directed to the loading platform most nearly opposite the ship's hatch where the cargo is being worked. The movement of cargo is limited to the short distance across the wharf. This plan is especially successful where a wide, sheltered platform is provided at truck-bed height.

In ports where railroad lines extend to the waterfront, it is usual for the tracks to parallel the ship's berth. This arrangement permits railroad cars to be placed under the ship's hooks, and for their contents to be transferred directly into the hold of the vessel. Conversely, the goods carried in the ship may be discharged into waiting railroad cars. Terminal management techniques for operations of this sort are the same as for lighters bringing cargoes to the offshore side of the ship. Delivery of cargo by rail cars to be "spotted" at ship's side must be scheduled to conform to switching schedules of the railroad. Many terminals have their own means of moving cars after they are spotted on the tracks. Where no method of moving cars is provided by the terminal, the railroad company must be requested to make a switch locomotive available. In the port of New Orleans, the Public Belt Railroad has two deliveries and two pickups of cars each day; the terminal manager arranges his car-loading and car-discharging schedule accordingly.

Some ports are so situated geographically that much of the cargo brought to the water's edge in railroad cars is transferred from these land conveyances to waterborne lighters that are towed to and secured on the offshore side of the ship. This releases the rolling stock promptly, while expediting delivery to the waiting

vessel. If the requisite cranes are available, the lighter's load may be taken aboard concurrently with goods being lifted from the dock-side platform.

Break-bulk cargo terminal operations are more efficient, and port turnaround time is reduced when the pier or wharf has a wide apron between the shed and the ship's side. Railroad and motor vehicles can be brought directly under the ship's hook for delivery or receipt of goods. Terminal equipment can be used effectively to spot or pick up cargo drafts. Ships' booms extend over the apron to deposit or pick up their loads. Two or more drafts can be deposited on the apron and still be within reach of a single ship's hook. If the terminal provides shoreside cranes, as is true in most European ports, all the above listed advantages apply, with the addition of a larger radius of operation. Either several trucks or railroad cars can be served concurrently, or a number of piles of cargo can be built up while the materials-handling units are working to and from the interior of the transit shed. These factors are recognized in modern terminal design, and aprons varying in width from 15 to 50 feet are found in the new structures.

The value of cargoes piled in a transit shed, or in the area immediately adjacent to the entrance, is great, and the terminal manager must provide for its safety. If terminal operations are on a scale sufficient to justify the expense, a security director may be assigned to the staff of the terminal manager. If the terminal is small, a foreman ("roundsman") is in charge of the guards. There always is a gateman on duty at the entrance, and no person is permitted to enter the premises without the knowledge and approval of that functionary. Just inside the shed, where it will be clearly visible at all times to the gateman, is the locker for especially valuable cargo. In this space, commonly referred to as the "crib," are stored items like bales of furs and valuable skins, perfumes, liquor, watches and watch movements, cigarette lighters, and costume jewelry. When the terminal is in full operation, the most reliable and trustworthy guard is assigned to duty inside the crib. He is to make sure that only authorized persons remove anything from the enclosure. Patrolmen make frequent inspections of the entire terminal area, seeking evidence of theft and irregularities which suggest actual or contemplated pilferage, and indications of fire. Alertness, willingness to take a second look at anything, and

facilities for prompt communication with the roundsman are essential to satisfactory performance of the duties of the patrolman.

The minimum staff for the security of any transit shed is a gateman, a patrolman, and the roundsman. During the working day, it may be necessary or desirable to increase the staff by a large number of men who will be assigned posts in accordance with the plan proposed by the security director or the roundsman. If pilferable cargo is to be worked in a ship, a uniformed guard in the hold has a deterrent effect upon would-be thieves. If passengers are to be embarked, a number of guards must be assigned to assure that baggage is not stolen, that passport routine can be followed without disturbance, and that traffic of both vehicles and people is directed properly. If a ship is moored alongside for some time, a gangway watchman is provided to limit access to authorized persons. All these extra men are ordered from the agency holding the contract for guard service and are assigned by the roundsman or the security director, in accordance with the capabilities and experience of each guard. In the United States, security personnel are unionized and their working hours and conditions are set forth in appropriate detail in the agreement between the employers and the union.

The terminal manager depends upon the chief stevedore to supervise the actual labor of handling cargo to and from ships; upon the security director to take care of cargo while it is in the transit shed or the area immediately adjacent; and upon the receiving and delivery clerks for accurate performance of their duties. Without these assistants, it would be impossible to operate the terminal.

CONTAINER TERMINALS

To adjust to the container revolution has required major changes in the physical plant of the marine terminal and in the techniques of handling the big boxes in which the cargo is stowed. It also has involved the investment of billions of dollars, worldwide, to provide the necessary facilities and equipment.

For a break-bulk cargo terminal, a transit shed at least as long as the ship to be worked, plus an "upland area" of perhaps ten acres for outdoor storage of goods and some provision for a truck waiting line, was adequate for all but the largest operators. In dramatic

contrast, a modern container terminal may cover between 75 and 400 acres of asphalt-coated space enclosed by a high chain link fence. Instead of a headhouse, truck dock, and transit shed, there is a gatehouse with a minimum of ten lanes and possibly as many as twenty-five. Because so much land is essential, older port facilities in built-up sections like New York's Manhattan Island or San Francisco's Embarcadero have been deserted in favor of nearby regions where sufficient waterfront acreage with good access for highway and railroad vehicles is available. The cost of this property is enormous; its development into a container terminal requires installation of at least two container-hoisting cranes entailing an investment of about four million dollars each. Other equipment is equally high priced; to purchase a single straddle truck to move and stack containers within the terminal meant, in 1985, a disbursement of \$560,000 while maintenance amounted to \$40,000 a year.¹

Container terminals have been built concurrently with the growth of computer systems. Although the principles of management do not differ from those described for the break-bulk terminal, the actual process by which a container is handled in the modern terminal involves sophisticated systems of control using the latest computer technology to keep track of several hundred, or perhaps a thousand or more, over-the-road vehicles moving through the terminal each day.

A typical program is sketched here to demonstrate how the established principles of management are observed.

When the truckman brings the container to the gatehouse, a small camera photographs the driver, his operator's permit, and his vehicle registration. A clerk accepts the dock receipt, notes the name of the owner and the serial number of the container, as well as the date and number of the dock receipt tendered by the driver. These data are transmitted electronically to the home office computer, which verifies the authenticity of the dock receipt and that this container is expected. The gatehouse clerk fills out a "trailer (or equipment) interchange receipt," which shows when responsibility for the container is transferred to the steamship line, time-stamps this document, and gives it to the driver as he moves to the receiving office. The dock receipt is sent by pneumatic tube to the receiving clerk, who assigns a checker to meet the truck in the inspection line. Following the inspection, a computer signal indi-

cates the slot in the marshalling yard where the container on its chassis is to be parked. The driver takes the container to this point, and after releasing the trailer, returns to the gatehouse, surrenders the interchange receipt, and picks up the processed dock receipt as he leaves the premises. The interchange receipt has been time-stamped at each stopping point for purposes of control. Finally, the various data relating to the container are transmitted to the computer, which determines the exact cell in which the box is to be placed aboard the containership.²

It is to be noted that the principles of terminal management have been followed, especially in the following respects.

1. Strict accountability for each package of cargo is demanded of the terminal. When a container is delivered to the terminal, its identifying data are noted with utmost care. In many cases, the container bears on each side an intricate color code that is scanned electronically for appropriate information as to number, size, tare weight, and ownership. These data are transmitted instantaneously to a computer. An error in identification is intolerable because of the value of the "package"—in this case, the whole container.

2. Layout of the terminal area (or marshalling yard) is as critical as ever. Containers destined to a certain port are assigned a space to which access is gained over routes planned so that the tractors can move rapidly with the huge boxes, many of which are forty feet long. The square footage of space in the marshalling yard and the number of slots allocated to that destination port will vary with each sailing, depending upon the quantity of cargo.

3. Control of traffic within the terminal enclosure remains a problem but is more complex than in a conventional break-bulk transit shed. The great number of containers constantly moving in and out of the yard requires careful and systematic supervision to eliminate the possibility of collisions. In a very busy operation, there will be an almost uninterrupted flow of containers arriving from inland points of origin, containers being sent to consignees' warehouses, and containers being moved to and from the ship's side. Special attention to the establishment of "flow lanes" must be given by terminal supervisors who will be guided by the same principles that rule in the transit shed: one-way traffic, adequate width of

roadways, good fields of vision, and, insofar as space permits, the elimination of cross traffic.

4. The use of machines has reduced the number of longshoremen employed by the terminal, but has not changed the loyalty of these workers to their unions, or their insistence upon punctilious observance of the agreement with management. The terminal manager and his staff have the same responsibility for understanding and complying with the terms and conditions of the union agreement that they have in a break-bulk installation.

5. The balance of machines and men in the tasks of the terminal has become more important than ever, and the skill of machine operators has assumed a new and larger dimension in the scheme of activity.

6. Security of the terminal, despite the fact that all cargo is enclosed in weatherproof containers, continues to be a major responsibility of the operator. The greatly increased expanse of the terminal area, the uniformity of containers, the constant movement within the terminal enclosure, and the continuous flow of traffic have contributed to the complexity of the problem of providing security. Illumination of the marshalling yard is essential because so much of the activity goes on during night hours. Adequate, well maintained, constantly patrolled fences are mandatory, and deep setbacks from the fence line must be enforced to prevent thefts. Pilferage, although reduced in comparison with break-bulk terminals, still occurs; whole containers are stolen. Vigilance, intelligent curiosity, and frequency of patrols by the watchmen remain vital elements of good terminal management.

7. Breakdowns of machinery affect the entire terminal. As costs of ship operation and terminal equipment have mounted, senior management regards these failures as intolerable. Responsibility for the programs of preventive maintenance and regular overhauls has been made a major function for supervisors reporting to the terminal manager.

8. Stowage of containerized cargo aboard ship no longer requires the wide range of knowledge of the characteristics of individual commodities which was essential for break-bulk stevedoring. Much of the burden of determining how to distribute weight within the ship is carried by the computerized systems which are characteristic of modern container terminals, but the wisdom of the

terminal manager still is required to specify what the computer programs must encompass.

9. "Time is money" always has been an axiom in terminal operation. More than ever is this true as ship turnaround time has been shortened and the pressure increased upon the terminals to find ways to unload and reload the ships in the least possible number of hours. The reasoning is unchanged from earlier days, but the demand for expedited operations has been magnified as the burden of the enormously costly ships, the multimillion-dollar dock-side cranes, and the vast investment in containers, chassis, and tractors has become more and more difficult to bear. Constant improvement in productivity is demanded by every echelon of management, and the margins for error are narrowed constantly in the effort to balance costs and revenues.

In summary, the operation of a container terminal is based on the application of the same principles by which the break-bulk terminal has been managed for decades. The introduction of computer technology to handle the more complex details of terminal routine has been the most obvious development in day-to-day supervision. In no way, however, has the need been reduced for the terminal manager to possess wide experience, good judgment, keen intelligence, great integrity, a desire to find better ways to accomplish any task, and a ceaseless concern for the safety of employees, ships, cargo, and machinery.

The Stevedore Contract

AMONG the documents enumerating the duties and responsibilities of shipowners and shoreside specialists, none is of greater importance than the contract between the steamship company and the stevedore. Fundamentally, this is an agreement delineating the charges, terms, and conditions under which the arduous, skilled, and highly essential labor of loading and unloading ships is to be performed.

The usual contract to handle break-bulk cargo at a conventional terminal contains a list of the commodities that will be handled and the charge per ton or other unit of each of the listed items. The fee is predicated upon normal working hours of the longshoremen, and includes all necessary handling of goods, clerical work, checking cargo into or out of the ship, and routine "recoopering" (sewing torn bags, securing loose boards in boxes and crates, and similar minor repair work), and watchman service. A sample of the listing of different items handled in a break-bulk terminal follows:

Large unboxed automobiles	Charged per unit
Compact unboxed automobiles	Charged per unit
Bagged cargo stowing at more than 60 cubic feet per weight ton	Charged per 2,240 lbs.
Bagged cargo stowing at less than 60 cubic feet per weight ton	Charged per 2,240 lbs.
Cocoa beans in bags	Charged per 2,240 lbs.
Wet hides, in bundles	Charged per 2,240 lbs.
Dry skins, in bales	Charged per 2,240 lbs.
Copper, slabs	Charged per 2,240 lbs.
Cement, in bags or strapped to pallets	Charged per 2,240 lbs.

The method by which these different rates are set is worthy of study. Many of the "old-timers" in the stevedoring business used intuition, experience, and the barest minimum of rudimentary

calculation to determine what they should charge to handle a particular commodity. Today, rates are made somewhat more scientifically and are computed to the fourth decimal place, reflecting with greater accuracy the actual cost of performance under the contract.

The stevedore's efforts to obtain a contract to work conventional break-bulk ships at a terminal in the United States normally commence by learning, as accurately as possible, the details of the trade of the carrier he proposes to serve. First, he must know how frequently and in what form (bundle, bale, bag, carton, slab, or whatever packaging is customary) each item moves, and approximately how many tons are included in each ship's load. Access to, and careful study of, a number of inward and outward cargo manifests will provide this essential information. Inasmuch as vessel characteristics, particularly hatch sizes and the capabilities of cargo gear, affect the rapidity with which the vessel may be worked, the stevedore must become acquainted with the ships of the fleet he aspires to handle. The terminal where he will load and unload the vessels merits expert scrutiny since the physical features of the plant can influence the speed with which longshoremen are able to transfer cargo from shore to ship and vice versa.

At this point in his investigations, the contracting stevedore must talk to the shipowner to reach agreement on what he will be required to furnish. If the contractor is to provide the equipment and gear used in the operation, his bid must reflect his investment. If, on the other hand, the shipowner is to supply the machines, slings, nets, and other items of equipment, the stevedore's bid should be correspondingly lower.

A number of unchangeable factors, however, must be taken into account by the contractor as he makes his computations. For one thing, the size of the gang of longshoremen to be employed cannot be altered by an individual contractor since it has been fixed by negotiation between the union and all the employers of longshoremen in the port. The wages to be paid, and the working conditions in which the laborers perform their tasks, have a decided effect upon final rates for individual commodities. Necessarily, the stevedore's bid is the result of determining all the hourly costs of labor (including "fringe" benefits paid by the employer as well as actual wages, insurance premiums, and related taxes), and estimating as accurately as possible the number of tons of the commodity

the standard gang of longshoremen can load or unload in one hour. An error of a half-ton an hour in the performance (or "production," as it is known in the trade) of the gang can be financially disastrous to the stevedore, should the commodity under study begin to move in quantities exceeding expectations.

The following clauses, quoted from a standard contract used in a major United States port, set forth the responsibility which each of the two parties to that contract must assume:

Commodity Rate Inclusions. As part of the foregoing specified rates, the Contractor agrees to include in the commodity rates the following described services:

a. Transport Contractor's gear and equipment to and from the pier where the ship is berthed, excepting to locations that are inaccessible to motor trucks.

b. Provide all necessary stevedoring labor, including winchmen, hatch tenders, tractor and dock crane operators, also foremen and such other stevedoring supervision as is needed for the proper and efficient conduct of the work.

c. Adjust rigging of booms and guys, etc., at hatches where work of discharging and/or loading will be conducted and unrigging when completed; also removing and replacing beams and hatch covers.

d. Discharge cargo from or load cargo into vessel's holds, 'tween decks, on deck, shelter or bridge spaces, deep tanks, cargo lockers and lazarettes, also temporary bunker spaces, but excluding fore and aft peaks and bilges.

e. Shift gangs as required between inshore and offshore, also from lower to upper floor (or vice versa) on double-deck piers. Shift lighters into working position after they have been placed alongside vessel, when this can be done without tugs.

f. Sort (by longshoremen) and stack cargo man-high on pier upon discharge of vessel or break down cargo from man-high upon loading of vessel.

g. Perform such long trucking as required within the limits of the pier where the vessel is berthed; limited to the section occupied by the vessel, should the pier have multiple sections.

h. Load and lay dunnage board (except freighted dunnage lumber) as required during loading for proper stowage of cargo.

i. Work two gangs simultaneously in hatches when required and when practical to do so, provided necessary additional booms, falls, and winches are supplied by vessel or from shore facilities.

Vacation, Welfare and Pension Allowances. Assessments charged to

the Contractor to carry out the requirements of paid vacations, welfare benefits and pension fund plan under the prevailing Wage Agreements with the International Longshoremen's Association are not included in the commodity rates quoted herein, unless otherwise stated, and such assessments shall be charged for by the Contractor at the percentage factors prevailing in the Port. When charges are to be rendered by the Contractor on the basis of the labor wages incurred, these assessments shall be added at cost.

Equipment. The ship is to supply booms, adequate winches, in good order and with sufficient steam or current for their efficient operation; blocks, topping lifts, guys; wire or rope falls of sufficient length and strength, hatch tents, lights for night work; tugs, derricks or cranes for such heavy lifts as exceed the capacity of the ship's gear, and cranes in the absence of ship's winches. The ship is also to supply dunnage, paper, and all material for shoring and lashing cargo as well as grain bags and separation cloths.

The Contractor is to supply all other cargo handling gear and equipment, such as hooks, pendants, save-alls, nets, trays, bridle chains and slings (except slings for heavy lifts when hoisted by heavy-lift floating or shore derrick), also hand trucks, mechanical trucks or tractors, also dock tractor cranes as needed for efficient stevedoring work.

Insurance. The rates quoted include the cost of, and the Contractor agrees to maintain in full force and effect, Workmen's Compensation Insurance covering the Contractor's statutory and common law liability for injury or death sustained by its employees and Public Liability Insurance covering the Contractor's liability for bodily injury, including death, sustained by third parties, arising out of operations performed hereunder, such Public Liability Insurance to be in the amount of \$50,000 for death or injury of one person and \$150,000 for death or injury of more than one person in a single accident.

The rates specified herein also include Social Security Taxes and Unemployment Insurance as presently payable by the Contractor. Whenever labor wages are to be charged for by the Contractor under this agreement, the Social Security Taxes and Unemployment Insurance incurred thereon shall be added to charges for Workmen's Compensation and Public Liability Insurance, and all such charges shall be termed "Insurance."

Responsibility for Damage or Loss. The Contractor will be legally liable for damage to the ship and its equipment, and for damage to cargo, or loss of cargo overside, through its negligence. When such damage occurs to ship or its equipment, or where loss or damage occurs to

cargo by reason of such negligence, the Ship's Officers or other authorized representatives will call this to the attention of the Contractor at the time of accident. With respect to claims for loss or damage to cargo and/or baggage, the liability of the Contractor shall be limited to physical damage, caused by the negligence of the Contractor and to such claims that result from fraud on the part of employees of the Contractor engaged in the delivery, receiving, and watching of cargo.

Detentions, Waiting, Lay Time. Whenever work is interrupted after starting and detentions of not over 20 minutes' duration occur, the Contractor will make no charge for reimbursement therefor. Should such detention time exceed 20 minutes' duration, the Contractor will charge for the full detention time at cost plus insurance. When men are employed and unable to work through causes beyond the Contractor's control, or when men are to be paid for a minimum working period in accordance with the wage agreement, the cost of such waiting or idle time will be charged for by the Contractor at cost, plus insurance.

Overtime. When overtime hours are worked, the additional wages thereby incurred and paid to all labor and other stevedoring personnel so employed will be charged for by the Contractor at cost, plus insurance.

In the event that, under any Government Order or final determination by a court of competent jurisdiction, labor is required to be paid wages in excess of the wages paid under the Federal Fair Labor Standards Act as presently interpreted throughout this port, such wages plus insurance and social security and unemployment taxes together with any additional amount other than wages for which the Contractor may be legally liable under the Act, shall be reimbursed to the Contractor by the Owners, Agents, or Charterers, at cost.

Increase or Decrease in Wages. All rates specified are based on and subject to the employment of present longshore labor at the wage scale and working conditions existing in the port in the month of _____ 19____ under the International Longshoremen's Association Agreement. In the event of an increase or decrease in such wage scale or change in the present longshore labor or working conditions, the rates specified herein shall, as a consequence, be proportionately increased or decreased.

Rehandling or Shifting of Cargo. The rates specified herein apply to one handling of cargo. When rehandling, resorting, or shifting of cargo is necessary through no fault of the Contractor, the time required for such work will be charged for at cost, plus 10% for overhead and gear, plus insurance.

Damaged Cargo. When handling cargo damaged by fire, water, oil, etc., and where such damage causes distress or obnoxious conditions, or in all cases where the men are called upon to handle cargo under distress conditions, the Contractor's charges are to be based on the cost in accordance with the International Longshoremen's Association Agreement, plus insurance and 20% for overhead, depreciation of gear, and profit, in lieu of the rates specified herein, together with the cost of the gear destroyed and the cost of the equipment for the protection of the men as may be required.

This agreement shall not include the furnishing of terminal facilities by the Contractor to accommodate damaged cargo or in connection with any services for which wages are the basis of charges, nor where a contract of affreightment exists which interferes with the carrying out of this contract.

Condition of Cargo. If the condition of the cargo or packages is other than in customary good order, thereby delaying prompt handling, special arrangements shall be agreed upon in lieu of the rates herein specified.

Acts of God, War, Etc. No liability shall attach to the Contractor if the terms of this agreement cannot be performed due to the Act of God, War, Governments, Fire, Explosion, Civil Commotion, Strikes, or other labor difficulties.

It must be remembered that the stevedore contractor is selling labor, usually reinforced with machines and equipment but often just strong arms and skilled hands. The contract merely mirrors wages, working conditions, and legal requirements imposed upon the employer by union agreements or governmental decree. An indication of this fact is found in the clause on "Increase or Decrease of Wages," in which the contractor binds himself to the contract only so long as the wage scale and working conditions existing on the effective date of the contract continue in force.

There is much more to the stevedoring process than can be accomplished by the men who comprise the longshore gangs that work aboard and alongside the ship. This complementary activity is known as "extra labor," and usually is performed by additional personnel obtained by the contractor for the purpose. The definition of the tasks constituting "extra labor" is set forth in the contract in these words:

Extra Labor Services. When required to supply extra labor, the Contractor will render its charges therefor at cost plus 10% and insurance

for the following described services: (a) handling ship's lines and gangways; (b) cleaning ship's holds; (c) discharging excess dunnage or debris; (d) tiering cargo on pier above man-high upon discharge of vessel or breaking down cargo on pier to man-high upon loading of vessel; (e) loading or discharging ship's stores, material or equipment, mail, baggage, specie, bullion, livestock, animals, live poultry and birds; (f) carpenter or coopering work of any nature; (g) handling and placing flooring or timbers for heavy lifts or for use by carpenters; (h) services of Harbormaster for the berthing and unberthing of lighters; (i) lashing and shoring cargo; (j) bolting and unbolting tank lids; (k) battening down hatches when called upon to do so upon completion of the vessel; (l) rigging and unrigging heavy lift booms and hatch tents; (m) supplying extra labor for any other services when authorized.

Overtime work, when performed, always is compensated at a rate—usually 100 percent on weekdays and 150 percent on holidays—higher than that paid in normal working hours. In setting his commodity rates, the stevedore does not allow for the higher wages of labor in the overtime hours but agrees to perform any overtime work authorized and directed by the shipowner, on the basis of actual wage cost plus taxes and insurance. "Actual wage costs" means the monetary difference between the cost of labor in the straight time period and the compensation paid laborers (plus insurance and taxes directly related to earnings) in the overtime hours. If the contractor, for example, were to determine that his wage costs for a gang, including the foreman, were \$492.99 per hour, the overtime cost would be double that, or \$985.98. The differential would be passed along to the shipowner, without any markup except for the added cost of taxes and insurance.

Taxes are assessed by the government and apply to social security benefits and unemployment compensation, as set forth in the contract. Insurance is designed to provide for workmen's compensation, public liability, and property damage. The insurance premium is a function of the wages paid to the laborers and varies in cost as a result of factors which might contribute to accidents, for example, frequency of claims against the assured, the hazardous nature of certain commodities, and the environment in which the work is to be performed. The percentage figure used by stevedore contractors to estimate the cost of insurance may be as little as 11 percent of the wage cost and as high as 45 percent, depending upon

the location of the port in the world. In some large ports, the rates charged contractors by underwriters may differ as much as ten percentage points, depending upon their loss experience.

These provisions of the contract are reasonable since the stevedore computes all cargo-handling expenses and allows an appropriate margin of profit on the assumption that all wages will be paid at the straight time rate. So long as the differential between straight time and overtime wages and related costs is for the account of the shipowner, it is immaterial to the stevedore when the work is performed. This arrangement permits the shipowner to control the use of overtime.

The commodity charges stipulated in the contract are the result of careful consideration of a number of interrelated matters. Each element in the rate is identified, and a cost for that element, prorated to the freighting unit (e.g., weight ton, measurement ton, or individual unit), is calculated as accurately as possible. There is no prescribed method for accomplishing this task; each contractor follows his own scheme of computation predicated upon the wages paid at any given time.

The contractor's estimated profit must be sufficiently large to make the contract worthwhile, but at the same time not so great that it takes the proposal out of competition. There is no rule for the percentage of profit, which ranges generally between 12 percent and 25 percent of all costs.

Labor is, by all odds, the most expensive single element in the commodity rate. It is, moreover, an element that consists of both wages and those items tied directly to wages. The components of the labor cost are illustrated by the following hypothetical schedule:

Labor cost		
18 men	@ \$14.00 per hour	\$252.00
1 foreman	@ 14.50 per hour	<u>14.50</u>
	Subtotal	266.50
Insurance	@ 45.8% of wages	122.06
Social security	@ 7.5% of wages	19.99
Workmen's compensa- tion, state and federal unemployment taxes	@ 14.0% of wages	<u>37.31</u>
	Subtotal	<u>179.36</u>
Total direct labor costs per hour		\$445.86

Fringe benefits		
Vacation	@ \$2.10/man/hr	39.90
Welfare	@ 1.05/man/hr	19.95
Pension	@ 1.50/man/hr	28.50
Guaranteed income	@ 0.22/man/hr	4.18
Port security	@ 0.05/man/hr	.95
Subtotal		<u>\$ 93.48</u>
Total labor costs per hour		\$539.34

Using this determination of the total hourly wage cost for a gang of longshoremen, the contractor must forecast what his production will be; i.e., how many tons of a given commodity this gang of workmen can handle in one hour. Having established this figure, it is relatively simple to put a price on each of the components of the rate, somewhat in the following manner:

Commodity: Coffee beans, in bags	
Handling rate: 28 tons per hour	
Basic labor cost: \$539.34 per gang hour	
Labor cost (\$539.34 divided by 28 tons)	\$19.2621
Checker's services @ \$28.38, prorated per ton	1.0138
Equipment costs (2 forklift trucks @ \$14 per hour, supporting equipment @ \$8 per hour), prorated per ton	1.2857
Supervision @ 10% of labor and checking costs, prorated per ton	2.1562
Overhead @ 15% of all above prorated costs	3.5577
Profit @ 8% of all above prorated costs	<u>2.1821</u>
Commodity rate, per ton	\$29.4576

From this illustration, it may be seen what factors must be taken into account, and how carefully each commodity must be considered in establishing this rate schedule. In deciding which stevedore shall be awarded the contract, the shipowner devotes especial attention to those rates applying to the commodities transported in largest quantities. Suppose, for example, that the records show that of all cargo carried, four commodities accounted for the major tonnage lifted in the ships: bagged cargo, 32 percent; automobiles, 23 percent; copper slabs, 11 percent; and dry skins, 4 percent. The shipowner's concern for obtaining the lowest possible bid for han-

dling these particular items is understandable. If the remaining 30 percent of the goods handled were distributed among twenty or thirty commodities and none individually accounted for more than 2 percent of the total tonnage, the price of handling these assorted items would be comparatively unimportant in the final selection of a stevedore contractor.

The container revolution has simplified the computations involved in establishing stevedore contract rates. Because goods now can be delivered to the terminal in a large van preloaded and sealed by the shipper, the stevedore is freed of the need to determine the characteristics of the items moving, the customary stowage practices, and the number of tons of each commodity that his laborers can handle in an hour. Instead, the entire computation is directed toward the rapidity with which the available cranes, whether on land or aboard ship, can hoist, transport laterally, and lower containers. Basic wage conditions and overtime provisions are unchanged. Should the stevedore contract to provide the complete container terminal, he would be required to calculate the huge investment in cranes and real estate as well as ship berths. Relatively few stevedore firms have assumed this financial burden, and therefore the contract generally covers only the supply of labor, crane operators, and such supervisory and security personnel as may be dictated for the operation. Consequently, in bidding to service a container terminal, the stevedore's labor costs will outweigh other considerations even more than they do for break-bulk cargoes.

Many containership operators offer their customers the opportunity to send goods to the terminal in break-bulk fashion, and to have them packed into containers for a certain fee per ton of each commodity. In this event, the stevedore will be asked to provide the information on which the carrier may base his charge to the shipper. The determination of the labor cost would involve the same process as was followed in fixing the individual rates for break-bulk operations, and would entail considering some of the elements that affected the break-bulk rates. For instance, it would be necessary to know the rapidity with which the laborers could stuff the commodity into a van, or strip the container, measured in terms of tons per hour per gang, or per man, depending upon how the laborers were employed. The characteristics of the commodity that influence handling (fragility, awkward shape, heavy weight, or similar fac-

tors) would have to be established. Other items of expense would include details like the cost of separating small lots to assure correct delivery to consignees; the disbursements incurred in checking goods into the container; the charges paid to move containers to and from the loading dock; and the investment in a safe and weatherproof loading shed. Overhead, insurance, taxes, and fringe benefits must also be computed as essential elements in the fee per commodity ton.

The sophistication introduced by containers is paralleled by the widespread use of electronic computers which permit much more elaborate calculation of costs than has been illustrated in these pages. The basic principles of identifying every item of expense and prorating it against the unit, whether that unit be a ton, a pallet, or a container, remain unchanged. The major benefits resulting from the use of computers are the significantly greater detail of calculations, the enhanced awareness and appreciation by management of all those elements that contribute to the true cost of terminal operation and service, and the ability to identify immediately those areas to which the attention of management should be directed for greater efficiency and economy.

In a highly competitive industry, partially dominated by factors beyond the control of the contractor, the methodical determination of stevedore costs is of immense importance to both the shipowner and the contractor. The more each party to the contract understands the problems of the other, the more likely it is that the final agreement between the two will be fair and reasonable.

Procurement of Vessel Stores and Supplies

IT IS difficult to overemphasize the importance of proper procurement of stores, spare parts, and supplies for ships. The seaworthiness of a ship depends, in part, upon having on board at the outset of the voyage adequate quantities of replacements, maintenance materials, and consumable stores such as paint, manila or nylon line, and food—especially the last. These items cumulatively are costly and represent a significant portion of vessel operating expense. Close control over both their procurement and expenditure therefore is a major responsibility of management.

Procurement of all ship needs normally is centralized in the purchasing department, the manager of which often is called the *purchasing agent*. He has the same status as the heads of the deck, engine, and commissary departments, and reports to the vice-president, operations, through the operations manager. Along with these supervisors, he participates actively in the establishment of standards of quality and quantity of the stores to be placed aboard ship. Of equivalent magnitude is the obligation to procure all material called for in requisitions in accordance with company policy, at the best available prices. Good management dictates that the heads of the engine, deck, and commissary departments, *not* the purchasing agent, make all decisions as to what shall be deleted from, or added to, requisitions from the ships. Only by strict enforcement of this principle can the procurement policy function effectively and efficiently.

For generations, the chief engineer of a ship has served as her maintenance and repair officer, and all requisitions for spare parts or material used in repairs of any sort are channeled through him to the offices of the cognizant departments ashore. In many cases, the superintendent engineer and the marine superintendent,

drawing upon their own seagoing experience, have followed the unwritten but time-honored policy expressed in the phrase, "If the chief wants it, give it to him." In recent years, however, this uncritical acquiescence has come under attack because of the greatly increased cost of spare and replacement parts and the considerably improved performance records of present-day shipboard equipment. More attention is being given to the history of breakdowns to ascertain whether there is any justification for the fears of seagoing engineers that failure to carry a particular part, or a certain quantity of parts, will result in unsatisfactory ship operation. Shoreside management also has begun to be concerned with the problems associated with the physical possession of a large inventory of equipment. Much of what is placed on ships as original spare parts is not used; one estimate is that, in terms of monetary value, approximately three-quarters of the investment in original spare parts is wasted because the parts never are used.

The formation and enforcement of policy concerning the quantities and locations of spare parts to be purchased and held either aboard ship or in warehouses ashore are functions of the highest echelons of the operating department. The logistical support policy is developed only after considering such matters as the availability of parts from suppliers, the frequency with which mechanical units break down, the advantages of replacing the entire unit compared with substituting new components for old ones, and the cost of maintaining substantial inventories of spares in one or more ports in contrast to acquisition only when an item is needed and for which a premium price may be charged.

Modern steamship management demands that attention be given to every level of cost control and to the development of a system by which the needs of the ships are met efficiently and with appropriate consideration of the impact upon the financial stability of the shipowner. To illustrate this point, a typical (but hypothetical) situation involving the availability and use of replacement parts is offered.

A special survey of the *Bounding Main*, as directed by the classification society and requiring a number of days to complete, was planned originally for September. An unexpected reduction in cargo offerings forced a reduction in the frequency of sailings by the owners, and made it possible to withdraw the *Bounding Main* in

June. The survey was ordered immediately following the release of the ship from her commitments, and only one week's advance notice was given to concerned officials. No procurement of replacement parts had been initiated in June, although the superintendent engineer intended to order a new set of blades for the rotor of the turbine powering the ship's electrical generator. Operations records from the *Bounding Main* indicated that while the blades probably might pass the survey, they were losing efficiency as a result of normal wear. When the turbine casing was lifted during the course of the survey, it was found that the blading was lacy, but still within the minimum standards of the classification society. Everyone agreed that the blading could be kept in service for another year, but at some cost in the efficiency with which steam was utilized. At the prevailing and anticipated price of fuel oil, it was calculated that continuing with the old blades would add about eight thousand dollars to ship operating expense. New blades could be obtained in three days, but only by paying a premium of \$3,500 above the normal price. The decision was to pay the premium, and to install the new blades. At the same time, the vice-president, operations, launched an inquiry to discover why a spare set of these blades had not been purchased and held in the warehouse against the probability that it would be needed by the ship. He also wished to establish whether the frequency of replacement and the lead time required for most economical procurement justified the cost of keeping so expensive an item in the spare parts inventory.

This example demonstrates that whenever policy is made relative to the acquisition and retention of expensive spare parts, it is the function of wise management to have each of the executives concerned—marine and engineer superintendents, repair specialists, accountants, and purchasing agent—contribute from their knowledge and experience. When all analyses are examined, it may be found that the cherished notion that “if the chief wants it, he gets it” is justified in some cases but in others there is no reasonable basis for investing that amount of money in certain parts and becoming responsible for their proper care and custody. For example, the pumps installed in six ships of a company fleet are shown to have performed without a breakdown for ten consecutive years. That same make of pump is to be installed in a new ship. It is deemed unnecessary to acquire a spare pump, either to keep in the shore-

side warehouse or to put aboard the vessel. The compressor on a refrigeration unit, however, has an operating life of about six months before it must be replaced. The chief engineer's request for a replacement to be carried by the ship is granted because it is supported by the facts.

In January 1979, the Maritime Administration (then part of the United States Department of Commerce) revised its specifications concerning the requirements for spare parts on ships built with construction differential subsidy. The shipbuilder was directed to furnish, and to put aboard ship, the quantity of repair parts required to meet the minimum dictates of the regulatory and classification bodies. The shipowner was authorized to select, in addition, those spare parts it wished to have the builder supply as part of the original outfit of the vessel. For the owner's guidance, a monetary limit for such parts was fixed, rather than a stipulated number of items. Every item of equipment was assigned to a category, and each category had a fixed percentage of the equipment's base cost which could be used to pay for spares. The categories, in ascending order of percentages allowed, were:

Limited to one percent (of base cost of equipment):

- Engineer's workshop
- Condensers
- Boilers

Limited to two percent:

- Ventilating and heating equipment
- Hull piping
- Cargo oil system

Limited to three percent:

- Air conditioning machinery
- Main engine
- Feed heaters and other heat exchangers

Limited to four percent:

- Fuel oil service piping
- Steam piping
- Feed, condensate, circulating, and drain piping
- Lubricating oil piping

Limited to five percent:

- Electricity generation and distribution equipment
- Electronics
- Deck machinery

Limited to six percent:

Shafting and propeller(s)

Limited to seven percent:

Saltwater evaporator system

Miscellaneous auxiliaries

Limited to thirteen percent:

Pumps

Limited to fifteen percent:

Instruments and gauges

When a new ship is delivered to her owners by the builder, she has on board a year's supply of spare parts, based partially on the manufacturer's estimate of need. The Maritime Administration conducted a study in 1979 of the procurement of spare parts by a number of United States-flag steamship operators.¹ It found that the manufacturers' estimates of the requirements for spare parts could not be relied upon without any question, because they were predicated upon fragmentary information coming back from ships actually in service. A contributing factor to this unreliability was the latent desire of some manufacturers to sell spare parts. In addition, the study suggested that other manufacturers were reluctant to admit that their products were subject to breakdown after a relatively short time. Ideally, the manufacturer of each item of equipment would be informed every time a repair was made and would be able to compile from such notices an accurate history of operational experience and the need for replacement of components. The impracticality of this ideal is obvious, especially in view of the many occasions when ships are sold, often from one national registry to another, or converted into new types of transportation capability with demands upon shipboard equipment totally different from the original purpose for which it was installed.

At the time they are delivered by their builders, new ships have on board a library of manuals that give complete information on each and every piece of machinery. The purchasing department should assure that all "name-plate data" and appropriate additional information from the manuals are recorded in each ship's file and kept readily available to furnish to suppliers whenever replacements or spare parts are ordered.

It is important to note that the regulatory agencies (like the United States Coast Guard or the British Department of Trade) set

certain minimum standards of spare parts and reserve supplies that must be carried by ships. The classification societies likewise establish requirements for spare parts. These stipulations, however, have little impact upon the actual number of spare parts carried because each steamship operator has established its own levels for these items. This disparity results from the fact that the regulatory and classification agencies are interested primarily in assuring that the ship will be able to make port in the event of an emergency, whereas the owner is concerned that the ship remain in full operating condition when away from the home port.

An indispensable element in company policy concerning the quantity of spare parts to be placed aboard ship is the "allowance list." There is no uniformity in the way this list is developed by individual companies; each one reflects the experience of the owner in a particular type of service. The Maritime Administration study determined that the following considerations, of which the first three are predominant, influence the formation of the list:

1. Engineering experience
2. Fleet and vessel experience for this class of ship
3. Manufacturers' and equipment suppliers' recommendations
4. Shipbuilders' allowance list
5. Usage data, wear, and failure rates
6. Regulatory and classification society requirements
7. Loss of ship time incident to failure of a given component
8. Availability of spare parts (i.e., lead time required for procurement)
9. Cost limitations
10. The trade route on which the ship will be employed
11. Where the ship was built (i.e., foreign or domestic yard, with foreign or domestic components)
12. Types of equipment installed
13. Reliability records provided by manufacturers of shipboard equipment

It is almost axiomatic that spare parts for machinery can be procured most cheaply if they are ordered at the time the machinery is being manufactured and assembled. When these same parts are purchased years later, it may be necessary for the manufacturer

to stop normal production routines in the factory and to tool up expressly to fabricate the required item. In this connection, the Maritime Administration learned, during its study of procurement of spare parts, why replacements cost so much when ordered years after the initial run of manufacturing had been completed.

Most of the manufacturers surveyed stated that they did not cease production of a particular item, but instead they either upgraded that item or provided a satisfactory substitute. One fabricator reported with pride that it continued to put out spare parts for older equipment, some of which was found in ships built thirty years earlier. These statements, however, could not be interpreted as indicating that these individual components were kept constantly in stock and were available immediately, but rather that the company retained the capability to manufacture the product on demand, by interrupting current output. An example will make this clear.

A wholesale supplier purchased a large stock of coupling pins during a three-year period when many ships of similar type were under construction. At the time of acquisition, the price was \$2.50 per pin. Twenty years later, a shipowner ordered six of these pins and was charged \$110 for each one. In explaining the tremendous increase in price, the wholesaler said that the manufacturer had informed him that every minute and every operation involved in researching and producing these six pins were charged against this single order. It was necessary to hunt for the engineering drawings, to identify the material used, to procure the very small quantity of the specialized material at a high price, to set up the turret lathes to make the six pins, and then to reset these lathes for the job from which they had been diverted for the special order.

A less extreme situation exists with those manufacturers who do keep a small reserve of older spare parts in stock. Whereas at the time of original fabrication, the output might have been in terms of five or six hundred units, the restocking order would be for twenty or thirty. Even if there were a sufficiently steady demand to keep the research and procurement procedures current, there still would be the cost of interrupting the production cycle on current orders.

A major concern for those dealing with the logistics of providing spare parts is the fact that some equipment installed in new

ships is of an obsolete design, and spare parts may not be available throughout the operating lifetime of the vessels. The reason for this often is that an improved model of that piece of equipment is under design and test but is not yet ready for production. To meet the builder's time schedule, the obsolescent part is supplied. Production of this older piece is terminated perhaps five years after the ship has been placed in service, with resultant problems for those associated with the working of the ship. It is desirable, but not always practical, to demand from the shipbuilder a warranty that spare parts for all component machinery in the ship will be available for not less than twenty years after delivery from the shipyard. The advantages of obtaining a statement of this kind of logistical support are obvious.

If the shipbuilder cannot provide the assurance of spare parts availability for the life of the vessel, the alternative of buying sufficient spare parts to meet all demands for that period of time poses the twin problems of making a very large investment of funds that will be frozen for upwards of two decades, and of properly caring for and safeguarding this large inventory.

When a ship is purchased secondhand and brought into a different fleet, it is the part of good management to take a thorough inventory not only of spare parts and consumable supplies, but also of all equipment installed in the ship, in order that the purchasing department will have complete name-plate data for future reference. The results of this scrutiny should establish what must be acquired to have the levels of stores, spare parts, and supplies brought into conformity with the new owner's standards. Taking an inventory on this scale is a major task; one operator estimated that it took a team of ten experienced persons one full week to complete this task. Thereafter, ideal ship husbandry would repeat the process annually, but this is very nearly impossible in the circumstances of fast turnaround that surround most ship operations today.

The Maritime Administration requires that inventories be taken when an American-flag ship is placed in subsidized service, when the operating differential subsidy contract expires, when the accounting period ends for determining the amount of subsidy subject to recapture, and when the vessel is withdrawn permanently from subsidized service. These inventories are required for inclu-

sion in reports on the financial operations of the company. They must make reference to the stores levels fixed in the *standards book*, which regulates the expenditures of spare parts. The regulations pertaining to these inventories also stipulate that the condition be reported of any equipment for which spare parts are provided. For instance, a refrigerator compressor is part of the original equipment installed by the builder and is estimated to have a working life of five years. Annually, therefore, the condition of this compressor must be listed in a manner that would permit its elimination from the records when it was worn out and a replacement had been made.

Because of the complexity and detail associated with inventories of spare parts, the availability of a computer and its proper use to keep constant and updated records of procurement, storage, expenditure, and replacement of each item is of major benefit to all concerned. One operator of ships registered in the United States has developed a highly automated, computer-based inventory control system that is extremely simple in its fundamentals. Every spare part is tagged individually, and from those tags the needed information can be supplied to the computer, which then provides the following:

1. An inventory of what each vessel in the fleet has on board;
2. An inventory that combines the spare parts on the ships with the spare parts in the company's warehouse;
3. A report showing the usage (consumption) of parts by vessel and by voyage; and
4. An automated purchase order for procurement of replacements that have been used.

All tags on spare parts are coded for the computer program and show the company's identification number for the part, the exact location of the place where the part is stored, and the name of the ship(s) on which the part is carried. Spares kept in the warehouse ashore are tagged in the same manner, except that the name of the ship is left blank. The warehouse itself is divided into five sections, one for each of the four ships in the fleet, and one for general storage. Spares purchased while a ship is on a voyage are placed in the designated area to await the return of that vessel to the home

port. Storage shelves are indexed to the ship's boxes in which those spare parts will be stowed when they are turned over to the custody of the chief engineer.

Whenever spare parts are needed by the ship, the chief engineer removes and retains the tags. At designated ports of call, he gives the local agent of the line a list of the spare parts that have been issued from his stock. This list is sent to the home port by the agent. At the end of the voyage, the chief engineer personally delivers the tags to the supply office. In this way, a careful and up-to-date inventory is maintained, and resupply material is available in ample time for the next scheduled voyage.

Aboard ship, spare parts, as already indicated, are stored in appropriately marked boxes, which then are sealed until there is a requirement for whatever is in the box. This greatly facilitates control of inventory; it also makes it comparatively easy to verify the supply of spares in the custody of the chief engineer. Boxes that have been unsealed are inspected and their contents, if any, counted. The quantity used should correspond to the number of tags retained by the chief engineer.

Once all the data have been assembled and the computer has processed the material, the superintendent engineer is in a position to decide how to bring the inventory up to standard. The process may involve inter-ship transfer, shifting from warehouse to ship, or direct purchase from a supplier. If either of the first two steps is taken, the tags are marked accordingly, and the computer record is updated immediately. If procurement is ordered, the computer is programmed to prepare the requisite purchase documents.

At the beginning of each outbound voyage, the ship's chief engineer is furnished with a report showing the status of the spare parts aboard his ship, including the exact location of each part and each box. This report replaces the one used on the preceding voyage, and eliminates any possibility of error from omission or transcription.

A system of this sort has significant advantages from the viewpoint of management. In the first place, there is a continuous record of shipboard requirements for each and every spare part. The superintendent engineer is able to order replacement parts for the shipboard stock on the basis of the experience of each ship. Thus, if an air compressor runs 1,400 hours during a voyage, and

requires a new drive belt after 350 hours of operation, the minimum support level can be set accurately at four belts. At the same time, if a new brand of belt is tested in service, the relative merit may be determined on the basis of actual performance, which is duly recorded and is available for later study. Another benefit is that supply personnel can keep an accurate list of the lead time required for procurement of replacements. Additionally, because there is continuous control over the parts, noticeable economies can be effected in the logistical support of the ships.

The basis for all procurement, as well as for long-range planning for acquisition, storage, and control of both spare parts and consumable stores, is the requisition submitted by the chief engineer. If the vessel is assigned to a long voyage—two or more months between calls at the home port—it is customary to process requisitions so that everything needed for the forthcoming trip will be placed aboard ship well in advance of sailing time. If, however, the ship returns to her home port every two weeks, good management would suggest that the ship submit one requisition every third or fourth voyage, and while the ship is on the fourth or fifth trip, the appropriate examination of and adjustments to the requisition may be accomplished and procurement completed.

A fairly standardized pattern of procurement is followed by steamship operators. The cognizant department analyzes the ship's requisition, makes changes as required by company policy or records of experience, and then passes the requisition to the purchasing department. Replacement of "proprietary" items, meaning parts unique to a particular manufacturer, is accomplished by ordering the desired quantity from that source. Material that, like tools, valve packing, and interchangeable parts, can be acquired from a number of suppliers, is procured only after bids from several (a term generally interpreted to mean three or more) sources have been solicited and compared. In each case, the goal is to meet the needs of the ship most efficiently, promptly, and economically.

A problem of significant dimension is encountered in the procurement and warehousing of what may be called "insurance items." These are items that are expensive, usually require long lead time for delivery, and are needed only rarely. Typical of this category are propellers, tail shafts, turbine rotors, major propul-

sion-unit spares, reduction gears, electric motors, turbogenerator sets, pumps, and valves. Where two or more steamship operators have the same port and their ships use the same equipment, it is quite possible that a sort of informal "pool" may be set up. If a ship suddenly needs a replacement tailshaft and the owner has expended the one in his inventory and has not yet received the replacement ordered from the manufacturer, the other shipowner(s) will be asked to "lend" a shaft until the replacement is delivered. The advantage of this arrangement is that these expensive and rarely required items may be stocked as single, rather than multiple, units in a warehouse. If the fleet owner attempted to keep a complete complement of these spare parts for his ships in his own warehouse, it might represent an investment of approximately two million dollars for each large containership. For smaller and less complex vessels, the cost of a complete collection of spare parts could be close to a half-million dollars per ship. The importance of keeping the number of insurance items as low as possible is self-evident when the amount of money invested in them is taken into account.

A noteworthy facet of spare parts procurement was cited in the Maritime Administration's study of the subject. This is the procurement of insurance items on what might be termed a speculative basis. Often a part, and sometimes a complete unit, will be offered for sale by a manufacturer at a bargain price. For example, imagine that a shipowner signed a contract to build two ships and had an option to build two more. The shipyard, at its own risk, constructed three sets of reduction gears, on the chance that at least one of the two ships under option would be ordered. Because of adverse business conditions, however, the shipowner waived the options, leaving the shipyard with one complete set of reduction gears for which it had no prospective need. Rather than store this surplus item in its warehouse for an indefinite time, the shipyard offered it for sale at approximately two-thirds of its normal cost. On learning of this action, an operator of ships equipped with these gears immediately purchased the set, because his records showed that one in ten ship-years a replacement was required. The combination of the lower acquisition price and the advantage of having the item on hand to meet a predictable demand made it worthwhile for the shipowner to assume the burden of possession.

It is not always necessary that the part required by a ship be new. Many pieces of equipment outlast the ships in which they are installed, and when those vessels are dismantled the used parts dealers acquire these items, and offer them for sale at prices which invariably include some premium markup. Shipowners who need this equipment, however, often find that the dealer's price results in a cost which overall is lower than if the material had been bought new and warehoused for a period of time. Additionally, as ships get older and the original suppliers of machinery components either cease production or go out of business altogether, the only source of replacement parts may be the secondhand dealers.

A notable consideration in procurement is whether to insist upon the use of "genuine" replacement parts—that is, parts manufactured by the original maker of the major component. These parts, as might be expected, are precisely machined to the dimensions and tolerances fixed by the designer and specified in the engineering drawings. They make perfect replacements, but they also may be more expensive than "non-genuine" parts made by an independent. In most cases, non-genuine parts are copied slavishly from genuine originals, and will perform satisfactorily in the equipment for which intended. Because they are copies, rather than made from the engineering drawings, there always is the possibility that some clearances or cuts may be slightly different from the genuine part, and could cause operational difficulties. Some operators, however, believe that the difference in cost more than offsets the possible damage that might result from a rare instance of poor machining. Ideally, good management would insist upon the use of genuine parts, because the buyer gets the correct design, proper material, and expert fabricating procedures. In addition, by using genuine parts, the buyer deals with a source that maintains engineering and design departments, a quality assurance group, and an inspection team, all of which combine to give to the buyer exactly what has been ordered.²

Some steamship owners have elected to follow what might be designated as the "zero-based spare parts policy." Rather than maintaining fixed levels of spare parts, these companies keep on hand, both afloat and ashore, as little reserve material as experience justifies. They make a determined effort to anticipate the need for

replacement parts and to procure these early enough to forestall breakdowns or delays to ships necessitated by waiting for the required items. Perforce, procurement officers must keep track of the availability of equipment, both old and new, to meet unexpected failures. This often involves the practicality of substituting equipment similar to, but not identical with, that which is installed in their ships. Additionally, the purchasing department must be alert to the possibility of using parts that can be modified to fit the needs of a particular repair job. It is obvious that this zero-based policy can be effective only if there is a sustained review of requirements from the ships and a systematic maintenance and repair program.

An essential element in purchasing department routine is the establishment and continuous updating of an active and accurate list of spare parts, both on board ship and in company storage places, supplemented by a current list of manufacturers and distributors who have various types of equipment readily available. Purveyors of surplus and used machinery are indispensable sources of both replacement and spare parts for those items no longer in the manufacturers' stocks. Great reliance is placed upon distributors of equipment who have been authorized by the manufacturers to handle their products. The advantage of dealing with distributors is that items obtained from them meet specifications, both as to fabrication and material. Although not necessarily typical, the practice of one fleet operator's procurement division indicates, by the percentage of resupply acquired from each source, the relative importance of each:

Authorized distributors	50%
Original equipment manufacturer	20%
Wholesalers	15%
Jobbers	10%
Custom manufacturers	5%

In liner service, where the same voyages are repeated again and again on closely monitored and rigidly followed schedules, experience and established rates of consumption make preliminary planning for purchasing relatively simple. The principles to be observed in these circumstances are:

I. Length of the proposed voyage. The quantity of consumable stores and spare parts to be placed on board is regulated by the length of time the ship will be away from the home port.

II. Route and ports of call of the proposed voyage. Because of the critical situation prevailing in many nations in regard to international balances of payments, the tendency in many ship operating companies is to purchase as much as possible in the country in which the operator is domiciled and of which it is a citizen. Vessels engaged in the cross-trades, which do not return to their home ports, are stored in the areas where the most satisfactory arrangements can be made after quality, variety, quantity, and dependability have been evaluated. Ships operated under subsidy contract with the Maritime Administration are required to obtain all stores in the United States, except for perishable foodstuffs needed by ships on protracted voyages. The ports where supplies of these commodities (especially meat, milk, vegetables, and fruit) are available should be made known to the purchasing agent in time for him to establish the procurement program for the voyage. Passenger ships assigned to long cruises (60 to 100 days) usually are scheduled to take on fresh fruits and vegetables at a number of ports of call; meats, cheese, wines, and staples like flour normally are stored for the entire cruise at the port of origin. It is mandatory, regardless of whether the ship be a passenger or a cargo carrier, that the quality of supplies obtained along the route be acceptable. Many American owners have found it advantageous to obtain the greater proportion of their needs in United States cities rather than to expect foreign sources to meet their standards. One operator, for instance, discovered that canned fruit juices met with greater favor among his seafaring employees than did frozen concentrates, because the potable water available along the route varied too greatly in flavor.

III. The number and nationalities of crew and passengers. In procurement of food supplies, it is necessary that the purchasing department be well informed as to the number of persons, both passenger and crew members, to be fed, and insofar as possible their nationalities and racial backgrounds. Dietary customs and restrictions can become very significant in personnel management, and consideration must be given to these facts. The following examples represent typical cases:

A ship with her home port in New York and manned by persons from that part of the United States will have to provide a substantial quantity of potatoes. A ship sailing from New Orleans, and carrying a crew from Gulf communities, must offer rice in preference to potatoes.

A German-flag vessel must cater to her people's appetites by providing such specialties as sauerkraut, knockwurst, and rye bread; a vessel with an Italian crew would require quantities of spaghetti, macaroni, and similar products as well as supplies of tomatoes and seasonings.

Chinese and Indian seamen traditionally are fed diets in which rice and fish play a prominent part.

Passenger ships carrying large numbers of Jewish people must not only offer kosher food but must be able to assure the travelers that Jewish dietary laws are observed in the preparation of food.

Incidental to planning commissary offerings is the time of the year in which the voyage is to take place. If the ship is to run from northern to equatorial latitudes, it would be desirable for an American ship to stock abundant salad-making materials, fruits, and the lighter foods. On the other hand, if the ship is to proceed to extremely cold areas, the heavier, more warming foods will be in demand. Should the voyage take the ship through various seasons, then procurement of food must be designed to meet the changing tastes and demands of the personnel aboard.

From time to time, operational delays will be encountered by a vessel employed in liner service. To make up for the days lost during the voyage, and to sail the ship on the advertised departure date, frequently the turnaround time in the home port is reduced drastically. If the decision is to meet the scheduled departure at the expense of time in port, those responsible should notify the ship of the curtailment of turnaround time, and should instruct the master to send estimates of required stores and spare parts by airmail or radio, as circumstances dictate. The alternative is for the purchasing agent, in consultation with the superintendents of the engineer, deck, and catering departments, to prepare tentative requisitions, and to use these as the bases for soliciting bids from suppliers. The successful bidders are notified of their selection, but exact quantities to be delivered are not made known until the ship arrives and details are cleared up. When this is accomplished, the purveyors

receive final instructions by telephone, and deliveries are effected in accordance therewith.

As already noted in discussing procurement of spare parts, it is the part of good management for requisitions to be compared with the standards established for the ship to be supplied, and to make whatever adjustments may be appropriate to the anticipated employment of the vessel. In the liner services, which entail close supervision of ship movements, the actual quantity of goods purchased for each voyage becomes almost a routine, and normally should be the difference between what is on board and what is needed to bring the stores level up to company standards.

The goal of good stores procurement is to provide everything for the ship that reasonably may be expected to be needed and used, without stockpiling large inventories either afloat or ashore. Items deemed necessary for the efficient operation of the ship and the personal comfort of the ship's company should be supplied in adequate but not excessive quantity. To put too much of any consumable item on board a ship is to promote wasteful extravagance while working capital is tied up unprofitably.

Spare parts for shipboard machinery, maintenance supplies like paint and housekeeping equipment, and other nonperishable items sometimes are ordered periodically in sufficiently large quantities to supply the entire fleet. While this may result in discounts because of the wholesale lots in which procurement is made, it is important that the investment in these large inventories be compared with alleged savings to assure that working capital is being used efficiently. Perishable foodstuffs must be purchased with especial care to eliminate, insofar as possible, the losses from spoilage resulting from long periods of storage.

Items on requisitions are divided by the purchasing department into the appropriate categories (fruits, vegetables, meats, hardware, cleaning gear, paints, etc.) and interested suppliers are invited to submit proposals. Ordinarily the lowest responsive bid consistent with good quality and economy over the long run will win the business for a purveyor of a single category, even though prices of some items within the category may be a trifle higher than those of a competitor. Quality is under constant scrutiny, and whenever a particular brand is found to be uneven or otherwise undependable, it is dropped from the list of approved products. Dealers—whether

manufacturers, agents, or wholesalers—are held to exact compliance with specifications, and failure to meet the terms of the contract as to brand and quality may be judged sufficient reason for cancellation.

A major exception to the practice of purchasing by categories exists in the procurement of meats, especially for passenger ships. The cost of meat represents a substantial proportion of total food expense, and quality is of critical importance. To assure minimum cost and highest quality for a particular grade of meat, proposals from suppliers are analyzed to determine which merchant offers the lowest price, consistent with the quality stipulated, for individual cuts such as rib roasts and filets mignons.

To encourage competition both in price and in service, some purchasing agents follow the practice of dividing food purchases to give the lowest bidder about 85 percent of the total, and to award the remaining 15 percent to the second lowest bidder.

Purchases sometimes are made from large-scale shippers whose products can be used by the fleet. This “reciprocal buying” is a decided asset to the traffic department in maintaining the goodwill of these customers, who frequently pay more in freight charges than the value of the purchases made by the procurement office. For example, a carrier earns \$5,000 a year in freight revenues from handling the cargo of a varnish manufacturer, and buys in return only \$750 worth of that company’s product. So long as the varnishes are of good and consistent quality, and the prices are competitive with those of other offerers, the fact that the supplier is a valuable customer may tip the scale in favor of that firm. Occasionally there may be a slight premium paid for the merchandise procured from these large-scale shippers, but in most cases the tender made by the shipper is at the market price. In any event, the quality and long-term economy of the merchandise must be equivalent to that available from other bidders.

Procurement of stores and supplies, other than spare parts, is from manufacturers, authorized distributors, jobbers, and ship chandlers. Generally, purchasing agents of the larger steamship companies attempt to deal directly with the manufacturers of those items used in large quantities. Canvas, paints, oils, and greases are typical of such items. Should the manufacturer set his prices at the same level as those quoted by the jobber (a wholesale merchant who

buys in very large lots and sells in quantities to retailers and heavy consumers), it is simpler to order from the jobber who has offices in the home port. In addition to the convenience of location, dealing with a jobber facilitates purchases in smaller quantities than the manufacturer might require. Another, less visible and therefore less measurable, advantage of dealing with the jobber is that business contacts are kept at the local level, with resultant goodwill toward the shipowner.

Ship chandlers are called upon to supply the miscellaneous small lot items needed by a ship. Large steamship companies do not buy paint, pipe, canvas, rope, or meat through the chandlers. They do place orders with these merchants, however, for those articles required in lots of one, six, or ten units, and thus simplify their own bookkeeping, receiving, and inspection procedures. Some small shipowning companies do not maintain purchasing departments, and use the services of the chandlers in lieu thereof. An alternative is to engage a "purchasing house" to perform the functions of the supply section. In either case, the burden remains with the shipowner to provide very explicit specifications and instructions because there will be no personal acquaintance with the ships or their personnel, and no access to owners' records.

Ships, especially those engaged in the tramping trade, often are supplied through the offices of the agents who represent the owners in the various ports of call.³ Customarily, these people are engaged in the routines connected with the arrival, working, and departure of the ships, and are not necessarily experienced in supply procedures. Should the ship require stores or supplies of any sort, the various department heads (chief engineer, chief officer, chief steward) should consult with the agent to assure that the requisitions are understood and that the exact nature of whatever is to be procured is explained fully. This is of particular importance where the requisition must be translated into a different language. Exceptionally careful inspection of all goods so procured must be made at the time of delivery, and shipboard personnel should be thoroughly instructed by the owners prior to the commencement of the voyage as to the procedures to be followed, especially if some deficiency or shortage is discovered. Because ship chandlers are in the business of filling ship requisitions, it may prove to be more satisfactory for ships engaged in tramp operations to deal directly

with some reputable chandlery than to attempt to work through the agent.

Ship chandlers, especially in the larger ports, offer an extensive range of goods, usually in the form of a carefully prepared catalog issued once a year. Changes in prices often are made by means of a separate sheet distributed periodically. An alternative is for the chandler to submit bids on the basis of the catalog price minus a discount or plus a surcharge. If no catalog is supplied by the chandler, offerings may be made to shipowners by referring to the manufacturer's catalog and quoting a fixed price or the manufacturer's price minus a discount.

Purchases for vessels employed in the long trades usually are made in large quantities and often in containers of the largest size provided by the manufacturer or processor. It is poor economy, however, to acquire everything in these very big containers, because wastage is almost inevitable. This is true in food supplies; for example, rice bought in one-pound bags really is less costly than rice bought in one-hundred-pound sacks, because the cook will use one, two, or five pounds for a meal, will not spill any, and will not tend to overestimate liberally what is the designated quantity. Although the unit price of rice in one-pound packages may be somewhat higher than if the sack is purchased, the saving from the reduced wastage more than offsets the greater cost.

To establish an efficient supply program for a fleet of ships requires attention to many details. It is essential, as has been noted previously, that the purchasing department be furnished by the requisitioning department with complete information (for example, name-plate data for machinery and equipment) concerning anything that is to be purchased. To eliminate mistakes in ordering, one carrier devised a system of preprinted cards for every item on the ship's allowance lists. The cards had three parts, each of which contained the complete name-plate data, manufacturer's information, and engineering drawing numbers, together with any other pertinent data. Whenever an item was withdrawn from the storeroom or taken off the shelf, the card was annotated to show the date and reason for its use. The chief engineer retained one part of the card when the item was put into service, and at the first opportunity mailed the other two parts to the home office. The purchasing department was able to order replacements with the assurance that

it was acquiring exactly what was needed, and retained one of the two parts of the card. The other part was sent to the data processing unit to be worked into the computerized record system.

Preparing a procedure of this sort is a major undertaking. If the ship is acquired from the shipyard, full details accompany the vessel when she is delivered, and should be made available to the purchasing and data processing departments as soon as possible. If the ship is purchased from another operator and does not have full documentation aboard when title is transferred, a stem-to-stern and keel-to-truck inventory may be needed to establish the basis for logistical support. Ships taken on bareboat charter should be inspected in the same way that an old ship is examined, so that appropriate records can be set up, ideally before the commencement of the first voyage.

The hull plan and vessel layout must be studied with care, since procurement of supplies will be affected by structural features of the ship. In particular, the purchasing department must determine the location, size, accessibility, and interrelation of storerooms of all types (paint and other deck stores, engine room spares, commissary dry stores, refrigerated foodstuffs, and hotel equipment). It must also ascertain what facilities have been built into the ship which simplify or complicate the handling of stores. This information will be useful in selecting sizes and weights of packages. For example, if the boatswain's storeroom is located near a hatchway through which delivery can be made with the ship's cargo-hoisting gear, it may be feasible to order manila or nylon line on large spools. If the storeroom is accessible only through a narrow doorway, and everything must be man-carried down a ladder, the obvious limitation on spool sizes is to that which can be handled by one or two men.

A steamship line engaged in transporting large numbers of passengers, whether on cruises or overnight ferry runs involving feeding, must procure, among other items, large stores of linens, china, glassware, and silverware house-marked according to company practice. This means that orders for these articles must be placed months in advance because of the time required to weave or mark the material. Sheets and towels can be obtained in one or two months, whereas china and glassware will not be delivered for a half year or longer. Stocks of these long-lead items usually are kept in shoreside storerooms under the supervision of the purchasing

agent, and ship requisitions are filled immediately. This method of replenishing what has been lost or broken during the voyage is both convenient and efficient; at the same time, it justifies procurement in very large quantities with the resultant economy of scale.

It is worth noting that just as hotels have found it to be almost an invitation to souvenir hunters to carry off house-marked items, so ships have encountered the same problem. A predictable result has been the abandonment of the time-honored practice of having the house-flag appear on every item used by the travelers. Insofar as the purchasing agent is concerned, this has all but eliminated the waiting time for these items.

Cargo liners with a capacity of twelve or fewer passengers usually are supplied with standardized table services; linens in some instances may be marked with the house-flag. Depending upon the frequency of sailings, the length of the voyages, and the policy of the owners, procurement may be in lots large enough to supply the entire fleet, or it may be on a ship-by-ship basis. If ordinary restaurant and hotel supplies are used by the vessels, and no problems are encountered in obtaining the items requisitioned by the ship, it is convenient to purchase whatever is needed for each vessel as she returns to her home port.

Perishable foodstuffs (meats, fruits, vegetables) are obtained on contracts which, in New York, are let each week. The suppliers submit their bids on Tuesday to cover the week ending with the close of business the following Monday. The lowest overall bidder, as previously explained, normally is awarded the contract for the week. As requisitions are received from ships during that seven-day period, the successful bidder is notified of the items and quantities desired, and these stores are delivered in accordance with pertinent instructions.

The quality of food offered to passengers, especially on luxury cruises, is of major concern to the operator. Competition in the cruise trade is so intense that any deviation from the highest standards may have a disastrous effect on the prestige of the operator. As a result, some companies keep foodstuff buyers in the market constantly, seeking the finest quality of fruits, vegetables, and meats. Meats purchased in the United States are graded by the Department of Agriculture and inspected by the buyer to assure that cuts and weights correspond to specifications and invoices.



Passengers' tastes are reflected in this cruise ship's buffet table. Photo: Cunard Line.

Passengers normally are served those meats that are graded "prime," while those provided to crew members are graded "good." If inspections are made at the purveyor's warehouse, the delivery trucks may be loaded and sealed in the presence of the shipowner's representative, to eliminate a final inspection when purchases are taken aboard ship. An incidental benefit from this warehouse inspection is that the longshoremen who handle the foodstuffs are not held up by the need to have each item examined for quality and quantity.

Ships registered in the United States are subject to the terms of the agreements between shipowners and the maritime labor unions. Among other provisions of these negotiated arrangements is the stipulation that the meat proffered to crew members shall be equal to the Department of Agriculture standard for the classification of "good." Those cargo ships that carry twelve passengers are permitted by those same union agreements to serve only one menu for all persons embarked. This simplifies the procurement, inspection, and delivery process to a notable degree.

Deliveries to the ship of commissary and other stores and supplies are made pursuant to schedules set up and made known to purveyors by the purchasing department. In many ports of the United States, as already indicated, stores are placed aboard ship by longshoremen, and therefore delivery schedules must be coordinated with cargo operations to cause as little interruption as possible in the handling of cargo. In terms of the working day, this means that the terminal manager should be consulted prior to instructing the various suppliers as to the time they should have their goods at the ship's side. The shipboard department heads also must be included in the planning and scheduling of deliveries, so that storerooms and verifying officers may be ready to receive the orders. It has been found to be good practice to have the supplier furnish his truck driver with manifests in quadruplicate. The original is signed by the responsible ship's officer when the count, quality, and condition of the goods have been determined to be satisfactory in all respects. The truckman returns this original to the supplier as evidence that the delivery has been made and accepted; eventually, it becomes the supporting document for the bill sent to the steamship company's disbursing office. The duplicate copy goes to the ship, to provide an exact record of what has been placed

in the storerooms. The triplicate is retained by the purchasing department and filed with the other papers relating to procurement for that voyage, and the quadruplicate is routed to the accounting office for verification that all charges are in accordance with the contracts and are consistent with that which has been delivered. This system assures that everyone concerned is informed immediately of any deviation from the requisition as submitted by the ship and the quantity ultimately stacked in the storeroom.

Although complete inventory of a vessel's stores at the end of each voyage is ideal, it has been found impractical when turn-around time has been reduced to a matter of hours. To offset the possibility that elimination of the inventory may result in waste and extravagance, various expedients have been devised by senior management. Among these "checks and balances" are the following:

1. Engine room stores, particularly spare parts for individual machines, are inventoried, boxed, and sealed. Until the seal is broken, or there is reason to suspect that the contents have been tampered with, it is not necessary to do more than note that the box is in the proper place. As parts are used, the required replacement is obtained and put into the box, which is resealed and the updated packing list is displayed conspicuously. In this manner, unreported expenditure of spare parts is prevented.

2. Inventories may be made by ship's personnel during the voyage and mailed to the home office from an intermediate port of call. When this is done, stores-control supervisors determine whether the expenditure of stores is consistent with their anticipations. If usage exceeds expectations, and is appropriately justified, procurement standards can be adjusted to meet the facts. At the same time, preliminary plans for "spot inspections" and partial inventories may be formulated.

3. Spot inspections of vessel stores are made at irregular and unannounced times, but always in the presence of the responsible officer.

4. Requisitions are compared with vessel standards, and any unexplained deviations become the subject of investigations and inquiry by the appropriate department head. Where the standards are found to have been based on unrealistic assumptions, they should be adjusted to conform to actual experience.

5. File folders are prepared for each ship, and into them are inserted all requisitions, purchase orders, approved and signed invoices, and delivery receipts. These records show the actual expenditures for each class of supplies, and the cost of procurement of all items authorized by the appropriate managers. These records furnish the basis for comparisons between standards and expenditure of supplies, and also for planning future procurement.

Operating differential subsidy contracts between the United States Maritime Administration and operators of ships of that nation's registry prohibit purchases of stores or supplies in foreign countries except on long voyages that require replenishment of fresh stores. Emergency procurement of parts or stores is accomplished as the need arises, but must be explained fully by memoranda from all ship's personnel concerned. In any case, subsidized operators must maintain careful records of every purchase made outside the United States both to satisfy the Maritime Administration and to justify applications for exemptions from customs duties.

Some carriers do not procure stores abroad because quality is variable, there are health hazards, and the supply is not dependable. If meat is bought outside the United States, it must be sealed in refrigerated lockers before entering the territorial waters of the United States, or be inspected by the United States Public Health Service to ascertain that it is free of disease.

Some large steamship operating companies have concluded that because several of their ships may happen to be in the home port at the same time, or their turnaround time is very short, or they arrive at odd hours, it is neither convenient nor practical to rely upon the normal routine of procurement from merchants and manufacturers. To meet the demands of the ships without delaying their movements, these companies have established warehouses and buy in wholesale quantities. Requisitions from the ships are filled from stocks maintained in the warehouse. The warehouse, in turn, follows normal procedures for resupply.

The warehouse must be distinguished from the storeroom maintained by each division of the operating department. The storeroom contains, in the case of the marine superintendent, charts and other aids to navigation, and a few of the most critical pieces of navigational equipment. Similarly, the superintendent

engineer will have in his storeroom a small supply of those parts that experience has shown are most likely to be needed as soon as a ship returns to the home port. The commissary division's storeroom, especially for cargo liner companies, will contain sufficient glassware, crockery, and linen to meet the immediate demands of a ship.

There are certain definite advantages that accrue from operating a warehouse; the principal ones are:

1. Ship requisitions may be filled in accordance with vessel schedules, regardless of the business hours of the purveyors of goods.

2. The exact item desired or needed may be furnished the ship without the delay associated with normal purchasing routine.

3. By procurement in large quantities, greater discounts may be obtained from the suppliers, with resultant enhanced savings.

4. Articles difficult to acquire, or in short supply on the market, may be stockpiled against future requisitions.

5. Seasonal purchases may be made at considerable savings, and occasionally greater savings may be effected by the ability to take advantage of fluctuations in market prices.

6. Wholesale procurement of linens, china, glass, and silverware, marked or designed to company specifications, often results in lower costs of those items.

7. Savings from wholesale purchases may offset all or a major portion of the cost of operating the warehouse.

8. A "pedigree card" showing the history of each major part kept in warehouse stock may be maintained. Items like tailshafts, turbine rotors, generators, gears, and propellers would be recorded on these cards to show the source of procurement, name of manufacturer, original certificate or forging number, the ships on which the component (if rebuilt or refurbished) had been installed, and the date when that component was removed from the last ship, together with the name of the activity that performed the repair or refurbishment. Attached to the pedigree card are the applicable certificates from the classification society. This is especially valuable if the certificate shows any restriction for the installation or use of the item. The advantages of having this kind of information to pass along to those concerned aboard ship are self-evident.

There are some disadvantages to the warehouse system of vessel supply, of which the following are significant:

1. A complete warehousing operation must be established, with the accompanying problems of inventory, records, and security of goods.

2. Qualified warehouse operatives must be added to the company payroll.

3. The investment in inventory is high. Depending upon the size of the fleet supported, the capital required for a warehousing program will range from a million dollars to fifteen or twenty million dollars.

4. The warehouse must be bought or rented (or, alternatively, space in a commercial warehouse must be leased from its operator), and some mechanism must be devised to send the goods requisitioned to the ship. Few warehouses are located right on the waterfront.

5. Warehouse stocks, in some of the fifty United States, are subject to taxation either by the state or the community in which the warehouse is sited. The impact of this potential burden must be measured carefully before a decision concerning the establishment of a warehouse can be reached.

One containership operator devised a modification of the warehouse system to fit the peculiar needs of his ships and their short turnaround schedules. Several forty-foot containers were fitted as storerooms, and supplies and spare parts were packed on the shelves and placed in the bins and lockers as they were received from the purveyors. Complete control over every phase of the procurement process was achieved. When it was time for the ship to sail, the loaded container-storerooms were hoisted aboard and placed where they would be convenient for all concerned to have access to them. There were obvious savings in the actual process of transferring stores from shore to ship; what was more significant was the complete storing of the ship which was accomplished by this development.

Whether procurement practices be based upon using a ship Chandler, a purchasing house, or a division of the operating department of the steamship company, the shipowner carries a heavy

burden of responsibility. He must assure that, at the outset of a voyage, the ship is adequately supplied with everything that will be needed while she is at sea. It is, as has been brought out in these pages, a matter of good management that careful control be exercised over all facets of the procurement program. It also is a matter of assuring that the lives of the mariners who take the ship across the trackless oceans are protected by having at hand the material that experience and foresight indicate will be required while the vessel is "off soundings."

Containerization: The Beginning

APRIL 26, 1956, was a rainy, cool day in Port Newark, New Jersey, not unlike many other spring days in the New York area. The departure for Houston, Texas, of the partially converted World War II vintage T-2 tanker *Ideal X* was very much of a routine affair, except for one thing. That one thing was destined to change deep-sea shipping practice around the world, but like other strokes of commercial genius it was not recognized at the time as being of any real significance.

As Malcom P. McLean waved farewell to his ship, he could not have been aware that he was initiating a revolution in shipping which would have as great an impact upon that business as did the coming of steam. For what McLean was watching was the first movement of cargo in which carriage by truck and ship was purposefully combined to form an integrated transportation system. As in other demonstrations of commercial genius, the achievement of McLean lay not in the mechanics of the operation but in the wholly new approach to relating diverse modes of transportation to each other and to the shipper. What had been a series of disconnected haulings by rail, highway, and waterborne equipment was converted into a coordinated and mutually supporting procedure which made the most efficient and economical use of the potentials and capabilities of the different modes.

Very briefly described, what McLean had done was to modify the standard 35-foot-long highway trailer used by trucking firms in the United States to permit the cargo-containing box (which almost immediately came to be called simply the "container") to be separated from the chassis. He then placed these trailers at the inland warehouses of shippers who filled them with their goods and sealed them in the same way that railroad boxcars were sealed. A trucker was summoned to haul the loaded trailer directly to the side of the

oceangoing ship, where a crane lifted the container from the chassis and deposited it on the especially constructed "spar deck" of the tanker. At the port of discharge, the process was reversed, and the trailer was delivered to its consignee in an inland city. Most important to the whole scheme was the fact that a single contract of affreightment covered the entire movement from the shipper's loading dock to the consignee's warehouse.

The "container revolution," as the McLean conception of an integrated transportation system was denominated, originated in the mind of a man who had spent over twenty years in the intercity trucking business, building up his enterprise until his big trucks linked many cities in coastal states between New York and Florida.¹ He was well aware that a large portion of what was hauled in his vehicles came from shippers who filled trailers with goods consigned to single addressees. Systematic analysis of this one-shipper-to-one-consignee business revealed that there was a substantial proportion moving between those cities along the Atlantic and Gulf coasts which had good facilities for handling oceangoing ships. Furthermore, it was learned that a considerable quantity of other cargo came to or went from points some distance inland from these port cities. It was clear, therefore, that if the means were developed to carry the trucks by ship along the coasts, and simultaneously to simplify the paperwork involved in a multimedia system of transportation, at least some of the complexities which bedeviled the interstate movement of merchandise by truck might be alleviated.

As a possible answer to the dilemma, McLean remembered earlier proposals to put trailers aboard ships built expressly for the purpose. The ships themselves resembled in many details a single-deck garage. Once loaded, the relatively high-speed ships were to proceed to major coastwise ports where truck-tractors would be waiting to haul the trailers to their destinations. Preliminary considerations of these schemes intrigued the veteran truckman, who directed his staff to make detailed studies, including careful financial analyses, of the proposals. After determining accurately the characteristics of the flow of cargo—the main cities of origin and destination, the tonnages, the commodities carried, the frequency with which lots of cargo moved, the length of time needed to effect delivery, and the freight revenue earned—it became evident that a ship running between New York and Charleston, South Carolina,

would serve the route very well. The consulting naval architect was commissioned to draw preliminary plans for the "floating garage" ship and to ascertain its performance capabilities. As the conception was studied in greater depth, it seemed desirable to consider the economic practicality of using the alternative ports of Providence, Rhode Island, in the north, and Wilmington, North Carolina, in the south. To meet the demands of the anticipated trade, it was determined that four of these floating garages would be required.²

Concurrently, the economic analyses were disclosing some startling facts which dampened the enthusiasm of the champions of the garage ships. Although the trailers could be rolled on and off the vessels in a very short time, and port turnaround therefore was at the optimum, too much of the available space in the ship was absorbed by the nonrevenue-producing chassis of the trailers. Only the goods *inside* the container were charged for transportation. A standard 35-foot highway trailer might carry forty-one tons of cargo measuring forty cubic feet to the weight ton, but mounted on the chassis the space occupied was equivalent to about ninety-four measurement tons. Identifying the problem suggested the solution. The containers must be separated from the chassis and loaded aboard ship in no more time than the roll-on process required. Additionally, the number of container-carrying spaces in the ship had to be great enough to earn both operating expenses and a reasonable profit.

McLean redirected the energies of his staff to devise the means to accomplish the goal envisioned by the economic analysis. The first move was to initiate technical discussions with the trailer manufacturer, who was asked to create a new type of highway trailer. This new unit was to consist of a flat-bottomed box with vertical strength to withstand stacking, longitudinally reinforced so that it could be lifted by a crane when fully loaded, and capable of being separated quickly from or connected to the chassis. The next step was to put aside the proposal for a garage ship and to concentrate upon the design of a ship to accommodate the especially constructed container.

It became obvious at this point that a full-scale test had to be conducted to determine the reaction of the shipping public to the innovation. This meant acquiring ships to transport the containers.

Because they would be operated in the coastwise trade, certificates of public convenience and necessity from the Interstate Commerce Commission were essential. To design and build container-carrying ships was a multiyear project, entailing a very large investment, while the process of obtaining original certificates from the commission was a tedious and protracted undertaking. With what proved to be characteristic boldness, McLean solved the problem by purchasing the seven-ship fleet of the Pan-Atlantic Steamship Company, a subsidiary of the Waterman Steamship Corporation, of Mobile, Alabama, which held valid certificates for operation in both the coastwise and the intercoastal trades.³

The conventional break-bulk cargo carriers obtained in his bold move were not adapted to McLean's new idea. Rather than delaying the test until the ships could be converted, four standard T-2 tankers built during World War II were bought from their operators, and spar decks were installed above their main decks. The containers were secured to the spar deck for the ocean voyage between New Jersey and Texas. McLean and his technical staff thus capitalized upon a wartime expedient devised to transport fighter aircraft to the European theatre of combat. A steel skeleton deck was erected above the tanker's main deck, and the airplanes were lashed to this structure. The only difference between the two conceptions was that the McLean spar deck was solid, with sockets inserted at appropriate locations to accept the legs of the containers, and to lock the boxes to the deck by means of devices built into the socket-leg combination. The payload of each ship's "suit" of fifty-eight containers was about 1,160 tons.⁴ Because the revenue from this tonnage was insufficient to pay all costs incident to initiating the new service, McLean planned to carry full loads of petroleum products from Texas to New Jersey to augment vessel earnings.⁵ The United States Coast Guard negated this scheme, citing regulations prohibiting the mingling of dry cargo and bulk petroleum. Undeterred by this setback, McLean filled the cargo tanks with ballast water, and carried out his plans while accepting the losses as unavoidable "start-up" expenses.

The two potential obstacles to development of this idea having been removed by the acquisition of the certificates of convenience and necessity and by the purchase and modification of the four tankers, McLean was free to devote his energies to working out the

details of moving the containers to and from shippers and consignees. It was simple enough to get the container to the ship for loading; it was more of a problem to assure that there would be a chassis and a truck-tractor awaiting each container when it was lifted off the ship by the dockside crane. By a series of contracts with truckers, these details were brought completely under McLean's control. At last, he was able to solicit business on the basis of a single bill of lading which applied to the whole movement from the point of origin to the final destination of the goods regardless of the number or type of participating carriers.

To Malcom McLean, therefore, must go full credit for conceiving and developing the integrated transportation system. An examiner on the staff of the Interstate Commerce Commission included this comment in his recommended decision submitted to the full Commission on November 27, 1956:

Malcom McLean is pioneering in the integration of sea-land transportation and in the application of the latest technological developments. A man of vision, determination, and considerable executive talent, he is making a valuable contribution.⁶

McLean placed a chassis-mounted container at the inland warehouse of the shipper, where that individual loaded (or "stuffed," as the term came to be) his goods into the container and sealed its doors shut. A truck-tractor then hauled the trailer directly to the ship's side, where the box holding the cargo was separated from its chassis and hoisted aboard the vessel by a dockside crane. At destination, another crane lifted the container off the ship and lowered it to a waiting chassis, following which it was towed to the consignee's warehouse. The consignee broke the seal, emptied (or "stripped") the container, closed its doors, and released the trailer to the truckman. A single bill of lading covered the entire movement.

Anticipating at this point the developments which came about some years later, it is appropriate to observe that, in time, a more complex interchange was arranged. The container was stuffed and sealed by the shipper who directed the truckman to haul the trailer to the railroad yard. There the container was transferred to a flatcar on which it moved from the inland city to the seaboard. When the flatcar reached the seaport, the container was shifted to a

chassis and towed by a truck-tractor to the ship's side. In this manner, most efficient use was made of three distinct methods of transportation, without costly and time-consuming rehandling of the goods. This combination eventually was designated as "inter-modal transportation."

As often has been the case when a totally new procedure is adopted, those casual observers who happened to be present at the moment when the action occurred saw nothing of overwhelming interest. The magnitude of what was transpiring on that cloudy, cool April day lay not so much in the visible scene but rather in the uncounted hours of thinking, experimenting, and establishing the most effective means by which to accomplish the objective.

So it was that, on April 26, 1956, the sailing of the modified tanker *Ideal X* with fifty-eight loaded containers stowed on her recently installed spar deck, seemed very little out of the ordinary. The waterfront watchers saw only the unusual feature of a slotted deck erected above the tanker's main deck, with its maze of pipes and valves, to which were secured the special boxes that had been placed on the spar deck by shoreside cranes. The enormous importance of the procedure could not be detected because much of it was represented by what *did not* happen to the cargo stowed in the boxes.

Traditionally, the over-the-road trailers would have been unloaded in the transit shed of the terminal, and the separate packages would then have been moved to a consolidation point before being taken to the side of the ship to be hoisted aboard and then stowed in the appointed cargo space. McLean's procedures bypassed this time-honored routine by bringing the container directly under the crane and hoisting the entire box, unopened, aboard ship. In five minutes, some forty thousand pounds of revenue cargo would have been shifted from a highway vehicle to a seagoing vessel. To the farseeing observer, the implications were tremendous. Quite literally, a new era in transportation of goods by ship had dawned.

Fully to appreciate the loftiness of this stroke of commercial genius, it is necessary to sketch, in an admittedly highly oversimplified manner, the whole procedure involved in what was known technically as "break-bulk carriage," and what later was called "conventional" handling of seaborne goods. For convenience, a single

crate of furniture will be followed as it is moved from Grand Rapids, Michigan, to the inland city of Tours, France, about one hundred kilometers from Paris. The same pattern of activity would be followed for a thousand packages, regardless of their contents, so long as each package could be identified.

The shipper crated the furniture with great care to assure its safe delivery despite the numerous handlings that lay ahead. The first movement was by truck from the warehouse to the railroad yard in Grand Rapids, where the crate was transferred to a boxcar for the trip to New York. On arrival at that point, the crate was removed from the boxcar and placed in a truck to be carried to the marine terminal. The truck was unloaded at the steamship operator's transit shed, and the crate was shifted to a "place of rest" where it waited until time to be loaded in the ship. At the designated moment, the crate was picked up by a forklift truck and brought to the ship's side. It was hoisted aboard the vessel and lowered into the hold. Longshoremen set it securely in its predesignated place where it remained until the unloading process began at the port of Le Havre. What transpired until the crate was delivered to the consignee in Tours was the reciprocal of what has occurred between Grand Rapids and New York. A minimum of twelve handlings was needed from the moment the crate was placed in the shipper's truck until it was taken into the consignee's warehouse. It is unnecessary here to expand upon the expenditure of time and money, as well as the possible damage to the crate and its contents, which these multiple handlings entailed.

Certain steps required by the shipowner for his own protection against unwarranted claims, but which did not involve additional handling, have been omitted from the foregoing skeletonized account. These steps, however, took the time and effort of well-paid employees, and consisted of examining the crate for damage at the time it was received, and recording what, if anything, was discovered then and at the time it was placed at the ship's side, when it came off the ship, and when it was turned over to the consignee's truckman. Individually, this series of examinations was justifiable and represented only a small amount of paperwork, but cumulatively, for a ship being loaded with ten or fifteen thousand tons of heterogeneous cargo, it escalated into a major aspect of terminal management.

McLean, in a flash of genius, saw a means of simplifying the whole process. When the shipper loaded the container and then sealed it, he filled out a bill of lading and thereby certified to the carriers concerned that certain items, and only those items, were within the box. So long as the seals were not broken, McLean reasoned, the shipper's declarations were binding, and no verification was needed. All documentation required by the steamship operator therefore could be in terms of the big box, considered as a single package. Instead of four inspections of each individual package, there would be one when the container was received at the marine terminal and one when it was released to the consignee's truckman. Labor, money, and, above all, precious ship's time would be saved.

What distinguished McLean's thinking from previous efforts to improve the procedures followed in break-bulk shipping was this appreciation of the whole process as a single distribution system in which every part was interrelated and had a definite function. Another American shipowner had come close to this idea but never brought it to the full fruition which occurred under McLean's guidance.

On January 12, 1929, Graham M. Brush placed in service a seagoing railroad-car carrier which he appropriately named *Seatrain*.⁷ The ship was devised by Brush, while the technical details of design and engineering were worked out by William Spofford, vice-president and chief engineer of Brush's organization, known initially as Over-Seas Railways, Inc., and later renamed Seatrain Lines. What made the *Seatrain* unusual, if not unique, was the manner in which her hull was arranged to accommodate the railroad cars. One very large hold, extending from the forward collision bulkhead to the engine-room bulkhead near the stern, had three decks, each fitted with four sets of parallel railroad track. The hold and the lower deck each could accept twenty-six cars; the upper deck had space for thirty cars, while the superstructure deck, because of its lighter supporting elements, was restricted to a maximum of thirteen empty cars. Altogether, the ship could carry from ninety to ninety-five cars on a trip.

An essential part of the Seatrain operation was the ingenious crane which Brush, a civil engineer by training and background, had designed. The cables of the crane were secured to a massive

steel "cradle" which served as the means of lifting the railroad car. When the cradle had been raised to the level of the superstructure deck, it was moved athwartships to the designated "slot" and lowered to the appropriate deck. Here the car was connected to a wire rope linked to a steam winch and was pulled in this manner from the cradle to the place where it was chocked securely for the voyage.

Brush set up his steamship as an adjunct to the railroads. His thinking was set forth in an advertising booklet issued by the company in 1932, and appropriately entitled, *Out to Sea on Rails*. Seatrain Lines sought out those shippers who normally would use railroads, and offered a financial saving if the car were routed so that a portion of the total transportation was performed by the ship. This was a very successful procedure until competition from McLean began to assert itself. Seatrain, by this time no longer controlled by Brush, attempted to adapt to the new methods, but was too conservative. Unable to survive in the new climate of competition, the company sold its fleet and name to a group of investors who followed McLean's lead and carried all cargo in containers.

In an earlier chapter, the complex and expensive operation of a break-bulk marine terminal has been described. The great expense of this activity always was of major concern to senior executives, terminal managers, stevedore superintendents, and ships' officers, and over the years many suggestions were offered to improve the traditional practices. There was, for instance, the logical proposal that cargo piled on pallet-boards should be strapped to permit each loaded pallet to be placed aboard ship as a single unit. Out of this idea developed the thought of placing certain types of cargo, especially fragile items, in large boxes which would be transported to the overseas destination without rehandling the contents.

Immediately following World War II, a Great Lakes businessman named Leathem D. Smith devised a steel container approximately eight feet square and almost eight feet high. Smith was concerned with the persistent problem of containers: the return voyage, usually without any cargo, involved substantial expense to the owner of the container. To reduce this back-haul cost, Smith's containers were intended to be folded so they would measure only eight feet square by approximately three feet high. Unfortunately, the rough handling given to the boxes made the folding feature unworkable, and the idea was dropped.

Along the waterfront, meantime, gradual improvements in the techniques of cargo handling were being tried. Traditionally, packaged goods were received at the terminal by longshoremen who loaded the packages on hand trucks and trundled them to the place of rest to await moving to the ship's side. The actual process of hoisting cargo from dockside platform to ship's deck was no different from that used by the Phoenicians, except that steam or electric power had been substituted for human muscles. The individual packages were stowed by hand into the appropriate spaces in the ship. The whole operation was infinitely laborious and consumed much time for which the ship was not paid: it is axiomatic that a ship earns money only when at sea actually moving cargo from one point to another.

About 1935, the forklift truck was introduced without fanfare to the marine terminal. Its use required that packages be piled systematically on pallets so that the truck could fit its forks under it and lift it before transporting the load to any designated point. So successful was this development that the hand truck disappeared from the waterfront within a few years, and a whole new technique of materials handling was adopted. It was the fulfillment of a prediction when a pallet load of packages destined to the same port was strapped with steel bands and loaded aboard ship, eliminating the tedious process of individually stowing every box.

The use of containers in the movement of goods did not originate with McLean. As far back as 1911, several commercial enterprises engaged in the packing, crating, and shipping of household goods began to stow these valuable items in heavy steel boxes for intercity movement by rail. Overseas shipments of household goods in these steel boxes occasionally was accomplished, but in each case it was a matter of a series of disconnected movements arranged for and supervised by the shipper. Insofar as the carriers were concerned, the container was nothing more than an oversized and sometimes excessively heavy unit that was handled as any other package would be. The number of these shipments was small and cannot be considered as anything more than a minor part of conventional break-bulk carriage.

Following the end of World War II, the idea of stowing certain commodities in containers for transportation by ship began to spread rather widely. Great Britain, Denmark, Belgium, the Neth-

erlands, Germany, and France were the western European nations where the practice received greatest attention; concurrently, there was some increase in the use of containers in the United States. The boxes varied in dimensions according to the notions and experience of the carriers who owned the equipment and made it available to shippers. Generally, the containers were small, ranging in capacity from about five to fifteen cubic yards (four to ten cubic meters). Shippers who had sufficient goods to fill a container sometimes were given the option of having the container sent to their premises for loading, or delivering their cargo to the marine terminal where the carrier's personnel stuffed the container. By 1949, it was not unusual for a ship to sail with the square of one or more hatches filled with these steel boxes. An immediate advantage to shippers was that these containers were the last packages to be placed aboard ship, and the first to be taken off, which thereby shortened total transit time for the goods.

In the United States, the Alaska Steamship Company and the Bull Line, operating respectively out of Pacific northwest and Atlantic northeast ports to noncontiguous United States possessions, were among the leading advocates of containers to reduce ship turnaround time. Alaska Steamship used metal boxes in 1949, added what it called "collapsible cargo cribs" in 1952, and began carrying loaded trailers aboard its seagoing ships in 1953. The "cribs" were developed from the standard wooden pallet and consisted of a base with lattice sides and a plywood top; capacity was 1.68 cubic meters. Cribs were made available, on request, to shippers who could pack and strap them. If they were received by the terminal in this condition, they were delivered to the consignee in that same condition.

The Bull Line commenced its experiment with containers about the same time as Alaska Steamship. Because shippers to Caribbean destinations used packaging for their goods that was too fragile to withstand the rough handling involved in movement by ship, the carrier was interested in methods to reduce the inevitable damage. Large steel boxes in which the fragile cartons could be stowed appeared to be one solution. In time, the company developed a box of 19 cubic meters capacity. Somewhat later, a larger unit was devised that was 4.6 meters long, 2.4 meters wide, and approximately 2.4 meters in height. These dimensions were suggested by

the need to fit two boxes on a standard flatbed truck-trailer, which was 9.75 meters long. These containers, which Bull Line called "vans," were stowed in the holds of conventional break-bulk ships. Their handling below decks was facilitated by casters permanently fitted at the corners, and also by rugged frames which permitted lifting by forklift trucks.⁸

While it is undeniable that there were steamship operators who placed cargo in large containers, as described above, it cannot be deduced from this fact that the container revolution antedated the first sailing of Malcom McLean's *Ideal X* in 1956. The practices noted in the preceding pages were the results of efforts to improve the techniques of marine terminal operations and thereby to reduce the time required to discharge and reload a conventional break-bulk carrier. Concurrent with this objective was the desire to bring about better outturn of cargo in order to cut down on costly claims by shippers for lost and damaged goods. The containers used by shipowners prior to McLean's day rarely, if ever, left the terminal premises. Shippers apparently preferred to send their offerings to the terminal where, at the discretion of the superintendent, they might be stowed in a container or they might be put aboard ship as individual packages. The essential component in the structure of the container revolution was McLean's introduction of the integrated transportation system. It was the flowering of this conception which made the sailing of the *Ideal X* on April 26, 1956, of lasting importance in the development of marine transportation.

A fully integrated distribution system, in theory, puts under one management all the participating modes of transportation and handles the business under a single contract of carriage. This makes possible the most efficient employment of each component of the system, and produces the most economical operation. A practical, but not insuperable, problem is to assemble in one corporate structure the diverse talents required to manage the different types of transportation. The day-to-day practices of railroading are not necessarily similar to those of long-distance trucking, and the supervision of inland waterways tugs and barges varies in many details from that of oceangoing vessels. The senior executive must possess an unusual background of experience to provide rational and evenhanded administration, including allocating financial resources among the carriers. In the United States, existing law

regulating interstate commerce forbids the proprietor of one mode of transportation to own a competing method. Until recently, no railroad, for instance, was permitted to control an intercity trucking enterprise. A coastwise steamship company may not own a railroad that parallels its route and competes for the same business. The ideal of a single corporate unit controlling and using most efficiently the various kinds of transportation available in the United States therefore remains more of a dream than a reality.

McLean saw the potential of his conception of the shipping industry, and in September 1955 divested himself of the trucking enterprise. Thereby he freed himself to devote his energy and his vision to the development of the new system of transportation. History has shown that his decision to channel all his interests toward maritime matters was correct, for his new method of sending goods between seaports spread throughout the world and influenced, in the most profound manner imaginable, the carriage of merchandise by sea.

Shippers responded with enthusiasm to the new development, and provided the empirical proof that was required if further growth and evolution were to occur. To be profitable, however, at least two hundred loaded containers had to be transported on each trip. This was beyond the capabilities of the spar deck tankers. New construction was deemed unwise at this stage of the fledgling steamship company's existence, and therefore conversion of ships was the only alternative.

Anticipating this turn of events, McLean completed negotiations on May 5, 1955, to acquire the remaining assets of the Waterman Steamship Corporation for the sum of \$41,567,000. Among the properties purchased were thirty standard C-2 cargo ships built between 1942 and 1945.⁹ Only ten of these vessels were retained; all the other Waterman holdings were sold within the next four months.¹⁰

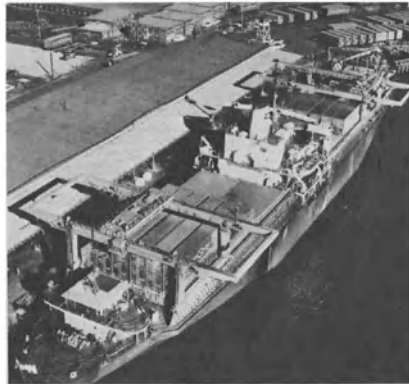
George G. Sharp, an outstanding American naval architect, planned the conversion of the ships acquired from Waterman. At a cost of approximately three and a half million dollars each, the ships underwent a major rebuilding. All interior bulkheads and decks were removed, and weather deck hatch openings were enlarged to a uniform length to accommodate the 35-foot-long containers. The cargo spaces were fitted with an elaborate steel frame-

work to form cells into which the containers were lowered and stacked one atop another to a depth of five units. Two additional tiers of containers could be carried on deck. By installing a 30-ton gantry crane forward of the midships house, and another abaft this structure, the ships were made self-sustaining, and could be sent to any port which had a dockside apron wide enough to permit a truck-tractor to maneuver a trailer on which to place the container. The cranes traveled back and forth on tracks laid on sponsons which were built on each side of the hull for a total additional width of nine feet. A diesel driven generator provided electric power for each crane, which had arms that, in port, projected about twenty feet over the side of the ship and were folded down to touch the hull when the vessel was ready to sail.¹¹

When the ships were placed in service, initially in the coastwise run from New York to Houston, Texas, they astounded observers by the rapidity with which a full cargo of inbound containers could be removed and replaced by a complete set of outbound boxes. In 1958, a newspaper report noted that only fourteen hours of ship time were needed for this procedure, and altogether forty-two longshoremen were hired, compared with about 126 for a break-bulk vessel.¹² Admittedly, the ship operator was able to economize on cargo-handlers' wages, but the more important saving was in ship time. It is axiomatic that the hours a ship spends in port working cargo are not revenue-producers. By reducing cargo handling time from the eighty-four hours needed for break-bulk operation to the fourteen used to handle the containers, the number of ships required to maintain a specified frequency of service on a given route was reduced significantly.

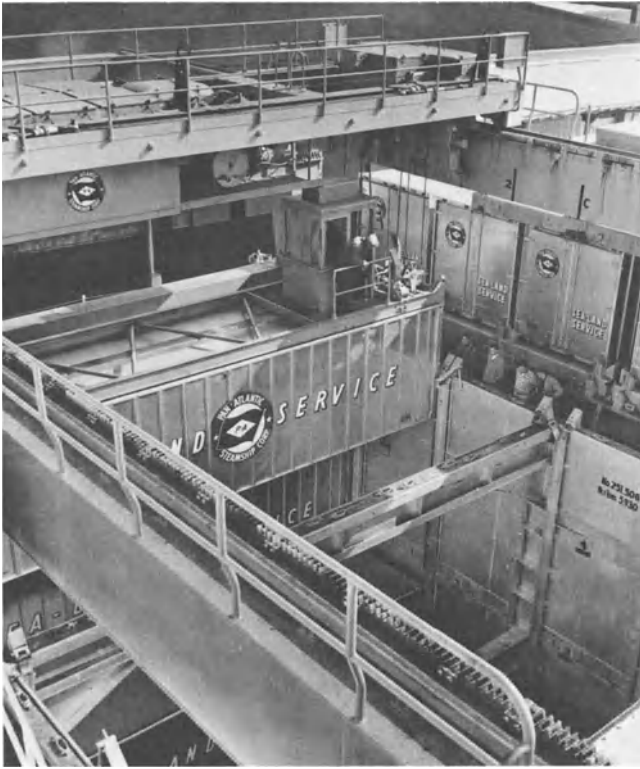
The following comparison between the operation schedule of the *Gateway City*, the first McLean ship to be converted to carry containers exclusively, with a conventional break-bulk ship of the same basic design and an identical power plant, will demonstrate how the saving in stevedoring time benefitted the shipowner.

The *Gateway City* was assigned to sail between Port Newark and Houston, a distance of 1,928 miles. At her normal cruising speed of 14.5 knots, the one-way passage required 133 hours, without allowing for possible delays and the inevitable slow transit through inner harbor channels. In Houston, she used 14 hours to discharge and reload, and then spent another 133 hours steaming back to Port



Top left: A container goes aboard the *Ideal X*. Note the “feet” to fit into securing sockets. *Top right:* Spar deck of the *Ideal X*. *Bottom left:* Containers are secured to the spar deck. Note sockets for securing containers’ “feet.” *Bottom right:* The *Gateway City* was the first ship converted to carry containers in cellularized spaces. Her gantry cranes made her independent of port facilities. Photos: Port Authority of New York and New Jersey.

Newark. Port time in the New Jersey terminal was 14 hours, bringing the total time for the round trip (from departure on Voyage No. 1 to departure on Voyage No. 2) to 294 hours. To assure against breakdowns, it was established that the *Gateway City* would be withdrawn for maintenance purposes at the end of each twenty-five weeks of operation. During the fifty weeks of the operating



A container is lowered to its cell in the *Gateway City*. Photo: Port Authority of New York and New Jersey.

year, the ship was able to complete 28.56 round voyages. If every one of the 226 containers carried a full load of twenty revenue tons, ship earnings would be based on 4,520 tons per one-way passage.

The *Fair Isle*, a conventional break-bulk ship comparable in all respects to the *Gateway City*, was operated on the same route. Her sea time was identical with that of the containership, but discharging and reloading required 84 hours in Houston and a similar period in Port Newark. Total voyage time came to 434 hours. In fifty weeks, the *Fair Isle* completed 19.35 voyages. Her earnings also were based on carrying 4,520 tons per one-way passage.

Translated into fleet size, the calculations reveal that two containerships had the theoretical capability of maintaining sailings every 6.14 days through the 351 days of their operating year. The longer turnaround time for the break-bulk ships meant that three

vessels would be required to make possible a sailing every 6.05 days. Because no more revenue would be earned by the three break-bulk ships than by the two container ships, it is obvious that the reduction in total voyage time was of major financial importance.

(It must be emphasized that the above calculations are entirely theoretical and do not reflect actual ship operations. The purpose of the computation is to show the disparity between the performance potential of the two types of ships and to suggest the relative earning power of the vessels.)

With experience, the McLean management team learned that equipping each ship with gantry cranes, while it made possible calls at ports which lacked the facilities needed to handle containers, resulted in more drawbacks than advantages. The cranes were idle while the ship was at sea; they were very heavy and therefore reduced the tonnage of cargo which could be carried; they added greatly to the cost of ship building; their presence aboard ship restricted the load of containers to two tiers on deck; and the maintenance of the cranes was an expensive aspect of ship operating cost. Dockside cranes, on the other hand, could be used continuously, serving one ship after another. They also provided a much wider outreach than that of ships' cranes and thus increased the efficiency with which containers were handled alongside the ship. Although the land-based cranes were purchased and installed by the operators of the ports (usually public bodies like the Port Authority of New York and New Jersey or the Board of Commissioners of the Port of New Orleans), their great cost—the first ones were acquired for one million dollars each, and later versions were priced two or three times higher—was reflected in the rentals charged to ships for the berthing spaces they occupied. These increased charges, however, were less than the investment required to provide cranes aboard every ship. All of McLean's later ships, both conversions and new construction, were designed to depend upon dockside cranes. Worldwide, containership operators have followed this precedent almost without exception.¹³

As an inducement to shippers to use the new method of transportation of goods by sea, McLean made the specially designed containers available to his customers. This expensive expedient was necessitated by the fact that the standard over-the-road trailer used by American trucking fleets did not have a detachable box, nor was

it built with the reinforced corners required for stacking the boxes aboard ship. McLean vessels, perforce, loaded and carried only McLean-owned containers. In this way it came about that initially the containership operators assumed full responsibility for procurement, maintenance, and control of containers, and only those boxes that belonged to, or were leased by, a particular steamship company were transported by that company. The financial burden imposed upon the shipowner may be gauged by the fact that when McLean sold his interest in the shipping company in 1969, the inventory listed 27,000 containers which cost about \$3,500 each—an investment conservatively estimated at nearly one hundred million dollars.

It was not long after the maiden voyage of the *Gateway City* that other operators began cautiously to explore the new method. This attitude was based in part upon consideration of the capital investment required, changes in terminal operating practices and the resultant conflict with the longshoremen's unions over reduced employment opportunities, and possible legal problems connected with extending the carrier's liabilities beyond the marine terminal. It also reflected the conservatism characteristic of much of the shipowning fraternity which accepted change only after proof that what was proposed was both practical and necessary.

In fairness to operations in the international overseas trades, it must be conceded that conditions encountered on these routes were very different from those prevailing in the coastwise service between two United States ports. If the overseas area were well developed with internal highway and railroad networks, and the ports were equipped to handle the big boxes (which might weigh twenty-five or more tons), it was feasible to adopt the new mode of integrated transportation. If, however, the overseas nation were still in the process of internal development, it would be a major error in judgment to convert to containers. Some operators were quick to perceive the long-range benefits of an integrated system and ventured into the new era by modifying one or more hatchways to accommodate containers, while retaining the capability to handle large amounts of break-bulk cargo.

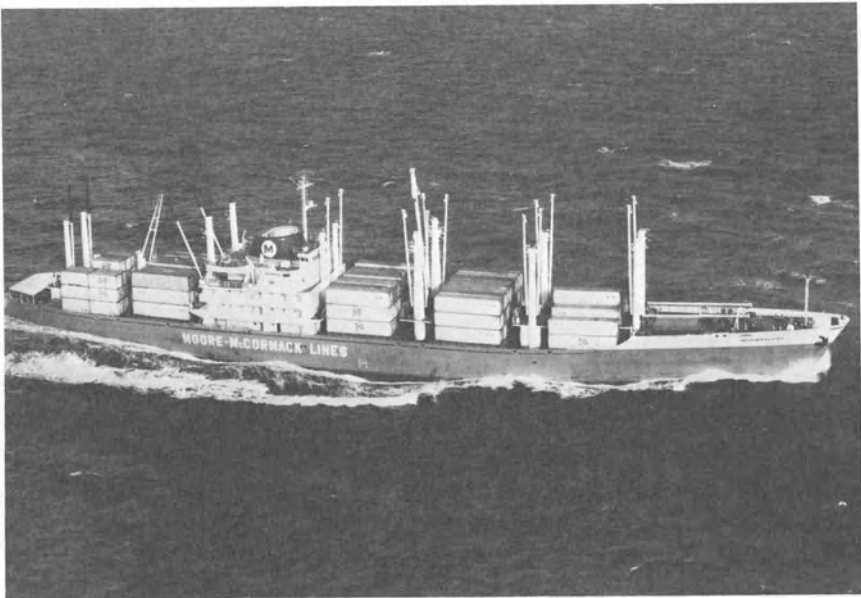
To understand the attitudes toward the novel system, it is appropriate to examine some aspects of the economics of containerization.

First, but not necessarily most importantly, the process of stowing merchandise in the container was a logical step on the part of carriers in their never-ending search for more efficient and more economical ways to handle cargo. The increasing number of uniform-sized packages offered for shipment made the use of the pallet feasible, and the introduction of the forklift truck made it possible to move the loaded pallets economically and conveniently from place of rest in the transit shed to the ship's side.

It took some years for the next step to be adopted rather widely, but the practicality of the notion eventually won adherents. Lots of uniform-sized packages destined to either the same consignee or the same port were assembled, under the watchful eyes of a cargo checker, on a pallet, and then secured with steel straps so that thereafter the whole collection could be handled as a single unit. This had several advantages: (a) it was more difficult to pilfer the individual package; (b) checking on the condition of individual cartons was reduced; (c) stowage in and breakout from the ship's hold were more expeditious and efficient; and (d) the goods themselves were less susceptible to breakage and damage. From the use of the steel-banded palletboard to the all-weather container was a simple and almost predictable step.

Second, containerization transferred the task of stuffing and stripping the container from the carrier to the shipper and consignee, and thereby reduced significantly the shipowner's cost to check the condition of cargo when received and delivered.

Handling merchandise in the marine terminal, prior to placing it aboard ship and after it was removed from the vessel, represented a major portion of terminal expense. By transferring this charge to the shipper and consignee of the goods, the ocean freight rate could be restated to reflect more accurately the actual cost of port-to-port transportation. Additional benefits in which all parties shared were the allocation to the shipper of responsibility for proper stowage of the goods in the container, a marked reduction in the total transit time from warehouse to warehouse, and the elimination of numerous rehandlings of the merchandise. When the container was turned over to the ocean carrier, all that was required was to inspect the exterior for evidence of possible damage and to establish that the original seal attached by the shipper was intact.



The 12,750-tons deadweight break-bulk ship *Mormaclynx* was adapted to carry containers. Her cargo gear made her independent of port facilities.
Photo: Moore-McCormack Lines.

Third, containerization saved time by having cargo brought to the ship's side in boxes that could be loaded directly into the vessel. The benefits to the shipowner of the shortened stay in port already have been delineated. To the shipper, the reduction of the time needed to deliver the goods to the ultimate consignee meant receiving payment with less delay.

Fourth, marine terminals were made much more productive and efficient when they were completely mechanized and operated around the clock whenever a ship was at the berth. This shortened the ship's turnaround time and thereby effected savings for the ship operator. Relocating terminals to areas convenient to major highways and railroad lines helped to eliminate delays in delivery and pickup of containers, with resultant economies to the shippers.

The operation of a modern, well-outfitted container terminal requires fewer longshoremen, checkers, and coopers, but remains a major item of shipowners' costs because of the large number of

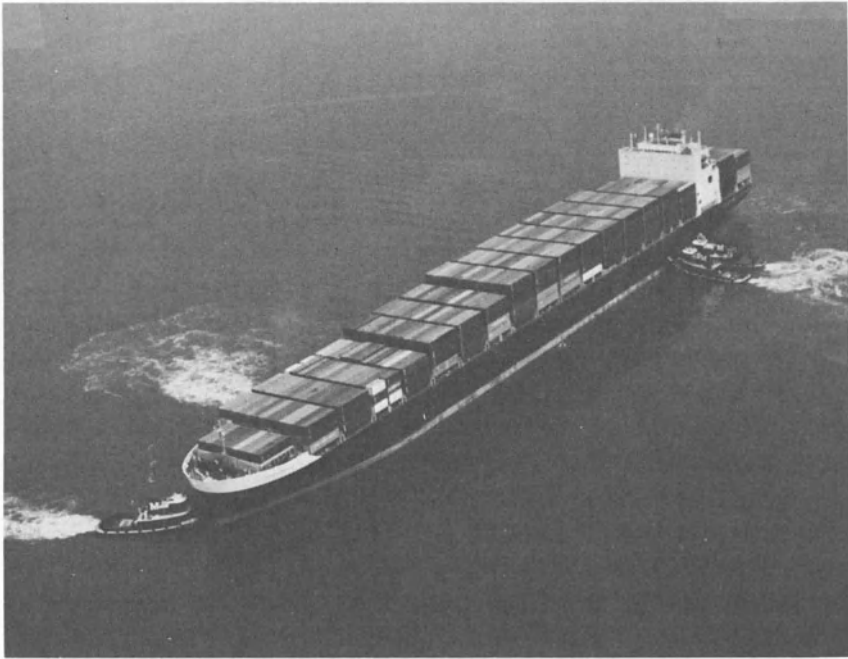


A partial view of the enormous area of Port Elizabeth's containerized cargo terminal. Photo: Port Authority of New York and New Jersey.

expensive machines needed to handle the containers. Straddle trucks, very high capacity forklift trucks, mobile cranes, and similar equipment are indispensable. Their operators, being especially skilled, command premium wages.

Finally, the containerized shipment of goods has resulted in better outturn of cargo, with a consequent reduction in claims for loss and damage. There also has been less pilferage of cargo since containers have come into general use.

Up to this point, little attention has been devoted in these pages to the question of the sustained sea speed of containerships. One reason for this seeming neglect of an important subject is that the velocity of vessels is of less importance to both shippers and consignees than reducing the length of time the cargo has to be in the custody of the shipowner. When the *Gateway City*, which had a maximum speed of fourteen and a half knots, made her first voyage as a carrier of containerized cargo exclusively, she traveled at ap-



United States Lines' *American New York* had eleven sisters. These 950-foot long ships were built in Korea, and each had a capacity of over 4,000 containers. Photo: Moran Towing Co.

proximately the same speed as did all other cargo ships. What gave McLean a major advantage over his competitors was that his method reduced the time required to load the cargo aboard, remove it from the ship, and make it available to the consignee. It was as though he had raised the speed of a conventional break-bulk ship by at least 25 percent.

It was not until ships were designed and built to handle all their cargo in containers that significant increases were made in their cruising speed. As pointed out earlier, the fewer the days needed for a single voyage, the more revenue-producing trips a ship can complete in a year. There are, however, only two ways in which to reduce the duration of a voyage. One is to cut the time to discharge and reload. The other is to decrease the number of hours spent in transit between ports. To justify the cost of greater speed, it must be shown positively that the revenue earned by the additional voyage

exceeds the cost involved. The speed of the container-carrying ship therefore must be related to the integrated transportation system and evaluated for its contribution to that system.

To anticipate later developments, it may be observed that the cost of building the newer containerships, which were much larger, faster, and more complicated in structural details than the pioneering originals, escalated during the second decade of the container era, and reached almost incredible levels. As a natural consequence, the necessity to reduce to the minimum the number of units required to maintain the acceptable frequency of sailings became a matter of survival. Sea speeds were raised, and cargo working schedules were established on a round-the-clock basis.

Concurrent with these developments, there have been continuous modifications in terminal management ever since that day when the *Gateway City* inaugurated fully containerized shipping service. In those early days of the revolution, the problem of providing an adequate terminal was of comparatively minor importance. The *Gateway City* and her sisters were equipped with their own cranes and therefore could be berthed at any wharf or pier with a wide apron. As techniques improved and awareness increased of the potential of container transportation, it was perceived that specialized terminals equipped with appropriate cranes were essential.

Traditionally, berthing for ships has been made available by the government authority having jurisdiction over the waterfront. Piers and wharves, with supporting transit sheds and head houses, have been constructed by this authority and leased to steamship companies serving the port. Almost invariably, features requested by the oceangoing carriers have been incorporated into the facilities built for their use.

The role of these public agencies has not changed, and owners of container ships still look to them to design and construct satisfactory terminals. The enormous expense of creating container terminals requires the financial resources of port authorities, which can issue bonds backed by the full faith of the state or municipality. Shipowners have been hard pressed to pay the high prices of their ships without having to assume in addition the burden of costly terminal construction.

At the time the completely cellularized ships were introduced into McLean's operation, computer technology was in its earliest stages. It was clear to the management team guiding the growth of the integrated transportation system that only by taking advantage of the computer and exploiting it to the maximum could the goal envisioned by McLean be attained. What emerged, after appropriate study and testing, was a computer-oriented technique of control which provided the information essential to good management. It is true that the emergence of computer technology made possible the fruition of the container revolution. It also must be acknowledged that this achievement was owed in part to the willingness of the McLean team to abandon traditional methods in favor of the computer procedures that were being introduced, and to adapt the "electronic brain" to the requirements of the McLean enterprise.

An almost classic example of the application of progressive management techniques may be found in the solution to the problem of keeping track of the containers controlled by McLean. When the *Ideal X* sailed in 1956, the data relating to the fifty-eight containers aboard had been assembled and posted by hand. This method was still in use when the *Gateway City* made her inaugural voyage in 1958. When, however, she was joined by five more ships, each outfitted with at least three sets of 226 containers, and boxes were scattered the length of the Atlantic and Gulf coasts of the United States, it was apparent immediately that only a completely computerized record system would serve to prevent these expensive units from going astray through either delay in getting information or clerical error in using it. As the McLean fleet grew and expanded into worldwide operation, the value of such a control device was magnified.

A retrospective examination of Malcom P. McLean's role in the development of the containership and its essential function in the integrated transportation system reveals that he was bold but never reckless or foolhardy. The consequences of every step, insofar as it was possible to do so, were evaluated methodically and pragmatically before any action was taken. It will be recalled that the first idea was to load wheeled trailers aboard ships, and to roll them off at destination. The economics of this scheme were found, after exhaustive analysis, to be unfavorable. The alternative suggested by

the study was to carry the boxes without the chassis. McLean approved the idea and forthwith acquired the necessary certificates of convenience and necessity by purchasing the Pan-Atlantic Steamship Company, an established coastwise operator. The full-scale laboratory experiment then was carried out with four modified tankers.

McLean proved, by the initial voyage of the container-carrying tankers, that the theory of the containership was valid, and that its potential would be achieved when an integrated transportation system was developed around the indispensable vessel. The almost storybook success that crowned McLean's pioneering efforts is now history and justified in every respect the basic premises on which he had proceeded.

Looking back to April 26, 1956, it is easy to comprehend that one of the great technological events of modern times, the beginning of the container revolution, occurred on that date. It is much more difficult to establish the moment when McLean first had the inspiration of an integrated transportation system using existing railroads, trucks, ships, containers, and contracts of carriage to obtain the greatest possible efficiency and economy. Whenever the inspiration was translated into a cohesive idea, it became the driving force that led to the practical success of the new method. That McLean had both the imagination and the financial resources to convert his dream into hard reality is one of the fortunate twists of history. Had it been otherwise, there is basis for endless speculation as to when the container revolution would have come about—if ever.

What is of paramount importance is to realize that without the combination of practical wisdom, commercial genius, and unwavering audacity, the events which today are described as the container revolution might not have occurred. That wisdom was joined to genius, and that audacity supported them both is, in final judgment, the reason that Malcom P. McLean may and must be hailed as the initiator and the leader of a revolutionary business system which he put together and made function perfectly the day it was unveiled to the world.

The Ramifications of Containerization

ONCE the feasibility of the container movement had been proven by Malcom McLean's success in the U. S. coastwise trade, shipowners in the international services began to look with interest on the new method. For many reasons, including the entirely pragmatic one that there was a shortage of the proper containers, it was not until early 1966 that any regular transportation of containerized cargo was offered to shippers on the busy North Atlantic route. In February, Moore-McCormack Lines began to dispatch containers in their fast break-bulk ships, emphasizing as an inducement to exporters and importers that these boxes were the last to be loaded aboard and the first to be taken out of the ship, thereby saving a considerable amount of time over and above the reduction in conventional handling. The United States Lines followed suit the next month by assigning four break-bulk ships to carry containers. Fully cellularized containerships transporting all cargo in the big boxes were introduced to the North Atlantic by McLean's Sea-Land Service in late April of the same year, when the *Fairland* and three sister ships were placed on berth to operate between Baltimore (Maryland) and Port Newark (New Jersey) in the United States and Rotterdam, Grangemouth, and Bremerhaven/Bremen in Western Europe. From this date onward, the conversion of international shipping on the main trade routes to full use of containers was rapid and dramatic.

This revolution in transportation was almost indescribably expensive. In the period between 1958 and 1973, shipowners, terminal operators, and port agencies in the United States alone invested seven and a half billion dollars in ships, containers, and port facilities. The concurrent growth of container operations throughout the world caused similarly high levels of expenditures. This was pointed out by the New York *Journal of Commerce* report of June 20,

1977, publishing the results of a survey of the expense involved in converting to the new method. These facts were emphasized:

1. Containerships cost twenty-five to thirty dollars per cubic foot to build, compared with seven to eight dollars per cubic foot for a conventional break-bulk carrier;

2. A standard steel container, twenty feet long by eight feet wide and eight and a half feet high, represented an investment of \$3,500, while a fully refrigerated container might be priced four times higher;

3. A thirty-ton-capacity container-handling crane involved a minimum outlay of \$1,750,000 before it lifted its first box; and

4. Waterfront property for marine terminals and marshalling yards could be obtained only by paying between \$250,000 and \$300,000 an acre.

Added to these formidable figures was the harsh economic fact that the new mode of transportation was forced upon shipowners only a few years after they had commenced a very expensive replacement program for their obsolescent break-bulk vessels. These new ships embodied the latest and most efficient ideas of cargo handling and stowage and had been acquired in the expectation that their costs would be amortized in from fifteen to twenty-five years. Furthermore, a number of operators had expended very sizable sums in modernizing and equipping their break-bulk terminals both in home ports and overseas. It is easy to understand their reluctance to write off somewhat abruptly the many millions of dollars (or pounds, or yen, or other appropriate currency) they had put into what had been intended as a long-range fleet modernization plan. The tide of change, however, was irresistible, and adaptation to the container method was a matter of simple survival. It is a sad commentary upon the remorseless pressures of economics that a number of reputable but small-scale operators found it necessary to withdraw from the shipping business and to liquidate their holdings.

The burden thus suddenly thrown upon shipowners by the technological advance was immense. New containerships not only cost much more to build than did the break-bulk vessels they were displacing, but they were capable of higher speeds and had signifi-

cantly greater capacity. The duration of sea voyages was shortened and port turnaround time was reduced radically, contributing to the heightened performance of these big ships. In theory, one modern cellularized container carrier of about 26 knots speed and a deadweight of approximately 29,000 tons, carrying close to 2,000 twenty-foot containers, could transport the same quantity of cargo in a year that five or six conventional break-bulk ships could handle. Had it been possible to effect a simple reduction of ship numbers according to that ratio, it would have resulted in a reasonably satisfactory solution to many of the problems spawned by containerization. Shippers had become accustomed, however, to a certain frequency of service, such as one sailing a week by a favored operator, and looked askance upon any suggestion to change this to one sailing every third or fourth week. To harassed shipowners, this intractable attitude might have seemed most unfriendly, but in the end they acknowledged that, as always, the requirements of the shipper determined the nature of the trade. It was a case of satisfy the customer or go out of business.

To provide that satisfaction, shipowners were forced to accept the idea of building and operating whatever number of large, fast, high performance, and extremely expensive carriers might be needed to provide the service demanded by their customers. It worked out, in practice as opposed to theory, that the new fleet would replace the older vessels in the ratio of one to three. This meant that shipowners were compelled to build almost twice as many ships as the trade would support financially. In terms of potential revenue tonnage, there was only one interpretation of this fact: there was a great deal more capability (or cargo space) than there was cargo.

The obvious solution was the one adopted by many operators. They joined forces with one or more of the other carriers on the same trade route to form a sort of composite management of that number of ships which could be justified economically while still meeting the demands of shippers and consignees. In effect, what happened was that four owners might band together and each supply one ship to the consortium. Every week there would be a sailing, thereby satisfying the shipping public while at the same time assuring revenue cargo adequate to pay the expenses of the voyage. This arrangement between carriers required prior approval by the

Federal Maritime Commission before any operators participating in the foreign commerce of the United States could become active in the association.

An example of a consortium was the formation in 1967 of the Atlantic Container Line. Three Swedish firms (Swedish American Line, Wallenius Shipping Company, Transatlantic Steamship Lines), the Holland America Line (a Netherlands organization), the Compagnie Generale Transatlantique (a subsidiary of the French Compagnie Generale Maritime), and the Cunard Steam Ship Company (part of Trafalgar House, an English conglomerate) each contributed one or two ships of approximately the same size, speed, and characteristics to a new company in which the six operators had equal holdings. Denominated the Atlantic Container Line, this organization operated between European and British Isles ports and United States Atlantic Coast cities from Boston to Norfolk, Virginia. The first sailings were made in the fall of 1967, but it was not until some months later that the planned number of ten ships was in full service. When the entire fleet was functioning, each ship was placed on a 21-day cycle; not all ships called at all ports, but dependable schedules for the different routes were set and maintained. A computer-oriented system controlling the loading and discharging of containers assured that consignees could obtain their boxes with a minimum of delay. Full containers for a single consignee were ready for pickup not later than 30 hours after the ship arrived, while cargo shipped as part of a container load and requiring more attention was available to the consignee in less than 72 hours after the ship was berthed.

The essence of the consortium was to substitute cooperation for competition, and to form a single unit in which the good of each participant was the predominant principle, rather than a frantic scramble by individuals for the greatest possible share of the available business. An association of this sort could be successful, however, only if the members provided equipment of substantially the same capability, and a single management handled the enterprise. Carriers were disposed favorably toward this idea, in view of the enormous cost of setting up a complete container service (ships, containers, acquisition of appropriate terminal facilities, and the necessary supporting network of agents) to say nothing of the problems inherent in the training and oversight of qualified salesmen, office personnel, and supervisors at every level.

Success crowned the idea of the consortium, if results are the proper criteria by which to judge. It proved eminently practical on trade routes having a sound economic base and a reasonable balance between incoming and outgoing cargoes. Not every prospective consortium, it must be noted, was able to achieve its goals. Some associations were proposed, but died aborning; others were put together with every prospect favoring the scheme but had to be dissolved in the face of insurmountable obstacles, many of which were not of the carriers' making. Still others began their existence with a number of operators but in short order were narrowed down to as few as two partners.

Two concerns of management assumed unprecedented importance as containerships became relatively standardized, very large, extremely expensive, and somewhat inflexible units.

The first of these concerns dealt with what accountants designate as "variable costs." In the steamship business, these are the items of expense related directly to the specific voyage on which a ship is engaged at a given time. Wharfage and dockage and other fees charged by the port authorities, pilotage fees, tugboat hire, stevedoring expense, watchmen's services, and agents' commissions are examples of these variables. To illustrate how they fluctuate in break-bulk shipping, consider the hypothetical example of a vessel which called at the same port on two consecutive voyages. On the first visit, the ship lifted only a thousand tons of cargo, and incurred total charges of twelve thousand dollars. On the second call, she loaded fifteen thousand tons of cargo and spent \$109,000. Variable costs in break-bulk shipping usually represent about one-half of total voyage expense; the other half is allocated to capital and overhead charges, which must be met regardless of what the ship carried or even whether she earned any revenue.

In containership service, it soon became apparent that several items of costs which had been considered as variable had become so uniform and regular that they no longer had any flexibility, and had little relationship to vessel employment. Genuinely variable costs, it was perceived, had been halved by the very nature of containership operation. For instance, the dockside cranes charged the same to hoist empty containers as they did to handle filled ones; the draft of the ship on arrival and departure was almost always the same, so the pilot fees (which were based on the draft of the ship) scarcely ever changed significantly. The same ports were visited in

the same rotation voyage after voyage, and except during times of adverse weather conditions at sea, fuel charges were very nearly identical for every trip. These expenses, although technically variable, in fact had become so constant that they were as predictable as capital costs of amortization, interest, depreciation, insurance, and overhead.

The other area of special interest to management was related to the dawning awareness that the container was in reality an integral part of the ship, and therefore it was possible to "put the ship alongside the cargo" even when that cargo was in a warehouse a thousand miles from the sea. A most significant corollary of this new definition was that geographical limits for cargo solicitation were extended enormously. It also quickly became clear that the differences between inland and deep sea transportation were substantial enough to preclude adoption of any truly uniform contract of carriage applicable to all parties involved in intermodal transportation. The solution to the problem was pragmatic: the shipowner furnished the shipper a bill of lading stipulating that he accepted full responsibility for any loss or damage sustained in transit. By separate agreements negotiated between the carriers, the liabilities of inland carriers for loss or damage were established, and appropriate remedies were set forth to protect the shipowner in cases where the inland carrier might be found at fault. This somewhat ponderous method did accomplish what was required, and made it possible for the shipper to receive a single document and to have one carrier accept complete responsibility for the care and custody of the cargo while it was in transit.

One of the noteworthy developments of the container age, spawned at least in part by this new theory of placing the ship alongside the cargo, was the emergence of an "infrastructure" for container shipping. This term implies a whole catalog of functions and responsibilities on the part of the shipowner which explains, at least in part, the complexities of the container "system." Of primary importance in this supporting organization was the indispensable network of agents, freight forwarders, and cargo brokers to represent the water carrier throughout the inland territory, and to serve as its local sales and delivery staff. On their diligence, enterprise, and judgment depended, to a great extent, the ultimate success of the shipowner. The best representatives required complete sup-

port if they were to perform to maximum capability. Convenient and well-supplied container depots had to be established, together with satellite pickup and delivery points. Transfer stations at major junctions of highways and railways had to be set up so that containers could be placed on railroad cars, or removed from flatcars and put aboard trucks, for onward carriage. In some parts of Western Europe, where the railroad network was very comprehensive and the movement of goods by rail correspondingly heavy, it was found both desirable and necessary to set up "railroad container terminals." Of necessity, these were located at those points where railroad cars were switched from one routing to another, or where trucks converged either to bring or to take away loaded containers. When such terminals were put into operation, they often were expanded as the need arose to serve as "pools" where customers were able to pick up containers after they had served their purpose. Finally, this infrastructure included facilities for maintaining and repairing containers, as well as the means by which every pool could be supervised and adequately stocked with serviceable equipment.

INTERMODALISM

As the complexities and tremendous potential of containerization came to be understood, there was a concurrent movement within the industry to capitalize on the inherent advantages of other modes of transportation to create an economical and effective system. This movement was given the name of "intermodalism."

Intermodalism is a technique of transportation that envisions the uninhibited movement of individual containers by two or more different kinds of carriers. That is to say, a shipper located a thousand miles from the seacoast may pack his goods in a standardized container and send the box by truck, barge, ship, and railroad to an overseas destination under a single contract of carriage which adequately protects all parties concerned at any given moment during the period the goods are in transit. Admittedly, there are no technical or mechanical difficulties associated with the process of shifting the container from one form of transportation to another as long as the container will fit into the designated cell aboard ship. The problem is entirely a legal matter.

In any long-distance transportation of goods, the relationship between shipper and carrier is set forth in the contract of carriage. Unlike the conventional contract, which is the result of a meeting of minds following negotiation, the maritime contract of carriage is drawn up unilaterally by the shipowner and is not subject to negotiation except in unusual circumstances. It becomes binding upon the shipper when it is accepted and does not require his signature. The act of acceptance is construed as agreement to all the terms and conditions of carriage.

The maritime contract of carriage reflects the experiences of the individual shipowner who issues it. There is no necessary similarity between contracts issued by different operators on the same route, except insofar as legal prescriptions apply. That the contracts drawn up and used by two competing carriers on the same route do contain a number of identical phrases is both a fact and a coincidence, because only very rarely will there be consultation between shipowners on the wording of their individual contracts.

As has been set forth earlier in this chapter, the ocean carrier involved in intermodal transportation was responsible to the shipper for loss or damage to the goods while in transit. The water carrier's agent was authorized to issue the bill of lading when the shipment was accepted at the point of origin.

In the movement of goods by containers, it has become standard practice for the shipper (rather than the carrier) to stow the goods in the container. When it is filled, the shipper closes the container and secures it with a seal. This seal is practically identical with the one used by railroads, and consists of a thin metal band fitted through a slot in the locking mechanism of the door. It terminates in a weighted knob into which the band is threaded and held by a type of cement. It has a serial number, which becomes part of the permanent record of the movement, and thereby assures that no substitution can be made without detection. No key is used because of the practical impossibility of the consignee possessing this key when the container is delivered.

Containers filled by the shipper ordinarily are not subjected to inspection of the goods inside when custody is accepted by the carrier. The description of the goods provided by the shipper must be accurate, under penalty of legal action. Unless the container is received in a condition which shows that some damage probably has

been sustained, the carrier and the shipper agree that they will be bound by the contractual stipulation that the "package," i.e., the container, is "in good order and condition." The consignee is obligated to set forth in detail any damage discovered when the container is opened, and may be invited by the shipper to prove that this damage was not the result of faulty stuffing.

Participants in an intermodal movement which crosses one or more national boundary lines must take cognizance of the rules and regulations applicable to their business. The shipowner, as the party responsible to the shipper, must be well informed on these points, and of necessity depends upon the inland carriers for notification of any changes which affect the intermodal arrangements.

The ideal of intermodal transportation is to give the shipper those routings which will result in the fastest and least costly carriage of his goods. A typical case might require the services of an intercity trucker to haul the container to a feeder-barge collection point, transit by feeder barge to the ship selected to take the container to a foreign port, and a railroad trip to an inland point where a trucker would accept the container for transportation to the ultimate consignee. The shipper is interested only in three details: the party to whom claims for loss or damage during transit should be presented; the rapidity with which the container is moved from origin to destination, and the total cost of transportation. Intercarrier relationships are of no concern to the shipper, which in part explains their pressure for a single, comprehensive bill of lading.

One of the obstacles to developing an intermodal contract of carriage acceptable to all participating carriers has been the question of financial responsibility for loss or damage to the goods. Participants in the intermodal movement understandably are reluctant to become involved in claims actions when there is no evidence of their culpability. Added to this reluctance is the fact that in most circumstances the container is not opened while it is in transit; so the true condition of the contents at any given time is unknown to any of the carriers.

The following description of a hypothetical intermodal shipment of a container loaded by the manufacturer of high-grade porcelain tableware will illustrate the problem.

A consignment of 106 cartons of finest quality tableware, valued altogether at approximately \$200,000, and with some cartons

worth as much as \$5,240 each, was loaded by the manufacturer into a new and carefully inspected container supplied by direction of a steamship line operating from Oakland, California, to Honolulu, Hawaii. The purchaser demanded that the goods be sent by the fastest possible surface means. Pursuant to this instruction, the shipper's traffic manager decided to move the container by intercity trucking service from Trenton, New Jersey, to the railroad yards in Newark and thence overland to Oakland, where the transfer to a fast containership would be effected. Transportation was accomplished accordingly, with no external evidence that the container had been subjected to any forces that might cause damage, or had been the victim of pilferage. Inspection of the exterior of the container confirmed the original statement by the initiating carrier that the container was in good condition, and the contents in apparent good order and condition.

When the container was delivered to the consignee in Honolulu, it was discovered that one carton, worth \$1,000, was missing, a void space near the door of the container giving incontrovertible evidence that the cargo had been loaded. Two other cartons, each with a declared value of \$5,240, had been crushed into rubble. They were loaded at the forward end of the container, and even had the doors been opened by a carrier for an inspection, their condition would not have been detected. The total claim by the consignee amounted to \$11,480, a sum sufficiently large to make it important for those not involved to prove their lack of responsibility for the loss and damage.

Inquiries were initiated by the steamship line as the delivering carrier. In the order of their contact with the shipment, these facts were assembled by the claims agent:

1. The manufacturer in Trenton stated in an affidavit that everything listed in the manifest had been placed in the container, that expert packers handled the stowage, that no careless or rough treatment was accorded to any of the cartons, and that the container had been sealed after a final inspection to assure that all was in good order within the container.

2. The intercity trucker declared, also under oath, that delivery had been effected to the railroad yard in Newark without incident. The driver of the truck was identified by name and license number,

and his record for safe driving was confirmed by the state police. Furthermore, the driver testified that he personally had examined the seal at the time custody was transferred to the railroad, and the seal had not been damaged in any way.

3. The railroad set forth in its notarized comment that the container had been lifted off its chassis by a skilled and dependable operator of a mechanically perfect crane, and there was no record of improper handling at the time of loading or unloading of the railroad flatcar. Examination of the train crews' reports did not lead to any conclusions that those cars making up that particular train had encountered any abnormal circumstances or been treated in an unusual manner.

4. The steamship operator had its own records of the container from the time it was brought by the truckman who accepted it from the railroad until it was turned over to the consignee's drayman and he was cleared for departure from the Honolulu container yard. The external appearance of the container was satisfactory, and the condition of the seal corroborated the truckman's statement; there had been no basis for questioning the accuracy or veracity of the shipper's description of the contents, and therefore the container had not been opened for inspection.

The determination of who was responsible for the loss and damage clearly posed a problem. Because of the high value of the claim, and the unusually complete data relating to the transfers, it was impossible to place responsibility on any one carrier. Negotiations between all the participants were conducted by the steamship company's claims department, and it was agreed that the consignee would be reimbursed by assessing each of the carriers in direct proportion to the amount of freight money which had been earned.

The illustration has been simplified deliberately by keeping the entire transaction under U. S. jurisdiction, which permitted one of two solutions: (a) to initiate legal action in a U. S. court, pursuant to laws and regulations known to and binding upon all participants; or (b) to permit negotiations within the framework of the pertinent regulations. It takes only a little imagination to comprehend the practical difficulties resulting from an intermodal shipment in which, for instance, an American railroad was the initiating carrier, a Dutch ship provided the ocean transportation, a German barge

service moved the container inland and turned it over to a Swiss truckman for delivery to the consignee.

True intermodal transportation is still in the process of evolution. A major step toward realization of the ultimate goal of intermodalism—the most efficient use of the available means of transportation without the requirement for special intercarrier contracts—has been taken in the development of the “landbridge” system. More detailed description of this procedure will be presented in later pages; what is important is that the single bill of lading has become a reality. The ocean carrier is willing to issue the bill of lading because that carrier controls the inland movement. That is to say, the shipowner already has contracted for use of the inland carriers, and knows exactly what the relationships are with them. Drawing up the bill of lading for shipments involving landbridge is, therefore, a comparatively simple legal exercise.

The essence of the intermodal contract is an agreement among carriers of different types, such as barge lines, railroads, and trucking firms to perform certain carefully described and defined services, for agreed rates of compensation. Ideally, the ocean carrier offering landbridge service should be free to negotiate the most favorable rates, terms, and conditions for the type of service required. The resultant intermodal rate is the total of what must be paid to connecting carriers plus the shipowner's compensation. There are persuasive reasons why the carriers in the same trade (for example, Japan to the U. S. Pacific coast) would find it advantageous to negotiate with the landbridge operators to establish the charges for inland transportation. Although this appears to be a logical element in the conception of intermodal landbridge transportation, the Shipping Act of 1984 bars any association of ocean carriers domiciled in the United States from negotiating “on any matter relating to rates or services provided to ocean carriers within the United States by those non-ocean carriers.” Ocean carrier groups, however, are permitted to set “joint through rates” for cargo destined to inland points. From the text of the law, it would appear that the intent is to permit the individual ocean carrier to negotiate a rate for landbridge service and then to meet with the other carriers in the group to agree upon a combined sea and land rate which would be common to all members. The problem of how to establish an acceptable joint rate when all the component parts cannot be revealed has not yet (1986) been resolved.

A major growing pain of the container age has been the conflict over the size of the containers. When Malcom McLean began his operation with the sailing of the *Ideal X*, he modified for shipboard stowage the standard thirty-five-foot trailer used by the U. S. trucking industry. As other steamship owners cautiously entered the new phase of seaborne commerce, they designed containers that seemed most appropriate for the trades they served. Before many years had passed, but well before there was the almost complete conversion of liner-service shipping to container transportation, there was a profusion of containers. All were eight feet wide and approximately eight feet high, but in length they ranged from ten to forty feet, although not in uniform increments. There were, for instance, boxes that were twenty, twenty-four, twenty-seven, twenty-eight, thirty-five, and forty feet long. Each was excellent for one particular route, but as shippers began to accept the new method and to demand that their goods be moved in the big boxes on many trade routes, shipowners faced the gigantic problem of making containers interchangeable in the same way that railroad cars in the United States, Canada, and Mexico are moved across international boundary lines without mechanical hindrance.

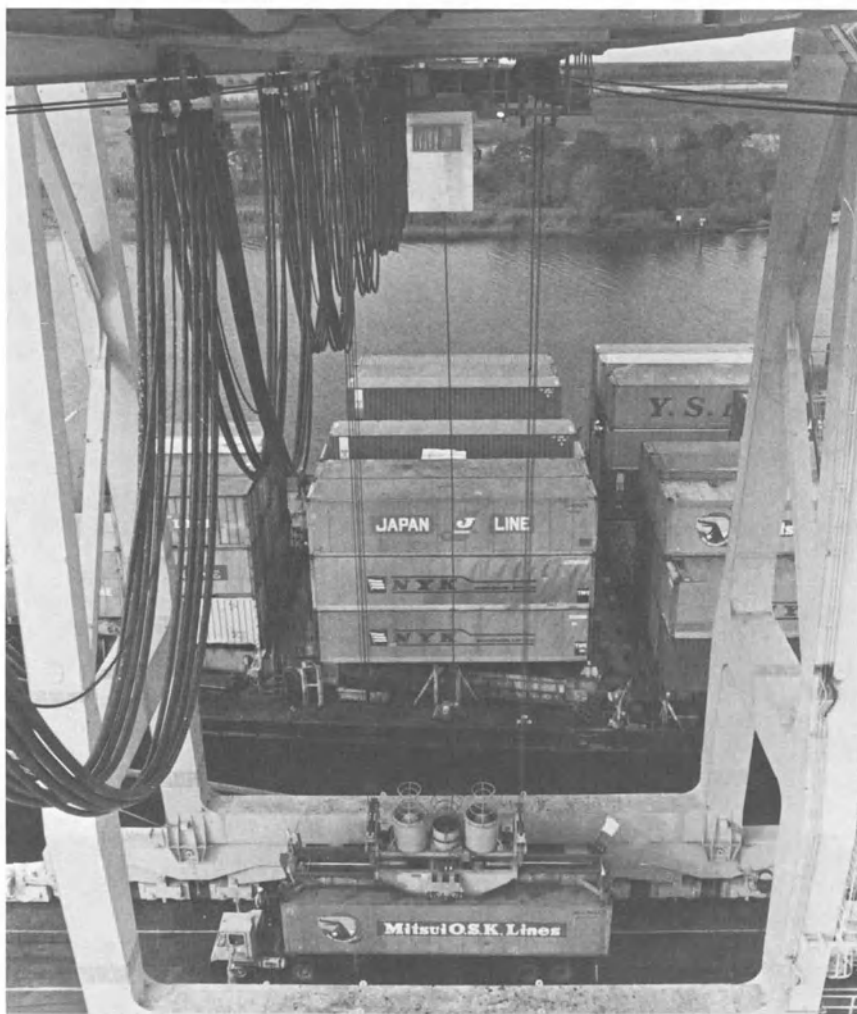
The International Standards Office (ISO) strongly recommended that containers be designed and built to "standard" specifications. Many carriers adopted the idea. The basic ISO module was ten feet; approved containers therefore were ten, twenty, and forty feet long and eight feet wide by eight feet high. McLean stubbornly clung to the thirty-five foot boxes for the reason that they fit the chassis used by the American trucking industry and therefore could be handled most efficiently and effectively to any point in the nation. He built into his ships cells for this size unit regardless of ISO proposals. Over the intervening years, the battle of size appears to have been won by the proponents of twenty- and forty-foot units, and most of the containerships built today are designed to accommodate either of these sizes. The worldwide yardstick for determining the capacity of containerships as well as expressing the statistics of container traffic is the "twenty-foot equivalent unit," universally abbreviated to TEU. Even Sea-Land Service modified many of its ships to handle forty-foot boxes as well as its own thirty-five-foot containers. Its newest ships were designed for both twenty- and forty-foot boxes.

The question of making containers interchangeable between



A container is attached to the lifting frame of the massive shoreside crane.
Photo: North Carolina State Ports Authority.

steamship lines (an essential element of intermodalism) has resisted rapid settlement. Pioneer containership owners had to buy their own boxes and make them available to shippers. Inevitably, they tended to load their ships only with goods packed in these company-provided containers. In overseas and foreign trades, the dispersion of containers to destinations quite remote from the ports at



Crane operator's view of a container ready to be hoisted aboard ship.
Photo: North Carolina State Ports Authority.

which the ships called quickly assumed the dimensions of a major problem. Significant losses of boxes were sustained by some operators, and elaborate measures had to be devised for tracing them. Some were discovered to be serving as warehouses for consignees, who explained that it was cheaper to take the goods out of the containers as their counters needed replenishment than to empty



Containers are protected from boarding seas by the forward location of the bridge house of this United States-flag containership. Photo: Sea-Land Service.

the big boxes into warehouses and then shift the goods to shelves and sales racks. Other owners learned that the containers were used not only as warehouses but as the stores themselves. In many instances, in the more impoverished regions of the world, containers were converted into houses by cutting windows and doors into them and placing them on hard stands.

As a result of these findings, complex and expensive systems for keeping track of containers were developed, but the high cost of buying, maintaining, and tracing the boxes has encouraged the formation and growth of companies with no interest except to buy containers and to lease them to shipowners. The emergence of these entities has brought closer to fruition the dream of interchangeability, and attention now is directed toward forming “neutral pools” of containers. The term is a shorthand reference to those boxes owned by leasing companies which are used for one-way movements, and then turned over to an agent. This functionary places the containers in central collecting points where they form a pool from which individual units can be drawn to meet the needs of

shippers within geographical range of that central point. Because they are specialists in the procurement and management of containers used in worldwide trade, the leasing companies have devised elaborate, computer-based systems by which they are able to keep informed of the location of every one of their trailers at any time.

The probability is that shipowners will withdraw gradually, as their investments in containers are amortized, from owning the boxes, and will depend entirely upon the leasing organizations. There are two compelling reasons for this trend. The first is that the new containerships are so expensive that little capital is left to the shipowner with which to purchase the required thousands of containers. The second reason is that the need for interchange is growing day by day, and the folly of attempting to force shippers to use only carrier-owned containers is increasingly evident. One major user of containers complained that he had to return "empties" to the shipowner within a fixed number of hours, and then found himself waiting a day or longer for the same shipowner to bring him a container in which he could pack the next outbound consignment. Said this traffic manager, "Sometimes my returning 'empty' passes the incoming container, which also is empty. It's a very wasteful and extravagant way of doing business."

With the advent of the neutral pool of containers, the old system by which Deep Sea Line would send a container to Consignee X in Sacramento, and then bring it back empty to Oakland, while Blue Water Line was dispatching a container to Sacramento for Trader Q to load with goods for Australia, will come to an end. Instead, Deep Sea Line will move a container to Sacramento, and simultaneously notify the leasing company that Box 123 is now in the custody of Consignee X. The leasing company becomes responsible for keeping track of Box 123 when it is unloaded and turned over to the neutral pool. Blue Water Line, meantime, will have informed the pool that Trader Q needs a container for the shipment to Australia, and the pool manager delivers Box 123 to that shipper's warehouse. When Trader Q accepts the box, he becomes responsible for notifying the pool when the container is delivered to the steamship company. In turn, the carrier tells the pool agent in Australia when the consignee has taken custody. The consignee is obligated to return the container to the pool depot.

The advantages of this system are obvious. Intermodal transportation envisages the adoption of a uniform-sized container that can be loaded with equally productive use of space in ships, aircraft, barges, trucks, and railroad cars. Already wide-bodied, all-cargo aircraft accept one or two twenty-foot boxes. Railroad flatcars are in existence which handle two forty-foot containers, and trucks habitually employ flatbed trailers to haul containers for which they do not have chassis. Physically and mentally, there are no obstacles to using all types of carriers for containerized cargo.

Looking ahead, future growth and acceptance of the container as a means of packaging goods for transfer from one point to another depend upon the establishment of true intermodal systems. Before this can be achieved, all the different carriers who would be involved in the system must agree on the following points:

1. The standardization of container sizes.
2. Complete interchangeability of containers between carriers.
3. Establishment, maintenance, and control of container depots by an agency or agencies which would be neutral (unaffiliated with the carriers) and dedicated to working only with containers.
4. The formulation of a uniform contract of carriage binding upon all participants. This is necessary to handle claims and to facilitate negotiations concerning clauses of the contract.
5. Reduction in the present number and complexity of documents required for cargo shipped in containers. The ultimate goal must be one document that will be accepted by commercial and governmental interests as meeting legal and other stipulations relating to origin, value, ownership, and routing instructions.
6. The introduction and widespread use of highly sophisticated computer systems to support the interchange program. This will have to be a cooperative enterprise, with all carriers and container owners contributing essential and timely data to the centralized pool of information.
7. Efficient, economical collection and distribution of containers from and to the marine terminals, monitored by all parties concerned. In the United States, this may not be possible without legislation exempting the cooperation inherent in the intermodal principle from existing antitrust law.

8. Uniform rules concerning responsibility for the condition of containers while in the custody of a given person.

9. Agreement on a uniform method of processing and adjudicating claims.

LANDBRIDGE

It is true that, although containerization depends upon fast, efficient ships for the ocean passage, the growth of the idea is rooted in the land. The shipowner has no choice but to become involved in every phase of transportation if there is to be a smooth, uninterrupted flow of cargo to the ships.

As the use of containers for international shipping expanded following the first voyages in 1966, it was logical for both operators and shippers to think of ways in which to obtain even greater benefits. A look at the map showed that from Tokyo to New York there were two routes: one went through the Panama Canal, an all-water trip which might take up to thirty days, depending upon the speed of the ship, while the other proceeded to a port on the west coast of the United States and overland to the eastern metropolis. With the goods already packed in a container, the time and risk factors involved in transferring from ship to railroad were minimized and between seven and ten days could be saved.

Seatrains generally is credited with being the initiator of the program, with the first shipments completed in 1971. The idea, however, was neither new nor original with Seatrain but rather revived what had been a successful operation during the 1920s when a very interesting service was offered by steamship operators. Known informally as the "silk express," the service connected Yokohama and New York via Seattle, carrying bales of raw silk as primary and premium cargo. Express steamers loaded their valuable consignments in Japan, followed the Great Circle course to Seattle, and transferred the silk to waiting railroad baggage and express cars that were dispatched at high speed to the market in New York. This coordinated arrangement was offered to shippers whenever the need arose and therefore was not a regular feature of transpacific steamship operations. It was popular and lucrative and was terminated only as economic conditions and international rivalries created circumstances inimical to its continued existence.¹

The silk express was a coordinated operation of the steamship carriers and the transcontinental railroads. Two separate contracts of carriage were issued, one by the water carrier and one by the land carrier. The only reason for the service was to reduce transit time between Japan and the New York silk market. Each transporter functioned in its own area; there was mutual support without any attempt to establish single control. Both the steamship owner and the railroad company benefitted, but aside from adjusting schedules and assigning equipment so that there would be no delays in transshipment, the silk express was the result of two distinct activities.

What differentiated the Seatrain scheme from the earlier arrangement was that Seatrain assumed and retained full responsibility for all transportation and made its own subcontracts with the transcontinental railroads. The shipper dealt only with Seatrain which added to the attractiveness of the new method. As the number of containers booked to this service increased, Seatrain arranged for "unit trains" to transport the boxes across the United States. A sixty-car train, loaded with two twenty-foot containers on each car, was dispatched not more than forty-eight hours after the ship docked in a west coast port, and then proceeded under special orders to Weehawken, New Jersey, the Atlantic coast container terminal maintained by Seatrain.

There was one illogical feature about this operation. Containers might originate as far west as Pittsburgh or Chicago but had to be sent to Weehawken, at the expense of the shipper, before the water-rail arrangement and freight rate became effective. It was a waste of time and railroad car space to insist on this routing, but it was necessary for the reason that the unit train was made up in Weehawken and was taken as an unbroken element directly to the Pacific coast. Stops were made only for the convenience of the railroad to relieve crews or to refuel or change locomotives. A special contract was negotiated between the railroad and Seatrain for this transportation; no exceptions were permitted or requested.

In 1979, as part of its strategy to expand its hinterland, American President Lines initiated a "linertrain" service from Seattle to Chicago and New York. Compared to the all-water route, this saved seven days of transit time. Capitalizing on the success of this venture, the company in 1984 introduced its individually designed

railroad flatcars which carried containers stacked two high. A contract was negotiated with the railroad to provide unit train service transporting a total of 200 40-foot containers, with two departures a week in each direction. So effective was this operation that two additional "linertrains" per week were scheduled from Los Angeles to Chicago and New York. A freight forwarding company, Transway International Corporation of New York, served as the company's representative in the midwest and northeast.

As often occurs in business, a successful venture encourages others to emulate the pioneer. In the case of American President Lines, the port authorities in Seattle purchased sufficient double-stack flatcars to commence a common carrier unit train operation from Seattle to Chicago and New York. This was available to any steamship company using the port of Seattle. By the middle of 1985, the Danish-owned Maersk Line had duplicated in every detail the pattern of service from Seattle by American President Lines. Orient Overseas Container Line announced in June 1985 that it was initiating linertrain service from the west coast to Houston and New Orleans.

Any survey of the extension of landbridge service must include reference to the activities of the Russians in developing the trade between Germany and Japan, using the trans-Siberian railway as the link to the ships. In March 1971, the Russians sent a few containers from Moscow to Nakhodka, where Japanese-flag ships were waiting to transport the boxes to Yokohama. A year later, the service was extended to Germany in what became known as the Trans-Siberian Container Service. This trade grew significantly, and not only was the port of Nakhodka modernized and enlarged, but a nearby harbor, Vostochny, was converted into a container terminal. Both Japanese and Russian ships handled the water phase of the movement. Despite problems inherent in moving containers through the extremely low temperatures of a Siberian winter, the saving of seven to ten days in transit time and also a lower cost for freight were approved by the shipping public. The only difficulty of any magnitude resulted from the greater number of loaded containers moving toward Germany as compared to the traffic flow toward Japan.² Returning the containers to their respective owners has been a problem, however, since the dawn of the container age and promises to remain a matter of concern well into the future.

In the United States, landbridge operations have been the source of controversy. Atlantic and Gulf coast ports that formerly handled cargo destined from the coastal states to the Far East complained that the new system was "diverting" business from regions that were "naturally tributary" to those ports. They contended that for decades before the advent of the container revolution, exporters domiciled within a certain radius of a port had sent their goods to that city's waterfront to be loaded into ships that then proceeded through the Panama Canal to ports of the Orient. With the introduction of containers and the adoption of the landbridge scheme by Seatrain and other carriers, the containerized cargo was placed on railroad flatcars that went directly to Los Angeles or Oakland, California, or to Seattle, Washington, for transfer to the containerships. This disrupted the economy of these ports. The dispute was brought before the Federal Maritime Commission in 1973, and after prolonged hearings involving many facets of administrative law, a decision was promulgated on August 8, 1978 which held that landbridge service did not violate any laws.

In commenting on the long-held doctrine of "cargo naturally tributary" to a given port, the commission laid down two "general principles" for the future guidance of all concerned.

The first of these principles was that "certain cargo may be naturally tributary to a port, but any naturally tributary zone surrounding a port is constantly changing." In a particular case, the tributary zone would be determined by considering (1) the flow of traffic, including the points of origin or destination of the cargo, through the port prior to the introduction of the new practices of the carriers; (2) the relevant freight rates for inland transportation; (3) the natural or geographic patterns of transportation and the most logical and efficient use of those patterns; and (4) the needs of the shippers and the characteristics of the goods being moved.

The second principle was that "a carrier or port may not unreasonably divert cargo which is naturally tributary to another port." Judging by the factors listed under the first guideline, certain cargoes may be designated correctly as "naturally tributary" to a particular port. Should any or all of this cargo be diverted from that port, the logic and reasonableness of that diversion would have to be established. "Reasonableness" of an individual routing of cargo is to be determined by considering (1) the quantity of cargo being

diverted, and whether substantial injury is being sustained by the bypassed port; (2) the cost to the steamship operator of providing direct service to the complaining port; (3) any practical and demonstrable difficulties in ship operation or other factors, such as the small quantity of cargo offered for scheduled sailings, or inadequate facilities for handling the proffered cargo, which would influence the water carrier's ability to provide direct service; (4) the competitive conditions existing in the trade; and (5) the fairness of the method or methods employed to effect the diversion, including, for example, the absorption of certain costs or the techniques of soliciting business.

The commission further noted that the contention that a region was naturally tributary to a given port "cannot be extended to the point where a port or a range of ports can claim a multi-state inland region as its exclusive 'territory.' This, however, is precisely what the complainants are attempting to do in this case."

Cargo was found to be originating in states far removed from the Gulf coast. Observed the commission in its ruling:

Even if it were assumed that all minibridge [mini-landbridge] cargo originates in Texas and Louisiana, the Gulf Coast ports all lay equal claim to those areas and no individual port has established an area locally tributary to it alone. . . . The theory that an entire region of the country might "belong" to a range of ports is not a tenable basis upon which to build a regulatory framework of fair competition between the interests of ports and carriers. Historical movements of cargo are not without some relevance, but it cannot be seriously maintained that Congress intended [the port-protecting] Section Eight of the 1920 Merchant Marine Act to freeze international transportation movements into their 1920 patterns.

An indication of the sense of financial injury that a port sustained by the diversion of cargo resulting from landbridge operations may be gained from that portion of the commission's decision dealing with the claim by Houston (Texas) that three-quarters of a forty-million-dollar bond issue had been assigned to build facilities to handle containers in the expectation that Seatrain would continue to provide direct port-to-port service from Houston to Europe.

The commission held that:

Absent clear proof to the contrary, it must be assumed that a local investment decision of this magnitude was dependent upon a number of factors other than the unsecured assurances of continued vessel calls by a single containership operator. It has long been recognized that, absent unique circumstances, the Shipping Act does not require ocean carriers to provide service to a particular port.³

A reexamination of this decision several months later prompted these comments from the vice-chairman of the commission:

Historical and geographical considerations [relating to naturally tributary cargo] may still be applied, but they should not serve as rigid criteria in defining any port's naturally tributary zone. Competitive conditions existing in the trade, shipper needs, and the variety of transportation services available are equally valid criteria in determining naturally tributary cargo. . . . By questioning the notion that certain ports have proprietary rights to certain cargo, we are increasing the need for ports to compete for their traffic and . . . increasing their incentive to develop innovative services to attract enough cargo to remain profitable.⁴

A basic principle of traffic management is to find the most economical routing for a consignment of goods. In this context, "economical routing" always has had a twofold meaning, applied both to the cost of transportation and to the time the goods must spend in transit. If using landbridge service would expedite delivery to an overseas market by two or three days, it must be assumed that the faster method will be selected. This would be true especially if the shortened transit time cost no more than the longer, all-water route.

Another problem encountered fairly early in the new era of containerization was the liability of the water carrier for loss of a container or damage to its contents. Almost all containerships carry some of their boxes on deck and therefore subject them to the effects of storms and heavy seas, to say nothing of temperature changes on passages through different climatic zones. For many years, steamship practice held that the maximum amount for which the carrier would be liable in the event of loss of a package was \$500, unless a higher valuation had been declared at the time the bill of lading was issued. With the advent of the container, however, the question arose, "What is a 'package'?"

The answer was provided in the 1978 court decision which held that, in those instances where goods stowed in the container had been packaged in cartons or wrappings which clearly separated each item or each quantity contained within the package from similar items or quantities held in other cartons or wrappings, liability of the carrier was \$500 for each and every one of the individual packages. If the goods were not segregated—as would be true for a container filled with bulk grain—and therefore no separate mark or number might be ascribed to each unit being transported, the liability of the carrier would be fixed at \$500 for the entire container, which the court considered as the “package.”

FEEDER SERVICES

As the interests of the shipowner have been extended deep into the land mass behind the ports, two developments have occurred since the initial long-distance movement of containers in international commerce. The first was the reassertion, as a practical precept of containership management, of the old ideal that an oceangoing ship is economically most efficient when operated from one loading port to one discharging port. The costly and time-consuming practice of sending the ship after the cargo therefore had to be abandoned. This precept recognized that containerships depended upon shore facilities for loading and unloading, and consequently had to be routed to appropriately equipped ports. The enormous daily cost of the big ships required that turnaround time be as short as possible. A complete cargo of containers therefore should be assembled in the marshalling area before the ship arrived.

It was recognized, however, that all the containers needed to fill the ship might not be available at a single port. Quite often, several hundred boxes would be held at a marine terminal within a comparatively short distance from the main port, while additional containers would be offered at outlying points requiring at least a full day of steaming to reach. The question to be answered, quite literally on a port-to-port basis, was this: “Is it cheaper and more efficient to pay trucking costs to bring the containers from Port Y to the ship’s berth at Port X, or to send the ship to Port Y?” Clearly, the number of containers and the ocean freight rates payable on their

contents would have much to do with the answer, but as a general rule it may be asserted that where the number of containers is small and the highway distance from Y to X is approximately one hundred miles, it would pay the owner to keep the ship at Port X and to absorb the trucking costs. Where the quantity of containers is substantial and the road distance significant, as from Norfolk, Virginia, to Newark, New Jersey, or from Los Angeles to Oakland, California, the usual decision is to move the ship.

In shipping circles, the matter of "vessel cost" invariably is related to total performance, which means that the time required to make a coastwise voyage to deliver or pick up cargo often becomes the compelling factor, surpassing even the very measurable fuel and port fees. To send a ship from Port A to Port B may be approved because it involves only eight to ten hours of steaming time. To get from Port B to Port C, however, twenty-eight to thirty hours of steaming time are needed. The total cost in ship time must be determined because it affects the number of voyages completed in a year as well as many other aspects of efficient management of the vessel and the fleet. Every operator must balance the advisability of moving the ship against the cost of assembling the cargo at a single port. Where this results in a net reduction of expense to the shipowner, the idea of naturally tributary cargo vanishes in the harsh glare of realistic cost analysis.

Containers were built to be weatherproof and easy to handle mechanically. They were planned for easy shifting from one carrier to another—water, rail, or highway—in a matter of minutes at an acceptable level of cost. When it was proposed to take advantage of coastwise sea routes and inland waterways by using smaller craft (either barges or deep-sea vessels) to distribute and pick up cargo at outlying ports, the response was immediate and affirmative, as demonstrated by the actions of a towing company based in New York City. It set up a common carrier barge service linking New York with New Haven and Boston, departing every Tuesday from each terminus. Containers destined to these cities were transhipped to the barges, and a fixed fee was paid for their onward transportation. If no containers were offered to the feeder carrier, no payments were made. The success of these efforts was owed to the regularity of schedules. Shippers and consignees knew when their containers would be picked up or delivered, and the deep-sea

carrier could determine in advance what their feeder service would cost.⁵

Because the big containerships were so expensive, and owners desired insofar as practicable to restrict them to one-port-to-one-port operation, the question of distributing and picking up containers in the Western European area, including the United Kingdom and Ireland, demanded attention at the inauguration of service. "Was it feasible," asked the operators, "to send the ship from New York to Rotterdam only, and to bring to that port containers originating as far away as Bergen, Oslo, Stockholm, Glasgow, and Southampton?" The answer lay in the existence of a rather large fleet of "short-sea traders," smaller ships designed to run between Southampton and Dunkirk, for instance, or from Oslo to Bremen. There also were somewhat larger coastal vessels that were operated between Rotterdam, Le Havre, Marseilles, and points in the eastern Mediterranean. If these ships could be used, and their schedules synchronized with the arrivals and departures of the oceangoing containerships, the objective of one-port-to-one-port might be attained while still providing container service to an extensive area of Europe.

It became apparent that, while the existing short-sea traders and coastwise vessels could be used, they lacked the capability to handle containers rapidly, efficiently, and economically. This deficiency was remedied in some cases by modifying the little ships for the trade, and in other instances bold entrepreneurs designed, built, and offered for charter vessels intended expressly to serve as feeders for the transoceanic carriers. In the years since 1966, an increasing fleet of purpose-built feeder ships has been introduced into these waters and has accomplished two objectives: it has extended the network of container services over an enormous area without adding unreasonably to total transit time for import and export cargoes, and it has made possible the most efficient use of the big containerships, with resulting economies for all concerned.⁶

It was to be expected that the advantages of feeder service would be appreciated in other parts of the world, especially in the Orient. Because the ports in this area are separated by greater distances of open ocean, compared to either Europe or the United States, the feeder ships had to be significantly larger than the vessels employed in those two regions. A notable variation from the Euro-

pean experience was the number of ships owned by the transpacific carriers and assigned exclusively to feeder operations. A third distinction lay in the establishment not only of base (or principal transshipment) ports, but in setting up major collecting ports along the feeder routes. From these collecting points a moderate-sized support fleet was brought together to handle container traffic to the smaller ports ("outports") within the hinterland of the collecting points.

A case history is provided by American President Lines. This company selected Yokohama, Japan, and Kaohsiung, Taiwan, as principal ports for its mainline transpacific container service. It assigned six smaller, but fully containerized, ships to run from Japan and Taiwan to major collecting points at Hong Kong, Singapore, Colombo, and Fujaira (a United Arab Emirate). Containers destined to, or originating in, either these points or their outports were handled by the vessels of the support fleet which consisted of chartered ships and also common carriers working under contract with American President Lines. Transfers between ships were accomplished at the major collecting points. When the company-owned containerships returned to Taiwan and Japan, the containers they had collected were loaded aboard the mainline ships for ultimate delivery to the United States. Although two transshipments were involved, the combination of rapid and dependable service, along with company insistence on meticulous care in the handling of the boxes resulted in customer satisfaction, and an enlarged territory in which American President Lines could seek business.

Generally speaking, there are two reasons why the deep-sea carriers charter feeder vessels rather than purchasing them. First, the investment in containerships is so great that little capital is available for acquiring the small craft. Second, by taking the feeder ships on time charter, the management of ships engaged in the highly specialized coastal trade is left in the hands of experts, with corresponding reduction in the responsibilities of the transocean carriers. Regardless of whether the feeder ships are owned or chartered, the shipper of goods receives a single bill of lading covering the entire movement from the moment the carrier accepts the container until it is delivered to the consignee.

LEASING *VERSUS* OWNING CONTAINERS

Again and again in these pages reference has been made to "containers," but there has been very little comment concerning their construction, maintenance, dimensions, or ownership. Fully to appreciate the growth in the use of containers on the sea routes of the world, it is necessary at this point to return to the beginning of the era.

On that April day in 1956 when McLean's *Ideal X* sailed from Port Newark with the first load of containers, the boxes secured to the spar deck were modifications of the standard highway trailer used by truckmen in the eastern and southern United States. They were thirty-five feet long, eight feet wide, and eight feet high and therefore fit on the generally available flatbed trailers as well as on the chassis provided by McLean. The containers were owned by the steamship company and furnished to the shipper. No other ship-owners on the coast at that time carried containers; no thought was given to the possibility of interchanging with cooperating organizations.

Several years later, while still engaged in hauling railroad cars from New Jersey to Savannah, Texas City, and New Orleans, Seatrain Lines decided to compete for some of the truck cargo. It devised a container that had the same height and width as McLean's but was only 27 feet long. Its reason was that this length permitted two containers to be loaded on a standard fifty-five-foot railroad flatcar, whereas only a single thirty-five-foot box could be transported on that flatcar. When Matson Navigation Company introduced containers into its California-Hawaii trade, it had to choose a size that could be maneuvered around the Hawaiian roads. The approved length was twenty-four feet. Meanwhile, European ship-owners, cautiously entering the container transportation business, selected boxes that, like all American units, were eight feet high and eight feet wide, but they varied in length. There were ten-foot, twenty-foot, and forty-foot units. Explaining their rationale for such dimensions, they pointed out that this gave shippers great flexibility, while making certain that the forty-foot cells of the containerships would be used efficiently.

Inasmuch as each operator furnished containers to its custom-

ers and loaded only those containers, the issue of interchangeability did not arise immediately. It was only after the successful entry of several carriers into the transatlantic services and the sudden expansion of demand for containerized transportation, that any consideration was given to standardizing exterior dimensions. The International Standards Office (ISO) urged that the steamship business worldwide accept a uniform height and width of eight feet and lengths of ten, twenty, and forty feet. Unfortunately for the proposals, American operators had invested many millions of dollars in their individualized boxes and were adamant against change. Despite the fact that Sea-Land, the largest single operator in the world, strongly advocated the thirty-five-foot length and refused to adapt its ships or its tariffs to the suggested ISO dimensions, the trend was inexorably against individuality and toward standardization. The width of eight feet was accepted universally; the height was questioned as early as 1967 and still is under discussion. The twenty- and forty-foot lengths have gained worldwide acceptance.

A major reason for approval of some of the ISO proposals was the fact that steamship owners found it financially almost impossible to purchase the large number of containers needed to support the ships. Increasingly they turned to organizations that specialized in building and renting containers. In the effort to meet the requirements of the maximum number of deep-sea operators, these leasing companies concentrated upon those boxes most in demand, the twenty- and forty-foot units. These were rented for whatever length of time suited the customer; it was possible to obtain the use of a box for two weeks or eight years, or for any period in between. Furthermore, these lessors maintained a number of depots around the world where containers could be returned to owners at the end of the agreed term of lease.

Two trends, one in container design and one in ownership, are evident. The first is that containers must be of standard dimensions, either twenty or forty feet long, and the cells in container-ships must be fixed at forty feet to accommodate either one long box or two short ones.

In terms of numbers, the twenty-foot container has attained worldwide approval. Evidence of this was furnished by a survey made by the Maritime Administration which disclosed that at the beginning of 1977 shipowners in the United States preferred the

short unit by a margin of 51.9 percent to 48.1 percent. The popularity of the forty-foot unit reached its peak in 1973, when 34 percent of the American-owned containers were of that size. By 1977, the percentage had declined to 31.8 percent.⁷

The second trend became noticeable as early as October 1970. This is the marked increase in leasing containers from corporations specializing in this business and having no links with the steamship carriers other than as suppliers of equipment. One leasing company, at the end of 1978, had an inventory of 213,000 twenty-foot equivalent units (TEU), a total said to be one-fifth of all containers available for lease.⁸

The Maritime Administration's survey of 1977 also set out the following facts:

1. About 85 to 90 percent of the available containers were in constant use throughout the world.
2. There were 450 containerships in service, with an aggregate capacity of a half-million twenty-foot equivalent units.
3. Between 80 and 85 percent of the cargo moved between industrialized nations of the world was in containers.
4. Of the total cargo handled by the combined facilities of the port of New York, approximately 75 percent (or one million boxes) was in containers.
5. About two-thirds of the containers used by American-flag steamship operators were owned by leasing companies.

The development of the neutral pool idea depends upon the gradual elimination of the present practice of steamship companies owning (or leasing on a long-term basis) a sufficiently large number of containers to meet the needs of their various trade routes. As has been pointed out earlier in this chapter, there are significant and recurring difficulties in keeping track of these boxes, and it has been necessary for shipowners to set up expensive and highly sophisticated systems to protect their investment.

In broad outline, this is the plan for the neutral pool.

All steamship lines would obtain their boxes from the pools maintained by companies specializing in owning and renting containers. Each container would be acquired by the carrier for a definite, individual movement, and when the contents had been

delivered to the ultimate consignee, the container would be returned to the control of the pool. This would be accomplished by the consignee notifying the representative that the container was empty and requesting instructions as to the disposition of the box. If the consignee needed a container for an outgoing shipment, and the just-emptied unit were satisfactory for the contemplated load, it would be assigned immediately for that purpose. This would eliminate the present wasteful procedure of returning the empty box to the marine terminal. As noted earlier, many export traffic managers have complained of the time lost under the present system of releasing to the oceangoing carrier every container as soon as it is emptied.

The neutral pool would eliminate the miniature pools set up by two or more operators who agree to interchange their equipment. These smaller pools have limited resources and encounter problems in tracing, cleaning, repairing, and positioning the boxes. Operators of the neutral pools are experts in these areas and have the organization needed to handle them effectively and economically.

Neutral pools would supply standardized equipment that would be interchangeable among all ships. Procedures for handling containers would become uniform in all parts of the world, to the benefit of shippers and receivers of goods. Local shortages would be ended; should there be an unforeseen demand for boxes, the neutral pool would have its huge resources on which to draw.

That the existing container-leasing companies already possess most of the capabilities required for the introduction of neutral pools is demonstrated by their growth and the continuing expansion of their inventories. As the use of containers increases and boxes are distributed more widely, the desirability of neutral pools becomes more explicit.

Finally, in analyzing the trends that support the growth of intermodalism, landbridge, and neutral equipment pools, it must be noted that the ocean carrier has come to depend increasingly upon supporting activities such as the leasing companies and the international freight forwarders doing business in cities far inland. These important members of the import-export team provide rapid and accurate information on the progress of a shipment and make possible the effective intelligence network on which the struc-

ture of containerization has been erected. Until recently, the value of these support operations in forecasting demand for shipping capacity, both in terms of containers and ship spaces, was not appreciated. Lacking a scientific basis for estimating their needs, shipowners in the early days of conversion from break-bulk operations depended entirely too much on the anticipated economies of scale, and built more container-carrying capacity than could be justified. This produced the crisis of over-tonnaging which afflicted the major trade routes, especially the North Atlantic and the North Pacific. Inevitably, destructive freight rate wars occurred, and there were many allegations of unethical and illegal practices adopted in the scramble to find cargoes for the big, expensive, and unfortunately superfluous ships. As the crisis eased, everyone involved in the movement of containerized cargo became aware of the new interdependence of all concerned with the new procedures, an interdependence which could improve the whole business of liner shipping.

THE BENEFITS OF CONTAINERIZATION

When international traffic in containerized cargo was introduced in 1966, there were many statements about the great benefits that would redound to the shipper. There also were predictions of new prosperity for shipowners. That these anticipations were voiced in sincere optimism and hope cannot be denied, but in the light of subsequent events, it must be admitted that the reality fell short of the dream.

There were obvious benefits to the shipper. There was, for instance, a marked reduction in pilferage of valuable cargo like television sets. Heavy boxes and crates for goods were eliminated, with welcome savings in the cost of packing for export. In those cases where containers were stuffed properly, a substantial improvement in the percentage of goods delivered in perfect condition was noted. Ship speeds were raised, and until the fuel crisis of 1973 ocean passages at previously record breaking levels became routine. Because containers were handled by machines, the time required to put them aboard ship or to unload them was cut to hours in place of the days formerly needed. Shippers who dispatched small lots (quantities sufficient to fill only a portion of a

container) were encouraged to bring their consignments to the marine terminal, where these lots were consolidated and stowed in the big containers. A wide variety of boxes was developed to meet the different needs of shippers. One company advertised that it had sixteen specialized containers available for its customers.

On the negative side, there were disappointments. Many containers were carried on the weather deck, and sustained damage from heavy seas washing over them. In some cases, whole containers were swept overboard. The broiling sun encountered in summer passages often caused condensation damage to the goods stowed inside the unventilated metal boxes. Insurance rates, which had been expected to decrease, remained relatively stable as the underwriters evaluated their responsibilities under the new system of transportation. Ocean freight rates, which shippers expected would come down because of the lower stevedoring costs, were kept at the old levels, partly as a result of the enormous cost of the new ships. In a period of inflation, however, many rates were not changed, which was, in a sense, a saving to the shipper.

Many lessons have been learned in the years since 1966. Among the most important are these:

1. Container services are of maximum benefit to exporters and importers when there is a two-way movement of goods. Only when the greater proportion of boxes is filled in each direction may the promised advantages of containerization be enjoyed.

A corollary of this principle is that the benefits of container service accrue only on trade routes joining areas ready and able to receive and ship goods in the big boxes. To use containers for one-way movements and to demand that the box be returned to the place of origin without additional charge is to invite the water carrier to establish a freight rate for the container which is high enough to compensate for the expense of the homebound trip. Some exporters have discovered that in these circumstances, it is cheaper to use established break-bulk transportation.

2. Maximum efficiency and hence greatest economy in transporting containers can be attained only by employing ships built and operated for that purpose.

There is a noticeable trend in some trades for new ships to be designed so that two-thirds of the cargo capacity is devoted to

cellularized stowage of containers and the remainder of the ship's space is intended for odd-sized or wheeled cargo. A number of these combination types exists today, and their productivity and profitability are under constant scrutiny.

3. Containers used in intermodal movements must be of uniform, standardized design and sizes which can be interchanged worldwide as the flow of commerce dictates.

4. Essential to realizing the potential of containerization is the growth of neutral pools. Because intermodal movements are economically of great benefit to the commercial world, the demand for a practical, inexpensive, and convenient arrangement for picking up or returning containers can be met most effectively by the establishment of these pools.

5. Computerized control of the entire container transportation system is mandatory. As the neutral pools become more common and use of their equipment grows more general, the need stands out for carriers to have not only the kind of data the computer can provide, but also to interchange information concerning the location, condition, and employment of individual containers.

6. Contributing to the effectiveness of this computer-based system are the elements of the infrastructure supporting container services: the inland freight forwarders and cargo brokers, container depots, maintenance and repair units to keep the containers in good condition, and the other associated activities.

7. Intermodal transportation is demanded by importers and exporters who seek the fastest and least expensive means of getting their goods across the oceans. They want not only the convenience and the rapidity of the landbridge, but they also seek a standardized contract of carriage enunciating the responsibilities of both shipper and carrier at every stage of transportation.

In summary, it may be said that the idea underlying the use of the container in the transportation of goods over the world's sea routes has spread because it met a need, and facilitated international commerce. At the same time, it must be acknowledged that it created many new and unforeseen problems, not least of which was the enormous financial cost of the transformation of ships, ports, and techniques of business. The change has been as profound as that resulting from the development of efficient and economical

steam engines for ships. Equally undeniable is that the methods and patterns of sea-trade have been altered irreversibly, and in every sense a new era has dawned for all who use the sea as an avenue of commerce.

The Ocean Bill of Lading

AS A device of commerce, the ocean bill of lading is of comparatively recent origin. During medieval times, the merchant accompanied his goods and selected his buyers on the spot, exchanging his property for gold or other considerations of value. In these simple circumstances, there was no need for documents that set forth the terms under which the merchandise was transported or would permit transfer of title from one person to another. Methods of dealing through agents, who took the place of the traveling merchant, developed during the heyday of the Hanseatic League (fourteenth and fifteenth centuries), and ways of making loans to finance the movement of goods were formulated by the banking families and cities.

The first document to be used was the receipt given by the carrier to the shipper in which possession of the goods was acknowledged, and it was agreed that delivery would be made only to the person (or agent) designated by the shipper. The most notable early example of a statute regulating these relations is the French Maritime Code of 1681, promulgated by Louis XIV.

The next step in the development of the bill of lading was to establish a contract of carriage. When the merchant dispatched his goods unaccompanied by his personal representative, he and the carrier had to agree upon the terms under which the cargo would be transported and delivered. At first, these terms were those of the trade, but gradually the conditions were written into the codes of commercial law. The English common law principle was that the carrier was liable for right and safe delivery unless he could show that he had been prevented from this by an act of God, an act of the king's enemies, or the inherent vice of the goods. Later it was argued that the contracting parties should be free to negotiate any terms they wished, and the freedom of contract developed in the

second half of the nineteenth century to a point where the carrier accepted the goods to be transported when and how he liked.

Shippers became discontented with the continuing efforts of the carriers to avoid any culpability for damage or loss and agitated successfully for the enactment of legislation to protect themselves and their interests. Today, the laws of sea-carriage in most nations of the world are quite definite in their delineation of the responsibilities assumed by both carriers and shippers.

In tracing the gradual growth of the bill of lading in use today, it is appropriate to cite its importance as a document of title by which ownership of cargo can be transferred from party to party. Whereas the medieval merchant accompanied his goods and sold them in overseas markets, today the cargo often changes ownership several times while in transit. The bill of lading has become identified with the cargo, and because the document is negotiable, title may be transferred from one person to another with little difficulty. The significance of this was not fully appreciated until the advent of fast mail-carrying steamships made it possible for shipping documents to arrive at a market center well in advance of slow cargo carriers. Now that airmail has reduced transmission time to a matter of hours, the convenience of the negotiable bill of lading has greater value to traders. The right of possession of the cargo can be transferred from local seller to domestic buyer and from him to the overseas purchaser without the intermediary seeing the cargo. The clauses of the bill of lading, which affect the insurance coverage as well as the financial aspects of the transactions, therefore become especially important.

THE COMMON CARRIER

A common carrier holds itself out to the public as one which, for a reasonable price, is ready, willing, and able to transport goods for anyone, without discrimination. Essential to the understanding of the status of the common carrier by water are the following facts:

1. The common carrier is required to serve all shippers alike; the carrier may not choose its shippers and must accept, within the limits of the facilities and capabilities of its vessels, the legitimate business of all applicants.

2. The common carrier may limit its business to a certain type of goods to be transported; for instance, refrigerated cargo in a refrigerated ship.

3. The common carrier normally operates over a definite route in repetitious service between stated termini.

4. The common carrier under the general maritime law of the United States (prior to the enactment of the Harter Act of 1893 and the Carriage of Goods by Sea Act of 1936) was required to deliver the cargo at the stipulated destination in the same good condition in which it was received from the shipper unless the carrier could show that the damage or loss incurred by the cargo resulted from one of the following five exceptions: (a) The act of God; e.g., lightning, floods, storms; (b) The acts of public enemies; e.g., a bombing attack upon a ship by an enemy during a period of hostilities; (c) The acts of public authorities; e.g., the confiscation of tainted food by the health authorities of a nation; (d) The inherent nature of the goods; e.g., the perishability of fruit like grapes or bananas; (e) The acts of the shippers; e.g., faulty packaging or improper marking of the shipment.

Even with these exceptions, which ordinarily serve to relieve the carrier of responsibility for loss or damage traceable to these causes, it would be incumbent upon the carrier to show that no negligence on its part had contributed directly to the loss or damage. For example, a consignment of grapes moving from Argentina to the United States in refrigerated stowage might spoil because of over-ripeness at the time of loading. So long as the carrier could prove that the refrigerator compartments in which the grapes were transported had been kept at the prescribed levels of temperature and humidity, and that adequate care had been given to the cargo, there would have been no negligence and the loss therefore would not be deemed to have resulted from any fault of the carrier. The general rule is that when customary stowage results in damage to the cargo, the burden of proof lies with the cargo interests which must demonstrate conclusively that the stowage and care given to the cargo by the carrier were inadequate.

It is noteworthy that the courts, in determining the liability of carriers for damage to or loss of cargo, have extended maximum consideration, under the law, to the welfare of shippers. For exam-

ple, when cargo was damaged in a storm, the carrier was held responsible because the worst of the disturbance could have been avoided by a reasonable deviation from the ship's initial course. In a similar ruling, the court decided that the failure of a master to take adequate precautions to meet a weather emergency constituted negligence within the meaning of the statute. The carrier consequently was held liable for the loss sustained by the cargo.

THE PRIVATE CARRIER

The private carrier, as the name implies, transports the goods of a single person and is privately employed. It is neither ready nor willing to accept cargo from the general public. Some private carriers are owned by the corporation using the transportation; others are under charter or some other form of contract. A tramp vessel chartered to a steel manufacturing company, transporting iron ore to be used by a steel company in fabricating its metal products, is a private carrier. A barge line owned by a common stock corporation, but carrying only the output of a particular mill or refinery, is a private carrier. There is no legal reason why two or more shippers, by mutual consent, may not use the same ship to transport their goods under separate, privately negotiated contracts (or charters) and still have the vessel retain the classification of a private carrier.

The private carrier is not protected by the provisions of the Carriage of Goods by Sea Act, 1936. It is liable to the shipper only for that damage or loss attributable directly to the negligence of the vessel. The private carrier is, however, under obligation to furnish a seaworthy vessel, unless the charter or contract affirmatively lessens this requirement. These provisions of liability would apply only to chartered or contracted tonnage; they would have no significance if the carrier were owned by the same company that held title to the goods being shipped.

LIABILITY OF THE COMMON CARRIER UNDER UNITED STATES LAWS

The liability of the common carrier by water has been limited by two general federal statutes in the United States. The Harter

Act of 1893 was passed to establish the responsibility of the common carrier to the shipper when the goods were transported under a uniform bill of lading. It applied to both the foreign and domestic commerce of the United States and was mandatory upon the "manager, agent, master, or owner of any vessel transporting merchandise from or between ports of the United States and foreign ports." The provisions of the law cover the movement of cargo from the beginning of the voyage (defined as the moment when the anchor breaks ground, or the last line is let go) until the goods are delivered to the consignee when the carrier-shipper relationship is dissolved.

In 1924, a convention was adopted at Brussels, Belgium, by the major maritime nations to serve as the basis for a uniform statute governing, on a worldwide basis, the relationship between shipper and carrier under the terms of the ocean bills of lading. To bring American law substantially into harmony with this convention, the Carriage of Goods by Sea Act was passed by the Congress in 1936.

The Carriage of Goods by Sea Act (often referred to as COGSA) did not fully supplant, nor did it repeal, the Harter Act of 1893. The older law applies, if the carrier does not elect the coverage of COGSA, to the American coastwise and inland waterways trades, and to the period during which the carrier has custody of the goods before they are loaded on the ship and after they are unloaded from the ship. The 1936 law applies to all shipments in the foreign trade of the United States, but only from the time the ship's cargo hook takes hold of the goods until discharge at the end of the ship's tackle in the port of destination ("hook to hook" coverage). Domestic operators in the coastwise service may place themselves under the protection of COGSA by stipulation to that effect in their bills of lading.

Other limitations to the liability of the carrier normally are included in the bill of lading by reference to the appropriate statutes. The principal restrictions are those stating that there is no responsibility for damage by fire, unless caused by the design or neglect of the carrier; the limitation of the liability of the shipowner to the value of his interest in the ships plus pending freight; and the provision that if the owner's interest is insufficient to meet all losses, the shippers will accept a proportionate settlement. It is significant that the operator of a vessel under bareboat charter is given all rights and privileges of a shipowner.

Both the Harter Act and COGSA hold the carrier responsible for loss or damage arising from negligence in proper loading, stowage, custody, keeping, care, and delivery of the goods if they have been packed adequately. Rough handling that causes damage is a fault of the carrier. Poor stowage resulting in the cargo shifting or coming into contact with other commodities that produce taint or odor is considered the shipowner's responsibility. Failure to deliver the stipulated quantity of cargo for which receipt was given becomes the basis for a collectible claim against the carrier.

Liner-service companies fill their vessels with miscellaneous cargo, and often find it advantageous to send ships into ports not on their advertised routes. COGSA permits a "reasonable" deviation to save human life or property at sea. The shipowner, however, must be able to prove that his change of routing for the purpose of loading or delivering cargo or passengers is not "unreasonable." For example, if a vessel were loaded partially at Norfolk and then called at Baltimore for additional cargo, without having included Baltimore in its advertised schedule, the delay of a day or two probably would be construed as reasonable. In one actual instance, a ship advertised to sail directly from Calcutta to North Atlantic ports called at an Indian outport to take on a cargo of castor beans. The loading of the beans required two weeks, thereby delaying by that length of time the ship's arrival in the United States. The court held this to be unreasonable deviation.

The law fixes the shipowner's limitation of liability for goods at the value of the vessel plus pending freight, regardless of the value of the cargo lost or damaged. Should the vessel be lost, liability is limited to the amount of the freight earnings. If it can be shown that the loss or damage occurred with the shipowner's knowledge of factors leading to that loss or damage, the carrier's petition to limit his liability will not be granted by any court.

Pursuant to the provisions of the 1936 act, the carrier is responsible for the value of the individual package or other customary shipping unit up to a maximum of \$500, unless a higher valuation has been declared in the bill of lading, and correspondingly higher freight rates have been paid by the shipper. The claimant must be able to prove that the damage or loss occurred while the package was in the custody of the carrier. If the packages were shipped in apparent good order and condition and the con-

tents were found to be broken or damaged upon opening the containers at destination, the claimant must prove that the goods actually were in sound condition when they were delivered to the carrier. If the carrier can show that the damage was not due to his negligence, he may be absolved of liability. Explosive and flammable cargo may be destroyed or landed short of destination without the carrier incurring any liability, if, in the opinion of the master, this action is necessary to assure the safety of the ship.

COGSA prohibits shipowners from phrasing their bills of lading to reduce liability under that act. The shipowner may not limit responsibility for the negligence of ship's personnel with regard to loading, stowage, custody, care for, or delivery of the cargo. A list of exonerations from liability, however, is included in both COGSA and the Harter Act, and these exonerations have been sustained by the United States courts. The net effect has been to change the status of the common carrier in American waterborne commerce. No longer is that carrier the absolute insurer of the goods, subject to certain specified exemptions set forth in the general maritime law. Instead, the carrier has been placed in the category of one who sells only careful transportation, governed by special and very definite laws of sea-carriage.

THE BILL OF LADING AS A CONTRACT OF CARRIAGE

The bill of lading, as a contract of carriage, sets forth the terms of the agreement between carrier and shipper under which certain goods are accepted for transportation between named ports in a designated ship with a stipulated sailing date in exchange for a financial consideration. The contract becomes effective when it has been signed by the master (or his deputy) and accepted by the shipper.

Pursuant to the provisions of both the Harter Act and COGSA, the carrier is required to issue a bill of lading as a receipt for the goods, and the bill constitutes prima facie evidence of such receipt. The bill must contain information as to the marks and numbers of the packages, the quantity of packages or the weight or measurement of bulk cargo, and any exception to the basic statement that the goods were received "in apparent good order and condition."

The carrier must provide a “shipped” or “on board” bill of lading (meaning that the cargo has been stowed aboard ship) on demand of the shipper.

In promulgating the terms of the contract, the carrier often includes many provisions not covered by existing statutes but recognized as valid contractual conditions. For example, in a clause describing the ways in which goods may be stowed in the vessel, the term “under-deck stowage” is extended to cover the use of poop, forecabin, deckhouse, shelter deck, or other covered space. Cargo will be given ordinary stowage unless the bill of lading stipulates that it be refrigerated or heated; failure of the shipper to demand this care exempts the carrier from liability for resultant damage.

The master is granted authority to discharge cargo short of the agreed port of destination when, in his judgment, there is the reasonable apprehension of danger to his ship or cargo. This apprehension cannot be based on rumors or imagined dangers; it must be founded on actual or substantial peril. Once cargo is discharged under this provision of the contract, the shipowner has no further responsibility. The shipper or consignee must accept delivery and assume any expense which may be incurred as a result of this delivery short of destination.

Another example of the clauses included in the contract is the reservation to the carrier of the right to effect transshipment to other carriers of cargo destined to ports not on the ship’s route; this “on carriage” is to be executed pursuant to the terms of the connecting carrier’s bill of lading, regardless of whether those terms are easier or more onerous than those of the originating carrier’s contract.

Only a few of the clauses included in a bill of lading have been singled out for comment. They are sufficient to indicate the nature of the contract. While each carrier will write his own contract, there will be similarity in the basic provisions of those bills that are issued to shippers in the foreign commerce of the United States.

It is important to observe that the information and statements contained in the bill of lading are accepted in these three particulars:

1. The accuracy (truth) of the loading tally; the statement that the cargo has been “shipped” in a named vessel; or the declaration

that the cargo is “on board” a particular ship. There is no allowance for shortages. A deficiency (shortage) in one lot may not be made up by a surplus (overage) in another lot of identical goods. The quantity of goods delivered to the consignee must correspond to the tally stated in the bill of lading.

2. The truth of the statement that, unless otherwise attested by the carrier’s notation on the bill of lading, the cargo gives the external appearance of being in good order and condition.

3. The truth of the date of loading. A “shipped” or “on board” bill of lading issued by the carrier states that on the date of the bill of lading, the cargo actually was stowed aboard the vessel. This may be significant in proving that the seller complied with the terms of the contract of sale, which often stipulate that shipment must be made by a certain date.

THE BILL OF LADING AS A RECEIPT

Current shipping practice in the United States provides that, when the shipper delivers the goods to the terminal for loading aboard a vessel, he be given a certificate known as a “dock receipt” (in some trades this is called a “mate’s receipt” or a “boat note”) which acknowledges that the cargo has been received and is in the custody of the carrier. This receipt is furnished to the shipper after the individual packages making up the consignment have been examined by one of the carrier’s inspectors (checkers) to detect external or superficial damage or shortages, or other exceptions to the statement of “apparent good order and condition.” The shipper provides the description of the goods, the numbers of packages, and the weights. On the reverse side of the receipt, the checker inserts the measurements of the packages and the “exceptions” he has noted. At the bottom of this sheet, the checker writes his initials, and then passes it to the designated official (usually the receiving clerk) for signature and return of the original to the shipper. The dock receipt is the carrier’s permanent record of the exact quantity and condition of the shipment at the time it came into his custody. It serves as a temporary receipt until the freighted and signed bill of lading is obtained from the carrier; at that time, the original dock receipt is surrendered by the shipper and is filed with the other papers relating to the shipment.

The bill of lading is completed by the shipper on forms supplied by the carrier. COGSA requires that the shipper furnish in writing certain data concerning his goods, and current practice dictates that the shipper insert these specifications in that section of the bill of lading headed, "Particulars Supplied by the Shipper of Goods":

1. The exact name of the commodity, i.e., granulated sugar, dried kidney beans, industrial refrigerator cabinets, etc.
2. The details of the shipment, showing the types of packages (bags, bales, boxes, crates, or whatever is appropriate), including the number and identification marks of the packages.
3. The gross weight of the goods.

The shipper "shall be deemed to have guaranteed to the carrier the accuracy at the time of shipment of the marks, number, quantity, and weight, as furnished by him; and the shipper shall indemnify the carrier against all loss, damages, and expenses arising or resulting from inaccuracies in such particulars" (COGSA, Section 3 (5)).

Normally, the contents of each package are not examined by the carrier; the shipper is obligated by law to provide an accurate description. Most carriers, however, reserve the right to open the containers for verification should there be suspicion that the contents are not as described in the bill of lading. In accepting the goods in "apparent good order and condition," the carrier is bound to declare what is the external appearance of the cargo. Where bulk cargo weight is ascertained neither by carrier nor by shipper (as in the case of bulk grain loaded at a public grain elevator), the master may insert that fact in the bill of lading and so be freed of the obligation to discharge the weight stated in the bill of lading.

Since the shipper guarantees to the carrier the accuracy of the marks, number, quantity, and weight of the goods, in case of a claim for nondelivery of a part of the shipment, he must be able to prove among other details that the quantity stated on the bill of lading actually was shipped in good condition. One of the defenses open to the carrier is to show, in his behalf, that the packages were delivered in the same quantity and condition as shipped, and that there was no negligence on his part in the loading, handling, stowage, carriage, care for, and discharge of the goods. Under COGSA, the

statute of limitations is fixed at one year; notice of loss or damage must be given not later than three days after the cargo has been delivered to the consignee, and suit for recovery of damages must be filed by the claimant within one year after delivery of the goods.

THE BILL OF LADING AS EVIDENCE OF OWNERSHIP

In medieval and early modern times, when the merchant accompanied his goods, he always could assert his ownership. As this arrangement yielded to a less direct contact with the cargo, some method of transferring title had to be devised. The simplest means of accomplishing this was to send the consignee a copy of the bill of lading which had been issued by the ship to the merchant. This might be done in a special envelope entrusted to the master, or it might be dispatched by a faster vessel so that the consignee would be awaiting arrival of the ship with the goods. If there were some desire to protect the goods against the possibility of an impecunious purchaser taking possession, the merchant's agent in the port might be the addressee of the bill of lading. The carrier agreed to transport the goods from Port A to Port B and there to deliver them to the shipper, or to the agent of the shipper, or to the person designated by the shipper to receive them, or to the order of the consignee.

The bill of lading in this manner came to represent the goods in respect to which it had been issued. The possessor of the bill of lading could pass authority to accept delivery to a third party by a statement written on the bill that he transferred his interest to that party. Endorsement and transfer by the owner of the bill of lading actually transferred legal ownership and the bill of lading became a document of title to the goods.

THE NEGOTIABLE OR ORDER BILL OF LADING

In foreign commerce today, the great bulk of bills of lading is issued by carriers to the order of the consignee which means that the carrier, shipowner, charterer, or master will deliver the goods at the port of destination not solely to the named con-

signee, but to any person designated by him. Through the use of the word, "order," the bill of lading becomes more than a receipt from the ship for the goods, and more than the contract of carriage. The bill, by the insertion of this word "order," possesses a legal and commercially important characteristic in that it becomes a transferable document of title. The law governing bills of this type expressly rules, in Section 3, that:

A bill in which it is stated that the goods are consigned or destined to the order of any person named in such a bill is an order bill.

Any provision in such a bill, or any notice, contract, rule, regulation, or tariff that it is non-negotiable shall be null and void and shall not affect its negotiability within the meaning of this Act unless upon its face and in writing agreed to by the shipper.

The legal ownership (the property interest in the goods described in the bill of lading) can be transferred from the consignee named in the bill to any other persons, and by them to still other persons, without any of them seeing the goods or having physical possession thereof. This transfer of possession is accomplished initially by the consignee signing his name on the bill of lading, thereby converting the bill into a "bearer" document. A refinement of this is for the consignee to direct that the goods be delivered to a particular person, and then to affix his signature to the endorsement.

The bill of lading thus becomes in practice a negotiable instrument. The endorsees and holders of the bill are entitled legally to rely upon the tally and upon the statements in the bill of "apparent good order and condition." In the eyes of the United States courts, the bill has been fully negotiable since 1916 when the Federal Bills of Lading Act was passed.

There are advantages to the seller who uses the order bill of lading. For instance, the merchant who ships his property under a negotiable bill of lading can insist on payment for the goods before endorsing and turning over the bill of lading to the purchaser. Similarly, by retaining possession of the bill, a bank which advanced money against goods represented by the order bill has security for its financial interest so long as it retains possession of the bill. Order bills of lading should be cancelled or repossessed by the carrier at the time of delivery to avoid claims by those who might have

Lykes Bros. Steamship Co., Inc.

has been informed that one or all of San Francisco Bills of Lading SFR. 1, SFR. 2, SFR. 3 issued in Mombasa for the S.S. Leslie Lykes, Voyage 94, have been stolen, and further it has been reported that attempts have been made to sell these negotiable documents in the Nairobi area. The public is hereby advised that the above indicated Bills of Lading stand null and void and therefore of no value.

Advertisement in *New York Journal of Commerce*, September 30, 1985, p. 1-B.

purchased the bill of lading in good faith from an unscrupulous person.

THE STRAIGHT (NONNEGOTIABLE)
BILL OF LADING

Markedly in contrast with the order bill of lading is the "straight" or nonnegotiable bill, under which the carrier accepts cargo for delivery to the named consignee only. Transfer of title to the property cannot be effected. Possession of the goods can be obtained by the consignee on presentation of a copy of the bill together with satisfactory identification of the agent. It is preferable to have the original bill presented at the time delivery is requested.

Except for the fact that the words "or order," following the name of the consignee, are deleted, the physical appearance and the text of the two bills of lading are identical. Some steamship companies use a paper stock of different color for ready separation of the straight from the order bills. Not infrequently, a conspicuous

heading on the straight bills will be imprinted to prevent confusion on the part of the user.

Because of the limitations on the use of the straight bill of lading, normally it is employed by shippers who likewise are the consignees (e.g., a manufacturing company sending components to its assembly plant overseas; or consignments dispatched to purchasers who have paid in advance for the goods).

THE BILL OF LADING IN FOREIGN TRADE FINANCING

Payment for goods shipped in export trade is more complicated than in domestic commercial arrangements and often requires the services of a financial agent like a bank. Without attempting to describe the intricacies of payment for goods moving in international commerce, the following sketch will indicate the role of the negotiable bill of lading in the financing of foreign trade transactions.

Assume that a consignment of ten typewriters, worth \$250 each in New York, is sold to a purchaser in Lima, Peru. Shipping costs come to a total of \$100 and the insurance premium on the whole transaction is \$50. The amount of money involved in a "cost, insurance, and freight" quotation is \$2,650. To assure prompt payment in American currency to the seller, especially on the part of buyers who reside in countries where collection in dollars may be difficult, it is customary for the buyer to arrange with his bank for a letter of credit to be issued in favor of the seller for the total value of the shipment (in this case, \$2,650). The letter of credit is a certificate that the amount of money stipulated therein has been placed on deposit with an American bank and is available to meet the seller's charges.

To collect the value of the shipment, the seller prepares a "bank draft," a form of check against the deposit established for the use of the seller and witnessed by the letter of credit. He attaches this draft to the other documents needed to support the transaction: the three original bills of lading freighted and signed by the carrier attesting to the fact that the goods have been loaded in the designated ship prior to a specified date; the certificate of insurance, protecting the purchaser against loss in the event of disaster during

transportation; the seller's invoice, showing the value of the merchandise (\$2,500), the cost of insurance (\$50), and the ocean freight (\$100); and the letter of credit.

The order bill of lading, to simplify the process of collection for the seller and also to safeguard the ownership of the goods, would be made out so that the name of the seller would appear on the lines for both shipper and the person to whose order the shipment is consigned; the actual purchaser's name would appear on the line designating the person to be notified when the ship arrives at her destination. The seller would be required by the bank to endorse the three original bills of lading to the order of the bank, and thus to transfer title to the goods to that financial institution. All the documents necessary to support payment to the seller likewise are surrendered to the bank. The seller, on receipt of the money, drops out of the transaction and the bank assumes temporarily the role of owner of the goods.

The seller's bank in New York, after taking possession of the bills of lading, insurance certificate, invoice, and bank draft, would endorse the bills of lading to the order of the buyer's bank in Lima. All documents then would be sent to Peru, and the Lima bank would in turn assume the role of temporary owner of the goods. As a matter of routine, and to explain the existence of three original signed bills of lading, one copy is sent by airmail, one follows by steamship mail or a second airmail carrier, and one remains in the sender's file for use in the event that both bills should be lost in transit.

On the arrival in Lima of the documents from New York, the bank notifies the purchaser that he can come to the office and receive the bills of lading endorsed to his order. The Lima bank, having completed its role in the transaction, drops out of the picture, and the buyer is free to proceed to the carrier's office. Here he will exchange the original bill of lading for a delivery order and, on the strength of this last document, will present himself at the marine terminal to accept delivery of his property.

The carrier must insist that the original order bill of lading be surrendered at the time the goods are turned over to the consignee. The bill should be cancelled, and filed along with the executed delivery order, as proof that the terms of the contract have been fulfilled by delivery. If either of the other two originals should be

presented at a later date, the carrier properly can refuse to honor them and support its position by producing the cancelled bill against which delivery was effected.

It is obvious that the nonnegotiable bill of lading, because it does not permit transfer of title, cannot be used in a transaction of the nature just described.

How Freight Rates Are Made

AN AMAZING assortment of goods is moved over the world's ocean trade routes. Of necessity, the carriers charge for the service they render. The charges vary as widely as the cargoes since they mirror both the shipowner's costs and the special conditions prevailing on the trade routes the ships traverse.

THE THEORY OF FREIGHT RATES

Ocean freight rates may be described as the prices charged for the services of water carriers. Each ship operator develops his own rates, usually without consultation with the shippers. The charges reflect the cost of providing the carriage, the value of this service to the owner of the goods, the ability of the merchandise to support the expense of transportation, and economic conditions in general.

There are no exact pricing formulae applied uniformly to the various items moving in transoceanic commerce. In the international movement of goods, there is little interference from artificial factors, such as regulation by governmental agencies. To a noticeable extent, ocean freight rates are the result of the workings of the law of supply and demand. In tramp shipping, especially, it is possible to observe how these factors influence the rise or fall of freight rates from day to day and from cargo to cargo.

Tramp ships transport, in shipload (or "full cargo") lots, commodities which, like coal, grain, ore, and phosphate rock, can be moved in bulk. The fact that usually only one shipper and one commodity are involved simplifies the establishment of a freight rate. All the costs of running the ship, handling the cargo, and paying port fees and harbor dues must be added to the capital charges of ownership and the expenses of administration and over-

head. Against this total is set the number of tons to be hauled, and the resultant figure is what must be earned by the tramp to break even on the contemplated voyage. There is no necessary relationship between the rate on one route and the rate on another; the circumstances of every individual voyage and each commodity to be carried will fix the amount of money the shipowner should demand for the use of his vessel if his rate is to be compensatory. A margin for profit may be added if competitive conditions permit.

Fundamentally, the tramp owner strives for a rate low enough to attract the shipper but high enough to pay the cost of service and to allow a reasonable profit. It may be an advantage to him that the rate quoted applies to just one voyage and is binding only for the duration of that voyage. If he finds it economically impractical to continue in operation, he has the privilege of retiring his vessel to a quiet backwater and waiting until the financial storm clouds are swept from the commercial horizon.

The tramp operator has no stake in the long-term goodwill of those who might send the same type of cargo over the same stretches of the ocean time and again. He accepts those offers which seem to him profitable at the moment and dispatches his vessels accordingly. Ever mindful of the existence of competition, his rates exemplify the workings of the law of supply and demand. When business conditions permit, he selects that cargo which assures the largest profit. When adversity threatens, he accepts those charters which minimize anticipated losses. When he has the choice of evils, he contrasts the cost of temporary lay-up against the loss which operation might produce and selects the less costly alternative. He bows to the inevitable with what grace he can muster and hopes that tomorrow will bring more favorable circumstances.

Liner-service companies, on the other hand, depend for their financial prosperity upon the accumulated goodwill of shippers who, through the years, come to rely upon the regular and continued operation of the company's fleet. Temporary withdrawal of service, whenever economic conditions are not favorable, is unthinkable.

The liner will sail on her regular run, whether full or not. She will carry a wide variety of commodities, each with its own peculiarities, in quantities which can be estimated in advance more or less accurately, but never with complete certainty. The ports of call are

known far in advance of sailing, and the total expense of working the ship throughout her voyage can be calculated with acceptable precision. Since, however, the exact distribution of tonnage, commodity by commodity, varies with every trip, it is not possible to establish a rate that reflects the cost of transporting a single ton of a particular commodity as closely as does the tramp owner's computation.

This is not to suggest that the liner-service operator cannot compute with nicety the costs of owning and operating his ships. He knows to a fraction of a cent his daily costs for amortization and interest on borrowed capital, and what administrative expenses he must charge to individual voyages. In the same manner that his counterpart in the tramping trade is able to fix individual rates, the liner owner can determine what he should charge per ton to carry a single commodity when it is offered in lots sufficient to fill one of his ships. The difficulty arises when the liner operator is forced to compile a list of charges for transporting hundreds of different commodities not in the ship sailing tomorrow, but in ships departing at weekly intervals during the next year or even over a longer period.

Because the ship is committed to sail on a fixed date, there may be a tendency among some shipowners to say, "It makes little difference what I charge for individual commodities, so long as the aggregate revenue is enough to cover my expenses." The contrary view is, "Unless I know that each ton of cargo will pay its full cost and earn a profit, I will not accept it."

Is there an intermediate point in making freight rates for liner-service shipping? There is.

From his experience, the liner-service operator knows approximately what is going to move, voyage after voyage, and has a good idea of the tonnage that he can expect. He must estimate the overhead to be charged against each commodity and the out-of-pocket costs of handling them at ports of loading and discharge. There must be an apportionment of revenue to defray the expense of vessel operation. Finally, a small profit should be added to compensate the owner for the risks he assumes as well as for his skill and enterprise, for by providing transportation he enhances the value of the goods. He is justified in assigning a reasonable value to this real, albeit intangible, contribution.

Underlying these general principles are certain factors that influence, in one way or another, the establishment of freight rates for individual commodities moving in liner-service vessels.

The first of these factors is that freight rates should be reasonable to shipper and to carrier. The shipper should be satisfied that the money he pays for transportation will not drive the price of his goods above the competitive level of the markets where he will trade. There are times when it becomes difficult to convince the shipper that any freight rate is reasonable. If the exporter or importer is trying to compete with goods from sources closer at hand, he will consider that the cost of transportation, no matter how low, is nothing less than a barrier to trade.

In determining what is reasonable as a charge for transportation, the shipper's complete indifference to the financial condition of the carrier must be remembered. The sales appeal of a given article often is set by the price which must include the expense of transportation. Should the margin between the seller's total costs and the market price be too narrow to leave a profit, the seller attempts to convince the carrier to reduce his freight rates. His argument always is, "If you don't come down, you will lose all my business. If you will help me to keep my price at the competitive level, you will benefit by my continued patronage. After all, your ship is going to sail on this route anyhow, so why not make this concession?"

The second factor which influences the establishment of freight rates is competition. If a carrier sets rates higher than those of his rivals, he may lose his patrons to shipowners whose services are offered at lower prices. If, however, he should quote a figure which nets no profit, he may be too successful and be overwhelmed by the volume in which this commodity is offered. Whereas he could afford to handle a few tons on each voyage, he cannot permit this item to crowd out other commodities on which the rate is remunerative.

There also is the rivalry between tramps and conventional break-bulk liners for certain items that can be loaded on either type of ship. These are commodities like grain which move in both shipload and part-shipload lots. The rates technically are "open," meaning that there is no uniform charge by the liners. Instead, the "regular traders" compete with the tramps by making bids on the open market. Tramp operators complain, with considerable justice

in their argument, that liners seeking these "berth cargoes" depress the market rates on these items to an unprofitable level. They also contend, again with some basis in fact, that liners in need of ballast, which is expensive to stow and to unload, often seek these berth cargoes as substitutes for ballast, and will accept any freight rate that will bring in enough revenue to defray the actual cost of handling this form of ballast.

Another type of competition is the rivalry between ports. In their never-ending search for cargoes to move through their facilities, the various ports of an area stress modernity of piers and wharves, their intraport systems of roadways and railroads, and frequency of sailings to all parts of the world. The shipper seeking the best bargain may be intrigued by these arguments but is hardly likely to make a change unless he stands to gain financially by doing so.

A good deal of cargo for overseas markets originates with shippers in inland cities who have the option of using one of several ports. The rail or truck routing that gives him the lowest cost of transportation normally is preferred. He may develop the habit of moving all goods from City X to Seaport A because the inland freight rate is a few cents per hundred pounds cheaper than to Seaport B. Rail or truck lines operating only between City X and Seaport B must compete for the business by bringing their rates to the level of those applying to Seaport A. This "port equalization rate" applies only on cargo with foreign destinations and actually represents a reduction in revenue on the part of the domestic carrier(s). Where the steamship line maintains service only from Seaport B, it is possible that it may develop special "export" rates in conjunction with the inland carriers, under which each of the modes of transportation offers a reduction provided the business is routed through Seaport B.

Steamship lines serving only the smaller ports in the area may seek to induce shippers to use their service by setting rates somewhat below those applicable to the major cities. In recent years this practice has become exceptional because freight rates to a given destination have been equalized, in many instances, for all ports in a large geographic area. The fact is mentioned here, however, because a new operator, striving to attract patronage, might fix his rates with this thought in mind.

Some commodities are found naturally in widely separated

geographic locations. The sellers of these commodities compete with each other all over the world. Buyers seek the lowest cost. If the combination of foreign price plus transportation produces a figure higher than the quotation from a rival area, the buying pattern may be changed to take advantage of the more favorable price. These examples make this clear:

Tin originates in Malaya, Bolivia, Indonesia, and the Congo. When exporters of Bolivian tin raise their prices, or the carriers of tin from Bolivia to the consuming market establish a new freight rate, the users of tin compare the new total with that of the mineral from the other sources and place their orders accordingly.

Copper is imported to the United States from Chile, Canada, and Central America. The difference of a cent a pound in the landed price in New York often determines whether the movement will be from one area or another.

No factor of competition is more difficult to assess in the analysis of freight rates than that which results from the struggle of different geographic areas for the same market. This emerges when the trade from two widely separated manufacturing nations comes into direct conflict in a consuming market thousands of miles from both producers. For instance, the United States and Japan both send large quantities of the same kinds of machinery to Chile. The Chilean buyers consequently are in the position of pitting Americans and Japanese against each other and taking advantage of whatever bargains they can obtain. While ocean carriers serving the two supplying nations are not faced directly with a demand to meet a freight rate established by the purchasers, they are aware that they are in competition. The pressures are felt from both suppliers and importers, and therefore the carriers must establish (or maintain) a rate acceptable to the consignees. That the freight rate may not be profitable to the ship operator is of no concern to the buyer, who is interested only in obtaining his goods at the lowest possible total cost.

From the foregoing, it might be deduced that freight rates for particular commodities are the same regardless of whether the goods are carried from A to B or from B to A. Logical though this might be, it is not always true in international shipping. In proof of this, the Federal Maritime Commission announced in 1970 that freight rates for a number of items moving from the United States

to Germany were from 23 percent to 200 percent higher than for the same items being transported from Germany to the United States.¹ Several years before this, freight rate analysts from the United Kingdom told a Congressional committee that disparities existed because, on the route from Europe to the United States, many tramp steamers had empty space which they offered to shippers of package cargo at very low rates. To hold on to their business, the liner operators had to reduce their rates to the competitive level. In the outbound trade from the United States, however, the tramps were not available because they had obtained full loads of commodities moving in bulk (principally grain). The lack of competition therefore gave the liner companies the opportunity to set their freight rates at levels noticeably higher than was true in the westbound ships.²

While it may be true that tramps and liners do carry the same types of cargo, and therefore compete with each other, it is important to observe that this argument is subject to question in the container age. To an ever greater extent, shippers are using large containers to send their goods across the seas and are requiring that these containers be transported only in fast ships especially designed for their safe carriage. The typical tramp steamer, even of the newest type, is not able to meet these demands. It is unlikely that the argument advanced by the European experts will be supported in fact for the indefinite future.

While the rivalry between containerships in liner service and break-bulk dry cargo tramps may be declining, competition remains very keen between the operators of the big containerships. The two most important trade routes—transpacific, linking the Orient to the west coast of the United States, and transatlantic, connecting Western Europe and the United Kingdom with ports of the United States on the Atlantic and Gulf coasts—have become very attractive to the big operators, and both routes in mid-1985 were suffering from the resulting excess of capacity compared to the quantity of cargo moving. A contributing factor to the problem was the high value of the United States dollar relative to other currencies. This meant that imports from abroad were comparatively cheap, and good sales were made easily, whereas exports from the United States were expensive (in terms of the foreign buyers' currencies) and therefore not in great demand. The imbal-

ance of trade was reflected in the number of empty containers which had to be transported at minimal freight rates or free of charge from the United States back to the overseas areas. It was a logical step, therefore, for carriers to add whatever they could to the freight rates for cargo destined to the United States in the hope thereby to defray some of the expense of repositioning the empty containers.

The aspiration of steamship operators to run their vessels between a single loading port and a single discharge port is not feasible on a long route, such as the round-the-world service maintained by both the United States Lines and the Taiwan-based Evergreen Lines. The practical compromise adopted by these carriers was the selection of a limited number of ports serving a productive hinterland, and limiting calls to those ports, denominated in the trade as "load centers." Containers were brought to and assembled at these centers by rail, truck, and barge. Inbound containers were distributed by the same means to destinations within a specified radius of each load center. The cost of pickup and delivery was included in the ocean freight rate. Because the introduction of the load center idea was effected very recently, the benefits or drawbacks had not been evaluated fully at the time these lines were written.

Other problems confront the rate makers. A consistent demand exists for certain commodities which are essential for many purposes in the world economy. Individually, these commodities do not move in quantities large enough to obtain the low cost of transportation associated with shipload lots. To put these items into the hands of their users, they must be dispatched by liner-service ships, but their value is so low that they cannot be charged the minimum cost of their transportation. Facing this dilemma, the rate maker has no choice but to set rates that permit their movement even though it means the carriage is performed at a financial loss to the shipowner. To offset this difficulty, the only solution is to set rates for items of high value which will absorb the loss incurred on the low value commodities.

"We charge what the traffic will bear," observed the steamship rate maker as he determined that electric washing machines would be assessed \$2.50 per cubic foot and bagged cement would be charged \$1.00 per cubic foot for transportation from San Francisco

to Valparaiso, Chile. The landed cost of the washing machine would be increased by about twelve percent; the price of cement after discharge from the ship would be exactly double the United States figure. Doubtless those who would buy washing machines would not be deterred by the comparatively small markup, but the steep rises in the landed price of cement would take it out of the market. If it actually cost \$1.25 per cubic foot to carry these two items, the excess above cost earned from the washing machine will offset the loss incurred in handling the cement. Illogical and superficially indefensible as this system may appear to be, it is the only workable method of providing the transportation for those commodities that must move, but cannot afford to pay the cost of liner service.

It follows (quite logically, in this case) that if "cement" is to move at all, it must be accompanied by a reasonable tonnage of "washing machines," so that the total "mix" of cargo will produce a profitable load. Ideally, if there is an equal balance between the two commodities, there will be a profit of \$1.00 per cubic foot, or \$40.00 per measurement ton loaded. When the balance is destroyed, and twice as much "cement" as "washing machines" is stowed in the ship, the shipowner loses substantially. The constant concern is that the ship be operated at a profit; this is matched by the ever-present obligation to transport nonprofit-producing commodities typified by the "cement" of this illustration. "To charge what the traffic will bear" is to discover ways by which to meet the conflicting demands of commerce without unduly burdening any one segment.

Because exploitation of the shipper seems to be implied in the understanding of "charging what the traffic will bear," it is preferable, given the economic conditions existing in the last quarter of the twentieth century, to think in terms of the value of transportation service in the interchange of goods. It is almost axiomatic that this interchange, or trade, takes place only when the object traded commands a higher price at the market than it does at the point of production. A corollary of this is that the market price is fixed by the competition of similar goods from different sources. It follows, therefore, that in modern commercial practice transportation has become an integral part of the distribution process established by the manufacturer or exporter. An efficient process reflects cooperation and coordination in which every participant is aware of the

impact of transportation upon the final demand for the object offered for sale.³

In recent years, as shipping capacity has exceeded the demand for cargo space, the ability of the carrier to assess rates on a unilateral basis has been eroded. Shippers have more influence on associations of carriers, and have become more insistent that their ability to pay freight rates be considered. An example of shipper demand was furnished in June 1985 when the Transpacific Westbound Rate Agreement carriers announced that they would raise the rate by \$100 on 40-foot containers moving from the Orient to the United States. This was to be effective on July 1; on September 1, the rate would be increased again by \$200. In a meeting held in Hong Kong, shippers protested so vigorously that the carriers first postponed action until a later date, and then cancelled the proposed new rates.⁴

Shippers are concerned that their cargoes be delivered in good condition and on time. They have indicated, both by their actions and in expressions of opinion, that the price of transportation is of less significance than the quality of service. This was made known in a survey of some eight hundred shippers which was conducted in 1971. Those who were sending goods of high value declared that the most important aspects of ocean transportation, in order of priority, were: (1) date of ship's sailing and scheduled arrival at port of discharge; (2) duration of the voyage from loading port to discharge port; (3) the reliance which shippers could place on the carriers' published schedules; (4) quality and availability of containers; (5) the reputation of the carrier, based on previous experience or history; and (6) the freight rate charged. Shippers of goods with low value understandably gave higher consideration to the freight rate, but placed it second to the date of sailing and arrival, which was considered most important in the selection of a carrier. In third place was the reputation and history of the ship operator, followed by the reliance which the shipper could place on published schedules. Transit time was in fifth place, and the availability of containers was given lowest priority.⁵

The shipowner is faced with the necessity of earning maximum profit from the goods offered on his trade route, while interfering as little as possible with their flow. In theory, he seeks to carry a minimum of those cargoes that pay either noncompensatory or

break-even rates, and he looks always for those items on which there is assurance of good profit. All the unprofitable commodities, even in theory, cannot be excluded. Frequently the low-grade raw materials are essential to the manufacture of goods which, in their turn, offer handsome profits when later transported across the seas. Since the carrier depends for much of the corporate profit on continued trade in these manufactured articles, he must transport the unprofitable item at whatever rate the shipper of that commodity can afford.

Carriage by water still is the cheapest form of transportation ever devised by man. In terms of the cost per pound of most commodities moved across the seas, the charges assessed by the liner operator are insignificant in proportion to the sales price at the overseas market. It is substantially correct that carriers would be happy if profits from freight rates averaged one and a half to two cents a pound. For those commodities moved in tramp ships, the return to the ship, in cents per pound, is almost ludicrously low. A recent sampling of charter rates during a period of comparatively good business showed that it was possible to send a pound of sugar a quarter of the way around the world for less than half a cent. One authority computed that to raise tramp rates on grain by \$1.50 a ton, which might be very welcome to many owners, would increase the cost of the flour used in a one pound loaf of bread by approximately one-sixteenth of a cent.

When freight rates are considered for what they really are—the earning power of a ship engaged in hauling the vital commodities of the world—they are seen to be aids, rather than barriers, to international trade.

FACTORS IN OCEAN FREIGHT RATE MAKING

The ideal of the ocean carrier is to foster international trade and to build up the tonnage of cargo carried by his ships. Although more attention may appear to be given to the large-scale shippers of goods, the small businessman actually is not ignored because the rates quoted by liner companies are the same for all shippers, regardless of the quantity of goods offered. In practice, the small-scale shipper benefits from the ability of the large-scale

shipper to charter tonnage to handle his goods whenever liner rates become excessive. Should tramps be brought into the trade by the big shippers, the only recourse for the liner operator is to reduce his rates sufficiently to make the employment of chartered tonnage unnecessary. The small shipper, of course, gains by the resultant lowering of liner-service freight rates as much as does the big user. Incidentally, this fact illustrates the method of controlling rates by the use of tramps, and how tramps become the great regulators of ocean freight rates.

The liner-service operator may increase the cost of sea-carriage only with extreme caution. Though he may need more revenue, the shipowner cannot take the risk of raising rates to levels which might throttle trade or invite new competition by making profits so attractive. The usual method is for the carrier to increase rates gradually, by increments of five to ten percent, and to be guided step by step by the reaction of the shipping public.

It cannot be stated too often that the economic survival of the carrier necessitates that the rates he establishes be adequate to cover the expenses he incurs. These expenses include both the investment (capital charges) and the operating charges. For example, a ship was purchased for \$8,500,000 with the expectation that she would have an operating life of twenty-five years. She must earn \$340,000 each year for amortization, in addition to the cost of interest on the money, plus a reasonable contribution to the operator's fixed costs, arbitrarily assumed to equal a further \$500,000 annually. The ship was scheduled to make one round voyage each month. Fixed charges assessed against each voyage were \$70,000. The ship regularly carried 15,000 revenue tons, and the capital charge against the vessel was fixed at \$4.67 per ton. Operating costs varied from voyage to voyage, but provision had to be made for wages, fuel, stevedoring, pilotage, and port dues. The aggregate of these items came to \$30.00 per ton. The minimum rate to break even therefore had to be \$34.67 per ton before computing any profit.

Certain criteria have especial importance in setting rates. Foremost among these criteria is the value of the article. This can be illustrated by comparing the value and carrying charges of two typical commodities moving in American foreign commerce. One is cement, which has a low intrinsic value. The other is a household

washing machine, which has a relatively high intrinsic value. It would be unsound to attempt to assess a freight rate for both of these items based solely on an identical percentage of value. For instance, let it be assumed that the cement has a sales price at the ship's side in New York of \$40.00 a ton, and the washing machine is worth \$450.00. The general cargo rate is \$1.08 per cubic foot for transportation from New York to Buenos Aires. Cement stows at 27 cubic feet to the weight ton; the washing machine weighs 300 pounds and requires 28 cubic feet of ship space. If the general cargo rate were applied to both items, the cement would be charged \$29.16 or 73 percent of its value; the washing machine would be carried for \$30.24, or 7 percent of its value.

If the percentages were reversed, the shipowner would be unable to accept cement at \$2.80 per ton, or 10.4¢ per cubic foot; the exporter of the washing machine would find the rate of \$328.50 (equal to \$11.73 per cubic foot) an impossibly high tariff. Some adjustment reflecting the intrinsic value of the goods would be necessary if all parties were to be satisfied.

The compromise was to carry the cement for \$12.15 per weight ton (equal to 45 cents per cubic foot of space used), and to charge the washing machine \$56.00 (equal to \$2.00 per cubic foot of space used). The cement was delivered in Buenos Aires with a value at ship's side of \$52.15; the washing machine had a landed value of \$506.00. Ship operating cost was \$1.05 per cubic foot; the 95 cents excess from the washing machine was applied to the 61 cents deficiency from the cement, and a profit of 18.9 cents per cubic foot was credited to the shipowner.

Should the owner be challenged by the shipper of washing machines because of the disparity in rates, his only reply would have to be, "You, the shipper of washing machines, can afford to pay more, whereas the owner of the cement cannot. So I charge you the higher rate to make it possible for the cement to be transported."

Inasmuch as the shipowner must have cargo in sufficient quantity to provide adequate loads for his vessel, any commodity that moves in substantial and regular volume from a given port is important to his economic well-being. It is reasonable, therefore, that the rate on that commodity be somewhat lower than on items moving only occasionally and in small lots. For example, cotton is a major export cargo from Houston, and because it can be depended

upon month after month to provide half the ship's load, the rate may be lower per ton than on wool, which moves through this port in very much smaller volume.

Perhaps the next most significant point in making a freight rate is the amount of space that a weight ton of a commodity will require when stowed in the ship. Theoretically, all rates are based on a weight ton of 2,240 pounds (or 2,000 pounds in certain trades) which occupies 40 cubic feet of space. (As the metric system gains worldwide acceptance, the metric ton of 2,204 pounds and the cubic meter, rather than the measurement ton, are becoming the basis for quoting freight rates.) Cargo which stows in less than forty cubic feet is "deadweight cargo," and habitually is freighted by weight, usually at a certain charge per one hundred pounds. Cargo requiring more than 40 cubic feet for stowage of a weight ton is known as "cubic cargo," and is freighted by volume, usually at a particular amount per cubic foot. Since packaging varies, the carrier often offers two rates, one per hundredweight and the other per cubic foot, but reserves the right to charge "by weight or measurement, whichever yields the greater revenue." A vessel loaded so that all space is filled and the hull is immersed to the load line is said to be "full and down." A ship with 400,000 cubic feet of cargo space and a cargo deadweight of 6,000 tons might be "down to her marks" with 6,000 tons of a heavy (dense, or deadweight) commodity stowing at 17 cubic feet per ton, and still have 298,000 cubic feet of unfilled space. If she carried cargo that occupied 40 cubic feet per weight ton, only 240,000 feet of space would be used for a full load of 6,000 tons. If, however, the cargo averaged 60 cubic feet per weight ton, the revenue tonnage would be 6,667, with all space filled.

In making rates, the handling and stowage characteristics of a commodity are taken into account. For example, an unboxed automobile requires a great deal of ship space, far in excess of the actual measurement of the vehicle. The rate therefore includes a charge for this empty space. Fragile cargo must be protected; and to do so may take cubic footage beyond the measurement of the package. Steel beams need little care and occupy a small amount of space in comparison to their weight. They may be used to offset electric light bulbs, which are exceedingly bulky and very fragile. The rates for these items are adjusted in accordance with their varying characteristics.

The perishability, fragility, bulkiness, odor, or dangerous nature of a commodity, and the resulting responsibility imposed upon the carrier to preserve and care for it while he is its custodian exert a significant influence on freight rates. Since certain commodities are damaged easily, or may be especially attractive to pilferers, the losses sustained from these causes and for which the carrier is responsible will have to be made good when the claim is settled. Freight rates on items of this sort consequently include an allowance to cover the anticipated loss.

Some commodities can be loaded only at special berths to which the ship may have to be moved at considerable expense. For example, explosives normally are loaded in a remote area of a harbor, and the ship must be taken to this isolated spot before any of this cargo can be worked. No other cargo is available at the explosives-loading facility. The very high rate for this commodity reflects these facts.

Another vital factor in making rates is the actual cost of handling cargo at the ports of origin and destination. The charges for stowing the commodity in the ship and the expenses of receiving, checking, watching, cooperating, shifting around the terminals, and delivery vary with the items. Refrigerated cargo, for example, is expensive because of the high compensation demanded by the longshoremen who must work in subfreezing temperatures. Coal and grain are loaded and unloaded mechanically, and therefore the charge per ton is low. No uniform fee for handling can be assessed against all cargo moving through a particular terminal because the cost varies with the individual commodity.

In general, it is true that the possibility of obtaining return cargo influences freight rates. If the ship always returns to her home port in ballast, the revenue from the outbound passage has to be sufficient to defray all costs of the round voyage. Contrariwise, where cargoes are available in both directions, the average rate per ton probably will be lower than in the trade which offers business only in one direction.

Exceptionally heavy packages, defined in the break-bulk services as those over three tons, traditionally have been freighted at the regular rate for the particular commodity, with a scale of surcharges based on the actual weight. This practice reflected the fact that, until relatively recently, ships' cargo gear had a safe working load of only three tons, and to hoist anything heavier

required that the heavy-lift, or “jumbo,” boom be used. Depending upon how the ship was rigged, working the jumbo might entail using the winches for four booms. This would idle one or more gangs of longshoremen for whatever time the jumbo might be in service. An alternative to rigging the heavy-lift boom was to hire a barge-mounted crane of very great capacity. This saved much work and time on the part of the longshoremen, but added noticeably to the cost of handling the overweight parcel. Unavoidably, the expense so incurred had to be charged to the shipper. Even where modern ships have cranes or booms handling up to ten or more tons without rerigging, there still is much arduous, time-consuming labor involved in stowing a very heavy piece of cargo, and compensation to the ship is justifiable.

Extra-long pieces (timbers or steel beams, for instance) cannot be handled quickly and often interfere with the process of stowing other cargo. While there is no uniformity in the actual basis on which the charge is computed, many carriers have set a limit of 35 feet as the maximum to be accepted without assessing additional fees. It should be understood that the operator rarely derives a profit from handling these overweight or over-length items; usually the added charge offsets the out-of-pocket expenses incurred.

One of the important expenses of a voyage is the premium which must be paid for insurance of the vessel. Many considerations affect the underwriters' computations, not least of which is the geographical area in which the ship is to be operated. Some regions present greater dangers to ships than do others; seasons of the year also have a bearing. Insurance coverage will not be valid, except at a higher premium rate, if the ship is sent to regions not included in the description and limitations of risk. Few policies include war risk and perils of navigation in ice-clogged waters as part of normal coverage. When a carrier is offered a consignment of cargo that requires the ship to depart from the insured area to effect delivery, or will take the ship into a war zone, it is appropriate for the shipper to be asked to pay the higher insurance fee. Sometimes this is accomplished by adding a surcharge (usually a fixed percentage) to the freight rate. In other cases, where “off limits” ports are visited only occasionally, the carrier will impose an arbitrary charge. From this procedure, the shipping industry has come to refer to these addenda as “arbitraries.”

Arbitraries also may be assessed against cargo destined to ports where facilities are antiquated, primitive, or nonexistent, and where the ship will face the risk of delays in handling the cargo. Regardless of the title by which the extra charge is denominated, it is standard practice for carriers to compute, as an integral part of the rate, the expense of putting into ports which do not assure quick turnaround. An interesting variant of this was the imposition of a 25 percent surcharge on cargo carried by the transatlantic superliners. The explanation was that heavy expense was incurred in working cargo at the passenger ship terminals in England.

Freight rates in international trade have to be quoted in the currency of some country. For many years, the United States dollar served this purpose. When the dollar lost its stability, shipowners were troubled because their freight tariffs were more rigid than the international money market. To protect carriers who accepted payments in a foreign currency and then converted to their national funds at a loss, a "currency adjustment factor" was adopted. This could be modified as often as needed to keep ship revenues approximately in balance with the prevailing rate of exchange.

Only after the foregoing items have been taken into account in setting freight rates does the steamship rate maker consider costs of vessel operation. This was demonstrated when the stability of fuel oil prices ended in 1973, and shipowners had to devise means to adjust their revenues quickly without disrupting trading patterns. Two possibilities existed. One was to establish new rates whenever the cost of fuel oil escalated beyond an agreed level. This was rejected because of its adverse impact upon long-range commercial commitments. The alternative proposal was to impose a "fuel oil surcharge" on every revenue ton of cargo whenever the price was raised. This scheme maintained the basic freight tariff while providing the needed additional revenue to shipowners. As the cost of oil was raised, the surcharge reflected this almost immediately. When the price was reduced, the surcharge was lowered without delay.

Shippers accepted the surcharge idea with reluctance, expecting that it would be temporary. When it was evident that there would be no return to the lower levels of earlier years, it was appropriate to devise a different system. Freight rates were recomputed to reflect fuel oil costs on a certain date, and increases or

decreases, as the fuel oil market fluctuated, were tied to this "base date." If over an agreed period of time the cost of fuel oil increased significantly, a surcharge to reflect this increase was to be added. If the price decreased, all freight rates were to be lowered correspondingly. The merit of this plan was that freight rates were given some stability, changes were made only after appropriate notice, and were effective for a predictable length of time.

Other costs of ship operation do not exert so great an influence in making freight rates. Supplies, spare parts, repairs, maintenance, crew wages, and vessel time need to be calculated in addition to the dues exacted in ports of call, canal tolls, pilot fees, and the other miscellaneous expenses of the voyage. While these are not negligible, it is impractical to prorate them against individual commodities, for the reason that liner-service operators cannot predict infallibly what the composition of a ship's load will be on any single voyage.

A ship carrying 20,000 manifest tons of cargo on each 15-day leg of a round voyage lasting 30 days might have direct operating expenses, other than fuel, of \$10,000 a day. Assessed against the total lift, this would mean that each ton of cargo should contribute 50 cents a day, or \$7.50. Divided by 40 cubic feet, this would equal 18.75 cents per foot; divided by twenty hundredweight, the charge would be 37.50 cents per unit. Given the size of this contribution, it is understandable that direct operating expense is the least significant of the criteria for setting freight rates.

TYPES OF OCEAN FREIGHT RATES

Ocean freight rates are divided into two categories, *class rates* and *commodity rates*. In general, the class rate may be described as that which is assigned to a large number of unrelated commodities that have been studied individually and found to require the same revenue for their transportation. The commodity rate is a charge for carrying a designated item like, for example, granulated sugar in bags.

Class rates vary in number, but many water carriers establish from six to nine classes, depending upon the variety needed in the rate structure. The most expensive is Class D, assigned to dangerous cargo (explosives and corrosive acids). The next most expensive

is Class 1, and the least expensive is Class 8. The higher the number of the class the lower the rate will be. There is no mathematical proportion or relation among these various classes.

Examples of classification of cargo are shown below:

<i>Article</i>	<i>Class</i>
Flowers, artificial	1
Formaldehyde	2
Fuses, hazardous	D
Glass, sheets, frosted	7
Grilles, iron or steel	4
Graphite, not otherwise specified	3
Graphite, in bulk, in bags or barrels	6
Gypsum	8
Heaters, coal burning	5

The corresponding rates would be

	D	1	2	3	4	5	6	7	8
Per 100 lbs.	\$4.45	\$3.50	\$2.90	\$2.25	\$2.00	\$1.78	\$1.65	\$1.56	\$1.35
Per cu. ft.	2.50	1.95	1.65	1.25	1.12	1.00	.92	.88	.75

Commodity rates are fixed by water carriers usually as a result of pressure from shippers who find it competitively impossible to pay the freight rate of the class to which the commodity otherwise would be assigned. In many cases, the difference between one class and the next lower class represents a reduction in freight revenue which is greater than the carrier is willing to grant. In that event, to avoid establishing a new class, a special rate will be assigned to the item. For example, frosted glass, in sheets, is placed in Class 7 and the rate is 88 cents per cubic foot, or \$1.56 per hundred pounds. The Class 8 rate is 75 cents per cubic foot, or \$1.35 per hundred pounds. The carrier, in response to a request for special consideration, decides that an acceptable rate for frosted glass, in sheets, could be 80 cents per cubic foot or \$1.45 per hundred pounds. This is a compromise between the two classes. On the one hand, the shipper is convinced that he cannot pay the higher rate; on the other hand, the carrier is unwilling to come down to the next lower class rate. The result is a "commodity" rate that is mutually acceptable.

Commodity rates take precedence over class rates, so that if the same item were listed in both parts of the tariff, the commodity rate would be applied. There is no necessary relationship between commodity rates since they are made after special consideration of each item. The description of every listing is very definitive:

	<u>Freighted Per</u>
Citrus fruit, fresh, viz.: oranges, lemons, limes, tangelos, tangerines, and grapefruit, under refrigeration, in cartons or boxes not exceeding 42 lbs. each	Carton
Nails, iron or steel, plain, galvanized, or cement coated (except shoe or horseshoe)	100 lbs.
Paper, wrapping, not corrugated, other than cellulose film (Cellophane), cloth-line, glass-line, gummed, laminated, oil, parchment, tissue, transparent, or waxed	100 lbs.

Regardless of how an item is freighted, the carrier always incurs some expense in processing the papers related to the shipment. It is customary, therefore, for shipowners to require a minimum payment on every bill of lading issued. One operator, for instance, fixed this fee at \$15.00.

“Through rates” are charged for shipments originating with one ocean carrier but transferred to connecting carriers at intermediate ports. By way of illustration, a consignment was sent from Hamburg (Germany) to Antofagasta (Chile). It was sent in a United States Lines’ vessel from Hamburg to New York, and there transhipped to one of the Chilean Line’s steamers for the voyage to South America. The combined operation is handled under one bill of lading, and therefore has come to be known as a “through route.” The freight rate for cargo moving in this manner is designated as a “through rate.” Usually the originating carrier issues the bill of lading, collects all charges, and subsequently divides the revenue with the other carrier(s) in the proportions established by the through rate agreement.

In many cases, the through rate is lower than the combination of the rates of each of the participating carriers. For example, a ton of coffee may be sent from Buenaventura (Colombia) to New York in

a Grancolombiana Line's ship for \$22.00. The freight rate for coffee carried from New York to Livorno (Italy) by a Prudential Lines' vessel is \$36.80. The through rate from Buenaventura to Livorno, by agreement between the carriers, is \$54.00. Each carrier reduced its rate by \$2.40 in order to meet the competition of an Italian operator whose ships sail between Colombia and Italy, and whose charge for the direct haul is \$54.50.

Not all through rates represent concessions in the rates charged by participating carriers. In some cases, two ship lines will offer the benefits of through service, but each will charge the full tariff rate for its portion of the total transit. This situation eventuates either from lack of direct sailings between countries of origin and destination or from significantly less desirable schedules by the only carrier linking the two ports. Some shippers will use the through service even though it is more expensive because they consider the time in transit to be worth the difference financially. There are other cases where no direct link exists between the exporting and the importing countries. Lacking the pressure of competition, the through rate is the total of the connecting carriers' charges plus any accessorial fees. Until 1981, for example, there was no line sailing directly between Australia and Chile. If an Australian importer wished to bring Chilean wine to Melbourne, it would have to be carried to San Francisco and there transshipped to a transpacific operator for the onward movement to Australia. The through rate would be constructed in this manner: From Valparaiso to San Francisco, the rate was \$32.00 per ton. Transfer costs in San Francisco from one marine terminal to the other were \$11.00 per ton. The freight charge from San Francisco to Melbourne was \$40.00, bringing the through rate to \$83.00 per ton.

FREIGHT RATES FOR CARGO IN CONTAINERS

Although vast changes in shipping have been occasioned by the container revolution, there is no difference between the principles by which freight rates are made for cargo carried in containers and those by which rates are established for break-bulk cargo. It is probable, however, that eventually the new techniques of handling cargo will bring about changes in the methods of mak-

ing rates for commodities handled exclusively in containers. If altogether new ideas concerning rates do not emerge, at the least there will be significant modifications in the way the traditional principles are applied to containerized cargo.

Certain facts must be accepted. Containers are of uniform size, and are loaded and unloaded by mechanized and almost completely automated methods. Regardless of their contents, these big boxes are handled at the same speed, using the same number of longshoremen and the same equipment. To position and secure an empty container takes the same care as it does to stow a box filled with delicate precision instruments of high value. Certainly not to be overlooked is the fact that most containers are stuffed by the shippers, who thereby become responsible for the efficient and economical use of shipping space.

Among the time-honored factors influencing break-bulk freight rates are demands for high levels of stevedoring skills in handling and stowing various commodities and products. The susceptibility to damage and pilferage is another basis for computing cost. The amount of ship's space necessary to accommodate a revenue ton must be taken into account. When, however, goods are received at ship's side in huge weatherproof boxes that will be stowed in the vessel by men and machines that give all containers, regardless of what is inside, exactly the same care, the applicability of these justifications for a freight rate may be questioned. If it costs no more to put a filled container into its shipboard cell than it does to deposit an empty container in the adjoining cell, how can the shipowner defend a rate structure built on theories and practices displaced by technological developments?

The first argument in support of the existing system is that the possibility of sustaining a major loss is much greater in the container age than in the days of break-bulk shipping. A container loaded with washing machines might be swept overboard during a storm, and the claim would amount to many thousands of dollars. Formerly, a single washing machine might be lost at a cost of a few hundred dollars.

There is a modicum of truth in this argument; many containers have been carried off by boarding seas. The freight rate, which has been assessed against all the washing machines in the container, includes an allowance to cover this contingency. The argument therefore loses much of its force.

The second point advanced in defense of existing freight rates is that the introduction of the container did not end the need to transport those low value commodities moving in small quantities, and therefore dependent upon liner service. Some method of continuing the subsidy of these commodities must be devised, and nothing better than the break-bulk principles has been brought forth. One school of thought has advocated the adoption of a single rate for all containers, regardless of the commodities being dispatched. This may be logical in view of the technology of container shipping, but it poses as many problems as it solves. If the rate favors the movement of cement, the revenue to the ship will be inadequate. If the rate is predicated on washing machines, this will stop the movement of cement. It also will reduce correspondingly the tonnage of cargo offered to the ship.⁶

Some years ago, an operator in the transatlantic trade instituted a "container rate," and explained the method by which cargo would be freighted. A shipper of microwave ovens, for example, would be charged a gross figure of \$1,500 for the container. The shipper was responsible for stowing his packages to get the most efficient use of the space within the box. Other merchandise could be mingled with the ovens so long as it had approximately the same value and was subject to the same freight rate.

In this case, the ship operator was substituting for the traditional commodity rate schedule a new listing for containers loaded at the shipper's option with one or more types of cargo which were "compatible" and therefore moved at the same freight rate. The shipper was obligated to furnish the carrier with an accurate manifest showing exactly what had been placed in the container. Regardless of how much or how little space was used by the shipper, the rate for the container remained the same.

Another carrier plying the trade route between New York and South Africa inaugurated its service with a standard charge for all containers without regard to what might be in the boxes. This "freight all kinds" rate was set arbitrarily by the carrier: initially it was \$4,150 for a 20-foot box and \$7,300 for the 40-foot container. Because of intense competition and the reduction in the volume of cargo flowing through this trade route, the company announced in mid-1985 that it was reducing these rates to \$2,500 for the smaller box and \$5,000 for the larger container.⁷

Steamship rate makers for years have been searching for a

logical and simple system by which freight rates can be applied to cargo moving in preloaded containers. As late as 1985, they had not found an entirely satisfactory substitute for the existing (and frequently criticized) rate structure, which did produce adequate revenue without unduly antagonizing shippers.

Several tentative suggestions have been propounded by both carriers and shippers as a basis for discussion. From these offerings it has been possible to detect the first glimmering of what may be the future. The outline of possible developments is set forth in the following paragraphs.

Rates for containers would be established for designated groupings of goods. High value merchandise, typified by the washing machines mentioned earlier in this chapter, would be charged a fixed fee for a container load. Less highly valued merchandise would be assigned appropriately lower rates. The problem commodities, typified in these pages by cement, which cannot pay fully compensatory liner rates, would be identified and a minimum charge fixed for their transportation.

Once the system became effective, shippers would be charged a single fee for the use of a container from origin to ultimate destination. The fee would be predicated upon the shipper's declaration of value of the goods that have been loaded into the container. To illustrate, a hypothetical steamship line engaged in the New York-Rotterdam trade would set charges for completely filled containers according to this scale:

Valuation under \$200 per metric ton or cubic meter: \$500

Valuation of \$201 to \$500 per metric ton or cubic meter: \$800

Valuation of \$501 to \$750 per metric ton or cubic meter: \$1050

Valuation exceeding \$751 per metric ton or cubic meter: \$1400

The shipper would be free to load any number and variety of commodities in the container, so long as the value did not exceed that stipulated when the reservation was made with the carrier.

To protect against abuses, the carriers would have the right to open and examine containers and would be supported by these simple rules:

1. Only one shipper would be permitted to load goods in a container.

2. A single bill of lading would be issued for each container.

3. If the valuation of any of the merchandise loaded in the container exceeded the agreed dollar limitation, the rate for the whole container would be that which applied to the highest valued goods.

4. All containers would be considered as having been fully loaded at the time they were delivered to the carrier. The fee for containers would not be prorated for less than container lots.

The Traffic Study

AMONG the responsibilities of the executives charged with developing sources of revenue for a steamship company, none is more exacting than preparing the *traffic study*. This is a systematic compilation of data concerning one or more of the many problems related to selling the service the ships are intended to provide. It may be concerned with major aspects of traffic management, such as commencing service on a new trade route; it may be limited to a simple matter of determining the most attractive color scheme to be used in redecorating the passenger spaces. In either case, it is a methodical procedure by which to reach a decision affecting the earning capability of the ships.

The traffic study is a forecast of future trends of the business of the carrier, based upon present practices and past experience in the same field. It depends to a great extent upon the history of the steamship company, as reflected in its records. Equally important is the personal knowledge and expert opinion of the persons invited to participate in the study.

Two traffic studies conducted by American shipowners typify the extremes of total disaster and complete success which can result from these efforts. The first shows how even the most punctilious examination of all aspects of a new idea may be upset by a single feature appreciated by none of the executives involved in the study. The second demonstrates how success is achieved when all participants share the vision of what can be attained.

The first study was conducted several years before Malcom McLean demonstrated the practicality of handling cargo in pre-loaded containers, rather than in the conventional manner used in break-bulk carriage. At the time the study was initiated, the project was of unusual complexity because it dealt with a wholly new, indeed a revolutionary, procedure involving the delivery in foreign

ports of cargo encased in large, sealed boxes which could be placed directly on a waiting truck or trailer without being opened. Only enough longshoremen were needed to hook the hoisting gear to the boxes and to release it.

Among the elements that had to be taken into account were the best size for the containers; the feasibility of obtaining return cargoes that could be stowed in the boxes; the problem of recovering boxes from inland points remote from the ports; the desirability of making the ship self-sustaining through the installation of very expensive gantry cranes as compared with developing special facilities to handle the containers at each port; the potential savings in stevedore costs which might be used to offset the expense of starting the program; and the reactions of waterfront labor groups to the new technique.

One of the critical items in the study was the size of boxes, because local trucking capability in the foreign countries to be served had to be determined. After establishing the nature of the support that might be obtained abroad, the condition of the highways, especially the arc of road curves, had to be studied. The recommended solution was quite ingenious: it capitalized on the trucking industries of all the different nations involved. Three of the proposed boxes could be fitted on an American trailer; two could be loaded aboard a foreign trailer, or one box could be carried easily on the bed of a medium performance truck in any area of the ship's proposed operation. Once these determinations were made, the traffic study, except for the labor aspects, was completed with relative ease.

Trouble was encountered in measuring the extent and strength of the reaction from foreign longshoremen. As it turned out, the best available advice from experts in the evaluation of labor attitudes appeared to be conservative and realistic, but in fact was colored by unwarranted optimism. In the light of the available data and conclusions, the costly project was approved, and millions of dollars were spent. Two ships were converted, containers were purchased, and much publicity was given to the new system. The ship was loaded with no labor problems in the American port, but at the first overseas point the longshoremen refused to work the ship except under terms and conditions that negated the whole idea. Eventually, arrangements were made to deliver that initial cargo,

but the project had to be abandoned and the ships laid up.

The second traffic study was dedicated to an analysis of the characteristics to be built into new ships planned for a particular trade route. It was typical of the major effort required and epitomized the range and depth of specialized knowledge required on the part of those directing the study.

The shipowner was developing a replacement program for its fleet of transatlantic cargo ships. Major changes in cargo packaging procedures had occurred shortly before the study was begun. In addition, a substantial movement of refrigerated produce from the United States to two overseas ports had developed. Finally, the competing ships on the route were being replaced with vessels capable of significantly greater speed. The combination of obsolescent vessels that were only marginally suitable for the variety of commodities offered and the modern, purpose-built, fast ships operated by the competition made it imperative that radically new ideas in ship design be evolved.

First consideration was given to speed. A canvass of both the sales staff and representative shippers proved that there was a strong demand for drastic reduction in transit time. Intensive study of port working schedules resulted in a finding that, given the cargo-handling techniques of the time, only a few hours could be saved by improving production on the part of the stevedores. Sustained sea speeds would have to be increased by about one-half, as compared to the existing fleet's normal cruising speed of fourteen knots. Engineering computations by the naval architect indicated that it was operationally and economically feasible to build the new ships with that greater capability.

The details of ship characteristics, primarily those related to a floating warehouse that had to be emptied and refilled in the shortest possible time, were studied by ships' officers, stevedores, terminal managers, and operating executives. It was learned that what was required was a hull with three hatches abreast over each hold, rather than the single hatch in the old ships. To hoist the cargo, new, very flexible, and extremely fast gear was needed. A major proportion of the cargo space was insulated, and refrigeration machinery would have to be installed to bring temperatures down to 0° Fahrenheit. "Reefer" spaces in particular were to be made very easily accessible to expedite discharge.

The naval architect took the results of these elaborate studies by the composite group of traffic specialists, cargo-handling experts, stevedores, ships' officers, cargo-gear riggers, and specialists in handling perishable foodstuffs, and converted them into plans for a fleet of large, high-speed, versatile, and economical ships that proved to be outstanding in their responsiveness to the requirements of the trade.

In this instance, the investment to be made as a result of the traffic study was so great that those in charge were justified in using every resource available, and in devoting thousands of man-hours to assure that every detail had been examined with painstaking thoroughness, and that every recommendation had been weighed in relation to all others before final decisions were reached.

A traffic study, then, is an essential tool of management. It systematizes thinking related to problems of every type of steamship traffic. It requires much imagination and complete freedom from preconceived notions on the part of those who direct the study. Finally, it places the responsibility for each step and each recommendation squarely on the shoulders of the individual participants.

Take, as an illustration, the action of a traffic manager who has heard many complaints that the ocean freight rate on stainless steel kitchen sinks is so high that American exporters of these items are priced out of the overseas markets to which the ships are sent. To acquaint himself with the general economic situation relating to stainless steel sinks in these countries, the traffic manager requests information from the appropriate governmental agencies. His out-bound manifest section will determine exactly how many tons of steel sinks moved during the past year. The terminal manager will supply data on the costs of loading and discharging, and any special problems that should be considered in analyzing the freight rates. From the shippers themselves will be obtained data relating to the sinks sold in overseas markets in direct competition with their products. When all the material has been assembled and is available for study, the traffic manager is in a position to review the entire problem intelligently and systematically, and to decide whether a reduction in the ocean freight rate is in order. Furthermore, he will be able to determine whether the reduction would help the exporters to sell more sinks in the foreign markets, with the obvious

benefit to the steamship carrier of obtaining more cargo.

As a normal routine of traffic management, it is customary for periodic analyses to be made of cargo manifests. These analyses often disclose changes in tonnage and revenue which suggest that a traffic study be undertaken to find answers to questions like these: What are the most important commodities, in terms of tonnage and ship revenues, moving this year, and what changes are to be observed when compared with last year's cargo lift? What are the current freight rates? What were the freight rates a year ago? Have shippers made any formal applications for adjustments in freight rates, and if so, what disposition was made of those applications? Are there any obstacles to the movement of these important commodities which exist today but were not in existence a year ago? Have any barriers been removed? What is the cost of stevedoring for each of the major commodities? What is the total cost to the company of transporting each of these commodities? What is the frequency of claims? How does the cost of claims settlement compare with freight revenues from each commodity? What is the true margin of profit? Are these commodities carried more as a convenience to shippers than as contributors to company profits? If there has been a loss of business this year as compared to last year, in the light of the information provided to answer the foregoing questions, is it worthwhile to make an effort to regain this tonnage?

Earlier in this chapter the story was recounted of a steamship company's successful fleet replacement program. To illustrate how so complex and critically important a study might have been conducted is to demonstrate the true nature of this tool of management. The board of directors probably would direct the vice-president, traffic, to be the coordinator of the study and would authorize him to request information and assistance from all departments. When all data had been gathered, the coordinator would make final recommendations as to the character of service needed, the special features deemed essential in the design of the ships, and the maximum acceptable cost of the vessels.

The first step in the traffic study would be to evaluate the characteristics of the ship that would be most satisfactory as a replacement for the existing type of vessel. This would involve not only determination of size, speed, capacity, and special features of the ships, but also the ports to be served, the approximate schedule

each ship would have to follow to meet the demands of shippers, and the types and tonnages of cargo to be carried. The coordinator would order the managers of the inbound and outbound freight traffic divisions to compile statistics showing the fluctuations in the amount of business during the previous decade. The operating department would furnish data on the facilities and limitations of the ports along the route, the comparative cost of operating newer and larger ships at substantially increased speeds, and the number of tons of cargo handled in an hour by longshoremen in each port visited. The accounting department would supply information concerning the average revenue from each of the various commodities carried, as well as the approximate net profit from a typical voyage.

As these data come to the desk of the coordinator, they would be collated with information from shippers and consignees as to their ideas, desires, and anticipated needs for marine transportations during the next decade. Various governmental agencies meanwhile would have supplied more elaborate and extensive information, statistical and otherwise, than company records could provide. Forecasts and predictions as to the economic growth of the region where the ships would operate also would have been solicited from international banking houses and foreign commerce associations.

When the accumulation of facts, theories, records, and statistics had been completed, the coordinator would be able to determine with considerable accuracy what they signified in terms of ship speed, schedules, capacities, earnings, and the other details to support his final recommendations for a certain number of ships of a particular design to operate on a set frequency of sailings.

The needs of the service, the peculiarities of the cargoes, and the economic justification of the improved design would all be recorded in systematic and convenient fashion. The completed traffic study might well fill hundreds of pages and represent a large investment in the form of executives' time, clerical labor, and other costs.

There are no fixed limits to the subject matter of a traffic study. It is precisely what the words imply: a tool of management. It has only one purpose, and that is to provide a systematic method of analyzing a problem and giving to management the best possible answer. In form, therefore, the traffic study can be extremely

varied. Depending upon its nature, it may take the shape of a learned researcher's written report, or it may be only an oral statement. It could take months of work on the part of practically the entire organization, or it might require only eight or ten hours of one man's time. Since it often contains many secrets of the company's operations, it never is released to the public, but its substance often is revealed in statements to government agencies, applications for route franchises, or announcements of new developments by the company.

The very nature of the traffic study requires that the person making or directing it have a wide knowledge of company operations. The more varied the experience, and the more comprehensive the understanding of company activities and problems, the more likely it is that the conclusions will be accepted. By the same token, the responsibilities placed upon the person who must make the final recommendations in the traffic study are tremendous. Almost any other duty in the traffic department becomes, by comparison, less significant. A traffic manager whose predictions have proved to be correct most of the time is of incalculable importance to the success of any steamship organization.

Steamship Conferences

FROM the early days of transoceanic shipping until the present, an international flavor has characterized the business of moving goods from one land to another. Merchants of many nations brought their wares to overseas markets, seeking the most profitable transactions. They cherished their own ideas of what constituted proper business practice, and resisted vigorously any effort of one sovereign power to impose its code of commercial morality upon the citizens of another country. This almost instinctive hostility toward regulation was absorbed early by steamship men who had inherited from their predecessors a sturdy independence and a willingness to assume great risks in exchange for great returns. Despite this attitude, however, there was a dim awareness among shipowners that uninhibited competition could be disastrous not only to them but to their customers. The alternative to government control had to be an industry-generated and industry-managed system of supervision and regulation.¹

Awareness changed into concern about the time that improvements in steam engineering, particularly as they applied to marine installations, made it possible for powered vessels to undertake long voyages. Bitter rivalry between the steamers and the sailing ships became the normal condition of business and forced the growth of the idea that shipowners who faced the same problems on the same trade routes and scrambled for a share of the available business had much to gain by establishing some form of mutually acceptable regulation. By surrendering a fraction of their rights to independent action in exchange for protection from predatory competition, the shipowners hoped to benefit themselves as well as their clients.

The soundness of this idea was put to the test in 1875 when steamship owners in the London-Calcutta trade became involved

in a freight rate "war" with the operators of sailing ships going between these two ports. In their search for a way to end this unhappy situation, the owners called a conference at which was advanced the idea of an industry association to police the trade. Approval of the idea resulted in the formation of the Calcutta Conference.² Although they were written in 1961, the following words from a report submitted to the United States Congress, after noting that the investment in an individual ship is larger than a corresponding investment in the component units of land or inland waterways transportation, described the rationale of the conference:

Because of the peculiar mobility of ships, it is entirely possible for a shipowner to place his vessel in a particular service on relatively short notice and without hindrance from any law or regulation of any nations. As a result, the only method that has proved practical to assure continuity of service on a particular route with a degree of stability of rates, in view of the very large investment required in the establishment of a regular service, is by providing specific inducements to shippers to utilize the services of the particular line or lines regularly serving that route. The conference system, with an appropriate tying device, has proven to be the most effective method, both from the point of view of the carrier and the shipper.³

The significant events in the development of the Calcutta Conference are of interest.

Just at the time the Suez Canal was opened in 1869, steamships were becoming not only more numerous and more efficient, but they were replacing sailing ships on one route after another. Because they did not depend upon the wind, the powered vessels could take advantage of the shorter distance to India made possible by the canal, whereas the sailing ships were thwarted by adverse wind conditions in the Red Sea and had to use the route around the Cape of Good Hope, almost 4,000 nautical miles longer. The steamers not only made the voyage more quickly, but they could predict with reasonable accuracy the day of arrival at their destination. To offset these disadvantages, the sailing ship owners reduced freight rates, a move met by even greater concessions from the steamship operators. The destructive nature of this struggle immediately became apparent.

The major steamship operators (Peninsular and Oriental Steam Navigation Company, British India Company, and T. & J. Harri-

son) and some of the smaller carriers met in the 1875 conference and jointly agreed to end concessions to shippers, to adopt uniform freight rates, and to offer identical terms of carriage. It was expected that these arrangements would stabilize shipping practice and assure more regularity in schedules, less frequent adjustments of rates, and greater convenience for shippers. The importers and exporters, however, resented the loss of concessions, and threatened to transfer a larger proportion of their business to the sailing ships, which were offering more favorable terms. The prospect of losing cargoes prompted the conference to develop the "exclusive patronage" contract, which would commit the shipper to give his business only to carriers belonging to the conference. The shippers rejected the proffered contract, demanding some guarantee of preferential treatment before they would bind themselves to the conference. Responding to this demand, the conference proposed in 1877 to establish the "deferred rebate," which would be paid after a designated period during which loyalty to the conference had been demonstrated.

The arrangement required that shippers desiring to receive the rebate agree to give their business only to the lines that were members of the Calcutta Conference. In return for this exclusive patronage contract, the conference members bound themselves to return one-tenth of all freights paid during a six-month period, contingent upon the continuing exclusive patronage of the conference carriers during a further period of six months. Failure to live up to the terms of the contract automatically cancelled both the earned refund and the credits being established toward a second refund.

The penalty for "disloyalty" was set forth in the following words, taken from a circular issued in 1885 by the England-China Conference:

Shipments for London by non-Conference steamers, at any of the ports of China or Hong Kong, will exclude the firm making such shipments from participation in the return, during the whole six-monthly period within which they have been made, of any portion of the freight charged, even though the firm elsewhere may have given exclusive support to the Conference lines.⁴

The working of the new system was described very clearly in this

quotation from the report of the Royal Commission on Shipping Rings submitted to Parliament in 1909:

. . . if at the end of a certain period (usually four or six months) they have not shipped goods by any vessel other than those despatched by members of the Conference, shippers will be credited with a sum equivalent to a certain part (usually 10%) of the aggregate freights paid on their shipments during that period, and the sum will be paid over to them if, at the end of a further period, usually 4 to 6 months, they have continued to confine their shipments to vessels belonging to members of the conference. The sum so paid is known as a "deferred rebate." . . . If, during any period, a shipper sends any quantity of goods, however small, by a vessel other than those despatched by the Conference lines, he becomes disentitled to rebates on any of his shipments by conference vessels during that period and the preceding one.⁵

Acceptance of the exclusive patronage contract was a voluntary act on the part of the individual shipper. The alternative was to pay the rate quoted in the tariff without expectation of receiving a partial refund. Inevitably, this meant that the landed cost of the goods at destination would be greater for the noncontract signer than for its competitor. This economic pressure brought many shippers under the contract; by the close of the century the deferred rebate scheme prevailed in practically all the conferences of carriers operating out of the United Kingdom. In many of these trade routes, the conference system with its rigid hold on shippers produced the closest approach to a monopoly that is possible in ocean shipping.

The Royal Commission found that deferred rebates had proven effective in gaining and holding the loyalty of shippers, as evidenced by the large sums the carriers owed to individual shippers. To control this limited monopoly, the commission recommended that conferences be directed to deposit with the Board of Trade copies of all conference agreements, rebate circulars and claim forms, all understandings with lines outside the conference, and all agreements with shippers' associations which had been recognized by the Board of Trade. To satisfy complaints that customers did not have ready access to tariffs of conference carriers, it was further recommended that tariffs be published and at least one copy be filed with the Board of Trade.⁶

In 1923, the British Imperial Shipping Committee concluded a further study of steamship conferences and found that, to some extent, deferred rebates were necessary in most trades. The system, however, did deprive the shipper of much of his freedom of choice and removed the corrective effect of free competition.

The contention is valid that the deferred rebate is an effective shield from the competition of both independent liner operators and tramps. Few shippers will risk the loss of benefits accrued under the deferred rebate system to take advantage of the comparatively small savings to be gained by employing either the independents or, when the volume of cargo justifies movement in shipload lots, by chartering a tramp. As a useful tool in building up dependable patronage, the value of the deferred rebate was acknowledged in both the English and the American investigations.

American attitudes toward monopolies and competition are quite different from those of both Western Europe and Japan, where monopolies and agreements in restraint of trade are acceptable business practices. In the United States, these arrangements among commercial firms are outlawed and discouraged by the stringent provisions of the antitrust laws. Aware of the restrictive nature of steamship conferences, and consistent with the antitrust sentiment prevailing in the United States, a committee of the House of Representatives was appointed in 1912 to investigate the conference system and its effect upon American foreign commerce.

Under the chairmanship of Representative Joshua W. Alexander, of Missouri, a subcommittee of the Committee on Merchant Marine and Fisheries conducted the investigation and held hearings over a two-year period. In 1914, it published its four-volume report, and made a number of recommendations for legislative action. Chief among them was that the conference system be exempted from the provisions of the antitrust law, but made subject to continuous scrutiny by a regulatory agency. Two practices of European conferences were condemned: the deferred rebate and the use of the "fighting ship."

The Alexander Committee considered that the deferred rebate was intended to establish a near-monopoly of ocean transportation on the trade route in which the carriers operated. The fighting ship, defined as a vessel used on a sea-lane by a group of operators (for example, a conference) for the express purpose of excluding,

preventing, or reducing competition by driving a nonmember carrier off that route, was viewed as the weapon of monopoly. Evidence had been presented proving that this practice had been followed, particularly in the North Atlantic, to the detriment of nonconference carriers.

A flagrant example occurred in 1908. German passenger ship operators reacted to competition for immigrant passengers on the Hamburg-New York run by using fighting ships to destroy this rivalry. Suitable steamers were chosen from the conference members' fleets and were sailed on the same days and between the same ports as the independents' ships. Fares were reduced enough to attract most of the passengers to the fighting ship. When more passengers bought tickets than the fighting ship could accommodate, they were transferred to the conference members' ships at the same low fares. Losses resulting from this activity were prorated among the conference members in proportion to the number of their ships. The committee summarized the situation in these words:

It was thus a case of all the lines, united in conference, opposing every sailing of a single opposition line. By distributing the loss over the several members of the conference, each constituent line would suffer proportionately much less than the one line which was fighting the entire group, and which would inevitably soon exhaust its resources in the conflict with the combined power of the large lines with their superior speed and better third class accommodations.

This type of predatory competition also existed in cargo service, as the committee took pains to illustrate. The case in point was that of a German "fighting corporation" formed in 1905 by six cargo liner companies. This corporation had as its sole purpose the destruction of rivals on its members' routes. Four ships were owned and others were chartered. These vessels sought the bulk cargoes normally transported by the independents, at freight rates which were intended to bankrupt the independent carriers.

The Alexander Committee's recommendations furnished the solid basis for what became the Shipping Act of 1916. Steamship conferences organized in the United States were to be exempted from the provisions of the Sherman Antitrust law, on condition that all joint actions be subject to some form of government supervision.

The tools of monopoly—deferred rebates, fighting ships, and discriminatory practices—were disapproved. To support its recommendations, the committee declared that trade with foreign nations required that uniform rates be available to any and all shippers, that they be comparatively stable, and that they take into consideration all the elements of transportation. Those carriers which elected to operate outside of the conference framework were not subject to government control unless discrimination were charged by a shipper.

It is important to note that the attitude and thinking of the Alexander Committee, which opposed cartels and other restrictions on trade but recognized that international shipping required special consideration and treatment, have been accepted and supported by the Congress. In a report written in 1961 from the House Committee on Merchant Marine and Fisheries, the need for a carefully controlled exemption from antitrust legislation was set forth in these words:

The conference, fixing as it does rates and practices, is permitted to exist only as an exception to the antitrust laws of the United States, and such exception is granted only because of the peculiar nature of ocean transportation, and provided certain conditions are met. In order to secure the benefits of such immunity from the antitrust acts, the conferences are required by Section 15 of the Shipping Act of 1916 to file all their agreements with the appropriate regulatory body, which at present is the Federal Maritime Board, and obtain approval of that body. The Board has considerable jurisdiction under the terms of that act to regulate the conferences and see that they observe the restrictions upon which their immunity from the operation of the antitrust acts is predicated.⁷

Perusal of the various laws enacted by the Congress of the United States to regulate steamship conferences reveals the persistence of the nineteenth century hostility toward monopolies of any sort. This attitude may be discerned in the provisions of the Shipping Act of 1984, even though by this date the influence of these associations of carriers had been curtailed very sharply. Intra-conference competition, for example, was evident in the frequent announcements to the shipping public that rates on certain commodities were “open”—that is, each carrier in the conference was

free to set rates at any level that seemed appropriate. Significant and unilateral improvements in the quality of service also demonstrated the rivalry between conference members. In combination, these actions signalled that conferences did not pose the threat of monopoly.

An analysis of the liner services of nine major maritime nations was conducted in 1957, and pointed out that the fleets of the individual carriers were small. Forty percent of the operators owned twenty or fewer ships each; an equal percentage claimed ten or fewer vessels. The largest single-owner fleet embraced only two percent of the total number of ships assigned to liner service. From these facts the conclusion was drawn that no single operator, nor any combination of two or more operators, attempted to dominate a trade route through control of the largest fleet. The conferences, consequently, could be described as having neither the capacity nor the intention to establish a monopoly.⁸

As the conference system evolved after 1875, the exclusive patronage contract (also called the loyalty contract) and the accompanying dual freight rate structure were considered indispensable to the existence of the associations. The purpose of the contract was to provide the members of each conference with a steady flow of cargo from shippers whose support would be forthcoming over a reasonably long period of time. In exchange, the members of the conference agreed to reduce the nominal (or "tariff") freight rate by a stipulated percentage—a benefit granted only to those shippers who signed the contract.

In the United States, from the effective date of the Shipping Act of 1916 until 1948, the regulatory authority exercised only the most cursory type of supervision over the contracts on the premise that the basic agreement having been approved, issuance of the contract by the conference automatically was approved. In 1948, the Maritime Commission carefully examined the details of the dual rate system, and prescribed modifications to be made by one conference. A major attack on the dual rate system was initiated in 1949 and culminated in 1952 in a decision by the United States Supreme Court that the differential between contract and noncontract rates was arbitrary and consequently unlawfully discriminatory. The use of the dual rate structure by two North Atlantic conferences therefore was permanently enjoined. Further legal action culminated in

the Supreme Court's ruling, announced on May 19, 1958, in the case of *Federal Maritime Board et al. vs. Isbrandtsen Company, Inc., United States of America, and Secretary of Agriculture*, which held that the use of contract/noncontract rates proposed by the Japan-Atlantic and Gulf Freight Conference violated that provision of the Shipping Act of 1916 prohibiting restrictions to trade and unjust discrimination (in this case, setting up a retaliatory device to stifle competition). The dual rate system was declared illegal.⁹

Congressional hearings conducted between 1958 and 1961 examined all aspects of the conference system. Testimony offered by both carriers and shippers led to the conclusion that conferences were essential in the foreign trade of the United States. The loyalty contracts, with the associated dual rate structure, were given general approval. As the conferences existed at that time, these factors supported their continued operation:

1. The "ease of market entry"—any ship on the ocean may stop at any port and discharge or load cargo. It costs a great deal to establish a liner service, but nothing beyond the investment in a single ship for a tramp owner to participate, as opportunity affords, in a specific trade.

2. The rigid fixed and operating cost structure of the steamship industry. It costs almost as much to sail a ship one-quarter full as completely loaded. This may encourage the independent to cut rates as a means of obtaining cargo.

3. No governmental control over the level of ocean freight rates.

4. A marked difference in the operating costs of similar ships flying diverse flags.

5. Usually an imbalance between inbound and outbound cargo. For example, on Trade Route 21 (from U. S. ports on the Gulf of Mexico to the United Kingdom and European continent) the disparity was 4 to 1 in favor of outbound loadings in 1959. This places enormous pressure on the carrier to find homeward cargoes.

6. Overtonnaging of ocean trade routes, which tends to accentuate the already severe competition existing among ocean carriers.¹⁰

Conferences were intended to provide, through the carriers' own efforts, the stability demanded by international trade but not

GENERAL RATE INCREASE **EFFECTIVE FEBRUARY 1, 1986**

The Member Lines of the United States and Gulf Ports/Eastern Mediterranean and North African Freight Conference, announce that spiraling operational expenditures necessitates the implementation of a General Rate Increase, for both Port to Port and Intermodal Rates, effective February 1, 1986, as follows:

Five percent (5%) on rates upto \$165.00 W/M or M

Five percent (5%) on all weight rated commodities

Five percent (5%) on all container rates,
rounded upto the highest \$25.00.

Minimum Bills of Lading, Supplemental Bills of Lading and
Optional Bills of Lading will also be increased by \$10.00

Any questions you may have relative to this announcement, should be directed to the Member Lines, their Agents or the Conference Office.

Constellation Line
Farrell Lines, Inc.
Lykes Lines

Prudential Lines, Inc.
Waterman Steamship Corp.

R. H. Cabrera
Chairman

furnished by national governments. The dependence of liner companies upon regular and repeated patronage is well known and is corroborated by the findings of a recent study of a trade route originating in the United Kingdom. Of the 3,900 different clients of the liner services on that route, only three shippers accounted for 25 percent of the total tonnage carried. The next 25 percent was generated by fourteen customers, and the third 25 percent came from 82 patrons. Only 2.53 percent of the shippers using these liners contributed 75 percent of the cargoes. Another line's experience was similar. On the route between northwest England and the Middle East, between 25 percent and 40 percent of the business originated with twelve shippers.¹¹

THE "8900" RATE AGREEMENT WISHES TO ANNOUNCE:

DUE TO STEADILY INCREASING OPERATIONAL COSTS, THE "8900" RATE AGREEMENT HAS FOUND IT NECESSARY TO INCREASE RATES AND CHARGES IN F.M.C. NO. 9, 10 & 11 FREIGHT TARIFFS. EFFECTIVE MAY 1, 1985, AN INCREASE OF FIVE PER CENT WILL BECOME EFFECTIVE WITH THE DATE THE VESSEL SAILS FROM THE PORT OF LOADING OR THE DATE THE CARGO IS RECEIVED BY THE CARRIER, WHICHEVER BENEFITS THE CARGO.

THE "8900" RATE AGREEMENT COVERS TRADE FROM POINTS IN THE UNITED STATES MOVING THROUGH U.S. ATLANTIC AND GULF PORTS AT INTER-CHANGE TO MIDDLE EAST PORTS, WEST OF KARACHI AND NORTH EAST OF ADEN, BUT EXCLUDING ADEN AND KARACHI AND TO INLAND POINTS IN BAHRAIN, IRAN, IRAQ, KUWAIT, OMAN, QATAR, SAUDI ARABIA AND THE UNITED ARAB EMIRATES.

**Very truly yours,
THE "8900" LINES
ANTHONY A. DE GIGLIO
VICE CHAIRMAN**

It is significant that the Senate Committee on Commerce, in its report written in 1961 on the proposed amendment to the Shipping Act of 1916, quoted with approbation the comments of the Alexander Committee in 1914:

It is the view of the committee that open competition cannot be assured for any length of time by ordering existing agreements terminated. The entire history of steamship agreements shows that in ocean commerce there is no happy medium between war and peace when several lines engage in the same trade. Most of the numerous agreements and conference arrangements discussed in the foregoing report were the outcome of rate wars, and represent a truce between the contending lines. To terminate existing agreements would neces-



COMPANIA PERUANA DE VAPORES (CPV)
(Peruvian State Line)

We regret the decision by members of The West Coast of South America Conference to put into effect 7/8/85 a general rate increase of 8%, both north and southbound to and from Peru and Chile..

CPV considers this increase inopportune at this time and hereby wishes to notify its customers, shippers and freight forwarders that we have already notified the Conference we intend to take "independent action" to and from Peru and Chile with current rates.

We hope this action will induce members of the Conference to reconsider their position and decline the general rate increase.

A member takes independent action against an increase in conference freight rates (*New York Journal of Commerce*, June 18, 1985, p. 18-A).

sarily bring about one of two results: the lines would either engage in rate wars which would mean elimination of the weak and the survival of the strong, or to avoid a costly struggle, they would consolidate through common ownership. Neither result can be prevented by legislation and either would mean a monopoly fully as effective [as], and it is believed more so than, can exist by virtue of an agreement.¹²

Three types of steamship conferences exist in the world today: closed conferences which apportion the capacity of their members' ships to the volume of cargo offered (rationalization); closed conferences which do not practice rationalization; and open conferences which do not limit capacity of member fleets. Only the United States requires as a matter of law that conferences admit to membership any common carriers prepared to serve the trade routes covered by the associations.

Proponents of the closed conferences with rationalization argue that it is only by limiting competition between carriers for the available business to a few dependable performers that the standards insisted upon by shippers will be maintained. Restricting membership in the conference to that number of carriers which will provide transportation for the proffered cargo is efficient and economical. Although conferences offer lower rates to those shippers who sign loyalty contracts, most of the cargo actually is controlled by a small number of shippers who are more concerned over the service they receive than with the relatively small savings af-

forded by the loyalty contracts. (In this connection, it is noteworthy that the Shipping Act of 1984 prohibits the use of loyalty contracts and dual freight rates.)

Records of two closed conferences which practiced rationalization and offered loyalty contracts supported these comments. Over a twenty-year period, marked by times of political instability and operational difficulties, load factors for these conference members came to almost ninety percent of capacity.

More common than the above-described conferences are the associations which have closed memberships, but do not attempt to rationalize tonnage. While freight rates and other matters governing transportation of goods are agreed upon jointly, conference members continue to compete with each other for the available cargo. From time to time, because of economic conditions, some restrictions may be imposed, such as sharing cargo tonnage, allocating ports to be served, and arranging schedules to enhance the efficiency of ship employment. These coordinated efforts, even without rationalization, have been beneficial. One conference, for instance, reported that over a span of five years its load factors ranged between 75 percent and 90 percent in one direction and between 85 percent and 92 percent in the other direction.¹³

Advocates of the closed conference recognize the desirability of having the concerns of the shippers given full consideration, and therefore recommend that so-called shippers' councils be formed to negotiate with the associated carriers. In support of their recommendation, these advocates point to the success of the Australian Meat Shippers' Council, which forced the conference covering the trade between Australia and the United States to lower its rates. A parallel case was that of the Australian wool-shipping interests which formed the Joint Wool Commodity Group, and exerted sufficient pressure on the Australian Northbound Shipping Conference to bring freight rates between Australia and Japan down by approximately ten percent.¹⁴

The Australian Overseas Transportation Association is a major council whose functions are: (1) to obtain economies in shipping by rationalizing tonnage; (2) to approve freight rates; and (3) to approve agreements between shippers and shipowners. Loyalty contracts and rebates, discounts, and the like are permitted. Founded in 1929, the association includes producers and exporters. Boards

having significant influence on conference matters are the Dairy Products Control Board, the Wine Overseas Marketing Board, and the Meat Board. The goal of the association when it was established was rationalization through cooperation, which left considerable strength in the hands of the shippers. The closed Australian conferences, based on strong loyalty contracts and rationalization with pooling of cargoes, have not led to the monopolistic abuses feared by some theorists.¹⁵

Under the closed conference system, internal pressure upon members is exerted to operate only that number of ships required to satisfy the demands of the trade. To maintain a certain frequency of service without putting additional ships on the berth, a number of conference members in recent years contracted for space in each other's vessels. Company Alfa, to cite a hypothetical case, sails on alternate Fridays, Company Kappa schedules its departures on the same route for the intervening Fridays. Alfa has built up a strong clientele of loyal shippers who control a weekly total of 250 containers which they insist upon dispatching every Friday, but want the Alfa bill of lading and Alfa service at the port of discharge. To keep this business, Alfa contracts with Kappa for 250 container spaces on each sailing. (The transaction is known in the trade as "space chartering.") Alfa and Kappa retain their individuality. They do not merge any of their activities, nor do they reveal any carrier-shipper relationships. There is no reduction in intercompany and intraconference rivalry. The number of ships serving the trade route is maintained at the level which gives reasonable assurance of survival for conference members and forestalls disastrous freight rate wars. A significant, and certainly not coincidental, benefit is that the load factor of the conference members' ships is kept at satisfactory levels.

A British report on conferences contained these observations on the relative merits of open and closed conferences:

... the "open" conference appears least likely to serve the interests of shippers. It is also least likely to serve that of shipowners; in their evidence to us they agreed that such a conference arrangement typically resulted in low load factors, low profits, and rising freight rates. . . .

The "closed" conference with fully rationalized sailings therefore appears to us most likely to serve the best interests of both shippers and shipowners. We appreciate that we have here come to a different

conclusion from those reached by some others who have studied conference arrangements; we believe that they may have given too little weight to the full range of the needs of shippers and to the practical results of unrestricted competition in all fields other than price. We identify the opportunity for providing a planned systematic series of sailings as the feature of shipping conferences which is potentially most beneficial. The full exploitation of that opportunity requires a "closed" conference.¹⁶

Conferences which may be joined by any common carriers serving the trade routes are precluded from setting up any restrictions on the number of ships or the total carrying capacity of vessels operated by conference members. Agreement is reached on a common tariff of freight rates, and some adjustments may be acceptable for sailing schedules. Competition within the ranks of the conferences, however, remains keen, often manifesting itself in unilateral revocation of published freight rates for particular commodities.

Advocates of the open conference assert that competition between carriers but held within the framework of the association results in the lowest cost of transportation and the highest levels of service. In theory, this may be true, but in the harsh realm of ship management the theory loses vitality. It is axiomatic that when too many ships seek too few cargoes, two results are inevitable: the revenue to the ship will be reduced to the unprofitable level, and the quality of service will suffer. Shippers have no institutional loyalties, and use those vessels which meet their needs most effectively without regard to previous performance. As has been noted repeatedly in this volume, owners of cargo are concerned only that what they send across the oceans shall arrive in good condition and on schedule. Whether the carrier makes a profit or sustains a loss is not a matter of interest to either shipper or consignee.

Conferences, especially those which are governed by the United States law requiring open membership, have been losing their economic power for decades, and no longer pose any threats to monopolize a trade. In 1953, Daniel Marx, Jr., identified six factors which restricted the monopoly powers of the conference:

1. Intraconference competition.
2. Actual or potential competition from other lines which may or may not intend to join the conference.

3. Alternate sources of supply or markets.
4. Actual or potential competition from tramps.
5. The bargaining strength of shippers.
6. Government regulation or intervention.¹⁷

These factors still exert powerful constraints on the conferences. An important addition to Marx's list could be made in 1985. This was the significant competition from airlines, which have siphoned off a very large percentage of the highest paying cargo. The economic effect on the waterborne carriers has been to distort the time-honored rate structure under which the revenue from commodities that paid high freight rates actually subsidized the movement of those items which could not afford to pay the full cost of sea carriage. It is not an exaggeration to assert that this competition from the airlines has undermined permanently the financial stability of many steamship companies.

The Shipping Act of 1984 directed that all agreements of conferences domiciled in the United States contain these provisions: (1) a statement of the purpose of the association; (2) specific assurance that any common carrier serving the trade route would be admitted (or readmitted) to membership upon application; (3) permission for members to withdraw, upon reasonable notice to the conference, without penalty; (4) establishment of an independent neutral body to police the obligations of the conference and its members; (5) a mechanism for consultations to resolve intercarrier disputes other than by judicial process, and to cooperate with shippers to prevent and eliminate malpractice; (6) procedures to consider promptly shippers' requests and complaints; and (7) authorization for any conference member, upon advance notice of not more than ten days, to take independent action on any rate or service stipulated in the tariff filed with the Federal Maritime Commission.

In addition, the law specified that conferences might not issue loyalty contracts nor set up dual rate structures; that they could not make any refunds not provided for in the published tariffs, nor deny cargo space to any shipper who had filed a complaint or had patronized a competing carrier; and that they might not employ fighting ships or offer deferred rebates. Discrimination between shippers or ports also was categorically forbidden.

A change from established procedure ended the role of the Federal Maritime Commission in enforcing the law against conferences. Instead, the commission shall investigate reports of alleged malpractice, and if sufficient evidence is uncovered, shall refer the matter to the United States District Court of the District of Columbia for action. If the allegations are sustained, the district court could enjoin the conference(s) permanently against the questioned conduct. A carrier found in violation of the law or an order of the commission could be fined \$5,000 per day for each violation. If the court determined that the violation(s) had taken place "willfully and knowingly," the penalty could be increased up to \$25,000 per day per violation.

Experience with self-policing of the conference system by means of a neutral body had been favorable since the system was established on January 1, 1979, pursuant to a directive from the Federal Maritime Commission, which included these stipulations:

1. The basic agreement forming the conference must show how complaints against conference members are to be handled.
2. The self-policing body must have no relationship to the conference members, and must be able to examine, and to acquire, whatever information and documents it needs by making appropriate demands upon the carriers concerned.
3. There must be provision for adjudicating disputes and for levying penalties. Accused parties must be afforded the opportunity to rebut charges made against them.
4. The self-policing agency must file two reports each year with the Federal Maritime Commission.
5. The self-policing bodies may not limit the carriers they police from producing documents and information demanded by the Federal Maritime Commission, nor must the self-policing agency deny access by the commission to any data it may have accumulated in connection with an investigation.

In a clarifying statement, the commission announced that these "neutral bodies" would be expected to make "self-initiated, onsite investigations . . . regularly . . . into the activities of each member line." These investigations were to be handled

with reasonable discretion, and did not have to be identical in nature. As a matter of normal routine, the agencies were to reveal the identi-

ties of complaining parties, but this would be waived in those circumstances that might encourage retaliation by or against members or . . . when it would unfairly prejudice the member's ability to rebut any material allegation made against it.

An example of the authority which was granted to a neutral body was set forth in a report published by the *New York Journal of Commerce* on September 15, 1983. A conference had contracted with a neutral body for policing service, and included in the agreement the power to assess fines. Acting on complaints, the neutral body determined that a carrier member had been guilty of malpractice (although the offense was not described, it was said to have involved rebates from the published freight tariffs), and assessed a fine of over \$9 million.

Historically, there has been criticism that the conferences do not prevent disastrous freight rate wars. This charge is true. Numerous wars have occurred on all trade routes. One of the more interesting (and one of the longest) was the war between the United States/South and East Africa Conference and the Seas Shipping Company (operating under the trade name of Robin Line) from 1935 to 1937. This war started in June 1935 when Robin Line announced its entry into the South and East African trade, and applied for membership in the conference. The application was denied on the grounds that there already was sufficient conference tonnage available to meet the needs of the American shipping public. Robin persisted in its plans and announced sailings to South Africa. The conference replied by reducing rates, and in the course of the war, rates fell to four dollars a ton, regardless of commodity. By 1937, all exclusive patronage contracts between shippers and the conference had expired. A major American automobile manufacturer was asked to renew its contract with the conference but was given no assurance concerning the level to which the rates would go at the end of the war. The manufacturer refused to sign the contract, and the conference retaliated by refusing to allot space to the automobiles of this shipper. The manufacturer appealed to the Robin Line for assistance, and an agreement was reached to transport the vehicles to South Africa for eight dollars a measurement ton. Robin Line also promised that after the war ended, the freight rate would be no higher than it had been in 1935. This had the effect of stifling further gains from the war, which came to an abrupt end on July 1,

1937. It was characterized throughout by bitterness on both sides. It is noteworthy that from the end of the war until the sale of Robin Line to other owners in 1955, there was no dual rate system in the South African conference.¹⁸

The introduction in 1969 and thereafter of new, very large, and extremely fast containerships on the routes from Japan, Hong Kong, and Taiwan to both Pacific and Atlantic coasts of the United States precipitated new rivalries which very nearly destroyed the entire conference structure in the Pacific. Containerships were placed in service as quickly as individual owners could take delivery, and in a short time a major problem of overtonnaging existed. The fierce struggle for available cargoes was characterized by rate cutting, secret rebating, and wholesale resignations from the conferences followed by recisions of the resignations when a modicum of agreement was established. As the volume of cargo dropped substantially during the worldwide business recession that began in 1971, turmoil on the route increased significantly, but under pressure from numerous governments a period of relative calm was reached in 1975. The transpacific trade has been plagued intermittently since then by a series of skirmishes between the carriers, precipitated again by adverse economic conditions in the entire Pacific basin. Several carriers were forced into bankruptcy, conference memberships were reduced, and sailing frequencies were curtailed. In mid-1985, some semblance of peace prevailed, but the basic difficulty of too many ships looking for too few cargoes persisted.

An insight into the actions and reactions involved in a freight rate war was granted by the extensive hearings conducted by the Federal Maritime Commission into the friction between the American-flag Grace Line and the foreign-flag Viking Line during 1960 and 1961. The commission's findings were not made public until November 1962. The following excerpt is revealing:

When Conference rates were opened, it was the policy of Grace not merely to meet Viking's rate, but to go down immediately to the minimum rate which Grace considered it could charge. Thus when rates were opened, Grace's rates were not decreased by stages but generally in one big cut.

The Conference rate, effective on and after December 14, 1955, on agricultural implements, was \$27 per ton. It was opened March 7,

1960 and Grace made its rate \$12 on that date. Viking's rate prior to the rate war was \$24 except for one shipment at \$20.25 early in 1960. On household washing machines, the normal Conference rate of \$20 was opened with a minimum of \$15, effective February 22, 1960, which was the rate Grace made effective on that date. The Conference opened the rate without any minimum effective March 7, 1960, and Grace's rate on that date became \$11. Viking's rate prior to that date was \$18. On toys, with a value of less than \$350 a freight ton, Viking normally charged \$31.50 and the Conference charged \$35. During August 1960, both Viking and Grace charged \$13.50. Grace intended to go as low as Viking's breakeven point in setting its rates, but even such low rates were met, and Grace went to even lower rates. On individual rates, both Grace and Viking undoubtedly reached non-compensatory levels.

When the rate war ended, although many rates went up to normal, others, while raised, did not rise to their previous levels. Commitments to shippers kept some rates from returning all the way up to their normal levels. Viking's policy during the rate war was to cut its rates, so long as it obtained enough revenue to cover loading, discharging, and commissions, plus "something for the ship," such as \$1 or \$2 a ton.¹⁹

The rate war lasted throughout 1960, in which year there was a major decline in trade with Venezuela. The war gradually ended early in 1961, and all rates slowly increased during that year.

Every discussion of conferences stresses that they give stable rates to the shipping public. This beneficial result is obtained in part from the conferences' practice of making rates which all members are able to accept. Reaching agreements on rates often is a tedious process, and during these discussions the shipping public gains from the status quo being maintained. Furthermore, proposed rate changes must be filed with the Federal Maritime Commission, and advance notice of intended increases in rates must be given to concerned parties.

As a normal function, conferences from time to time initiate reviews of their rate structures, usually as a result of developments which necessitate earning greater revenues. Typical of this situation was the drastic rise in the cost of bunker fuel oil in 1973, which most conferences countered by adding a surcharge to the rates. Shippers do not have to wait on this kind of conference action to have rates examined; they may submit applications for adjustments in freight rates at any time. These requests are handled by the rate

IMPORTANT NOTICE TO CONTRACT SIGNATORIES

BUNKER SURCHARGE

The Member Lines of the U.S. Atlantic & Gulf/Australia-New Zealand Conference wish to advise the shipping public that the previously announced Bunker Surcharge of 20% per revenue ton due to become effective December 15, 1981 has been reduced to 18%.

The Conference has also published further precautionary increases which will be subject to review and adjustment based on actual cost experience.

Schedule of Surcharges follows:

January 15, 1981	20% Per Revenue Ton
February 15, 1982	22% Per Revenue Ton
March 15, 1982	25% Per Revenue Ton

committee, composed of a small group of traffic experts appointed by the conference chairman from the membership. The shipper furnishes data relating to the value of the commodity, its selling price at point of importation, the price of competing articles offered on the same market, the number of tons of the commodity that the shipper exports (or imports) in a year, and the freight rate suggested by the shipper as fair and reasonable. Comparison of this information with similar data supplied by the statistical section of the conference staff leads to a recommendation from the committee to the whole conference. If the recommendation is approved, the change must be forwarded to the Federal Maritime Commission for filing. Eventually, new pages showing the revised rates are published and distributed to holders of the conference tariff.

Conferences have sustained a number of buffetings in recent years in addition to pressures from shipper organizations such as those in Australia. The overtonnaging of trade routes is a worldwide problem, and as already mentioned has resulted in the withdrawals from membership of several carriers. Freight rate wars have become almost cyclical in the more important trade routes, a situation not ameliorated by the growing use of intermodal routings, especially those overland movements competing with the all-water transportation of goods to or from the Orient to the Atlantic and Gulf coasts of the United States.

Whereas it may have been true earlier in the twentieth century that conference carriers dominated, but did not monopolize, the major trade routes of the world, the situation has changed significantly. Economic pressures caused by the prolonged recession which has afflicted shipping of all nations since 1973 have forced many carriers either to suspend operations or to sell out to larger and stronger organizations. The enormous increase in the cost of building and operating containerships had encouraged the formation of consortia of carriers. Space chartering between carriers which otherwise are intense rivals has proved to be another way to maintain frequency of service without the necessity to operate ships carrying only partial loads. Concentrating vessel voyages between one or two ports at each end of the route has made possible reduction in fleet sizes. International balances of payments and overvalued currencies have created unbalanced movements in which vessel revenue is derived disproportionately from one direction, rather than from the round voyage.

It remains true, however, that conferences serve a useful purpose. They still attract members from the stable group of carriers, and they continue to set and to maintain standards of service which are desired by shippers. If they ever posed a threat to monopolize the movement of cargo on trade routes, that day has passed into history, and conferences today are more likely to be seeking ways to assure that their members will survive the difficult period which has overtaken them. That they can and do perform a useful role has been demonstrated in the past, and it seems likely that as they march into the second century of their existence, they will continue to provide certain standards of conduct and service which will be welcomed by those who use the seas.

The Logic of Steamship Scheduling

TO THOSE who frequent the waterfront of any major port city, a special thrill is to see one of the huge containerships come into her berth precisely on time. Equally exciting is to watch the departure of this proud queen of the seas exactly at the advertised hour. What appears to be a simple, almost routine process involving a minor amount of calculation actually is a complex exercise in coordinated management. To pierce the veil normally masking the details of this activity is essential if this phase of ship operation is to be appreciated.

Drawing up a schedule for a steamship service is much more than establishing the time at sea between ports by dividing the number of miles to be traversed by the speed of the ship, and then adding enough hours to unload and reload the vessel. The determination of the basic rules to be followed in a liner-service schedule requires a combination of empirical knowledge and interpretive ability. Men experienced in the ways of ships and ports and the vagaries of the sea collaborate with those who are wise in the ways of traffic and can convert the demands of shippers into the practicalities of ship operation. Their common goal is to meet the needs of shippers while achieving efficient and profitable employment of the fleet.

Schedules are inseparable from the idea of liner service. One or more ships plying a fixed route and making port calls at regular intervals provide the transportation essential to the commercial enterprise of the region. In time, if ship movements have been dependable over a protracted period, shippers come to expect that vessels always will be available according to the established pattern, and any disruption in that pattern upsets their business practices. Merchants are especially concerned that the ships sail on the advertised day because many international trade transactions are

financed by letters of credit stipulating that the goods be dispatched no later than a certain date. Failure to meet this requirement can interfere with the financing of the deal.

In particular, therefore, the more frequently a fleet can offer a sailing, the more likely it will be to gain the good will of shippers and prosper from their patronage. This premise is valid so long as there is no question concerning the time the goods spend in transit.

The Blue Peter Line, for a hypothetical example, owns a number of twenty-knot ships. One of the routes it serves is between Landport and Seahaven, which are separated by 4,800 miles of ocean. The vessels traverse the distance in ten days, and require four days in each port to discharge and reload. The aggregate of these times is exactly four weeks. A fortnightly service is maintained, using two cargo liners. If a shipper is exceedingly fortunate in timing the movement of his cargo, it is possible for his goods to arrive at the loading berth only a few hours before the ship sails, and to be discharged at Seahaven on arrival day, so that the transit time of his goods is ten days. Another shipper, however, misses the sailing from Landport by a matter of hours, and his goods must wait two weeks for the next departure. This consignment is the first loaded aboard the ship, and the last to be discharged. Not only must the goods wait fourteen days, but at Seahaven they will not be released until the end of the two days needed to empty the ship. Total transit time for this lot of cargo is twenty-six days (fourteen days of waiting, ten days of steaming, and two days for discharge). Blue Peter Line's customers must be prepared to accept a maximum transit time of twenty-six days, but can obtain a minimum of ten days. If no objections are voiced by shippers, management would be correct in assuming that the service offered within these limits is adequate.

As cargo offerings improve, Blue Peter Line assigns a total of four ships to the route, making possible weekly departures from both termini. Maximum transit time is reduced to nineteen days (seven days of waiting, ten days of steaming, and two days for discharging) while the minimum transit time is maintained at ten days. Although each ship earns a small profit, there is space available for additional cargo. Calls therefore are ordered at Harborside and Transitown, which lie on the direct route to Seahaven and require no increase in steaming time. At Harborside, a day and a

half are used to handle the cargo, while at Transitown working the cargo takes two full days. The duration of the round trip has been lengthened to thirty-five days. To continue to provide weekly sailings, Blue Peter must place a fifth ship on the route.

The new schedule confronts management with two significant problems. The first is the complaint of the shippers that, although weekly departures are available, transit time has been increased by the calls at Harborside and Transitown. The minimum now is thirteen and a half days, compared to ten formerly, and maximum transit time has been extended from nineteen to twenty-four and a half days. For those exporters who are paid on delivery of their goods to the importers, the additional time represents a significant economic burden. Management concedes that the service offered by Blue Peter Line between Landport and Seahaven is no longer adequate to the needs of the shippers.

The second problem is comparable in magnitude. The tonnage of cargo offered on the route did not increase despite the fact that to maintain weekly departures on the route by way of Harborside and Transitown, it was necessary to assign a fifth ship to what economically is a four-ship service. If freight rates are raised to the point where all ships earn a profit, there is great likelihood that the higher charges will force some shippers to withdraw from the market. If ship revenues are not augmented, however, the company will be pushed into a money-losing operation.

The requirement originally presented by the shippers for a weekly service was justified by the additional cargo revenue that resulted. Attempting to enhance ship earning power by adding the calls at Harborside and Transitown undermined the adequacy of the service and imposed heavy financial burdens on the carrier. While it was technically possible to have the ships call at the two intermediate ports, the decision proved to be faulty because it did not adequately consider the impact on the shippers of the longer transit time.

The first step in making a schedule is to convert the broad statements of what is desired by the traffic department into the particulars of how many tons of cargo must be handled in terminal ports and in primary and secondary ports of call; the physical characteristics of the ports of call which affect the movements of ships into and out of their waters; the hours during which these

ports provide cargo working arrangements; and critical times of arrival of the ships at certain intermediate primary ports. As these details are assembled and appropriately related to the expressed desires of the traffic department, the outline of the schedule begins to emerge. Many suggestions will be put forward as the various versions of tentative schedules are examined. Of major importance to the operating department is the designation by the traffic experts of the primary ports along the route which will be visited on every trip outbound and homebound, and what ports will be considered as secondary, to be served only when sufficient cargo is offered to justify the expense of the call.

Secondary ports present a problem to the liner service that is operated with little spare time in its schedules. If, for example, calls are made at three primary ports en route, and turnaround time at the outbound terminus is to be accomplished in forty-eight hours, a six-hour call at a secondary port can be troublesome. One solution lies in the operational and traffic histories of the line. If calls at one or more secondary ports have been included on every voyage, it is good management to provide sufficient time for those visits. This arrangement will eliminate the unavoidable disruption to schedules which must follow unprogrammed demands on ship time. Another solution is to develop the schedule at a cruising speed that can be increased as necessary to compensate for the time required to make the calls at the secondary ports. In either instance, the value of close coordination between the operating and traffic departments is evident.

Cargo ships are designed to operate at a certain speed, and barring unusual events beyond the control of the owner or of the master, will be run at that speed. No allowance is made for the possibility of unfavorable weather interfering with the completion of the appointed voyage, unless there are reliable data showing that at certain seasons of the year it is impossible to maintain maximum speed. This is true for those vessels assigned to year-round service on the North Atlantic, where winter storms reach fantastic dimensions, and require significant reductions in speed. In other regions, for example along the southeastern coast of South Africa, winter storms make slow speeds necessary if major damage to vessels is to be avoided. In still other areas, like the Caribbean Sea and the Gulf of Mexico, weather usually is excellent but the threat of hurricanes

is known to all who voyage in those waters. They are, however, unpredictable in every way: when they will occur, where they will reach dangerous dimensions, what course they will follow, and how long they will menace the sea lanes. Operators sending ships into those areas have found over the decades that they can meet the problem best by alertness and flexibility at the time the storm approaches. If a hurricane delays a voyage for several days, this is an isolated phenomenon that does not justify altering schedules for the duration of the hurricane season.

A basic rule in making schedules, therefore, is that the ships will be operated at designed speeds, and from sea buoy at the port of departure to sea buoy at the port of arrival, the performance of the ship will be up to standard.

Next in order of consideration, and complicated by their variability, are the type and tonnage of cargo to be transported over the entire length of the intended voyage. Not all items moving by sea can be handled with the same rapidity (computed in tons per hour): cotton bales are stowed aboard ship at one speed, while agricultural machinery is placed in the hold at a markedly different pace. Similarly, elevators in New Orleans deliver grain to ships at a very great number of tons per hour, but in India the laborers shovel it out in an agonizingly slow tempo. The variety of commodities, the quantity of each, the skill and rapidity with which the stevedore performs his tasks, and the ability of a designated berth to handle the flow of break-bulk cargo are vital pieces of information on which to base the forecast of time needed in each port of call. The accurate determination of what will be carried, and in what quantity, is a factor in developing the schedule. Study of current and past manifests of actual loads carried in the ships assigned to the route provide much of this essential information. Supplementing these data will be the traffic department's estimates of anticipated trends in cargo movement, by commodities and ports of loading and discharge, and often by seasons of the year.

The transformation of these data into a proposed sailing schedule is accomplished by the combination of the accrued experience and pragmatic wisdom of shipmasters, terminal superintendents, and stevedores who have worked with these goods, ships, and ports, and know what can be accomplished at every point where the ship calls. The importance of accuracy in fixing the number of hours

needed to unload the inbound cargo and to take aboard the outbound tonnage cannot be overestimated in the break-bulk trades, where performance of longshoremen varies greatly from port to port. Errors in the estimate of time needed to turn the ship around are cumulative and can disrupt the regularity that is the essence of liner service.

To establish a workable schedule that will assure maximum performance for the big, fast, and very expensive containerships now operated on major trade routes between industrialized countries is at least as crucial to ultimate profitability of the steamship enterprise as it is in the break-bulk trades. The aggregate hourly costs of these huge carriers come to hundreds of dollars, and delays, other than those occasionally and unpredictably arising from causes beyond human control, are intolerable.

Few, if any, of these vessels are what the trade calls "self-sustaining," meaning equipped with cranes to handle the containers in which their cargo is loaded. The critical element in setting up the port working schedule is the availability of dockside cranes to lift the containers off and on the ships. Especially is this true for those ships that transport upwards of two thousand twenty-foot boxes and require from two to five cranes for optimum dispatch. To bring two ships into a berthing area where there is space for only one, or where insufficient cranes are available, is economically unacceptable practice, and every effort must be made to assure that this type of mistake does not occur.

Schedule makers for containerships may be freed of concern for the capability of individual gangs of longshoremen and for the difficulties of stowing certain commodities in a break-bulk ship, but in exchange must assume the burden of coordinating port time of the ships with the complex operation of the container terminal. Weather plays little part in the planning, inasmuch as the containers themselves are watertight and otherwise weatherproofed. Responsibility for the multiplicity of detail relating to terminal operation belongs to the terminal superintendent who must inform the schedule makers of every step that needs special consideration in the formulation of the definitive timetable.

The structure and equipment of break-bulk carriers require special consideration from the scheduling team. Few fleets in the merchant service consist of identical vessels; far more likely is the

heterogeneous collection of ships built in different yards to different plans, and often intended for trades different from those in which they now are employed. The individual characteristics of these ships have profound influence upon the rapidity with which longshoremen can handle the cargo, whether they are loading or unloading.

In the hypothetical Blue Peter Line, the common characteristic of the ships was their twenty-knot speed. It would be realistic to imagine that two of the five ships assigned to sail between Landport and Seahaven had six hatches, each rigged with two sets of cargo booms, and equipped with high-speed winches. The other three ships had seven hatches, five of which had individual fast-acting cranes, while the No. 1 and No. 7 hatches each had one conventional mast and boom assembly. At first glance it would appear that the rapidity with which cargo could be handled would differ significantly between the types of ships. Experience brought out the fact that all worked cargo at about the same speed. This important information would facilitate greatly the development of the timetable for the fleet.

Despite the remarkable success of naval architects in giving a balance of cargo-handling capability to all hatches in a break-bulk ship, the age-old problem of the "limiting hatch" has not been eliminated. To some extent, the difficulty of having more cargo in one hold than in any other is structural, and to some extent it is the result of the quantity or characteristics of the cargo stowed in a hatch. The Liberty ships had a disproportionately large hatch, and if the whole load were to be taken off in one port, the working schedule for the ship had to be keyed to the big hatch. The time in port was determined by that one hatch. Those who plan the loading of ships attempt to distribute cargo throughout the ship to mitigate the impact of the limiting hatch. Their efforts are an outgrowth of the number of hours allotted to each port.

Containerships affect timetables through the complexity of the terminal operation that must support them. When a number of ports of call must be considered by the schedule makers, the actual experience of the ships becomes of paramount importance in seeking the most efficient employment of the vessel's time. Ideally, all terminal operations are standardized, and only minor deviation from approved activities should occur. In practice, many variances

from the ideal are encountered for a large number of reasons. Routine analyses of terminals along the route are supplied to the schedule makers in order that adequate, but not excessive, time be assigned to each port. Although it is economically imperative that the ship get to sea as quickly as good management and proper care of the cargo will permit, it is equally important to control the movement of the containership to assure that there will be a berth at the next port, and that the terminal will be ready in every respect to service the ship on her arrival.

Despite the mechanization of the terminal which has accompanied the introduction of containerships on the principal trade routes, local labor customs and working hours must be respected by schedule makers in precisely the same way that they are for the break-bulk ships. For example, in the port of New York, work may be continued through the meal hour, provided the longshoremen are paid penalty overtime wages. In the observance of Labor Day, however, the holiday is complete, and only in the direst of emergencies will men report for work. Longshoremen in certain foreign ports will not work during the meal hour regardless of the financial inducement, but will violate holidays for premium compensation.

Calculating the length of time the ship must spend in port to work her cargo is a major challenge in making liner-service schedules. The ship earns no revenue for her owners while in port, and therefore every effort must be made to reduce this unproductive period. The rapidity with which cargo is worked by longshoremen varies from port to port, and often from terminal to terminal within a port. The more the schedule makers know about the cargo and the workers, the more closely can they estimate the hours required to unload or load the ship. Other time factors that must be considered are related to the movements of the ship: Is tide significant in assuring that the ship can use the channels? When do pilots work? What restrictions apply after dark?

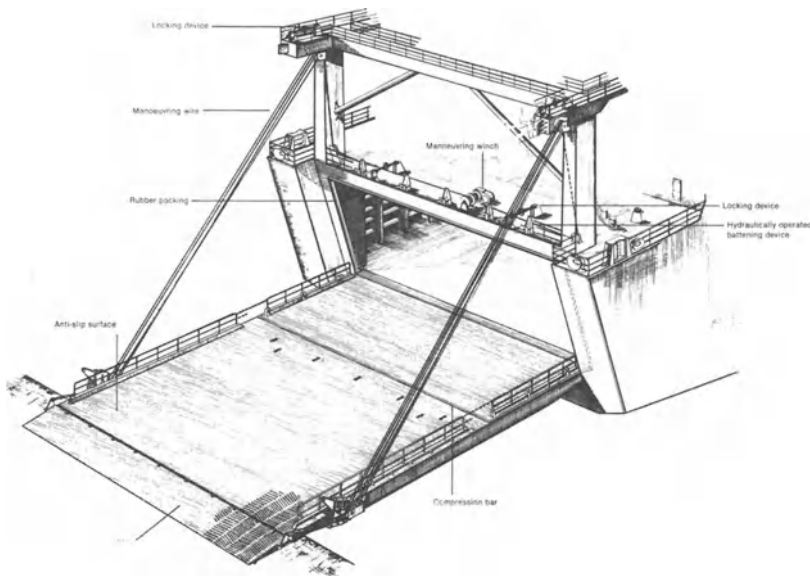
Underlying the idea of liner service, whether by break-bulk or container-carrying ships, is regularity—the dependable arrivals and departures of ships at the ports listed in the itinerary and the timetables. To assure this regularity, many expedients are practiced. The amount of cargo accepted for a port may be limited to that quantity which can be handled in the time allotted for the call. For example, a break-bulk ship works cargo at the rate of fifty tons



Automobiles, palletized break-bulk cargo, heavy equipment, and roll-on/roll-off cargo move in two directions over this huge ramp of a modern multipurpose cargo liner. Photo: Barber Blue Sea.

an hour and is scheduled to spend only fourteen hours in port. To maintain the schedule, a maximum of 700 tons will be booked into and out of that port. Again, the ship may be worked around the clock on the theory that overtime wages are less costly than ship time. Improving port turnaround means more voyages per year for the ship, and hence greater revenue will be earned. Intimately related to this thought is the fact that in many instances two fast ships can do the work of three slower vessels. The economy to the shipowner is obvious.

Certain assumptions must be made as the plan for steamship service is converted into the reality of the schedule. These assump-



Structural details of the mechanism of a stern ramp installed in a roll-on/roll-off ship. Photo: Kvaerner-Navire International.

tions are predicated upon the experience of the shipowner in the operation of seagoing vessels, and are intended to achieve the dependability that is implicit in liner service.

First, it is assumed that the cargo liner will be loaded with a representative variety of goods, stowed according to the line's approved stevedoring practices. This representative load will be based upon analysis of cargo manifests of enough voyages to define the trends in commodity movements.

Next, it is assumed that ports of call, and the sequence in which they are visited, will remain unaltered during the projected lifetime of the timetable. Working conditions in each of these ports are considered to be stable and not subject to significant change.

Then, it is assumed that port operations will not be disrupted by strikes, riots, or civil commotion, and that there will be no major breakdowns of cargo-handling equipment in the ports.

Finally, no allowance is made for unfavorable weather because of the unpredictability of storms and their intensity, or the effect they might have on port cargo operations.

It is important to note that ship schedules reflect those facts which affect the practicality of the ships' movements. Pertinent, therefore, is the analysis of harbor depths and the range of the tides. If a port of call can be navigated only at high tide, the schedule must be sufficiently flexible to accept the unavoidable variations in arrival and departure times. The flexibility may be assured by including an adequate number of unprogrammed hours, and authorizing the master to adjust his speeds to assure arrival at high tide. This same device may be used in connection with sailing times, depending upon the problems that may be confronted at the next port. If several ports of call are affected in this manner by the tides, failure to make appropriate allowance for the changes in the hours of high tide can lead to total disarray of schedules.

Some ports have strict rules about the hours when ships may move within the limits of the harbor. In Capetown, for instance, pilotage from the sea buoy into the ship's berth is compulsory and is available only during daylight hours. Pilots take the ships from the sea buoy in strict observance of their times of arrival—first come, first served. Some shipowners, in the effort to avoid long delays, plan arrivals well before dawn, even if this means an excessive consumption of fuel oil because of the high speed required.

Longshoremen observe different working hours from port to port. These idiosyncracies must be taken into account in preparing the schedule, so that adequate time will be allowed to handle the cargo. Some of these local peculiarities involve no night work, no work on Saturdays after five o'clock in the evening, lunch hours extending from eleven o'clock in the morning until one o'clock in the afternoon, and no work during certain religious holidays.

Holiday work, when local custom permits it, offers some problems for ship operators. Much of the commercial activity of the port is suspended, consequently no cargo is delivered to the terminal or taken away. Supporting services (tugs, heavy-lift cranes, shifting of lighters) must be arranged in advance and are very expensive. Many longshoremen, often the best workers, elect not to report for work, and overall efficiency is reduced. It is, therefore, routine for schedule makers to adjust schedules for individual voyages to avoid having ships in port over holidays.

Break-bulk cargo liner service, concerned as it is with handling thousands of tons of hundreds of commodities, requires somewhat

flexible schedules. It is not unusual for a ship to be considered as being on time if she arrives during working hours on the scheduled day, regardless of what time had been set in the schedule for her docking. This is accepted practice during those seasons of the year when the final two or three days of the voyage may be spent fighting storms which slow the ship. A realistic schedule will include an allowance of time to assure departure at approximately the advertised hour. If these liner-service vessels sail at intervals of two or three weeks, no serious difficulties result from these unavoidable delays. Shorter intervals call for managerial attention to prevent port congestion of the company's fleet.

Delays in themselves may not be especially serious, particularly when they are only of two or three hours duration. In their cumulative effect on fleet operations, however, they may cause numerous difficulties, not least of which is the overlapping of schedules. This overlapping, known as "bunching," is the obvious and cumulative result of a number of delays (or "slips") along the route. Ships of a single fleet which have bunched up in a port overtax the facilities available to the operator and may cause at least one ship to miss her next outward sailing from the home port.

To prevent bunching is one of the major objectives of schedule makers. Depending upon the relative importance to the carrier of the outbound and homeward trades, and the cargo commitments made, it may be possible to compensate for lost time by omitting ports on one leg of the voyage. Speeding up cargo operations is practical only to a certain extent, especially if port time already has been figured at approximately the maximum capability of ship and port. It is not always possible to transship cargo to avoid the necessity of calling at smaller ports.

Throughout this discussion of schedule making, there has been no mention of the effect on the schedule of the operational necessity to make repairs and to put ships into drydock at periodic intervals. Routine voyage repairs are accomplished while the ship is in port on her regular schedule. The proper performance of this work within the allotted port time is supervised by the staff of the superintendents of the deck, engineering, and catering departments. Drydockings for normal hull inspection and repainting are set months in advance, and adjustments are made to permit the ship to be withdrawn from what otherwise would be normal port

activity for the relatively short period required in drydock. As the vessel gets older, and the magnitude of repairs increases, it may become mandatory to set aside several successive days for the required work. This presents no insurmountable problem so long as it can be foreseen and appropriately scheduled. Emergency repairs are effected on an emergency basis, and any changes brought about as a result are made in the most satisfactory manner possible.

Schedules for voyages of tramp ships in the dry-cargo trades are predicated upon the same considerations that govern schedule making for cargo liners. The major difference is that no repetitious trips have to be taken into account for vessels on single-voyage charters. Time chartered vessels, if employed on the same route for the duration of the charter, are handled as though they were cargo liners. If they are assigned to different voyages during the life of the charter, the planning would be the same as that for single-voyage chartered ships.

Tanker schedules follow much the same routine as do those of cargo liners, except that often the ship does not repeat a voyage immediately. The schedule may be made for a number of single voyages within the framework of employment over a period of several months. Since tanker cargoes are liquid and are pumped rather than handled by longshoremen, that aspect of scheduling is eliminated. In its stead are substituted several problems peculiar to tankers: the location of the tanks into which the cargo is to be pumped (i.e., the distance and gradient of the pipeline through which the oil must be pumped by the force of the ship's equipment); the diameter of the pipes (i.e., the quantity of oil which can be moved in a given period); and the season of the year, because heat or cold affects the speed with which the liquid flows through the pipes. Port time for tankers always is set in terms of hours, and much effort is expended to turn the ships around within the limited time.

When one considers the extent of the sea routes, the variations in winds and currents, the almost complete lack of uniformity in the methods of handling ships in different ports of the world, and the amazing amount of detail that must be taken into account in making up the schedule, it is remarkable that the ships come into port and depart as near to the predicted hour as they do.

Scheduling and Bunkering

ONCE the owner decides on the employment of a ship, it is necessary immediately to develop a sailing schedule that shows the date of departure, the ports of call and the dates and times when those calls will be made, and the anticipated time of arrival at the destination. In direct support of this schedule a “bunkering plan” must be formulated to show where the ship will replenish her fuel supplies and the quantities to be taken aboard from the suppliers named in the plan. In essence, the bunkering plan is the result of studying the fuel needs of the vessel, her earning capability, the various ports along the route where fuel may be procured, and the cost in money and in ship time to buy fuel in the selected outports. The planning studies also take into account the requirements for the safety of the ship imposed by international treaties, the regulations promulgated by a number of quasi-official bodies, and the insistence of the owner that the voyage return a profit (or, in bad times, that the losses be kept to a minimum).

For the operator or charterer of tramp ships, preparing the bunkering plan is an integral part of the computations preceding every proposed voyage, and has a direct bearing on the earning power of the vessel. This is especially noticeable when the commodity to be transported is heavy (“dense”), and every ton of fuel means one less ton of revenue cargo.

In liner service, once the operation has become stabilized and the needs of the ships can be predicted with accuracy, the bunkering plan becomes a standard routine to be followed voyage after voyage.

The sailing schedule is the foundation of the voyage. Until the proposal for the voyage has been converted into definite movements over a selected route, with certain ports to be visited for a predetermined number of hours, no estimates can be made as to

the fuel needs of the ship, and no recommendations can be offered as to the most desirable refueling points. It is evident, therefore, that the sailing schedule reflects the owner's decisions where the ship is to go, and the bunkering plan (which automatically becomes the bunkering schedule when it is approved) is the indispensable supporting program that designates the ports where refueling is to be effected.

If economic considerations justify the action, and the sailing schedule can be adjusted to allow the time for the call, it is acceptable practice for the ship to be sent to a port for the sole purpose of taking on fuel. If, however, the schedule is rigid, and adherence to the advertised timetable is of paramount importance, the choice of fueling points must be limited to those ports to which the ship already is committed to meet the demands of the shippers. Whatever the circumstances, every voyage of a powered vessel depends upon both the sailing and bunkering schedules. Coordination is essential between the traffic department, which must please the customers, the operating department, which seeks optimum performance from the ships, and the treasurer's department, which looks for a profit from the ships' employment.

Speed is a tangible asset to steamship operators, but there are definite limits to the amount of speed that can be offered to shippers. To increase service speed, not only must the power plant be enlarged enormously (to double the speed requires eight times the horsepower), but the entire hull design has to be changed to permit the vessel to move efficiently through the water. When to these considerations is added the necessity to find space for the larger engine installation and tankage to carry the greater quantity of fuel oil, the complexities of speed become apparent.

Schedules, whether for a new service or to modify an existing ship operation, are the product of coordinated activity often involving the highest levels of management. Inevitably, when any tentative schedule is submitted for approval, affirmative answers must be forthcoming to these questions:

1. Does the proposed schedule meet existing or anticipated competition? Have the demands of shippers for minimum transit time been measured against the capabilities and performance of competing carriers?



The ore-bulk-oil carrier *Eric R. Fernstrom* (built 1971), 101,850 tons deadweight, receives diesel fuel from two barges. Photo: Jeff Blinn, courtesy Moran Towing Co.

2. Does the proposed schedule reflect improvements (or continuing deficiencies) in cargo-working capabilities of the ports along the route? As part of the inquiry into this area of operation, cognizance must be taken of the status of programs to install labor- and time-saving equipment and to negotiate better working hours (from the viewpoint of saving ship time) with labor groups.

3. Does the proposed schedule take into consideration those factors that are detrimental to efficient operation? Is there congestion from too many ships attempting to deliver cargo at the same time through a single, overworked port facility? Are alternate ports available to avoid those which are congested?

4. Does the proposed schedule make the best use of the ship's speed and cargo-working characteristics? The cost of speed has to be taken into account in the schedule, but at the same time the earning power of the ship cannot be overlooked. For instance, if a dawn arrival at a port will assure no delay in working cargo, and to effect this time of arrival it is necessary to cruise slightly above the speed that results in the most economical fuel consumption, the

cost of ship's time, the convenience of the shippers and consignees, the competitive advantage, and the impact upon the whole voyage must be worked into the schedule. The cardinal rule is that every aspect of the proposed employment of the ship must contribute positively to her revenue-earning potential.

5. Is maximum economy, consistent with the requirements of the service, achieved by the cruising speeds necessary to maintain the schedule? Has the earning potential of the vessel been measured against the proposed procurement plan for fuel to be taken aboard during the voyage?

Formulating the bunkering plan after the tentative sailing schedule has been approved involves observance of these rules:

1. Every bunkering plan must comply in all respects with international load line treaties and also with the requirements of marine insurance underwriters concerning reserves of fuel oil.

2. The bunkering plan must assure the maximum earning power of the ship. (This applies with particular force to tramp shipping and to the shipload transportation of any commodity which weighs as much as or more than the fuel.)

3. The ship may not be deviated significantly from her assigned route solely to acquire fuel at a low price. (Deviation may take the form of protracted delays as well as actually departing from the prescribed ocean course. Fuel normally is procured along the route, in the most satisfactory ports.)

4. Before the ship is ordered to a named port to take on fuel, there must be assurance from the provider that supplies are adequate to meet the needs of the ship.

Compliance with load line requirements is basic to any bunkering plan, under penalty of having the ship declared unseaworthy. Without attempting to trace the history of the load line agreements in detail, suffice it to recall that in 1875 Samuel Plimsoll, a member of the British Parliament, led a successful campaign to require ships to bear markings on their sides showing the depth to which they could be loaded safely in various areas of the world and seasons of the year. These rules in time became international conventions and the world now is divided into zones designated by the self-explana-

tory terms of Tropical, Seasonal Tropical, Summer, and Seasonal Winter. There is a further marking to allow for the difference in the buoyancy of fresh and salt water. Ships of not more than 100 meters (328 feet) in length are required to have a special load line for voyages into the North Atlantic in winter. This line (appearing on the ship's side with the distinguishing letters WNA) is determined so that the ship will have the maximum freeboard, and therefore imposes the greatest restriction on the amount of cargo that can be carried.

American shipowners were placed under the obligations of the International Load Line Convention of 1930 when the Congress ratified the agreement in 1931. The actual load lines are determined by the classification societies when the designs are submitted for their approval. Enforcement of load line regulations has been assigned to the United States Coast Guard. Courts of law have held repeatedly that overloading makes a ship unseaworthy, and consequently the owner is deprived of any legal protection for loss or damage sustained by the ship. Marine underwriters have resisted successfully any claims for restitution of loss when a ship has been loaded deeper than the assigned marks and thereby been rendered unseaworthy.

When a vessel crosses from one zone to another, as in the case of a ship going from New Orleans to Liverpool in December, she must be loaded so that when she crosses the winter zone line, she will have the prescribed load line. Arrival in Liverpool with a draft exactly at the winter load line would be *prima facie* evidence that she had less than the mandatory freeboard when she entered the zone. The most scrupulous observance of applicable load lines and zone areas is required of all those involved in determining how much cargo to put into a ship.

For example, imagine that a small tanker of 17,178 tons deadweight on her summer draft is to sail on December 15th from Bahrein Island, in the Arabian Gulf, for Bantry Bay, Ireland. Bahrein lies in the year-round tropical zone, and therefore the ship can load to the minimum freeboard, taking on a total of 17,498 tons deadweight. The ship is to proceed through the Suez Canal into the Mediterranean, where her deadweight at the winter load line cannot exceed 16,282 tons. On passing Gibraltar and entering the North Atlantic, a further restriction would be encountered, and

maximum tonnage would be only 15,962. In planning the voyage, therefore, paramount consideration must be given to the permissible deadweight of the ship when she rounds Gibraltar. Her load at Bahrein would be determined by that limitation plus the amount of fuel oil that would be burned in transit from the Suez Canal to Gibraltar, plus the fuel necessary to take the ship from Bahrein Island to the Mediterranean end of the Suez Canal.

Another aspect of the demands for safety of life at sea is the underwriters' mandate that all ships carry a reserve supply of fuel equal to not less than approximately one-quarter of the fuel to be burned between scheduled ports of call. This means that a vessel proceeding from New York to Panama cannot carry just enough fuel to reach her destination plus 25 percent of the amount needed from New York to some intermediate, but unscheduled, port where fuel is available. The ship must take on 125 percent of what will be consumed on the passage to Panama. If she is fueled in New York for the round voyage, she would carry enough oil for the trip plus 25 percent of the amount needed from Panama back to New York.

The second rule in drawing up the bunkering plan is to assure the maximum earning power of the ship. This requires not only a knowledge of the ship's routing, the possible ports where oil may be purchased, and the load line restrictions, but also awareness of the freight rate payable per ton of cargo loaded, the daily costs of the ship in addition to fuel, and the characteristics of the cargo being transported. In liner service, it is generally true that what is carried is bulky and fills the ship's holds before the vessel is down to her load line marks. Bunkering in this service involves selection of the ports along the route where fuel can be obtained in adequate quantity at a good price without delaying the ship. Tramp ships often carry commodities which, per ton, weigh at least as much as fuel oil. Every ton of fuel put aboard the ship which exceeds the anticipated requirements therefore reduces the potential profit by the revenue from one ton of cargo.

When the great expense of owning and operating a ship is appreciated, the reason for the third rule in preparing the bunkering plan becomes clear. Just because a fueling point distant by two days of steaming from the shortest course between origin and destination is selling oil at a reduced price does not justify diversion

of the ship from her routing. Not only must the intent of the contract of carriage be met, and all reasonable speed be made to traverse the sea distance, but the cost to the owner in ship time cannot be overlooked. If the ship is in liner service, the ripple effect of the added time of sailing may not be overcome for the remainder of the voyage. For a tramp, the ripple effect might be less conspicuous, but it could have serious consequences in the ship's ready availability for subsequent employment.

The reason for the fourth rule in preparing the bunkering plan is found in the aftermath of the oil crisis of 1973, when shipowners began to demand assurance from suppliers that the requisite quantity of the desired grade of fuel would be available when a ship arrived at the fueling point. Since 1973, many changes have been made in the techniques of oil refining, with deleterious effects on the quality of marine fuel oils. Ship operators consequently have placed great emphasis on the acceptability of the oil provided by suppliers. In many instances, what has been delivered to the ship has failed to meet minimum standards for that grade of oil. Significant mechanical difficulties, especially with diesel motors, have been experienced. Procurement of fuel that meets the owners' specifications is a matter of serious concern to senior executives as well as ships' engineering officers. Deliveries must be monitored carefully, and samples taken throughout the period of delivery. These samples should be tested both aboard ship and at especially equipped laboratories.

The Baltic and International Maritime Council, which represents a large number of owners whose ships are always under time and voyage charter, proposed in December 1981 that all the council's standard charter parties contain a "bunker quality clause." An appreciation of the comprehensiveness of this clause may be gained from these excerpts:

1. Charterers shall provide fuels suitable to the vessel to enable the main propulsion and auxiliary machinery to operate efficiently and without harmful effects, normal wear and tear excepted.
2. Fuels supplied shall conform to specifications mutually agreed between Owners/Charterers.
3. Representative samples of the fuel supplied shall be taken at the vessel's bunker manifold during bunkering and sealed in the presence of competent representatives of the charterers and the vessel.

Such samples shall be retained by the user and the charterer for 90 days after the date of delivery of the fuel. In the case of a prior dispute, samples shall be retained.

4. Any dispute as to whether the bunker fuels conform with the agreed specification and/or with the warranty set out in paragraph 2 shall be settled by analysis of the sample(s) by _____ or by another mutually-agreed fuel analyst, whose findings shall be conclusive evidence between the parties as to conformity or otherwise with the bunker fuel specification.

5. Any delay . . . caused by difficulties in obtaining bunkers meeting the agreed specifications shall be for Charterer's account.

6. Any warranty of performance given by Owners is conditioned upon the fuels supplied meeting the agreed specification.¹

The application of these rules can be seen in the following example of how a bunkering plan is prepared:

A modern bulk carrier of 63,500 tons deadweight has a speed of 15 knots. She consumes 37 tons of oil per day at sea and 10 tons when in port. With all tanks filled, she can carry 2,870 tons of fuel and 250 tons of water. Stores normally weigh an additional 100 tons. The minimum cargo capacity therefore is 60,280 tons. Vessel earnings can be increased by reducing the fuel load to that which is needed for the planned voyage.

Assume, for the purposes of this illustration, that the ship is assigned to trade for a year between two ports approximately 6,300 miles apart. Routed nonstop from loading point to place of discharge, the voyages require 17.5 days at sea plus four days to work cargo. On each voyage the engines consume 648 tons of oil and also need 40 tons to provide services while in port. The reserve requirement comes to 162 tons, and water and stores amount to 350 tons. The cargo lift is 62,300 tons. The alternative routing is to go by way of three fueling points, which would lengthen the voyage by three days. The longest leg between fueling points is nine days, requiring 333 tons of oil. Adding the reserve of 84 tons and the 40 tons of fuel to satisfy port needs, the total weight of oil comes to 457 tons. Together with the 350 tons of water and stores, the reduction from the ship's deadweight leaves a cargo lift of 62,693 tons. The shorter voyage, however, makes possible the completion of 9.36 round trips delivering 583,128 tons in one year, while the longer passages leave time only for 8.3 deliveries totalling 520,352 tons.

The ship always must have on board sufficient fuel to make the prescribed voyage, regardless of how much cargo may have to be sacrificed. Once the minimum fuel requirement has been determined, the question of whether to load additional oil or more cargo will be answered only after studying the prevailing freight rates, the costs of fuel along the route, and the potentials for profit (or reductions in cost) which may exist.

Ships in liner service, repeating their voyages, usually take on the same quantities of fuel, water, and stores in the same ports on every trip. A pattern thus is established which makes bunkering a matter of procurement rather than of planning. For example, a vessel operating from New York to Valparaiso via several South American ports might find that Cristobal (Panama Canal) is the cheapest point in which to buy oil. The bunkering pattern would call for the ship to sail from New York with sufficient fuel, plus reserve, for the voyage to Cristobal. At that port, oil would be taken aboard for the voyage from Cristobal to Valparaiso and return. The reserve requirement would be computed on the basis of the fuel needed for the last leg of the northbound passage. At Cristobal, homeward bound, the ship would be supplied with enough oil, including reserve, for the passage to New York and then back to Cristobal, where the procedure would be repeated.

As soon as the ship has been assigned to a voyage, the bunkering staff must determine what ports are available or desirable as possible fueling points. These are some of the questions that must be answered in the process of developing the bunkering plan:

Are supplies of fuel in this port plentiful?

Is the fuel of the desired type and quality, and is it available in the quantity required?

What is the price of fuel? Does this price include delivery charges if lighterage is involved?

What are the physical arrangements for bringing oil to the ship (i.e., pipeline, barge, harbor tanker)?

Can the ship take on fuel while working cargo at her berth?

If a barge is used for delivery, will it be brought to the ship after dark? How many tons of oil can the barge carry, and what is the pumping rate per hour from the barge into the ship's tanks?

If fueling must be accomplished only at a special fuel pier, what fees are involved? How much time is needed to shift to this facility,

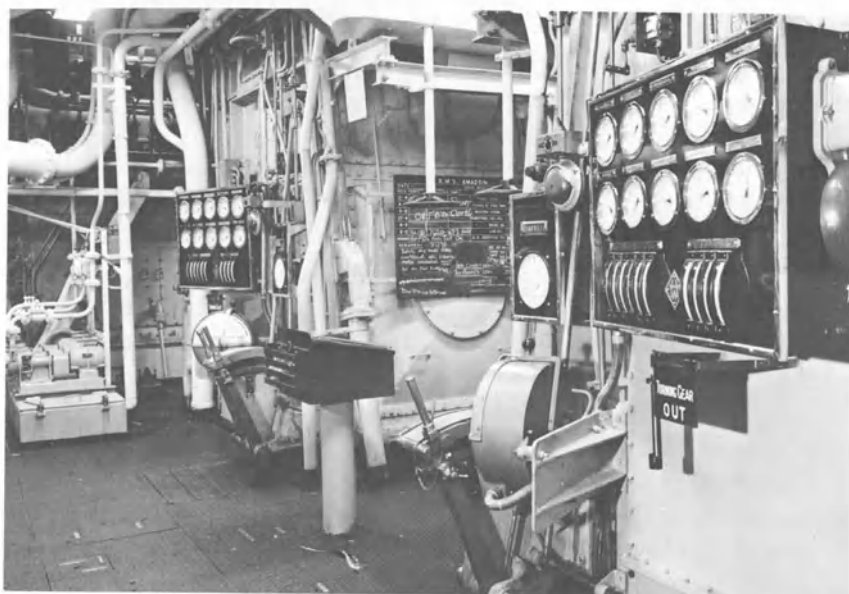
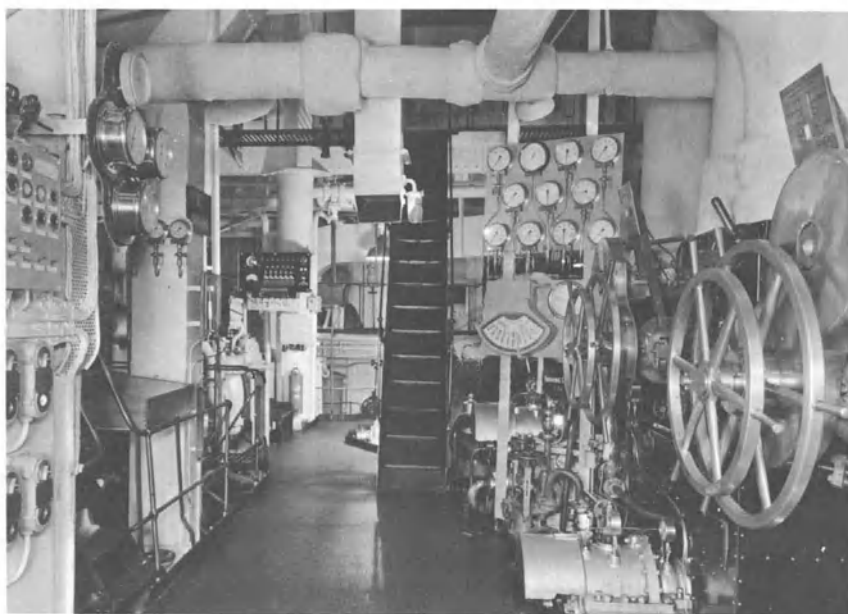
and how fast is fuel loaded aboard ship?

What regulations govern the fueling of a ship at anchor in the harbor?

Assuming that both the schedule and the bunkering plan are approved, and orders are issued to make them effective, the operating department instructs the agents in the ports of call. Among the points to be verified are that pilots actually perform according to contracts, that tugs can and will be available to berth and unberth the ships, and that longshoremen, checkers, security guards, operators of dockside equipment, and any other essential working or supervisory personnel will be on hand when the ship arrives. It also requires assurance from the port authorities, if the steamship company does not have an exclusive lease on the terminal, that appropriate pier or wharf space with the supporting transit shed and upland area will be clear and ready for the vessel. For shipboard personnel, the working schedule must make certain that officers and unlicensed mariners needed during the port call will be on duty, and that no delays will result from failure to have the "right man in the right place at the right time."

As soon as possible after a fueling port has been designated, the agent must be notified. The information to be transmitted to this representative, at the minimum, must include the name of the ship, the estimated time of arrival, the estimated quantity and grade of fuel oil required, the length of time the ship is scheduled to be in the port, and the amount of cargo, if any, which will be handled during the port call. Two days before the scheduled arrival, the master must send a radiogram to the agent giving the time of arrival of the vessel, and the approximate number of barrels of oil his ship will require. This notification permits the agent to make local arrangements to receive and service the vessel. A confirming message usually is sent by the master twenty-four hours before arrival to assure that the vessel will be accorded the treatment desired. It comes as no surprise to anyone that the agent is waiting at the berth when the ship is being moored.

Oil companies, particularly major organizations with many foreign affiliates and bunkering stations, offer contracts to shipowners on a yearlong, worldwide basis. Should there be no contract, the operator must purchase his fuel oil at the local ("spot") price, which



Top: Starting platform of a twin-screw turbine-driven steamship. Photo: Union Castle Line. *Bottom:* Control station for the two diesel engines of a modern twin-screw motorship. Photo: *The Motor Ship*, courtesy Royal Mail Line.

is subject to daily fluctuation as a result of changes in the cost of crude oil, tanker rates, and the normal working of the laws of supply and demand. The contract price, on the other hand, may be increased only after advance notice to the purchasers. Reductions in price always are made effective immediately since they are advantageous to purchasers and serve to retain goodwill. By signing a contract, the shipowner obtains a certain degree of stability in fuel oil costs. This permits computation of voyage costs to be made with a satisfactory margin of accuracy at the outset of the voyage.

Because no single oil supplier has sales outlets in every port of the world, shipowners trading in several geographical areas usually sign contracts with more than one provider. When this is done, the different suppliers are responsible for certain ports or for definite geographic regions. For instance, a liner-service company operating in a number of different regions might contract with M Company for ports of the United States and the east coast of South America, N Company for West African ports, O Company for ports in South and East Africa, and P Company for the Mediterranean. By spreading the business among the various suppliers, the shipowner is assured that all vessels will obtain fuel of acceptable quality at the published contract price. When supplies of oil are adequate, ships usually are bunkered on a first come, first served basis, without regard to whether the vessels are covered by contracts. In the event of shortages, however, priority would be given to vessels under contract.

Into the bunkering schedule are written the names of each port on the ship's itinerary, the places where she will take on fuel oil, the estimated quantity to be loaded, and the amount of oil the ship is expected to consume between bunkerings. If the vessel is engaged in conventional liner service, and her refueling points remain practically unchanged voyage after voyage, the responsibility of preparing the bunkering schedule may be assigned to a junior member of the operating department, usually in the office of the superintendent engineer. Should the ship be assigned to tramping operations, especially under voyage charters, the senior executives may either prepare the bunkering plan themselves, or insist on approving it before it becomes effective. This results from the fact that the earning power of the ship is affected directly by the ports selected and the quantities and prices of the oil purchased.

In the tanker trades, a basic fact is that the quantity of fuel taken aboard invariably reduces the payload. The impact of this upon bunkering practice has been explained earlier in this chapter. It also is important to appreciate that the ship delivers her cargo to ports where oil is not naturally available. To make the tanker as useful as possible, therefore, the proposed bunkering plan should not envision taking any fuel out of the destination port except in special circumstances. If, for example, the tanker is bringing a cargo of aviation gasoline, and maximum tonnage is needed for airline operations, the shipowner would be justified in buying ship fuel at this port of discharge. If the cargo were the crude oil from which ship fuel was derived, purchase of bunkers would produce little economic advantage.

One of the reasons for using very large crude oil carriers (250,000 tons deadweight or larger) in preference to ships of about 70,000 tons deadweight is that the big ships deliver 28 barrels of cargo for every barrel of fuel they burn, whereas the smaller ships deliver only 11 barrels of cargo for each barrel consumed. The same reasoning applies in comparing diesel engines and steam turbines; diesels use less oil per horsepower output than any other form of seagoing power.

In making the bunkering plan for a tanker, therefore, the ideal is to deliver the greatest possible quantity of cargo, in order to earn maximum revenue, while removing from the discharge port the minimum amount of fuel oil. Achievement of this ideal may be facilitated if an intermediate and low-priced bunkering port can be visited without significant deviation from the shortest sea distance, and the time spent taking on the required fuel is offset by the increased earnings of the ship.

Making the schedule for a tanker follows the same processes used for dry-cargo carriers, except that the tanker is self-sufficient in her ability to discharge her cargo. Regardless of the hour of the day or night when she arrives at the receiving terminal's facility, a tanker commences cargo operations immediately. If she is to load at this port, the delivery of cargo begins as soon as the hoses from shoreside tanks are connected. If the vessel is to discharge, her pumps are started when all hoses have been matched with shore piping systems. No large crews of longshoremen or cargo checkers are needed; usually permanently assigned terminal personnel han-

dle all details. Relatively few tankers are sent to more than one port to discharge, although multiple port unloading is not an isolated phenomenon. Hardly ever does a tanker deliver a cargo and then take on an outgoing load, simplifying the problem of port turn-around time to that extent.

The dry-cargo tramp trade is governed in its bunkering and scheduling practices by the same principles that apply to the liner and tanker operations. There is still the need to load cargo as quickly as possible. In the negotiations for a voyage charter, the number of days allotted for loading and discharge is fixed, with corresponding bonus and penalty for reducing or exceeding these limits. The master must supervise operations and keep an accurate record of time used and activities pursued. Bunkering for a trip under a voyage charter entails the same considerations as in the liner trade: if the cargo is bulky, then the port where the appropriate fuel can be obtained at least cost is chosen, while for dense cargoes the revenue from each ton of cargo carried is compared with the price of fuel oil to determine where and in what quantity the fuel is to be acquired. To illustrate the process, the imaginary voyage of the *Lucky Lady* will be followed.

The *Lucky Lady*, a diesel-powered vessel of 18,150 tons dead-weight, has been chartered for a voyage from Hampton Roads to San Antonio, Chile, carrying coal at a rate which barely meets operational expenses. The terms of the charter permit the owner to load not less than 15,000 tons and not more than 17,000 tons of coal. Northbound, the *Lucky Lady* has been fixed for a part load of 6,000 tons of slab copper from San Antonio to Baltimore, and all the bagged sugar the ship can accommodate from Callao, Peru, to New York City. The freight rate on the copper is at the break-even level, but the rate on sugar is very profitable.

The essential facts influencing the decisions required for the bunkering plan were: (1) the ship consumed 21 tons of fuel per day at sea, and 7 tons per day when working cargo in port; (2) there was no diesel fuel available in San Antonio; (3) fuel could be purchased in Callao, but at a high price; and (4) the lowest priced fuel was sold in Panama.

To reduce the cost of the voyage, it was determined to buy only enough fuel in Hampton Roads to reach Panama. There, fuel for the voyage from Panama to San Antonio and return would be



Computer-oriented engine controls are centralized in a soundproof room aboard a modern containership. Photo: Seatrain Lines.

purchased. When the *Lucky Lady* came through Panama enroute to Baltimore and New York, she would be supplied with oil for that passage.

Steaming time from Hampton Roads to Panama was 4.75 days; from Panama to San Antonio, 7 days; from San Antonio to Callao, 3.5 days; from Callao to Panama, 3.5 days. The passage from Panama to Baltimore would require 5.2 days, and from that city to New York approximately a half-day. Total fuel consumption at sea was calculated to be 565 tons; the ship's tanks had a capacity of 800 tons.

The quantity of revenue cargo was established by the tonnage of fuel, stores, and water that had to be aboard when the ship departed Panama southbound, and from Callao northbound with the combined loads of copper slabs and bagged sugar. Fuel needs were 415 tons for steaming, 56 tons for reserve fuel, and 19 tons for cargo work, for a total of 490 tons; water and stores added another 350 tons to the ship requirements, and left 17,310 tons for cargo. Pursuant to terms of the charter, 17,000 tons of coal were loaded in Hampton Roads. Northbound from Callao, the ship had to allow 109 tons for the trip to Baltimore, plus 28 tons reserve; four days of

cargo work required 28 tons of oil, and the short distance from Baltimore to New York needed only 11 tons. Stores and water remained at the constant figure of 350 tons, so the maximum tonnage of sugar that could be loaded came to 11,624 tons.

The bunkering plan and schedule were developed with all aspects of the ship's employment in mind. Full advantage was taken of the flexibility granted by the charters for both the coal and sugar cargoes, and fuel was purchased at the least expensive point. By these devices, the greatest revenue was earned from the ship's employment.

In summary, it may be stated that the successful activity of any ship depends upon the skill with which the schedule is developed, the manner in which advantage is taken of the characteristics and capabilities of the ship, and the nature of the cargo as well as the prevailing freight rates. Customer desires for the shortest possible transit times and the requirement by the owner that the ship be used effectively and efficiently usually may be combined to satisfy all parties concerned. In the tanker and many dry-cargo tramping trades, deadweight cargo is carried almost as a matter of routine. Because only cargo produces revenue, the accurate determination of how to fuel the ship is vital to owner and operator. That the intricacies and possible pitfalls of planning the scheduling and fueling of the ship have been mastered is one of the signs of professionalism in ship management.

Planning for a New Ship

IT IS almost an axiom of the steamship business that the best ship for any trade is one that has been designed expressly to meet the particular needs of the route to which the vessel is assigned. This belief is not vitiated by the fact that many ships have been built in assembly line fashion to a single pattern, and have been used in widely different services. Standardized ships are accepted only because the cost of planning, designing, and building ships intended particularly for certain trades is so great. Every shipowner cherishes a dream of the perfect ship for one special type of employment, and every ship built to individual specifications embodies as much of those dreams as the vision and skill of the naval architect can transform into reality.

Dreams shatter on contact with reality, and dreams of perfect ships often vanish when faced with the harsh facts of economics. Dreams must yield to the newest developments in marine engineering and naval architecture as well as in packaging and cargo-handling techniques. The dreamer also must accept the unavoidable mandate that the ship earn her way and pay a profit. It becomes the part of owner, naval architect, marine engineer, traffic manager, and operations chief to combine their capabilities in the endeavor to make the ship an asset throughout her economic life.

Converting dreams into the tangible reality of a ship follows an orderly and systematic process that begins with examination of the trade route to be served. This involves study of the cargoes that are, and may be expected to be, carried, the extent of existing and anticipated competition, and the efficiency of the ships now on the route. The purpose of this examination is to describe with nicety the mission of the new ship. Competition must be evaluated, for on the capability of the new ship to offer better transportation than any other vessel depends the economic success of the venture. Of



Lykes Lines' barge carrier ("Seabee") can transport inland waterways barges as well as containers. Photo: Lykes Lines.

almost equal importance is the determination of what cargoes will be carried, and the methods of packaging which are now in use and may be adopted in the next ten or twenty years. The wrong decision, as for instance the selection of a cellular containership in preference to a more versatile combination type, may doom the vessel even before she sails on her first voyage. To make the correct choice is especially critical when it is recalled that a break-bulk cargo

carrier costs at least twenty million dollars, and a containership cannot be obtained for less than fifty million dollars.

The following list demonstrates the areas of concern and the depths of inquiry necessary to assure that the ship will fulfill her mission:

Basic Data

1. Anticipated years of operational life of the ship.
2. Ports of call, length in miles and duration in days of each voyage, and number of voyages planned for each year.
3. Estimated average tonnage of cargo to be loaded on outbound and homeward passages of each voyage. Types and quantity and the maximum weights of extra-heavy-lift items must be indicated separately.

Traffic

4. Total deadweight tonnage and cubic capacity required in the ship. If refrigerated cargo is to be carried, the tons of this kind of cargo, and the number of cubic feet of space needed, must be determined separately.
5. Liquid cargo capacity (i.e., number and characteristics of deep tanks and other liquid cargo spaces) essential in this trade.
6. Cubic capacity for dry bulk cargo and the various types of this cargo to be carried (number and special features of below-decks spaces allocated to dry bulk cargo, or usable for this kind of cargo).
7. Cubic capacity and deadweight tonnage of special cargo lockers. If special features are required, they must be identified.
8. Provision for stowage of containers (size of boxes, number, weight, stowage plan, number of containers to be carried under decks and number to be stacked on deck).
9. Provisions for handling wheeled vehicles (square footage, by decks; access arrangements; stowage and securing of vehicles).
10. Provisions for handling heavy-lift cargoes.

Operations

11. Location of bunkering ports on the trade route(s) to which the ship normally will be assigned.
12. Propulsion equipment, consumption per day, and type of fuel.

13. Harbor and port restrictions (depth of channel at low tide in the port with the least depth of water; berthing arrangements; depth of water at the berth; canal locks, including dimensions of lock chambers; dock entrance widths, and depth of water on the sills; tide ranges).

Port Cargo Handling Facilities

14. Equipment in ports of call (availability, number, type, and location within the port of fixed dockside equipment to unload and load bulk cargoes; container terminals; floating or land-based heavy-lift cranes).
15. Supporting facilities (provision for storage of dry and liquid bulk cargoes; railroad and highway vehicles' access to ship's side).
16. Height of booms of container-handling cranes which project over the ship's berthing space.

Regulatory Agency Requirements

17. Government regulations affecting ship construction.
18. Classification society requirements.
19. Load line rules.
20. Requirements relating to subdivisions of the hull.
21. Limitations affecting hull admeasurements.¹

Once the mission has been determined, the owner can translate it into proposals on which the naval architect can base suggestions for a preliminary design, and can commence a series of technical studies leading to specifications of length, beam, depth of hull, draft, speed of the ship and resultant shape of the underwater body, and horsepower. As these ideas take on more substance, it becomes feasible to prepare outlines on which to base an initial estimate of construction costs. When this is available, the financial and traffic managers evaluate the revenue-producing capability of the ship in contrast to the cost.

To ascertain the maximum price which the prospective owner can afford, it is satisfactory to make pro forma calculations which show what the vessel must earn per ton of cargo carried if she is to pay operating expenses, construction cost, and interest. For example, it is estimated that a tanker of 30,000 tons deadweight will cost

\$42,400,000. She is to operate between Texas and New York, and her normal load will be 28,200 revenue tons. To determine what she must earn, these assumptions are made:

Operating days per year	350
Length of voyage, in days	10.1
Number of trips to be completed in a year	34.7
Cargo delivered in one year, tons	978,840
Crew wages for 26 persons, per year	\$1,470,000
Subsistence of crew	50,000
Stores, supplies, and equipment	95,000
Maintenance and repair	350,000
Insurance	190,000
Other expenses	35,000
Port fees and charges	682,000
Fuel costs	1,980,000
Capital cost (construction amortized over 20 years; interest payable at 16 percent per year)	<u>4,980,000</u>
Total annual costs	\$9,832,000 ²

The freight rate required to meet these costs is \$10.04 per ton. It is derived by dividing the total annual cost by the number of tons of cargo carried in a year. The senior traffic executive must determine whether the freight which can be charged will equal or exceed the required freight rate.

Notwithstanding the mass of statistical, mathematical, and economic information which the decision-making process generates, there is no substitute for the experience and accumulated wisdom of the senior executives. The vice-president, traffic, draws on his knowledge of the vagaries and vicissitudes of the trade, its peaks and valleys, and the intensity of competition, in his examination of the proposed design to determine if it meets the demands of shippers for improved service. The vice-president, operations, scrutinizes the design to be sure the ship can perform efficiently in her intended trade. Acceptance of the idea of a new ship and final approval of all the detailed studies and computations ultimately are the responsibilities of a single individual who, while analyzing all that has been provided, in making the critical decision must rely upon his personal knowledge, experience, and wisdom.

Once the idea of the new ship has been approved by all concerned, the architect is able to start developing the preliminary design. It is at this point in the process that the physical characteristics of the ship are fixed, and thereafter no substantive changes in the length, beam, horsepower, and deadweight tonnage are made. Only after the preliminary design has been approved can the immensely detailed and costly plans and specifications be prepared for the use of the shipbuilders who seek to construct the vessel.

To understand the complexity of this procedure it is helpful to outline the historical development of ship design during the last hundred years.

Growth of both commerce and merchant shipping since the last quarter of the nineteenth century onward has been spectacular. Transportation of goods of every description over the sea routes exceeds in ton-miles those of all other forms of carriage combined.³ On January 1, 1983, the need for ships was sufficient to justify the existence worldwide of almost 25,500 power-driven vessels of all types—and this number included only those units exceeding 1,000 gross tons in measurement.⁴ Among the cargoes moving in international trade in the decade of the 1970s, the principal ones, in descending order of magnitude, were crude oil and oil products, iron ore, grains, forest products, coal, iron and steel products, chemicals and fertilizers, bauxite and alumina, and general cargo.⁵

A survey of dry-cargo ships constructed during the half century from 1910 to 1960 revealed that there was a striking similarity in vessels throughout the period. There were significant advances in engineering and shipbuilding which made it feasible to increase ship speeds from an average of approximately ten knots to about fifteen, and welding replaced riveting in the fabrication of hulls. The basic techniques of cargo handling, however, remained substantially unchanged, and although the capacity of the cargo carriers was expanded almost three-fold during this half century, the pattern of ship design did not have to be altered to any great extent. Because the variations that did occur were evolutionary, it was possible for shipowners to plan new vessels which represented genuine improvements on the past but showed no radical changes. Based on past experience and on what, in retrospect, seems to have been only elementary economic analysis, new ships were ordered

from builders who competed for the business by submitting bids; the lowest was deemed the most satisfactory. The idea had yet to be appreciated of combining purchase price and full-life operating cost to determine whether a certain kind of ship was economically desirable.

About 1960, several developments occurred which had the effect of revolutionizing both the business of hauling goods across the seas and the techniques of management of ships and shipyards. The first and probably most important was the introduction of the electronic data processing machine, now known worldwide as the computer. This calculator suddenly made it possible to obtain an assortment of data which permitted analysis in depth of many different proposals, suggestions, and alternative ideas. The computer transformed ship design into a many-sided scientific operation involving economic studies, financial analyses, and systematic appraisals of not one but many possible types of vessels for a single route.

It was probably only a coincidence of history, but at about the same time that computer technology became an accepted tool of management, there was an upsurge in the demand for raw materials of all sorts. From 1919 to approximately 1960, the dry bulk cargoes of world commerce moved predominantly in ships of not more than 15,000 tons deadweight capacity. In the petroleum trades, the switch to tankers of 25,000 tons began shortly after World War II, and by 1960 these transporters had reached, and in some cases exceeded, 50,000 tons deadweight. The beginning of the decade of the 1960s marked the start of the sudden growth in dry-cargo ship capacities. In the bulk trades, especially to transport ores, the trend was toward ships of 40,000 to 50,000 tons deadweight. Liner-service ships were converted to handle containers, and in numerous instances they were enlarged. At the end of the decade, tankers of over 300,000 tons deadweight no longer were news; "bulkers" able to lift from 100,000 to 150,000 tons were numerous; break-bulk ships in liner service were almost uniformly of 15,000 tons deadweight, and container ships varied from 20,000 to 40,000 tons. For most of these big ships, an increase in speed accompanied the growth in size.⁶

Throughout this time, the capability of the computer to perform an extraordinary variety of analytical calculations was being

applied to both the shipowning and the shipbuilding industries. There was abundant reason to capitalize upon every advantage offered by the computer as the cost of new construction soared to unprecedented heights and the expenses of operation increased year by year. To forecast with greatest accuracy the potential revenue which a ship could earn was essential to financial stability in the steamship business. Many owners turned to the computer. A new atmosphere began to prevail as computer experts were called upon to assist management personnel, and novel techniques, starting with analyses of projected shipbuilding programs, were introduced.

To understand the role of the computer and the specialists responsible for their activity, it is necessary to look briefly at the steamship business as it exists today.

In the dry-cargo trades, ships are employed either in liner service or in trampng. Liner service implies repeated voyages on the same sea route, with calls at specified ports. If the owner is well established, with long experience in this particular operation, and plans to build a ship to replace an obsolescent unit in the fleet, it is relatively simple to distinguish those features of the trade which influence the design of the ship. If, however, a completely new route is to be served, the owner must conduct whatever research is required to determine why ships now operated on that route are designed as they are, and how those ships accomplish their mission. If it appears that the existing ships are deficient in one or more respects, the prospective owner must determine how to improve the design of the proposed vessels to assure that they will meet the requirements of the trade efficiently and economically.

A cargo liner in all probability will transport a rather well defined variety of items in predictable quantities and packages. Study of the specific design of the ship therefore may be concentrated upon arrangements for cargo stowage and the means of handling that cargo. If booms or cranes are to be installed, their capacity must be examined in the light of the shipper's existing and anticipated practices. To minimize the time and cost of loading and unloading ships, the sizes of hatches have been increased greatly over the years, reaching what may be almost the structural limits in the *Delta Argentina* class of ships built in 1968. These vessels had four rows of hatches, each row consisting of three parallel open-

ings, plus one row of three openings. The effect was to give direct access to almost every part of the stowage area below decks. As an added feature, the ships of this class had high-speed cranes to serve the hatches, plus a 75-ton Stuelcken boom erected between two sets of hatches.

Equally important in the consideration of cargo handling and stowage capability is the amount of usable space within the hull. Ideally, cargo holds and 'tween decks should be as nearly rectangular as possible and completely free of obstructions. Very fast ships usually have fine lines which create virtually unusable areas in the forward and after holds. Not only is cubic footage lost, but dunnage must be used to fill the voids into which cargo cannot be placed. Where the design of the ship produces such areas, they should be converted to cargo deep tanks or fuel oil capacity, or to tonnage-exempted ballast spaces.

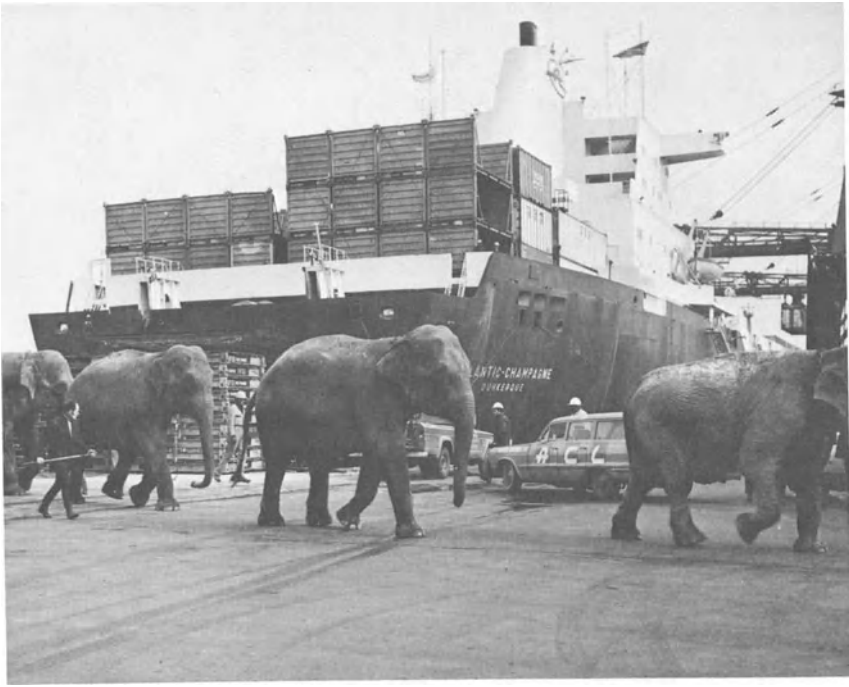
Should the owner contemplate a ship for worldwide tramping, several decisions must be made before the essential analyses can be initiated. Is the ship to be put into a specialized trade, such as the carriage of either iron ore or grain? If so, the dominant characteristics of the commodities to be handled and the various loading and discharge ports—the potential employment, in other words—must be fixed in order that the vessel may have operational flexibility. If the ship is to be a general trader, she would be equipped with the minimum outfit of cargo booms or cranes to enable her to work in any port. This illustrates the principle that the shallowest and least well equipped port of call has a disproportionate impact upon design.

In contrast to the tramp owner, the liner-service operator is committed for the long term to a particular route. As ports are developed and variations are experienced in the flow of goods, the ships can be adapted to these changes. When new construction is appropriate, the experience of the recent past has a significant influence upon ship design. For example, the rapid conversion of major trade routes from break-bulk to container carriage resulted in the withdrawal of a number of comparatively new, well-equipped, and expensive break-bulk vessels and the substitution of purpose-built containerships. The installation of shoreside container-hoisting cranes in the ports between which these ships operated made it feasible to omit hoisting gear from the ships but limited them to sailing only to those highly developed ports.



The 160-ton capacity centerline cargo ramp of the *Hellenic Innovator* can be positioned to port or starboard to work cargo. Photo: Jeff Blinn, courtesy Moran Towing Co.

Sea-Land Service commenced transoceanic operations with fully equipped (“self-sustaining”) ships. It soon proved to be more efficient and profitable to construct dockside cranes and to design ships with no cargo-handling capability. So long as the ships plied only between equipped ports, there were no handicaps to trade. In 1980, however, Sea-Land opened a new service to Middle Eastern ports which had no container-handling cranes at dockside. To meet the demands of the shippers for container transportation, the company equipped four ships with the requisite gantry cranes, having calculated that the cost of procuring, installing, and maintaining the shipboard cranes, as well as the loss in cargo deadweight incurred by the presence of the heavy cranes, would be more than offset by the greater freight earnings.



The versatility of a roll-on/roll-off ship was proven by this walk-on/walk-off cargo. Photo: Atlantic Container Line.

If a roll-on/roll-off ship is envisioned, present practice is to include a massive ramp at the stern which can be lowered to the apron of the pier or wharf to which the ship is moored. The flexibility thus conferred upon the ship's employment is a distinct advantage in establishing and maintaining service to those ports to which are destined extremely heavy trailer-mounted items and large numbers of wheeled vehicles.

Recent developments in the more heavily patronized routes have produced ships which combine the features of the container-ship and the roll-on/roll-off carrier. A rather well publicized photograph of a file of elephants walking ashore from such a combination ship dramatized the flexibility and adaptability of this type of design. Cargo which otherwise would be shipped on break-bulk vessels because it could not be containerized would be loaded on trailers to eliminate handling at the loading and unloading terminals. Heavy, usually over-size, pipe exemplifies this cargo. The



Delta Lines' *Del Oro* had many hatch openings to improve cargo-working characteristics. Her service speed was 18.6 knots, and her deadweight was 13,500 tons. Photo: Delta Lines.

shipper secures it to a trailer, often built expressly for the purpose, and the consignment is handled as a single unit from reception by the carrier until delivery to the consignee at destination.

For ships engaged in the break-bulk trade, a full set of cargo booms and their supporting equipment must be provided if one of the ports to be visited with some regularity does not operate dock-side cranes. Inasmuch as heavy-lift cranes, either fixed or floating, are available in most ports, owners may elect not to install the expensive and cumbersome “jumbo” boom in their ships.

In considering how to adapt the ship to carry cargo, it is essential to take into account the nature and quantity of whatever is offered for transportation. In the break-bulk liner trade, there is so much variety in the normal load that emphasis is placed primarily on the ease and efficiency with which cargo can be worked. As was pointed out in a treatise on naval architecture:

It is well to agree on the fact that what constitutes an efficient layout from the point of view of cargo carrying and cargo handling is greatly dependent upon the special requirements of each trade. It is obvious

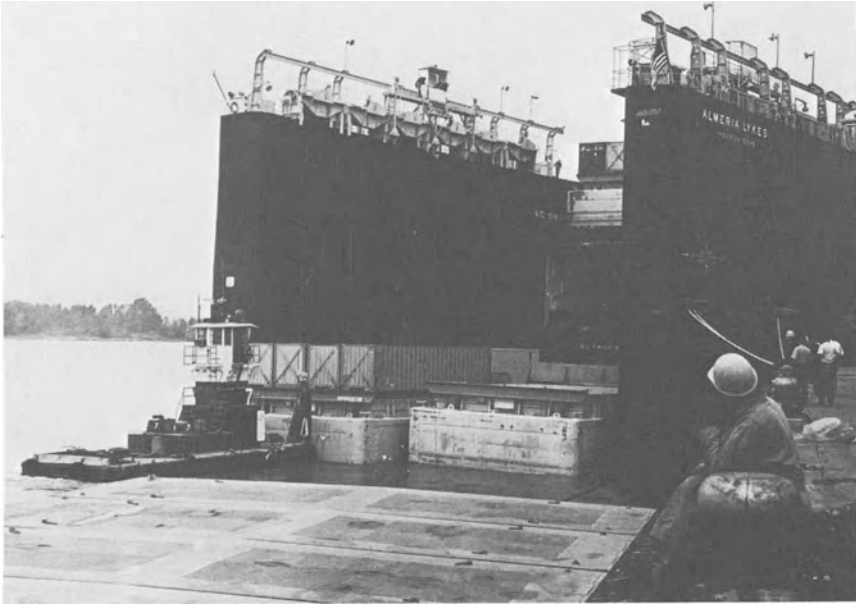


The 22-knot "lighter aboard ship" (LASH) *Benjamin Harrison* was built in 1980 to carry 80 lighters and 119 containers. Photo: McAllister Bros.

that if a great preponderance of the cargo moving in a given trade route consists of long rails, piling, pipes, etc., the optimum ship for that trade should have extra long and deep holds, long hatches, centerline pillars, special rig to facilitate handling extra long drafts of cargo in and out of the holds, etc. Extra long holds result in very large bale cubic per cargo gear; extra long hatches result in a smaller number of hatches, and a smaller amount of cargo handling gear, per ship. A ship designed with the object of being optimum for handling and stowing long pieces of cargo would not be an optimum ship to carry general cargoes. The ability of the ship to survive damage involving flooding also may be lessened.

An operator who wants to move general cargoes in and out of a large number of ports, on the other hand, has no special interest in extra long holds and hatches. He will prefer to have as many hatches, and cargo handling gears, as he can get satisfactorily in the ship, in order to facilitate reaching and moving the cargo that he must handle in each port, with a minimum of interference. The requirements of the majority of long trade routes will average somewhere between these extremes.⁷

In those trades where the volume of cargo offered justifies the installation, refrigeration or specially ventilated spaces may be included in the design of the ship. Vessels operated in the dry bulk



Two loaded barges each weighing 1,000 tons are hoisted by an electric elevator at the stern of the *Almeria Lykes*. Photo: Lykes Lines.

trades usually depend upon shore facilities for both loading and discharge; on the Great Lakes, however, many ships built after 1975 have been self-unloaders. To obviate at least to some extent the unprofitable ballast voyages that characterize both dry bulk carriers and tankers, the combination ore-bulk-oil ship was developed. On separate voyages, these vessels can carry full cargoes of metallic ores or grain or oil. Although a number of these ships has been built, many have not been able to exploit their versatility on a consistent basis.

Speed is important to all parties concerned with ocean transportation. To the shipper, it reduces the time before payment for his goods can be expected. To the consignee, it is the basis for planning the level of inventory that must be maintained until the next shipment arrives. To the shipowner, it fixes the number of ships needed to maintain a certain frequency of sailings. Over the years, the consistent demand by users of steamships has been for higher

speeds. The ultimate answer was made in 1972 when Sea-Land Service introduced its big 33-knot containerships. On all the sea routes to which these big carriers were assigned, new records were established for cargo ships' passages, and in many instances the records held by the fastest passenger vessels were eclipsed. Following the oil crisis of 1973, which made the cost of fuel the major operating expense, the cruising speed of these ships was reduced drastically as an economy measure. In 1981-1982, the eight ships of this class were withdrawn from commercial service and sold to the United States Navy for conversion to amphibious operations support vessels.

The prospective owner of a new ship must determine what kind of power plant is most effective, economical, dependable, and serviceable on the route to which the ship will be assigned, and the type of trade in which the ship will be engaged. Diesel motors have significant advantages up to about 50,000 brake horsepower. Steam turbines are most efficient in the higher power ranges. Nuclear power has been used successfully aboard naval vessels but has not been found economical for commercial ships. Fuel has become the most costly item of operating expense, and the choice of propulsion equipment is critically important in the total economic life of the vessel.

Competition has been called the soul of trade. It is undeniable that competition does bring about change and improvement in service. Evaluating not only existing competition but also the type and level thereof to be faced during the anticipated lifetime of the new ship is an essential procedure in planning. If the judgments are correct, the investment in the ship will prove profitable; if the forecasts are faulty, the penalty may be bankruptcy.

Although it is a basic principle of transportation economics that the larger the vessel, the lower the unit cost of carriage, the maximum size of ship for a particular route must be predicated upon the quantity of cargo actually moving from port to port. The intensity of competition, which might limit the size of a shipload, must be taken into account in determining the revenue-producing potential of the ship.

If, for example, there is a sustained offering of 25,000 tons of manufactured steel per week from a single United States Atlantic port to the Havre-Hamburg range, and four steamship companies

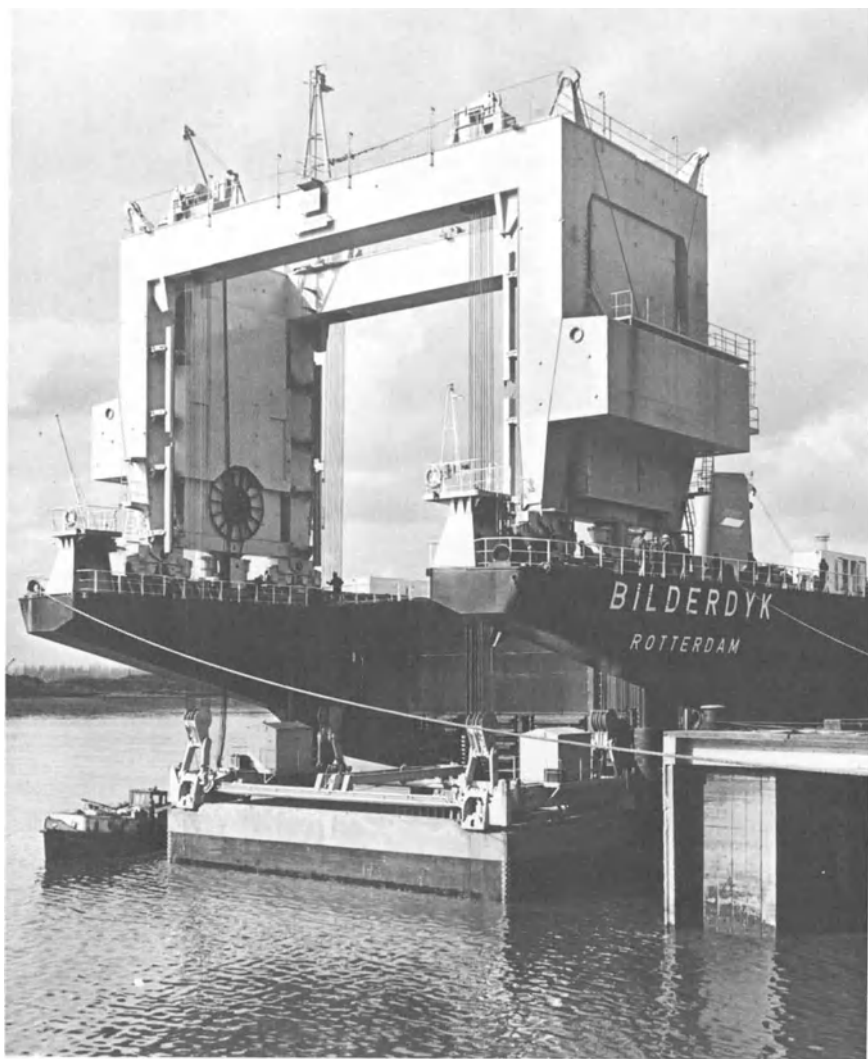
normally compete for that business, the maximum load to which any one of them reasonably should aspire would approximate ten thousand tons. Although it is possible, in theory, for one very large vessel to lift the entire week's production, the reality of ship operation is different. Some of the facts which assail the theory are:

1. The cost of the ship's time which would be dedicated to loading and unloading that quantity of goods, compared to the freight revenue earned;
2. The questionable wisdom of relying upon this type of movement for so large a portion of vessel earnings;
3. The demands of shippers and consignees for the shortest possible time that their goods must be in transit. "In transit time" includes every minute from the moment of delivery to the carrier until the consignee takes charge of the goods at destination. The average shipper or receiver of oceanborne cargo has little interest in the performance of the vessel, so long as his goods are delivered quickly and in good condition.

If speed is crucial to successful competition, the cost of providing record-breaking speed must be determined to the nearest dollar. Speed is not just adding horsepower to the engines; it entails many refinements in hull design which affect significantly (and often adversely) the cargo stowage capabilities of a ship. The plodding Liberty ship of World War II had one enormous advantage: her hull was almost rectangular, which made possible efficient use of all the space in the holds. By way of contrast, a postwar cargo liner with twice the Liberty's speed had such fine hull lines that the forward end of the hatchway in No. 1 hold was almost useless for cargo stowage. The full effect of increased speed must be determined before any decision is reached; failure to consider the less obvious aspects can be economically disastrous.

Finally, in the preliminary planning for the new ship, the computer permits analysis of the economic forecasts for the entire life of the ship. A number of contingencies can be postulated, and the computer will provide the answers.

For instance, if the ship is to be used in the tramp trade, the projected cost of building can be made the basis for determining the minimum freight rate or charter hire required to defray all



The 500-ton capacity traveling gantry crane of a lighter-carrying ship (LASH) is prepared to hoist a lighter aboard the *Bilderdyk*. Photo: LASH Systems, Inc.

operating costs and repay the investment. As part of the calculation, the best of circumstances can be assumed: each year the ship will make six voyages, will require only two weeks of shipyard time, will be forced to travel in ballast a total of twenty days, and will

spend a maximum of ten days idle in port between charters. A "worst case" assumption also can be made: the ship would make six voyages, but would lose four weeks for shipyard repairs; would steam in ballast a total of forty-five days; and be idle in port for another thirty. Many possibilities can be programmed into the computer to obtain a realistic picture of the ship's profitability over the full term of her working life.

It is evident, from these examples, that the computer makes it possible for the shipowner, the naval architect, and the shipbuilder to perform elaborate and deeply penetrating economic studies of the earning power of the proposed ship. Before the advent of this electronic marvel, the calculations were either so laborious and so slow that their use was discouraged, or the data available were so scanty that many decisions had to be based on empirical knowledge, experience, and (for lack of a more accurate description) intuition.

It is correct to deduce that the process of planning for a new ship involves concurrent action in several areas of concern. At no time may any participant in the decision-making process overlook the fact that the ship has to earn all expenses and pay at least a modest profit. At the same time, the new vessel must possess certain physical characteristics that assure efficient and effective operation in her assigned trade.

Up to this point, major attention has been devoted to economic matters, because until these have been explored, the idea of a new ship cannot become a reality. The naval architect should share in many of these economic studies if his preliminary specifications and sketches are to be included in the solution. Not until the economic studies are finished can some basic decisions be made, of which the size, speed, and characteristics necessary to achieve the ship's mission are fundamental.

These decisions become binding once the intent to build is approved. Suggestions for development of the design within the approved limitations are studied and sketched by the architect so that the management team can appraise their worth. It often transpires that examination of a drawing prompts improvement and justifies reworking earlier proposals in order to accommodate the new ideas. While these design programs are in progress, a concurrent series of technical analyses of the evolving plan, often performed by carefully developed computer techniques, must be undertaken to assure that these requirements are satisfied:



Centerline cranes and three hatches above each hold of the *Del Oro* permit rapid and economical handling of cargo. Photo: Delta Lines.

I. Trading pattern. Does the design meet the needs of the established pattern of operation, including the ability of the ship to serve all the ports on the route, to carry the quantity and variety of cargo offered (or anticipated for the future), and to provide rapidity of transit satisfactory to the shippers?

II. Speed of the ship. Does the ship have sufficient speed at least to equal or surpass the competition on the route? Is the proposed speed economically sustainable given existing freight rates and predicted prices for fuel?

III. Dimensions of the ship. Is the proposed size correct for the ports the ship is to enter and the type and quantity of cargo she is to carry?

IV. Hull form and displacement. Does the design incorporate the best hydrodynamic qualities possible within the limitations imposed by the requirements of the intended employment of the ship? Do the hull lines produce cargo stowage spaces that are acceptable and economical to work?

V. General arrangement of the ship. Does the design result in maximum return from the interior layout of the ship, especially with reference to cargo loading and discharge, but not discounting in any way the need to stow cargo efficiently and safely? Are cargo handling devices correctly assigned to satisfy the requirements of the ship's employment?

VI. Cubic measurements within the hull. Are the capacities of the cargo holds distributed appropriately, with due appreciation of the need to balance working times for each compartment? Is cargo handling facilitated by the internal arrangements of space?

VII. Choice of machinery. Has the most effective and reliable mode of propulsion been selected, in view of the projected employment of the ship? Is the recommended mode the most economical, in terms of sustained performance in circumstances of maximum strain as well as in ideal situations? If a diesel motor is being considered, does the study evaluate the problems which can arise from using fuel that barely meets minimum specifications?

VIII. Outfit of the ship. Are the weight and cost of different types of cargo gear and supporting equipment proportionate to the capabilities of the ship? What is the difference in cost between the ideal installation and one that meets minimum specifications? What effects can be predicted from the selection of less than ideal outfitting?

IX. Cargo capacity and ship's total deadweight tonnage. What is the net revenue-producing capability of the ship after fuel, water, and stores have been placed aboard?

The answers to these questions can be found only by careful examination of every proposal, not only for its immediate applicability and practicality but also for the "second order" effects of these ideas. If a containership is to be built, the desirability of installing

gantry cranes to make the vessel independent of port facilities must be analyzed to establish "first order" effects like increased costs of building, reduced cargo deadweight tonnage, and more expensive ship maintenance. At the same time, second order effects like greater flexibility in scheduling ports of call, elimination of crane-hire expense, and the opportunity to work cargo at an anchorage, must be taken into account. Only by systematic evaluation of all effects can the correct decisions be reached. It follows that once a decision has been made for a component part of the ship, that decision must apply to all future developments in the design. Inasmuch as many of these early stage decisions involve computations and mutations brought about by a variety of contingencies, the importance of the computer in the analyses and forecasts is difficult to exaggerate.

In some instances, the analysis takes the form of elaborate comparisons of two possibilities. A rather obvious subject for this type of computer-assisted scrutiny is the selection of the propulsion system for the ship. Within specific limits of horsepower, the usual alternative to the geared turbine is a diesel motor. There are advantages to the turbine, not least of which is its sturdiness and dependability under adverse operating conditions, and the rather large number of ship repair activities around the world which can provide assistance if required. Compared to the diesel engine, however, the turbine burns more oil per horsepower per hour, and second order impact is noted in the consequential reduction in the cargo deadweight of the ship. Diesel fuel costs more than the oil burned under boilers to produce steam, but the diesel's lower rate of consumption per horsepower per hour usually offsets the price differential. If the cheapest and lowest grade of oil is used as a matter of routine, there may be a noticeable impact upon the operating efficiency of the motor. Primary effects which must be studied therefore include the mechanical modifications to permit burning low-grade oil and the anticipated higher maintenance expenses. Second order considerations involve loss of ship time, and hence reduced earnings, consequent upon the need for repair work; erratic schedules, with inevitable loss of shippers' and charterers' confidence; and lowered efficiency in overall ship performance caused by undue cylinder wear from using inferior fuel. By instructing the computer to take into account all these contingencies as essential elements of the analysis of the two types of



Energy Independence was the first seagoing coal-burning steamship built in the United States since 1921. She loaded her first cargo, 32,366 long tons of coal, at Hampton Roads on August 8, 1983. Photo: General Dynamics.

propulsion equipment, a valid basis for decision can be established.

In the wake of the oil fuel crisis of recent years, there has been a resurgence of interest in the use of coal to generate steam. Mechanical stoking, improved draft, and innovative methods of burning the coal have reduced the problems of operating the furnaces, but significant second effects remain to be studied. There is, for instance, the availability of coal in ports of call and suitable equipment to deliver it to the ship's bunker spaces. Additionally, somewhat larger hulls may be required for the same amount of enclosed space for cargo because of the bulkiness of the coal and the requirement that it be adjacent to the fireroom. A third factor is the time needed to take the coal aboard and the interruptions, if any, in working cargo caused by this fueling procedure. That solutions exist was borne out by the construction (or conversion) in 1982 and 1983 of eight coal-burning ships. These vessels were produced in the United States, Italy, Japan, Korea, and Spain, and ranged in size

from 33,000 tons deadweight to 151,500 tons deadweight. They have been operated successfully by Australian, United States, and Spanish owners.⁸

In preparing the analysis of full-life economy of the proposed vessel, a number of assumptions must be made and these become major elements in programming the computer. It is an absolute requirement of this analysis that the assumptions be as realistic as possible, based upon the recent experience of the operator and the latest available information, both statistical and otherwise. The amount of cargo that a ship actually lifts for a voyage is a critical matter; to assume that she sails completely full on every trip when in fact the average load occupies not more than 80 percent to 85 percent of the available space is to create a totally false picture of the earning power of the ship. Seasonal variations in cargo tonnage also must be taken into account, especially if the new ship is to be larger than the one to be replaced. Another assumption which bears on the full-life capability of the vessel is speed. Some operators insist upon the same speed being maintained on every voyage and achieve this result by increasing the horsepower output as the hull accumulates barnacles and marine growth. Other operators stipulate that the engines may produce a certain maximum horsepower, and hold to that output regardless of the effect on ship speed. If the sea route passes through areas which are notorious for adverse weather conditions at certain times of the year, and voyages take longer because of the need to safeguard the ship, this fact also must be included. Finally, as equipment ages, it often is less capable of optimum performance than when it was new. To instruct the computer, therefore, that all voyages for the life of the ship are to be calculated at a standard speed is to invite major errors.

The economic study of ship design is an extremely complex matter, especially as computers explore many avenues previously inaccessible to management. Many mistakes can be avoided by subjecting both the preliminary intentions to build a new ship and the proposals for actual construction to a methodical and systematically developed program of analysis. Excessive reliance on the "wisdom" of the computer, however, is contrary to the best interests of the prospective operator. The experience of veteran members of the staff, the accrued knowledge of business trends which has been built up by actual participation in ocean transportation, the performance history of existing units of the fleet, and records of

achievement are as much in demand today as they ever were. The computer's results will be only as good as the instructions given to the machine, and those instructions necessarily are derived from the combined contributions of all the persons involved.

The introduction of computer technology to steamship management coincided with some of the major changes in transoceanic transportation, of which containerization was but one facet. Ship capacities and cruising speeds were increased, and a wide variety of improved cargo handling systems became available. Simultaneously, shipbuilding costs escalated noticeably. The old procedure of improving existing models of ships, without any radical departures from established practice, no longer was satisfactory. By linking the computer and ship managers' need for a better basis for decisions, it became possible to project ten or twenty years ahead to establish the profit-making potential of a ship under a wide range of contingencies. That previously elusive figure, the freight rate (or charter hire) required to break even, was given both substance and credibility when realistic assumptions and actual experience could be supplied to the computer.

Steamship owners, despite the incursion of the "thinking machine," still must make decisions affecting the future welfare of their organizations. They continue to consult with the naval architect and the marine engineer in seeking the proper interpretation of data supplied by the computer. Aided by economic analyses, and especially by full-life forecasts of costs and revenues, the actions taken by owners and senior managers can be based on a depth and range of information never before available. Although it is possible for owners to become so engrossed in the financial aspects of management that they consider the ships they direct on their courses only as cogs in the complex mechanism called international trade, it remains true that it is the ships that must do battle with the elemental forces of nature, or lie in remote and perchance exotic harbors while being loaded or discharged, and do the bidding of the skillful and dedicated people who make up their complements. Computers may aid in the improvement of ship design, but everyone concerned must be aware at all times that the ship is the only vehicle of transportation in which a number of human beings live, work, and play for protracted periods of time as they exercise final control over the actual performance of that admirable creation.

Passenger Cruises

VACATION CRUISES—defined as the transportation of pleasure-seeking travelers on ocean voyages offering one or more glamorous ports of call—are the only elements that have survived from the once flourishing passenger steamship business. The importance of this type of service may be gauged by the fact that the port of Miami, Florida, embarked approximately two million travelers during 1983. Twenty-one cruise ships made that southern city their home port. Nationwide, the cruise business generated revenue of five billion dollars.¹

Although not a post-World War II innovation, for the history of passenger cruises reaches back at least to 1844, the cruise business as it exists today began in 1946. Its growth since then has been phenomenal. Scores of ships, many built especially for this type of operation and representing the investment of hundreds of millions of dollars, sail from United States ports. More and more cruises are originating in the ports of Western Europe, the British Isles, and Australia, and the demand seems insatiable. Ships of Norwegian registry dominate the scene, both in the size of the fleet and in the number of beds offered for sale.² The vessels themselves, regardless of nationality, are floating resort hotels, equipped with elaborate facilities for passenger recreation, and offering well-publicized professional entertainment for the amusement of the travelers.

Although the enormous popularity of the cruise business is a relatively recent development, the idea of vacation sea travel can be traced back more than a century. The first cruise from the United States was immortalized by Mark Twain in *Innocents Abroad*. This humorous masterpiece describes the experiences of a group of Americans who voyaged to the Mediterranean in the paddle wheel steamer *Quaker City* in 1867. It is worth the trouble to read the sales

promotion material quoted by Mark Twain and to compare it with the modern versions. Only the phraseology has changed; the intent to allure the prospective passenger with both the glamour and romance of far-off places and pleasure of the ship has not changed one whit.

Exactly when cruising began on the other side of the Atlantic is not readily to be determined. A "special Mediterranean tour" was operated by the Peninsular & Oriental Steam Navigation Company for its British customers in 1844. In 1891, Albert Ballin, the guiding genius of the Hamburg-America Line, sent the *Augusta Victoria* on a cruise around the Mediterranean rather than have her sail half-empty on her regular transatlantic route during the winter season.³

Until the disappearance of regular passenger liner service during the late 1960s and early 1970s, almost all the ships that engaged in cruising out of United States ports were assigned for the major portion of the year to shuttle operations between fixed ports. During the winter months, when the volume of passenger traffic dropped to very low levels, it was common practice to divert all but the very largest liner vessels to cruise operation, even though only a portion of their accommodations could be considered acceptable for warm weather travel. In the years antedating World War II, as many as ninety ships were scheduled for cruises during the colder months. Year-round cruising was not feasible until air-conditioning was installed on ships, and passengers could be assured of comfortable quarters regardless of outdoor temperatures.⁴

Cunard Line introduced the *Caronia* in 1949 as a ship intended primarily for cruising, but adaptable to liner service whenever there was sufficient demand. This dual purpose design proved satisfactory, and other steamship owners followed Cunard's lead. Interchanging ships between liner and cruise service ended when jet-propelled aircraft began carrying passengers across the oceans. International steamship service was overwhelmed by the competition, and gradually faded away during the years between 1960 and 1972.

Driven off their regular routes by the airplane, many passenger liners were placed in year-round cruise service. Newer ships like the *Caronia* had been planned for this type of employment, and were fitted with the amenities demanded by the cruising public: outdoor swimming pools, broad decks for strolling and playing games, large

public rooms with good views to seaward, complete air-conditioning, and private baths for every cabin. These vessels made the transition to their new employment with considerable success. Some of the older ships were rebuilt completely, while others were sold for scrap. Reconstruction was justified on two counts: it took much less time than to build a ship, and it cost about half as much.

As work weeks have been shortened and vacations (or holidays) have become universal in the industrialized nations, the demand for recreation has intensified. As early as 1954, Greek shipowners sensed this and began to offer short voyages from Piraeus to the nearby islands of the Aegean Sea. These cruises varied in length from three to seven days and proved immensely popular, affording employment to two dozen ships.⁵

Holiday voyages were offered from England, France, West Germany, Russia, Indonesia, and Australia. Surprisingly, about one hundred thousand Russians enjoyed the pleasures of a seagoing vacation. At least one ship, the *Gruziya*, with space for about 480 passengers, was operated between Crimean and Caucasian resorts. Russian flag ships also participated for several years in the cruise trade from the United States, offering rates significantly below those of the competition. This operation ended in 1979 when the United States government reacted against the Russian invasion of Afghanistan and barred the ships from American ports. The Holland-America Line assigned a ship to Indonesia during the European winter season, and found it a profitable enterprise when combined with scheduled airline services bringing passengers from distant lands to the port of embarkation. Pearl Cruises of Scandinavia since 1981 has based its 425-passenger *Pearl of Scandinavia* in the Orient, sailing from Singapore, Hong Kong, and Kobe to ports of China. Alternate routings took the ship to the Philippines, Thailand, and other Pacific points.⁶

West Germany, as the most prosperous nation in Western Europe, presented a very attractive market for cruise operators. Hapag-Lloyd, the largest German shipowner, bought the *Kungsholm* (built for the Swedish American Line in 1952) in 1972, and under the name of *Europa* placed her in full-time cruise service. In 1977, bookings for the ship were at the level of 70 percent to 80 percent of capacity; the occupancy rate required to cover expenses was 63 percent to 64 percent at the fare structure then in effect.

Prices charged for cruises in the *Europa* were about 30 percent higher than those of the average cruise ship, but were justified by the superb condition in which she was maintained, the unceasing attention of her owners to assuring that the ship met the highest standards of the maritime authorities, the excellence of her cuisine, and her outstanding wine cellar, which was said to have been superior to the best luxury hotels.⁷

Hapag-Lloyd in November 1978 ordered a replacement for the aging *Europa*, awarding the contract to Bremer-Vulkan in West Germany at a price of \$85,000,000. The new ship, also named *Europa*, was delivered in 1982. She had accommodations for 600 passengers, an unusually small number for a 27,000-gross ton ship. Her design differed from other cruise vessels in that the rather spacious passenger cabins were in the forward (therefore quieter) area, and the public rooms were spread over several decks in the after part. She also had an outdoor pool which could be enclosed by a glass superstructure. The crew, numbering 275, was quartered in 182 rooms, each with a shower and toilet. Special recreation facilities including a swimming pool also were provided for the staff.⁸

The decision to build a new ship for the cruise trade is of the first magnitude. Between the signing of the building contract and sailing on the first voyage, a period of about three years must elapse. The owners must be optimistic that the world's economy will encourage thousands of people to spend millions of dollars (or pounds sterling, or kroner, or drachmas) on pleasure voyages. This optimism, although based on the most careful and conservative forecasts and calculations, still represents an enormous gamble. Fortunately, the gamble has paid handsome rewards during most of the years since 1945.

On May 4, 1984, the 46,000-ton *Fairsky* arrived in Los Angeles on her delivery voyage from the shipyard at La Seyne sur Mer, France, bringing to a climax three years of construction and, according to the *San Francisco Examiner* of June 10, 1984, ten years of planning. The owners of the \$150 million vessel adopted a novel procedure when they hired four interior design firms—from England, Finland, Italy, and the United States—to plan different areas of the ship. While coordination and consultation among these four were necessary, each one was permitted to carry out individual schemes for their assigned sections. The American designer, Joszi



The *Queen Elizabeth 2* (built 1969) was designed for regular transatlantic service in summer, and cruise operation during the other seasons. She carried 1,870 passengers; the crew numbered 940. Photo: Cunard Line.

Meskan, commented in an interview that her awareness of the larger physical dimensions of Americans had influenced the design of seats throughout the ship. She also remembered that many American women habitually use hair dryers, and therefore had the lamps on every dressing table equipped with an outlet for 120-volt electrical equipment.⁹

Operating a passenger carrying ship in any trade is expensive, but cruise ship owners also must pay for the entertainers and social activity directors who are essential to success. In 1978, for instance, the Holland-America Line let it be known that the service and entertainment staffs on its five cruise ships cost over \$9 million per year.¹⁰ In 1981, Royal Caribbean Line paid well-known singers and other entertainers between \$1,500 and \$3,500 a week for four to eight shows. On board the *Nordic Prince*, the cruise director had a staff of fifteen assistants, described as singers comedians, masters of ceremony, or simply as good mixers.¹¹

In regular liner service, operating costs are offset by revenue from mail, express, and whatever cargo may be carried as well as from passenger fares. In cruise service, however, only the proceeds from travelers' tickets are available to defray the expenses of running the ship. The *Queen Elizabeth 2*, according to a 1978 report, had a daily operating bill of \$130,000, a figure which required a very large number of passengers to overcome. Because cruising is a



Constructed in 1976, the 20.5-knot *Cunard Countess* typified the moderate-sized, purpose-built cruise ships. She carried 750 passengers. Photo: Cunard Line.

vacation activity, and therefore subject to many factors conducive to cancellation or postponement, it is extraordinarily difficult to predict months in advance the number of persons who will board the ship on sailing day. If patronage is good, and the voyage is made exactly as planned without any unpredicted major expenditures, the cruise can be a financial success. Ticket revenue for the *Queen Elizabeth 2*'s three-month, round-the-world cruise of 1975 amounted to \$15 million. Disbursements came to only \$12½ million. The unfavorable side of the ledger was displayed three years later, when the ship had to undergo a major overhaul which forced cancellation of three successive cruises, with consequent loss of income.¹²

As the passenger liners were phased out of scheduled shuttle service and cruising became a year-round activity, shipowners set-

tled into fixed patterns of operation. Some restricted their business to a certain area, as did the Norwegian Caribbean Line, which concentrated on the Caribbean. Other companies chose to establish what amounted to shuttle voyages between the home port and one or two ports of call. Still others allocated their vessels to the longer voyages, perhaps of fourteen to twenty or more days. From this last group came the ships placed in extended cruises of fifty to one hundred days. Because the pool of travelers who might be interested in such protracted vacations is very limited—for the most part they are independently wealthy or retired—only about a half dozen ships were so employed. Possibly the longest round-the-world cruise in recent years was that of Hapag-Lloyd's *Europa*. Originating in Germany, the 155-day, 30,000-nautical mile voyage terminated in Genoa on June 4, 1984. Although about two thousand passengers traveled in the ship, only forty-two persons actually made the entire cruise.¹³

Beginning about 1970, the major cruise operators shifted their base of operation from New York to the Florida ports of Miami, Port Everglades, and Fort Lauderdale. There were two reasons for this decision: (1) distances to the islands of the Caribbean, the north coast of South America, and the Panama Canal are comparatively short, and passengers may be landed in foreign territory very soon after departing from the United States; and (2) travelers domiciled in the northeastern states can fly to Florida in a matter of a few hours and therefore are able to enjoy the tropical climate for the maximum period. The transfer to the southern ports proved to be popular with the patrons of both long and short voyages.

Longer cruises, exemplified by those from the east coast of the United States to the eastern Mediterranean or around the world, represent a challenge to the managerial skill of the operator while presenting him with a very real gamble. To win the favor of a notoriously fickle and highly selective public, the traffic managers must devise itineraries to allure those planning shipboard vacations. They must determine how many hours ashore will satisfy the desires of the passengers for sightseeing and shopping, how many days should be spent at sea between ports so that the travelers will not become too fatigued, and how long the cruises should last from start to finish. Their goal is to come as close as possible to the ideal set forth in this quotation from a column by Georgia Hess in the

travel section of the *San Francisco Examiner and Chronicle* on September 21, 1980:

Where does Supership sail? When I'm aboard, it will be in waters where there are fascinating ports of call for two or three days running, then a day at sea, then more ports, then another rest day, etc. Others might prefer the vastness of midocean where day follows day at sea, allowing adaptation to the rhythm of nature and the leisure to do nothing but sleep.

When Supership is in port, no other ships are there to disgorge their hundreds of shop-seekers. Taxis are available as well as tour buses, but of course Supership always docks within walking distance of midtown, like on the Grand Canal in Venice.

Supership calls at a judicious combination of tried-and-true destinations and rarely visited ports: in the Greek Islands, for instance, she never skips Santorini, but might add Samothrace.

To meet the expectations of people like the newspaper columnist quoted above, the experts begin their planning by canvassing by mail, by telephone, and in person the travel agents who are the best producers of cruise business, have the greatest contact with the travelers, and therefore know what their current desires may be. From these analyses of the travel trade and the reactions of their patrons to experiences on cruises, surprisingly accurate data may be obtained. Additional information may be procured from resort hotels and conventions of business and professional associations where wealthy travelers congregate. Assessing all these bits and pieces of knowledge, the cruise manager can determine rather definitely what will be acceptable to the public.

Occasionally, even the most highly qualified experts are frustrated by events altogether beyond the control of ship operators. Incidents in the recent past illustrate the variety of these circumstances.

In late 1973, the Greek Line proposed to use its *Olympia* for a series of week-long voyages from Piraeus to eastern Mediterranean ports. The first trip was scheduled for early March 1974. As the New Year dawned, friction between Israel and its Arab neighbors was so threatening that the cruise program had to be cancelled, and for want of other employment the *Olympia* was laid up.¹⁴

Eastern Steamship Lines, operators of the steamer *Bahama Star* in cruises between Florida and the Bahama Islands, announced in

January 1975 that it was terminating the service because of the extremely high cost of fuel oil and foodstuffs. Whereas in 1973 only 150 passengers were needed to pay expenses of the voyage, by the end of 1974 it was necessary to carry at least 350 to break even. The ship was laid up following the owners' determination that the required quota of travelers could not be found every week.¹⁵

The *Oriental Empress*, owned by C. Y. Tung, of Hong Kong, left Los Angeles on December 7, 1973, on a projected one-hundred-day cruise around the world. The international crisis resulting from fuel shortages caused abrupt cancellation of the voyage when the ship reached Hong Kong on January 8, 1974. The ship had exhausted her supply of oil, all efforts made en route to obtain fuel having been unsuccessful. Her passengers were flown back to the United States, and the unused portion of their ticket money was refunded.¹⁶

When England undertook military action in the Falkland Islands in April 1982, the P & O Line's cruise ship *Canberra* and the Cunard Line's *Queen Elizabeth 2* were requisitioned on very short notice to serve as troop carriers for an indefinite period. Both ships had been scheduled for extensive cruise operations, and thousands of would-be passengers suddenly had to make other plans.¹⁷ The Costa Line, an Italian carrier which annually made a cruise to Argentina and included a call at the Falkland Islands, cancelled the voyage of the *Eugenio* less than a week before her scheduled departure because of "uncertainties related to the Falkland Islands crisis."¹⁸

Competition for the available patronage is very keen. Traffic departments constantly seek ways to induce new segments of the general public to embark on cruises. When the seven-day voyage became a standard offering of the industry, many persons who could not take longer vacations discovered that a shipboard holiday was a genuine possibility. Others who had been interested in cruises but had been deterred both by the high daily rate and the length of the voyages found that they could afford the shorter trips. To attract those who had evidenced no previous enthusiasm for these "pleasure jaunts," promoters developed "special interest cruises," with a wide variety of inducements: a jazz festival, with a well-known artist as the principal exponent; an educational activity featuring a prominent scholar; an investment forum and symposi-



Top: The *Sea Goddess I* and her twin, *Sea Goddess II*, accommodate only 116 passengers in unusual luxury. *Bottom:* The dining room of the *Sea Goddess* ships. Photos: Sea Goddess Cruises.

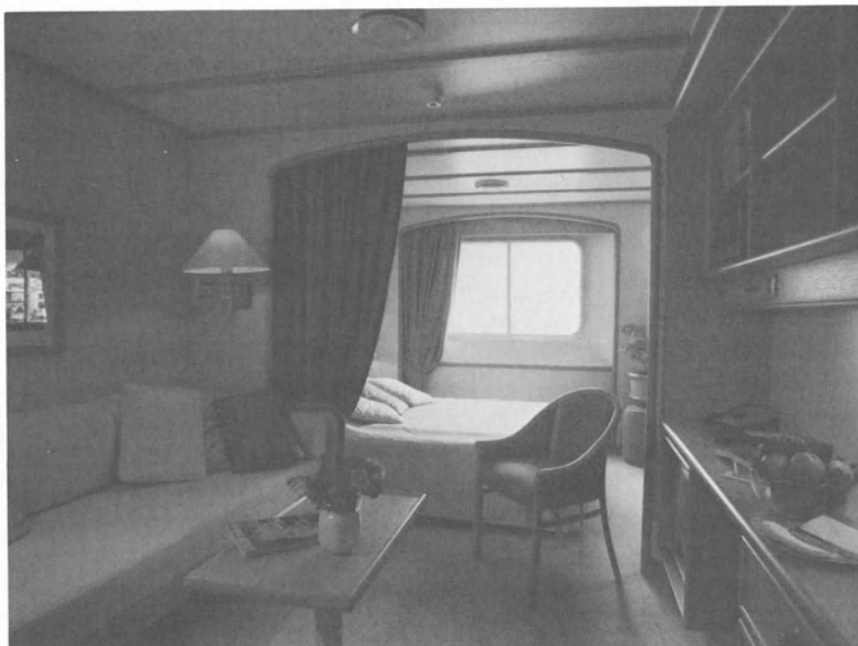


Top: Passengers' main lounge aboard the *Sea Goddess I*. *Bottom:* The reception area of the *Sea Goddess I* is enhanced by soft, indirect lighting and Persian rugs. Photos: Sea Goddess Cruises.

ium conducted by a team of financial counsellors; an athletic colloquium chaired by world famous sports coaches and commentators; a musical festival centered on a distinguished symphony orchestra; and a gourmet's special emphasizing various national cuisines. Perhaps the most adventurous innovation in cruise incentives was to invite 98 persons to take part in the first passenger ship voyage through the Northwest Passage. Sailing from St. John's, Newfoundland, Canada, on August 20, 1984, the *Lindblad Explorer* completed the voyage through these northern waters on September 12. Only 35 ships have traversed this route across the northern rim of Canada since 1906. Capitalizing upon the successful outcome of the 1984 venture, a larger ship, the *World Discoverer*, carrying 140 passengers, sailed on August 23, 1985 from Nome, Alaska, to Halifax, Nova Scotia, on what was said to be the first eastbound crossing of the Northwest Passage. The trip was completed successfully on September 14.¹⁹

Shore excursions are made available, usually at extra cost, to passengers on almost all cruises. Normally, the shipowner specifically and categorically disavows any connection with or responsibility for these sightseeing trips, and all its promotional literature contains carefully worded statements to that effect. It is true, however, that the disavowal is somewhat technical, in that the shipowner nearly always has selected a single concessionaire at each port of call to provide the tours, tickets for which are sold on board the cruise vessel. The alternative is to select a travel agency with a worldwide organization and have it responsible for all shore trips. The shipowner, in either case, must assure that the contractor or concessionaire performs to specification. Obviously it does the shipowner's reputation no good if passengers are displeased by their experiences ashore. After all, they do not have the opportunity to choose among the competing excursion operators. In realization of this ultimate managerial burden, steamship companies exercise great care in the selection and appointment of concessionaires.

An interesting exception to the general rule that shore excursions are optional at extra cost was observed during 1980 in the voyages of the motorship *Aquamarine*, which sailed fortnightly from Hong Kong or Kobe on cruises to China. Because of restrictions imposed by the government of China, only those passengers who had made reservations for shore trips were allowed to leave the ship



All fifty-eight suites in the *Sea Goddess I* are identical. Photo: Sea Goddess Cruises.

for any purpose. Shore excursions therefore were made an integral part of the cruise “package.”

Sun Line, advertising the *Stella Maris*’ two projected fourteen day “grand gala” cruises in the Mediterranean during the fall of 1985, announced that the voyage would offer thirty-five ancient sights “with all land arrangements complimentary.” Pearl Cruises of Scandinavia, as did the *Aquamarine*, included the cost of “nonoptional” shore excursions in China.

Just as ships engaged in liner service were designed expressly to meet the needs of a particular route and clientele, so vessels intended for cruise operation must be planned to meet the demands of this specialized employment. Whether an existing ship is to be purchased, or a new ship is to be built, the prospective owners must fix definitely the trade to which the ship will be assigned, and the special features needed to fit her for this assignment. The Norwegian Caribbean Line, for instance, established to its satisfaction that there was a good opportunity to operate a very large ship in short



When named in 1980, the *Norway* was the world's largest cruise ship, accommodating 2,400 passengers. Her crew numbered 800. Photo: Jeff Blinn, courtesy Moran Towing Co.

cruises from Florida to the Virgin Islands. The superliner *France* was available for purchase and seemed to meet the ideas of the potential buyers. Exhaustive studies were conducted to determine the cruising speed desirable on this route, the adaptations and modifications essential if the passenger accommodations were to be acceptable in the subtropical environment, and the earning capability of the ship. The computations indicated that the proposal to acquire the *France* and put her into the seven day cruise service was economically sound. In June 1979, ownership was transferred from the Compagnie Generale Maritime, of Paris, to the Norwegian Caribbean Line. A multimillion dollar conversion, which included changing the ship's name to *Norway*, preceded her entry in

May 1980 into year-round cruise operation. The wisdom of the decision to acquire the largest cruise ship in the world has been borne out by the owners' experience.²⁰

The Royal Viking Line, on the contrary, required more flexibility and long-distance sailing capability from the three ships it built. Annually, at least one of these vessels was assigned to a long, perhaps a world-circling, cruise. The ships therefore were designed with large fuel tanks, adequate distillation equipment to provide all the fresh water desired by the passengers, and accommodations that would be comfortable regardless of the climatic zone in which the ship happened to be. Their draft assured that they could enter practically any harbor in the world, from well-known commercial meccas like Hong Kong and Copenhagen to the out-of-the-way islands of Majuro in the Marshalls and Saipan in the Marianas, and to the ever popular Hamilton, Bermuda.

Built and operated to cater to very wealthy passengers who demand (and will pay for) the utmost in luxury and personal service, the *Sea Goddess I* and *Sea Goddess II* carry 116 passengers on seven- and fourteen-day cruises from base ports to off-the-beaten-track and relatively unknown harbors in the Mediterranean and Caribbean seas, and along the Atlantic coast of South America. Delivered by Finland's Wärtsilä shipyard in 1984 and 1985, the ships cost \$34,000,000 each, equivalent to \$293,100 per passenger bed—the most expensive cruise ships ever built. Instead of cabins, the ships offered 58 two-room suites, complete with full bath and equipped with remote control television, radio, stereophonograph, videocassette recorder, refrigerator, and bar stocked with liquor chosen by the occupants. In 1985, a uniform rate of \$3,600 per person, double occupancy, was charged for Mediterranean cruises. The rate was increased in 1986 to \$4,000 for the seven-day cruises. Every passenger was considered a guest, so no charges of any sort were made for personal services or beverages at the bar. Tipping was discouraged. Meals were served at the convenience of the passengers, and every dish was prepared individually.²¹

The owners' advertisement offered these inducements to those who could afford to indulge their tastes:

A cruise ship that takes exception to all the clichés and regimented activities of traditional cruises, she has the ambiance of a regal yacht, the amenities of a stable liner, . . .

Long before you sail, the Captain will write to ask your preferences in every detail, from the competitive sports you want in your visits ashore to the complimentary spirits you wish in your personal bar.

And the moment you come aboard, you will sense that a *Sea Goddess* is indeed a different world. The Persian carpets warm your welcome. The European staff knows your name . . .

Come live the *Sea Goddess* life.

Selecting the ports to which a cruise ship is to be sent is a very demanding task. There must be enough of the popular cities to satisfy both first time and well-experienced travelers, while there also must be new places which will stimulate the interest of those who have patronized this carrier on earlier cruises. For short voyages off the Atlantic coast of the United States, the favorite destinations are Bermuda, the Virgin Islands, the Netherlands Antilles, and Trinidad-Tobago. Long cruises have been most successful when the itinerary includes the traditional attractions of Bangkok, Capetown, Casablanca, Copenhagen, Hong Kong, Honolulu, and Sydney.

An interesting trend in the sales management of cruises emerged about 1975. Instead of stressing the glamorous nature of the ports of call, increased emphasis was given to the luxury and comfort of the ships, the pleasures to be enjoyed in passage from one harbor to another, the professional entertainment provided each evening at sea, the varied nature of shipboard activities available to passengers, and the elegance of the meals served in dining rooms. The unique feature of the cruise—a vacation afloat while living in a deluxe hotel—became a major selling point, and if the number of persons embarking on cruise ships each year was any criterion, it was what the traveling public wanted.

The ship which, because she was unique and also because the luxury of her accommodation possibly exceeded that of any commercial vessel afloat, epitomized the appeal of the ship herself rather than any itinerary, was the glamorous *Sea Cloud*. Built in 1931 as a four-masted, steel-hulled bark, this magnificent yacht embodied every feature that exquisite taste and unlimited wealth could provide. She was 316 feet long, and had a beam of 49 feet, and was the largest square-rigged private yacht ever constructed.

When she was rebuilt in 1979, at a cost of six and a half million dollars, the *Sea Cloud's* passenger capacity was expanded from the



The beautiful barque-rigged *Sea Cloud* has thirty-one sails. She is unique in the world's cruise ship fleet for her oceangoing capability as well as for the luxury of her passenger quarters. Photo: Travel Dynamics.

original twenty-three berths in two owner's suites and eleven other rooms by adding twenty-six double and two single cabins, while retaining the standards of luxury set by the first owner. In 1985, she was employed in Mediterranean cruises.

Some idea of the sales appeal of the ship may be gained from these paragraphs in a recent advertising circular:

The original owner's suites and staterooms present features and ambience not found on any other ship. Large and comfortable, they are wood-paneled, featuring antique furniture, fireplace and Italian marble bathroom. The new cabins were constructed in keeping with the original style. Although somewhat smaller, they are equally comfortable. The dining room, in the manner of a Scottish country-house, and the adjoining lounge with its beautiful wood-carved walls, together accommodate all passengers at one sitting. The lounge also serves as a lecture room. The food and service aboard are equal to the ship's elegance with complimentary wine served at every meal. The ship is managed by fifty-five skilled German officers and crew. Life aboard is relaxed and informal, and passengers are welcome to help with the operation of the sails. There is plenty of open space for sunning and relaxing.

The *Sea Cloud* is a ship that meets the highest safety standards. With a hull constructed of steel, the vessel is classified by German Lloyd as 100A-4 Yacht — the highest classification. Large sums of money have been spent to install the latest safety equipment which includes fire detectors and automatic sprinkler fire-fighting systems in all staterooms and public areas.

The *Sea Cloud* is unique in every respect. Stately, elegant, and comfortable, she provides a cruising experience which, until now, was available only to a few privileged individuals.

Regardless of the wonders of the ship carrying the cruising public, it remains true that the itinerary must include places that have the power to intrigue the fancy of the travelers. Promotional literature still features the exotic nature of foreign cities and the fascination of the areas the shore excursions will explore. It is no accident that, from the days of Mark Twain to the present, the brightness of tropical stars, the brilliance of phosphorescent seas, the magic of the unceasing trade winds, and the glamour of far-off places continue to be extolled.

The lure of the unusual, in contradistinction to the sybaritic luxury of the ship, was exploited to the full by at least one promoter.



Built as a yacht to suit the taste of a multimillionaire, the owner's suite of *Sea Cloud* was spacious and sumptuously outfitted. Today, this suite is the most luxurious accommodation offered by a ship in commercial service.

Photo: Travel Dynamics.

Lindblad Travel, Inc., built a ship of 2,481 gross registered tons in 1969 to carry 98 passengers to truly out-of-the-ordinary destinations. Voyages were made, for example, to the Palmer Peninsula in Antarctica, the Galapagos Islands, and Easter Island. In 1972, the *Lindblad Explorer* was sold to a subsidiary of the Swedish American Line, and in 1985 was purchased by an American enterprise, Society Expeditions, Inc., which continued the program of unusual voyages.

Peak demand for cruises, both in the United States and in Western Europe, occurs during the winter months when those who can arrange vacations seek respite from the rigors of cold weather and its attendant discomforts. In recent years, shipowners serving the United States cruise market have found it advantageous to base their ships in southern ports: Miami and Fort Lauderdale are favored on the Atlantic Coast, and Los Angeles is preferred on the Pacific Coast. Several steamship operators run their ships out of

San Juan, Puerto Rico. On the other side of the world, the Holland-America Line stations a ship in Indonesia in a successful attempt to cater to a developing clientele. Passengers fly from their homes to the embarkation port, thereby spending the maximum number of days in the warm weather area.

Selection of a base port from which cruise ships will operate is conditioned primarily on the convenience with which passengers may be transported to that point. For decades, almost all cruises from the United States Atlantic coast originated in New York, in large measure because of the network of railroads which fed that metropolis. The obvious drawback to New York was that a winter voyage to the Caribbean always began and ended with at least twenty-four hours of steaming through cold weather and often stormy seas. When high-speed jet-propelled aircraft put all cities east of the Mississippi River within two to four hours flying time from Miami, the picture changed dramatically. Most of the ships serving the Caribbean area were assigned to the Florida ports of Miami, Port Everglades, and Fort Lauderdale. An incidental benefit to the shipowners was that the airplane widened enormously the region from which cruise patronage might be obtained. It is no longer remarkable that residents of California are among those embarking on one- or two-week cruises from Florida.

Economic and geographical logic, however, do not always produce the result envisioned by traffic managers. In 1974, the Royal Viking Line determined that midwest and southeastern United States customers would find New Orleans attractive as a port from which to start their cruises. The *Royal Viking Star* was selected for this service, and during the three winter seasons from 1974 to 1977 a series of cruises was operated from this city. The response of the passengers was disappointing despite excellent airline connections and schedules, and the attraction of the city itself. At the beginning of the 1977-1978 fall cruise season, the ship was returned to the base port of Fort Lauderdale, from which it operated with greater success.

A significant expansion of this alliance between airlines and cruise ships occurred about 1970. Passengers who desired to spend twelve of their fourteen days of vacation cruising through the Mediterranean or to points along the North and Baltic Sea coasts flew from all parts of the world to international gateways like

London, Marseilles, and Athens. In these cities, they made connections with surface transportation directly to the sides of the ships which would take them on their cruises. Beginning in 1981, the round-the-world cruise of the *Queen Elizabeth 2* was offered in whole or part, depending on the desires of the traveler. The practice quickly spread, and identical inducements were offered by competing operators. Cunard countered by making special arrangements by which travelers in the *Queen Elizabeth 2* would be flown by supersonic jet aircraft from designated cities to seaports where they would join the ship. The cost of air travel was included, in whole or part, in the cruise price, or was deducted from the ship's fare.

Because of the keen competition between cruise operators, the most careful attention is given to advertising, promoting, and sailing individual cruises. In the case of the ships which, like the *Norway*, sail every week on a fixed route, the emphasis is placed upon the ship and the ports of call, rather than on any particular date. If the ship is assigned to a long cruise, such as a voyage around the world, the unusual features of the trip receive special stress. The Cunard Line's folder describing the world cruise for 1986 contained these words:

Discover new parts of the world on the most talked about ship of our time—*Queen Elizabeth 2*.

You'll experience a round-the-world voyage never before accomplished—in the true spirit of the great adventurers. The *QE2* will visit 32 enchanting ports on 4 continents, including 9 maiden arrivals. Sail . . . to the tip of South America, transit the Strait of Magellan (a world cruise first), and explore the coasts of Africa, Asia, and the Indian subcontinent.

And all in the unparalleled splendor of the *QE2*.

Longer cruises represent a special effort on the part of the operator, and in most cases are planned for a selected ship and a particular time. The names of the ships and their owners often serve to attract patrons; either these travelers are "repeaters," who have made previous cruises with the ship or the company, or they are persons who have been impressed with the reputation of the operator. Often the reservation books are opened twelve to fifteen months before the departure date; always there is as much publicity and advertising as can be arranged to attract attention and to provide impetus to the sales campaign.

To establish some rules and thereby to forestall cutthroat competition, thirteen cruise ship operators came together in 1970 to form the Caribbean Cruise Association. As was true of the conferences established in other trades, the aim of this group was to adopt a code of ethics concerning promotion and sale of cruise services, and to fix certain areas of responsibility for the welfare of the traveling public. Five years later, after an interim merger in 1972 into the International Passenger Ship Association, the Cruise Lines International Association came into existence with administrative offices in New York and San Francisco. This body consisted of the twenty-five major cruise ship operators catering to the tourist trade from the United States, functioned in a manner similar to other steamship conferences, and was subject to the scrutiny of the Federal Maritime Commission by reason of its relation to American foreign commerce.

The more desirable the accommodation aboard ship, the higher the cost to the person taking the cruise. The most modern ships in 1981 featured cabins which, by categories, were almost identical in size, decor, facilities, and amenities. About three-quarters of all rooms were "outside," with portholes looking out to sea. Suites and oversize rooms, which could be combined to form a suite, were in the best locations. Inside rooms cost less than outside, and double rooms were priced lower than single-bedded quarters. While the newest ships provided a bath for every room, tubs were available only by paying a premium rate; the standard bathrooms offered only a shower. The older ships had many cabins without a bath; a substantial increase in the fare was demanded for any room with private bath.

Indicative of the differentials charged for space in a modern ship designed with "modular," i.e., standardized, accommodations were the following categories of rooms. The rates shown are for each person, and are listed below by categories ranging downward from the most desirable rooms in the ship (the suites) to the least desirable cabins, located on the waterline deck:

- | | | |
|---|---|---------------|
| I | A suite, consisting of a bedroom with twin beds, separate sitting room, and tub/shower bath, on the promenade deck, looking forward | Highest price |
|---|---|---------------|

II	A suite, similar to the above but without the view forward; also located on the promenade deck	\$500 less than I
III	Deluxe single room, outside; all decks except the promenade deck	\$280 less than II
IV	Twin bed room, outside, with shower, on promenade deck	\$182 less than III
V	Twin bed room, outside, with shower, one deck lower than rooms in Category IV	\$196 less than IV
VI	Twin bed room, outside, with shower, two decks below Category IV space (one deck above waterline deck)	\$126 less than V
VII	Twin bed room, outside, with shower, waterline deck	\$56 less than VI
VIII	Room with double bed, outside, with shower; all decks except promenade deck	\$70 less than VII
IX	Room with double bed, inside, with shower, two top decks	\$238 less than VIII
X	Room with double bed, inside, with shower, same deck as Category VI	\$126 less than IX
XI	Room with double bed, inside, with shower, waterline deck	\$42 less than X

Modular construction, as noted above, was common on many of the newest ships. This prompted one operator to adopt the practice of making reservations for beds in a category of rooms rather than in a room selected by the passenger at the time of booking. The advertising folder issued by this operator contained this statement:

The modular design of the ships [in our fleet] provides for a standard of amenities and quality by deck location within each room type. These ships are equipped with stabilizers to provide maximum stability in all cabins. You will receive a confirmation of the room type of your choice at the time of booking. Specific stateroom number will be assigned at the time of ticketing or shortly prior to embarkation.

Seating capacity in the ship's dining room frequently was limited, and two sittings were required to handle the total number of embarked passengers. There is a perennial argument about the desirability of the "first sitting" or the "second sitting." Those travelers who have children or who follow early schedules usually elected the earlier hours; others who considered that dining at late hours somehow reflected greater social status, or who normally led lives starting later in the day, chose the second sitting. The food was identical, but might be a bit fresher for the first serving. There was some pressure on diners to finish their meals and to leave the dining room to be prepared for the later group. The *Queen Elizabeth 2* was designed as a three class, shuttle-service passenger liner, and had three different dining rooms. Fares were keyed to both the desirability of the individual rooms and the dining room to which cruise passengers were assigned. There was an inherent but undeniable "snob appeal" about having a table reservation in the same room and at the same time as the passengers occupying the highest priced suites.

In addition to the revenue from the sale of tickets, cruise operators looked to the ship's bar (or bars, in many cases) and shops for significant contributions to voyage profits. On short trips, bar profits could make the difference between a break-even operation and a profitable voyage. On long trips, where there was a great deal of social activity and many private parties, the bar profits could be substantial. Shops of all sorts became more and more prominent aboard cruise ships. The "International Promenade" on the *Norway* had seven shops offering luxury goods at prices rivalling those in the most exclusive stores ashore. Jewelry, porcelain, watches, clothing, sportswear, and children's clothing and toys were featured. An exotic "boutique" offered Chinese silk robes, jade flowers, carved ivory goddesses, and exquisite little tea sets. Passengers were lured into these places by their presence and also by the fact that on the high seas there were no duties on imported goods. (Whatever was purchased at sea had to be declared to customs officials, and might be subjected to duty when brought into the United States if the value exceeded the "free allowance" accorded by the government to returning residents.)

Some ship operators have elected to contract with specialists to handle all details connected with the stores. If this were done, the

concessionaire procured the merchandise, hired the necessary sales personnel, maintained the required inventories, and assured that stocks were replenished before the commencement of each voyage. Usually the contract required that the seagoing sales-persons be quartered and subsisted either as cruise staff members or at the same level as junior officers of the ship.

To reduce the operational responsibility of the shipowner while making certain that qualified personnel were assigned to those duties, some shipowners made contracts with food-service organizations to take over the entire catering activity of the cruise ship. When this was done, the contractor agreed to purchase, deliver, store, prepare, and serve all food during the cruise, and to hire, train, and supervise the dining room waiters and helpers and the necessary kitchen staff in its many specialized subdivisions; to plan and execute appropriate menus for each day; to be ready to meet individual orders by passengers for parties of any and every kind; and to operate the crew's galley in all particulars. A fixed fee per passenger per day might be charged, or the arrangement might call for a service fee over and above the actual cost of all purchased stores. The contractor assigned a person considered qualified to manage this complex operation who was responsible to senior management ashore for satisfactory performance under the contract. The shipboard representative of the owners who supervised the contractor's activities usually was the ship's purser. One company took a new approach to solving the problem of good supervision and placed on each of its ships a highly qualified specialist known as the hotel manager. He had exactly the responsibilities that his title suggested, and he reported directly to the master of the ship. Although this violated the traditional position of the purser, it accomplished two objectives: it permitted the purser to concentrate upon the activities peculiar to his office by relieving him of major distractions from the efficient operation of his department. It also assured that the standards of service to cruise passengers would equal, at the minimum, those of the finest hotels ashore.

The practice of bringing concessionaires aboard ship to operate the dining rooms and bars did not meet with universal approval. Many complaints were registered by passengers, who stated that the personnel provided by the contractor often were poorly trained, inefficient, and ungracious in manner. Maritime labor

unions alleged that the concessionaires procured catering staff on the basis of the least possible cost, without regard to the capabilities of those workers. They also charged that the treatment of these persons was unsatisfactory, in that they were crowded into inadequate quarters and required to work excessively long hours.²²

Whatever the truth might be, it must be acknowledged that using concessionaires places the reputation of the shipowner in serious jeopardy. In the first place, passengers assume that everyone working aboard the ship is employed directly by the steamship operator, and therefore condemn the company for any shortcomings or mistakes. In the second place, the concessionaire may have experience in catering ashore, but not in shipboard activity. Unless the owner has a large and very well trained staff of supervisors to keep the concessionaire's performance up to the desired or stipulated standard, there can be no certainty that the promises made by the traffic department will be carried out. In the third place, the contractor does all the purchasing and inspection of foodstuffs, and the quality of what is procured may not be ascertained by the ship's representatives until it is offered to the passengers. If the meat or vegetables are of inferior grade, the criticism levied against the vessel could have serious effects upon the efforts of the traffic department to find customers.

A few cruise operators have attempted to capitalize on the popularity of casino gambling by installing full-scale casinos aboard their ships. To avoid any conflicts with territorial laws governing gambling, the seagoing casinos were open only when the vessel was in international waters.²³ Professional personnel to staff the casinos in some instances were recruited by the shipowners from the ranks of those working in casinos ashore; in other cases, the whole operation was placed under contract to a concessionaire who shared the profits on an agreed basis. On January 15, 1985, the *New York Journal of Commerce* reported that a state-controlled company in Austria had active contracts to operate the casinos aboard nineteen cruise ships around the world. The magnitude of each ship's gambling activity may be gauged by the fact that the Royal Caribbean Cruise Line's facilities, which were handled by this Austrian concessionaire, comprised eight blackjack tables, an Austrian roulette table, and one hundred computerized slot machines.

Adding this type of amusement to the recreational program of

the cruise ships was acclaimed by many travelers, but it had to be regarded by the ship operators with great caution. Regardless of whether the casino were run by the ship or by the concessionaire, the reputation of the carrier always was in jeopardy. Complaints by irate losers could become explosive and probably would become the subject of gossip among cruise passengers, to the ultimate detriment of the steamship company. Supervision of all gambling activity therefore must be exemplary, and the fact of constant surveillance given publicity in advance of any possible complaints.

For those operators who send their ships on a variety of voyages, some long and some short, some repeating earlier experiences and some new in many respects, the normal procedure is for the traffic department to conjure up the itinerary that will be most attractive to the public, and then to have the marine operations department determine what is feasible, given the characteristics and capabilities of the ship. Of major importance is the depth of water available in the suggested ports of call; unless it exceeds the draft of the ship, it will be necessary to bring passengers ashore in tenders. If tenders are required, local conditions of wind, surf, height of waves, and landing area must be scrutinized. If the ship is to be anchored some distance offshore, and a heavy swell is experienced throughout the year, it means that passengers may have to jump into the tender when the swell is at the peak. This requires both agility and good timing; serious injury can result from hesitation or lost footing. In all probability, the recommendation from the operations department would be to delete that place from the itinerary.

Each proposed port of call is given the same methodical examination, and only those which are acceptable from the viewpoint of the navigators and ship handlers are approved. The traffic department must adapt its plans to the conditions discovered by the operating department. Once the ports of call are set, the engineering department works out the schedule of the ship's movements to achieve maximum economy and efficiency. Speed is of relatively minor importance to cruise passengers, but is of great significance to management. Slower speeds require very much less fuel than the maximum speed of which the vessel is capable. Whenever speed can be reduced without thereby increasing the duration of the voyage to an uneconomical point, it is standard practice to slow the ship. While most of the purpose-built ships have speeds of approximately 22 to 23 knots, these maximum levels are reached only

occasionally. One itinerary for a round-the-world cruise of a 23-knot ship required a single leg of sixty-two hours' duration when the speed was 21.7 knots; all other legs were accomplished at lower (and hence more economical) speeds ranging between 15 and 20 knots.

Certain basic principles apply to making cruise schedules. One is that the ship should arrive either at the anchorage, if tenders are to be used, or at the berth, if wharf or pier is available, not earlier than eight o'clock in the morning. Time for the passengers to enjoy the attractions ashore must be proportioned to the physical size of the place being visited and the interest it may have for tourists. Generally speaking, an average stay of four hours for an island like Saipan has been found satisfactory; for a well-known city with numerous attractions like Honolulu or Copenhagen, a call of at least sixteen hours is proper. In a city where shopping is the great lure, as in the case of St. Thomas (Virgin Islands), a full business day must be allowed for the pleasure of the passengers.

Departure time poses something of a problem because of the possibility of stragglers. Standard procedure is to notify all members of the cruise party that the ship will sail at the appointed hour, and anyone who is not on board will have to find his or her way to the next port of call at personal expense. This notice is published in the advertising leaflets, the ship's newspaper, and on bulletin boards in the ship. It is accepted practice in the cruise industry to fix the exact time to leave a port to accommodate the ship's arrival at her next scheduled stop. If the distance to be covered is short, as from San Juan, Puerto Rico, to Christiansted, St. Croix (Virgin Islands)—104 miles—departure time might be as late as 2:00 A.M., and cruising speed would be set at 17 knots to assure arrival at eight o'clock that same morning.

As already mentioned, the cost of ship's fuel is so great that only rarely is a cruise ship operated at full throttle. Instead, the lowest speed that will meet requirements is dictated. From Barcelona, Spain, to New York, for instance, is 3,728 miles. At 22 knots, this distance can be traversed in seven days and two hours; at 20 knots, in seven days and nineteen hours; and at 19 knots, in eight days and five hours. Very likely, the decision would favor 20 knots, which would permit departure at 7:00 P.M. and arrival at 2:00 P.M., an acceptable hour for the end of the cruise.

Ideally, arrival at every port of call would be in the early morn-

ing and departure would be effected after dark, so that passengers might spend the entire day ashore. In the real world of winds, tides, and distances, however, the schedule must be made to fit the circumstances regnant in the port at the time of the proposed call. If tides regulate the hours of entry and departure, or pilots take ships strictly in the order in which they have arrived at the pilot station, the schedule must take these facts into account.

Once the itinerary and schedule for the cruise have been determined, the supply division of the shipowner organization begins its work. The deck and engineering departments of the ship must requisition enough materiel and spare parts for every reasonably foreseeable need, and must make sure that delivery schedules can and will be met. This is particularly critical where the ship is completing one fairly long cruise and is to start on another of equal duration within a maximum of twenty-four hours after arrival.

The catering (or hotel, as it sometimes is called) department must order the foodstuffs that will be required for the cruise, based upon the menus which have been prepared. Some appreciation of the quantities involved may be gleaned from the following report of purchases made for the *Fairsky*, a 1,200-passenger cruise ship, assigned to a ten-day voyage from Los Angeles to the "Mexican Riviera":

15,186 pounds of meat	22,200 pounds of assorted fresh fruit
3,547 pounds of lobster	9,500 pounds of flour
2,720 pounds of assorted prawns	2,700 pounds of assorted pasta
1,150 pounds of crabmeat	960 gallons of fresh milk
4,242 pounds of assorted fish	1,960 cans of assorted fruits
18,600 pounds of fresh vegetables	7,900 pounds of poultry
4,850 pounds of frozen vegetables	1,800 pounds of coffee
7,900 pounds of potatoes	1,460 pounds of sugar

Bar supplies were ordered in similar large quantities, as evidenced by this list:

286 bottles of French champagne	38 bottles of gin
690 bottles of American wine	27 bottles of rum
215 bottles of French wine	155 bottles of vodka
585 bottles of Italian wine	95 bottles of assorted liqueurs
40 bottles of cognac	3,940 bottles of assorted beers
60 bottles of whiskey	5,976 bottles of soft drinks
55 bottles of bourbon and rye whiskies	

When the *France* made a round-the-world cruise lasting ninety-one days (from January 10 to April 10, 1972), her executive chef promised that there would be no repetition of the same recipe. Fresh vegetables, fresh meat, and fish were procured at various ports of call, and advance notice of what would be required had to be furnished to the local purveyors so that they could assemble the designated quantity of each item. Wines and spirits of all types and vintages also were stocked against the demands of the passengers.

Although it is an extreme case because of the length of the cruise and the approximately 1,200 passengers carried, the manner in which laundry was handled for the *France* is a lesson in itself.

It was impractical to put aboard the ship enough napkins, tablecloths, bedspreads, sheets, and towels for ninety-one days. To keep the accumulation of soiled linen in bags stowed in closets would have created serious housekeeping problems. The solution was to ship ahead to strategically located ports of call quantities of clean linen, which could be exchanged for the soiled items. The huge collection of dirty linen was to be sent back to France in a connecting cargo ship. At the same ports, city laundry facilities were placed under contract to perform the well-nigh impossible. In Rio de Janeiro, for instance, about ninety thousand pieces of laundry were washed, dried, ironed, and returned to the ship within twenty-four hours, while a simultaneous exchange of clean for dirty was made of an additional 150,000 pieces. At Valparaiso, Bombay, and Capetown, similar arrangements were made.²⁴

While some operators believe that the seven-day cruise is the "bread and butter" of the industry,²⁵ others have put their trust in the three- and four-day voyages which have become popular in the United States. Whatever the length of the voyage, the pattern of operation has become standardized: depart in the early evening, and return to the home port no later than midmorning. Only eight to twelve hours therefore are available to make the vessel ready for the next sailing. It takes little imagination to comprehend the coordination required between the management staff ashore, the suppliers, the service agencies located in the port city or its immediate environs, and the shipboard personnel. A careful schedule of deliveries must be set up, and rigidly adhered to; adequate inspection personnel must be available to assure that what is being put aboard the ship is what the purchase contract called for, and in the quantity ordered. The problem is complicated by the fact that

after the ship has been engaged in this seven-day voyage routine for several months, there will be some rotation of key personnel every time the vessel comes into port. The arrangements are made somewhat more difficult in those instances where the ship's turnaround day is Sunday, which means that everyone is on the overtime payroll, and some emergency procurement may be impeded because the supplier has closed for the weekend.

No matter how capacious the tanks may be, a ship sooner or later will have to be refueled. Petroleum supplies have been troublesome and expensive to procure ever since 1973, and the greatest care and attention must be given to placing orders that assure the commitments will be honored. The quality of oil used by diesel-powered ships has deteriorated since 1973, and shipboard engineers must be alert to test that which is to be pumped into the tanks to ascertain that it meets minimum specifications. Scheduled time in those ports where fuel is purchased must be long enough to permit the oil to be pumped aboard.

An important function of the operating department is to arrange to bring the ship into port and to care for her until she departs. That means a pilot must be waiting to direct the cruise vessel without delay to a waiting berth, for passengers do not accept apologies when they are delayed in getting ashore. Tugs must be available to berth and unberth the ship promptly on schedule unless she is fitted (as many of the newly constructed cruise ships are) with bow thrusters to facilitate maneuvering without assistance. Launches or tenders, if needed to ferry passengers to and from shore, have to be at the ship's side when the anchor splashes into the water. Since cruise vessels normally are accorded "yacht privileges" in the ports they visit, the formalities of entrance and clearance are reduced to a minimum. It is important, however, that the ship's officers be informed concerning all local regulations so that no embarrassment will be experienced by the passengers. These responsibilities are among those entrusted in each port to the ship's agent, who must be reliable and resourceful as well as reasonable in his charges for services rendered.

It is only after the itinerary receives final approval by the operating and traffic departments that the actual choice of agents can be made. In one port, the decision may be to use the branch office of a liner-service company. Among the reasons influencing this deci-

sion might well be that the liner company is accustomed to handling ships on fixed schedules, and has established certain routines to be followed by its office staff. This kind of experience and "know how" eliminates much of the chance that something may go wrong. Then, too, the liner company is already established in the area, and has the proper contacts and friends to dissolve difficulties if and when they arise.

It is the responsibility of the operating manager to notify the agent of his appointment. Full information on the ship's movements must be supplied, and the staff of the cruise ship must be instructed when and how to communicate with him. The traffic department also must be informed, so that it can list the names of port agents for the information of passengers.

Employment aboard a cruise ship is very exacting, especially for members of the catering staff who are in direct and constant contact with the passengers. It is axiomatic that the master, purser, chief steward, hotel manager (if there is one aboard), head waiter, and senior wine steward be selected not only on the basis of technical competence but also on their records for tact, diplomacy, suaveness of manner, and patience. On the long cruises, these people, as well as room stewards, bellboys, bartenders, and table waiters, are subjected to almost unending demands for special attention. They are expected to render whatever service is asked promptly and courteously, regardless of how long they may have worked or the circumstances in which this work has been performed. The senior medical officer of the *Queen Elizabeth 2* stated in an interview that he had many patients from the steward's department who were suffering from ulcers, hypertension, anxiety, and insomnia. In part these health problems resulted from the length of the voyage, but the principal reason was the constant striving always to please even the most arbitrary and unreasonable passengers.

On the short voyages, some of the same problems exist among shipboard personnel. While the ship completes a cruise in seven days, a new cruise is scheduled to begin on the same day the ship arrives in the home port. The eight to twelve hours in that port are barely sufficient to accomplish the enormous amount of work required. The ship must be cleaned thoroughly. Every cabin must be stripped, wiped down, refitted, and made to look as though it never had been used. Damaged fittings and scarred bulkheads must be

repaired. Stores must be taken aboard and stowed properly. Deck chairs must be inspected and replaced if necessary. The list is long, and involves tedious and exhausting manual labor as well as alertness. Then, about two hours before sailing time, stewards must make themselves ready to receive the new passengers. Because many of the persons making the seven day cruises never have been aboard ship before, or consider that having paid a comprehensive fare they are going to "get their money's worth," the demands for service may be anywhere from unending to degrading.

Selecting, training, supervising, and holding desirable and qualified seagoing personnel for cruise ships is a major responsibility of shoreside management. This is being recognized by some operators, who make special efforts to assure that working conditions are the best possible in existing circumstances, and that personnel are rotated between sea duty and shore leave in accordance with established schedules.

Although not a professional seafarer, and not in the chain of shipboard command, the cruise director is important to the success of every cruise. This person—there is no tradition that the director be a man or a woman; both sexes have been employed with marked success—is responsible for running all the shipboard games and entertainments. Every activity is scheduled through the director's office, so that no conflicts arise in the use of space or equipment. Events involving passengers are listed by this office and published in the ship's daily news sheet. To encourage cruise participants to take part in those games, sports, meetings, sessions, or other forms of amusements or instruction that take place during a typical day at sea is the basic reason for having the director on board. The larger ships have a director and several assistants, all of whom are qualified as recreation specialists.²⁶ Administratively, the director and staff are supervised by the purser, but perform none of the technical duties of that official.

Beside the cruise director, other "supernumeraries" in the ship's complement include musicians, professional entertainers (often hired directly from well-known nightclubs or variety shows, and at the same compensation they receive for shoreside engagements), lecturers, photographers, and in some ships a clergyman who serves as the ship's chaplain. Lecturers and chaplains normally travel as guests of the ship, exchanging their professional services for the cruise.

More and more, as other opportunities to experience the pleasure of a sea voyage as part of international travel become less available, cruises are gaining in popularity. Not only do they carry the passenger to overseas points, but they offer a completely different style of life for the duration of the voyage. As floating vacations, cruises demand specialized management, and to meet this demand corporations have been formed for the express purpose of designing, building, owning, and running ships that cater to that fortunate group of people who find a way to make a cruise and to enjoy the “thrill of a lifetime.”²⁷

Industrial and Special Carriers

INCLUDED in the thousands of vessels that collectively compose the mercantile marine of the world are many fleets which, because of highly individualistic employment or very specialized design, belong to one of two categories of shipping. The first of these, the *industrial carriers*, serve the needs of particular industrial enterprises, while the second, the *special carriers*, meet the demands peculiar to certain types of cargo.

Industrial carriers are those ships that exist as part of the plant and transportation properties of large industrial organizations and are operated exclusively for them. The huge fleet of tankers maintained by Exxon Corporation, the largest oil company in the world, typifies this category. The only purpose of this globe-girdling aggregation of ships is to provide the transoceanic transportation essential to the life of the parent concern. There are no other users to be considered by the owners, and no obligations to be met except those laid down by Exxon Corporation, which has full responsibility and control.

Other ships, like the refrigerated fruit carriers or the automobile transporters, are owned by steamship companies that are not affiliated with any industrial activity but are built to carry a certain cargo with greatest efficiency and economy. These ships are the special carriers; they are made available, either under charter or in regular berth service, to those users who must send their goods across the oceans. For reasons of their own, these users do not desire to become owners and operators of ships and are satisfied to benefit from the capability without having to accept the responsibilities and burdens of ownership.

It is apparent that many ships appropriately designated as industrial carriers are designed to transport a single commodity and therefore may be considered as both special and industrial carriers.

There is nothing contradictory in this finding, so long as the principles underlying the reasons for the existence of these ships are understood.

INDUSTRIAL CARRIERS

The industrial carrier is defined as the marine transportation link in the process of manufacture or distribution (or both) of the materials used or produced by an industrial organization. Some carriers handle nothing but raw materials, and others transport only the finished goods, while still others may be used interchangeably.

It must be pointed out that some industrial carriers serve a one-way trade, and as a consequence the parent concern may place the ship in berth service, or on voyage charter, for the return trip. The economic reason for this action is to reduce the cost of ship operation, and to eliminate, if possible, the need to sail ships in ballast on nonrevenue producing voyages.

The Broken Hill Proprietary, of Australia, operates a fleet of eighteen bulk carriers and two specialized roll-on/roll-off steel carriers. The bulkers haul raw materials to company steelworks; occasionally they are dispatched to Southeast Asia. The roll-on/roll-off ships move semifinished and finished steel products from plants to customers in Australia and across the Tasman Sea. An understanding of the magnitude of this operation is derived from the fact that the annual cost thereof exceeds one hundred and fifty million dollars.¹

Crown-Zellerbach Corporation, the paper manufacturing and distributing organization based in San Francisco, California, has two specialized ships under long-term charter to haul newsprint from Scandinavia and Canada to California ports. Serving only the charterers, these ships are classified correctly as industrial carriers.

Exxon Corporation, already mentioned, at the end of 1984 operated a fleet of 104 ships of its own and about fifty additional units chartered from other sources. Ranging in size from 3,800 tons to 516,423 tons deadweight, this is one of the largest tanker fleets in the world. A complete vessel operating department exists within the Exxon corporate structure to control and administer this fleet.

United Brands, the banana-growing and distributing company domiciled in the United States, owns a fleet of eleven refrigerated

ships for the sole purpose of providing the transportation required to serve its markets.

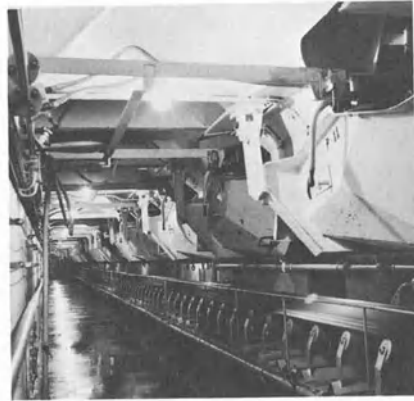
The principal purposes served by the industrial carrier are:

1. To control delivery schedules of raw materials or finished products (or both) handled by the parent concern.
2. To reduce the cost of ocean transportation by elimination of the "middleman's" profit.
3. To provide efficient transportation to the destinations required by the parent company for the type of cargo of particular interest to it.

There is a heavy burden of expense and responsibility inherent in the ownership and operation of a fleet of oceangoing vessels. Only those corporations with a continuing need for transportation of goods in shipload lots, or between ports not served by common carriers, have found it profitable to establish and maintain industrial carrier service. The large investment of capital may deter some companies from buying their own ships; instead, they resort to



A lumber carrier, *Star Columbia* depends upon her own gantry cranes for fast dispatch. Photo: Munck International.



Top left: Melvin H. Baker, a self-unloading bulk carrier. The port for the conveyor is at the stern. Top right: Discharging conveyor extended through stern port. Bottom left: Discharging coal to inland waterways barges. Bottom right: Conveyor systems below cargo compartments. Photos: Nordstroms Marine Systems.

long-term chartering. It is the use to which the ship is put, rather than the design or the ownership, which determines the classification of a vessel as an industrial carrier.

It is worth repeating that industrial carriers may be of a specialized design, but only when this suits the needs of the industry they serve. The fleet of Broken Hill Proprietary is an excellent exponent of this principle. The roll-on/roll-off ships were designed for the express purpose of transporting the long lengths and prefabricated pieces of steel sold by the parent company to purchasers in different parts of Australia. No other ships could perform this task so well or so economically.

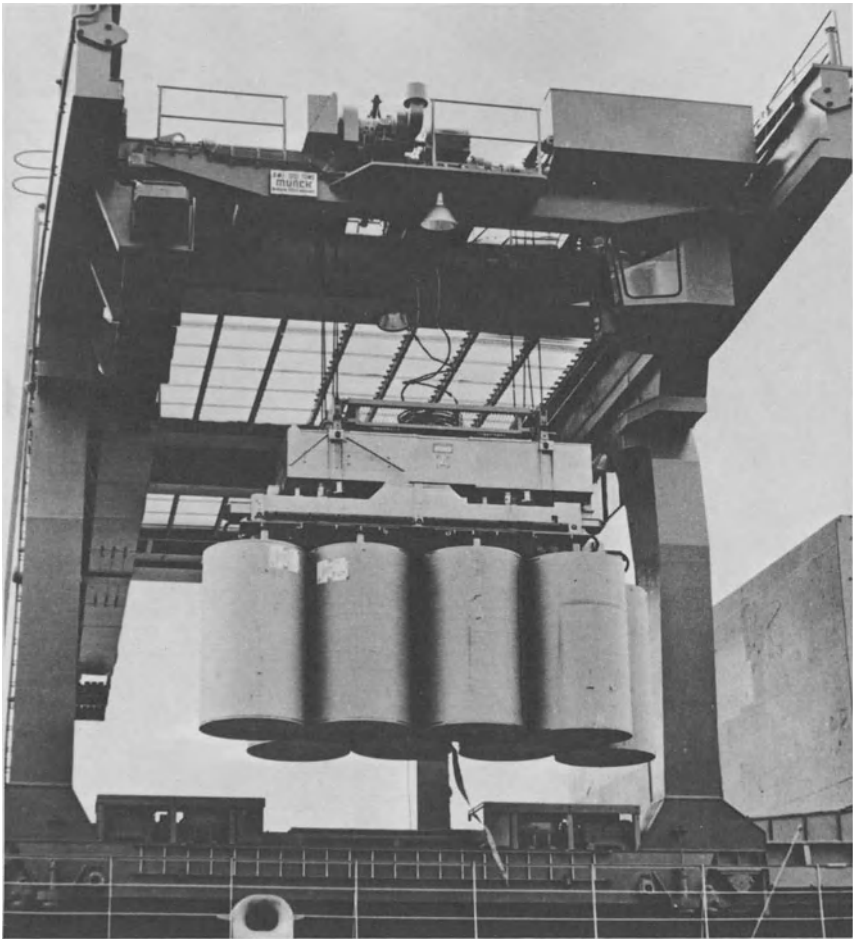


The *Rondeggen* carried rolls of newsprint in her holds and packaged lumber on deck. Three traveling gantry cranes assured fast cargo handling. Photo: Brenner, courtesy Crown Zellerbach.

Considerations of cost also may influence a company initially to charter one or more ships of either conventional or specialized type, depending upon what is required and what is available on the market that is satisfactory for the intended purpose. As experience is gained, and the economics of the situation may justify, ships of appropriately specialized character may be acquired.

It has been noted above that one advantage of the industrial carrier is that it gives to the parent company control over delivery schedules. Vessels are dispatched in accordance with the needs of the industry; there is no concern for possible customers who might be upset by sudden changes in sailing times, ports of call, or space allocations within the ship. To companies like Exxon, United Brands, Crown Zellerbach, and Broken Hill, which must depend upon ships to bring in materials indispensable to their operations, this factor alone justifies the maintenance of controlled fleets. The fruit company also has flexibility in routing its ships to those ports where the demand for bananas is at the maximum. In this way, it satisfies the demands of its retail merchant-customers as well as capitalizing on opportunities to sell the bananas at the highest prices.

Two other advantages accrue to the owner of an industrial carrier. One is the lower cost arising from elimination of the profit-seeking shipping organization, while the other is the independence of the ship charter market conferred upon the parent industry by having a fleet already under its control. The aggregate savings from



Gantry crane with specialized lifting device and clear plastic overhead panels handles newsprint safely in any weather. Photo: Munck International.

these factors, tangible and otherwise, often outweigh the cost of owning and managing the fleet.

Finally, it was noted that an advantage to the parent concern of the industrial carrier is that it provides efficient and careful transportation for the proprietary cargoes. As experience is gained by the operator of an industrial carrier in handling one or more commodities, many cost-saving and profit-producing techniques



The diesel-powered ore-bulk-oil carrier *Furness Bridge* (built 1971) had a deadweight of 166,064 tons and a speed of 15.5 knots. Piping for oil cargoes lies outboard of the hatches with their thwartships sliding covers.

Photo: Ralston, courtesy Furness Withy Group.

are certain to be developed, eventually culminating in highly effective, efficient, and economical methods.

For these reasons, industrial carriers are justified. The principles are not violated by using chartered ships, so long as the charterer maintains the same complete control over the ships' activities as if he had title to those vessels.

SPECIALIZED CARRIERS

One of the more interesting, and certainly significant, trends in mercantile shipping during the twentieth century has been the development of specialized carriers, which are found today on practically all sea routes. Their numbers and variety increase annually, and bear witness to their ever growing importance.

The specialized ship is one designed for the express purpose of carrying, usually in shipload lots, a particular type of cargo, either in bulk or in containers of stipulated size. A specialized ship thus may be constructed for the exclusive carriage of alumina; a completely different design would be needed for a vessel hauling automobiles and trucks.

Among the many types of specialized carriers are ships built to transport bauxite, bulk cement, chemicals in liquid form, coal, iron ore, liquefied natural gas, newsprint and other paper products, petroleum and its by-products, refrigerated fruits and meats, wood chips, and wood pulp.

The significant feature, by which a ship employed in this manner is assigned to the category of a specialized ship, is the design, as distinguished from the industrial carrier, which is classified according to the use to which it is put.

Although transportation by water is the most economical means of movement ever devised by man, there is no single commodity for which the individual cost of carriage does not constitute, in some measure, a barrier to its ready sale in the markets of the world. The carrier is entitled to be compensated for his services, but the shipper demands that this compensation be kept at the lowest possible level. To satisfy this requirement has called for the exercise of amazing ingenuity on the part of the shipowner, the naval architect, and the shipbuilder. Particularly in the second half of the twentieth



The starkly utilitarian design of the Japanese automobile/truck carrier *Nada V* is her dominant characteristic. Photo: Moran Towing Co.

century have the fruits of these endeavors become important in a wide range of shipping operations, as one specialized design after another has been created to transport commodities once considered immovable, or to reduce the costs of carriage to levels previously believed not to be feasible. Extraordinarily complex ships have been built to provide safe and efficient, as well as economical, transportation. Paralleling these achievements, and often contributing to the success of the project, have been the successful attacks on that age-old problem of the shipowner: to reduce the time required to work the ship in port and send her back to sea.

Designed as she is to transport most efficiently a single commodity in shipload lots, the specialized carrier often succeeds in areas where the conventional ship would be totally unsatisfactory. An interesting demonstration of this thesis may be found in the transoceanic movement of liquefied natural and petroleum gas. Efforts to transport these by-products of the oil fields which were then, and still are, in great demand, began in the 1950s, but it was not until 1960 that a commercially practicable method of handling these

cargoes was brought to perfection. The first full scale experiments involved the shipment of methane, which becomes liquid when cooled to -162° Celsius (-265° Fahrenheit). One of the many distinctive features of liquefied gas, insofar as its carriage by ship is concerned, is its bulkiness; one naval architect estimated that it occupied about four times the space of an equivalent tonnage of petroleum. Special ballast tanks to bring the ship down to her normal draft were needed.²

A small dry-cargo ship was converted into a liquid methane carrier in 1958. The *Methane Pioneer* used a "wool" made from balsa wood as the principal insulation for the cargo tanks. The experiment was so successful that about two years later the *Bridgestone Maru*, a tanker of 25,626 tons deadweight, was built particularly to haul liquefied gas. She was assigned to the supply route from Indonesia to Japan. Her performance validated many theories relating to the technique of moving liquefied gases.³

As experience was gained with the pioneering ships, significant advances were made in the design, metallurgy, and insulation of the tanks in which these cargoes were carried. Beginning in 1962, a succession of improved and enlarged gas carriers was put into service. As if to prove that the movement of liquefied gas could become an entirely routine and uneventful activity, the *Polar Alaska* and the *Arctic Tokyo*, each with a capacity of 71,645 cubic meters of liquid gas, completed more than 110 round trips between Alaska and Yokohama in the six years that ended on May 31, 1976.⁴

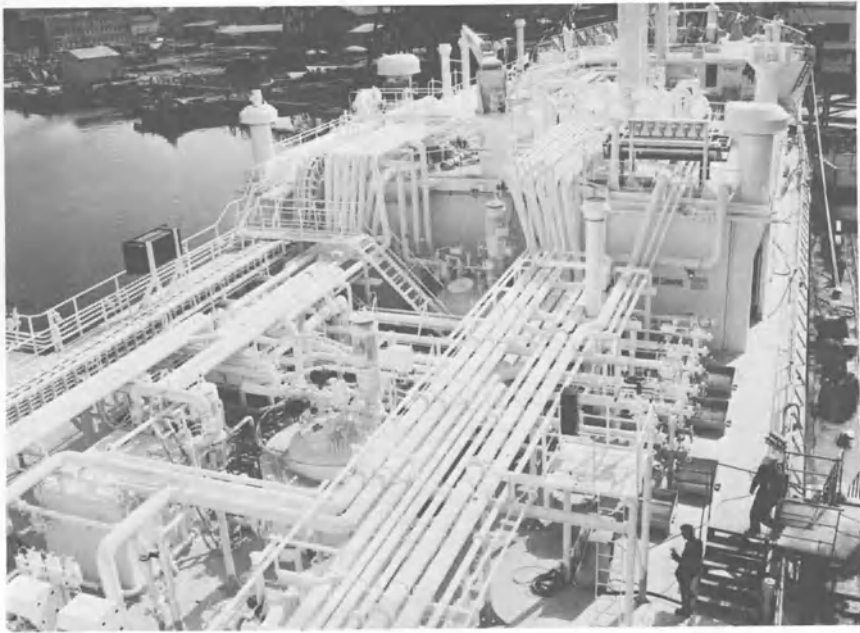
In September 1975, the 125,000-cubic meter *El Paso Paul Kayser* was delivered by her builders, Chantiers de France - Dunkerque. This twenty-knot, turbine-driven vessel was 280.62 meters long, 41.60 meters in beam, and had a draft of 11 meters. The six cargo tanks were encased with heavy insulation to hold the cargo at -165° Celsius. Boilers supplying steam to the 45,000-horsepower turbines were equipped not only to burn oil but also to accept the gas that "boiled off" from the cargo. To provide maximum safety, each cargo tank was isolated by cofferdams at the forward and after ends, by wing tanks on each side, and by a double bottom. The wing tanks could be used for ballast water, and provision was made to heat this water in the event of cargo leakage to prevent the excessively cold gas from damaging the intrinsic properties of the steel in the hull structure.⁵



The packaged-lumber carrier *Emma Johanna* was equipped with six centerline cranes to assure rapid cargo-handling. Photo: Jeff Blinn, courtesy Moran Towing Co.

Considered to be among the most complex of ship designs, these liquefied natural gas carriers were enormously expensive to build. The contracts for the first vessels of this type to be built in the United States were signed in 1972, and the price was just over 100 million dollars for each of three. By the fall of 1980, the price had risen to approximately 155 million dollars. There was sufficient demand for the gas, however, to justify construction worldwide of a fleet which, at the end of 1980, numbered 65 liquefied natural gas carriers of 5,400,000 cubic meters total capacity, and 568 liquefied petroleum gas carriers, with an aggregate of 6,500,000 cubic meters of cargo tank space.

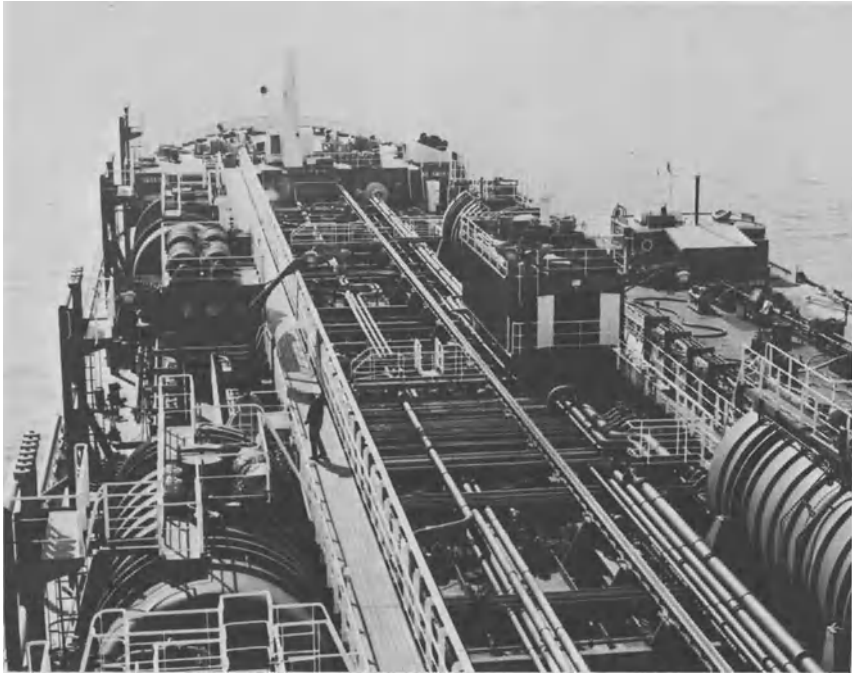
An example, although somewhat old, of developing a special type of ship to meet the peculiar demands of an industry was presented in 1956 with the entry into service of the *Duncan Bay*. This 20,000-ton deadweight capacity tanker was built to carry wood pulp which had been loaded in a wet state (approximately one-third pulp to two-thirds water), and after the water was pumped out of the tanks, was transported from British Columbia to California. There the pulp was resaturated to a consistency of 98 percent water



To assure uncontaminated handling of many different types of chemicals at one time, an intricate web of cargo pipes is essential. Photo: Moss-Rosenberg Shipbuilders.

and 2 percent pulp, so that it could be pumped into “desaturation” tanks before being stored. The ultimate paper product was improved through elimination of the process by which the pulp was dried at the mill before shipping. At the same time, the cost of handling was reduced by the mechanized loading and discharging arrangement. Special features incorporated into the ship’s design were turbine-driven pumps for the dilution water, agitators to blend and keep the pulp in solution during the unloading operation, and nine high capacity pumps to transfer the pulp from the twelve cargo tanks.⁶

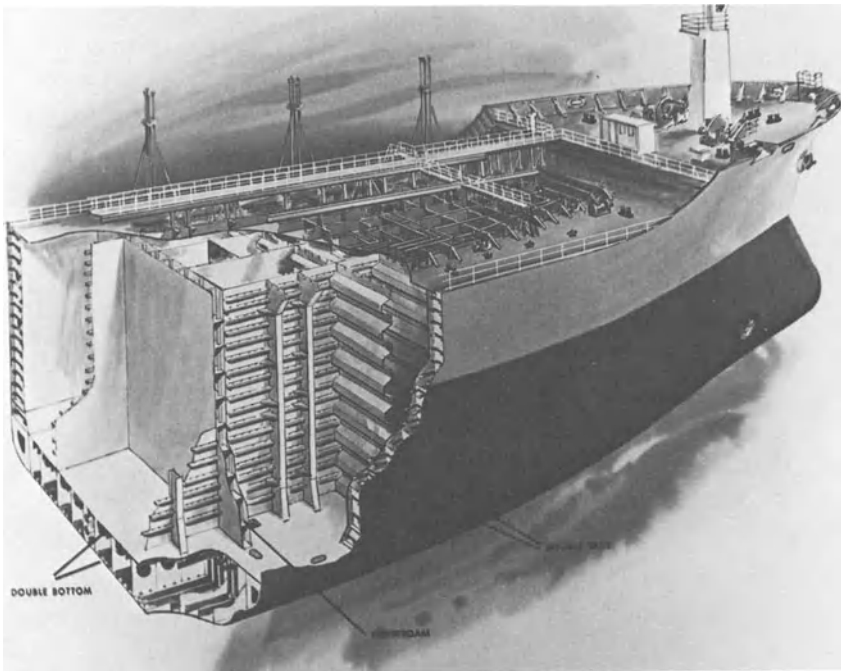
In the construction and long-term charter of the Norwegian motorships *Bessekken* and *Rondeggen* are seen examples of the procedure by which an industrial shipper will obtain the use of a specialized ship without investing a substantial amount of his own capital or establishing a marine department to operate the ships. Crown Zellerbach Company wanted a ship to carry 5,500 tons of



This modern chemical tanker has thirty-five piping arrangements. Four deck-mounted tanks accommodate small lots of specialized cargo. Photo: Stolt-Nielsen, Inc.

newsprint in rolls stowed below decks and 1,000 tons of packaged lumber on deck. High-speed loading and discharging in any kind of weather was required. A Norwegian shipowner agreed to build the ships with the stipulation that they be time chartered for a major part of the anticipated life of the vessels.

Acting in concert, the charterer, owner, designer, and builder produced a noteworthy solution to the problem. The engine and bridge were located at the after end of the ship, to provide the deck space for the packaged lumber and to facilitate operation of three gantry cranes installed on rails extending the length of the weather deck. The cranes were equipped with hydraulically operated, hinged jibs to provide long outreach over the ship's sides when loading or discharging. A rack-and-pinion driven trolley traveled athwartships on rails welded to the main girders and jibs, and could function despite noticeable list or tilt of the ship. A permanent,



The chemical tanker has a complex piping system, double bottom, double skin, and cofferdams separating the numerous cargo tanks. Photo: Stolt-Nielsen, Inc.

semitranslucent weather cover over the main girders provided the mandatory protection against rain and snow, and the sides of the cranes had curtains to complete the “all weather” protection. The cranes weighed 126 tons each, and had a lifting capacity of 30,000 pounds. Special yokes were devised to accommodate eight rolls of newsprint simultaneously, and another attachment was developed to handle the lumber. The cyclic rate of cargo work was 1,000 tons per hour; experience showed that about 750 tons per hour represented the practical maximum. A minimum number of longshoremen was required for the cargo-handling operation.

These two highly specialized ships proved to be both economical and efficient. Much of their success could be attributed to the cranes, which were of unique design and utility, very heavy, and quite expensive. It was reported that they cost \$196,000 apiece when the ships were delivered at the end of 1962; an investment

which would make the general service carrier look critically at any benefits which were expected to redound to him.⁷

In many parts of the world, there is a continuing demand for cement in very large quantities. For decades, it was impossible to ship this commodity in bulk and to discharge it with the self-unloading equipment installed in coal and ore carriers. Cement is extremely fine and powdery in its composition, packs easily, and has a very high angle of repose. Mechanical discharge required that the cement flow to the conveyors, a movement eventually achieved by pumping compressed air into the mass. Using this principle, engineers developed several self-unloading mechanisms by which ships carrying cement in bulk could discharge their cargoes efficiently and economically.

When she was delivered to her owners in 1981, the *Castillo de Javier* was the largest self-unloading bulk cement carrier in the world. She was a vessel of 44,000 metric tons deadweight, and had six holds. Two of these holds were exceptionally large and were intended for the cement cargoes. Transverse screw conveyors moved the cargo to each side of the hold where the cement fell through valve-operated trap doors into pipeways laid in the double bottoms. Vacuum pumps sucked the cement out of the ship in two cycles, one from the hold to a reloader chamber, and the other from that chamber to the shoreside silo. By means of this system, the ship could be emptied at the rate of 600 tons an hour. The builders claimed that the mechanism they devised made it possible to carry not only bulk cement but also low density commodities like flour or any granular or pulverized material. A further advantage of the pipeway system was that the tank tops were flat and easy to clean, thereby making it feasible to carry different types of cargo on the return voyage. The ship's utility was enhanced by the three twenty-ton cranes so mounted that they served all six hatches.⁸

There is a significant movement to all parts of the world of unusually large or extremely heavy articles, of which railroad locomotives and electrical generator components are typical, which cannot be transported in the cargo liners and tramps used in general trade. To meet this demand, some steamship operators designed their liner-service ships so that they had the necessary heavy-lift capability, their decks would support the weight, and the hatches were long enough and wide enough to permit these over-size items to be stowed under cover.



The specialized heavy-lift motorship *Uhenfels* uses both her 275-ton booms to hoist a 500-ton heat exchanger. Photo: Hansa Line.

Typical of the cargo liners with high capacity booms was the *Benarty*, of the Ben Line, an English company serving the routes from the United Kingdom to the Far East. She was fitted with a Stuelcken derrick of 180 tons capacity which was mounted between two hatches 12.19 and 15.92 meters long, respectively. Until its liquidation in 1980, the German shipping company, Hansa Linie, for many years operated a fleet of cargo liners with Stuelcken booms of 350 tons which could be paired to lift a total of 700 tons. American Heavy Lift Shipping Company, of Pittsburgh, Pennsyl-

vania, owned the only two purpose-built heavy-lift ships to fly the United States flag. These vessels, the *John Henry* and *Paul Bunyan*, each had two 216-ton cranes, bow and stern ramps strong enough to accept weights of 1,000 tons, and a shallow draft of 5.15 meters. They were employed in worldwide tramp-type service.⁹

One of the reasons that cargo liner operators put specialized ships into their services is that it assures them of a segment of the business which, but for this capability, would be denied to them altogether. Often the special capability applies to cargo moving only in one direction, as in the case of meats, bananas, and other perishable fruits, and vegetables brought to consuming countries in ever increasing quantities. On the return passage, these specialized carriers offer to carry any kind of general cargo that will not damage the expensive equipment of the vessel or interfere unduly with the schedules of the homebound perishables. It must be noted, in passing, that the stipulation by the owner or operator that only the specialized cargo will be accepted does not change the common carrier status of the line.

A large fleet of refrigerated ships has been built since the end of World War II, both by liner companies and by independent owners who seek long-term charters from importers of refrigerated goods. Most of the independent (tramp) "reefers" are owned by European, and particularly by Scandinavian, operators. The liner companies running reefers are of many nationalities and are found on numerous trade routes.

A further example of the responsiveness of shipowners to demands for specialized tonnage may be found in the automobile carriers operated from Europe and Japan to the United States. A considerable variation between designs manifested itself; some ships, like the Wallenius Lines' *Madame Butterfly*, were designated as "pure" vehicle carriers, because they did not attempt to carry bulk cargoes on the return passage. Other ships were equipped with portable 'tween decks which could be stowed under the main deck to permit full use of the holds for loads of grain. Most of these vehicle ships were designed with side or stern ports for drive-on loading and drive-off discharge.¹⁰

Sea-carriage always has been responsive to the needs of shippers, and the development of the numerous types of specialized carriers during the twentieth century merely emphasizes this char-

acteristic of the shipping industry. The only limits to the possibilities appear to be the imaginations of the users and the naval architects. Only a few of the specialized designs have been mentioned in this chapter because of the wide range of the subject. Economic reasons of the strongest sort have undergirded the construction and operation of ships with unique capabilities to support a particular industry. Equally valid have been the justifications found by common carriers to tailor their ships to fit a certain subdivision of the trade routes they serve. As has been pointed out, economic demands often result in a merging of industrial and specialized carriers, and it is important to recall that it is the manner in which the capabilities of the ship are held out to the user that determines whether a ship properly may be assigned to one category or the other.

Tanker Management

RANGING in size from enormous to almost tiny, transporting more tons of cargo of direct concern to more people in more parts of the world than any other type of ship afloat, tankers have become indispensable to the modern international industrial economy. As vital components of the merchant marine, tankers require many of the same managerial skills needed by any other type of ships engaged in moving cargoes across the seas. Dedicated to the movement of liquid commodities in bulk, however, these vessels demand specialized knowledge and techniques both from the personnel who take them to sea and those on shore who supervise their activities.

In the age-old history of shipping, transporting oil in bulk is a relatively new enterprise. The first steamship designed and built to carry petroleum in bulk, rather than in wooden barrels or metal drums was put into service in 1886.¹ This was the *Glückauf* ("Good Luck"), built in England to the order of Wilhelm A. Riedemann, a merchant of Hamburg, to serve the German-American Petroleum Company.² She was a small ship, 300 feet long with a deadweight of 3,020 tons—hardly noticeable alongside today's gigantic crude oil carriers. Her place in history, however, derives from the fact that she was the prototype of the world's tankers, with steam driven pumps segregated from the cargo tanks by steel bulkheads, and engine-room at the after end, separated from the cargo tank area by a cofferdam formed by twin bulkheads. This pattern of construction, it is appropriate to note, has not altered significantly over the years.

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Since the *Glückauf* was launched, bulk oil carriers have multiplied in numbers, increased in speed and, most conspicuously, grown enormously in size. Change was slow at first; as late as 1920, the standard tanker had a deadweight of only some 12,000 tons, although there were a few vessels of about 20,000 tons.³ In 1939, a small group of 16,000-tonners was in operation, but the majority of tanker-owning companies still leaned toward the "handy-sized" ships of 12,000 tons. It is a minor footnote to history that the average speed of all these ships was about eleven knots, compared to the *Glückauf*'s ten.

The demands for transportation of petroleum during World War II greatly exceeded the capability of the prewar fleet and called forth new standards for tankers. Bigger and faster ships were essential, and the United States built 532 oil carriers of the T-2 class. With a deadweight of about 16,750 tons and a speed of 14.5 knots, these ships proved incontrovertibly that large, fast oil carriers were economical, efficient, and as versatile as the handy-sized tankers of 12,000 tons.⁴

Once the utility of the T-2 had been demonstrated, growth became characteristic of the tanker-operating industry. In 1947, the first of the really big ships were built. These "colossal" vessels of 25,000 to 27,000 tons deadweight were relegated quickly to lesser positions by the prompt advent of tankers of 32,000 tons and, very shortly thereafter, of 45,000 tons. By 1979, the peak had been reached with the completion of the *Seawise Giant*, a crude oil carrier of 569,783 tons deadweight owned by the C. Y. Tung interests of Hong Kong. Owing to a reduction in consumption of oil worldwide, the demand for the very large crude carriers decreased markedly, and no building programs including ships as large as the *Seawise Giant* have been initiated since then.⁵

Paralleling the enlargement of the ship hulls was the proliferation of tankers in the world's merchant fleets. On January 1, 1984, there were in existence 5,548 tankers, including: 74 specialized vessels designed to carry asphalt, molasses, wine, and other unusual liquid cargoes; 752 chemical transporters; 47 liquefied natural gas carriers; and 448 liquefied petroleum gas carriers.⁶ The big carriers (generally those of 200,000 tons deadweight or larger) are used exclusively to move raw petroleum from the oil fields to refineries. Distribution of refined products traditionally has been



Top: The 14.5-knot, 16,650-tons deadweight T-2 tanker proved her worth in war and peace. *Bottom:* First of the American-flag supertankers, *Ezzo Zurich* (built 1949) had a deadweight of 26,555 tons, and was 628 feet long. Photo: Standard Oil Co.

the province of the smaller ships. The considerations limiting size have been related to depth of approach channels and restricted storage facilities ashore. As a result, a sizeable fleet of tankers of 3,500 to 10,000 tons deadweight was employed in this service. In recent years, however, to take advantage of the economies inherent in larger ships, and to capitalize on advances in naval architecture, tankers up to 42,000 tons deadweight, along with a few broad-

beamed, shallow-draft petroleum carriers of nearly 80,000 tons deadweight capacity, have been built for these trades.

As the size of ships increased, tanker owners showed more interest in obtaining optimum performance, but had to wait for the acceptance of computers as standard tools of management before realizing significant progress. To illustrate how the capabilities of the computer might be used by an operator, a study was conducted in 1970 of the hypothetical employment under time charter of a 20-knot, 500,000-tons deadweight tanker making a round voyage from the Arabian Gulf to Northwest Europe, a total distance of 22,614 miles. Fuel consumption for the big ship varied from 200 tons per day at a cruising speed of 14 knots to 790 tons per day at the maximum of 20 knots. The cost of fuel oil, at the time the computation was made, was \$17.50 per ton. At the slowest speed, the round voyage would take 67 days and would require 13,460 tons of fuel. At maximum speed, twenty days would be saved, but fuel consumption would jump to 37,209 tons. Calculating the possibilities for different cruising speeds, the computer showed that ordering the ship to travel at 16.5 knots for the round voyage of 57 days, with a total consumption of 20,355 tons of oil would be the most effective employment of the tanker. Assuming that the ship were registered under a flag of convenience and had a crew paid minimum wages, the owner could cover all expenses and have a profit equal to 10 percent on the investment if charter hire were fixed at \$1.15 per deadweight ton per month. This equated to a freight rate of \$3.05 per ton of oil delivered.⁷

During the years that the number and size of tankers in the world fleet was increasing, a concurrent change in the pattern of ownership became evident. Shortly after the end of World War II in 1945, half of the world's tanker tonnage was owned and operated by the oil companies. Three decades later, operators not affiliated with the petroleum producing and refining corporations (and therefore usually designated as the "independents") controlled nearly 75 percent of total tanker tonnage.⁸ In the five years which ended in June 1983, the major oil companies (BP, Chevron, Exxon, Gulf, Mobil, Royal Dutch Shell, and Texaco) reduced their fleets by 40 percent, compared to a worldwide decline in tanker tonnage of 10 percent. During that same half-decade these seven companies cut transportation in company vessels from about 37 percent to 23



Top: The 32,650-tons deadweight *Cities Service Baltimore* (built 1956) carried 18 grades of petroleum products. *Bottom:* The 16-knot *Sealift Arctic*, on a draft of 10.7 meters, has a deadweight of 27,225 tons, and can enter almost any harbor. Photo: Jeff Blinn, courtesy Moran Towing Co.

percent.⁹ This shift in capability was deliberate, as the producers sought both to allocate to different purposes the funds invested in tankers and to divest themselves of the burdens of fleet ownership.

There has been no diminution of interest on the part of the oil companies in the capabilities and economics of tankers, whether

owned or chartered. Pressure on the operating and marine traffic departments, which are concerned directly with the ships, remains as great as ever. Indeed, there is a good argument that the pressure is greater as the duration of voyages has been shortened by higher sea speeds and concurrent reduction in port turnaround time resulting from the capabilities of ships' pumps to discharge their cargoes in fewer hours. There is, consequently, less time to accomplish the many duties involved in transporting oil across the seas. Also, the availability of adequate storage tanks ashore to accommodate the huge quantities of oil delivered by the big carriers is much more of a problem than in earlier days. Operating expenses of modern tankers are extremely high, making every minute the ship is not moving between ports a matter of concern. Refineries set their schedules on the assumption that tanker deliveries will not be delayed for any reason, thereby augmenting the pressure on the operating department to assure that vessels, both owned and chartered, are in peak condition.

Maintaining the ships to the highest standards is essential if optimum performance is to be obtained. The current practices of substituting sophisticated computer-controlled machinery for skilled seamen may improve performance in one direction, but at the same time it makes possible much less maintenance work on the part of the ships' crews. This results in greater dependence upon shore-based technicians who are brought aboard at loading and discharging ports to accomplish the necessary work.

As the cost of fuel escalated after 1973, owners evinced greater interest in methods of keeping the hulls of their ships free of marine growth without the necessity of dry-docking the ship at comparatively frequent intervals. Several years ago, a device was perfected which made it possible to clean the hull while the ship was working cargo, thereby eliminating the time and expense of dry-docking. At a cost of about \$20,000, and in a period of eight hours, a tanker of 100,000 tons deadweight could have both sides and the bottom cleaned of all marine growths. The saving to the operator as a result of the clean hull would be on the order of fifty barrels of fuel per steaming day, not including the time otherwise required for dry-docking.¹⁰ Self-polishing copolymer paints, which reduce the friction generated by passage of the hull through the water, also inhibit marine growths. These benefits make possible longer inter-

vals between dry-dockings, with a corresponding increase in productive voyage time.

Another facet of the practice of installing machinery to permit fewer personnel to operate the tanker is that a great deal of what formerly was classified as "ship's work" to be performed only by experienced seamen now is done by computer-controlled devices or is transferred to shore-based artificers. The challenge and romance of seafaring have been cut back, and those men who once took pride in the skills they had mastered over a period of time are finding that the work they are assigned aboard ship no longer is interesting or demanding. This means that the crews' morale is of much greater significance to management than in former years, and supervisors must make more opportunities for personal contact with the mariners who, in the final analysis, are the indispensable element in the operation of tankers.¹¹

The problems of tanker management listed above bear witness that operation of these ships has become more complex through the years. It would be wrong, however, to conclude that these problems make the transportation of petroleum and its derivatives a unique type of shipping. It is specialized, but it is only one aspect of the whole business of moving goods across the seas. The following facts support this premise.

Tankers, like other ships, must be built, maintained, and repaired. Seamen must be obtained from the ranks of available seafarers, and in competition with other ships which may offer more attractive inducements to crewmen. Sailing, bunkering, and dry-docking schedules are based on the same fundamentals as for any other merchant vessels. As tankers have become larger and faster and have been equipped with more complicated machines and control devices, there has been a corresponding increase in the demand for better-trained seamen. Dry-cargo ships, especially container carriers, have concurrently been built with the same types of complex mechanical installations, and they have just as much need for the best and the most competent mariners they can hire. Shipyards, especially those with repair docks able to accommodate the enormous tankers of today, are available only in a comparatively few ports of the world and are equally in demand by huge dry bulk carriers and very large containerships.¹² Time is money to the owner of any ship, whether small coaster or very large crude car-

rier. Any delay to any ship is very nearly intolerable, and the complaints of tanker managers are no more heartfelt than those of other members of the shipowning fraternity.

It cannot be denied, however, that there are some aspects of ship management which are peculiar to the tanker business. Among these distinctive characteristics may be cited the nature and inherent properties of petroleum and its derivatives, the specialized terminals required for loading and discharging the ships, the normally one-directional flow of traffic, and the integration of the schedules of operation of refineries, distribution centers, and tankers.

The fact that petroleum traffic flows only in one direction is of major economic importance in analyzing the operation of tankers. Except in unusual circumstances, the ships are engaged for a significant portion of their lives in expensive and genuinely nonproductive voyages in ballast back to their loading ports. Various expedients have been examined in the search for ways in which to reduce the cost, both in time and in the usefulness of the ship. One of the more successful was the combination carrier that could haul dry bulk or ore or oil, depending upon the demand for transportation of these commodities. The theory was that, by judicious scheduling, after delivering a cargo of oil the ship could be chartered to lift a cargo of grain or iron ore to a port not far from the oil loading point, thereby virtually eliminating the ballast leg. The *San Juan Pioneer* was one of the first ships of this type to be employed in alternate trades. A typical voyage involved carrying oil from Sumatra to Los Angeles; sailing in ballast to San Juan, Peru, to pick up a full load of iron ore for Japan; and then proceeding in ballast to Sumatra to start the cycle again. The passages from Los Angeles to San Juan and from Japan to Sumatra were relatively short, and the revenue from the ore cargoes improved the ship's annual profit-making capability.¹³

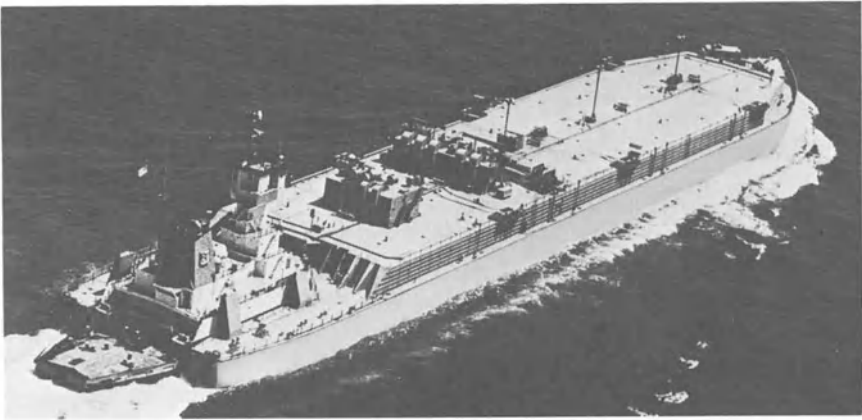
The vitality of this theory of combining trades may be observed in the number and size of these ships. On January 1, 1983, there were 479 oil-bulk-ore (OBO) and ore-oil ships in existence.¹⁴ Many of these "combis," however, were used only in one trade, being chartered either for carriage of dry bulk cargoes for an extended period of time, or being employed exclusively to haul oil.

The primary reason for an oil company to own its tankers

rather than to depend upon chartered tonnage is to assure that transportation for its petroleum will be available when needed. A secondary, and equally important, reason is to exert a moderating influence upon the charter market in times of ship shortages, and to prevent exploitation by the independents of a producer's obligations to meet delivery schedules at all costs. Experience has shown that an oil company can satisfy requirements if its proprietary fleet can lift approximately 60 percent of the petroleum tonnage that must move in one year. The third reason to own ships is to provide transportation for special cargoes such as many grades of lubricating oils, petrochemicals, and numerous small lots of a variety of fuels, or to make deliveries to shallow draft ports. These sometimes unusual assignments often cannot be met by the vessels available on the charter market, and therefore owning the appropriate types of tankers is the only solution for the oil company.

Once the decision has been reached to own ships, it is logical for the oil company to design its tankers. It is true that, in the years immediately following World War II, the ready availability of T-2 tankers at bargain prices made it worthwhile for many companies to operate these ships. With the passage of time, the T-2 was superseded by newer designs that met company needs more effectively. Some of the considerations that entered into planning replacements were depth of channel required when the ship was fully loaded; capacity of pumps; number of pumping systems necessary to deliver several grades of petroleum at the same time; numbers and capacities of cargo tanks; limitations on length and breadth of hull imposed by certain ports, canals, or berths at refineries; and economical cruising speeds. Size has varied from 20,000 tons to 390,000 tons deadweight for ships registered in the United States, and up to 569,763 tons deadweight for vessels registered elsewhere.¹⁵

In the coastal trades primarily, but also in some transoceanic operations, very large "integrated tug-barges" have proved to be practical and helpful in the constant effort to control the cost of transportation. A number of major oil producers in the United States as well as independent operators have built these units and placed them in service between Texas and the New England states, as well as in shorter routes. Typical of the largest integrated tug-barges is the *Belcher Port Everglades*, owned by The Belcher Company of New York, a distributor of petroleum. The unit is a 55,000-



The integrated tug-barge *Belcher Port Everglades - Belcher Barge #102* was built in 1981 to carry 55,000 tons of oil. The 15,000-horsepower tug fits into a 35-meter notch at the stern of the 195-meter-long barge. Photo: Aurora Photography, courtesy Belcher Oil Co.

ton deadweight, double-hulled barge 195.6 meters (640 feet) long, 32 meters (105 feet) wide, and 14.6 meters (48 feet) deep. Equipment of the barge includes five cargo pumps, two ballast pumps, a cargo heating system with an output of ten million BTU, and a bow thruster to increase maneuverability. Propulsion is by the single-screw tug that fits into a “pushing notch” at the stern that is 16.8 meters (55 feet) wide and 35 meters (115 feet) deep.

As previously noted, oil companies look to the independents to augment the capabilities of their proprietary fleets. Sound economic justification exists for this. The demands for capital to support activities related directly to the production of oil have become so great that little money is available for either construction or purchase of ships. If there is need for one or more ships of a design especially well adapted to meet the oil company’s requirements, often a long-term time charter equal to the anticipated life of the tanker will be negotiated with an independent. The charter-party serves as collateral for the independent to borrow the funds with which to build the ship. This financial arrangement was devised just after World War II and has proved satisfactory to all concerned.

Chartered tankers, regardless of whether they are on time charter or on single or consecutive voyage charter, are integrated

into the operation of the proprietary fleet, and their movements are controlled by the marine traffic department.

Short-term, single voyage, or consecutive voyage charters are used to meet immediate requirements which exceed the capability of the ships either owned or on long-term contracts. "Spot" coverage is a vital, but marginal, activity arising out of the need for one or more ships for a voyage or two to provide transportation demanded by the producing or marketing divisions. If a temporary surplus of cargo exists, or a shortage of company-controlled ships develops, the prompt chartering of one or more tankers will ease the crisis without taking on the obligation to keep the ship employed once the pressing need has been met. Even though the price of the voyage-chartered ship may be comparatively high when measured against the company's average time charter rate, the convenience of having a tanker at the time required, and only for that period, is worth a great deal to the oil company. The independent operator tries to anticipate these demands and to position his ships to take maximum advantage of the situation.

Some independents have built their fleets entirely on the basis of medium to long-term time charters. While the return is not so great as it would be if all the ships were chartered on the spot market, the stability of the longer contracts has proven, in many cases, to be more profitable than the earnings on the highly volatile spot market. Regardless of what type of charter is being sought, the reputation of the independent tanker owner is of paramount importance in determining whether he will be favored with the commitment.

Tankers generally are engaged in one of two distinct types of service, distinguished as "clean" and "black" (sometimes also called "dark oil" or "dirty") trades. Ships in the latter trade carry only crude, residual, and darker oils up to the diesel grade. Many of the very heavy crude oils require heating to reduce viscosity before they can be pumped ashore.

The clean trade, as the name implies, demands that the ship be very carefully prepared before any cargo oil is pumped into the tanks. Certain jet engine fuels must be carried only in tanks coated with inorganic, nonferrous compounds to eliminate contamination through corrosion of the tank walls. Lubricating oil may be emulsified if any water is left on the tank walls, and therefore all these

surfaces must be wiped by hand to assure that no trace of moisture remains to pollute the cargo. It is customary to fill ("press up") the tanks to capacity so that there is no void space in which condensation can form. In those instances where several grades of lubricating oil, varying in color from grade to grade, must be pumped through the same piping system, the lightest colored oil is pumped first. This assures against contamination from any residue encountered in the pipes.

Black oils are heavier than clean products, and therefore fewer barrels of the former may be loaded for a given deadweight tonnage. This fact is important in planning the load of the ship, inasmuch as distribution of the load must be made in appreciation of the stresses imposed by filled and void tanks. It also has a bearing on the number and size of ships needed to move a given quantity of oil. Normally, the capacity of a tanker is computed in terms of barrels of gasoline and total deadweight tonnage. The T-2 tanker, for instance, had a nominal barrel capacity of 140,000 and a deadweight of 16,750 tons. Fully loaded, fueled, and stored for a voyage from Port Arthur, Texas, to New York, the T-2 was able to lift 140,000 barrels of aviation gasoline but only 90,000 to 110,000 barrels of the heavier black oils.

Safety is another managerial responsibility of paramount importance not only on board ship but around terminals where tankers load and discharge. At sea, in the approaches to ports, at berth when handling cargo as well as when waiting for orders, and throughout the normal routines of cleaning tanks, performing maintenance work, and standing watches, the utmost vigilance must be exercised by all personnel if accidents are to be avoided. Fire, obviously, is the ever present hazard, and elaborate precautions habitually are taken to reduce this danger. Terminals also must be supervised continuously to assure that safe practices are followed. Most tanker operating companies devote considerable attention to the study of work practices and the analysis of accidents to use in training programs for both seafaring and shoreside employees. Closely related to these aspects of safety are the provisions for safe navigation of the vessel: selection of routings, use of pilots, choice of adequate berths, proper communication procedures, and supervision of electronic aids to navigation.

To reduce the possibility of explosions set off by static electricity

in the enormous cargo tanks of the very large crude carriers, installation of inert gas systems has been made mandatory in ships of 40,000 tons deadweight or larger; this requirement will be extended in the near future to vessels exceeding 20,000 tons. The inert gas may consist of treated flue gas from main or auxiliary boilers, gas from a gas turbine's exhaust, or gas from a separate inert gas generator. Oxygen content of the gas must not exceed 5 percent by volume. In essence, this system pipes an inert gas into the tanks as the cargo is pumped out, thereby avoiding the mixture of air and petroleum vapor which can become critical and subject to explosion from a single spark.

Concurrent with the inert gas system has come the development of crude oil washing of cargo tanks, which has been found safer than the time-honored method of washing tank walls with streams of water under high pressure. One of the benefits of the crude oil washing system is that cargoes are delivered with less contamination, at the same time that ballast water is clean and nonpolluting.

The new techniques necessitated the establishment of training centers to instruct ships' personnel in the proper use of the inert gas installation and the approved methods of washing tanks with crude oil. This program fitted well into the existing effort of major tanker operators to assure that their seafaring employees were able to cope with the problems of managing (in all its varied aspects) the large ships which are part of the world's tanker fleets. Simulators, using computers and elaborate scenic effects, are providing extensive training in ship handling in crowded channels and other demanding circumstances. The latest methods of fighting shipboard fires, a constant threat, are under unending study. Seafarers are sent to highly specialized fire-fighting schools on a periodic basis to assure that they have the most up-to-date knowledge of this essential skill.

Victualing, storing, scheduling, repairing, and bunkering a fleet of tankers in worldwide service are other aspects of management responsibility. Few tankers are employed in repetitious voyages between designated ports, and therefore long absences from the home port often occur. To provide adequate supplies of food and stores of good quality, and to procure these at reasonable prices for scores of ships in oil-handling ports around the world, is an activity of impressive dimensions. Full appreciation of the principles of bunkering is mandatory for management of a globe-gir-

dling fleet of tankers if maximum capability of the ships is to be attained. Management never is a routine matter when voyages are subject to unforeseen deviations, reroutings on short notice, and unexpected delays beyond control of either the ship or her owners.

Personnel administration in tanker operation is a concern of increasing magnitude. To recruit, train, assign, and supervise the performance of licensed and unlicensed seamen, and to assure that they receive adequate consideration in handling pay and allotments, delivering mail to the ship, granting vacations when due, and imposing disciplinary measures if necessary, are formidable undertakings, particularly when the ships are scattered over the oceans.

The current practice of granting tanker seamen extensive vacations has resulted in the enlargement of company personnel pools and in the installation of complex record systems. Many owners have ships under the registry of one nation but staffed with crews from a different country. A Liberian-flag vessel, for instance, might have a full crew of Spaniards. By the terms of employment, the Spanish seamen are entitled to vacation in Spain each year, regardless of where the ship may be trading. Transportation to Spain and back to the ship is the responsibility of the owner. Vessels with multinational personnel have a correspondingly more complex problem of repatriation. American ships trading along the coast of the United States, however, are privileged to adopt a comparatively simple plan of having relief contingents join the vessels periodically at designated ports, replacing the crew members eligible for vacation. Whatever the method, a program of rotation must be devised to assure both efficient operation of the tanker and contentment among the seamen. It is not coincidental that the increasing mechanization of ships, which facilitates turnaround in a matter of hours, has created problems of human relations requiring the exercise of vision and imagination in the search for solutions.

Differing quite markedly from dry cargo ship-operating procedures is the significance attached by tanker owners to vessel accounts and statistical reports. Ideally, the records of expenditures for wages, port dues, pilotage fees, tug hire, and the like, should be in the home office no later than one working day after the vessel's departure. The practice is somewhat at variance from the ideal, but every effort is made to transmit these data to the analysts as quickly

as possible. Ships are under orders promptly to provide detailed information concerning the time used to work cargoes, reasons for any delays encountered, the effects of weather experienced while in port, and the conventional operation reports of every completed voyage. Evaluation of the reports must be accomplished promptly if they are to have any value, and cognizant supervisors must be made aware of the findings in every case, whether routine or exceptional.

It must be emphasized that, from the viewpoint of the oil companies, tankers are as vital to the industry as are the pumps in the oil fields and the distillation units at the refinery. Failure of a tanker to deliver a cargo can range from an annoyance to the cause of a major shutdown. Until the advent of earth-orbiting communication satellites, the only links between shore staff and the ships at sea were radiotelegraph and radiotelephone. With the new technology, it is possible for written messages ("telex") to be transmitted instantaneously and inexpensively from ship to shore and vice versa. The advantages are obvious, especially in the establishment of permanent records.

There are additional benefits from the telex system. The ship can send detailed information concerning repair work to be accomplished at the next port of call, and can set forth the exact spare or replacement parts needed and the specialized technicians required to do the work. Payrolls may be prepared aboard ship and sent by telex a matter of a few hours before the ship's arrival in port, permitting full payment of crew wages. The list of additional functions that the new system can perform, especially when connected to a small computer on shipboard, grows almost daily as both managers ashore and seafarers afloat become accustomed to the closer relationship that has been made possible.

A most important feature of the telex is the ability it confers upon the ship's master to report on operating conditions that affect the efficiency of the ship. If the ship has been delayed in berthing because of port congestion, and demurrage is payable, the sooner the chartering staff is aware of this, the sooner steps can be initiated to collect (or pay) what is owed. If, for any reason, the tanker is unable to perform to normal standards, a written report of the event can be sent immediately to the cognizant authority, and detailed instructions can be issued as to the deployment of the ship

in these unexpected circumstances. The existence of documentary records not only facilitates discussions between concerned parties but also establishes a sound basis for evaluating at-sea and in-port performance. Increasing numbers of tanker operators, both oil companies and independents, are installing these satellite communication systems as essential tools of management.

There is no substitute for personal contact between the shore staff and seagoing personnel. When tankers return frequently to the home port, this contact can be maintained without difficulty. If the fleet is large, the opportunity is decreased significantly for the fleet manager to know personally the men aboard his ships. When the tankers are dispersed worldwide, the problem becomes acute. Of the possible alternatives to losing contact, the device most favored by oil company management has been the formation of subsidiary companies to which are assigned responsibility for a small number of vessels operating in a somewhat restricted geographical area.

For example, Exxon Corporation in 1984 owned nineteen American flag tankers used to transport petroleum in domestic coastwise service. Subsidiaries in Denmark, England, France, Germany, Italy, Norway, and other countries owned and operated tankers under their respective national registries. Although all these ships flew the same house-flag, management was delegated to comparatively small organizations. Worldwide fleet policy was set by the parent company, but details of operation of the ships were left to the staffs of the subsidiaries. In this manner, the burden of operation was divided, and personal touch with the seamen was more likely to be preserved.

No matter what the range of managerial responsibility, all tanker owners strive to reduce nonrevenue producing time in port. Unavoidable delays resulting from weather are accepted for what they are. The concern is over deficiencies in pumping capabilities by shore installations, mechanical failures and personnel errors aboard ship, problems of heating heavy oil to hasten its flow through the piping systems, lack of proper berthing spaces, and similar controllable impediments to maximum performance. One major oil company, in a successful effort to improve overall employment, fixed certain standards, depending upon the category of petroleum carried, to be met by the various types of tankers in its

fleet. Unexcused failure to meet these standards resulted in censure of the responsible supervisors afloat and ashore. Concurrent with attempts of this nature to achieve optimum results from existing practices, management must be alert to find new or improved ways to work the ships when they come into port. As costs of owning and chartering ships have increased, there has been a corresponding rise in the pressure placed upon supervisors to better previously accepted methods and to exercise greater ingenuity and imagination in solving problems.

Gauging exactly the number of barrels of petroleum cargo pumped into or out of a ship has become an important aspect of tanker management. In the normal course of operation, a 75,000 tons deadweight tanker may have one or two thousand barrels of oil remaining in the tanks after discharge. This residue clings to bulkheads and tank bottoms; it does not flow to pump intakes. Because the value of the oil is now so great, consignees demand full delivery of what was loaded, or a compensatory adjustment. It is critical, therefore, that the number of barrels loaded be determined with precision, and the outturn be measured equally accurately. This is accomplished by hiring specialists to monitor both loading and unloading by use of an assortment of equipment designed for these purposes.

Modern tankers, especially those built since 1973, are equipped with very efficient stripping pumps and extra pipelines intended to suck up the maximum quantity of cargo oil. It is when the pumping operation stops that the specialists begin their work of determining the amount of oil that remains in the tanks. The ship owner or operator has every reason to demand that the specialists perform competently, that their instruments be calibrated by qualified testing laboratories, and that reports be clear and unquestionable. A major responsibility for overseeing this phase of oil transportation has been assigned to both shipboard and shoreside supervisors.

Tanker employment is influenced markedly by the characteristics of the cargoes the ships transport. Few tankers (excluding the combination and specialized carriers) participate in anything except the one-way movement of a given grade of petroleum. Ships assigned to lift gasoline, kerosene, and other clean products are restricted to those cargoes which have very nearly the same characteristics and never are switched casually to the black oil trade. The

reason is easy to discover: it takes much effort, time, and money to convert a ship from carrying black oils to the condition requisite to accept clean products. Prudent management can effect the change economically and efficiently provided sufficient time is allowed to make a number of voyages, each one carrying progressively lighter grades of oil. Once qualified to work clean cargoes, the ship is retained in that activity as long as possible.

To prepare a black tanker for an immediate shift to clean service is a very expensive, time-consuming, labor intensive operation. Upwards of two weeks may be required for the extensive washing (which may involve the use of chemical solvents) of tank surfaces, and the removal and proper disposal of the sludge and slops produced. When the ship eventually is made ready to accept gasoline, she still may not be able to meet the specifications for carrying exotic liquids like jet fuel. To be acceptable for that type of cargo may necessitate several successive voyages with gasoline cargoes, each followed by special cleaning action, before the tanks are given approval for these sensitive cargoes.

Many tankers today have the cargo tank surfaces coated with special nonferrous compounds that facilitate cleaning and also neutralize reactions between the metal of the hull and the cargo. If the ship carrying black cargoes has been treated in this manner, the shift from black to clean is made less difficult. In these coated ships, the use of carefully formulated chemical solvents may be appropriate as a means of expediting the cleaning process.

Tankers have become so costly, both to acquire and to operate, that good management dictates that all nonproductive time be eliminated. One aspect of this effort may be observed in the industry-wide standards adopted to control the length of time allowed to charterers for loading and discharging vessels. Discharge time depends upon the tanker's pumps, and significant improvement in performance can be achieved only by construction of new ships with pumping systems of greater capacity. For example, the average pumping capacity of the prewar handy-sized tankers was about 200 tons per hour. The T-2 tanker was designed with larger-capacity pumps, and handled approximately 700 tons per hour. Present-day medium-sized tankers are able to discharge about 1,500 tons an hour, while the giant tankers, especially those over 225,000 tons deadweight, are moving a minimum of 4,000 tons an hour. Actual

discharge time depends upon the grade of cargo carried, the available shore facilities, and the ambient temperatures: cold weather slows the flow of oil and imposes heavy burdens upon the ship's pumps and the booster pumps along the pipeline.

The single greatest cause of loss of time in port is the delay in placing the tanker alongside its berth. If vessel schedules overlap and two or more ships, earlier expected to arrive in orderly sequence and to use the available docking space in turn, converge upon the port at one time, congestion ("bunching") occurs, and inevitably there are delays in operation.

Bunching results from an accumulation of minor, unanticipated troubles afflicting ships in loading, sailing over thousands of miles of ocean, and discharging. The generic term applied to these difficulties is "slip." It is caused by late arrivals, mechanical deficiencies, and many other failures to maintain the established schedule, regardless of whether they are within or beyond the control of managerial personnel. Despite a high degree of vessel maintenance, efficient personnel, careful management, and watching, cajoling, checking, praising, criticizing, and the best of intentions, vessels do not always keep to schedules and therefore present the operator with a substantial amount of slip. The contributing factors vary from port to port, company to company, cargo to cargo, season to season, and only constant attention by operating management keeps slip from becoming more of a grave problem than it is already.

The complexity of managing a fleet of world-circling tankers was brought into appropriate perspective in 1952 by a Norwegian scholar who analyzed tanker operation in that year. Except that there are more tankers, and some of the routes are longer, the patterns delineated by the study remain substantially unchanged. In 1952, the statistics showed that 2,500 crude oil carriers recorded 1,028,177,000,000 ton-miles and transported 325,106,000 tons of cargo. There were 290 important routes that fanned out from 18 exporting areas to 43 receiving regions. About 45 percent of the tanker traffic was assigned to haul oil from Arabian Gulf and Eastern Mediterranean oil fields to all parts of Europe. These passages were almost twice as long as the average worldwide distance of 3,163 miles. Only 53.7 percent of the time spent at sea was devoted to carrying oil; the remaining 46.3 percent was consumed in ballast trips.

A somewhat more specialized study of the employment of tankers of 200,000 tons deadweight or larger was conducted in 1970 by an English ship brokerage firm which analyzed a total of 107 voyages. Only 33 of these trips were between one loading port and one discharging port. Twenty-five sailings had involved loading at two ports and discharging at two ports, inevitably requiring long port time. Had it been possible to make all 107 voyages on a one-port-to-one-port basis, the saving to operators in the cost of port time alone would have amounted to £18,400,000.¹⁶

A combination of unfavorable events occurred in the years after 1973. Fuel oil prices were escalated suddenly in that year and have been maintained at high levels since then. For a variety of reasons, a surplus of petroleum developed around the world about 1977, with almost catastrophic effects upon the tanker industry. Charter hire rates dropped precipitately at the same time that ship operating expenses rose. To offset these developments, many owners with ships committed to voyage charters, and time charterers of tankers, directed masters and chief engineers to cruise their ships at the speed which would result in optimum mileage per barrel. While the length of passage was increased, ship time was slightly less costly than bunker fuel. "Slow steaming" became a pattern of operation, even though it was admittedly not the most efficient employment of an expensive resource. Diesel-powered tankers gained noticeably in popularity with charterers because they used less fuel than did the ships equipped with steam turbines. At the end of 1981, slow steaming also was practiced to avoid long waiting periods in the oil loading ports of the Middle East. Owners considered it was economically justifiable to take longer on the ballast passage if time at anchor, always unproductive, could be shortened.

Slow steaming was a worldwide practice in 1980 and 1981 when the reduction in demand for petroleum created the classic situation in the tanker chartering market of many ships competing for the few cargoes available, thereby driving charter rates down to non-compensatory levels. To avoid the great expense of laying up their ships, many owners accepted these unprofitable charters. The result was that very large crude carriers were loaded in the Arabian Gulf and ordered to steam at between six and ten knots throughout their passage to the United States. Charterers used the big ships for two purposes: they transported crude petroleum, and simultaneously provided much needed storage space for the surplus oil.



The 285,050-tons deadweight, 15.25-knot turbine-driven *Arietta Livanos* (built 1972) was 330.7 meters long, 51.8 meters wide, and had a draft of 28.4 meters. Photo: McAllister Bros.

There are no ports in the United States with sufficient depth of water to accommodate tankers of greater size than about 225,000 tons deadweight. To obtain the economy of scale from the huge ships, the oil producers arranged for smaller ships (up to 75,000 tons deadweight) to meet the big ships about sixty miles offshore, and to transship the cargo into the smaller, shallower-draft vessels. A complete technique for this operation was developed, and light-
erage of petroleum cargoes became standard practice.¹⁷



The 263,549-tons deadweight *Texaco Japan* (built 1975) was 377 meters long, 53.6 meters wide, with a draft of 20.4 meters. Photo: Foxboro Company.

To provide a more economical and more permanent method of using the very large crude carriers, a huge buoy was designed to serve as the mooring point for the ship as well as the offshore terminal for a submarine pipeline to shore facilities. The only one in the United States was put into operation in June 1981. It is emplaced about eighteen miles off the Louisiana coast, in water at least one hundred feet deep, and is capable of handling over one million barrels of petroleum per day. The tanker is moored by a cable over the bow ("single point mooring"), which permits her always to face into the prevailing wind and seas. Her cargo is pumped through a floating hose to the buoy, and there it flows into the pipeline. A number of these offshore moorings exists worldwide. They are satisfactory devices for delivery by the largest tankers afloat to areas where coastal water depths are too shallow to permit their entry into the harbors.

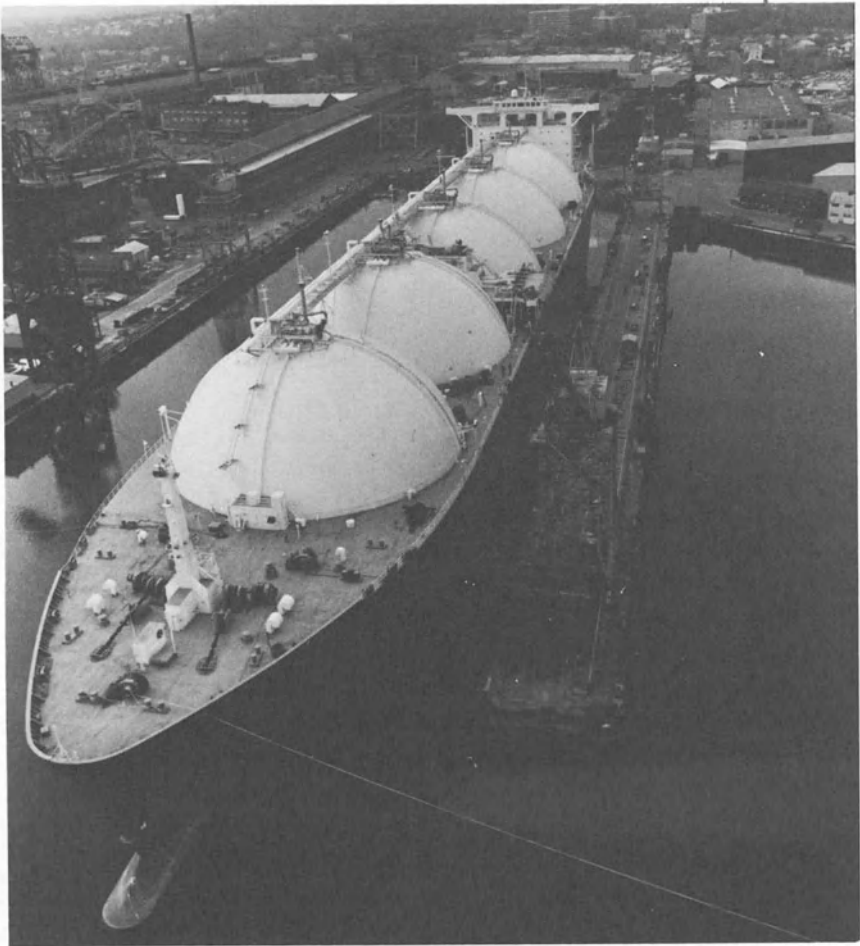
Tankers of 200,000 tons deadweight capacity and larger have become numerous in all fleets because they provide transportation

at a lower cost per barrel of petroleum delivered. They are less expensive to build; crews are only very slightly larger than for ships lifting 35,000 tons; fuel consumption is much more favorable, per ton of cargo, than in the smaller ships; and fewer hulls are needed to transport a given tonnage of petroleum. A 275,000-ton tanker delivers 28 barrels of oil for each barrel of fuel oil burned. At a uniform charter rate, a 60,000-ton tanker would earn £500 a day and have offsetting operating costs of £460. A ship of 100,000 tons deadweight would earn £1,120, with costs of £540. The 200,000-ton tanker would bring revenue of £3,200 per day, but her expenses would total £660.¹⁸

In actual operation, the very large crude carriers have encountered significant problems. The great draft of these ships, ranging from 18.29 meters for a 200,000 tonner to the 27.73 meters of the 569,783-ton *Seawise Giant*, restricts them to trade between a few ports in the world. Ships drawing over 22.80 meters, when sailing between the Arabian Gulf and Japan, must be routed via the Straits of Lombok rather than the thousand-mile shorter passage through the shallower Straits of Malacca. Dry docks capable of accommodating these very large bottoms are being built in many strategic locations, but still are not ubiquitous. Tank-cleaning is more hazardous than in smaller ships because of the vast size of the cargo areas and the augmented possibilities of generating static electricity that could cause an explosion.

Closely related in many respects to the movement of petroleum and its derivatives is the carriage of liquefied natural and petroleum gas. Begun on an experimental basis in 1947, it was not until 1964 that routine transportation of these liquefied gases was available. Many of the principles of management enunciated earlier in this chapter are applicable to the management of these specialized tankers. Certain aspects of operation understandably receive greater emphasis, especially those related to safety, ship handling, and the maintenance of equipment. Unceasing attention must be given to compliance with local rules governing the entry of these ships into busy ports, their berthing, and the techniques of working cargo. Adjustments to ship schedules to conform to these regulations are normal in this service.

A large fleet of liquefied gas carriers was built during the years between 1975 and 1980. Representative of these carriers was the



The *Lake Charles* (built 1980) carried 125,000 cubic meters of liquefied natural gas in her five spherical tanks. Photo: Eckstein, courtesy Moore-McCormack Resources.

LNG Capricorn, one of eight vessels constructed by General Dynamics at Quincy, Massachusetts. A characteristic of the cargo is that the gas must be cooled to -160° Celsius (-265° Fahrenheit) before it becomes liquid. To accommodate this cargo, five tanks, each of 25,000 cubic meters capacity, strengthened and insulated to very exacting specifications, were installed in the hull. The ship was 285 meters (936 feet) long, 44 meters (144 feet) in beam, and on a draft

of 11 meters (36 feet) had a deadweight of 64,620 tons. Turbines of 43,000 shaft horsepower gave her a cruising speed of 20.4 knots.

Whether the ships carry the relatively exotic liquefied natural gas or the more mundane liquefied petroleum gas of which butane is typical, the goal of management must be the safe completion of every voyage. Liquefied natural gas is delivered to public utilities with large storage tanks, while liquefied petroleum gas is distributed to local vendors with limited storage facilities. Consequently, the ships which transport "butane" are of much smaller size. Even granting that butane at -51° Celsius presents less of a problem in handling than liquefied natural gas at -160° Celsius, the burdens imposed upon management to assure that every voyage will be completed without incident are almost identical. It is a tribute to the skill and vigilance of the ship operators that hundreds of voyages are made each year, and millions of gallons of these exotic liquids are delivered in what appears superficially to be a completely routine activity. Only the constant exercise of the best supervisory techniques, coupled with expert ship handling and well-developed loading and delivery techniques, can produce these results. Those who are responsible for this transportation take understandable pride in the fact that they have achieved an unparalleled record of safety and customer satisfaction.

On January 1, 1984, 47 liquefied natural gas carriers and 448 liquefied petroleum gas transporters were in existence. That the stringent regulations concerning mandatory safety measures and precautions in navigation have been worthwhile is borne out by the fact that this large fleet, over a 17-year period from 1964 through 1981, incurred only 11 minor accidents, none involving escape of the gas cargo or a fire.

Liquid chemicals have been transported in bulk across the seas only during the period beginning about 1950. At that time, shippers of large quantities of animal fats, tallow, and vegetable oils who were in quest of lower transportation charges began to charter oil tankers. As the advantages of bulk carriage of nonpetroleum commodities became apparent, additional liquids, including a variety of chemicals, were offered. As could have been expected, it was found that some of the chemicals possessed characteristics which demanded more sophisticated techniques for bulk transportation than the standard oil tanker could offer. A number of tankers were



This floating dry dock, in Galveston, Texas, can lift a ship weighing 40,000 tons. Photo: Todd Shipyards Corp.

modified to the extent necessary to meet the new requirements. Multiple piping and pumping systems were installed. Cargo tanks were coated to present a smooth, uniform surface which would resist corrosion and also prevent contamination of the cargoes.

While a number of chemicals are no more dangerous to transport in bulk than is liquefied petroleum, other chemicals are genuine "high risk" cargoes, and present management, both ashore and afloat, with peculiar problems, of which the following have special significance:

1. The danger of fire may be enhanced by the nature and characteristics of the chemical. Carbon disulphide, for example, has a low flash point and a wide range of flammability; phosphorus ignites spontaneously when exposed to air; adiponitrile gives off poison gases when it burns.

2. Vapors from chemicals may irritate personnel, thereby necessitating that vents exhausting to the atmosphere be elevated above any crew passageways. The exact height varies with the

chemical(s) being transported. Additionally, vents must be at least forty feet distant from any opening to the crew's living spaces.

3. Vapors from, or contact with, certain chemicals cause health risks. Protective clothing, including face masks, must be available for the ship's company, and therefore have to be part of the vessel's original outfit.

4. Some chemicals react when mixed with other chemicals. Very definite regulations have been promulgated concerning the compatibility of these liquids and must be obeyed precisely. The operator of the ship and her master must be fully informed whenever a mixed load is to be carried. Appropriate steps then must be taken by those in charge of loading and by the master to assure that parcels loaded in adjacent tanks separated by a single bulkhead do not cause dangerous results if a leak should develop in that bulkhead.

5. Certain chemicals have the propensity to react violently when contaminated by water. The master must be notified of this characteristic of any chemicals his ship may be carrying, and every precaution must be exercised to assure that no water of any sort be permitted to reach the tank(s) containing these chemicals. This "family" of chemicals should be transported only in tankers that have double bottoms and non-water ballast tanks adjacent to the spaces in which these liquids are contained.

6. Many chemicals have the capability of polluting human and marine life if they are spilled. Heavy penalties may be incurred if a spill occurs. The most stringent precautions, therefore, must be observed to prevent any accidental discharge.

In time, the offerings of chemicals to be shipped in bulk increased sufficiently to justify building ships which had been designed expressly to transport a wide variety of these liquids, including the exotic as well as the extremely dangerous substances. A primary requirement was that a tanker of this type have a double bottom for the entire length of the hull. To isolate the different chemical cargoes one from another, inner skins in at least some of the tank areas had to be installed. Pumping and piping systems must be planned not only to prevent contamination of the cargo, but also to minimize the possibility that violent reactions could occur from mixing noncompatible chemicals. Although not an

integral part of the safety features of the chemical tanker, the multiple piping and pumping arrangements do contribute significantly to the efficiency of ship operations in the rapid loading and unloading of the assorted cargoes.¹⁹

All these matters were studied by the International Maritime Organization before it promulgated its "Code for Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk." Concurrently this organization prepared a list of hazardous chemicals for which specialized care was mandatory, as well as designating which chemical liquids were not hazardous. In agreement with a proposal by delegates from the United States, the organization directed that chemicals be carried only in tankers which belonged to one of these categories:

Type I. The most hazardous chemicals shall be limited to ships designed with double bottoms in way of the cargo tanks. The cargo tanks shall not be closer to the side of the ship than a distance equal to one-fifth of the vessel's beam.

Type II. Moderately dangerous chemicals shall be carried only in vessels with double bottoms and side cofferdams at least 760 millimeters inboard from the exterior plating of the hull.

Type III. Nonhazardous chemicals may be moved in ordinary tankers provided that precautions are taken to prevent loss of the ship when carrying high density cargoes. Normally this would result in selected tanks being left empty to distribute the weight evenly throughout the ship's structure.²⁰

Ships dedicated to the transportation of "parcels" (i. e., less than shipload lots) of liquid chemicals have to be designed in full awareness of the problems peculiar to this trade. No single vessel can handle every type of liquid chemical; a recent count showed over 1,300 different chemical products which were potential candidates for bulk transportation across the seas. Many of the chemicals now moving by ship have characteristics which must be considered in the selection of the steel to be made into bulkheads for the cargo tanks. Progressive parcel tanker operators seem agreed that most chemicals can be stowed satisfactorily in stainless steel tanks; a qualified number can be loaded in tanks fabricated of mild steel and finished with layers of organic and zinc coatings.²¹



The chemical-carrying tanker *Stolt Avance* has thirty-five cargo tanks and four deck-mounted cylindrical tanks for small lots. Photo: Stolt-Nielsen, Inc.

Capitalizing upon twenty-five years of experience in the operation of specialized chemical tankers, the Stolt-Nielsen Group (the largest operator of chemical carrying tankers) recently accepted delivery of the first of five ships constructed for worldwide use in the parcel trade. This prototype of the new units was the *Stolt Sapphire*, built in Korea in 1985, which met the highest standards of the Norwegian ship classification society, Det Norske Veritas, for "tankers for chemicals and tankers for oil." Other rules and regulations influencing the design and construction of the ship were: International Marine Office Chemical Bulk Code (Type I, II, and III), including 10th amendment; Safety of Life at Sea Convention (1981), including the 1978 amendment; Maritime Pollution Convention, 1973-1978, Annex I and II; United States Coast Guard rules for foreign-flag ships; Panama Canal rules; and Suez Canal rules.

Stainless steel was used for the bulkheads of the thirty-two center tanks on each of these ships. Stainless steel was three to five times more costly than mild steel, but when the expense of frequent recoating of the mild steel surfaces was computed, the economic justification for the more expensive metal was self-evident. Those tanks which were finished in mild steel were covered with organic or zinc coatings. These tanks would be used to transport only chemicals which would not react with the bulkhead surfaces.

To meet the needs of shippers of either small parcels of a specified chemical, or of highly specialized chemicals, a total of four cylindrical tanks was mounted on deck. With a capacity of about 250 tons, considerable flexibility was afforded the agents seeking cargo for these vessels.

These five ships were designed for employment in trading worldwide. Maximum freedom in vessel routings demanded that they be able to transit the Panama Canal and to enter most ports. As a result, the ships were 176 meters (580 feet) long, 32.25 meters (105 feet 9 inches) wide, and had a draft of 10.5 meters (34 feet 5 inches). The corresponding deadweight was 36,500 tons. Cruising at 15 knots, the ship required her diesel engine to produce 12,500 brake horsepower. Fuel consumption at cruising speed made possible operation for 48.7 days without having to refuel.

To assure comfort and to improve the morale of the crew, each member of the thirty-person complement had an individual cabin.²²

The American Shipping Subsidy System

INTERNATIONAL shipping involves not only the normal rivalry among carriers for the available cargo, but also the more subtle competition between nations with disparate standards of living. Often the pattern of trade between two countries reflects the significant differences in their economic status. Nation A, for instance, has a well-developed industrial economy with a high standard of living for its citizens. It exports manufactured goods to all parts of the world, and imports great quantities of raw materials. Nation B, in contrast, has an agricultural economy, and depends upon the products of its fields to provide the funds to pay for the manufactured items it requires. Its standard of living, compared to that of Nation A, is significantly lower, as evidenced by the differences in wages and prices prevailing in that country.

For many sound reasons, Nation A has encouraged its citizens to own and operate ships to serve its foreign trade, here typified by the interchange of goods between A and B. The costs of building, maintaining, and running ships under its registry are well above those of Nation B, which also has a merchant fleet plying from B to A and back again. If the principles of pure competition applied perfectly to this trade, the level of freight rates would be fixed by what the ships of Nation B must earn to assure an adequate profit. To remain in business and to obtain any of the cargo, the owner of Nation A's fleet cannot charge more than do their rivals.

There are valid reasons why each of the trading partners should have merchant fleets. One is that money paid to national-flag operators is retained almost entirely in the local economy, in much the same way that the proceeds of sales of domestic goods to citizens stay inside national boundaries. Another reason is that the earnings of the fleet from freights paid by foreign shippers are equivalent to exports of goods; in fact, they often are referred to as "invisible exports." Sweden, for instance, considers shipping as a major ex-

port industry which brings in about eight percent of its revenue from external sources.¹ *Per contra*, money paid to carriers of other nationalities affects negatively the country's trade balance.

It can be argued that Nation B has greater justification for a national-flag fleet than does Nation A. With its limited ability to earn foreign exchange, it must find ways to augment the revenue from sale of its agricultural produce. A merchant marine participating in world trade offers a means to increase national income. The money so realized is essential to help pay for the imports needed by Nation B. In particular, the freight derived from carrying goods bought from Nation A reduces the net disbursement from Nation B's treasury, while the freight collected from Nation A for transporting its imports from Nation B represents positive credits. Carried to its logical conclusion, this argument would seem to justify the elimination of Nation A's merchant fleet in order to strengthen that country's export sales. The counter to this conclusion is that Nation A experiences the same difficulties with its international balance of payments, and therefore its invisible exports are vital to its economy.

Another purpose of national ownership of a merchant fleet is to provide service on those trade routes which are of greatest economic importance to the nation. By their presence on a particular route, the ships of Nation A attract vessels from other countries which are seeking gainful employment. It is a fact of shipping that no trade route is so insignificant that there is not at least one competitor of the principal carrier. Even a small fleet from Nation A will attract ships from other nations, thereby providing adequate transportation to meet the needs of importer and exporter.

Finally, a corollary to the ability of ships to earn foreign exchange with which to pay for imports is the employment of vessels in the "cross trades." That is to say, ships of Nation A may sail regularly between Nations M and P, rarely returning to a port in Nation A. Money earned in this manner is a direct and positive addition to the financial strength of the shipowning nation.

These arguments touch upon the basic rationale for national-flag merchant fleets. They gloss over the problems of disparity of cost and difficulties of competition which arise out of that inequality. This has been suggested earlier in this chapter in the commentary upon differences in national living standards.

A comparison of the costs of operating ships under the regis-

tries of these two countries shows that Nation B, with its lower standard of living, enjoys a significant advantage over Nation A. The freight rates charged by the shipowners of Nation B reflect this; what produces a satisfactory profit for them is less than the minimum required for survival of the ships flying the flag of Nation A. It is a harsh fact of shipping, set forth repeatedly in this volume, that cost of transportation influences the shipper's decision as to which carrier will be favored with the cargo. If service and schedule are substantially equal, it is almost certain that the lower cost carrier, regardless of nationality, will receive the business.

The dilemma facing the owner of ships registered under the flag of high cost Nation A is, to put it very simply, "Where can I obtain the money I need to bridge the gap between my costs and the revenue I can earn from carrying cargo at the rates set by my lower cost competitor?"

There is only so much that efficient and economical management can do to reduce costs. Once these possibilities have been exhausted, the most obvious source of the indispensable supplementary income is the national treasury. Commercial interests will not pay higher freight rates simply to maintain a national-flag fleet. There is no patriotism among shippers anywhere; they give their business universally to the carrier who provides the best service at the lowest cost, without regard to the flag under which the ship sails. Philanthropists show no interest in assisting a commercial enterprise in which some corporation is seeking to make a profit. With no other avenue open, maritime activities around the world have turned to their governments for financial help.

In 1978, the Maritime Administration of the United States Department of Commerce conducted a survey of the types of aid available to shipping and shipbuilding companies in fifty-eight maritime nations, and learned that the principal forms of direct aid offered to their merchant fleets were these:

- Operating subsidies for ships
- Subsidies applied to ship construction
- Trade-in allowances for ships
- Low interest loans
- Subsidies on payments of interest on loans
- Government guarantees of loans to shipowners

- Accelerated depreciation of assets for purposes of taxation
- Tax free reserve funds
- Duty free imports of materials needed for ship construction
- Cargo preference schemes
- Restrictions on foreign-flag shipping in domestic trade (cabotage protection)
- Maritime research programs²

No single nation offered all the forms of assistance listed above, nor was any standardized pattern disclosed which emphasized one or another of these features. Instead, each nation adopted whatever schemes seemed most likely to assure the survival of their merchant fleets. In many instances, the survey brought out that a variety of indirect aid programs, the exact financial value of which was impossible to measure accurately, also had been established in one or more of the maritime nations studied. Among the less obvious forms of assistance were:

- Schools for merchant seamen
- Hospital and medical care for merchant seamen
- Social security payments to families of merchant seamen
- Laws directing that ships engaged in both the domestic and foreign trades of a nation be built in that nation's shipyards
- Laws dictating that domestically produced materials be used in construction and repair of national-flag ships
- Laws requiring that purchases of food, stores, and vessel supplies be confined to domestic markets.³

Of the maritime nations, the United States has the most comprehensive, most widely publicized, and most openly administered subsidy program. It evolved over generations of experience, starting in 1847 when the postmaster-general was authorized to enter into contracts with steamship operators to transport mail destined to foreign countries. The underlying purpose of this legislation was to encourage American investment in, and operation of, steam-powered vessels in direct competition with foreign-flag carriers. In 1840, with support from the British government, Samuel Cunard had inaugurated regular passenger and mail service between England and the United States. To compete with Cunard, some

method had to be devised to offset the higher costs of building ships in the United States and operating them under American registry. The mail contract appeared to be a satisfactory method. It called for lump sum payments for each voyage without regard to the amount of mail actually carried. A major deficiency in the law was the absence of a stipulation that the contractor set aside adequate funds to construct replacement tonnage.

In 1817, the Congress enacted a law which restricted the waterborne trade between the states to ships built and owned in the United States, and manned by citizens of the United States. Because California was admitted to the United States in 1850, the well-publicized Gold Rush voyages of the clipper ships from Atlantic coast ports around Cape Horn to San Francisco were legally interstate commerce, and therefore limited to ships meeting the criteria set forth in the 1817 law. Subsequently, an extensive coastwise trade, especially along the Atlantic and Gulf coasts, developed under this protective legislation and flourished until the advent of World War II. When the Panama Canal was opened to commercial traffic on August 15, 1914, it made possible an active intercoastal trade which prospered for the next quarter century. Aside from protecting against foreign competition the ships engaged in these trades, however, there was no government assistance.

This cabotage restriction described above applies today in the coastwise and intercoastal trade, and in the movement of cargo and passengers between the forty-eight contiguous states and Alaska, Hawaii, and the offshore commonwealths and possessions. An exception exists in the waterborne commerce between the mainland and the Virgin Islands, which were purchased from Denmark in 1917. Vessels of foreign registry are permitted to transport goods on this route despite the fact that both origin and terminus are on United States soil.

To assure an additional source of revenue to American-flag ships, which were experiencing difficulty in meeting foreign competition, the Congress enacted a law in 1904 which required that all cargo belonging to the United States military, or consigned to its overseas installations, be carried in ships either owned by the government or registered in the United States. In 1954, the "Fifty-Fifty Law" was passed. This legislation directed that at least one half of all cargoes financed by the United States government and destined to

foreign nations (primarily economic aid to less developed nations, as well as shipments of agricultural products to relieve hunger or to initiate self-help programs) be transported in ships registered in and operated under the laws of the United States for not less than three years.⁴

Direct subsidy of United States ships was not adopted until the Merchant Marine Act, 1936, was signed into law by President Franklin D. Roosevelt. This legislation terminated the ninety-year-old mail contract program, and for the first time established a systematic method by which government aid could be granted to the merchant fleet. Until the act was amended in 1970, the provisions of the law applied only to ships engaged in liner services. Dry bulk carriers and vessels working in the tramp trades had been excluded, largely because in 1936 the fleet did not include any ships of this type which were engaged in the foreign trade.

As part of the Reagan administration's policy to reduce the role of government in commercial affairs, the provisions of the Merchant Marine Act, 1936, relating to ship construction subsidy have been suspended since October 1, 1983. While no new operating differential subsidy contracts have been approved, none of the contracts in effect has been altered or cancelled. Despite the practical abandonment of that portion of the law which was designed to support an active American shipbuilding industry, and the lack of any encouragement to new organizations to seek operating differential subsidies, the law retains its vitality and can become once again an important factor in American shipping at any time.

Of all the methods by which governments worldwide make funds available to assist in the support of their national flag merchant fleets, the United States law is probably the most complete and most openly administered legislation of this type. It is worthy of careful study. The analysis which follows is directed only toward those sections establishing the construction and operating differential subsidy system. The analysis is intended to provide information; it does not pass judgment on either the fundamental premise of government assistance to shipping or on the manner in which the purposes of the Merchant Marine Act, 1936, have been carried out.

Section 101 of the law contains the declaration of national policy concerning the United States merchant marine. Although this declaration was hailed in 1936 as the "Magna Carta" of the merchant



Last of a long line of ships built under subsidy, the *Falcon Champion* was launched by Bath Iron Works on September 10, 1983. Photo: Bath Iron Works.

marine, in reality it was a repetition, with only minor modifications, of the statement of purpose contained in the 1920 Shipping Act, and later “confirmed” by the 1928 Shipping Act. In its entirety, this declaration of policy, which has not been amended in the decades since adoption, reads:

It is necessary for the national defense and development of its foreign and domestic commerce that the United States shall have a merchant marine (a) sufficient to carry its domestic water-borne commerce and a substantial portion of the water-borne export and import foreign commerce of the United States and to provide shipping service essential for maintaining the flow of such domestic and foreign water-borne commerce at all times, (b) capable of serving as a naval and military auxiliary in time of war or national emergency, (c) owned and operated under the United States flag by citizens of the United States insofar as may be practicable, (d) composed of the best-equipped,

safest, and most suitable types of vessels, constructed in the United States and manned with a trained and efficient citizen personnel, and (e) supplemented by efficient facilities for shipbuilding and ship repair. It is hereby declared to be the policy of the United States to foster the development and encourage the maintenance of such a merchant marine.

Proceeding logically, the act prescribes the basic responsibilities of the United States Maritime Commission, which was established to administer the law. An independent agency which reported directly to the President, the commission was assigned the dual role of promoting the merchant fleet and handling the details of subsidy agreements and also of regulating the waterborne commerce of the United States. In 1951, the independent status of the commission was terminated when the agency was placed under the Secretary of Commerce, and the regulatory functions were assigned to the independent Federal Maritime Board, of which the chairman was the Maritime Administrator. The agency was reorganized again in 1961, when the independent Federal Maritime Commission was set up for the sole purpose of regulating ocean transportation and enforcing the provisions of existing legislation (at that time, primarily the Shipping Act of 1916). The Maritime Administration remained in the Department of Commerce, and administered the subsidy programs and promoted the American merchant marine pursuant to the provisions of the 1936 act. In 1981, the Maritime Administration became a component of the Department of Transportation, and the Secretary of Transportation was named as the official spokesman on all matters relating to maritime policy. (In the following paragraphs, all references to the Secretary will be directed to the Secretary of Transportation.)

Pursuant to the requirements of Section 211 of the 1936 act, the Secretary is empowered to "investigate, determine, and keep current records of" certain types of activity:

(a) The ocean services, routes, and lines from ports in the United States, or in a Territory, district, or possession thereof, to foreign markets, which are, or may be, determined by the Secretary to be essential for the promotion, development, expansion, and maintenance of the foreign commerce of the United States, and in reaching his determination the Secretary shall consider and give due weight to the cost of maintaining each of such steamship lines, the probability

that any such line cannot be maintained except at a heavy loss disproportionate to the benefit accruing to foreign trade, the number of sailings and types of vessels that should be employed in such lines, and any other facts and conditions that a prudent businessman would consider when dealing with his own business, with the added consideration, however, of the intangible benefit the maintenance of any such line may afford to the foreign commerce of the United States, to the national defense, and to other national requirements;

(b) The bulk cargo carrying services that should, for the promotion, development, expansion, and maintenance of the foreign commerce of the United States and for the national defense or other national requirements be provided by United States-flag vessels whether or not operating on particular services, routes, or lines;

(c) The type, size, speed, method of propulsion, and other requirements of the vessels, including express-liner or super-liner vessels, which should be employed in such services or on such routes or lines, and the frequency and regularity of the sailings of such vessels, with a view to furnishing adequate, regular, certain, and permanent service, or which should be employed to provide the bulk cargo carrying services necessary to the promotion, maintenance, and expansion of the foreign commerce of the United States and its national defense or other national requirements whether or not such vessels operate on a particular service, route, or line;

(d) The relative cost of construction of comparable vessels in the United States and in foreign countries;

(e) The relative cost of marine insurance, maintenance, repairs, wages and subsistence of officers and crews, and all other items of expense, in the operation of comparable vessels under the laws, rules, and regulations of the United States and under those of the foreign countries whose vessels are substantial competitors of any such American vessel;

(f) The extent and character of the governmental aid and subsidies granted by foreign governments to their merchant marine;

(g) The number, location, and efficiency of the shipyards existing on the date of the enactment of this Act or thereafter built in the United States; . . .

(j) New designs, new methods of construction, and new types of equipment for vessels; the possibilities of promoting the carrying of American foreign trade in American vessels; and intercoastal and inland water transportation, including their relation to transportation by land and air.



A British tramp's bridge, *top*, is efficient but austere. The American cargo-liner's bridge, *bottom*, is both well-equipped and esthetically pleasing. Photo: Delta Lines.

Pursuant to the mandate of Section 212 (c), the Secretary is directed:

To collaborate with vessel owners and shipbuilders in developing plans for the economical construction of vessels and their propelling machinery, of most modern economical types, giving thorough consideration to all well-recognized means of propulsion and taking into account the benefits accruing from standardized production where practicable and desirable.

The authority so conferred was used initially to design the ships which served so well during World War II: the famous C-2, C-3, C-4, Liberty, and Victory dry-cargo ships and the T-2 tanker. After the war, the Maritime Commission's (later the Maritime Administration's) staff of naval architects designed and supervised the construction of the Mariner class, the world's first large high-speed cargo ships built on a mass production basis. In the years since 1960, the Administration has worked closely with designers of new ships to achieve significant advances in naval architecture.

To carry out the intent of the Declaration of Policy that ships of United States registry be manned by "trained and efficient citizen personnel," the United States Maritime Service was created as a "voluntary training organization for the training of citizens of the United States to serve as licensed and unlicensed personnel on American merchant vessels." This uniformed, federal service trained thousands of seamen during World War II. Section 216 (b) established the United States Merchant Marine Academy as the federal agency charged with educating citizens of the United States to serve as licensed officers of the national merchant marine.

CONSTRUCTION-DIFFERENTIAL SUBSIDY

During the colonial epoch of its national history, ships were built along the Atlantic Coast of what was to become the United States at prices substantially lower than those prevailing in the traditional shipbuilding centers of Western Europe. This economic advantage gradually diminished as stands of virgin oak were depleted and raw materials became more expensive. One of the factors influencing the enactment in 1847 of the ocean mail contract legislation was the disparity in American and foreign shipyard

prices. Unfortunately, there was no provision in this or subsequent laws to assure that aging ships would be replaced.

The Merchant Marine Act, 1936, embodied a bold new approach, continued and reinforced by the 1970 act, which systematized the building of ships under subsidy. In Title V are found the terms and conditions under which government funds will be made available to defray a portion of the cost of building ships in American yards. Section 501 (a) states that

Any proposed ship purchaser who is a citizen of the United States or any shipyard of the United States may make application to the Secretary for a construction-differential subsidy to aid in the construction of a new vessel to be used in the foreign commerce of the United States. No such application shall be approved by the Secretary unless he determines that (1) the plans and specifications call for a new vessel which will meet the requirements of the foreign commerce of the United States, will aid in the promotion and development of such commerce, and be suitable for use by the United States for national defense or military purposes in time of war or national emergency; (2) if the applicant is the proposed ship purchaser, the applicant possesses the ability, experience, financial resources, and other qualifications necessary for the operation and maintenance of the proposed new vessel, and (3) the granting of the aid applied for is reasonably calculated to carry out effectively the purposes and policy of this Act. The contract of sale, and the mortgage given to secure the payment of the unpaid balance of the purchase price shall not restrict the lawful or proper use or operation of the vessel except to the extent expressly required by law. The Secretary shall give preferred consideration to applications that will tend to reduce construction-differential subsidies and that propose the construction of ships of high transport capability and productivity.

Recognizing that from time to time it may be necessary or desirable to reconstruct or modify vessels used in the foreign trade of the United States, Section 501 (c) provides that any citizen of the United States may apply to the Secretary of Transportation for subsidy assistance. Pursuant to this authority, many ships were enlarged and adapted to carry containers following the worldwide shift to that mode of ocean transportation. The funds to defray the costs of this type of financial help are derived from the annual appropriations for ship construction.

The prospective purchaser of the ship (or the shipyard) must submit an application for construction subsidy, together with complete plans and specifications, to the Secretary. The plans are analyzed to ascertain that approved engineering and construction standards have been met. Concurrently, a determination must be made by the Secretary that the proposed ship will be satisfactory in all respects for the trade for which she is intended, and that she is competitive with the foreign-flag ships employed on that trade route. Assuming that the plans and specifications are approved by the Secretary of Transportation, they are forwarded, in accordance with the terms of Section 501 (b), to the Secretary of the Navy for his approval of the ship as a potential military auxiliary. The Navy's examination permits recommendations concerning the additional features which would make the ship more adaptable to service as a military auxiliary in time of war or national emergency. Typical of these "national defense features" would be increased shaft horsepower, enlarged generator capacity, greater lifting power for cargo booms and winches, limited refueling-at-sea capability, and provision for washdown after experiencing radioactive fallout. The applicant must approve the changes proposed by the Navy, as they affect the economic value of his ship. When the prospective owner has signified acceptance of the recommended modifications or additions, the Maritime Administration must determine which of these national defense features has commercial utility and which has only military value. Those items of strictly military value are paid for by the Maritime Administration, and are deducted from the cost of the ship when the calculations are initiated to establish the amount of the construction-differential subsidy.

This provision has not achieved the results envisioned by the framers of the legislation. Although the cost of installing the military features is paid by the government, no arrangement exists to subsidize the added expense of operation, maintenance, and repair of the unit installed at the request of the Navy. Because of this deficiency in the law, shipowners have been reluctant to accept the recommendations of the Navy, and have convinced the Maritime Administration of the validity of their complaints. Although national defense features have been built into all ships constructed under subsidy, they are minimal.

A theoretical example will illustrate the problem.

The operator of a ship stipulated that an electrical generator have a capacity of 100 kilowatts, but the Navy requested that this be increased to 500 kilowatts. The Maritime Administration paid for the difference between the price of the 100-kilowatt unit and the 500-kilowatt unit. In service, the operation of the generator proved to be inefficient and expensive because its capacity was greatly in excess of demand. There was more need for attention and repair, which added to the cost of operating the ship, but there was no assistance from government funds to defray these added expenses.

Once procedures relating to the national defense features have been concluded, and the Secretary of the Navy has certified that the ship will be acceptable as a military auxiliary, the shipbuilders enter the picture. The owner invites interested yards to submit proposals, and is obligated to accept the lowest responsive and responsible bid. The government, through the Maritime Administration, may pay the fair and reasonable differential between the cost of construction in the United States and building in a qualified foreign shipyard, but is limited by law to one-half the American price.

The Secretary of Transportation is directed by the law to estimate annually the cost to build in foreign shipyards each of the different types of ships included in the construction program for that year. In making these estimates, the Secretary uses American specifications. That is to say, the specifications on which the estimate is based are those called for by American practice, regardless of whether these specifications exceed those customarily followed by foreign builders. For instance, until very recently American shipowners ordered steam turbine propulsion plants, whereas foreign operators favored diesel motors. The comparison of costs between the two ships was a very complex matter, and inevitably ended up in a highly theoretical computation which the owners claimed did not give them the parity in costs which the act intended that they receive.

An example of how construction subsidy for a modern container ship ordered in 1979 would have been computed is presented:

American shipyard cost (lowest responsive and responsible bid received)	\$90,660,000
National defense features, of no commercial utility (to be paid by Maritime Administration)	850,000

Net cost of commercial ship	89,810,000
Estimated cost of building this type ship in a foreign yard	44,922,962
Dollar difference between United States and foreign construction costs	44,887,038
Foreign construction price as a percentage of United States construction price	50.02%
Construction-differential subsidy payable	49.98%

To assist prospective purchasers of ships to obtain money from conservative lending institutions, Title XI of the law authorizes the government to guarantee these loans, or to guarantee bonds issued by steamship companies for sale to the investing public. The fee for this guarantee on obligations incurred for construction or reconstruction of ships is fixed at a maximum of one-half of one percent per annum on the outstanding principal.

Vessels built under subsidy must be constructed in American shipyards, using American components insofar as practicable, and must remain under United States registry for twenty-five years or until paid for, whichever is the longer period. They must be operated (1) exclusively in the foreign trade of the United States, or (2) on a round-the-world voyage, or (3) from the Pacific coast of the United States to European ports on a round voyage which includes intercoastal ports of the United States, or (4) from the Atlantic coast of the United States to the Orient on a round voyage which includes intercoastal ports of the United States, or (5) in the foreign trade on a voyage including a call at Hawaii or an island possession or territory of the United States.

Section 506 provides that those owners of ships built under subsidy who operate these vessels in the coastwise or intercoastal service as a part of their foreign voyages shall return to the United States treasury a proportion of the construction differential. This proportion is based on the relation which earnings in the domestic (i.e., coastwise or intercoastal) trade bear to the total earnings of the vessel during the preceding operating year. This limitation was included by the Congress in the Merchant Marine Act, 1936, to protect American carriers serving the domestic routes exclusively, and was consistent with legislation dating back to 1817 restricting the waterborne commerce between the states to American-built and American-registered ships. Because they were free from com-

petition with lower cost foreign operators, there was no need to grant subsidies to ships in the interstate trade. It was appropriate, however, that these same ships be safeguarded from incursions by subsidized carriers whose costs were reduced to foreign levels.

The application of this theory may be observed by referring to the calculations incidental to granting a construction-differential subsidy. The government's portion was \$44,887,038, and the purchase price was \$44,922,962. In theory, both the government and the shipowner would set aside annually an amount equal to 4 percent of their respective contributions toward the total cost of the vessel. The government therefore would allocate \$1,795,481, while the owner would deposit \$1,796,918 to meet the amortization payments. The ship was assigned to the round-the-world trade, and in making her regular voyages carried intercoastal cargo. Her freight earnings in the domestic route came to \$750,000 for the year, while her total revenues aggregated \$7,500,000. The ratio of domestic to total earnings was one to ten. Under the terms of the law, the owner would have to deposit not only his normal amortization of \$1,796,918 but also would pay to the Department of Transportation an additional ten percent of that amount or a total of \$1,976,609. In other words, the amortization would be increased to the point where domestic operators suffered no harm from a less expensive ship being placed on their route.

As an incentive to owners of subsidized ships to replace them when they became economically obsolete even while mechanically still serviceable, the Secretary of Transportation was authorized to purchase old vessels when their owners contract to build new tonnage. These "traded-in" ships had to be at least twelve years old, employed in the foreign trade, and considered in need of replacement because they were obsolete or noncompetitive on the trade routes to which they were assigned. The value of these ships, when sold to the government, in the words of the law, "shall not exceed the cost to the owner . . . plus the actual cost previously expended thereon for reconditioning." A reasonable depreciation based upon a 25-year life expectancy was to be calculated in fixing the monetary value of the ship. The purchase price realized had to be applied by the seller to the construction cost of the new ship.

No appropriations have been made by the Congress for construction differential subsidy since October 1, 1983. The need to

replace obsolescent tonnage was satisfied by granting authority for two fiscal years to a limited number of steamship operators to build their ships abroad. Those owners who took advantage of this opportunity were required to make all financing arrangements through private channels. Permission also was given to buy existing ships from foreign owners or builders, with the stipulation that any work required to bring these vessels up to American standards would be divided equally between American and foreign shipyards. Despite many protests from the American shipbuilding industry, this modification of the intent of the Merchant Marine Act, 1936, was effective at least through September 1985.

Ships intended for operation in the domestic routes are not eligible for subsidy, but the mortgages given by the owners to lending institutions can be guaranteed by the Maritime Administration, pursuant to Title XI of the merchant marine law. The limit on any single ship is fixed at 87½ percent of the building price. Supported by this assurance, a number of tankers and bulkers has been built since 1970. Several ship operators have encountered overwhelming difficulties, and the Maritime Administration has had to pay off the creditors. In May 1985, for instance, the Phoenix Companies of Houston and New York defaulted on debt service payments of \$128,395,000, and the indentured trustee for the bondholders demanded and received full payment by the guarantors of the loan.⁵

OPERATING-DIFFERENTIAL SUBSIDY

Entirely separated from the program to assist in meeting the cost of building American ships in American yards is the system by which government financial aid is extended to ships registered in the United States and engaged in the nation's foreign trade. This separation was made necessary, if for no other reason, because of the complexity of the problem faced by the American operator. So long as the philosophy of government support is directed toward assuring that funds will be used only for designated purposes, the detailed specifications of the Merchant Marine Act, 1936, as amended by the act of 1970, are necessary. In the paragraphs that follow, an examination is made of some of the most significant principles by which American ship operating costs are subsidized.

Ships seeking to obtain financial assistance through subsidy contracts must be operated in the nation's foreign trade in an essential service. Deliberately, the definition of an "essential service" has not been set forth with finality; the intent of the Congress was to permit ships engaged in liner service as well as bulk carriers (primarily engaged in hauling dry commodities, but the movement of liquid bulk cargoes by tankers is not positively excluded) to obtain subsidies. The phrasing of the law permits assistance to tramp ships when they are operated in the foreign trade of the nation.

The details of the system are set forth in Title VI of the Merchant Marine Act, 1936, as amended and rewritten by the 1970 law. It is significant that the principles enunciated in the earlier legislation, which were unique and trailblazing, have not been overturned. Some of the details of administration have been changed, but the essential elements laid down in 1936 remain intact.

Any American citizen is eligible to apply for an operating-differential subsidy contract. In submitting an application, the citizen must show, in compliance with Section 601, that

(1) The operation of such vessel or vessels in an essential service is required to meet foreign-flag competition and to promote the foreign commerce of the United States . . . , and that such vessel or vessels were built in the United States . . . ;

(2) The applicant owns or leases, or can and will build or purchase or lease, a vessel or vessels of the size, type, speed, and number, and with the proper equipment required to enable him to operate in an essential service in such manner as may be necessary to meet competitive conditions, and to promote foreign commerce;

(3) The applicant possesses the ability, experience, financial resources, and other qualifications necessary to enable him [so] to conduct the proposed operations of the vessel or vessels as to meet competitive conditions and to promote foreign commerce;

(4) The granting of the aid applied for is necessary to place the proposed operations of the vessel or vessels on a parity with those of foreign competitors, and is reasonably calculated to carry out effectively the purposes and policies of this Act.

The contract between the government and the shipowner for an operating-differential subsidy may be for a period as long as twenty years. Before the contract is approved, the Maritime Subsidy Board must conduct an investigation, as required by Section

602, to establish the necessity for this form of assistance. The Board, by law, must find that a subsidy is needed to meet the competition of foreign-flag vessels in the essential service the applicant proposes to provide.

Section 603 (b) sets forth the terms of the operating-differential subsidy contract in these words:

Such contract shall provide, except as the parties should agree upon a lesser amount, that the amount of the operating-differential subsidy for the operation of vessels in an essential service shall equal the excess of the subsidizable wage costs of the United States officers and crews, the fair and reasonable cost of insurance, subsistence of officers and crews on passenger vessels, as defined in Section 613 of this Act, maintenance, and repairs not compensated by insurance incurred in the operation under United States registry of the vessel or vessels covered by the contract, over the estimated fair and reasonable cost of the same items of expense (after deducting therefrom any estimated increase in such items necessitated by features incorporated pursuant to the provisions of Section 501 (b)) if such vessel or vessels were operated under the registry of a foreign country whose vessels are substantial competitors of the vessel or vessels covered by the contract: *Provided, however*, that the Secretary of Transportation may, with respect to any vessel in an essential bulk cargo carrying service as described in Section 211 (b), pay, in lieu of the operating-differential subsidy provided by this subsection (b), such sums as he shall determine to be necessary to make the cost of operating such vessel competitive with the cost of operating similar vessels under the registry of a foreign country. . . .

In other words, compensation under the operating-differential subsidy equals the percentage by which the "fair and reasonable cost" to an American shipowner of operating a United States-registered vessel with an American crew exceeds the estimated fair and reasonable costs to a foreign owner operating the same ship with a foreign crew.⁶ The amount to be paid under the subsidy contract is determined by agreement between the applicant and the Secretary of Transportation, and consists of a percentage of the various items of expense enumerated in the passage quoted above. The applicant must agree that, throughout the life of the contract, his ships will be manned according to American practice, and that American living and working standards will be observed.

The whole philosophy of the subsidy system is epitomized in the calculations of the fair and reasonable difference in costs between United States and foreign operators. To show exactly how this works out in practice, a sample computation determining the percentage of wage costs to be paid by the government to the American shipowner is presented below. This sample reflects the provisions of the 1970 law, and illustrates the method of measuring the competition and identifying the components of wage expense that must be borne by the employer. To recognize the variations in cost among different nations as well as the impact of the competition experienced on the route from non-American operators, the participation of these foreign carriers in the trade was established in terms of percentage. This percentage figure then was applied as the "competition weight factor." In the calculation, shown below, the share of the Netherlands in the trade was determined to be 19.1 percent, that of Belgium was fixed at 22.1 percent, and Norway's proportion was calculated to be 39.2 percent. The percentage of American to foreign wages was multiplied by the competition weight factor to establish the weighted percentage. By combining the weighted percentages, the composite weighted percentage was derived, and this figure was subtracted from 100 percent (the American wage level) to establish the actual percentage of the seamen's wages to be subsidized. The table below demonstrates these steps in the computation.

	United States (a)	Nether- lands (b)	Belgium (c)	Norway (d)
1. Number in ship's crew	39	26	39	32
2. Base wages	\$ 53,687	\$ 23,127	\$ 24,779	\$ 27,257
3. Allowances	1,074	3,097	4,584	289
4. Vacation pay	35,681	9,499	13,009	11,976
5. Pension and welfare	33,407	3,923	2,065	124
6. Social security	6,608	4,584	7,227	10,118
7. Overtime and other variable costs	48,732	7,021	10,944	12,389
8. Repatriation	-	-	-	413
9. Total wage costs	184,189	51,251	62,608	62,566

	United States (a)	Nether- lands (b)	Belgium (c)	Norway (d)
10. Unweighted per- centage, United States to foreign wage costs	-	27.82%	33.99%	33.96%
11. Competition weight factor	-	19.1%	22.1%	39.2%
12. Weighted percent- age	-	5.31%	7.51%	13.31%
13. Composite weighted percentage (sum of b, c, and d) is	26.13%			
14. Percentage of wage costs subject to subsidy is	73.62% ⁷			

Similar calculations are performed for the other items of operating cost to which subsidy assistance is applicable.

The operating-differential subsidy contract requires that the operator maintain a designated frequency of sailings with a particular type of ship, that he operate his ships only in the essential service to which he is assigned, that he have no foreign-flag liner shipping under his operation, and that he meet certain requirements as to procedures of management. Ships under subsidy must be less than twenty-five years old, unless the Secretary of Transportation finds it in the public interest to waive this rule.

Operating-differential subsidy contracts ordinarily are granted only to one American operator in each essential liner service. Under the terms of Section 605 (c), however, if it can be shown that the subsidized American operator is not adequately meeting the needs of the American shipping public, and the Secretary of Transportation finds it is in the public interest to provide increased American-flag operations in the essential service, subsidy for one or more additional carriers may be authorized. In practice, the onus of showing that service is inadequate has rested upon the applicant. Very few approvals of this "double-tracking" of subsidized service have been granted.

Operators who fail to meet contract requirements are subject to cancellation of their contracts, or to substantial reductions in the level of subsidy payments. Those operators, however, who can demonstrate conclusively that it is economically impossible to main-

tain the essential services to which they are assigned, even with the subsidy, may be relieved of the contract. There are no cases on record of cancellation of contracts because the carriers did not meet requirements; release from their contracts was requested during the 1970s by several carriers serving routes including Great Lakes ports.

Mindful of the abuses of government supports under the ocean mail contract system, the Congress directed that operators holding subsidy contracts provide for the acquisition of replacement and additional ships needed to keep their services competitive with those of foreign carriers. Section 607 (a) stipulates that any citizen of the United States who owns or charters one or more eligible ships may enter into an agreement with the Secretary of Transportation to establish a capital construction fund to finance the replacement or reconstruction of existing ships, or to acquire additional tonnage. All these ships must be built in the United States, and documented under the laws of this country. They must be used in the foreign, Great Lakes, or noncontiguous domestic trade or in the fisheries of the United States. The incentive to set up and maintain the capital construction fund lies in the fact that deposits are not taxed as part of a corporation's income, and if these deposits are used to construct ships, lighters, barges, or containers, the tax liability is waived. All carriers in the foreign trade are eligible to participate in the advantages of this program; for the first time, therefore, construction subsidy was placed within reach of tankers as well as dry-cargo carriers.

In 1985, six steamship companies in liner service held operating differential subsidy contracts. The largest containership company in the world, Sea-Land Service, operates under the American flag, but never has applied for a subsidy. In part, this may be explained by the desire to be free of obligations to maintain unprofitable services and to be restricted to assigned trade routes. The company has built all its ships, except those used in the coastwise trade (which by law must be American built), in foreign shipyards, and has arranged its own financing. The largest subsidized operator was the United States Lines, which maintained service on a number of routes, not all of which were covered by the operating subsidy. The other liner service carriers holding subsidy contracts were American President Lines, Farrell Lines, Lykes Lines, Prudential Steam-

ship Company, and Waterman Steamship Corporation. In addition, a number of dry bulk carriers was granted subsidies on a voyage-by-voyage basis as they obtained cargoes destined to foreign ports.

The American steamship company which enters into an operating-differential subsidy contract with the Secretary of Transportation must assume these obligations:

1. The contractor must man his ships under wage scales and working conditions, and provide living quarters, as prescribed by the government. He must carry, in addition to the regular complement, those cadets and apprentices assigned by authority of the Secretary of Transportation.

2. All financial accounting must be in accordance with rules of practice laid down by the government. In actuality, the Maritime Administration issues the rules and prescribes the accounting practices to be followed. Reports on any phase of a contractor's business must be submitted to the Secretary of Transportation whenever he directs. All financial records are subject to periodic audit by government officials.

3. The contractor may not, except with written approval of the Secretary of Transportation:

- a. operate any vessel over twenty-five years of age;
- b. engage in any auxiliary services directly or indirectly connected with the operation of the subsidized ships, such as stevedoring, ship repairing, ship chandlery, towboat, or other allied activities;
- c. engage in the intercoastal or coastwise (protected) trades of the United States;
- d. effect any merger or consolidation, or directly or indirectly embark upon any new enterprise not connected with the business of shipping;
- e. own, charter, or act as agent or broker for, or operate, any foreign-flag ship that competes with an American-flag service determined by the Secretary of Transportation to be essential.
- f. dispose of any interest in the subsidy agreement.

From the foregoing commentary upon the Merchant Marine Act, 1936, and the important amendments made thereto by the

Merchant Marine Act of 1970, some understanding may be obtained of the procedures by which government aid is extended to American shipping. Tight control is evident in many aspects. Experience has shown that this supervision is required for the protection of the public interest as well as to assure that the United States-flag operator will be assisted in meeting the objectives of the law. It appears significant in the evaluation of these two laws that few changes were made in the basic principles of federal support of the United States merchant marine through construction- and operating-differential subsidies.⁸

The Business of Shipping

IN acquainting the reader with the intricate story of the management of a modern steamship company, as has been attempted in the preceding pages, some of the fascination and the deeply personal aspects of the business of shipping have received short shrift. The fascination exists, however much it has been covered over with operational detail. The rewards of the enterprise are understood by some, appreciated by others, and shared in by those fortunate ones of all ranks who lift their eyes above their desks and seek the far horizons beyond which sail the ships.

To show how the manifold activities and responsibilities of management actually can be combined in one man's experience, the career of an imaginary steamship company president is outlined below, together with this fictional executive's reactions to the challenges and rewards of shipping as a career. Except that no individual is the subject of the sketch, the career delineated is typical of that of many steamship company managers. The philosophical musings also are based more on memories of comments by friends and associates than on imagination.

It is a fall evening in a large seaport city. Outside the city lights are coming on one by one. Inside the big office the jangle of telephones has subsided, and the clatter of business machines has been stilled. A few men and women linger at their desks as they finish up some small task before leaving for the day.

In the president's office, a man sits at the desk beside a window commanding a view of the harbor. He looks down at the unending activity on the water, and his eyes dwell upon the shape of a ship standing out to sea. The president muses over what has gone into that ship's sailing, and his musings take him back over a lifetime of service to his fellow man in the transportation of goods and people across the oceans.

He looks around his office. There, in a special place of sentiment and honor, is his certificate of graduation from the merchant marine academy. He had been well-trained there; he had come to know something of the hard yet fascinating life he had chosen when only a boy. He also had had the veil of management and its problems lifted just a bit. It had been a good experience, learning to be a seaman and also finding out why many things were done in a certain way.

Just above the diploma was another framed document, this one written in an engraver's script and ornamented with pictures of three ships. This was a license authorizing the person named thereon to command any size ship on any ocean. His master mariner's "ticket"! What years of work and aspiration had preceded the wonderful day when he had been presented with that precious proof of qualification for that post of distinction and almost immeasurable responsibility! A year as an able-bodied seaman, four months as a quartermaster, a year as junior mate, followed by two years in tramp steamers plodding all over the world with cargoes of coal, grain, ores, fertilizer, and lumber had given him a firm knowledge of the vicissitudes of the seaman's life and a competence in his chosen profession that no amount of theoretical study could have produced. He had had a good time on the tramp; the ports were both familiar and exotic, the hazards of navigation with pilots who spoke no English and seemed to understand no sign language, had added a grey hair or two. He had made some good friends, too, real seamen who had no other ambition than to command ships sailing in international commerce.

Tramping was all right, but promotion required a variety of experience. So he had transferred to a liner company, even though it meant starting over again as a third mate. He had learned at first hand about precise schedules, difficult loadings of odd cargoes, round-the-clock operations in one port after another, endless paperwork, and always the need to keep the agents informed systematically of progress so that the tugs, pilots, lines-handlers, and supply people could be scheduled to avoid delays to the ship. It had been good duty. Promotion was swift, and soon he was chief mate. Chief mate! Memories of the never-ending demands upon his time, skill, and knowledge crowded in upon the president. Responsibility for safety of the ship's cargo, for proper loading of the ship to

assure seaworthiness, for efficient supervision of longshoremen, seamen, cadets, and shoreside workers swarming over the ship, as well as meeting senior company executives who came down to the ship—all this had been his. He had learned to be everywhere at once, to carry a hundred details in his head without becoming confused or excited. He had had to concentrate upon his daily duties, and he took real pleasure in knowing that each job had been well done.

Three years sped by for the chief mate, and after a total of only six years with that company he was transferred ashore to be cargo superintendent in a small foreign port. The euphonious title covered a hundred routine chores from arranging for lines-handlers to tie up the ship when she arrived to explaining for the hundredth time to unresponsive longshoremen how to sling up certain types of cargo. Six months of that, and he was moved to a major foreign port as assistant terminal superintendent. That had been a challenging job! He had had his first contact with the financial aspects of shipping, and began to relate ship time to money, efficiency on board to coordinated and systematic support from the shore staff, the flow of cargo to economical employment of resources of the cargo terminal.

Two years ashore, although crowded with interest and real accomplishment, still had not brought him to the pinnacle of his youthful ambition. He requested, and received, return to sea duty and rejoiced once again to feel a ship's deck underfoot. Six months as chief mate preceded that wonderful day when the vice-president, operations, had told him that the company had decided to recognize his outstanding performance by giving him a command of his own. How thrilling it had been to walk aboard *his* ship, and to realize that he had the final responsibility for the safety of the ship and all the scores of people on board. He also knew how much pressure would be placed upon him to hold to schedules in spite of fog and storm; how little understanding his problems would receive from landlubbers who thought of ships only as money producers.

He had enjoyed his responsibility as master, and watched the results of his efforts as he slowly shaped the ship to his own ideals of a well-run vessel. To his pleasant surprise, he received a letter—there it was, yellowed and creased, hanging alongside his license—from the company's board of directors expressing appreciation of

his unceasing and exemplary devotion to duty. That letter was precious; it covered all those achievements that had called for so much effort: rescuing a dozen men from a foundering fishing boat in a frightful North Atlantic storm, winning the annual award two years running for the safest ship in the fleet, earning commendations for having the finest and cleanest ship in the company's service. He had had a good crew, and they had worked together with pride and with a sense of purpose.

The president's eyes moved on, to stop for a moment in front of the naval reserve commission. What a wrench assignment to active duty had been, as he changed from master of his own ship to lieutenant, navigator in a destroyer! Responsibility sought him out, however, and before many months he had been promoted, assigned first as executive officer of a large ship and then commander of a small cargo ship. From that job he had moved up to command of a large transport and then a major shore station. In those years of naval service he had come to understand the reasons why merchant shipping was a vital component of a nation's seapower. He had never forgotten those lessons, for they had influenced his life and his thinking ever since he had returned to civilian pursuits.

After his military service, he had found employment as chief mate with the company he now headed. Two trips as mate were followed by assignment to a command of his own. The policy of the company was changing, and more attention was being given to capitalizing on sea experience. As an experimental move, he was brought ashore as a salesman in the outward cargo department. *That* had been a change! "Pounding pavements and ringing doorbells, working up a good sales promotion talk, and never being downhearted" were the maxims he learned to follow. He had enjoyed these contacts with the shippers. For years, he had served them impersonally when he was loading and transporting their goods, but now he was meeting them face to face. It gave a new dimension to his work at sea. He understood what the formerly anonymous "shipper" thought, and he was better able to meet his requirements. From salesman, he was detailed to the manifesting department for a few weeks, more for experience than for the value of his contribution to that type of work. That wasn't his "cup of tea," since he had been an active seaman, and he disliked the routine of checking figures and cargo descriptions all day long. He realized

how important it was that he know this part of the business, even though it lacked glamour. A transfer soon came to the inbound traffic department, where he met the consignees, the men and women who were waiting for the goods his ships were bringing to the port. He came to know many people, many government officials, and much about the business of keeping customers happy. He enjoyed this human contact, and he liked the idea of helping to generate patronage for the line through giving good service.

A new world was opened to him when he was promoted to general freight agent. He had become intimately familiar with the manner in which freight rates were made. He spent uncounted hours in sessions with executives of other companies, both carriers and shippers, discussing and making policies and seeking solutions to problems. He attended those all-important meetings of the Conferences (he always spelled the word with a capital, to distinguish those associations of carriers on the trade routes from just routine face-to-face talks with his fellows). He had been very happy when the chairman, acting on behalf of the membership of the Conference, presented him with a letter expressing appreciation for his substantial contributions to the work of the Conference, based on his extensive seafaring experience.

Traffic and all it meant were fine and a wonderful education, but operating ships was the president's real love, for he was a seaman first and last. He welcomed, therefore, the assignment overseas as operations manager. It had been good to get back to having direct contact with the ships and the seamen, the terminal personnel and the faithful watchmen, coopers, and tradesmen who worked so well and so anonymously! Long hours had been spent at his desk, and even longer hours in the terminal and around the ships, but they had been rewarding, instructive, constructive, and, above all, companionable hours.

Those happy two years passed all too quickly. Then came the cable summoning him back to the home office to take over the duties of vice-president, operations. What a tremendous thrill that had been, and how proud he felt! From cadet to vice-president in charge of all the ships! It had been quite a climb, and he had enjoyed every step of the way. He recalled those discussions of company policy on the proper operation of the fleet which he had

had with the former president, the positions he had taken and for which he had fought vigorously because he believed them to be right even though they had not been popular.

Two years as vice-president, and one day the directors called him into their meeting room. Everyone rose as he entered. The chairman greeted him and escorted him to the only empty chair in the room. He noted, as he walked toward the table, that a handsome card beautifully lettered with the word, "President," had been affixed to that chair. It didn't seem possible, but he was led to that chair, and when the directors burst into applause, he came out of the daze.

The president's gaze shifted to a frame in which were a photograph of himself, in the uniform of a quartermaster, standing beside the master of his first ship, and that red-lined page from an old desk calendar, with the notation in pencil: "I was elected president today."

Thirty-odd years of his life were shown on the walls of his office, mused the president, as he scanned the room quickly: Cadet, seaman, quartermaster, mate, lieutenant, captain, master, pier superintendent, salesman, general freight agent, operations manager, vice-president, and president. It had been a good life.

As the president reached for his hat, he glanced out of the window again. Another ship was standing out to sea, and her lights shone brightly in the gloom. He walked out of his office and noted, from force of habit, the board showing locations of all the company ships—*his* ships, their next ports of call, and their dates of arrival and departure. He always looked at that board each evening as he left the office; he liked to identify himself with the ships and the men on board. He remained a seaman, was proud to call himself by that term, and he loved the ships.

Outside the building, he walked briskly to the ferry landing and boarded the ferry. He went swiftly to his usual place up forward, where he could see the ships and their lights. Another big steamer was standing in from sea. In his mind, he envisioned the preparations being made even at that hour to receive the ship. The hoarse whistle of an outbound ship turned his thoughts to the people who had toiled over manifests and crew lists, cargo plans and repair specifications, stores lists and bills of lading, and to the strong men

who had stowed away the boxes and bales and assorted commodities, and had secured the huge containers, in their appointed places in the ship.

The big vessel slipped down the harbor and was lost in the dark. She was outward bound, and she carried with her not only the work of hundreds of men and women but the fortunes of dozens and dozens of shippers and consignees, the dreams of countless people who would use the great piles of goods that she carried, to say nothing of the lives of the fortunate men who were taking her to sea.

The echo of the big ship's whistle sounded through the night. It was lonely, brave, challenging, competent, cheerful. The president looked up and down the harbor, and said to himself, "It has been a wonderful life. I wouldn't exchange a minute of it for all the wealth of the Indies. It has been rewarding, exacting, exciting, vital, important, and, above all, necessary. What an honor to be able to say, 'My business is shipping!'"

Notes

Chapter 2. Tramp Shipping

1. I. L. Buxton, *Engineering Economics and Ship Design*, 2d ed. (Wallsend, England, The British Ship Research Association, 1976) pp. 12-15. (Hereafter cited as Buxton.)

2. M. J. Collins, "Wallenius Lines: The Youngest of Sweden's Major Ship-owners, Pioneers in Sea Transportation of Automobiles," *The Compass*, 45, no. 3 (1975), pp. 16-21.

3. William McFee, *Watch Below: A Reconstruction in Narrative Form of The Golden Age When Coal Took the Place of Wind and the Tramp Steamer's Smoke Covered the Seven Seas* (New York, Random House, 1940), pp. 48-53. Much of the material presented in this book is drawn from the author's service as an engineer officer in British tramps prior to 1914.

4. The Liberty ship hull was 134.12 meters (441 feet 6 inches) long, 17.37 meters (57 feet) in the beam, and had a summer loadline draft of 8.45 meters (27 feet 9 inches). Her deadweight was 10,400 tons. There were five hatches, each served by one set of cargo booms. Her oil-fired boilers generated steam at 220 pounds per square inch to actuate a three-cylinder, triple expansion steam engine of 2,500-shaft horsepower. In calm seas, the ship had a cruising speed of 10½ knots.

5. The "S. D. 14" was 140.87 meters (462 feet 6 inches) long, 20.42 meters (67 feet) in the beam, and on a draft of 8.84 meters (29 feet) had a deadweight of 14,910 tons. She had five hatches and five holds, each served by cargo derricks of five tons capacity. Her grain cubic capacity was 22,920 cubic meters (764,280 cubic feet). Speed in service under average fair weather conditions was 14.9 knots. A five-cylinder diesel engine rated at 7,500-brake horsepower was required to deliver only 6,750 horsepower (90 percent of maximum continuous rating) to maintain this speed. Fuel consumption for the main engine was about 25½ tons of heavy oil per day. In 1968, the price of the ship was £900,000. "The Austin and Pickersgill 'S. D. 14' Liberty-ship Replacement," special supplement to *The Motor Ship*, May 1967, includes a detailed examination of the design of this vessel. This issue was published at the time the builders were active in solicitation of orders.

6. "Cunard's Growing Bulk Carrier Fleet: Two Further Ships Named at Seville," *The Motor Ship*, October 1973, p. 320.

7. McFee's *Watch Below* is a semiautobiographical account of the working and living conditions aboard British-flag tramps in the period from about 1906 to 1914.

8. Roland H. Thornton, *British Shipping* (Cambridge, England, Cambridge University Press, 1939), p. 130. (Hereafter cited as Thornton.)

9. The term "lay days" is used, with different meanings, for both circumstances described in these paragraphs. The content normally indicates which meaning is intended.

10. A charter party embodying these stipulations sometimes is described as a "liner terms" or "berth terms" agreement. Generally speaking, this terminology implies that loading and discharging expenses are included in the charter hire, and are for the shipowner's account.

11. For a detailed analysis of both voyage and time charter parties, see Lars Gorton, Rolf Ihre, and Arne Sandeværn, *Shipbroking and Chartering Practice* (London, Lloyd's of London Press, 1980). The authors have varied backgrounds, being, respectively, a professor of maritime and commercial law, a legal advisor to a major Swedish steamship company, and the marketing manager of this company.

Per Gram's *Chartering Documents* (London, Lloyd's of London Press, 1981), is a "running commentary" upon specimens of standardized charter parties. Written in nontechnical language, the analyses of voyage, time, and bareboat charters clarify many aspects of these documents. A typical "contract of affreightment" also is given a clause-by-clause examination.

Chapter 3. The Management of Tramp Shipping

1. In 1979, the United States exported 120,227,000 metric tons of grain, while Canada shipped 16,005,000 tons, Argentina sent 15,023,000 tons, and Australia provided 14,549,000 tons to overseas destinations. Fearnley & Egers Chartering Co., *World Bulk Trades, 1979* (Oslo, 1980), p. 29.

2. A study of patronage of British liner trade revealed that one carrier listed 3,900 clients, but 75 percent of the tonnage carried was offered by only 99 shippers. Roy Pearson, *Containerline Performance and Service Quality* (University of Liverpool, Marine Transport Center [1981]), p. 23. (Hereafter cited as Pearson.)

Chapter 4. Chartering and Tramp Ship Operation

1. A concise but comprehensive survey of the workings of this scheme is available in Hector Gripiaios, *Tramp Shipping* (London, Thomas Nelson, 1959), pp. 74-84. See also B. N. Metaxas, *The Economics of Tramp Shipping* (London, Athlone Press, 1971), pp. 229-49, for a wide-ranging, critical study of the various stabilization schemes proposed from 1965 to 1963.

2. The procedure sketched in the text is derived from an untitled information booklet distributed in 1985 by the Baltic International Freight Futures Exchange (BIFEX). Included in the booklet is a detailed listing of each of the thirteen selected trading routes. Two of the more important routes are cited as examples:

From one port, United States Gulf to Antwerp, or Rotterdam, or Amsterdam, lifting 55,000 tons, plus or minus 5 percent, of heavy soya sorghum. Cargo to be loaded and discharged at charterer's expense (FIO). Eleven days allowed for working cargo, Sundays and holidays excepted. Tendering date, ten days forward

from date of freight index; cancelling date, thirty days forward. Brokerage commissions, 3.75 percent.

From one port United States North Pacific to one port in southern Japan, 52,000 tons, plus or minus 5 percent, of heavy soya sorghum to be loaded and discharged at charterer's expense (FIO). Eleven days allowed for working cargo, Sundays and holidays excepted. Tendering date, ten days forward from date of freight index; cancelling date thirty days forward. Brokerage commissions, 3.75 percent.

3. The specifications of a 52,000-ton deadweight bulker in service in 1980 were:

Length overall	218.85 meters
Length between perpendiculars	205.25 m.
Beam	30.50 m.
Draft, about	12.00 m.
Propulsion:	
7-cylinder diesel	11,900 bhp
Crew:	
Deck and petty officers	11
Engineering officers	6
Unlicensed personnel	<u>21</u>
Total	38
Tonnage:	
Gross register	29,900
Net register	20,600
Deadweight	52,000
Cubic capacity:	
Bale, cubic meters	61,000
Grain, cubic meters	62,500

4. "Prospects for Bulk Carriers," in "Ships on Order" supplement to *The Motor Ship*, July 1974, pp. 44-45.

5. "Worldscale Directory: Tanker Industry's Bible," *New York Journal of Commerce*, December 27, 1983, p. 3-B.

Chapter 6. Terminal Management

1. For a detailed analysis of terminal security procedures, see Warren H. Atkins, *Modern Marine Terminal Operations and Management* (Oakland, California, The Port of Oakland, 1983), pp. 184-90. (Hereafter cited as Atkins.)

2. *Ibid.*, pp. 190-94.

3. *Ibid.*, pp. 110-11.

4. The bistate port of New York and New Jersey reported that 80 percent of the general cargo moved through that port in 1983 was containerized. "Slump Held Dampening Cargo Security Efforts," *New York Journal of Commerce*, February 3, 1984, p. 24-B.

5. "Terminal Plan Aided in Keeping USL in NY," *ibid.*, April 19, 1985, p. 1-B.

For a detailed profile of four well-designed and equipped container terminals,

together with descriptions of the various machines used, and specimens of equipment interchange receipts, see Atkins, pp. 45-56.

John R. Immer, *Cargo Handling* (Washington, D. C., Work Saving International, 1984), pp. 271-94, provides a technical description of container terminals in Long Beach, California; Antwerp, Belgium; and Rotterdam, Netherlands.

6. American President Lines' container terminal in San Pedro, California, has a well-equipped freight station covering 122,000 square feet. It includes facilities for United States customs inspectors to carry out their functions.

7. An excellent comparison of the organizations of a container terminal and a break-bulk terminal is offered in Atkins, pp. 24-26.

8. Atkins, pp. 96-103, contains a full description of gatehouse and marshalling yard operations.

Chapter 7. Terminal Operation

1. Atkins, p. 45.

2. For a survey of operations at a container terminal, see *ibid.*, pp. 14-19, 23.

Chapter 9. Procurement of Vessel Stores and Supplies

1. U. S. Department of Commerce, Maritime Administration. *Spare Parts Provisioning for Merchant Ships*, 2 vols. (Washington, D. C., National Technical Information Service, 1979).

This comprehensive study was prepared for the Maritime Administration by Mystech Associates, Inc., of Mystic, Connecticut, after detailed examination of procurement practices of 22 ship operators and 26 manufacturers and suppliers of spare parts. It was the most detailed exposition of current procedures in this field available at the time this chapter was written.

2. J. V. White, "How Cost Effective Are Pirate Spares?" *The Motor Ship*, March 1982, pp. 39-42.

The author of this article supports strongly the argument that shipowners are well-advised to purchase replacement and spare parts from the original equipment manufacturer. He provides numerous examples of the expensive mistakes made by ship operators overly concerned with initial purchase price rather than reliability and compatibility with the equipment.

3. For a comprehensive explanation of the duties and responsibilities of the ship's agent, see J. D. Eadie's articles entitled "Ships' Agency," which were published intermittently in *Fairplay* between September 23, 1982 and February 3, 1983. The series later was reprinted and issued under separate cover with the overall title of "Ships' Agency."

Chapter 10. Containerization: The Beginning

1. McLean entered the long-distance trucking business in 1933, driving one truck. Over the next two decades, his transportation company grew into a multi-million dollar operation. Headquarters were maintained in Winston-Salem, North Carolina, the state in which McLean was born.

2. *New York Journal of Commerce*, February 17, 1954; *Marine Engineering*, April 1954, pp. 47-48.

3. On January 21, 1955, McLean announced that he had acquired all the capital stock of Pan-Atlantic, and at the same time had resigned the presidency of the McLean Trucking Company. (*New York Times*, February 13, 1955, Part V, p. 9.) The final separation of Pan-Atlantic from Waterman was accomplished on March 1, 1955. (*Ibid.*, March 3, 1955, p. 55.) McLean sold his stock in the trucking company in September 1955. (*Ibid.*, November 28, 1956, p. 70.)

4. The first tanker acquired was the *Marine Leader*, purchased for \$1,375,000 from Marine Navigation Co., of New York, on April 5, 1956. After conversion, she was renamed *Maxton* in honor of the North Carolina city in which McLean was born. (*New York Times*, April 6, 1956, p. 46.)

The *Ideal X* (ex *Potrero Hills*) and the *Almena* (ex *Whittier Hills*) were modified at the Baltimore shipyard of Bethlehem Steel Company, while the *Maxton* and the *Coalinga Hills* were adapted to the new service by Mobile Ship Repair, Inc., of Mobile, Alabama.

5. *New York Journal of Commerce*, August 18, 1955.

6. Quoted in *New York Times*, November 28, 1956, p. 70.

7. Later renamed *Seatrains New Orleans*, this ten-knot vessel was built by Swan, Hunter & Wigham Richardson, of Newcastle, England, and delivered on November 15, 1928. She was 130.15 meters long, 18.89 meters in beam, and had a draft of 7.85 meters. Her deadweight tonnage was 10,500. The single hatchway was 13.72 meters long and 16.15 meters wide. In 1938, the ship was lengthened to 137.97 meters, and her capacity increased to approximately 100 cars from her earlier maximum of 90 to 95.

The *Seatrains New Orleans* was operated in the New Orleans-Havana trade from 1929 until that service was abandoned in 1950. In that year she was sold, transferred to foreign registry, and renamed *Sealevel*.

8. International Cargo Handling Coordination Association, *Journal*, 2, Special Issue (September 1955), pp. 16-49, *passim*.

9. *New York Times*, May 7, 1955; August 16, 1955.

10. Among the properties sold were a shipyard, a hotel, and an office building in Mobile, Alabama.

11. See *Marine Engineering/Log*, December 1957, pp. 67-69, for a complete description of the *Gateway City*, the first converted C-2 to enter McLean's service.

12. *New York Times*, November 23, 1958.

13. When Sea-Land Service, Inc., placed four diesel-powered and crane-equipped ships in service during 1978, McLean had no connection with that organization. The reason advanced by the owners for installing cranes was that the ships were to serve ports in the Middle East which had no facilities for handling containers. *New York Journal of Commerce*, May 5, 1978, p. 32.

Chapter 11. The Ramifications of Containerization

1. As early as 1923, a shipment of 7,500 bales of raw silk worth \$8,250,000 was sent from Yokohama to Seattle, and was transshipped into two special trains of fifteen baggage cars each. These trains followed schedules equal to those carrying

passengers (*New York Times*, January 7, 1923, p. 1). In 1926, a shipment of 10,000 bales of raw silk, worth \$11,000,000 arrived in New York just 17 days after it was loaded in Yokohama. The steamer used was the *Arabia Maru*, of the Osaka Shoshen Kaisha fleet, which sailed from Japan on December 27, 1925. Two trains consisting of 11 and 12 baggage cars respectively were ready when the ship docked in Seattle, and departed on January 10, 1926, arriving in New York in the evening of January 13 (*ibid.*, January 14, 1926, p. 43). A record was claimed for the transcontinental run when a shipment valued at \$5,400,000 was moved from Seattle to New York in 73 hours and 25 minutes; express passenger trains required 95 to 100 hours for the same trip. The demand for speed had two bases: the raw silk deteriorated rapidly, and interest charges amounted to about \$1,000 a day. The "silk express" in 1925 handled 261,853 bales of new raw silk, valued at almost \$700,000,000 (*ibid.*, September 26, 1926, Section VIII, p. 14).

2. "It is said that westbound traffic is three to four times heavier than east-bound trade, and that there is 'always' a shortage of containers in Nakhodka." *New York Journal of Commerce*, June 23, 1980, p. 22-A.

3. "Land-Water Bridge Service Is Ruled Legal By the FMC," *ibid.*, August 9, 1978, p. 1.

4. "Tributary Cargo Interpretation of FMC Receives Support," *ibid.*, December 4, 1978, p. 9.

5. "McAllister Doubles Container Capacity," *ibid.*, February 24, 1977.

6. "Sea Containers' 'Strider' Vessels," *The Motor Ship*, September 1976, pp. 110-13. This is an excellent article describing these feeder ships, which could carry 330 twenty-foot containers, and had a deadweight of 6,500 tons.

7. "Firms Shift to 20 Foot Containers," *New York Journal of Commerce*, April 11, 1977, p. 10.

8. "Container Leasing Firms' Profits Good," *ibid.*, April 18, 1979, p. 2.

Chapter 12. The Ocean Bill of Lading

1. For a detailed, clause-by-clause examination of standardized bills of lading, see Per Gram, *Chartering Documents*, pp. 85-106. A number of standardized bills of lading is included in the appendices of this book.

Chapter 13. How Freight Rates Are Made

1. "Ship Lines Must Justify Lower Rates," *New York Journal of Commerce*, December 17, 1970, p. 3.

2. "Fixing of Freight Rates," *Fairplay*, April 9, 1970, pp. 10-11.

3. Gunnar K. Sletmo and Ernest W. Williams, Jr., *Liner Conferences in the Container Age* (New York, Macmillan, 1981), pp. 66-67. (Hereafter cited as Sletmo.)

4. "TWRA May Postpone July 1 Rate Hike," *New York Journal of Commerce*, June 6, 1985, p. 1-A. "Rate Hikes Cancelled by TWRA," *Ibid.*, September 30, 1985, p. 3-13.

5. Sletmo, p. 114.

6. *Ibid.*, pp. 185-89; 252-54. The problems inherent in the "freight all kinds" rate are described in these pages.

7. "Armada GLTL Announces Lower Rates," *New York Journal of Commerce*, June 11, 1985, p. 24-A.

Chapter 15. Steamship Conferences

1. The comments of the Senate Committee on Commerce, in discussing the need to amend the Shipping Act of 1916, are pertinent:

"Your committee has been constantly aware of the fact that further unilateral attempts by this Government to regulate the details of commercial transactions of shippers and carriers which take place and are documented abroad, are likely to have major deleterious effects and negligible counterbalancing benefits. First, they are likely to create further resentment among our foreign friends and, as a consequence, to cause reactions ranging from passive loss of confidence to active retaliation in foreign ports. In the process, they might well force foreign lines out of conferences, thereby destroying conferences and placing the high-cost American operator, even if subsidized, at a serious if not impossible competitive disadvantage. Second, and perhaps equally serious, to the extent such extraterritorial regulation cannot be as effectively enforced against their competitors, our American lines, we would be working at serious cross purposes with fundamental precepts of national maritime policy." (U. S. Congress, Senate, *Senate Report No. 860*, "Steamship Conferences and Dual Rate Contracts," 87th Cong. 1st sess., August 31, 1961, p. 3.) (Hereafter cited as *Engel Report*.)

2. An excellent short history of the origin of the conference system is contained in Inter-American Maritime Conference, *Report of Delegates of the United States* (Washington, D. C., Government Printing Office, 1941) pp. 159-63.

3. U. S. Congress, House, *House of Representatives Report No. 498*, "Providing for the Operation of Steamship Conferences," 87th Cong., 1st sess., June 8, 1961, p. 12. (Hereafter cited as *Bonner Report*.)

4. John A. Hobson, *The Evolution of Modern Capitalism*. Rev. ed. (London, Allen & Unwin, 1926), pp. 175-76.

5. Quoted in American Maritime Council, *Foreign Trade and Shipping* (New York, McGraw-Hill, 1945), pp. 25-26.

6. *Ibid.*, pp. 26-27; Daniel Marx, Jr., *International Shipping Cartels: A Study of Industrial Self-Regulation by Shipping Conferences* (Princeton, N. J., Princeton University Press, 1953), pp. 50-67. (Hereafter cited as Marx.)

7. *Bonner Report*, p. 5.

8. Sletmo, p. 200.

9. Federal Maritime Board *vs.* Isbrandtsen Co., Inc. (354 U. S. 481, 1958). For a commentary on this case, see Allen R. Ferguson, et al., *The Economic Value of the United States Merchant Marine* (Evanston, Illinois, The Transportation Center at Northwestern University, 1961), pp. 389-96.

The Shipping Board was in existence from 1916 until 1936, when it was superseded by the United States Maritime Commission. In 1950, the regulatory functions which had been performed by the commission were assigned to the newly established Federal Maritime Board. The chairman of the board was also the chairman of the Maritime Commission. President Kennedy proposed, and the Congress approved in 1961, a plan by which the Federal Maritime Commission

was set up as an independent agency with sole responsibility for regulating maritime commercial activities.

10. *Engel Report*, pp. 5-6.
11. Pearson, p. 23.
12. *Engel Report*, pp. 7-8.
13. Sletmo, pp. 306-7.
14. "Australian Wool Exporters Continue Rate-cut Efforts," *New York Journal of Commerce*, October 8, 1983, p. 24-B.
15. See Sletmo, pp. 278-86, for a comprehensive critique of the Australian experience.
16. Great Britain, Committee on Inquiry into Shipping (Chairman, The Rt. Hon. the Viscount Rochdale): *Report* (London, Her Majesty's Stationery Office, Cmmd. 4337, 1970), p. 132, par. 468-69, quoted in Sletmo, p. xxxi.
17. Marx, p. 240.
18. Personal interview, L. C. Kendall and Sylvester J. Maddock, vice-president, traffic, Seas Shipping Company, 1947. An excellent account of the rate war is contained in Robert Greenhalgh Albion, *Seaports South of Sahara* (New York, Appleton-Century-Crofts, 1959), pp. 120-27.
19. *Congressional Information Bureau*, November 15, 1962.

Chapter 17. Scheduling and Bunkering

1. Early in 1981, A British group concerned with the standards for marine fuels proposed the establishment of certain criteria by which fuel oil would be graded. ("Fuel Quality Standards: A British Proposal," *The Motor Ship*, April 1981, pp. 26-28.)

For an operational report on problems encountered with various types and grades of heavier oils used in motor ships since 1947, see "Burning High Viscosity Fuel in Diesel Engines," *ibid.*, December 1981, pp. 31-34.

Det Norske Veritas, the Norwegian ship classification society, set up an elaborate system to analyze fuel oil samples sent by ships from bunkering ports around the world. Immediately on their receipt in Oslo, these samples are tested, and the results are communicated to the owners by the most rapid means possible. ("Bunker Qualities: A Progress on DNV's Test Scheme," *ibid.*, April 1982, pp. 63-64.)

Chapter 18. Planning for a New Ship

1. Robert Taggart, ed., *Ship Design and Construction, Written by a Group of Authorities* (New York, The Society of Naval Architects and Marine Engineers, 1980), p. 11. (Cited hereafter as Taggart.)

2. *Ibid.*, p. 8.

3. Thorsten Rinman and Rigmor Linden, *Shipping—How It Works* (Gothenburg, Sweden, Rinman and Linden [1979]), p. 14.

4. U. S. Department of Transportation, Maritime Administration, *A Statistical Analysis of the World's Merchant Fleets, Showing Age, Size, Speed, and Draft by Frequency Groupings as of January 1, 1983* (Washington, 1984), p. v.

5. Buxton, p. 11.

6. M. Meek, "Developments in Merchant Ships Over the Last Decade," *The Institution of Mechanical Engineers Proceedings, 1980*, 194, no. 9, pp. 47-64, *passim*.

Nicholas Barber, "Box, Wheel or Carton?" *The Motor Ship*, February 1982, pp. 23-26.

7. V. L. Russo and E. K. Sullivan, "Design of the Mariner Type Ships," *Transactions of the Society of Naval Architects and Marine Engineers, 1953*, quoted in Taggart, p. 53. Quotation used by permission.

8. Rolf Glasfeld, *Coal Fired Modern Cargo Ships*, presented at Maritime America '83 Conference, Baltimore, March 29-31, 1983. (Published by General Dynamics Corp.)

Alkis Alexandrakis and Rolf Glasfeld, *Economic and Technical Aspects of Collier Design*, presented at Coal Transportation, Distribution and Policies Seminar, Houston, February 27-March 1, 1983. (Published by General Dynamics Corp.)

Chapter 19. Passenger Cruises

1. "Miami Cruise Ships Set to Carry 2.3 Million Passengers in 1984," *New York Journal of Commerce*, May 14, 1984, p. 6-C. "Call Issued to Strengthen U. S. Cruise Sector," *ibid.*, April 16, 1985, p. 14-B.

2. Norwegian Caribbean Lines is the largest cruise ship operator in the world, with a fleet of eight ships worth \$640 million. "Norwegian Line Buys Royal Viking," *San Francisco Chronicle*, August 16, 1984.

3. Thornton, pp. 208-9.

4. "Holiday in the Sun," *Via Port of New York*, December 1959. At least two ships were built for and operated in year-round cruise service prior to World War I. The Hamburg-American Line brought out the *Prinzessin Victoria Luise* in 1900. She had the sharp hull lines of a yacht, but could accommodate 400 passengers, all of whom slept in brass beds. Used exclusively in cruise service, the ship was wrecked off Kingston, Jamaica, on December 6, 1906. (William H. Miller, Jr., *The First Great Ocean Liners in Photographs, 1897-1927* (New York, Dover Publications, 1984), p. 42.) The British P & O Line introduced the *Vectis* in 1904, and kept her employed in cruises from England to the Canary Islands, the Baltic Sea, and the Mediterranean. (Thornton, p. 209.)

The *Stella Polaris* was built in 1927, for the Bergen Steamship Company of Norway. Except during the years of World War II, she was assigned year-round to deluxe cruise operation. She was sold to the Clipper Line, of Malmo, Sweden, in 1952, and registry was transferred from Norwegian to Swedish. She was withdrawn from cruise service in 1968, and later was sold to Japanese interests which renamed her *Scandia* and converted her into a floating hotel. She was still in existence in 1980. This handsome 160-passenger ship had a sharp clipper bow and exceptionally tall masts, which gave her a unique profile. Always painted white, she was 126.8 meters long, 15.5 meters in beam, and had a draft of 5.5 meters.

5. *New York Journal of Commerce*, October 23, 1973, p. 5.

6. "New Int'l Line Offers Cruises to China, Asia," *ibid.*, October 9, 1981, p. 24-B. Also see the company's advertising booklet issued in 1985.

7. "Hapag-Lloyd Confirms Passenger Ship Order," *ibid.*, November 6, 1979, p. 34.
8. *Marine Week*, March 2-8, 1979, p. 28.
9. "Cruise Ship Fairsky Will be Christened in Los Angeles," *Los Angeles Times*, May 4, 1984, Part V, p. 1.
10. "Ship-Shaping the Cruise Liners," *ibid.*, September 24, 1978, Part VII, p. 10.
11. "Cruise Ship Party—A Joyful Noise," *ibid.*, April 12, 1981, Part VII, p. 6.
12. "U. K. Shipowners Seen Placing Orders in '79 for Passenger Ships," *New York Journal of Commerce*, November 28, 1978, p. 34.
13. "Cruise Trade Booms in Australia," *ibid.*, October 11, 1984, p. 14-A.
14. "Outlook Seen Brighter for Cruise Ship Activity," *ibid.*, August 1, 1979, p. 10.
15. "Bahama Star Cruises Are Cancelled," *ibid.*, January 5, 1975, p. 24.
16. "Fuel Crisis Scuttles World Cruise," *Los Angeles Times*, January 1, 1974, p. 6.
17. "ACL Vessel Chartered for Falklands Crisis," *New York Journal of Commerce*, April 15, 1982, p. 12-A.
18. "Costa Cancels Cruise to Buenos Aires," *ibid.*, May 14, 1982, p. 1-B.
19. "Off to the Icy Waters of Antarctica," *Los Angeles Times*, November 25, 1984, Part VII, p. 6. "Northwest Cruises Being Discouraged," *New York Journal of Commerce*, April 4, 1985, p. 1-B.
20. "'France' into 'Norway' by Hapag-Lloyd," *The Motor Ship*, July 1980, p. 87.
21. The sister ships, *Sea Goddess I* and *Sea Goddess II*, are 344 feet long, 48 feet in beam, and draw 14 feet of water. Driven by two medium-speed diesel engines of 2,400 brake horsepower each, the vessels' cruising speed is 17.5 knots. Registered in Norway, the ships have crews of 80 persons for the 116 passengers carried at capacity. The contract for building the ships was signed in March 1981 by Marine Investors and Shipowners, Inc., an American holding company, and Wärtsilä shipyard (Finland). The *Sea Goddess I* was delivered in the fall of 1984, and her sister ship was christened in Monaco in May 1985. "U. S. Firm to Offer Fly-Cruise Service," *New York Journal of Commerce*, March 14, 1981; "The Grandest Tour of All," *Los Angeles Times*, October 7, 1984, Special Section: Traveling in Style; "The Sumptuous *Sea Goddess*," *Travel and Leisure*, February 1985.
22. "Cruise Ship 'Concession Catering' Hit," *New York Journal of Commerce*, April 5, 1974, p. 24; "Campaign Under Way for Seamen's Rights," *ibid.*, January 27, 1984, p. 24-B; "Cruise Lines Respond to Institute Pressure," *ibid.*, March 1, 1984, p. 1-B.
23. A number of American-flag passenger ships operating in both the Atlantic and Pacific oceans were outfitted with slot machines in the last years before World War II began. In 1961, the *Queen Mary* had slot machines aboard for several transatlantic round voyages. The Cunard Line, in 1966, "in response to many requests," contracted with an English firm to take over and operate the gambling concession on the *Franconia*. Games included roulette, blackjack, and *vingt-et-un*. *New York Times*, February 13, 1966, Part V, p. 20.
24. Joseph Wechsberg, "The Logistics of Glut," *Esquire*, December 1976, p. 191.

25. "Cruise Line Competition Heating Up," *New York Journal of Commerce*, February 21, 1985, p. 7-A.

26. "Charm is Part of Job," *Los Angeles Times*, Part VII, p. 23, June 24, 1984.

27. For a detailed, critical, and historical survey of the development of the passenger cruise business, see John Malcolm Brinnin, *The Sway of the Grand Saloon: A Social History of the North Atlantic* (New York, Delacourt Press, 1971), pp. 486-504.

Chapter 20. Industrial and Special Carriers

1. "Economy Drive by BHP," *The Motor Ship*, July 1981, pp. 49-50.

2. E. C. B. Corlett and J. F. Leathard, "Methane Transportation by Sea," *ibid.*, April 1960, p. 18.

3. For an account of the *Methane Pioneer*, see "First Large Liquid Gas Cargo Carrier," *ibid.*, January 1959, p. 505. The ship was retired from service when the larger *Bridgestone Maru* was built. A comprehensive commentary on this latter carrier is available in "Refrigerated L. P. G. Tanker *Bridgestone Maru*," *ibid.*, February 1962, p. 529. See also the article by Charles Kelby, "Bringing Cheap Energy From the Sahara," in Liverpool, England, *Journal of Commerce Annual Review*, 1963, pp. 223-39.

4. "LNG Tankers," in United States Naval Institute *Proceedings*, April 1977, pp. 61-73; 99-101. "Kockums Re-enters the Gas Tanker League," *The Motor Ship*, January 1977, pp. 62-65. "Large LNG/LPG Tankers from HDW's Kiel Yard," *ibid.*, November 1976, pp. 97-99. "World LNG Fleet Doubles Since 1976," *New York Journal of Commerce*, May 9, 1978, p. 34. "LNG Transport Needs Examined," *ibid.*, August 15, 1977, p. 32.

5. "El Paso Paul Kayser," *The Motor Ship*, November 1975, pp. 101-2.

6. The *Duncan Bay* was built in Kure, Japan, and was 178.9 meters long, 25.6 meters wide, and had a molded depth of 12.5 meters. She was owned by Universe Tankships, Inc., and chartered by Crown Zellerbach Company. "Unique Ship that Revolutionized Wood-Pulp Trade," *Marine Engineering/Log*, September 1957, pp. 100-1.

7. The ships were owned by Rederiet Storeggen A/S (Chr. Ostberg), of Oslo, and were designed by Robert Herbert, then a member of the naval architecture office of Philip F. Spaulding & Associates, in Seattle, Washington. They were built by Kaldnes Mek. Verksted, Tønsberg, Norway. The cranes were designed and built by Munck International A/S, a Norwegian manufacturer of mechanical handling equipment. There were six holds and six hatches, each 12.3 meters long by 14.04 meters wide, except for No. 1 hatch, which was 10.4 meters wide. The 15.5-knot ships were 139.9 meters long, 19.5 meters in beam, and on a summer draft of 7.7 meters had a deadweight of 9,300 tons. *Fairplay*, January 10, 1963, pp. 57 and 77.

See Robert Henry, "The Story of Designing and Building the First Open-Hatch Ship with 'Rectangular' Holds to Carry Newsprint," in Henri Kummerman and Robert Jaquinet, eds., *Ships' Cargo—Cargo Ships* (Hounslow [England], MacGregor Publications, 1979), pp. 131-41.

8. "World's Largest Cement Carriers from AESA's Sestao Yard," *The Motor Ship*, December 1980, pp. 103-4.

9. "180-Ton Lifting Gear on the *Benarty*," *ibid.*, March 1963, pp. 568-69. The ship was 154.8 meters long, 20.1 in beam, and had a deadweight of 12,420 tons on a 9.8-meter draft. She was powered by a 9,000-horsepower diesel engine and had a service speed of 17.5 knots.

H. C. Helms, "Heavy Lifts," *Ships' Cargo—Cargo Ships*, pp. 181-91.

"Heavy-Lift Ship *John Henry* to call at New York Port," *New York Journal of Commerce*, March 19, 1979, p. 9.

10. "'Madame Butterfly': A Large Pure Car/Truck Carrier for Wallenius from Kockums," *The Motor Ship*, September 1981, pp. 23-27.

The specifications of this ship were: Length overall, 198.12 meters; beam, 32.25 meters; depth of hull to upper (14th) deck, 31.20 meters; design draft, 9.50 meters; deadweight at design draft, 17,558 metric tons. A diesel engine of 18,400 brake horsepower was coupled to a single propeller, and gave the ship a speed of 20.3 knots on a fuel consumption of 65 metric tons per day. The ship could carry 6,120 passenger cars with gasoline tanks filled; the cargo was driven on and driven off. Unloading time on an early voyage was sixteen hours.

See also, in this same issue, the analysis of design and operating problems envisioned and overcome by the architects, builders, and the Japanese ship-classification society: "Wheels Around the World," pp. 28-29.

Chapter 21. Tanker Management

1. The first bulk oil-carriers were the products of the remarkable Norwegian team of Shipowner Gustave Conrad Hansen and Shipmaster Even Tollefsen, who converted the sailing ships *Jan Mayn* (258 registered tons), *Stadt* (377 registered tons), and *Lindesnaes* (674 registered tons) to carry bulk oil. The work was done at Tønsberg in 1877-1878. See Ragnar Schjøttelvig, "Norwegian Sailing Tankers," in United States Naval Institute *Proceedings*, February 1958, pp. 106-8. This brief article includes a reproduction of a picture of the *Lindesnaes* and the line drawings of her hull as converted.

2. "Of Shoes and Ships," *Fairplay*, March 19, 1970, p. 14. This sketch features a photograph of the *Glückauf*.

3. The motor-tanker *C. O. Stillman* was launched for the Panama Transport Company at the Bremer-Vulkan yard, Bremen, in 1928. She was 178 meters long overall, 22.86 meters in beam, and had a draft of 10.35 meters. Her deadweight was 24,200 tons, and her liquid capacity was about 200,000 barrels. The French tanker *Scheherazade* was contemporaneous with the *C. O. Stillman*; she was 167 meters long and her beam was 21.95 meters. She is reported to have had a capacity of 200,000 barrels.

4. The T-2 tanker was 159.56 meters long, with a deadweight of 16,750 tons on a draft of 9.16 meters. The barrel capacity was about 140,000. Sustained sea speed of 14.5 knots was obtained on a consumption of 285 barrels of fuel oil per day. Because of the shortage of gear-making facilities during World War II, all these ships were powered with turbo-electric propulsion units. Despite some complaints about the lack of economy of this type of power plant, the T-2 proved to be a valuable addition to the peacetime merchant marine. As late as 1981, a

number of these ships, many of them reconstructed and enlarged, was still in service.

5. Between 1953 and 1962, the average deadweight of the world tanker fleet increased from 13,800 tons to about 23,000 tons. Speed averaged 15.2 knots (*New York Times*, June 15, 1963). By November 30, 1970, the deadweight of the average ship in the world tanker fleet had risen to 47,100 tons. On that date, 179 tankers of 125,000 tons deadweight or larger were in existence (London, H. P. Drewry, Ltd., *Shipping Statistics and Economics*, December 1970), p. 27. As of January 1, 1983, the fleet had grown to 856 tankers of 125,000 tons deadweight, or larger. As of that same date, 60 ultralarge crude carriers were part of the world fleet. They averaged six years of age, and were capable of speeds between 14 and 17 knots (United States Department of Transportation, Maritime Administration, *A Statistical Analysis of the World's Merchant Fleets, January 1, 1983*, Washington, D.C., Government Printing Office, 1984), p. 192. (Hereafter cited as *Statistical Analysis*.)

6. United States Department of Transportation, Maritime Administration, *Tankers in the World Fleet as of January 1, 1984*, p. viii.

7. "Optimum Speed for a Mammoth," *Fairplay*, February 26, 1970, p. 8.

8. Sun Shipbuilding & Drydock Co., *Analysis of World Tank Ship Fleet, December 31, 1976* (Chester, Pennsylvania, 1978), p. 20.

9. "BP Tanker Fleet May be Cut Further," *New York Journal of Commerce*, October 25, 1983, p. 24-B.

10. "Underwater Hull Cleaning Jobs Make Millionaire of Briton," *ibid.*, February 26, 1981, p. 3-B.

11. Frank C. Seitz, Jr., "Shipboard Manning Reduction—How Few Will Do?" United States Naval Institute *Proceedings*, October 1981, pp. 50-54.

12. A survey by *The Motor Ship* in 1981 reported that graving dock facilities to accommodate ships of 300,000 tons deadweight or larger existed only in these countries: Japan, 8 docks, the largest accepted ships of 500,000 tons; West Germany, 4 docks, the largest handled ships of 700,000 tons; France, 4 docks, the largest with the capability of docking a ship up to 1,000,000 tons deadweight; and Portugal, with 3 docks, the biggest of which also could take a ship of 1,000,000 tons deadweight. Each of the following nations had one graving dock with the capability to handle a ship of not over 300,000 tons deadweight: Brazil, Canada, Greece, Italy, Korea, Malta, Philippines, Sicily, and Singapore. "World Ship Repair Facilities," a special survey of *The Motor Ship*, September 1981.

13. The steam-turbine-driven *San Juan Pioneer* was built in Japan in 1962, and when she entered service, was the largest combination ore-oil carrier in the world. She was 253.3 meters long, 32 meters wide, and 19.8 meters deep. Her 22,500 shaft horsepower engines gave her a loaded speed of 16.4 knots. To permit rapid conversion from ore to oil cargoes, three holds were fitted with especially designed oil-tight hatches (*Fairplay*, October 25, 1962).

Particular mention of the *Sinclair Petrolore* is appropriate; she was a ship-owner's unique attempt to eliminate ballast voyages. This combination ore-oil carrier had 65 cargo tanks with a capacity of 389,800 barrels of crude oil and 26 ore compartments with a total capacity of 31,907 cubic meters. A huge self-unloading mechanism, extremely visible because of the big A-frame and long boom which

were placed on and above the weather deck, was intended to make the vessel independent of shore-discharging facilities. She was 240.48 meters in length, had a beam of 32 meters, and her hold was 16.45 meters deep. On a draft of 12.34 meters, the ship's deadweight was 56,089 tons. On her builder's trials, her 12,500 shaft horsepower turbines gave her a speed of 15 knots. She was lost off the Brazilian coast in 1961 after an explosion and fire. No reports were published that the ship ever carried a cargo of ore or used her unloading machinery.

Combination ships historically have found employment in the petroleum trade for about 85 percent of each year. To derive maximum advantage from the varied capabilities of this type vessel, repetitious triangular voyages should be scheduled. Very few sea routes, however, offer ore or bulk cargoes in large enough lots consistently to fill the existing combination ships. *Shipping Statistics and Economics*, March 1973, p. 26.

14. *Statistical Analysis*, pp. 164-90.

15. The largest ships ever built in American shipyards were the *UST Atlantic* and her identical sister, the *UST Pacific*. These tankers, each of 390,800 tons deadweight, were completed in 1979 by the Newport News Shipbuilding & Drydock Company. They were 361.8 meters long, 69.49 meters in beam, and the hull was 28.96 meters deep. When fully loaded, the ship draft was 22.6 meters (74 feet).

The *Seawise Giant*, registered in Liberia but owned by Hong Kong interests, is the world's largest ship, with a deadweight of 564,763 tons. She was built in Japan in 1979, and entered commercial service in December 1980. She was 458 meters (1,502.24 feet) long, and almost 69 meters (226 feet) in beam. Her draft of 24.61 meters (80.7 feet) permitted her to enter some harbors which the other twelve ultra-large oil carriers but with deeper draft (up to 94 feet) were not able to use. ("Seawise Giant Opens 'New Age' in Seafaring," *New York Journal of Commerce*, December 26, 1980, p. 32.)

16. "Prospect of Tanker Shortage," *Fairplay*, September 17, 1970, p. 8.

17. "Large Oil Tankers Seen 'Indispensable Link, Cheap, Safe,'" *New York Journal of Commerce*, January 27, 1977, p. 32.

18. *Fairplay*, January 26, 1967, p. 45.

19. "Today's Parcel Chemical Tankers: A 25-Year Development," *Stolten*, 9, no. 3 (December 1984), pp 30-33; "Stolt-Nielsen: Four Generations," *Stolten*, 10, no. 1 (April 1985), pp. 1-7.

20. Edward Crowley, "The Carriage of Chemicals in Bulk," *Fairplay*, July 30, 1970, pp. 37-39.

21. "In Search of the Universal Cargo Tank Coating," *Stolten*, 10, no. 1 (April 1985), pp. 18-19.

22. "Stolt Sapphire Class: A Technical Overview," *ibid.*, pp. 21-25.

Chapter 22. The American Shipping Subsidy System

1. "Swedish Shipowners Seek Tax Exemption for Seamen," *New York Journal of Commerce*, October 29, 1981, p. 24-B.

2. United States Department of Commerce, Maritime Administration, *Maritime Subsidies, 1978* (Washington, D. C., Government Printing Office, 1978), pp. v, vi, and vii.

3. Lane C. Kendall, "The Purpose of Subsidy," United States Naval Institute *Proceedings*, October 1981, pp. 29-31.

4. For a brief but comprehensive analysis of cargo preference legislation, see Clinton H. Whitehurst, Jr., *The U. S. Merchant Marine: In Search of an Enduring Maritime Policy* (Annapolis, Maryland, Naval Institute Press, 1983), pp. 23-24.

5. "Title XI Default Is Biggest Ever," *New York Journal of Commerce*, May 3, 1985, p. 1-A. Guarantees have been made on mortgages and bonds on approximately 6,000 vessels, with an aggregate value of \$7.3 billion. Ships, barges, tugs, offshore supply craft, floating oil-well rigs, as well as river boats have been the beneficiaries of this program. The government charges a fee for the guarantee, and in May 1985 its resources for paying these guarantees amounted to \$142 million.

6. The pressures brought to bear on shipowners in international trade were demonstrated dramatically by the Swedish Shipowners Association, which compiled the following estimates. A Swedish ship of 20,000 tons deadweight had running expenses of about \$1,440,000 per year, excluding capital costs and fuel. A similar vessel under British registry cost about \$1,000,000, whereas a bulk carrier of the same tonnage flying the Liberian flag and paying wages below those set by the International Transport Workers Union would have had expenses of only \$630,000.

A further example was cited of a Swedish-flag bulk carrier that cost \$530,000 a year more to operate than an identical ship owned by the same Swedish firm but registered in Hong Kong and manned by a non-Swedish crew. "Swedish Shipowners Seek Tax Exemption for Seamen," *New York Journal of Commerce*, October 29, 1981, p. 24-B.

7. United States Department of Commerce, Maritime Administration, *Manual of General Procedures for Determining Operating-Differential Subsidy Rates* (Washington, D. C., Department of Commerce, 1977), p. 20.

8. For an exhaustive analysis of the 1936 law, see Samuel A. Lawrence, *United States Merchant Shipping Policies and Politics* (Washington, D. C., Brookings Institution, 1966), especially Chapters 6, 7, and 8. An excellent compilation of data relating to the programs of government aid in 58 nations (including the United States) which have merchant fleets is found in *Maritime Subsidies*, previously mentioned. Particular details, such as the exact amount of money expended in the subsidy program, the names of the companies which have held and those which currently do hold operating-differential subsidy contracts, the numbers and types of ships under construction with subsidy assistance, and the total strength of the subsidized fleet at the end of each fiscal year, are available in the *Annual Report of the Maritime Administration* (Washington, D. C., Government Printing Office).

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